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LEVETOP

LT32U03

32Bit Micro Controller

Data Sheet

V1.1

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1 Introduction

1.1 Overview



LT32U03 is a high performance 32-bit MCU. It contains a Cortex M4 core that can run up to 150MHz operating frequency. The LT32U03 includes high capacity 508KB flash memory, 256KB SRAM and 8KB high-speed instruction data sharing Cache, and also provides a variety of standard communication interfaces, including 3 sets of SPI, 3 sets of SCI (Uart), I2C, PWM, high precision 12-bit ADC, DAC output, USB interface, and multiple interrupt inputs and GPIO interfaces.

The Cortex M4 core is highly processed with IEEE754-compatible single-precision FPUs. Coupled with fast interrupt response, the built-in sleep mode enables extremely low power consumption. The 8KB Cache module supports group command and line command, as well as write-through and write-back modes, in line with the USB 2.0 high-speed (480 Mbps) functional standard. Multiple sets of SCI and SPI interfaces, support elastic interface configuration, suitable for external Bluetooth, WiFi and other modules. The internal AES/DES algorithm module supports standard encryption and decryption algorithms, as well as the cyclic redundancy check (CRC) module, watchdog module (WDT), RTC (Real Time clock) and other complete functional configurations.

LT32U03 has excellent computing power, sufficient Flash memory and SRAM, with multiple sets of SPI Interface, multiple sets of SCI Interface (Uart), USB 2.0 transmission interface, can be applied to all kinds of electronic instruments, equipment, meters, high-end household appliances, smart electronic toys, smart door locks, all kinds of electronic controllers, and the need for expansion modules or USB transmission of various electronic products.

1.2 Internal Block Diagram

Figure 1-1 is the interna; block diagram og LT32U03.

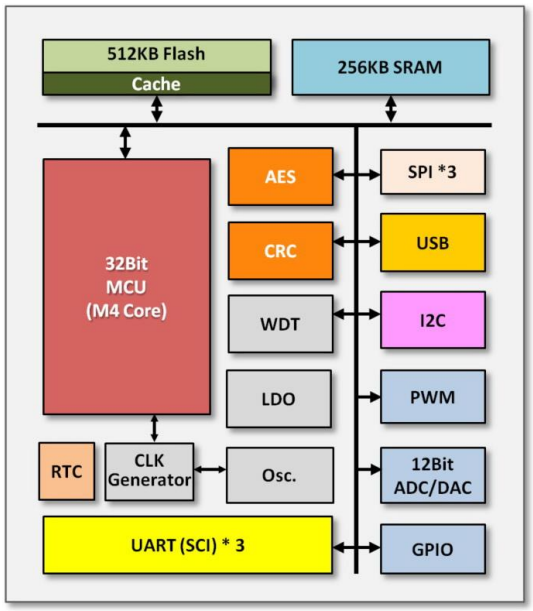


Figure 1-1: Internal Block Diagram

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1.3 Features

- The Cortex-M4 processor is a high performance 32-bit processor designed for the microcontroller market. It offers significant benefits to developers, including:
  - _ Outstanding processing performance combined with fast interrupt handling
  - _ Enhanced system debug with extensive breakpoint and trace capabilities
  - _ Efficient processor core, system and memories
  - _ Ultra-low power consumption with integrated sleep modes
  - _ Platform security robustness, with integrated memory protection unit (MPU).
  - _ IEEE754-compliant single-precision FPU
- Cache
  - _ Has two AHB bus interfaces, a master and a slave interface.
  - _ Has a 2-way set-associative organization.
  - _ Has an AHB bus interface to access its programmer's model.
- On-chip Embedded Flash (EFLASH)
  - _Memory Origanization: 128Kx38+128x38
  - _ Reads of bytes, aligned halfwords (16 bits) and aligned words (32 bits)
  - _ Automated program and erase operation
  - _ Optional interrupt on command completion
  - _Data Retention: 10 years under 85 degrees
  - _ 0.99~1.21V/1.5~1.98V dual power supplies
- CPM
  - _ Selectable system clock sources
  - _ Separate clock divider
  - _ Support for power saving mode
  - _ Module clock can be gated separately
- Programmable 32bit Interrupt timer :
  - _ 32-bit counter with modulus "initial count" register
  - _ Selectable as free running or count down
  - _ 32 selectable prescalers — 2^0 to 2^{31}
- Watchdog Timer :
  - _ 16-bit counter with modulus "initial count" register
  - _ Pause option for low-power modes
- Time Counter :
  - _ 16-bit counter with modulus "initial count" register
  - _ Pause option for low-power modes
- Reset :
  - _ Separate reset in and reset out signals
  - _ Sources of reset:

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- Power-on reset
- Software reset
- Watchdog timer
- Time Counter
- Power Attack Detect Reset (Low and High Voltage Detect Reset)

_ Status flag indicates source of last reset

● DMA Controller

_ Six independently programmable DMA controller channels
_ Data transfers in 8, 16, 32, 64bits
_ Support single transfer, Burst 4, 8, 16 transfer, and burst always under a speical case.
_ Support single cycle transfer
_ Support automatic transfer mode
_ Support LLI transfer mode
_ Follow a fixed priority rule

● EDMAC

_ Programmable transfer total number
_ Programmable read buffer address and write buffer address
_ Support read, write and write then read transfer

● CRC coprocessor

_ Support CRC32 / CRC16 / CRC8
_ Support DMAC Data from CRC

● External Interrupts Supported(EPORT) :

_ Rising/falling edge select
_ Low-level sensitive
_ Ability for software generation of external interrupt event
_ Interrupt pins configurable as general-purpose I/O

● I2C Controller

_ Supports 10 bit addressing.
_ Supports Standard Mode, Fast Mode and High-Speed Mode
_ Software option to select between High-Speed mode and Standard/Fast mode
_ Compatibility with standard and fast-mode of I2C bus version 2.1 standard.
_ Multiple-master operation.
_ Software-programmable for one of 64 different serial clock frequencies.
_ Software-selectable acknowledge bit.
_ Interrupt-driven, byte-by-byte data transfer.
_ Arbitration-lost interrupt with automatic mode switching from master to slave.
_ Transfer completion and read configure interrupt.
_ Start and stop signal generation/detection.
_ Repeated START signal generation.
_ Acknowledge bit generation/detection.
_ Bus-busy detection.
_ Option slave address receiving enable when system clock stop mode
_ SCL or SDA line gpio function supported

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- Serial Communications Interface(SCI):
  - _ Full-duplex operation
  - _ Standard mark/space non-return-to-zero (NRZ) format
  - _ The baud rate divisor is a 22-bit number consisting of 16-bit integer and 6-bit fractional part
  - _ Programmable 8-bit or 9-bit data format
  - _ Separately enabled transmitter and receiver
  - _ Separate receiver and transmitter central processor unit (CPU) interrupt requests
  - _ Programmable transmitter output polarity
  - _ Two receiver wakeup methods:
  - _ Interrupt-driven operation with eight flags
  - _ Receiver framing error detection
  - _ Hardware parity checking
  - _ 1/16 bit-time noise detection
  - _ General-purpose, I/O capability
  - _ Serial IR interface low-speed,IrDA-compatible(up to 115.2Kbit/s)
  - _ Separate 16x9 transmit (TX) and receive (RX) FIFOs to reduce CPU interrupt service loading
  - _ FIFO trigger levels of 1/8, 1/4, 1/2, 3/4, and 7/8
  - _ Efficient transfers using Direct Memory Access Controller (DMA)
  - _ Automatic hardware flow control supported
- Synchronous Serial Interface (SSI)
  - _ Serial-master operation.
  - _ DMA controller interface – Enables the SSI to interface to a DMA controller over the bus using handshaking interface for transfer requests.
  - _ Clock stretching support in enhanced SPI transfers.
  - _ Data item size (4 to 32 bits) – Item size of each data transfer under control of the programmer.
  - _ FIFO depth – Configurable depth of the transmit and receive FIFO buffers from 2 to 256 words deep. The FIFO width is fixed at 32 bits.
  - _ Enhanced SPI support.
  - _ Execute in Place (XIP) mode support.
- Serial Peripheral Interfaces (SPI) :
  - _ Master mode and slave mode configurable
  - _ Slave select output
  - _ Mode fault error flag with CPU interrupt capability
  - _ Separate transmit and receive FIFOs
  - _ Serial clock with programmable polarity and phase
  - _ Control of SPI operation during doze mode
- USB
  - _ Performs all transaction scheduling in hardware
  - _ Operates either as a function controller for a USB peripheral or as the host/peripheral in point-to-point communications with another USB function
  - _ Synchronous RAM interface for FIFOs
  - _ Certified compliant with the USB 2.0 standard for high-speed (480 Mbps) functions and with the On-The-Go supplement
  - _ Supports point-to-point communications with one high-, full- or low-speed device

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_ Supports Session Request Protocol (SRP) and Host Negotiation Protocol (HNP)
_ Supports Suspend and Resume
_ Configurable for up to 15 additional Transmit endpoints and up to 15 additional Receive endpoints
_ Configurable FIFOs, including the option of dynamic FIFO sizing
_ Support for DMA access to FIFOs
_ Soft connect/disconnect option
_ Support No VBUS Select when USB as peripheral role

- PWM
  _ Programmable period
  _ Programmable duty cycle
  _ Two Dead-Zone generator
  _ Capture function
  _ Pins can be configured as general-purpose I/O
- ADC
  _ 12-bit, 10-bit, 8-bit or 6-bit configurable resolution
  _ ADC conversion time: 1.0 μs for 12-bit resolution (1 MHz), 0.88 μs conversion time for 10 bit resolution, faster conversion times can be obtained by lowering resolution.
  _ Programmable sampling time
  _ DMA support
- DAC
  _ Left or right data alignment
  _ DMA capability
  _ External triggers for conversion
  _ Programmable internal buffer
  _ Input voltage reference, Vrefh
  _ FIFO Based operation
- AES module
  _ Support AES encryption/decryption algorithm
  _ Support AES algorithm with 128/192/256 bits key
  _ Support Electronic Code Book (ECB) mode operation and CTR(counter) mode operation
- DES coprocessor
  _ Support DES and Triple-DES encryption and decryption algorithm
  _ Support DES algorithm with 64(56) bits key
  _ Support Triple-DES algorithm with 128(112) bits or 192(168) bits key
  _ Support ECB mode and CBC mode
  _ Support MLBBUS Interface

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2 Pin Description

2.1 Pin Assignment

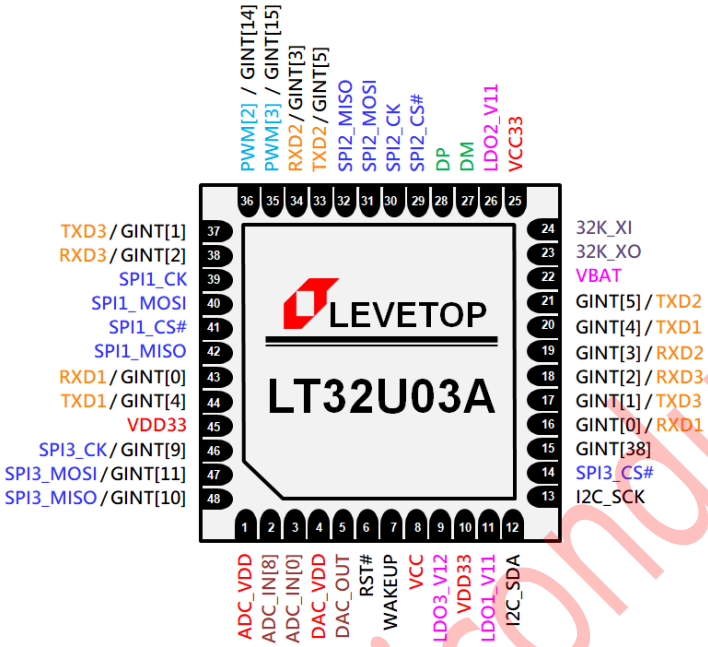


Figure 2-1: LT32U03A Pin Assignment

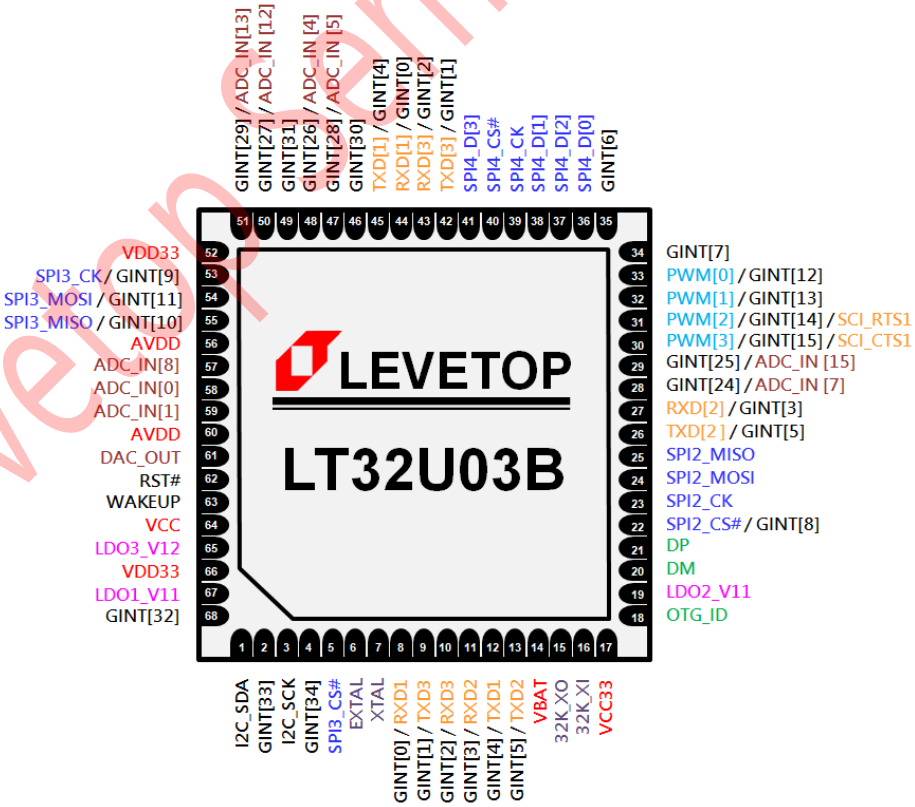


Figure 2-2: LT32U03B Pin Assignment

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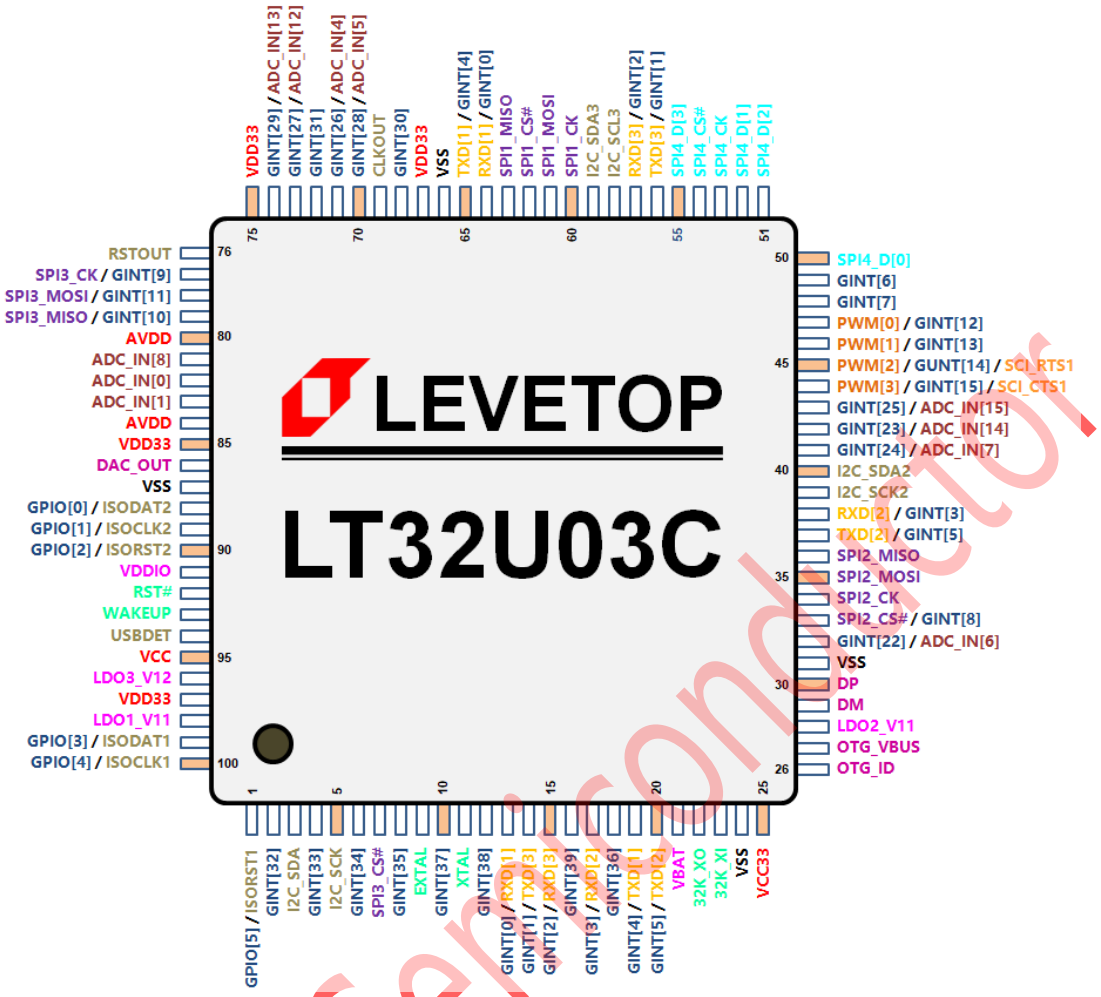


Figure 2-3: LT32U03C Pin Assignment

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2.2 Signal Description

2.2.1 SCI / Uart Signals

Table 2-1: SCI / Uart Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
16 43	8 44	13 64	RXD1	I	**SCI#1 Serial Communication Port (Uart) Receive Date Input** This is the data entry for the SCI#1 serial communication interface. You can optionally enter one of the Pin pins as RXD1.
20 44	12 45	19 65	TXD1	O	**SCI#1 Serial Communication Port (Uart) Transmittion Date Output** This is the data output of the SCI#1 serial communication interface. You can optionally enter one of the Pin as TXD1.
19 34	11 27	17 38	RXD2	I	**SCI#2 Serial Communication Port (Uart) Receive Date Input** This is the data entry for the SCI#2 serial communication interface. You can optionally enter one of the Pin pins as RXD2.
21 33	13 26	20 37	TXD2	O	**SCI#2 Serial Communication Port (Uart) Transmittion Date Output** This is the data output of the SCI#2 serial communication interface. You can optionally enter one of the Pin as TXD2.
18 38	10 43	15 57	RXD3	I	**SCI#3 Serial Communication Port (Uart) Receive Date Input** This is the data entry for the SCI#3 serial communication interface. You can optionally enter one of the Pin pins as RXD3.
17 37	9 42	14 56	TXD3	O	**SCI#3 Serial Communication Port (Uart) Transmittion Date Output** This is the data output of the SCI#3 serial communication interface. You can optionally enter one of the Pin as TXD3.
36	31	45	SCI_RTS1	O	**SCI#1 Request To Send**
35	30	44	SCI_CTS1	O	**SCI#1 Clear To Send**

2.2.2 I2C Signals

Table 2-2: I2C Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
13	3	5	I2C_SCK	O	**I2C#1 Clock Signal**
12	1	3	I2C_SDA	IO	**I2C#1 Data Signal**
--	--	39	I2C_SCK2	O	**I2C#2 Clock Signal**
--	--	40	I2C_SDA2	IO	**I2C#2 Data Signal**

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2.2.3 SPI Signals

Table 2-3: SPI Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
41	--	62	SPI1_CS#	O	**SPI #1 Chip Select** This is Chip Select output of SPI#1.
42	--	63	SPI1_MISO	I	**SPI #1 Data Input** This signal is the read data input for SPI#1.
40	--	61	SPI1_MOSI	O	**SPI #1 Data Output** This signal is the output data for SPI#1.
39	--	60	SPI1_CK	O	**SPI #1 Clock Output** This is SPI Clock output for SPI#1.
29	22	33	SPI2_CS#	O	**SPI #2 Chip Select** This is Chip Select output of SPI#2.
32	25	36	SPI2_MISO	I	**SPI #2 Data Input** This signal is the read data input for SPI#2.
31	24	35	SPI2_MOSI	O	**SPI #2 Data Output** This signal is the output data for SPI#2.
30	23	34	SPI2_CK	O	**SPI #2 Clock Output** This is SPI Clock output for SPI#2.
14	5	7	SPI3_CS#	O	**SPI #3 Chip Select** This is Chip Select output of SPI#3.
48	55	79	SPI3_MISO	I	**SPI #3 Data Input** This signal is the read data input for SPI#3.
47	54	78	SPI3_MOSI	O	**SPI #3 Data Output** This signal is the output data for SPI#3.
46	53	77	SPI3_CK	O	**SPI #3 Clock Output** This is SPI Clock output for SPI#3.
--	39	53	SPI4_CK	O	**SPI #4 QSPI Clock** This is QSPI Clock output for external QSPI device.
--	36	50	SPI4_D[0]	IO	**SPI #4 QSPI Data[0]** This is QSPI Data[0] for external QSPI device.
--	38	52	SPI4_D[1]	IO	**SPI #4 QSPI Data[1]** This is QSPI Data[1] for external QSPI device.
--	37	51	SPI4_D[2]	IO	**SPI #4 QSPI Data[2]** This is QSPI Data[2] for external QSPI device.
--	41	55	SPI4_D[3]	IO	**SPI #4 QSPI Data[3]** This is QSPI Data[3] for external QSPI device.
--	40	54	SPI3_CS#	O	**SPI #4 QSPI Chip Select** This signal is the chip select for external QSPI device.

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2.2.4 GPIO/INT Signals

Table 2-4: GPIO/INT Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
16 43	8 44	13 64	GINT[0]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[0].
17 37	9 42	14 56	GINT[1]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[1].
18 38	10 43	15 57	GINT[2]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[2].
19 34	11 27	17 38	GINT[3]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[3].
20 44	12 45	19 65	GINT[4]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[4].
21 33	13 26	20 37	GINT[5]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port. One of the Pins can be selected as GINT[5].
--	45	49	GINT[6]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
29	22	33	GINT[8]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
46	53	77	GINT[9]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
48	55	79	GINT[10]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
47	54	78	GINT[11]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	33	47	GINT[12]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	32	46	GINT[13]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.

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Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
36	31	45	GINT[14]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
35	30	44	GINT[15]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	--	32	GINT[22]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	28	41	GINT[24]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	29	43	GINT[25]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	48	71	GINT[26]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	50	73	GINT[27]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	47	70	GINT[28]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	51	74	GINT[29]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input, a General Purpose I/O port, or an ADC analog signal input.
--	46	68	GINT[30]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	49	72	GINT[31]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	68	2	GINT[32]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	2	4	GINT[33]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	4	6	GINT[34]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.

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Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
--	--	8	GINT[35]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	--	18	GINT[36]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	--	10	GINT[37]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
15	--	12	GINT[38]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	--	16	GINT[39]	IO	**Interrupt / GPIO** This signal can be used as an interrupt input or as a General Purpose I/O port.
--	--	90, 89, 88	GPIO[2:0]	IO	**GPIO** This signal can be used as a General Purpose I/O port.
--	--	1, 100, 99	GPIO[5:3]	IO	**GPIO** This signal can be used as a General Purpose I/O port.

2.2.5 PWM Signals

Table 2-5: PWM Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Signal Description
--	33	47	PWM[0]	IO	**PWM#0 Output**
--	32	46	PWM[1]	IO	**PWM#1 Output**
36	31	45	PWM[2]	IO	**PWM#2 Output**
35	30	44	PWM[3]	IO	**PWM#3 Output**

2.2.6 Alanog Signals

Table 2-6: Analog Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
3	58	82	ADC_IN[0]	I	**ADC Analog Input**
2	57	81	ADC_IN[8]	I	**ADC Analog Input**
--	59	83	ADC_IN[1]	I	**ADC Analog Input**
--	48	71	ADC_IN[4]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	50	73	ADC_IN[12]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.

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Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
--	47	70	ADC_IN[5]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	51	74	ADC_IN[13]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	28	41	ADC_IN[7]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	29	43	ADC_IN[15]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	--	42	ADC_IN[14]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
--	--	32	ADC_IN[6]	IO	**ADC Analog Input** This is an ADC analog signal input, and can also be used as a GPIO or interrupt input signal.
5	61	86	DAC_OUT	O	**DAC Analog Output** This is the DAC analog signal output and can be used as an audio output or other control signal.

2.2.7 USB Signals

Table 2-7: USB Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
28	21	30	DP	IO	**USB Data (Positive)**
27	20	29	DM	IO	**USB Data (Negative)**
--	18	26	OTG_ID	I	**USB Detect** This ID signal is used to indicate the ID status of the Mini USB plug, thus determining the master or slave device.
--	--	27	OTG_VBUS	I	**USB VBUS Detect** It connect to VDD as Host (master), and connect GND as slave.

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2.2.8 Miscellaneous

Table 2-8: Miscellaneous Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
7	63	93	WAKEUP	I	**Wake up Input**
6	62	92	RST#	I	**Reset Input** When RST# = 0, a reset action is generated on the internal MCU. With the exception of a few registers that can only be reset by POR, most registers controlled by the MCU will revert to their default values.
--	--	76	RSTOUT	O	**Reset Output** This is a reset signal output and can be used as an external component for synchronous reset.
--	--	94	USBDET	I	**USBDET Wakeup Input**
--	--	1 90	ISORST1 ISORST2	I	**Smart Card Reset Input**
--	--	99 88	ISODAT1 ISODAT2	IO	**Smart Card Data Signal**
--	--	100 89	ISOCLK1 ISOCLK2	I	**Smart Card Clock Input**

2.2.9 Crytal and Power Signals

Table 2-9: Crytal and Power Signals

Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
24	16	23	32K_XI	I	**RTC 32768Hz X'tal Input**
23	15	22	32K_XO	O	**RTC 32768Hz X'tal Output**
25	17	25	VCC33	PWR	**3.3V Power Input for USB**
8	64	95	VCC	PWR	**3.3V Power Input for System**
1, 4	56, 60	25	AVDD	PWR	**3.3V ADC/DAC Power Input**
22	14	21	VBAT	PWR	**2.7V~3.6V Battery Power for RTC**
10, 45	52, 65	67, 75, 85, 97	VDD33	PWR	**3.3V Power Input** This pin must connect an 1uF and a 0.1uF filter capacitor to ground.
11	67	98	LDO1_V11	PWR	**1.1V LDO Output for Internal Flash** This pin must connect an 1uF and a 0.1uF filter capacitor to ground.
26	19	28	LDO2_V11	PWR	**1.1V LDO Output for Internal USB PHY** This pin must connect an 1uF and a 0.1uF filter capacitor to ground.
9	65	96	LDO3_V12	PWR	**1.2V LDO Output for Core Logic** This pin must connect an 1uF and a 0.1uF filter capacitor to ground.

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Pin# (LT32U03A)	Pin# (LT32U03B)	Pin# (LT32U03C)	Pin Name	I/O	Description
--	--	92	VDDIO	PWR	**IO Power Output** This pin must connect an 1uF and a 0.1uF filter capacitor to ground.
--	--	24, 31, 66, 87	VSS	PWR	**GND Ground**
--	--	--	Thermal Pad	-	**GND Thermal Pad** The back thermal pad of the LT32U03A and LT32U03B. It must connect to ground directly (VSS/GND).

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2.3 Pin Attribute and Alternate Function

Table 2-10: Annotations of Pin Attribute

Item	Abbreviation	Definition
Pin Type	S	Power Pin
	I/O	Input/Output Pin
	I	Input Pin
	O	Output Pin
	ANA	Analog Pin
Output State	ST	Standard CMOS
	OD	Open-Drain Pin
Default State	I	Input
	O	Output
	PU	Pull-Up
	PD	Pluu-Down
	HIZ	High Impendance
Alternate/ Default Function	GPIO	Set via Peripheral Module Registers
	EPORT	Set via EPORT module Registers, Support Interrupt Input

Table 2-11: LT32U03A Pin Attribute and Alternate Function Table

Pin Name	Alternate	Type	Driving	Description	Default	Pin # (QFN48)
ADC_VDD	-	S	-	ADC Power	I	1
ADC_IN[8]	-	ANA	-	ADC Input CH[8]	I	2
ADC_IN[0]	-	ANA	-	ADC Input CH[0]	I	3
DAC_VDD	-	S	-	DAC Reference Vlotage	I	4
DAC_OUT	-	ANA	-	DAC Output	O	5
RST#	-	I	-	Reset Input	I PU	6
WAKEUP	-	I	-	Wake up Input	I PD	7
VCC	-	S	-	3.3V Powe	I	8
LDO3_V12	-	S	-	V1.2 LDO O/P	O	9
VDD33	-	S	-	3.3V Power	I	10
LDO1_V11	-	S	-	V1.1 LDO O/P	O	11
I2C_SDA		I/O	ST/OD	I2C Data	I PU	12
I2C_SCK		I/O	ST/OD	I2C Clock	I PU	13
SPI3_CS#		I/O	ST/OD	SPI#3 Chip Select		14
GINT[38]		I/O	ST	INT / GPIO	I PU	15
GINT[0]	RXD1(swap)	IO	ST/OD	INT / GPIO	I PU	16
GINT[1]	TXD3(swap)	IO	ST/OD	INT / GPIO	I PU	17
GINT[2]	RXD3(swap)	IO	ST/OD	INT / GPIO	I PU	18
GINT[3]	RXD2(swap)	IO	ST/OD	INT / GPIO	I PU	19
GINT[4]	TXD1(swap)	IO	ST/OD	INT / GPIO	I PU	20
GINT[5]	TXD2(swap)	IO	ST/OD	INT / GPIO	I PU	21

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Pin Name	Alternate	Type	Driving	Description	Default	Pin # (QFN48)
VBAT	-	S	-	RTC Power	I	22
32K_XO	-	ANA	-	32K OSC Output	O	23
32K_XI	-	ANA	-	32K OSC Input	I	24
VCC33	-	S	-	3.3V Power	I	25
LDO2_V11	-	S	-	V1.1 LDO O/P	O	26
DM	-	ANA	-	USB D- Data		27
DP	-	ANA	-	USB D+ Data		28
SPI2_CS#	GINT[8]	I/O	ST/OD	SPI#2 Chip Select	I PU	29
SPI2_CK	-	I/O	ST/OD	SPI#2 Clock	I PU	30
SPI2_MOSI	-	I/O	ST/OD	SPI#2 Master Out, Slave In	I PU	31
SPI2_MISO	-	I/O	ST/OD	SPI#2 Master In, Slave Out	I PU	32
TXD2	GINT[5] (swap)	I/O	ST/OD	SCI#2 TX Data	I PU	33
RXD2	GINT[3] (swap)	I/O	ST/OD	SCI#2 RX Data	I PU	34
PWM[3]	GINT[15] SCI_CTS1	I/O	ST	PWM Output	I PU	35
PWM[2]	GINT[14] SCI_RTS1	I/O	ST	PWM Output	I PU	36
TXD3	GINT[1] (swap)	I/O	ST/OD	SCI#3 TX Data	I PU	37
RXD3	GINT[2] (swap)	I/O	ST/OD	SCI#3 RX Data	I PU	38
SPI1_CK		I/O	ST/OD	SPI#1 Clock	I PU	39
SPI1_MOSI		I/O	ST/OD	SPI#1 Master Out, Slave In	I PU	40
SPI1_CS#		I/O	ST/OD	SPI#1 Chip Select	I PU	41
SPI1_MISO		I/O	ST/OD	SPI#1 Master In, Slave Out	I PU	42
RXD1	GINT[0] (swap)	I/O	ST/OD	SCI#1 RX Data	I PU	43
TXD1	GINT[4] (swap)	I/O	ST/OD	SCI#1 TX Data	I PU	44
VDD33	-	S	-	3.3V Power	O	45
SPI3_CK	GINT[9]	I/O	ST/OD	SPI#3 Clock	I PU	46
SPI3_MOSI	GINT[11]	I/O	ST/OD	SPI#3 Master Out, Slave In	I PU	47
SPI3_MISO	GINT[10]	I/O	ST/OD	SPI#3 Master In, Slave Out	I PU	48

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Table 2-12: LT32U03B Pin Attribute and Alternate Function Table

Pin Name	Alternate	Type	Driving	Description	Default	Pin # (QFN68)
I2C_SDA	-	IO	ST/OD	I2C Data	I PU	1
GINT[33]	-	IO	ST/OD	INT / GPIO	I PU	2
I2C_SCK	-	IO	ST/OD	I2C Clock	I PU	3
GINT[34]	-	IO	ST	INT / GPIO	I PU	4
SPI3_SCS#	-	IO	ST/OD	SPI3 Chip Select	-	5
EXTAL	-	A	-	USB Clock(12MHz)	O	6
XTAL	-	A	-	USB Clock	I	7
GINT[0]	RXD1(swap)	IO	ST/OD	INT / GPIO	I PU	8
GINT[1]	TXD3(swap)	IO	ST/OD	INT / GPIO	I PU	9
GINT[2]	RXD3(swap)	IO	ST/OD	INT / GPIO	I PU	10
GINT[3]	RXD2(swap)	IO	ST/OD	INT / GPIO	I PU	11
GINT[4]	TXD1(swap)	IO	ST/OD	INT / GPIO	I PU	12
GINT[5]	TXD2(swap)	IO	ST/OD	INT / GPIO	I PU	13
VBAT	-	S	-	RTC Power	I	14
32K_XO	-	A	-	32K OSC Output	O	15
32K_XI	-	A	-	32K OSC Input	I	16
VCC33	-	S	-	3.3V Power	I	17
OTG_ID	-	I	-	USB Detect	I	18
LDO2_V11	-	S	-	V1.1 LDO O/P	O	19
DM	-	A	-	USB D- Data	-	20
DP	-	A	-	USB D+ Data	-	21
SPI2_CS#	GINT[8]	I/O		SPI2 Chip Select	I PU	22
SPI2_CK	-	I/O	ST/OD	SPI2 Clock	I PU	23
SPI2__MOSI	-	I/O	ST/OD	SPI2 Master Out, Slave In	I PU	24
SPI2__MISO	-	I/O	ST/OD	SPI2 Master In, Slave Out	I PU	25
TXD[2]	GINT[5]	I/O	ST/OD	UART2_TX, or as INT / GPIO	I PU	26
RXD[2]	GINT[3]	I/O	ST/OD	UART2_RX, or as INT / GPIO	I PU	27
GINT[24]	ADC_IN[7]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	28
GINT[25]	ADC_IN[15]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	29
PWM[3]	GINT[15]	I/O	ST	PWM O/P, or INT / GPIO	I PU	30
PWM[2]	GINT[14]	I/O	ST	PWM O/P, or INT / GPIO	I PU	31
PWM[1]	GINT[13]	I/O	ST	PWM O/P, or INT / GPIO	I PU	32
PWM0	GINT[12]	I/O	ST	PWM O/P, or INT / GPIO	I PU	33
GINT[7]	-	IO	ST/OD	INT / GPIO	I PU	34
GINT[6]	-	IO	ST/OD	INT / GPIO	I PU	35
SPI4_D[0]	GINT[18]	I/O	ST/OD	QSPI(SPI4) Data[0]	I PU	36
SPI4_D[2]	GINT[16]	I/O	ST/OD	QSPI(SPI4) Data[2]	I PU	37
SPI4_D[1]	GINT[19]	I/O	ST/OD	QSPI(SPI4) Data[1]	I PU	38

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Pin Name	Alternate	Type	Driving	Description	Default	Pin # (QFN68)
SPI4_CK	GINT[20]	O	ST/OD	QSPI(SPI4) Clock	I PU	39
SPI4_CS#	GINT[21]	O	ST/OD	QSPI(SPI4) Chip Select	I PU	40
SPI4_D[3]	GINT[17]	I/O	ST/OD	QSPI(SPI4) Data[3]	I PU	41
TXD[3]	GINT[1]	I/O	ST/OD	UART3 TX Data, or as INT / GPIO	I PU	42
RXD[3]	GINT[2]	I/O	ST/OD	UART3 RX Data, or as INT / GPIO	I PU	43
RXD[1]	GINT[0]	I/O	ST/OD	UART1 TX Data, or as INT / GPIO	I PU	44
TXD[1]	GINT[4]	I/O	ST/OD	UART1 RX Data, or as INT / GPIO	I PU	45
GINT[30]	-	IO	ST/OD	INT / GPIO	I PU	46
GINT[28]	ADC_IN[5]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	47
GINT[26]	ADC_IN[4]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	48
GINT[31]	-	IO	ST/OD	INT / GPIO	I PU	49
GINT[27]	ADC_IN[12]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	50
GINT[29]	ADC_IN[13]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	51
VDD33	-	S	-	3.3V Power	I	52
SPI3_CK	GINT[9]	I/O	ST/OD	SPI3 Clock, or as INT / GPIO	I PU	53
SPI3_MOSI	GINT[11]	I/O	ST/OD	SPI3 Master Out, Slave In, or as INT / GPIO	I PU	54
SPI3_MISO	GINT[10]	I/O	ST/OD	SPI3 Master In, Slave Out, or as INT / GPIO	I PU	55
AVDD	-	S	-	ADC Power	I	56
ADC_IN[8]	-	A	-	ADC Input CH[8]	I	57
ADC_IN[0]	-	A	-	ADC Input CH[0]	I	58
ADC_IN[1]	-	A	-	ADC Input CH[1]	I	59
AVDD	-	S	-	ADC/DAC Power	I	60
DAC_OUT	-	A	-	DAC Output	O	61
RST#	-	I	-	Reset Input	I PU	62
WAKEUP	-	I	-	Wake Up Input	I PD	63
VCC	-	S	-	3.3V Power	I	64
LDO3_V12	-	S	-	V1.2 LDO O/P	O	65
VDD33	-	S	-	3.3V Power	I	66
LDO1_V11	-	S	-	V1.1 LDO O/P	O	67
GINT[32]	-	IO	ST/OD	INT / GPIO	I PU	68

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Table 2-13: LT32U03C Pin Attribute and Alternate Function Table

Pin Name	Alternate	Type	Driving	Description	Default	Pin # (LQFP100)
ISORST1	GPIO[5]	IO	ST/OD	Smart Card Reset, or as GPIO	I PU	1
GINT[32]	-	IO	ST/OD	INT / GPIO	I PU	2
I2C_SDA	-	IO	ST/OD	I2C Data	I PU	3
GINT[33]	-	IO	ST/OD	INT / GPIO	I PU	4
I2C_SCK	-	IO	ST/OD	I2C Clock	I PU	5
GINT[34]	-	IO	ST	INT / GPIO	I PU	6
SPI3_SCS#	-	IO	ST/OD	SPI3 Chip Select	-	7
GINT[35]	-	IO	ST	INT / GPIO	I PU	8
EXTAL	-	A	-	USB Clock(12MHz)	O	9
GINT[37]	-	IO	ST	INT / GPIO	I PU	10
XTAL	-	A	-	USB Clock	I	11
GINT[38]	-	IO	ST	INT / GPIO	I PU	12
GINT[0]	RXD1(swap)	IO	ST/OD	INT / GPIO	I PU	13
GINT[1]	TXD3(swap)	IO	ST/OD	INT / GPIO	I PU	14
GINT[2]	RXD3(swap)	IO	ST/OD	INT / GPIO	I PU	15
GINT[39]	-	IO	ST	INT / GPIO	I PU	16
GINT[3]	RXD2(swap)	IO	ST/OD	INT / GPIO	I PU	17
GINT[36]	-	IO	ST	INT / GPIO	I PU	18
GINT[4]	TXD1(swap)	IO	ST/OD	INT / GPIO	I PU	19
GINT[5]	TXD2(swap)	IO	ST/OD	INT / GPIO	I PU	20
VBAT	-	S	-	RTC Power	I	21
32K_XO	-	A	-	32K OSC Output	O	22
32K_XI	-	A	-	32K OSC Input	I	23
VSS	-	S	-	GND	I	24
VCC33	-	S	-	3.3V Power	I	25
OTG_ID	-	I	-	USB Detect	I	26
OTG_VBUS	-	I	-	USB VBUS Detect	I	27
LDO2_V11	-	S	-	V1.1 LDO O/P	O	28
DM	-	A	-	USB D- Data	-	29
DP	-	A	-	USB D+ Data	-	30
VSS	-	S	-	GND	I	31
GINT[22]	ADC_IN[6]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	32
SPI2_CS#	GINT[8]	I/O		SPI2 Chip Select	I PU	33
SPI2_CK	-	I/O	ST/OD	SPI2 Clock	I PU	34
SPI2__MOSI	-	I/O	ST/OD	SPI2 Master Out, Slave In	I PU	35
SPI2__MISO	-	I/O	ST/OD	SPI2 Master In, Slave Out	I PU	36
TXD[2]	GINT[5]	I/O	ST/OD	UART2_TX, or as INT / GPIO	I PU	37
RXD[2]	GINT[3]	I/O	ST/OD	UART2_RX, or as INT / GPIO	I PU	38
I2C_SCK2	-	IO	ST/OD	I2C Clock	I PU	39
I2C_SDA2	-	IO	ST/OD	I2C Data	I PU	40

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Pin Name	Alternate	Type	Driving	Description	Default	Pin # (LQFP100)
GINT[24]	ADC_IN[7]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	41
GINT[23]	ADC_IN[14]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	42
GINT[25]	ADC_IN[15]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	43
PWM[3]	GINT[15]	I/O	ST	PWM O/P, or INT / GPIO	I PU	44
PWM[2]	GINT[14]	I/O	ST	PWM O/P, or INT / GPIO	I PU	45
PWM[1]	GINT[13]	I/O	ST	PWM O/P, or INT / GPIO	I PU	46
PWM0	GINT[12]	I/O	ST	PWM O/P, or INT / GPIO	I PU	47
GINT[7]	-	IO	ST/OD	INT / GPIO	I PU	48
GINT[6]	-	IO	ST/OD	INT / GPIO	I PU	49
SPI4_D[0]	GINT[18]	I/O	ST/OD	QSPI(SPI4) Data[0]	I PU	50
SPI4_D[2]	GINT[16]	I/O	ST/OD	QSPI(SPI4) Data[2]	I PU	51
SPI4_D[1]	GINT[19]	I/O	ST/OD	QSPI(SPI4) Data[1]	I PU	52
SPI4_CK	GINT[20]	O	ST/OD	QSPI(SPI4) Clock	I PU	53
SPI4_CS#	GINT[21]	O	ST/OD	QSPI(SPI4) Chip Select	I PU	54
SPI4_D[3]	GINT[17]	I/O	ST/OD	QSPI(SPI4) Data[3]	I PU	55
TXD[3]	GINT[1]	I/O	ST/OD	UART3 TX Data, or as INT / GPIO	I PU	56
RXD[3]	GINT[2]	I/O	ST/OD	UART3 RX Data, or as INT / GPIO	I PU	57
RXD[1]	GINT[0]	I/O	ST/OD	UART1 TX Data, or as INT / GPIO	I PU	64
TXD[1]	GINT[4]	I/O	ST/OD	UART1 RX Data, or as INT / GPIO	I PU	65
VSS	-	S	-	GND	I	66
VDD33	-	S	-	3.3V Power	I	67
GINT[30]	-	IO	ST/OD	INT / GPIO	I PU	68
GINT[28]	ADC_IN[5]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	70
GINT[26]	ADC_IN[4]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	71
GINT[31]	-	IO	ST/OD	INT / GPIO	I PU	72
GINT[27]	ADC_IN[12]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	73
GINT[29]	ADC_IN[13]	IO	ST/OD	INT / GPIO, or ADC I/P	I PU	74
VDD33	-	S	-	3.3V Power	I	75
RSTOUT	-	IO	ST/OD	Reset Output	O	76
SPI3_CK	GINT[9]	I/O	ST/OD	SPI3 Clock, or as INT / GPIO	I PU	77
SPI3_MOSI	GINT[11]	I/O	ST/OD	SPI3 Master Out, Slave In, or as INT / GPIO	I PU	78
SPI3_MISO	GINT[10]	I/O	ST/OD	SPI3 Master In, Slave Out, or as INT / GPIO	I PU	79
AVDD	-	S	-	ADC Power	I	80
ADC_IN[8]	-	A	-	ADC Input CH[8]	I	81
ADC_IN[0]	-	A	-	ADC Input CH[0]	I	82
ADC_IN[1]	-	A	-	ADC Input CH[1]	I	83
AVDD	-	S	-	ADC/DAC Power	I	84

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Pin Name	Alternate	Type	Driving	Description	Default	Pin # (LQFP100)
VDD33	-	S	-	3.3V Power	I	85
DAC_OUT	-	A	-	DAC Output	O	86
VSS	-	S	-	GND	I	87
ISODAT2	GPIO[0]	IO	ST/OD	Smart Card Data / GPIO	I PU	88
ISOCLK2	GPIO[1]	IO	ST/OD	Smart Card Clock / GPIO	I PU	89
ISORST2	GPIO[2]	IO	ST/OD	Smart Card Reset / GPIO	I PU	90
VDDIO	-	S	-	IO Power	O	91
RST#	-	I	-	Reset Input	I PU	92
WAKEUP	-	I	-	Wake up Iput	I PD	93
USBDET	-	I	-	USB Wake up	I PD	94
VCC	-	S	-	3.3V Power	I	95
LDO3_V12	-	S	-	V1.2 LDO O/P	O	96
VDD33	-	S	-	3.3V Power	I	97
LDO1_V11	-	S	-	V1.1 LDO O/P	O	98
ISODAT1	GPIO[3]	IO	ST/OD	Smart Card Data / GPIO	I PU	99
ISOCLK1	GPIO[4]	IO	ST/OD	Smart Card Clock / GPIO	I PU	100

NOTE: Due to the difference in the packaging, some of the pins mentioned in this specification that are register-controlled are not elicited. Therefore, it is necessary to refer to the actual pin description in use.

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3 System Memory Map

3.1 Introduction

The address map, shown in Section 3.2, includes:

- 24K Bytes of internal read only memory (ROM)
- 256K Bytes of internal static random-access memory (SRAM)
- 512K Bytes Embedded Flash
- Internal memory mapped registers

3.2 Address Map

Table 3-1: Boot Modes

Mode[1]	Mode[0]	Boot mode	Aliasing
0	0	ROM	ROM is selected as the boot space
0	1	EFLASH	EFLASH is selected as the boot space
1	0	Reserved	Reserved
1	1	SPI FLASH	SPI FLASH is selected as the boot space

Table 3-2: Memory mapping vs Boot mode/physical remap

Address	Boot/Remap in ROM	Boot/Remap in EFLASH	Boot/Remap in SPI FLASH
0xA4000000~0xA7FFFFFF	Reserved	Reserved	Reserved
0x80000000~0x87FFFFFF	Reserved	Reserved	Reserved
0x90000000~0x90FFFFFF	Reserved	Reserved	Reserved
0x94000000~0x97FFFFFF	Reserved	Reserved	Reserved
0x60000000~0x60FFFFFF	SPI FLASH	SPI FLASH	SPI FLASH
0x20030000~0x20037FFF	SRAM3	SRAM3	SRAM3
0x20020000~0x2002FFFF	SRAM2	SRAM2	SRAM2
0x20010000~0x2001FFFF	SRAM1	SRAM1	SRAM1
0x20000000~0x2000FFFF	SRAM0	SRAM0	SRAM0
0x1FFF8000~0x1FFFFFFF	TCM(SRAMD)	TCM(SRAMD)	TCM(SRAMD)
0x10000000~0x10FFFFFF	SPIM1	SPIM1	SPIM1
0x08000000~0x0807FFFF	EFLASH	EFLASH	EFLASH
0x04000000~0x04005FFF	ROM	ROM	ROM
0x00000000~0x00FFFFFF	ROM(24K) Aliased	EFLASH(512K) Aliased	SPIM1(16M) Aliased

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Table 3-3: Register Address Location Map

0x4000_0000	4Kbyte	AHB_IPS1	IO Control Module(IOCTRL)
0x4000_1000	4Kbyte		Chip configuration (CCM)
0x4000_2000	4Kbyte		Reset (RESET)
0x4000_3800	4Kbyte		EFM
0x4000_4000	4Kbyte		CPM
0x4000_5000	4Kbyte		Watchdog timer (WDT)
0x4000_6000	4Kbyte		TC
0x4000_7000	4Kbyte		PIT32_0
0x4000_8000	4Kbyte		PIT32_1
0x4000_9000	4Kbyte		USI1
0x4000_a000	4Kbyte		EDMAC1
0x4000_b000	4Kbyte		Reserved
0x4001_0000	4Kbyte		SPI1
0x4001_1000	4Kbyte		SPI2
0x4001_2000	4Kbyte		SPI3
0x4001_3000	4Kbyte		SCI1
0x4001_4000	4Kbyte		SCI2
0x4001_5000	4Kbyte		USI2
0x4001_6000	4Kbyte		Reserved
0x4001_7000	4Kbyte		I2C
0x4001_8000	4Kbyte		PWM
0x4001_9000	4Kbyte		Edge Port (EPORT)
0x4001_a000	4Kbyte		Edge Port 1(EPORT1)
0x4001_b000	4Kbyte		I2C2
0x4001_c000	4Kbyte		I2C3
0x4001_d000	4Kbyte		SCI3
0x4001_e000	4Kbyte		Reserved
0x4002_0000	4Kbyte		ADC
0x4002_1000	4Kbyte		DAC
0x4002_2000	4Kbyte		MCC
0x4002_3000	4Kbyte		TSI
0x4002_4000	4Kbyte		Edge Port 2(EPORT2)
0x4002_5000	4Kbyte		Edge Port 3(EPORT3)
0x4002_6000	4Kbyte		Edge Port 4(EPORT4)
0x4003_0000	4Kbyte		LD
0x4003_1000	4Kbyte		True Random Number Generator (TRNG)
0x4003_2000	4Kbyte		PGD
0x4003_3000	4Kbyte		SEC_DET
0x4003_4000	4Kbyte		PCI
0x4003_5000	4Kbyte		PMU_RTC
0x4003_6000	4Kbyte	AHB_IPS2	Reserved
0x4003_7000	4Kbyte		CRYPTO
0x4003_8000	4Kbyte		SHA
0x4003_9000	4Kbyte		EDMAC0
0x4003_a000	4Kbyte		Reserved

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0x4004_0000	4Kbyte	AHB1(AHB_CLB)	Data Encryption Standard Module (DES)
0x4004_1000	4Kbyte		AES
0x4004_2000	4Kbyte		SM4
0x4004_3000	4Kbyte		ZUC
0x4004_4000	4Kbyte	AHB2	CRC0
0x4004_5000	4Kbyte		CRC1
0x4004_6000	4Kbyte		DMAC1
0x4004_7000	4Kbyte		DMAC2
0x4004_8000	4Kbyte		Reserved
0x4004_9000	4Kbyte		Reserved
0x4004_a000	4Kbyte		SPIM2
0x4004_b000	4Kbyte	AHB3	Reserved
0x4004_c000	4Kbyte		USBC
0x4005_0000	4Kbyte	AHB_APB	Reserved
0x4005_1000	4Kbyte		Cache_Config
0x4005_2000	4Kbyte		Reserved
0x4005_3000	4Kbyte		Reserved
0x4005_4000	4Kbyte		Reserved
0x4005_5000	4Kbyte		Cache2_Config
0x6000_0000	16Mbyte	AHB_SPIM1	SPIM1
Reserved	4Kbyte	Reserved	Reserved
Reserved	4Kbyte	Reserved	Reserved
0xe000_0000	4Kbyte	M4	M4 SYS

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4 Chip Configuration Module (CCM)

4.1 Introduction

The chip configuration module (CCM) controls the chip's configuration and operation mode.

4.2 Features

The CCM performs these operations.

- Selects the chip operation mode
  - Normal Master mode
  - Normal Single chip mode
  - Special Test Modes for EFLASH(Stb Mode)
  - Test Mode for EFLASH direct
  - Test mode for TRNG
  - Test mode for OSC
  - Test mode for phy bist
  - Test mode for phy direct
  - Test mode for memory bist
  - Test mode for PMU
  - Test mode for PMURTC
  - Test mode for LD and PGD
- Selects boot device
- Selects bus monitor configuration

4.3 Modes of Operation

The CCM configures the chip for following modes of operation:

- Normal Master mode
- Normal Single Chip mode

4.3.1 Normal Master Mode

In normal master mode, the boot device is internal eflash.

4.3.2 Normal Single chip Mode

In normal single chip mode, the boot device is internal rom.

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4.4 Signals Descriptions

Table 4-1 provides an overview of the CCM signals. For more detailed information.

Table 4-1: Signal Properties

Name	Function	Reset State
MODE[2:0]	Modes Selects	Internal weak pulldown device
TEST	Modes Selects	Internal weak pulldown device

4.5 Memory Map and Registers

This subsection provides a description of the memory map and registers.

4.5.1 Programming Model

The CCM programming model consists of the following registers:

- CCR — chip configuration register: controls chip configuration . (see Figure 4-1 Chip Configuration Register(CCR)).
- CIR — chip identification register: contains a unique part number. (see Figure 4-4 Chip Identification Register(CIR))

Some control register bits are implemented as write-once bits. These bits are always readable, but once the bit has been written, additional writes have no effect, except during debug and test operations .

Some write-once bits and test bits can be read and written while in debug mode or test mode. When debug or test mode is exited, the chip configuration module resumes operation based on the current register values. If the first write to a write-once register bit occurs while in debug or test mode, the register bit remains writable on exit from debug or test mode.**Table 4-2** shows the accessibility of write-once bits.

Table 4-2: Write-Once Bits Read/Write Accessibility

Configuration	Read/Write Access
All configurations	Read-always
Debug operation (all modes)	Write-always
Test operation (all modes)	Write-always
Master mode	Write-once

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4.5.2 Memory Map

Table 4-3: CCM Memory Map

Address Offset	31:16	15:0	Access[1]
0x0000	CCR — chip configuration register	FDCR — FD configuration register	S
0x0004	PHYPA — PHY parameter configuration register	CIR — chip identification register	S
0x0008	CTR — chip test register	Reserved[2]	S
0x000c	Reserved		S
0x0010	PCFG12		S
0x0014	PCFG3		S
0x0018	RTCCFG12		S
0x001c	RTCCFG3		S
0x0020	RTC_SR		S
0x0028	RTC_PR		S

NOTES:

- S = supervisor-only access. User mode accesses to supervisor only address locations have no effect and result in a cycle termination transfer error.
- Writing to reserved addresses has no effect; reading returns zeros.

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4.5.3 Register Descriptions

The Registers' Read-Write Properties and Attribute are description as following table.

Table 4-4: Registers' Read-Write Properties and Attribute

Read-Write Properties and Attribute	Description
read/write (rw)	Read and Writable
read-only (ro)	Read Only
write-only (wo)	Write Only
read-clear (rc)	Read only, and the bit is automatically cleared while reading.
read/write_1_clear (r/w1c)	Read and writable, clear this bit when writing 1, and write 0 has no effect.
read/write_0_clear (r/w0c)	Read and writable, clear this bit when writing 0, and write 1 has no effect.
read/write_1_only (r/w1o)	Read and writable, only 1 can be written, and writing 0 has no effect.
read/write_0_only (r/w0o)	Read and writable, only 0 can be written, and writing 1 has no effect.

4.5.3.1 Chip Configuration Register

Addr. Offset: 0x0002~0x0003 Reset Value: 0x0808

15	14	13	12	11	10	9	8
SWAPDIS	TESTDIS	BTLDDIS	JTAGDIS	CLKMODE_PDE	MODE2	MODE1	MODE0
ro	ro	ro	ro	rw	ro	ro	ro
7	6	5	4	3	2	1	0
Reserved	PERIPH_BRIDGE_PAE	PERIPH_BRIDGE_RAE	SHINT	BME	BMD	BMT	
ro	rw	rw	rw	rw	rw	rw	

Figure 4-1: Chip Configuration Register(CCR)

CLKMODE_PDE — Clkmode pad pulldown enable

if clkmode is set, choose internal oscillator. if clkmode is clear, choose external oscillator.

0 = pulldown disable, 1 = pulldown enable

BTLDDIS — Boot Loader Mode Disable Bit

1 = Boot Loader Mode is disabled

0 = Boot Loader Mode is enabled

JTAGDIS — JTAG DISABLE STATUS

1 = JTAG is disabled, 0 = JTAG is enabled

TESTDIS — TEST MODE DISABLE STATUS

1 = TESTMODE is disabled, 0 = TESTMODE is enabled

SWAPDIS — SWAP MODE DISABLE STATUS

1 = SWAPMODE is disabled, 0 = SWAPMODE is enabled

MODE[2:0] — Chip Configuration Mode Field

This read-only field reflects the chip configuration mode as shown .Descriptions (see **4.3 Modes of**

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Operation)

MODE[2:0]	CHIP MODE
000	Normal Single Chip(Rom boot)
001	Normal Master(Flash boot)
010	Normal Single Chip Swap Mode
011	Normal Master Swap Mode
1XX	Chip Test Mode

periph_bridge_pae — early absent slot error enable

1 = enable

0 = disable

periph_bridge_rae — powered down slot error and slave transfer error enable

1 = enable

0 = disable

SHINT — Show Interrupt bit

The SHINT bit allows visibility to any active interrupt request to the processor. If the SHINT bit is set, the RSTOUT pin is the OR of the fast and normal interrupt signals.

1 = Internal requests reflected on RSTOUT pin

0 = Normal RSTOUT pin function

NOTE: The FRCRSTOUT function in the reset controller has a higher priority than the SHINT function.

BME — Bus Monitor Enable Bit

The BME bit enables the bus monitor to operate.The bus monitor can monitor the internal bus operation.Once there is on reponse in in timeout period (according to the BMT) in internal bus, the bus monitor will terminate the bus operation .

1 = Bus monitor enabled

0 = Bus monitor disabled

Table 4-2 shows the read/write accessibility of this write-once bit.

BMD — Bus Monitor Debug Mode Bit

The BMD bit controls how the bus monitor responds during debug mode.

1 = Bus monitor enabled in debug mode

0 = Bus monitor disabled in debug mode

This bit is read/write always.

BMT — Bus Monitor Timing Field

The BMT field selects the timeout period for the bus monitor.

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Table 4-5: Bus Monitor Timeout Values

BMT	Timeout Period (in System Clocks)
00	256
01	128
10	64
11	16

Table 4-2 shows the read/write accessibility of these write-once bits.

4.5.3.2 PHY parameter configuration register.

Addr. Offset: 0x0006~0x0007 Reset Value: 0x8000

15	14	13	12	11	10	9	8
USB_REG_ENDIAN	PHY_RESUME_SEL	PHY_ID_PULLUP	PHY_ID_DIR	PHY_VALID_DIR	PHY_VBUSVALID	PHY_AVALID	PHY_SESSENDVALID
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
USBPHY_REG_EN_SEL	USBPHY_12M_EN_SEL	USBPHY_DIR	USBPHY_TERMSEL	USBPHY_TERMSEL_DIR	USBPHY_SUSPENDM	USBPHY_REG_EN	USBPHY_PLL_EN
rw	rw	rw	rw	rw	rw	rw	rw

Figure 4-2: PHY parameter configuration register(PHYPA)

PHYPA[15]: USBC Register Big Endian and Little Endian Select

1 = USBC Register Select Little Endian

0 = USBC Register Select Big Endian

PHYPA[14]: USB Resume Signal Select

1 = USB Resume generate by LineState and Power State

0 = USB Resume generate by LineState

PHYPA[13]: USBPHY ID PULLUP Configure,need to configure with PHYPA[12]

1 = USBPHY ID PULLUP Enabled

0 = USBPHY ID PULLUP Disabled

PHYPA[12]: Select USBPHY ID PULLUP Controlled By PHYPA[13]

1 = Select USBPHY ID PULLUP Controlled By PHYPA[13]

0 = Select USBPHY ID PULLUP Controlled By USBC Controller

PHYPA[11]: Select USBPHY VBUSVALID(PHYPA[10]), AVALID(PHYPA[9]), SESSENDVALID(PHYPA[8]) Controlled By Register

1 = Select USBPHY VBUSVALID AVALID SESSENDVALID Controlled By Register

0 = Select USBPHY VBUSVALID AVALID SESSENDVALID Controlled By USBC Controller

PHYPA[10]: USBPHY VBUSVALID Configure, need to configure with PHYPA[11]

1 = USBPHY VBUSVALID Enabled

0 = USBPHY VBUSVALID Disabled

PHYPA[9]: USBPHY AVALID Configure, need to configure with PHYPA[11]

1 = USBPHY AVALID Enabled

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0 = USBPHY AVALID Disabled

PHYPA[8]: USBPHY SESSENDVALID Configure, need to configure with PHYPA[11]

1 = USBPHY SESSENDVALID Enabled

0 = USBPHY SESSENDVALID Disabled

PHYPA[7]: Select USBPHY REG_EN Controlled By PHYPA[1]

1 = Select USBPHY REG_EN Controlled By PHYPA[1]

0 = Select USBPHY REG_EN Controlled By USBC Controller

PHYPA[6]: Control CLK_12M_EN Ouptut

1 = Enable USBPHY CLK_12M_EN Output

0 = Disable USBPHY CLK_12M_EN Output

PHYPA[5]: Select USBPHY PLL_EN(PHYPA[0]), SUSPENDM(PHYPA[2]), Controlled By Register

1 = Select USBPHY USBPHY PLL_EN and SUSPENDM Controlled By Register

0 = Select USBPHY USBPHY PLL_EN and SUSPENDM Controlled By USBC Controller

PHYPA[4]: USBPHY TERMSELECT Configure, need to configure with PHYPA[3]

1 = USBPHY TERMSELECT Enabled

0 = USBPHY TERMSELECT Disabled

PHYPA[3]: Select USBPHY TERMSELECT(PHYPA[4])

1 = Select USBPHY USBPHY TERMSELECT Controlled By Register

0 = Select USBPHY USBPHY TERMSELECT Controlled By USBC Controller

PHYPA[2]: USBPHY SUSPENDM Configure, need to configure with PHYPA[5]

1 = USBPHY SUSPEMDM Enabled

0 = USBPHY SUSPENDM Disabled

PHYPA[1]: USBPHY REG_EN Configure, need to configure with PHYPA[7]

1 = USBPHY REG_EN Enabled

0 = USBPHY REG_EN Disabled

PHYPA[0]: USBPHY PLL_EN Configure, need to configure with PHYPA[5]

1 = USBPHY PLL_EN Enabled

0 = USBPHY PLL_EN Disabled

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4.5.3.3 Chip Identification Register

The CIR register is a read-only register; writing to CIR has no effect.The chip's ID and Revision number can be read in this register.

Address Offset : 0x0004

Addr. Offset: 0x0004~0x0005 Reset Value: 0x3403

15	14	13	12	11	10	9	8
PIN							
ro							
7	6	5	4	3	2	1	0
PRN							
ro							

Figure 4-3: Chip Identification Register(CIR)

PIN[7:0] — Part Identification Number Field

This read-only field contains a unique version identification number for the chip.

PRN[7:0] — Part Revision Number Field

This read-only field contains the full-layer mask revision number. This number is increased by one for each new full-layer mask set of this part. The revision numbers are assigned in chronological order.

4.5.3.4 Chip Test Register

The chip test register is reserved for factory testing.

NOTE: To safeguard against unintentionally activating test logic, write $0000 to lock out test features. Setting any bits in the CTR register may lead to unpredictable results.

Addr. Offset: 0x000A~0x000B Reset Value: 0x0000

15	14	13	12	11	10	9	8
Reserved							TEST
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
Reserved							
ro	ro	ro	ro	ro	ro	ro	ro

Figure 4-4: Chip Test Register(CTR)

TEST--ips_test_access select bit

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4.5.3.5 PMU Configuration1/2 Register

Addr. Offset: 0x0010~0x0013 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 4-5: PMU Configuration1/2 Register(PCFG12)

4.5.3.6 PMU Configuration3 Register

Addr. Offset: 0x0014~0x0017 Reset Value: 0xFF013C00

31	30	29	28	27	26	25	24
GINT0_IE	GINT1_IE	GINT2_IE	GINT3_IE	GINT4_IE	GINT5_IE	GINT6_IE	GINT7_IE
ro	ro	ro	ro	ro	ro	ro	ro
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved	GINT37_PS	SS2_SWAP	Reserved	Reserved	TDO_PUE	RSTOUT_PUE	CLKOUT_PUE
ro	rw	rw	ro	ro	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved					PWM2_EN	PWM1_EN	PWM0_EN
ro					rw	rw	rw

Figure 4-6: PMU Configuration3 Register (PCFG3)

GINTx_IE — Pin GINTx IE select bit

1 = Pin GINTx input enable

0 = Pin GINTx input disable

TDO_PUE — Pin tdo pull enable bit

1 = Pin tdo pull enable

0 = Pin tdo pull disable

RSTOUT_PUE — Pin rstout pull enable bit

1 = Pin rstout pull enable

0 = Pin rstout pull disable

CLKOUT_PUE — Pin clkout pull enable bit

1 = Pin clkout pull enable

0 = Pin clkout pull disable

GINT37_PS — Pin gint[37] pull select bit

1 = Pin gint[37] pullup enable

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0 = Pin gint[37] pulldown enable

SS2_SWAP — SS2/GINT swap enable signal

1 = swap enable(SPI function)

0 = swap disable(GINT function)

PWM0_EN-PWM0 input and output enable bit

PWM1_EN-PWM1 input and output enable bit

PWM2_EN-PWM2 input and output enable bit

4.5.3.7 RTC Configuration1/2 Register

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				cap_trim_value[1]			
ro				rw			
23	22	21	20	19	18	17	16
Reserved						RTC_TM	Reserved
ro[2]						rw[3]	ro[3]
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 4-7: RTC Configuration1/2 Register(RTCCFG12)

NOTES:

- The register is loaded from eflash,and W/R the register must enter ips test mode by configurated the CTR register
- The register is loaded from eflash,andW/R the register must enter ips test mode by configurated the CTR register

rtc_en_interface — enable rtc interface

1 = rtc interface enable

0 = rtc interface disable

bias_trim_value — config osc bias trim value

day_int enable — PMU_RTC day interruput enable

hour_int enable — PMU_RTC hour interruput enable

minute_int enable — PMU_RTC minute interruput enable

second_int enable — PMU_RTC second interruput enable

day pulse inv — PMU_RTC day pulse invert

hour pulse inv — PMU_RTC hour pulse invert

minute pulse inv — PMU_RTC minute pulse invert

second pulse inv — PMU_RTC second pulse invert

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4.5.3.8 RTC Configuration3 Register

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				cap_trim_value[1]			
ro				rw			
23	22	21	20	19	18	17	16
Reserved						RTC_TM	Reserved
ro[2]						rw[3]	ro[3]
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 4-8: RTC Configuration3 Register(RTCCFG3)

NOTES:

- The register is loaded from eflash,W/R the register must enter ips test mode by configurated the CTR register
- W/R the register must enter ips test mode by configurated the CTR register

cap_trim_value — config osc c1/c2 trim value

NOTE: When before config bias_trim_value and cap_trim_value, plase set rtc_en_interface=1 first, then wait 2us,set bias_trim_value and cap_trim_value.

RTC_TM---enter rtc test mode

1 = enable rtc test mode

0 = disable rtc test mode

NOTE: Before set this bit, should set ips_test_access first

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4.5.3.9 PMU_RTC Status Register

Addr. Offset: 0x0020~0x0023 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				lvdt5v_src	lvdt1p8v_src	osc_128k_clk_src	osc_2k_clk_src
ro				rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 4-9: PMU_RTC Status Register (RTCSR)

osc_2k_clk_src — vdd 3p3v 2k clk source status when in test mode

1 = clk source value 1

0 = clk source value 0

osc_128k_clk_src — vdd 3p3v 128k clk source status when in test mode

1 = clk source value 1

0 = clk source value 0

lvdt1p8v_src — vdd1p8v flag when in vccq test mode

1 = vdd1.8v supply power

0 = not supply power

lvdt5v_src — vdd3p3v flag when in vccq test mode

1 = vdd5v work

0 = not work

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4.5.3.10 PMU_RTC Pulse Register

Addr. Offset: 0x0028~0x002B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
day pulse	hour pulse	minute pulse	second pulse	Reserved			
ro				ro			

23	22	21	20	19	18	17	16
Reserved							
ro							

15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 4-10: PMU_RTC Pulse Register (RTCPR)

day pulse - will assert every day and will last one second time.

hour pulse - will assert every hour and will last one second time.

minute pulse - will assert every minute and will last one second time.

second pulse - will assert every second and will last 16us.

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5 Cache

5.1 Introduction

The Cache module provides the processor with tightly-coupled processor-local memories and bus paths to all slave memory spaces.

A cache is a block of high-speed memory locations containing address information (commonly known as a tag) and the associated data. The purpose is to decrease the average time of a memory access. Caches operate on two principles of locality:

- Spatial locality — An access to one location is likely to be followed by accesses from adjacent locations (for example, sequential instruction execution or usage of a data structure).
- Temporal locality — An access to an area of memory is likely to be repeated within a short time period (for example, execution of a code loop).

To minimize the quantity of control information stored, the spatial locality property is used to group several locations together under the same tag. This logical block is commonly known as a cache line.

When data is loaded into a cache, access times for subsequent loads and stores are reduced, resulting in overall performance benefits. An access to information already in a cache is known as a cache hit, and other accesses are called cache misses.

Normally, caches are self-managing, with the updates occurring automatically. Whenever the processor wants to access a cacheable location, the cache is checked. If the access is a cache hit, the access occurs immediately. Otherwise, a location is allocated and the cache line is loaded from memory. Different cache topologies and access policies are possible.

However, they must comply with the memory coherency model of the underlying architecture. Caches introduce a number of potential problems, mainly because of:

- memory accesses occurring at times other than when the programmer would normally expect them,
- the existence of multiple physical locations where a data item can be held.

The Cache Controller (ahb_cache) is targeted for use with any 32 bit AMBA_AHB bus based application that desires a cache function. This cache function can increase performance by providing fast access to recently used code or data.

The local memory controller supports three modes of operation:

1. Write-through — access to address spaces with this cache mode are cacheable.

- A write-through read miss on the input bus causes a line read on the output bus of a 16-byte-aligned memory address containing the desired address. This miss data is loaded into the cache and is marked as valid and not modified.
- A write-through read hit to a valid cache location returns data from the cache with no output bus access.
- A write-through write miss bypasses the cache and writes to the output bus (no allocate on write miss policy for write-through mode spaces).
- A write-through write hit updates the cache hit data and writes to the output bus.

2. Write-back — access to address spaces with this cache mode are cacheable.

- A write-back read miss on the input bus will cause a line read on the output bus of a 16-byte-aligned memory address containing the desired address. This miss data is loaded into the cache and marked as valid and not modified.
- A write-back read hit to a valid cache location will return data from the cache with no output bus access.
- A write-back write miss will do a "read-to-write" (allocate on write miss policy for write-back mode spaces). A line read on the output bus of a 16 byte aligned memory address containing the desired write address is performed. This miss data is loaded into the cache and marked as valid and modified;

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and the write data will then update the appropriate cache data locations.

3. Non-cacheable — access to address spaces with this cache mode are not cacheable.

These accesses bypass the cache and access the output bus.

The ahb_cache has two AMBA_ AHB bus interfaces, a master and a slave interface. The master interface has decode logic to determine the cache mode of valid address phase accesses.

Master accesses will then access or bypass the cache depending on their cache mode. The slave interface is used for cache misses as well as accesses that bypass the cache.

The ahb_cache has a 2-way set-associative organization. The cache has 32-bit wide address and data paths and a 16-byte line size. The cache tags and data storage use singleport, synchronous RAMs. The ahb_cache controller has a variety of read and write data buffers to improve performance. Cache misses and line pushes generate 4-beat 32-bit wrapping burst accesses with the critical word accessed first for maximum performance.

The ahb_cache uses the positive edge of its single clock input. The single input clock must match the master bus clock. The master bus and slave bus operate at the same frequency as the platform bus.

The ahb_cache has an AMBA _AHB bus interface to access its programmer's model. The programmer's model provides access to cache control functions and resources as well as all cache tag and data storage.

5.2 Block Diagram

These Cache module provides zero wait state access to RAM and cacheable address spaces.

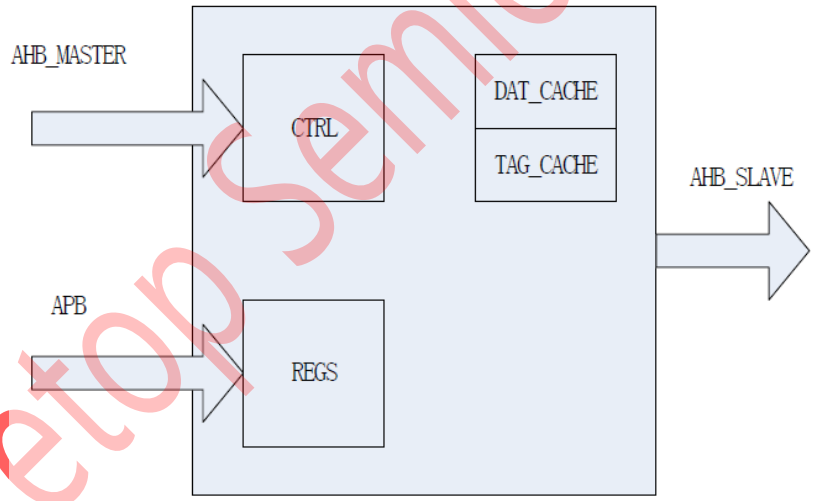


Figure 5-1: Cache Module Block Diagram

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5.3 Representative I/O List

5.3.1 Clocks

The hclk input should match the master bus clock. The ahb_cache uses the positive edge of this clock.

The slave bus operates at the same frequency as the master bus.

5.3.2 Resets

The ahb_cache controller uses a single asynchronous reset.

This is the active-low B asynchronous reset signal. This reset signal must be negated synchronously after a rising edge of the platform clocks.

5.4 Memory Map/Register Definition

5.4.1 Memory Map

Table 5-1: Cache Module Memory Map

Address Offset	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access
0x0000	Cache control register (LMEM_CCR)				S/U
0x0004	Cache line control register (LMEM_CLCR)				S/U
0x0008	Cache search address register (LMEM_CSAR)				S/U
0x000c	Cache read/write value register (LMEM_CCVR)				S/U
0x0020	Cache access register(LMEM_ACR)				S/U
0x0024	Cache region2_6_7 access register(LMEM_R267_ACR)				S/U
0x0040	Cache Region6 Segment0 Highaddress(LMEM_R6_S0_HADDR)				S/U
0x0044	Cache Region6 Segment1 Highaddress(LMEM_R6_S1_HADDR)				S/U
0x0048	Cache Region6 Segment2 Highaddress(LMEM_R6_S2_HADDR)				S/U
0x004c	Cache Region6 Segment3 Highaddress(LMEM_R6_S3_HADDR)				S/U
0x0050	Cache Region6 Segment0 Lowaddress(LMEM_R6_S0_LADDR)				S/U
0x0054	Cache Region6 Segment1 Lowaddress(LMEM_R6_S1_LADDR)				S/U
0x0058	Cache Region6 Segment2 Lowaddress(LMEM_R6_S2_LADDR)				S/U
0x005c	Cache Region6 Segment3 Lowaddress(LMEM_R6_S3_LADDR)				S/U
0x0060	Cache Region7 Segment0 Highaddress(LMEM_R7_S0_HADDR)				S/U
0x0064	Cache Region7 Segment1 Highaddress(LMEM_R7_S1_HADDR)				S/U
0x0068	Cache Region7 Segment2 Highaddress(LMEM_R7_S2_HADDR)				S/U
0x006c	Cache Region7 Segment3 Highaddress(LMEM_R7_S3_HADDR)				S/U
0x0070	Cache Region7 Segment0 Lowaddress(LMEM_R7_S0_LADDR)				S/U
0x0074	Cache Region7 Segment1 Lowaddress(LMEM_R7_S1_LADDR)				S/U
0x0078	Cache Region7 Segment2 Lowaddress(LMEM_R7_S2_LADDR)				S/U
0x007c	Cache Region7 Segment3 Lowaddress(LMEM_R7_S3_LADDR)				S/U
0x0080	Cache Region2 Segment0 Highaddress(LMEM_R2_S0_HADDR)				S/U
0x0084	Cache Region2 Segment1 Highaddress(LMEM_R2_S1_HADDR)				S/U
0x0088	Cache Region2 Segment2 Highaddress(LMEM_R2_S2_HADDR)				S/U
0x008c	Cache Region2 Segment3 Highaddress(LMEM_R2_S3_HADDR)				S/U
0x0090	Cache Region2 Segment0 Lowaddress(LMEM_R2_S0_LADDR)				S/U

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Address Offset	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access
0x0094	Cache Region2 Segment1 Lowaddress(LMEM_R2_S1_LADDR)				S/U
0x0098	Cache Region2 Segment2 Lowaddress(LMEM_R2_S2_LADDR)				S/U
0x009c	Cache Region2 Segment3 Lowaddress(LMEM_R2_S3_LADDR)				S/U
0x0180	cache page invalidation start address				S/U
0x0184	cache page invalidate size				S/U
0x0188	cache clock gate				S/U

NOTES:

- S = CPU supervisor mode access only, U = CPU user mode access only
- User mode accessed to supervisor-only address locations have no effect and result in a cycle termination transfer error

5.4.2 Register Description

This subsection provides a description of the Cache module.

5.4.2.1 Cache control register (CCR)

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000030

31	30	29	28	27	26	25	24
GO	Reserved			PUSHW1	INVW1	PUSHW0	INVW0
rw	ro			rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				Reserved			ENCache
ro				ro			rw

Figure 5-2: Cache control register (CCR)

ENCACHE — Cache enable.

1 = cache enabled

0 = cache disabled

reduce_cachemem_acc[1:0] - Configured '1' to reduce cache memory access when cache disabled.

cache_idle_ref- Determine whether cache status is idle. cache_burst_type- To make burst type from cache configurable.

1 = burst type is always wrap4.

0 = burst type is incr4 when start address is 16bytes aligned.

INVW0 — Invalidate way 0. If the PUSHW0 and INVW0 bits are set, then after setting the GO bit, push all modified lines in way 0 and invalidate all lines in way 0 (clear way 0).

1 = When setting the GO bit, invalidate all lines in way 0.

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0 = no operation

PUSHW0 — Push way 0

1 = When setting the GO bit, push all modified lines in way 0

0 = no operation

INVW1 — Invalidate way 1. If the PUSHW1 and INVW1 bits are set, then after setting the GO bit, push all modified lines in way 1 and invalidate all lines in way 1 (clear way 1).

1 = When setting the GO bit, invalidate all lines in way 1.

0 = no operation

PUSHW1 — Push way 1

1 = When setting the GO bit, push all modified lines in way 1

0 = no operation

GO — Initiate Cache Command Setting this bit initiates the cache command indicated by bits 27-24. Reading this bit indicates if a command is active. This bit stays set until the command completes. Writing zero has no effect..

1 = Write: initiate command indicated by bits 27-24. Read: cache command active.

0 = Write: no effect. Read: no cache command active.

5.4.2.2 Cache line control register (CLCR)

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				LACC	LADSEL	LCMD[1:0]	
ro				rw	rw	rw	
23	22	21	20	19	18	17	16
Reserved	LCR[2:0]			Reserved			TDSEL
ro	rw			ro			rw
15	14	13	12	11	10	9	8
Reserved	WSEL	Reserved		Cache_ADDRESS[9:0]			
ro	rw	ro		rw			
7	6	5	4	3	2	1	0
Cache_ADDRESS[9:0]						Reserved	LGO
rw						ro	rw

Figure 5-3: Cache line control register (CLCR)

LGO — Initiate Cache Line Command. Setting this bit initiates the cache line command indicated by bits 27-24. Reading this bit indicates if a line command is active. This bit stays set until the command completes. Writing zero has no effect.This bit is shared with CSAR[LGO].

1 = Write: initiate line command indicated by bits 27-24. Read: line command active.

0 = Write: no effect. Read: no line command active.

CACHE_ADDRESS — Cache address. CLCR[11:4] bits are used to access the tag arrays; CLCR[11:2] bits are used to access the data arrays.

WSEL — Way select. Select the way of the line commands.

1 = way 1.

0 = way 0.

TDSEL — Tag/Data select. Select tag or data for search and read or write commands.

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1 = Tag.

0 = Data.

LCMD[1:0] — Line command.

00 = Search and read or write.

01 = Invalidate

10 = Push

11 = Clear

LADSEL — Line Address Select. When using the cache address, the way must also be specified in CLCR[WSEL]. When using the physical address, both ways are searched and the command is performed only if a hit.

0 = Cache address

1 = Physical address

LACC — Line access type.

0 = Read

1 = Write

5.4.2.3 Cache search address register (CSAR)

The CSAR register is used to define the explicit cache address or the physical address for line-sized commands specified in the CLCR[LADSEL] bit.

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
PHYSICAL_ADDRESS[29:0]							
rw							
23	22	21	20	19	18	17	16
PHYSICAL_ADDRESS[29:0]							
rw							
15	14	13	12	11	10	9	8
PHYSICAL_ADDRESS[29:0]							
rw							
7	6	5	4	3	2	1	0
PHYSICAL_ADDRESS[29:0]						Reserved	LGO
rw						ro	rw

Figure 5-4: Cache search address register (CSAR)

PHYSICAL_ADDRESS — Physical Address. PHYADDR represents bits [31:2] of the system address.

CSAR[31:12] bits are used for tag compare

CSAR[11:4] bits are used to access the tag arrays CSAR[11:2] bits are used to access the data arrays.

LGO — Initiate Cache Line Command. Setting this bit initiates the cache line command indicated by bits 27-24. Reading this bit indicates if a line command is active. This bit stays set until the command completes. Writing zero has no effect.

This bit is shared with CLCR[LGO]

1 = Write: initiate line command indicated by bits 27-24. Read: line command active.

0 = Write: no effect. Read: no line command active.

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5.4.2.4 Cache read/write value register (CCVR)

The CCVR register is used to source write data or return read data for the commands specified in the CLCR register.

Addr. Offset: 0x000C~0x000F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DATA[31:0]							
rw							
23	22	21	20	19	18	17	16
DATA[31:0]							
rw							
15	14	13	12	11	10	9	8
DATA[31:0]							
rw							
7	6	5	4	3	2	1	0
DATA[31:0]							
rw							

Figure 5-5: Cache read/write value register (CCVR)

DATA — Cache read/write Data For tag search, read or write:

- CCVR[31:12] bits are used for tag array R/W value
- CCVR[11:4] bits are used for tag set address on reads; unused on writes
- CCVR[3:2] bits are reserved For data search, read or write:
- CCVR[31:0] bits are used for data array R/W value

5.4.2.5 Cache access register(CACR)

The CACR is used to control the attributes of 10 different memory regions such as write-through, write-back or non-cacheable.

Addr. Offset: 0x0020~0x0023 Reset Value: 0x000FFFFF

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				Reserved		R8_CacheABLE	R8WT_WB
ro				ro		rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-6: Cache access register(CACR)

ROM: 0x04000000 ~ 0x04005fff

ROM_CACHEABLE — Region is cacheable.

1 = cacheable.

0 = non-cacheable

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ROM_WT_WB — Region is write-through if cacheable.

1 = write-back

0 = write-through

In additional, there are cacheable regions such as R2, R3, R6 and R7. The base of these regions are:

R2: 0x08000000~0x0807ffff

R6: 0x10000000~0x10ffffff

R7:0x00000000~0x00ffffff

The attributes of these regions are defined in LMEM_DIV_ACRG.

5.4.2.6 Cache divide access register(CSACR)

The CSACR is used to control the attributes of different memory regions such as write-through, write-back or non-cacheable.

Addr. Offset: 0x0024~0x0027 Reset Value: 0xFFFFFFAA

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
R7_3_CachEABLE	R7_3_WT_WB	R7_2_CachEABLE	R7_2_WT_WB	R7_1_CachEABLE	R7_1_WT_WB	R7_0_CachEABLE	R7_0_WT_WB
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
R6_3_CachEABLE	R6_3_WT_WB	R6_2_CachEABLE	R6_2_WT_WB	R6_1_CachEABLE	R6_1_WT_WB	R6_0_CachEABLE	R6_0_WT_WB
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
R2_3_CachEABLE	R2_3_WT_WB	R2_2_CachEABLE	R2_2_WT_WB	R2_1_CachEABLE	R2_1_WT_WB	R2_0_CachEABLE	R2_0_WT_WB
rw	rw	rw	rw	rw	rw	rw	rw

Figure 5-7: Cache access register(CSACR)

The three region R2, R6 and R7 are divided into 4 sub-regions.

RX_Y_CACHEABLE — Region X the Yth(Y=0, 1, 2, 3) segment is cacheable. R2 means region2, R6 means region6, R7 means region7.

1 = cacheable.

0 = non-cacheable

RX_Y_WT_WB — Region X the Yth(Y=0, 1, 2, 3) segment is write-through if cacheable.R2 means region2, R6 means region6, R7 means region7

1 = write-back

0 = write-through

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5.4.2.7 Cache R6 segmentx high address register (CR6SXHA)

The CR6SXHA register is used to define the high bound address of segmentx of R6. x=0, 1, 2, 3

Addr. Offset: 0x0040~0x004C Reset Value: 0x80*(X+1)-0x1, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
R6_SX_HIGH_ADDRESS[8:0]							
rw							
15	14	13	12	11	10	9	8
R6_SX_HIGH_ADDRESS[8:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-8: R6 segmentx high address register (CR6SXHA)

R6_Sx_HIGH_ADDRESS[23:15] — Physical Address. The reset value is 0x80*(x+1) - 0x1, x=0, 1, 2, 3.

5.4.2.8 Cache R6 segmentx low address register (CR6SXLA)

The CR6SXLA register is used to define the low bound address of segmentx of R6. x=0, 1, 2, 3

Addr. Offset: 0x0050~0x005C Reset Value: 0x80*X, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
R6_SX_LOW_ADDRESS[8:0]							
rw							
15	14	13	12	11	10	9	8
R6_SX_LOW_ADDRESS[8:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-9: R6 segmentx low address register (LMEM_R6_Sx_LOW_ADDR)

R6_Sx_LOW_ADDRESS[23:15] — Physical Address Offset. The reset value is 0x80*x, x=0, 1, 2, 3.

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5.4.2.9 Cache R7 segmentx high address register (CR7SXHA)

The CR7SXHA register is used to define the high bound address of segmentx of R7. X = 0, 1, 2, 3

Addr. Offset: 0x0060~0x006C Reset Value: 0x80*(X+1)-0x1, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
R7_SX_HIGH_ADDRESS[8:0]							
rw							
15	14	13	12	11	10	9	8
R7_SX_HIGH_ADDRESS[8:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-10: R7 segmentx high address register (CR7SXHA)

R7_Sx_HIGH_ADDRESS[23:15] — Physical Address. The reset value is 0x80*(x+1) - 0x1, x=0, 1, 2, 3.

5.4.2.10 Cache R7 segmentx low address register (CR7SXLA)

The CR7SXLA register is used to define the low bound address of segmentx of R7. X = 0, 1, 2, 3

Addr. Offset: 0x0070~0x007C Reset Value: 0x80*X, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
R7_SX_LOW_ADDRESS[8:0]							
rw							
15	14	13	12	11	10	9	8
R7_SX_LOW_ADDRESS[8:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-11: R7 segmentx low address register (CR7SXLA)

R7_Sx_LOW_ADDRESS[23:15] — Physical Address Offset. The reset value is 0x80*x, x = 0, 1, 2, 3.

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5.4.2.11 Cache R2 segmentx high address register (CR2SXHA)

The CR2SXHA register is used to define the high bound address of segmentx of R2. X = 0, 1, 2, 3

Addr. Offset: 0x0080~0x008C Reset Value: 0x4*(X+1)-1, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				R2_SX_HIGH_ADDRESS[4:0]			
ro				rw			
15	14	13	12	11	10	9	8
R2_SX_HIGH_ADDRESS[4:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-12: R2 segmentx high address register (CR2SXHA)

R2_Sx_HIGH_ADDRESS[19:15] — Physical Address. The reset value is 0x4*(x+1)-1, x=0, 1, 2,3.

5.4.2.12 Cache R2 segmentx low address register (CR2SXLA)

The CR2SXLA register is used to define the low bound address of segmentx of R2. X = 0, 1, 2, 3

Addr. Offset: 0x0090~0x009C Reset Value: 0x4*X, X = 0, 1, 2, 3

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				R2_SX_LOW_ADDRESS[4:0]			
ro				rw			
15	14	13	12	11	10	9	8
R2_SX_LOW_ADDRESS[4:0]	Reserved						
rw	ro						
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 5-13: R2 segmentx low address register (CR2SXLA)

R2_Sx_LOW_ADDRESS[19:15] — Physical Address Offset. The reset value is 0x4*x, x = 0, 1, 2, 3.

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5.4.2.13 Cache page invalidation base address register (CPEA)

Addr. Offset: 0x0180~0x0183 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
INVAL_BADDR[27:0]							
rw							
23	22	21	20	19	18	17	16
INVAL_BADDR[27:0]							
rw							
15	14	13	12	11	10	9	8
INVAL_BADDR[27:0]							
rw							
7	6	5	4	3	2	1	0
INVAL_BADDR[27:0]				Reserved			START_INVAL
rw				ro			rw

Figure 5-14: cache page invalidate base address register (CPEA)

INVAL_BADDR[27:0] — cache invalidate start address for system memory. Lower 4bits is ignored because of line alignment.

START_INVAL — start to page invalidate. Cleared if the invalidation is completed. This bit is shared in bit 0 of CPES.

5.4.2.14 Cache page invalidation base size register (CPES)

Addr. Offset: 0x0184~0x0187 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
rw							
23	22	21	20	19	18	17	16
Reserved							
rw							
15	14	13	12	11	10	9	8
PAGE_SIZE[11:0]							
rw							
7	6	5	4	3	2	1	0
PAGE_SIZE[11:0]				Reserved			START_INVAL
rw				ro			rw

Figure 5-15: cache page invalidate size register (CPES)

page_inv_size[15:4] — cache invalidate s*ze* for system memory. Lower 4bits is ignored because of line alignment.

start_page_inv — start to page invalidate. Cleared if the invalidation is completed. This bit is shared in bit 0 of LMEM_PAGE_INV_BADDR_REG.

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5.4.2.15 Cache clock enable register (CCG)

Addr. Offset: 0x0188~0x018B Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							CLK_ENABLE
ro							rw

Figure 5-16: cache clock enable register (CCG)

CLK_ENABLE — cache clock is enabled. Must be ’1’ if cache is enabled.

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5.5 Cache Function

5.5.1 General Description

The caches on this device are structured as follows. Both caches have a 2-way setassociative cache structure with a total size of 8 KBytes. The caches have 32-bit address and data paths and a 16-byte line size. The cache tags and data storage use external single-port, synchronous RAMs.

For these 8-KByte caches, each cache TAG function uses two 256 x 22-bit RAM arrays and the cache DATA function uses two 1024 x 32-bit RAM arrays. The cache TAG entries store 20 bits of upper address as well as a modified and valid bit per cache line. The cache DATA entries store four bytes of code or data.

All normal cache accesses use physical addresses. This leads to the following cache address use:

CACHE - 8 KByte size = (256 sets) x (16-byte lines) x (2-way set-associative)

TAG:

- address[31:12] used in tag for compare (hit) logic
- address[11:4] used to select 1 of 256 sets
- address[3:0] not used

DATA:

- address[31:12] not used
- address[11:4] used to select one of 256 sets
- address[3:2] used to select one of four 32-bit words within a set
- address[1:0] used to select the byte within the 32-bit word

The caches use a one-cycle parallel TAG and DATA access structure. This structure is aligned in the two stage AMBA _AHB bus pipeline from the negative edge of the address phase to the negative edge of the data phase:

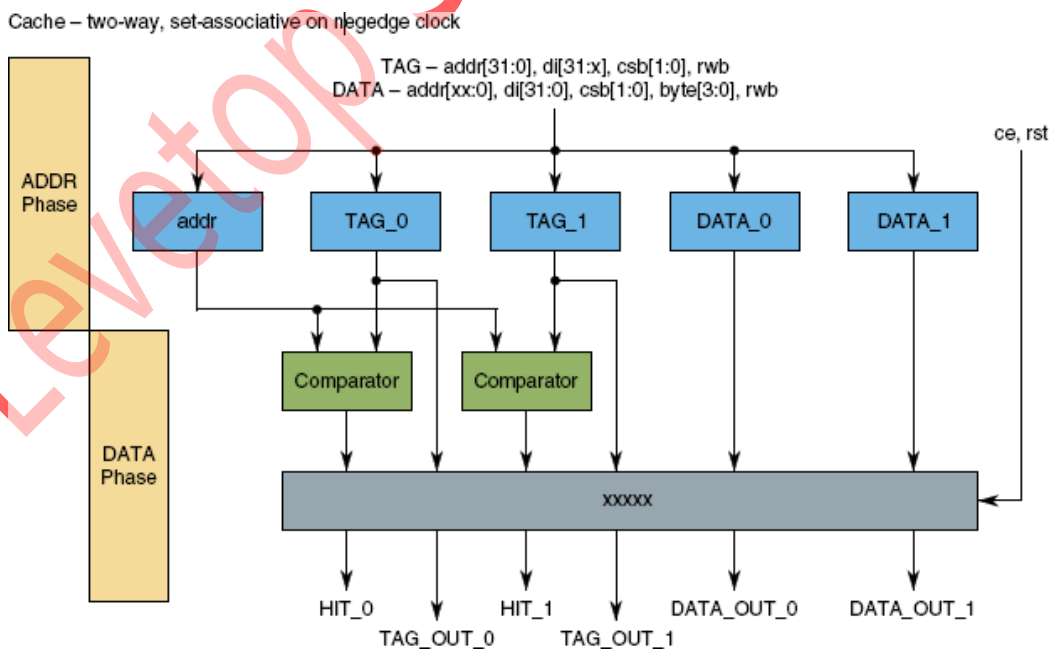


Figure 5-17: Cache Tag and Data access structure

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On a normal cacheable operation, the full address is registered on the negedge of the address phase in the cache control logic while address bits[11:4] are registered in the TAG arrays and address bits[11:2] are registered in the DATA arrays. During the full cycle from the negedge of the address phase to the negative edge of the data phase, the TAG and DATA arrays are accessed and cache hit analyzed. For hits, the desired read data is returned in the second half of the data phase and writes are done starting on the negedge of the data phase cycle. For misses, the data phase is held as needed while the cache uses its slave bus to access the desired cache line via a line-sized data transfer sent to the crossbar switch.

5.5.2 Cache control

The Code and System Caches are disabled at reset. Cache tag and data arrays are not cleared at reset. Therefore, to enable the caches, cache commands must be done to clear and initialize the required tag array bits and to configure and enable the caches.

5.5.3 Cache set commands

The cache set commands may operate on:

- all of way 0,
- all of way 1, or
- all of both ways (complete cache).

Cache set commands are initiated using the upper bits in the CCR register. Cache set commands perform their operation on the cache independent of the cache enable bit, CCR[ENCACHE].

A cache set command is initiated by setting the CCR[GO] bit. This bit also acts as a busy bit for set commands. It stays set while the command is active and is cleared by the hardware when the set command completes.

Supported cache set commands are given in **Figure 5-18**. Set commands work as follows:

- Invalidate − Unconditionally clear valid and modify bits of a cache entry.
- Push − Push a cache entry if it is valid and modified, then clear the modify bit. If entry not valid or not modified, leave as is.
- Clear − Push a cache entry if it is valid and modified, then clear the valid and modify bits. If entry not valid or not modified, clear the valid bit.

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Table 5-2: Cache Set Commands

CCR[27:24]				Command
PUSH W1	INVW1	PUSH W0	INVW0	
0	0	0	0	NOP
0	0	0	1	Invalidate all way 0
0	0	1	0	Push all way 0
0	0	1	1	Clear all way 0
0	1	0	0	Invalidate all way 1
0	1	0	1	Invalidate all way 1; invalidate all way 0 (invalidate cache)
0	1	1	0	Invalidate all way 1; push all way 0
0	1	1	1	Invalidate all way 1; clear all way 0
1	0	0	0	Push all way 1
1	0	0	1	Push all way 1; invalidate all way 0
1	0	1	0	Push all way 1; push all way 0 (push cache)
1	0	1	1	Push all way 1; clear all way 0
1	1	0	0	Clear all way 1
1	1	0	1	Clear all way 1; invalidate all way 0
1	1	1	0	Clear all way 1; push all way 0
1	1	1	1	Clear all way 1; clear all way 0 (clear cache)

After a reset, complete an invalidate cache command before using the cache.

5.5.4 Cache line commands

Cache line commands operate on a single line in the cache at a time. Cache line commands can be performed using a physical or cache address.

- A cache address consists of a set address and a way select. The line command acts on the specified cache line.
- Cache line commands with physical addresses first search both ways of the cache set specified by bits [11:4] of the physical address. If they hit, the commands perform their action on the hit way.

Cache line commands are specified using the upper bits in the CLCR register. Cache line commands perform their operation on the cache independent of the cache enable bit (CCR[ENCACHE]). Using a cache address, the command can be completely specified using the CLCR register. Using a physical address, the command must also use the CSAR register to specify the physical address.

A line cache command is initiated by setting the line command go bit (CLCR[LGO] or CSAR[LGO]). This bit also acts a a busy bit for line commands. It stays set while the command is active and is cleared by the hardware when the command completes.

The CLCR[27:24] bits select the line command as follows:

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Table 5-3: Cache Line Commands

CLCR[27:24]			Command
LACC	LADSEL	LCMD	
0	0	00	Search by cache address and way
0	0	01	Invalidate by cache address and way
0	0	10	Push by cache address and way
0	0	11	Clear by cache address and way
0	1	00	Search by physical address
0	1	01	Invalidate by physical address
0	1	10	Push by physical address
0	1	11	Clear by physical address
1	0	00	Write by cache address and way
1	0	01	Reserved, NOP
1	0	10	Reserved, NOP
1	0	11	Reserved, NOP
1	1	*xx*	Reserved, NOP

5.5.5 Executing a series of line commands using cache addresses

A series of line commands with incremental cache addresses can be performed by just writing to the CLCR.

- Place the command in CLCR[27:24],
- Set the way (CLCR[WSEL]) and tag/data (CLCR[TDSEL]) controls as needed,
- Place the cache address in CLCR[CACHEADDR], and
- Set the line command go bit (CLCR[LGO]).

When one line command completes, initiate the next command by following these steps:

- Increment the cache address (at bit 2 to step through data or at bit 4 to step through lines), and
- Set the line command go bit (CLCR[LGO]).

5.5.6 Executing a series of line commands using physical addresses

Perform a series of line commands with incremental physical addresses using the following steps:

- Write to the CLCR.
- Place the command in CLCR[27:24]
- Set the tag/data (CLCR[TDSEL]) control
- Place the physical address in CSAR[PHYADDR] and set the line command go bit (CSAR[LGO]).

When one line command completes, initiate the next command by following these steps:

- Increment the physical address (at bit 2 to step through data or at bit 4 to step through lines),

and

- Set the line command go bit (CSAR[LGO]).

The line command go bit is shared between the CLCR and CSAR registers, so that the above steps can be completed in a single write to the CSAR register.

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6 Clock and Power Module

6.1 Overview

The clock and power module contains:

- Internal high speed 120MHz oscillator
- Internal low speed 8MHz oscillator
- External crystal/resonator (12MHZ)
- Status and control registers
- Clock and Power Control logic

6.2 Features

Features of the clock and power module include:

- Two system clock sources
  - Internal high speed 120MHz oscillator
  - Internal low speed 8MHz oscillator
- Separate clock divider
- Support for power saving mode
- Module clock can be gated separately

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6.3 Clock Structure

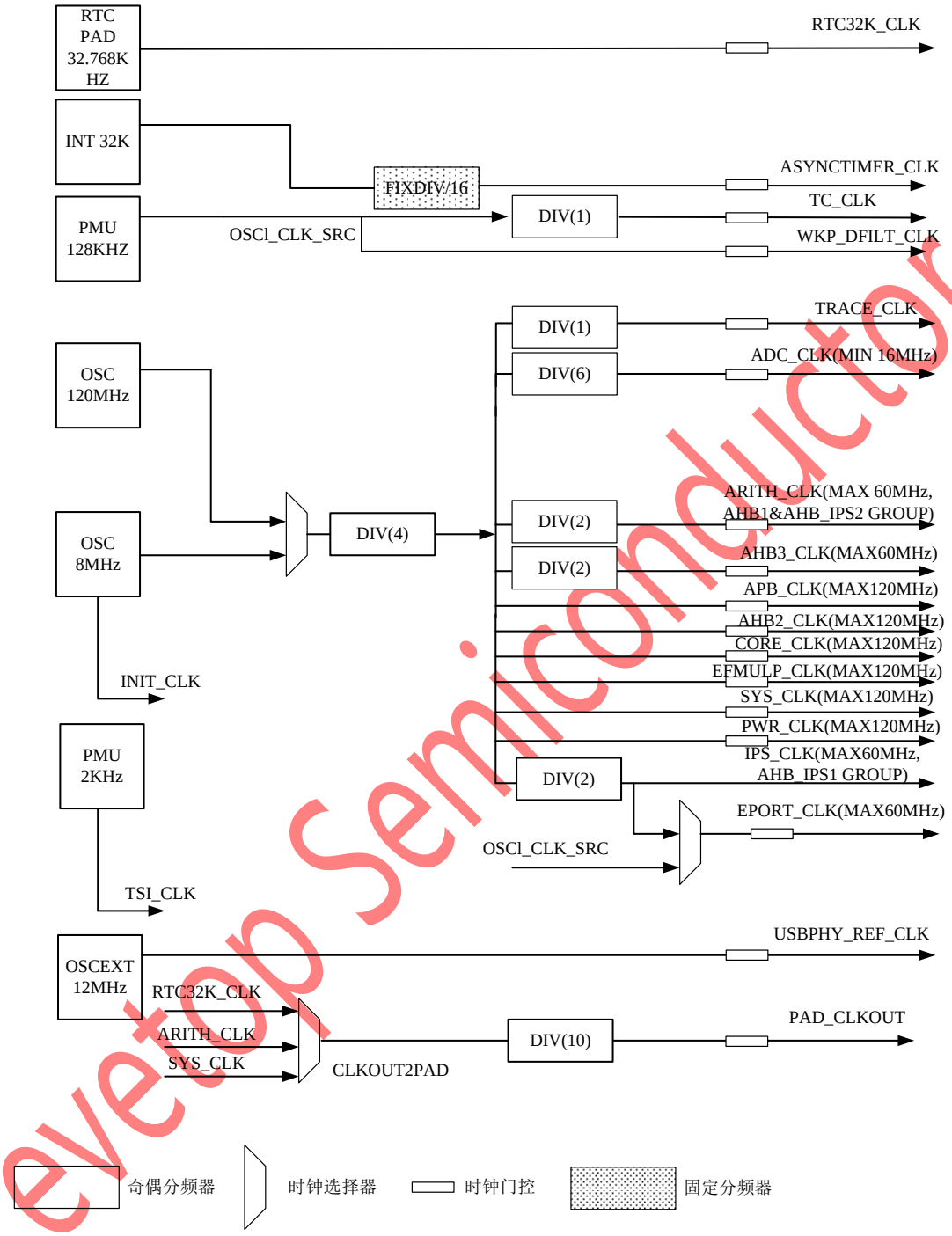


Figure 6-1: Clock Structure

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6.4 Operation Mode

System Clock Source can be internal high speed oscillator(120MHz) or internal low speed oscillator(8MHz). Clock source select is controled by **CSWCFGR** register of this module.To change clock source, config CSWCFGR and wait for switch status in **CSWCFGR**.

6.4.1 Power Saving Options

6.4.1.1 Wait and Doze

When the core executes WAIT and DOZE instruction, the system clocks to the peripherals and embedded-flash are still enabled, the clock to the core is gated.

6.4.1.2 Stop

When the core executes STOP instruction,the system will enter one of the sleep mode as specified in **SLPCFGR&SLPCFGR2**.

6.4.2 Detailed Power Saving Mode Description

Table 6-1: CPM Power Mode and Wakeup Source

Main system Power mode	Clock&PAD	Wakeup Source	Power*1
RUN	All clock work Clock gate configurable All PAD work	■ None	■ voltage scaling support (1.2V/1.15V/1.1V/1.05V/0.9V)
WAIT	Core M4 clock stop Other clock work All PAD work	■ All interrupt ■ WK PAD (level high) ■ USBDET (level high) ■ POR (reset) ■ PCI domain interrupt	■ AO domain ON ■ WK domain ON ■ PD domain ON ■ USB domain configurable ■ PCI domain ON
LOWPOWER	All clock stop (except Clock need by wakeup source and RTC32K) All PAD work	■ SPI (SS3 PAD, level low) ■ USI1 I2C1-3 ■ EPORT0-4 (level H/L, edge support) ■ USB RESUME (USB domain config ON) ■ TSI TOUCH ■ RTC (time pulse and alarm) ■ WK PAD (level high) ■ USBDET (level high) ■ POR (reset, level low) ■ PCI domain interrupt	■ AO domain ON ■ WK domain ON ■ PD domain ON ■ USB domain configurable ■ PCI domain ON
RETENTION (POWEROFF1.0)	All clock stop (except Clock need by wakeup source and RTC32K) All PAD state hold/latch*3	■ SPI (SS3 PAD, level low) ■ USI1 I2C1 ■ EPORT0 (level H/L, edge) *2 ■ USB RESUME (USB domain config ON) ■ TSI TOUCH ■ RTC (time pulse only) ■ WK PAD (level high) ■ USBDET (level high) ■ POR (reset, level low) ■ PCI domain interrupt	■ AO domain ON ■ WK domain ON ■ PD domain OFF ■ USB domain configurable ■ PCI domain ON

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Main system Power mode	Clock&PAD	Wakeup Source	Power*1
DEEP SLEEP (POWEROFF1.5)	All clock stop (except Clock need by wakeup source and RTC32K) All PAD state hold/latch*3	■ SPI (SS3 PAD, level low) ■ USI1 I2C1 ■ EPORT0 (level H/L, edge) *2 ■ USB RESUME (USB domain config ON) ■ TSI TOUCH ■ RTC (time pulse only) ■ WK PAD (level high) ■ USBDET (level high) ■ POR (reset, level low) ■ PCI domain interrupt	■ AO domain ON ■ WK domain OFF ■ PD domain OFF ■ USB domain configurable ■ PCI domain ON
HIBER (POWEROFF2.0)	All clock stop (except RTC32K) All PAD HIZ(tri-state)*4	■ WK PAD (level high) ■ USBDET (level high) ■ POR (reset, level low)	■ AO domain OFF ■ WK domain OFF PD domain OFF USB domain OFF PCI domain ON ■ C0_SUB domain supply independently
PCI	Independent power supply INT32K work PAD SDIO work	■ None	■ AO domain OFF ■ WK domain OFF ■ PD domain OFF ■ USB domain OFF ■ PCI domain ON

NOTES:

*1:

- AO domain modules include: CPM, CCM&RESET, IO_CTRL, IOMUX_AO, EPORT1, USI1, I2C1, TSI ;
- WK domain modules include: SRAMD (32KB, MEMORY IP only), KEY_CTRL ;
- USB domain modules include: USB ;
- PCI domain modules include: PCI ;
- PD domain includes: M4 +other modules;
- If domain power is ON, when exit the power saving mode, register&memory value in the specific domain is held ;
- If domain power is OFF, when exit the power saving mode, register&memory value in the specific domain is reset ;

Especially, core M4 uses RETENTION technique to hold register value in RETENTION mode, which makes M4 hold status in this power saving mode. When exit RETENTION mode, M4 continues to run without reset;

*2: Only support gint[5:0](default function only,not for alternate function) on chip level;

*3: Except gint[37] change to input pull down(can also be config to input pull up), gint[32], gint[33], tsi_ch[2] / gint[34], tsi_ch[3] / gint[35], tsi_ch[1] / gint[38], tsi_ch[10] / gint[39], gint[36] change to input pull up on chip level;

*4: Except WAKEUP, USBDET change to input pull down, POR change to input pull up on chip level;

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6.5 Memory Map and Registers

6.5.1 Module Memory Map

Table 6-2: CPM Memory Map

Address	31:16	15:0	Access[1]
0x0000	Sleep Config Register (SLPCFGR)		S
0x0004	Sleep Control Register (SLPCR)		S
0x0008	System Clock Divider Register (SCDIVR)		S
0x000c	Peripheral Clock Divider Register 1(PCDIVR1)		S
0x0010	Peripheral Clock Divider Register 2(PCDIVR2)		S
0x0014	Peripheral Clock Divider Register 3(PCDIVR3)		S
0x0018	Clock Divider Update Register(CDIVUPDR)		S
0x001c	Clock Divider Enable Register(CDIVENR)		S
0x0020	Oscillator Control and Status Register(OCSR)		S
0x0024	Clock Switch Config Register(CSWCFGR)		S
0x0028	Core Tick Timer Register(CTICKR)		S
0x002c	Chip Config Register(CHIPCFGR)		S
0x0030	Power Control Register(PWRCR)		S
0x0034	Sleep Counter Register(SLPCNTR)		S
0x0038	Wakeup Counter Register(WKPCNTR)		S
0x003c	Multiple Clock Gate Control Register (MULTICGTCR)		S
0x0040	System Clock Gate Control Register (SYSCGTCR)		S
0x0044	AHB3 Clock Gate Control Register (AHB3CGTCR)		S
0x0048	Arith Clock Gate Control Register (ARITHCGTCR)		S
0x004c	Ips Clock Gate Control Register (IPSCGTCR)		S
0x0050	VCC General Trim Register(VCCGTRIMR)		S
0x0054	VCC Lvdt Trim Register(VCCLTRIMR)		S
0x0058	VCC Vref Trim Register(VCCVTRIMR)		S
0x005c	VCC Core Test Mode Register(VCCCTMR)		S
0x0060	OSC8M Trim Register(O8MTRIMR)		S
0x0064	Reserved		S
0x0068	OSC120M Trim Register(O120MTRIMR)		S
0x006c	CARD LDO Trim Register(CARDTRIMR)		S
0x0070	OSCL Stable Time Register(OSCLSTIMER)		S
0x0074	OSCH Stable Time Register(OSCHSTIMER)		S
0x0078	OSCE Stable Time Register(OSCESTIMER)		S
0x007c	Power Status Register(PWRSR)		S
0x0080	Reserved		S
0x0084	Reserved		S
0x0088	Reserved		S
0x008c	RTC Trim Register(RTCTRIMR)		S
0x0090	PAD Wakeup Interrupt Control(PADWKPINTCR)		S
0x0094	Wakeup Filter Counter Register(WKPFILTCNTR)		S
0x0098	CARD Power On Counter Register(CARDPOCR)		S
0x009c	RTC32K Stable Time Register(RTCSTIMER)		S
0x00A0	Memory Power Down Sleep Control Register(MPDSLPCR)		S

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Address	31:16	15:0	Access[1]
0x00A4	Reserved		S
0x00A8	Reserved		S
0x00AC	Multiple Reset Control Register (MULTICGTCR)		S
0x00B0	System Reset Control Register (SYSCGTCR)		S
0x00B4	AHB3 Reset Control Register (AHB3CGTCR)		S
0x00B8	Arith Reset Control Register (ARITHCGTCR)		S
0x00BC	Ips Reset Control Register (IPSCGTCR)		S
0x00C0	Sleep Config Register 2(SLPCFGR2)		S
0x00C4	Reserved		S
0x00C8	Reserved		S
0x00CC	Reserved		S
0x00D0	Power Down Counter Register(PDNCNTR)		S
0x00D4	Power On Counter Register(PONCNTR)		S
0x00D8	PAD SS3 Control Register(PADSS3CR)		S
0x00DC	Wakeup Source Control Register(WKPSCR)		S

NOTE:

- S = supervisor-only access. User mode accesses to supervisor only address locations have no effect and result in a cycle termination transfer error.

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6.5.2 Register Description

6.5.2.1 Sleep Config Register(SLPCFGR)

Addr. Offset: 0x0000 ~ 0x0003 Reset Value: 0x1D6F8879

31	30	29	28	27	26	25	24
SLEEP_MODE[1:0]		Reserved	EPORT4_SLPEN	EPORT3_SLPEN	EPORT2_SLPEN	Reserved	
rw		ro	rw	rw	rw	ro	
23	22	21	20	19	18	17	16
Reserved	EPORT1_SLPEN	EPORT_SLPEN	OSCEXT_SLPEN	PMU128K_SLPEN	V33_HP_LPEN	HP_FLASH_LPEN	RTC32K_SLPEN
ro	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
CARD0PD	Reserved	CARD0PO		Reserved			
rw	ro	rw		ro			
7	6	5	4	3	2	1	0
VRFlashPD	Reserved		FLASH_IPSLP	Reserved			
rw	ro		rw	ro			

Figure 6-2: Sleep Config Register (SLPCFGR)

SLEEP_MODE — Sleep Mode for system to Enter When STOP

- 2'b00: Enter Low Power mode(with VDD_WK_SWOFF=0 & VDD_PD_RET=0 in SLPCFGR2).
- 2'b01: Enter Retention mode(with VDD_WK_SWOFF=0 & VDD_PD_RET=1 in SLPCFGR2)
- 2'b01: Enter Deep Sleep mode(with VDD_WK_SWOFF=1 & VDD_PD_RET=0 in SLPCFGR2)
- 2'b1x: Enter Hibernation mode(with VDD_WK_SWOFF=0 & VDD_PD_RET=0 in SLPCFGR2).

EPORT*_SLPEN — EPORT* Module Clock Sleep Enable When STOP

Specify if the EPORT* Module Clock is enabled and is switched to OSCL clock automatically in system sleep mode.

1 = enable

0 = disable

OSCEXT/PMU128K/RTC32K_SLPEN — OSCEXT/PMU128K/RTC32K Sleep

Enable When STOP

Specify if the OSCEXT/PMU128K/RTC32K is enabled in system sleep mode.

1 = enable

0 = disable

V33_HP_LPEN — VDD33/VDD33_FLASH LDO Enter Low Power State When STOP

Specify if the VDD33/VDD33_FLASH LDO enter low power state in system sleep mode.

1 = VRFlash 3.3V LDO enter low power enable

0 = VRFlash 3.3V LDO enter low power disable

HP_FLASH_LPEN — Internal FLASH HP LDO Power Down When STOP Specify if the Internal FLASH HP LDO enter low power in system sleep mode.

1 = FLASH HP LDO enter low power enable

0 = FLASH HP LDO enter low power disable

CARD0PD — Card 0 LDO Power OFF Config Specify if the Card 0 LDO is powered off.

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1 = power off

0 = power on

CARD0PO — Card 0 LDO Votage Out Config Specify the Card 0 LDO output voltage.

- 2'b00: 1.8V
- 2'b01: 3.0V
- 2'b10: 3.3V
- 2'b11: 5V

VRFlashPD — VDD33/VDD33_FLASH LDO Power OFF Config When STOP Specify if VDD33/VDD33_FLASH LDO is powered off in system sleep mode.

1 = power off, 0 = power on

FLASH_IPSLP — Flash IP Sleep Config When STOP

Specify if the Flash IP enter sleep mode in system sleep mode.

1 = enter sleep mode

0 = not enter sleep mode

6.5.2.2 Sleep Control Register (SLPCR)

Addr. Offset: 0x0004 ~ 0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
SLP_CFG_KEY[1:0]		SLP_CFG_MODE	Reserved				
			ro				

23	22	21	20	19	18	17	16
Reserved							
ro							

15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 6-3: Sleep Control Register (SLPCR)

SLP_CFG_KEY[1:0] — SLP CFG Mode Key.

If SLP_PROTECT_EN bit in **VCCVTRIMR** is set,the **SLPCFGR** register can not be changed,unless the correct sequence write in these two bits.The right sequence is

:2'b01->2'b10->2'b11.After write these two bits following this sequence,these two bits' value is 2'b11,then the **SLPCFGR** register can be changed at will.Only writes 2'b00 can clear these two bits when the value equals to 2'b11.Writes other value has no effect and returns 2'b11.

SLP_CFG_MODE — SLP CFG Mode

After write the SLP_CFG_KEY following the sequence,this bit will be 1.

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6.5.2.3 System Clock Divider Register (SCDIVR)

Addr. Offset: 0x0008 ~ 0x000B Reset Value: 0x00090003

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
CLKOUT_DIV[7:0]							
rw							
15	14	13	12	11	10	9	8
TRACE_DIV[7:0]							
rw							
7	6	5	4	3	2	1	0
SYS_DIV[7:0]							
rw							

Figure 6-4: System Clock Divider Register (SCDIVR)

*_DIV[n:0] — Clock Divider Divisor Configure

Specify the divide value of *_CLK. The divisor value is *_DIV[n:0]+1.

Write 1 to SYSDIV_UPDATE bit in **CDIVUPDR** will update the configured SCDIVR divisor to DIVIDER. read always return 0.

NOTE: If value of this register has been changed by flash info load, user need to set bit11 (OVERWR_SCDIV_TRIM) in VCCCTMR before CPU overwriting the old value.

6.5.2.4 Peripheral Clock Divider Register 1

Addr. Offset: 0x000C ~ 0x000F Reset Value: 0x50E21111

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
ARITH_DIV[3:0]				AHB3_DIV[3:0]			
rw				rw			
7	6	5	4	3	2	1	0
Reserved				IPS_DIV[3:0]			
ro				rw			

Figure 6-5: Peripheral Clock Divider Register 1(PCDIVR1)

*_DIV[n:0] — Clock Divider Divisor Configure

Specify the divide value of *_CLK. The divisor value is *_DIV[n:0]+1.

Write 1 to PERDIV_UPDATE bit in **CDIVUPDR** will update the configured PCDIVR1,PCDIVR2,PCDIVR3 divisor to DIVIDER. read always return 0.

NOTE: If value of this register has been changed by flash info load, user need to set bit10(OVERWR_PCDIV_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.5 Peripheral Clock Divider Register 2

Addr. Offset: 0x0010 ~ 0x0013 Reset Value: 0x0003E12B

31	30	29	28	27	26	25	24
TC_DIV[3:0]				Reserved			
rw				ro			
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 6-6: Peripheral Clock Divider Register 2(PCDIVR2)

*_DIV[n:0] — Clock Divider Divisor Configure

Specify the divide value of *_CLK. The divisor value is *_DIV[n:0]+1

Write 1 to PERDIV_UPDATE bit in **CDIVUPDR** will update the configured PCDIVR1, PCDIVR2, PCDIVR3 divisor to DIVIDER. Read always return 0.

6.5.2.6 Peripheral Clock Divider Register 3

Addr. Offset: 0x0014 ~ 0x0017 Reset Value: 0x02180EBE

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 6-7: Peripheral Clock Divider Register 3(PCDIVR3)

*_DIV[n:0] — Clock Divider Divisor Configure

Specify the divide value of *_CLK. The divisor value is *_DIV[n:0]+1

Write 1 to PERDIV_UPDATE bit in **CDIVUPDR** will update the configured PCDIVR1,PCDIVR2,PCDIVR3 divisor to DIVIDER. read always return 0.

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6.5.2.7 Clock Divider Update Register

Addr. Offset: 0x0018 ~ 0x001B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved						SYSDIV_UPD	PERDIV_UPD
ro						rw	rw

Figure 6-8: Clock Divider Update Register(CDIVUPDR)

*DIV_UPD — Clock Divider Update

Write 1 to DIVUPD bit,the clock divider and divider enable config specified in **SCDIVR**&**PCDIVR1 &PCDIVR2&PCDIVR3&CDIVENR** will take effect. Read *DIV_UPD bit returns 0.

6.5.2.8 Clock Divider Enable Register

Addr. Offset: 0x001C ~ 0x001F Reset Value: 0x001fffff

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CLKOUT_DIVEN	TRACE_DIVEN	TC_DIVEN	Reserved	Reserved	ADC_DIVEN	Reserved	Reserved
rw	rw	rw	ro	ro	rw	ro	ro
7	6	5	4	3	2	1	0
Reserved				ARITH_DIVEN	AHB3_DIVEN	Reserved	IPS_DIVEN
ro				rw	rw	ro	rw

Figure 6-9: Clock Divider Enable Register(CDIVENR)

*DIVEN — Clock Divider Enable

Specify the clock divider if enabled. When not enabled, no clock out from clock divider.Write 1 to *DIV_UPDATE bit in **CDIVUPDR** will update the configured enable bit to DIVIDER.

1 = Clock Divider enabled

0 = Clock Divider disabled

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6.5.2.9 Oscillator Control and Status Register

Addr. Offset: 0x0020 ~ 0x0023 Reset Value: 0x00000B6B/00006B6B

31	30	29	28	27	26	25	24
Reserved				TRNG_OSCEN[3:0]			
ro				rw			
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved	PMU2K_VALID	RTC32K_STABLE	OSCEXT_STABLE	OSC120M_STABLE	USBPHY240M_STABLE	PMU128K_STABLE	OSC8M_STABLE
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
Reserved	PMU2K_EN	RTC32K_EN	OSCEXT_EN	OSC120M_EN	USBPHY240M_EN	PMU128K_EN	OSC8M_EN
ro	rw	rw	rw	rw	rw	rw	rw

Figure 6-10: Oscillator Control and Status Register (OCSR)

*_STABLE — Clock Source Stable

1 = Clock Source Stable

0 = Clock Source not Stable

*_EN — Clock Source Enable

1 = Clock Source Enable

0 = Clock Source Disable

NOTE: If value of TRNG_OSCEN and *_EN this register has been changed by flash info load, user need to set bit9(OVERWR_OCSR_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.10 Clock Switch Config Register

Addr. Offset: 0x0024 ~ 0x0027 Reset Value: 0x10515201

31	30	29	28	27	26	25	24
CLKOUT_SEL_ST				Reserved		CLKOUT_SEL[1:0]	
ro				ro		rw	
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved						SYS_SEL_ST[1:0]	
ro						ro	
7	6	5	4	3	2	1	0
Reserved							SYS_SEL
ro							rw

Figure 6-11: Clock Switch Config Register(CSWCFGR)

*_SEL_ST — Clock Select Status Bit

These bits reflect which source the clock switch is being selected at the moment.These bits are in one hot pattern.For example,SYS_SEL_ST:

- 2'b01: OSC8M is being selected
- 2'b10: OSC120M is being selected

SYS_SEL — SoC Clock Source Select
When software want to switch the clock source to SoC, must ensure the current selected clock and the next selected clock source are active.

- 1'b0: Select the OSC8M as SoC clock source
- 1'b1: Select the OSC120M as SoC clock source

CLKOUT_SEL — Clock Output to PAD Select

- 2'b00: Select system clock
- 2'b01: Select arith_clk clock
- 2'b10: Reserved
- 2'b11: Select RTC 32K clock

NOTE: If value of CLKOUT_SEL/SYS_SEL has been changed by flash info load, user need to set bit29(OVERWR_CSWCFG_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.11 Core Tick Timer Register

Addr. Offset: 0x0028 ~ 0x002B Reset Value: 0x01000147

31	30	29	28	27	26	25	24
Reserved						NO_REF	SKEW
ro						rw	rw
23	22	21	20	19	18	17	16
CALIB[23:16]							
rw							
15	14	13	12	11	10	9	8
CALIB[15:8]							
rw							
7	6	5	4	3	2	1	0
CALIB[7:0]							
rw							

Figure 6-12: Core Tick Timer Register (CTICKR)

NO_REF — no alternative reference bit.

Indicates that no alternative reference clock source has been integrated.

1 = CORE_CLK is selected as reference clock source.

0 = RTC32K is selected as reference clock source.

SKEW — skew bit.

1 = not guarantee an exact multiple of 10ms.

0 = guarantee an exact multiple of 10ms.

CALIB — calibration bit.

Provides an integer value to compute a 10ms (100Hz) delay.

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6.5.2.12 Chip Config Register

Addr. Offset: 0x002C ~ 0x002F Reset Value: 0x3e8dfbf7

31	30	29	28	27	26	25	24
USBPHY_OSC_MODE[1:0]		USBPHY_CFG_SRM	USBPHY_PLL_SRM	Reserved	USBPHY_O_ISOEN	USBPHY_RSTMASK	USBPHY_PSWEN
rw		rw	rw	ro	rw	rw	rw
23	22	21	20	19	18	17	16
USBPHY_IP_SRM	M2S_INT	S2M_INT	S2M_INT_CLR	S2M_O_ISOEN	S2M_PST_ISOEN	ATIMER_H2L_ISOSEL	ATIMER_H2L_ISOEN
rw	rw	ro	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved	RTC1S_CLK_GTE	RTC1K_CLK_GTE	RTC32K_CLK_GTE	RTC32K_ISO_GTE	S2M_INT_EN	Reserved	
ro	rw	rw	rw	rw	rw	ro	
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 6-13: Chip Config Register(CHIPCFGR)

PCI_H2L_ISOEN — PCI H2L Isolation Enable.

1 = Enable

0 = Disable

PCI_H2L_ISOSEL — PCI H2L Isolation Select.

This bit indicates which isolation bit to select for PCI H2L isolation.

1 = Control bit PCI_H2L_ISOEN in CPM module is select

0 = Control bit in PCI module is select

USBPHY_O_ISOEN — USBPHY Output Isolation Enable.

1 = Enable

0 = Disable

USBPHY_RSTMASK — USBPHY Reset Signal Mask.

1 = Reset masked to assert

0 = Reset not masked

USBPHY_PSWEN — USBPHY Power Switch Enable.

1 = Enable

0 = Disable

*_SRM — Software Reset Mask.

Set *_SRM to reset corresponding circuit.

USBPHY_OSC_MODE[1:0] — Select USBPHY OSC Mode

00: Auto Dection Oscillator

01: Auto Dection Oscillator(only for fast simulation)

10: Select Internal Oscillator

11: Select External Oscillator

M2S_INT — Main System to Sub System Interrupt.

1 = assert the interrupt,

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0 = clear the interrupt

S2M_INT — Sub System to Main System Interrupt.

From this bit,Main System can read the status of S2M interrupt which is sent by S2M_INT bit of **SRSTMASKR** in CPM of Sub System.

S2M_INT_CLR — Sub System to Main System Interrupt Clear.

Set this bit to clear S2M interrupt until S2M_INT is clear.After S2M_INT is clear,clear this bit or S2M interrupt can not be set again.

S2M_INT_EN — Sub System to Main System Interrupt Enable.

Set this bit to enable S2M interrupt.

S2M_O_ISOEN — Sub System to Main System Output Isolation Enable.

1 = Enable, 0 = Disable

S2M_PST_ISOEN — Sub System to Main System Power Status Isolation Enable.

Clear this bit and read SUB_SYS_PWRRDY in PWRSR to get Sub System Power Status

1 = Enable, 0 = Disable

*_CLK_GTE — * Clock Enable.

1 = Enable, 0 = Disable

6.5.2.13 Power Control Register

Addr. Offset: 0x0030 ~ 0x0033 Reset Value: 0x2600008F

31	30	29	28	27	26	25	24
VCC_CLR_LATCH	VCC_SET_LATCH	VCC_RE_LVDT5	VCC_RE_LVDT18	Reserved	VCARD0_ISOEN	Reserved	
rw	rw	rw	rw	ro	rw	ro	
23	22	21	20	19	18	17	16
LVD5V_PD_CHIP	Reserved						
rw	ro						
15	14	13	12	11	10	9	8
VCC_IE_LVDT5	VCC_IE_LVDT18	Reserved	CARD0_IE_LVD	Reserved	CARD0_IE_EN_FAIL	Reserved	CARD0_RE_LVD
ro	rw	ro	rw	ro	rw	ro	rw
7	6	5	4	3	2	1	0
VCC_OE_LVDT5	VCC_OE_LVDT18	Reserved	CARD0_OE_LVD	VCC_EN_LVDT5	VCC_EN_LVDT18	Reserved	CARD0_LVD_EN
rw	rw	ro	rw	rw	rw	ro	rw

Figure 6-14: Power Control Register(PWRCR)

VCC_CLR_LATCH — VCC Clear IO latch.

Write 1 to this bit to clear IO latch.

VCC_SET_LATCH — VCC Set IO latch.

Write 1 to this bit to set IO latch.

VCARD*_ISOEN — VCARD0 interface Isolation Enable.

1 = Enable

0 = Disable

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LVD5V_PD_CHIP — LVD5V power down chip.

1 = When LVD5V detected by PMU, enter Power Off2 mode

0 = When LVD5V detected by PMU, keep current state of Chip

_RE_LVDT — Lvdt* Reset Enable.

1 = Enable, 0 = Disable

_IE_LVDT — Lvdt* Interrupt Enable.

1 = Enable, 0 = Disable

_OE_LVDT — Lvdt* Output Enable.

1 = Enable, 0 = Disable

_EN_LVDT — Lvdt* Enable.

1 = Enable, 0 = Disable

6.5.2.14 Sleep Counter Register

Addr. Offset: 0x0034~ 0x0037 Reset Value: 0xffffffff

31	30	29	28	27	26	25	24
SLPCNT0[7:0]							
rw							
23	22	21	20	19	18	17	16
SLPCNT1[7:0]							
rw							
15	14	13	12	11	10	9	8
SLPCNT2[7:0]							
rw							
7	6	5	4	3	2	1	0
SLPCNT3[7:0]							
rw							

Figure 6-15: Sleep Counter Register(SLPCNTR)

SLPCNT* — Sleep Counter.

Specify the delay time when the system enters sleep mode.

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6.5.2.15 Wakeup Counter Register

Addr. Offset: 0x0038~ 0x003B Reset Value: 0xffffffff

31	30	29	28	27	26	25	24
WKPCNT0[7:0]							
rw							
23	22	21	20	19	18	17	16
WKPCNT1[7:0]							
rw							
15	14	13	12	11	10	9	8
WKPCNT2[7:0]							
rw							
7	6	5	4	3	2	1	0
WKPCNT3[7:0]							
rw							

Figure 6-16: Wakeup Counter Register(WKPCNTR)

WKPCNT* — Wakeup Counter.

Specify the delay time when the system wakes up from sleep mode.

6.5.2.16 Multiple Clock Gate Control Register

Addr. Offset: 0x003C ~ 0x003F Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
MULTICLK_GTE[31:24]							
rw							
23	22	21	20	19	18	17	16
MULTICLK_GTE[23:16]							
rw							
15	14	13	12	11	10	9	8
MULTICLK_GTE[15:8]							
rw							
7	6	5	4	3	2	1	0
MULTICLK_GTE[7:0]							
rw							

Figure 6-17: Multiple Clock Gate Control Register(MULTICGTCR)

MULTICLK_GTE[31:0] — Multiple Clock Gate Enable Control Register

Specify the enable bit values of modules' clock gates. (see **Table 6-2 MULTICLK_GTE[31:0] Bits Corresponding Modules**).

1 = Module Clock Enabled, 0 = Module Clock Disabled

NOTE: If value of clkout in this register has been changed by flash info load, user need to set bit13(OVERWR_ARITHCGT_TRIM) in VCCCTMR before CPU overwriting the old value.

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Table 6-3: MULTICLK_GTE[31:0] Bits Corresponding Modules

GTE Bit	Corresponding Module
0	Reserved
1	EFM_BUS
2	Reserved
3	RSVD
4	Reserved
5	MCC
6	MCCADR
7	ADC
8	RSVD
9	MESH
10	TC
11	RSVD
12	RSVD
13	RSVD
14	RSVD
15	CLKOUT
16	KEY_CTRL
17	RSVD
18	EFM_IPS
19	CPM_IPS
20	EPORT
21	EPORT1
22	EPORT2
23	EPORT3
24	EPORT4
25	RSVD
26	TRACE
27	RESERVED
28	RESERVED
29	RESERVED
30	RSVD
31	RSVD

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6.5.2.17 System Clock Gate Control Register

Addr. Offset: 0x0040 ~ 0x0043 Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
SYSCLK_GTE[31:24]							
rw							
23	22	21	20	19	18	17	16
SYSCLK_GTE[23:16]							
rw							
15	14	13	12	11	10	9	8
SYSCLK_GTE[15:8]							
rw							
7	6	5	4	3	2	1	0
SYSCLK_GTE[7: 0]							
rw							

Figure 6-18: System Clock Gate Control Register(SYSCGTCR)

SYSCLK_GTE[31:0] — System Clock Gate Enable Control Register

Specify the enable bit values of modules' clock gates. (see **Table 6-3 SYSCLK_GTE[31:0] Bits Corresponding Modules**).

1 = Module Clock Enabled, 0 = Module Clock Disabled

Table 6-4: SYSCLK_GTE[31:0] Bits Corresponding Modules

GTE Bit	Corresponding Module
0	Reserved
1	DMAC1
2	DMAC2
3	RSVD
4	CRC0
5	CRC1
6	RSVD
7	Reserved
8	RSVD
9	RSVD
10	RSVD
11	AHB2_MUX
12	SRAMD
13	SRAM0
14	SRAM1
15	SRAM2
16	SRAM3
17	SSI4
18	SSI5
19	ROM
20	RESERVED
21	RESERVED
22	M2S_BUS_M

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GTE Bit	Corresponding Module
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

6.5.2.18 AHB3 Clock Gate Control Register

Addr. Offset: 0x0044 ~ 0x0047 Reset Value: 0xFFFFFFFF

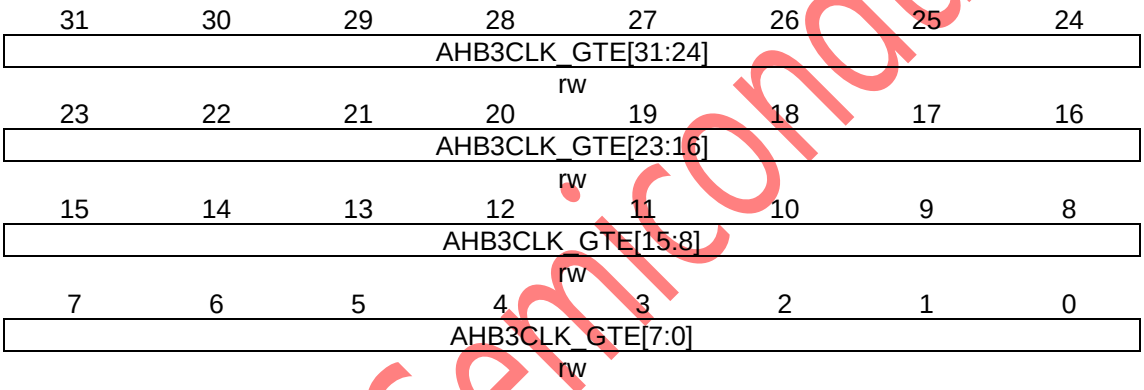


Figure 6-19: AHB3 Clock Gate Control Register (AHB3CGTCR)

AHB3CLK_GTE[31:0] — AHB3 Clock Gate Enable Control Register

Specify the enable bit values of modules' clock gates. (see **Table 6-4 AHB3CLK_GTE[31:0] Bits Corresponding Modules**).

1 = Module Clock Enabled, 0 = Module Clock Disabled

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Table 6-5: AHB3CLK_GTE[31:0] Bits Corresponding Modules

GTE Bit	Corresponding Module
0	RSVD
1	RSVD
2	RSVD
3	USBC
4	RSVD
5	AHB3_MUX
6	Reserved
7	Reserved
8	Reserved
9	Reserved
10	Reserved
11	Reserved
12	Reserved
13	Reserved
14	Reserved
15	Reserved
16	Reserved
17	Reserved
18	Reserved
19	Reserved
20	Reserved
21	Reserved
22	Reserved
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

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6.5.2.19 Arith Clock Gate Control Register

Addr. Offset: 0x0048 ~ 0x004B Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
ARITHCLK_GTE[31:24]							
rw							
23	22	21	20	19	18	17	16
ARITHCLK_GTE[23:16]							
rw							
15	14	13	12	11	10	9	8
ARITHCLK_GTE[15:8]							
rw							
7	6	5	4	3	2	1	0
ARITHCLK_GTE[7:0]							
rw							

Figure 6-20: Arith Clock Gate Control Register (ARITHCGTCR)

ARITHCLK_GTE[31:0] — ARITH Clock Gate Enable Control Register

Specify the enable bit values of modules' clock gates. (see **Table 6-5 ARITHCLK_GTE[31:0] Bits Corresponding Modules**).

1 = Module Clock Enabled, 0 = Module Clock Disabled

NOTE: If value of this register has been changed by flash info load, user need to set bit13(OVERWR_ARITHCGT_TRIM) in VCCCTMR before CPU overwriting the old value.

Table 6-6: ARITHCLK_GTE[31:0] Bits Corresponding Modules

GTE Bit	Corresponding Module
0	RESERVED
1	AES
2	SM4
3	RESERVED
4	RF for AES&SM4
5	CRYPTO
6	SHA
7	EDMAC0
8	DES
9	ZUC
10	AHB2MLB
11	AHB2IPS2
12	Reserved
13	Reserved
14	Reserved
15	Reserved
16	Reserved
17	Reserved
18	Reserved
19	Reserved
20	Reserved

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GTE Bit	Corresponding Module
21	Reserved
22	Reserved
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

6.5.2.20 Ips Clock Gate Control Register

Addr. Offset: 0x004C ~ 0x004F Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
IPSCLK_GTE[31:24]							
rw							
23	22	21	20	19	18	17	16
IPSCLK_GTE[23:16]							
rw							
15	14	13	12	11	10	9	8
IPSCLK_GTE[15:8]							
rw							
7	6	5	4	3	2	1	0
IPSCLK_GTE[7: 0]							
rw							

Figure 6-21: Ips Clock Gate Control Register (IPSCGTCR)

IPSCLK_GTE[31:0] — IPS Clock Gate Enable Control Register

Specify the enable bit values of modules' clock gates. (see **Table 6-6 IPSCLK_GTE[31:0] Bits Corresponding Modules**).

1 = Module Clock Enabled, 0 = Module Clock Disabled

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Table 6-7: IPSCLK_GTE[31:0] Bits Corresponding Modules

GTE Bit	Corresponding Module
0	IO_CTRL
1	WDT
2	RTC
3	PIT1
4	PIT2
5	USI1
6	EDMAC1
7	SPI1
8	SPI2
9	SPI3
10	SCI1
11	SCI2
12	USI2
13	RSVD
14	I2C1
15	PWM0
16	I2C2
17	I2C3
18	SCI3
19	RSVD
20	QADC
21	DAC
22	MCC
23	TSI
24	LD
25	TRNG
26	PGD
27	SEC_DET
28	PCI
29	PMURTC
30	AHB2IPS
31	CCM&RESET

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6.5.2.21 VCC General Trim Register

Addr. Offset: 0x0050 ~ 0x0053 Reset Value: 0x8E8E0331

31	30	29	28	27	26	25	24
V33_SW_ENB	DISCHARGE_EN	Reserved	TRIM_128KHZ[4:0]				
rw	rw	ro	rw				
23	22	21	20	19	18	17	16
2KHZ_GTE	Reserved		TRIM_2KHZ[4:0]				
rw	ro		rw				
15	14	13	12	11	10	9	8
ST_1V	Reserved	VCC_LATCH_AUTO_SET	VCC_LATCH_AUTO_CLR	Reserved		SAMPLE_DIV[1:0]	
rw	ro	rw	rw	ro		rw	
7	6	5	4	3	2	1	0
TEST_BIAS	BIAS1_RES_TRIM[2:0]			Reserved		BIAS2_RES_TRIM[1:0]	
rw	rw			ro		rw	

Figure 6-22: VCC General Trim Register(VCCGTRIMR)

Trim bits for PMC VCC.

V33_SW_ENB — V33 switch enable

when vdd5v<3.6V, set this bit to low will let vd33/vd33_flash=vdd5v.

DISCHARGE_EN — discharge enable

when this bit is set,discharge vd33 when chip switch to poff2 mode.

TRIM_128KHz — trim the 128KHz clk. TRIM_2KHz — trim the 2KHz clk.

2KHZ_GTE — 2KHz clk gate enable

1 = enable

0 = disable

VCC_LATCH_AUTO_SET — VCC auto Set IO latch when system enter sleep mode.

Write 1 to this bit to enable IO auto latch set.

VCC_LATCH_AUTO_CLR — VCC auto clear IO latch when system exit sleep mode.

Write 1 to this bit to enable IO auto latch clear.

ST_1V — when this bit is set, the core voltage is 0.9V.

SAMPLE_DIV — bg wake-up period trim.

TEST_BIAS — test the bias current enable signal.

BIAS1_RES_TRIM — triming bias1 current.

BIAS2_RES_TRIM — core voltage scaling function.

b'00 is 1.05V; b'01 is 1.1V; b'10 is 1.15V; b'11 is 1.21V

NOTE: User need to set bit23(OVERWR_VCC_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.22 VCC Lvdt Trim Register

Addr. Offset: 0x0054 ~ 0x0057 Reset Value: 0xC1001414

31	30	29	28	27	26	25	24
LVDT5V_TRIM_MSB[1:0]		Reserved					LVDT_COARSE_EN
rw		ro					rw
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved		LVDT5V_HYS_TRIM[1:0]		Reserved	LVDT5V_TRIM[2:0]		
ro		rw		ro	rw		
7	6	5	4	3	2	1	0
Reserved		LVDT18_HYS_TRIM[1:0]		Reserved	LVDT18_TRIM[2:0]		
ro		rw		ro	rw		

Figure 6-23: VCC Lvdt Trim Register(VCCLTRIMR)

Trim bits for PMC VCC.

LVDT5V_TRIM_MSB — trim bits for lvdt 5v

b'00 is~4.3V; b'01 is ~2.7V; b'10 is ~2.0V; b'11 is~1.6V

LVDT_COARSE_EN — coarse lvdt module enable signal

1 = enable coarse lvdt module

0 = disable coarse lvdt module

LVDT5V_HYS_TRIM — triming hystersis of lvdt 5v 20mv~40mv; 10mv one step

LVDT5V_TRIM — trim bits for lvdt 5v

When LVDT5V_TRIM_MSB=00,lvdt trig point is as below:

3'b000: 3.22V

3'b001: 3.30V

3'b010: 3.39V

3'b011: 3.48V

3'b100: 3.58V

3'b101: 3.68V

3'b110: 4.02V

3'b111: 4.15V

LVDT18_HYS_TRIM — triming hystersis of lvdt 1p5v 20mv~40mv; 10mv onestep

LVDT18_TRIM — trim bits for lvdt 1p8v

NOTE: User need to set bit22(OVERWR_LVD_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.23 VCC Vref Trim Register

Addr. Offset: 0x0058 ~ 0x005B Reset Value: 0x00070008

31	30	29	28	27	26	25	24
SLP_PROTECT_EN				Reserved			
rw				ro			
23	22	21	20	19	18	17	16
Reserved					WK_EN[2:0]		
		ro				rw	
15	14	13	12	11	10	9	8
Reserved		S2M_O_ISOEN_SRAM_MASK	S2M_O_ISOEN_VDD_MASK	VREF_FORCE_ST	VREF_TRIM_EN	VREF_TRIM_LOAD	VREF_STORE_EN
ro		rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved				VREF_TRIM_VAL			
ro				rw			

Figure 6-24: VCC Vref Trim Register(VCCVTRIMR)

Trim bits for PMC VCC.

SLP_PROTECT_EN — Sleep Cfg Reg protect enable bit

WK_EN[2:0] — wakeup source enable bits for PAD, POR,WAKE and USBDET respectively in poff2 mode,write 1 to VREF_TRIM_LOAD bit to confirm any change in these bits

S2M_O_ISOEN_SRAM_MASK — set this bit to disable c0sub to main system isolation for c0sub VDD_SRAM power domain signal

S2M_O_ISOEN_VDD_MASK — set this bit to disable c0sub to main system isolation for c0sub VDD power domain signal

VREF_FORCE_ST — let vref stable when encounter pesd VREF_TRIM_EN — used to trim the 1V vref

1 = enable vref trim

0 = disable vref trim

VREF_TRIM_LOAD — load the vref trimming value and wakeup source enable bits

VREF_STORE_EN — store the vref voltage value enable

1 = enable to store the vref voltage value

0 = disable to store the vref voltage value VREF_TRIM_VAL — trim the 1v vref

NOTE: If value of SLP_PROTECT_EN has been changed by flash info load, user need to set bit21(OVERWR_VREF_TRIM) in VCCCTMR before CPU overwriting the old value.

If user want to overwrite value of SLP_PROTECT_EN / WK_EN / VREF_TRIM_EN

/ VREF_STORE_EN / VREF_TRIM_VAL by cpu overwriting, user need to set bit21(OVERWR_VREF_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.24 VCC Core Test Mode Register

Addr. Offset: 0x005C ~ 0x005F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
CORE_TEST_KEY[1:0]		OVERWR_CSWCFG_TRIM	OVERWR_RTC_TRIM	Reserved	OVERWR_RTC_STABLE_TRIM	Reserved	OVERWR_CARDLDO_TRIM
rw		rw	rw	ro	rw	ro	rw
23	22	21	20	19	18	17	16
OVERWR_VCC_TRIM	OVERWR_LVD_TRIM	OVERWR_VREF_TRIM	OVERWR_OSC8M_TRIM	OVERWR_OSC120M_TRIM	OVERWR_OSCL_STABLE_TRIM	OVERWR_OSCH_STABLE_TRIM	OVERWR_OSCE_STABLE_TRIM
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved		OVERWR_ARITHCGT_TRIM	Reserved	OVERWR_SCDIV_TRIM	OVERWR_PCDIV_TRIM	OVERWR_OCSR_TRIM	OFF_MODE_WK
ro		rw	ro	rw	rw	rw	rw
7	6	5	4	3	2	1	0
TEST_MODE_EN	Reserved			SOFT_POR	OFF_MODE2	OFF_MODE	EN_LP
rw	ro			rw	rw	rw	rw

Figure 6-25: VCC Core Test Mode Register(VCCCTMR)

CORE_TEST_KEY[1:0] — Core Test Mode Key.

The **VCCCTMR** register can not be changed,unless the correct sequence write in these two bits.The right sequence is :2'b01->2'b10->2'b11.After write these two bits following this sequence,these two bits' value is 2'b11,then the **VCCCTMR** register can be changed at will.Only writes 2'b00 can clear these two bits when the value equals to 2'b11.Writes other value has no effect and returns 2'b11.

SOFT_POR[1:0] — Software POR Bit.

Set this bit the generate a software POR event to the Main System.

1 = soft por

0 = not soft por

OVERWR_CSWCFG_TRIM — Enable CPU to Overwrite the Trimmed Value for CSWCFGR.

NOTE: Only SYSCLK_SEL[1:0] being trimmed to 2'b00 is supported.

1 = enable

0 = disable

OVERWR_RTC_TRIM — Enable CPU to Overwrite the Trimmed Value for RTCTRIMR.

1 = enable

0 = disable

OVERWR_RTC_STABLE_TRIM — Enable CPU to Overwrite the Trimmed Value for RTCSTIMER.

1 = enable

0 = disable

OVERWR_CARDLDO_TRIM — Enable CPU to Overwrite the Trimmed Value for Card 0/1 LDO and Analog Wakeup Filter

1 = enable

0 = disable

OVERWR_VCC_TRIM — Enable CPU to Overwrite the Trimmed Value for VCC from FLASH

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1 = enable

0 = disable

OVERWR_LVD_TRIM — Enable CPU to Overwrite the Trimmed Value for LVD from FLASH

1 = enable

0 = disable

OVERWR_VREF_TRIM — Enable CPU to Overwrite the Trimmed Value for VREF from FLASH

1 = enable

0 = disable

OVERWR_OSC8M_TRIM — Enable CPU to Overwrite the Trimmed Value for OSC8M from FLASH

1 = enable

0 = disable

OVERWR_OSC120M_TRIM — Enable CPU to Overwrite the Trimmed Value for OSC120M from FLASH

1 = enable

0 = disable

OVERWR_OSCL_STABLE_TRIM — Enable CPU to Overwrite the Trimmed Value for OSC8M Stable Time from FLASH

1 = enable

0 = disable

OVERWR_OSCH_STABLE_TRIM — Enable CPU to Overwrite the Trimmed Value for OSC120M Stable Time from FLASH

1 = enable

0 = disable

OVERWR_OSCE_STABLE_TRIM — Enable CPU to Overwrite the Trimmed Value for External OSC Stable Time from FLASH

1 = enable

0 = disable

OVERWR_ARITHCGTR_TRIM — Enable CPU to Overwrite the Trimmed Value for ARITHCGTCR.

1 = enable

0 = disable

OVERWR_SCDIV_TRIM — Enable CPU to Overwrite the Trimmed Value for SCDIVR.

1 = enable

0 = disable

OVERWR_PCDIV_TRIM — Enable CPU to Overwrite the Trimmed Value for PCDIVR.

1 = enable

0 = disable

OVERWR_OCSR_TRIM — Enable CPU to Overwrite the Trimmed Value for OCSR.

1 = enable

0 = disable

TEST_MODE_EN — Enable CPU Core Test Mode of PMC VCC.

1 = enable

0 = disable

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OFF_MODE2&OFF_MODE&EN_LP are Direct control bits of the PMC VCC in core test mode only.Software should not change the value of these bits .

6.5.2.25 OSC8MHz Trim Register

Addr. Offset: 0x0060 ~ 0x0063 Reset Value: 0x00018779

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							OSC8M_TRIM[16]
ro							rw
15	14	13	12	11	10	9	8
OSC8M_TRIM[15:8]							
rw							
7	6	5	4	3	2	1	0
OSC8M_TRIM[7:0]							
rw							

Figure 6-26: OSC8MHz Trim Register(O8MTRIMR)

OSC8M_TRIM[16:0] — OSC8MHz Trim Value.

NOTE: User need to set bit20(OVERWR_OSC8M_TRIM) in VCCCTMR before CPU overwriting the old value.

6.5.2.26 OSC120MHz Trim Register

Addr. Offset: 0x0068 ~ 0x006B Reset Value: 0x0001877A

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							OSC120M_TRIM[16]
ro							rw
15	14	13	12	11	10	9	8
OSC120M_TRIM[15:8]							
rw							
7	6	5	4	3	2	1	0
OSC120M_TRIM[7:0]							
rw							

Figure 6-27: OSC120MHz Trim Register(O120MTRIMR)

OSC120M_TRIM[16:0] — OSC120MHz Trim Value.

NOTE: User need to set bit19(OVERWR_OSC120M_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.27 CARD LDO Trim Register

Addr. Offset: 0x006C ~ 0x006F Reset Value: 0x10700688

31	30	29	28	27	26	25	24
Reserved	WKP_DFILT_EN	WKP_DFILT_BYPASS	WKP_DFILT_GTE	Reserved			
ro	rw	rw	rw	ro			
23	22	21	20	19	18	17	16
EFM_RWSC_INIT[3:0]				Reserved			
rw				ro			
15	14	13	12	11	10	9	8
WKP_AFILT_BYPASS	Reserved		CARD0_REDUCE	WKP_AFILT_TRIM[3:0]			
rw	ro		rw	rw			
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 6-28: Card LDO Trim Register(CARDTRIMR)

WKP_DFILT_EN — Wakeup Digital Filter Enable.

1 = enable

0 = disable

WKP_DFILT_BYPASS — Wakeup Digital Filter Bypass.

1 = bypass

0 = not bypass

WKP_DFILT_GTE — Wakeup Digital Filter Clock Gate Enable.

1 = enable

0 = disable

EFM_RWSC_INIT[3:0] — EFM RWSC Initial Trim Value

WKP_AFILT_BYPASS — Wakeup Analog Filter Bypass.

1 = bypass

0 = not bypass

WKP_AFILT_TRIM[3:0] — Wakeup Analog Filter Trim Value.

CARD*_RREDUCE — When high will reduce 10% resistance of the current sensor.

1 = reduce

0 = not reduce

NOTE: If user wan to overwrite value of EFM_RWSC_INIT / WKP_AFILT_BYPASS / CARD0_REDUCE / WKP_AFILT_TRIM by cpu writing, user need to set bit24(OVERWR_CARDLDO_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.28 OSCL Stable Time Register

Addr. Offset: 0x0070 ~ 0x0073 Reset Value: 0x000301FF

31	30	29	28	27	26	25	24
Reserved				PMU128K_STIME[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
PMU 128K_STIME[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				OSC8M_STIME[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
OSC8M_STIME[7:0]							
rw							

Figure 6-29: OSCL Stable Time Register(OSCLSTIMER)

*_STIME — * Clock Source Stable Time.

Specify the stable time of * Clock Source when enable bit in OCSR is turned on.

NOTE: If user want to overwrite value of OSC8M_STIME by cpu wrinting, user need to set bit18(OVERWR_OSCL_STABLE_TRIM) in VCCCTMR before CPU overwriting the old value.

6.5.2.29 OSCH Stable Time Register

Addr. Offset: 0x0074 ~ 0x0077 Reset Value: 0x003f01ff

31	30	29	28	27	26	25	24
Reserved				OSC120M_STIME[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
OSC120M_STIME[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				USBPHY240M_STIME[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
USBPHY240M_STIME[7:0]							
rw							

Figure 6-30: OSCH Stable Time Register(OSCHSTIMER)

*_STIME — * Clock Source Stable Time.

Specify the stable time of * Clock Source when enable bit in **OCSR** is turned on.

NOTE: User need to set bit17(OVERWR_OSCH_STABLE_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.30 OSCE Stable Time Register

Addr. Offset: 0x0078 ~ 0x007B Reset Value: 0x000FFFFF

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				OSCEXT_STIME[19:16]			
ro				rw			
15	14	13	12	11	10	9	8
OSCEXT_STIME[15:8]							
rw							
7	6	5	4	3	2	1	0
OSCEXT_STIME[7:0]							
rw							

Figure 6-31: OSCE Stable Time Register(OSCESTIMER)

*_STIME — * Clock Source Stable Time.

Specify the stable time of * Clock Source when enable bit in **OCSR** is turned on.

NOTE: User need to set bit16(OVERWR_OSCE_STABLE_TRIM) in VCCCTMR before CPU overwriting the old value.

6.5.2.31 Power Status Register

Addr. Offset: 0x007C ~ 0x007F Reset Value: 0x0600000D

31	30	29	28	27	26	25	24
Reserved						VCARD0_ISOEN_O	Reserved
ro						ro	ro
23	22	21	20	19	18	17	16
VCC_LVDT5_F	VCC_LVDT18_F	Reserved	CARD0_LVD_F	VCC_LVDT5_RT	VCC_LVDT18_RT	Reserved	CARD0_LVD_RT
r/w1c	r/w1c	ro	r/w1c	ro	ro	ro	ro
15	14	13	12	11	10	9	8
Reserved	CARD0_EN_FAIL_F	Reserved	Reserved				Reserved
ro	r/w1c	ro	ro				ro
7	6	5	4	3	2	1	0
Reserved			Reserved	VCC_HP_READY	Reserved	VCC_CARD0_READY	Reserved
ro			ro	ro	ro	ro	ro

Figure 6-32: Power Status Register(PWRSR)

Status bits of PMC VCC.

VCARD*_ISOEN_O — VCARD* isolation enable output

VCC_LVDT5_F — VCC 5V Low Voltage detect flag

VCC_LVDT18_F — VCC 1.8V Low Voltage detect flag

CARD*_LVD_F — Card* Ldo low voltage detect flag

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VCC_LVDT5_RT — VCC 5V Low Voltage detect real time

VCC_LVDT18_RT — VCC 1.8V Low Voltage detect real time

CARD*_LVD_RT — Card* Ldo low voltage detect real time

CARD*_EN_FAIL_F — Card* Ldo fail flag

VCC_HP_READY — VCC High Power ready

VCC_CARD*_READY — VCC Card* Ldo ready

SUB_SYS_SRAM_REG[7:4] — Direct status of **PWR_INDIC_REG[7:4]** in **PWRINDICR** of C0sub system CPM section

M2S_BUS_S_RST — Main system to C0sub system bus reset release

1 = reset released

0 = reset not released

M2S_BUS_S_RDY — Main system to C0sub system bus ready

1 = ready

0 = not ready

SUB_SYS_PWRRDY — C0sub system power ready

1 = ready

0 = not ready

6.5.2.32 RTC Trim Register

Addr. Offset: 0x008C~ 0x008F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
RTC_TRIM_EN	RTC_TRIM_LOAD	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
RTC_TRIM_VAL[7:0]							
rw							

Figure 6-33: RTC Trim Register(RTCTRIMR)

RTC_TRIM_EN — RTC trim enable

1 = enable, 0 = disable

RTC_TRIM_LOAD — RTC trim load enable

1 = enable, 0 = disable

RTC_TRIM_VAL — RTC trim value

NOTE: User need to set bit28(OVERWR_RTC_TRIM) in VCCCTMR before CPU overwriting the old value.

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6.5.2.33 PAD Wakeup Interrupt Control Register

Addr. Offset: 0x0090~ 0x0093 Reset Value: 0x00FF0000

31	30	29	28	27	26	25	24
S2M_INT_RT	Reserved		PAD_SS3_RT	ATIMER_RT	Reserved	WAKE_RT	USBDET_RT
ro	ro		ro	ro	ro	ro	ro
23	22	21	20	19	18	17	16
S2M_INT_EN	Reserved		PAD_SS3_EN	ATIMER_EN	Reserved	WAKE_EN	USBDET_EN
rw	ro		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
S2M_INT_INTM	Reserved		PAD_SS3_INTM	ATIMER_INTM	Reserved	WAKE_INTM	USBDET_INTM
rw	ro		rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
S2M_INT_STAT	Reserved		PAD_SS3_STAT	ATIMER_STAT	Reserved	WAKE_STAT	USBDET_STAT
ro	ro		ro	ro	ro	ro	ro

Figure 6-34: PAD Wakeup Interrupt Control Register(PADWKPINTCR)

*_RT — wake up source real time value

1 = wake up source valid

0 = wake up source invalid

*_EN — wake up source wakeup enable

1 = enable

0 = disable

*_INTM — wake up source interrupt enable for wakeup interrupt request

1 = enable

0 = disable

*_STAT — wake up source interrupt status

1 = wake up source interrupt valid

0 = wake up source interrupt invalid

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6.5.2.34 Wakeup Filter Counter Register

Addr. Offset: 0x0094~ 0x0097 Reset Value: 0xFF0000F0

31	30	29	28	27	26	25	24
WAKEUP_FILTCNT0[7:0]							
rw							
23	22	21	20	19	18	17	16
WAKEUP_FILTCNT1[7:0]							
rw							
15	14	13	12	11	10	9	8
WAKEUP_FILTCNT2[7:0]							
rw							
7	6	5	4	3	2	1	0
WAKEUP_FILTCNT3[7:0]							
rw							

Figure 6-35: Wakeup Filter Counter Register(WKPFILTCNTR)

WAKEUP_FILTCNT*[7:0] — Wakeup Filter Counter. Specify the the wakeup filter counter value.

6.5.2.35 CARD Power On Counter Register

Addr. Offset: 0x0098~ 0x009B Reset Value: 0x000C0000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				POWAITCNT[19:16]			
ro				rw			
15	14	13	12	11	10	9	8
POWAITCNT[15:8]							
rw							
7	6	5	4	3	2	1	0
POWAITCNT[7:0]							
rw							

Figure 6-36: Card Power on Counter Register(CARDPOCR)

POWAITCNT[19:0] — CARD Power on wait Counter. Specify the the card power one filter counter value.

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6.5.2.36 RTC 32K Stable Time Register

Addr. Offset: 0x009C~ 0x009F Reset Value: 0x00000190

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				RTCSTIME[19:16]			
ro				rw			
15	14	13	12	11	10	9	8
RTCSTIME [15:8]							
rw							
7	6	5	4	3	2	1	0
RTCSTIME [7:0]							
rw							

Figure 6-37: RTC 32K Stable Time Register(RTCSTIMER)

RTCSTIME[19:0] — RTC 32K reference clock stable wait Counter.

NOTE: User need to set bit26(OVERWR_RTC_STABLE_TRIM) in VCCCTMR before CPU overwriting the old value.

6.5.2.37 Memory Power Down Sleep Control Register

Addr. Offset: 0x00A0 ~ 0x00A3 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
MEM_SLPPD[31:24]							
rw							
23	22	21	20	19	18	17	16
MEM_SLPPD[23:16]							
rw							
15	14	13	12	11	10	9	8
MEM_SLPPD[15:8]							
rw							
7	6	5	4	3	2	1	0
MEM_SLPPD[7:0]							
rw							

Figure 6-38: Memory Power Down Sleep Control Register (MPDSLPCR)

MEM_SLPPD[31:0] — Memory Sleep Power Down

Specify the control bit values of memorys' power saving mode signal in sleep mode. LS stands for Light Sleep,DS stands for Deep Sleep,SD stands for Shut Down.In LS&DS mode,the contents in memory array can be retained,while in SD mode,the contents are lost.(see **Table 6-7 MEM_SLPPD[31:0] Bits Corresponding Memory**).

1 = Memory enter Power Saving Mode in Sleep Mode

0 = Memory Not enter Power Saving Mode in Sleep Mode

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Table 6-8: MEM_SLPPD[31:0] Bits Corresponding Memory

SLPPD Bit	Corresponding Memory
0	ICACHE_DS
1	RESERVED
2	Reserved
3	Reserved
4	DCACHE_DS
5	Reserved
6	Reserved
7	Reserved
8	SRAM0_DS
9	SRAM1_DS
10	SRAM2_DS
11	SRAM3_DS
12	SRAMD_DS
13	Reserved
14	Reserved
15	CODE_ROM_LS
16	Reserved
17	Reserved
18	ARITH_RF_LS
19	ARITH_RF_DS
20	CRYPTO_RF_LS
21	CRYPTO_RF_DS
22	Reserved
23	Reserved
24	Reserved
25	ARITH_RF_SD
26	CRYPTO_RF_SD
27	Reserved
28	USBC_SRAM_DS
29	Reserved
30	Reserved
31	Reserved

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6.5.2.38 Multiple Reset Control Register

Addr. Offset: 0x00AC ~ 0x00AF Reset Value: 0x00000000

31	30	29	28	27	26	25	24
MULTIRST_MSK[31:24]							
rw							
23	22	21	20	19	18	17	16
MULTIRST_MSK [23:16]							
rw							
15	14	13	12	11	10	9	8
MULTIRST_MSK [15:8]							
rw							
7	6	5	4	3	2	1	0
MULTIRST_MSK [7:0]							
rw							

Figure 6-39: Multiple Reset Control Register(MULTIRSTCR)

MULTIRST_MSK[31:0] — Multiple Reset Mask Control Register

Specify the mask bit values of modules' resets. (see **Table 6-8 MULTIRST_MSK[31:0] Bits Corresponding Modules**).

1 = Module Reset Asserted

0 = Module Reset not Asserted

Table 6-9: MULTIRST_MSK[31:0] Bits Corresponding Modules

MSK Bit	Corresponding Module
0	Reserved
1	EFM_BUS
2	Reserved
3	RSVD
4	M2S_BUS_S
5	MCC
6	MCCADR
7	RSVD
8	RSVD
9	RSVD
10	RSVD
11	RSVD
12	RSVD
13	RSVD
14	RSVD
15	RSVD
16	KEY_CTRL
17	CCM&RESET
18	EFM_IPS
19	CPM_IPS
20	EPORT
21	EPORT1

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MSK Bit	Corresponding Module
22	EPORT2
23	EPORT3
24	EPORT4
25	RSVD
26	RSVD
27	RSVD
28	RSVD
29	RSVD
30	RSVD
31	RSVD

6.5.2.39 System Reset Control Register

Addr. Offset: 0x00B0~ 0x00B3 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
SYSRST_MSK[31:24]							
rw							
23	22	21	20	19	18	17	16
SYSRST_MSK [23:16]							
rw							
15	14	13	12	11	10	9	8
SYSRST_MSK [15:8]							
rw							
7	6	5	4	3	2	1	0
SYSRST_MSK [7:0]							
rw							

Figure 6-40: System Reset Control Register(SYSRSTCR)

SYSRST_MSK[31:0] — System Reset Control Register

Specify the mask bit values of modules' resets. (see **Table 6-9 SYSRST_MSK[31:0] Bits Corresponding Modules**).

1 = Module Reset Asserted

0 = Module Reset not Asserted

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Table 6-10: SYSRST_MSK[31:0] Bits Corresponding Modules

MSK Bit	Corresponding Module
0	Reserved
1	DMAC1
2	DMAC2
3	RSVD
4	CRC0
5	CRC1
6	RSVD
7	Reserved
8	RSVD
9	RSVD
10	RSVD
11	AHB2_MUX
12	SRAMD
13	SRAM0
14	SRAM1
15	SRAM2
16	SRAM3
17	SSI4
18	SSI5
19	ROM
20	RESERVED
21	RESERVED
22	M2S_BUS_M
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

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6.5.2.40 AHB3 Reset Control Register

Addr. Offset: 0x00B4~ 0x00B7 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
AHB3RST_MSK[31:24]							
rw							
23	22	21	20	19	18	17	16
AHB3RST_MSK [23:16]							
rw							
15	14	13	12	11	10	9	8
AHB3RST_MSK [15:8]							
rw							
7	6	5	4	3	2	1	0
AHB3RST_MSK [7:0]							
rw							

Figure 6-41: AHB3 Reset Control Register (AHB3RSTCR)

AHB3RST_MSK[31:0] — AHB3 Reset Mask Control Register

Specify the mask bit values of modules' resets. (see **Table 6-10 AHB3RST_MSK[31:0] Bits Corresponding Modules**).

1 = Module Reset Asserted

0 = Module Reset not Asserted

Table 6-11: AHB3RST_MSK[31:0] Bits Corresponding Modules

MSK Bit	Corresponding Module
0	RSVD
1	RSVD
2	RSVD
3	USBC
4	RSVD
5	AHB3_MUX
6	Reserved
7	Reserved
8	Reserved
9	Reserved
10	Reserved
11	Reserved
12	Reserved
13	Reserved
14	Reserved
15	Reserved
16	Reserved
17	Reserved
18	Reserved
19	Reserved
20	Reserved
21	Reserved

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MSK Bit	Corresponding Module
22	Reserved
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

6.5.2.41 Arith Reset Control Register

Addr. Offset: 0x00B8 ~ 0x00BB Reset Value: 0x00000000

31	30	29	28	27	26	25	24
ARITHRST_MSK[31:24]							
rw							
23	22	21	20	19	18	17	16
ARITHRST_MSK [23:16]							
rw							
15	14	13	12	11	10	9	8
ARITHRST_MSK [15:8]							
rw							
7	6	5	4	3	2	1	0
ARITHRST_MSK [7:0]							
rw							

Figure 6-42: Arith Reset Control Register (ARITHRSTCR)

ARITHRST_MSK[31:0] — ARITH Clock Reset Control Register

Specify the mask bit values of modules' resets. (see **Table 6-11 ARITHRST_MSK[31:0] Bits Corresponding Modules**).

1 = Module Reset Asserted,

0 = Module Reset Asserted

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Table 6-12: ARITHRST_MSK[31:0] Bits Corresponding Modules

MSK Bit	Corresponding Module
0	RESERVED
1	AES
2	SM4
3	RESERVED
4	RF for AES&SM4
5	CRYPTO
6	SHA
7	EDMAC0
8	DES
9	ZUC
10	AHB2MLB
11	AHB2IPS2
12	Reserved
13	Reserved
14	Reserved
15	Reserved
16	Reserved
17	Reserved
18	Reserved
19	Reserved
20	Reserved
21	Reserved
22	Reserved
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

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6.5.2.42 Ips Reset Control Register

Addr. Offset: 0x00BC ~ 0x00BF Reset Value: 0x00000000

31	30	29	28	27	26	25	24
IPSRST_MSK[31:24]							
rw							
23	22	21	20	19	18	17	16
IPSRST_MSK [23:16]							
rw							
15	14	13	12	11	10	9	8
IPSRST_MSK [15:8]							
rw							
7	6	5	4	3	2	1	0
IPSRST_MSK [7:0]							
rw							

Figure 6-43: Ips Reset Control Register (IPSRSTCR)

IPSRST_MSK[31:0] — IPS Reset Control Register

Specify the mask bit values of modules' resets. (see **Table 6-12 IPSRST_MSK[31:0] Bits Corresponding Modules**).

1 = Module Reset Asserted

0 = Module Reset not Asserted

Table 6-13: IPSRST_MSK[31:0] Bits Corresponding Modules

MSK Bit	Corresponding Module
0	IO_CTRL
1	WDT
2	RTC
3	PIT1
4	PIT2
5	USI1
6	EDMAC1
7	SPI1
8	SPI2
9	SPI3
10	SCI1
11	SCI2
12	USI2
13	RSVD
14	I2C1
15	PWM0
16	I2C2
17	I2C3
18	SCI3
19	RSVD
20	QADC
21	DAC

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MSK Bit	Corresponding Module
22	MCC
23	TSI
24	LD
25	TRNG
26	PGD
27	SEC_DET
28	PCI
29	PMURTC
30	AHB2IPS
31	RSVD

6.5.2.43 Sleep Config Register 2

Addr. Offset: 0x00C0 ~ 0x00C3 Reset Value: 0x0000F70A

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
S2M_INTM_SGL	Reserved		PAD_SS3_INTM_SGL	ATIMER_INTM_SGL	Reserved	WAKE_INTM_SGL	USBDET_INTM_SGL
rw	ro		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved				CORE_F_SLPEN	CLKOUT_SLPEN	CPM_IPS_SLPEN	TC_SLPEN
ro				rw	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved		VDD_WK_SWOFF	Reserved				VDD_PD_RET
ro		rw	ro				rw

Figure 6-44: Sleep Config Register 2(SLPCFGR2)

*_INTM_SGL — wake up source interrupt enable for each single interrupt request

1 = enable, 0 = disable

*_SLPEN — * Clock Sleep Enable When STOP

Specify if the * Clock is enabled in system sleep mode.

1 = enable, 0 = disable

NOTE: When system enters retention or deep sleep mode,CPM_IPS_SLPEN should be configed to 0.

VDD_WK_SWOFF — set with **SLEEP_MODE** in **SLPCFGR** to enter Deep Sleep Mode

1 = VDD_WK power domain switch off

0 = VDD_WK power domain switch on

VDD_PD_RET — set with **SLEEP_MODE** in **SLPCFGR** to enter Retention Mode

1 = VDD_PD power domain retention on(core M4 only)

0 = VDD_PD power domain retention off(core M4 only)

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6.5.2.44 Power Down Counter Register

Addr. Offset: 0x00D0 ~ 0x00D3 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
PDNCNT0[7:0]							
rw							
23	22	21	20	19	18	17	16
PDNCNT1[7:0]							
rw							
15	14	13	12	11	10	9	8
PDNCNT2[7:0]							
rw							
7	6	5	4	3	2	1	0
PDNCNT3[7:0]							
rw							

Figure 6-45: Power Down Counter Register(PDNCNTR)

PDNCNT* — Power Down Counter.

Specify the delay time when the system enters sleep mode.

6.5.2.45 Power On Counter Register

Addr. Offset: 0x00D4 ~ 0x00D7 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
PONCNT0[7:0]							
rw							
23	22	21	20	19	18	17	16
PONCNT1[7:0]							
rw							
15	14	13	12	11	10	9	8
PONCNT2[7:0]							
rw							
7	6	5	4	3	2	1	0
PONCNT3[7:0]							
rw							

Figure 6-46: Power On Counter Register(PONCNTR)

PONCNT* — Power On Counter.

Specify the delay time when the system enters sleep mode.

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6.5.2.46 PAD SS3 Control Register

Addr. Offset: 0x00D8 ~ 0x00DB Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
SS3_CTR_EN	Reserved			SS3_WKPSRC_MSK	SS3_DOUT	SS3_OBE	SS3_PUE
rw	ro			rw	rw	rw	rw

Figure 6-47: PAD SS3 Control Register(PADSS3CR)

SS3_CTR_EN — PAD SS3 Control Enable

Set this bit to override PAD SS3 control to **PADSS3CR** register.

1 = override

0 = not override

SS3_WKPSRC_MSK — PAD SS3 Wakeup Source Mask

1 = mask

0 = not mask

SS3_DOUT — PAD SS3 Output Data

1 = high

0 = low

SS3_OBE — PAD SS3 Output Enable

1 = enable

0 = disable

SS3_PUE — PAD SS3 Pull Enable

1 = enable

0 = disable

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6.5.2.47 Wakeup Source Control Register

Addr. Offset: 0x00DC ~ 0x00DF Reset Value: 0x0000FFFF

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
WKPSEN[15:8]							
rw							
7	6	5	4	3	2	1	0
WKPSEN [7:0]							
rw							

Figure 6-48: Wakeup Source Control Register(WKPSCR)

WKPSEN[15:0] — Wakeup Source Enable.

Specify if the wakeup source can wakeup the system in power saving mode. (see **Table 6-13 WKPSEN[15:0] Bits Corresponding Source**).

1 = Enable

0 = Disable

Table 6-14: WKPSEN[15:0] Bits Corresponding Source

WKPSEN Bit	Corresponding Module
0	EPORT[0]
1	EPORT[1]
2	EPORT[2]
3	EPORT[3]
4	EPORT[4]
5	EPORT[5]
6	USI1_ATR
7	USI1_RE
8	I2C
9	TSI
10	PCI_ASYNC_TIMER
11	PCI_SDIO
12	RESERVED
13	USBC
14	PMURTC_PULSE
15	RSVD
16	Reserved
17	Reserved
18	Reserved
19	Reserved
20	Reserved
21	Reserved

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WKPSEN Bit	Corresponding Module
22	Reserved
23	Reserved
24	Reserved
25	Reserved
26	Reserved
27	Reserved
28	Reserved
29	Reserved
30	Reserved
31	Reserved

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7 USBC

7.1 Introduction

USB 2.0 High-speed Dual-Role Controller provides a 'Dual-role' USB controller for use as either the host or the peripheral in point-to-point communications with another USB function (which may be either high-speed, full-speed or low-speed). Alternatively it can be used as the function controller for a high-speed/full-speed USB peripheral.

It is certified compliant both with the USB 2.0 standard for high-speed and full-speed functions and with the On-The-Go supplement to the USB 2.0 specification. .

7.2 Features

The USB 2.0 High-speed Dual-Role Controller includes the following features:

1. Performs all transaction scheduling in hardware
2. Operates either as a function controller for a USB peripheral or as the host/peripheral in point-to-point communications with another USB function
3. Synchronous RAM interface for FIFOs
4. Certified compliant with the USB 2.0 standard for high-speed (480 Mbps) functions and with the On-The-Go supplement
5. Supports point-to-point communications with one high-, full- or low-speed device
6. Supports Session Request Protocol (SRP) and Host Negotiation Protocol (HNP)
7. Supports Suspend and Resume
8. Configurable for up to 15 additional Transmit endpoints and up to 15 additional Receive endpoints
9. Configurable FIFOs, including the option of dynamic FIFO sizing
10. Support for DMA access to FIFOs
11. Soft connect/disconnect option
12. Support No VBUS Select when USB as peripheral role

7.3 System Clock

The USB controller is designed to take its system clock CLK from the AHB bus clock. This avoids any resynchronization logic between the USB controller and the AHB, allowing single cycle access to both the USB controller registers and the FIFOs.

The minimum frequency at which the core (and the AHB bus) can be clocked depends on the selected UTMI width. If an 8-bit UTMI is used (as is strongly recommended), the core can be clocked at speeds close to 30MHz. If however a 16-bit UTMI is used, the minimum frequency needs to be increased to over 48MHz to ensure that the bus turn-around time is met.

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7.4 USB RESET

7.4.1 IN PERIPHERAL MODE

When the USB controller is acting as a peripheral and a reset condition is detected on the USB, the device will perform the following actions:

- Sets FAddr to 0.
- Sets Index to 0.
- Flushes all endpoint FIFOs.
- Clears all control/status registers.
- Enables all endpoint interrupts.
- Generates a Reset interrupt.

If the HS Enab bit in the UCSR register (D5) was set, the USB controller also tries to negotiate for high-speed operation. Whether high-speed operation is selected is indicated by HS Mode bit (UCSR.D4).

When the application software driving the USB controller receives a Reset interrupt, it should close any open pipes and wait for bus enumeration to begin.

7.4.2 IN HOST MODE

If the Reset bit in the UCSR register is set while the USB controller is in Host mode, the USB controller will generate Reset signaling on the bus. If the HS Enab bit in the UCSR register (D5) was set, it will also try to negotiate for high-speed operation.

The CPU should keep the Reset bit set for at least 20 ms to ensure correct resetting of the target device.

After the CPU has cleared the bit, the USB controller will start its frame counter and transaction scheduler. Whether high-speed operation is selected will be indicated by HS Mode bit (UCSR.D4).

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7.5 USB 2.0 High-speed Dual-Role Controller Registers

7.5.1 Common USB Registers

Table 7-1: Common USB Registers

Mnemonic	Description	Register Address Offset
FAddr	Function address register.	0x000
UCSR	UCSR management register.	0x001
IntrTx	Interrupt register for Endpoint 0 plus Tx Endpoints 1 to 15.	0x002
IntrRx	Interrupt register for Rx Endpoints 1 to 15.	0x004
IntrTxE	Interrupt enable register for IntrTx.	0x006
IntrRxE	Interrupt enable register for IntrRx.	0x008
IntrUSB	Interrupt register for common USB interrupts.	0x00A
IntrUSBE	Interrupt enable register for IntrUSB.	0x00B
Frame	Frame number.	0x00C
Index	Index register for selecting the endpoint status and control registers.	0x00E
Testmode	Enables the USB 2.0 test modes.	0x00F

7.5.2 Indexed Registers

Table 7-2: Indexed Registers-Host Mode

Mnemonic	Description	Register Address Offset
TxMaxP	Maximum packet size for host Tx endpoint.	0x010
CSR0	Control Status register for Endpoint 0.(Index regiser set to Endpoint 0)	0x012
TxCSR	Control Status register for host Tx endpoint.(Index regiser set to Endpoint 1-15)	
RxMaxP	Maximum packet size for host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x014
RxCSR	Control Status register for host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x016
Count0	Number of received bytes in Endpoint 0 FIFO.(Index regiser set to Endpoint 0)	0x018
RxCount	Number of bytes in host Rx endpoint FIFO.(Index regiser set to Endpoint 1-15)	
TxType	Sets the transaction protocol and peripheral endpoint number for the host Tx endpoint.(Index regiser set to Endpoint 1-15)	0x01A
NAKLimit0	Sets the NAK response timeout on Endpoint 0.(Index regiser set to Endpoint 0)	0x01B
TxInterval	Sets the polling interval for Interrupt/ISOC transactions or the NAK response timeout on Bulk transactions for host Tx endpoint.(Index regiser set to Endpoint 1-15)	

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Mnemonic	Description	Register Address Offset
RxType	Sets the transaction protocol and peripheral endpoint number for the host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x01C
RxInterval	Sets the polling interval for Interrupt/ISOC transactions or the NAK response timeout on Bulk transactions for host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x01D

Table 7-3: Indexed Registers-Peripheral Mode

Mnemonic	Description	Register Address Offset
TxMaxP	Maximum packet size for host Tx endpoint.	0x010
CSR0	Control Status register for Endpoint 0.(Index regiser set to Endpoint 0)	0x012
TxCSR	Control Status register for host Tx endpoint.(Index regiser set to Endpoint 1-15)	
RxMaxP	Maximum packet size for host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x014
RxCSR	Control Status register for host Rx endpoint.(Index regiser set to Endpoint 1-15)	0x016
Count0	Number of received bytes in Endpoint 0 FIFO.(Index regiser set to Endpoint 0)	0x018
RxCount	Number of bytes in host Rx endpoint FIFO.(Index regiser set to Endpoint 1-15)	

7.5.3 Registers about FIFOs

Table 7-4: Registers about FIFOs

Mnemonic	Description	Register Address Offset
FIFOx	FIFOs for Endpoints 0-15.	0x020 ~0x05F

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7.5.4 Additional Control and Configuration Registers

Table 7-5: Additional Control and Configuration Registers

Mnemonic	Description	Register Address Offset
DevCtl	OTG device control register	0x060
TxFIFOsz	Tx Endpoint FIFO size	0x062
RxFIFOsz	Rx Endpoint FIFO size	0x063
TxFIFOadd	Tx Endpoint FIFO address	0x064
RxFIFOadd	Rx Endpoint FIFO address	0x066

7.5.5 DMA Registers

The DMA controller has one interrupt register which indicates which channels have a pending interrupt, and a set of three control registers for each configured channel.

Table 7-6: DMA Registers

Mnemonic	Description	Register Address Offset
INTR	Indicates pending DMA interrupts	0x200
CNTL (1)	DMA Channel 1 Control	0x204
ADDR (1)	DMA Channel 1 AHB Memory Address	0x208
COUNT (1)	DMA Channel 1 Byte Count	0x20c
-	Unused	0x210
CNTL (2)	DMA Channel 2 Control	0x214
ADDR (2)	DMA Channel 2 AHB Memory Address	0x218
COUNT (2)	DMA Channel 2 Byte Count	0x21c
etc.		

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7.6 Common Registers

7.6.1 FADDR

FAddr is an 8-bit register that should be written with the 7-bit address of the peripheral part of the transaction. When the USB controller is being used in Host mode (DevCtl.D2=1), this register should be set to the value sent in a SET_ADDRESS command during device enumeration as the address for the peripheral device.

When the USB controller is being used in Peripheral mode (DevCtl.D2=0), this register should be written with the address received through a SET_ADDRESS command, which will then be used for decoding the function address in subsequent token packets.

Table 7-7: FADDR

Address offset: 0x00 Reset Value:8'h00

Bit	Name	Function
D7	-	Unused, always returns 0
D6 ~ D0	Func Addr	The Function Address

7.6.2 UCSR

UCSR is an 8-bit register that is used for controlling Suspend and Resume signaling, and some basic operational aspects of the USB controller.

Table 7-8: UCSR

Address offset: 0x01 Reset Value:8'h20

Bit	Name	Function
D7	ISO Update	When set by the CPU, the USB controller will wait for an SOF token from the time TxPktRdy is set before sending the packet. If an IN token is received before an SOF token, then a zero length data packet will be sent.
D6	Soft Conn	If Soft Connect/Disconnect feature is enabled, then the USB D+/D- lines are enabled when this bit is set by the CPU and tri-stated when this bit is cleared by the CPU. (See Section 5.1)
D5	HS Enab	When set by the CPU, the USB controller will negotiate for High-speed mode when the device is reset by the hub. If not set, the device will only operate in Full-speed mode.
D4	HS Mode	When set, this read-only bit indicates High-speed mode successfully negotiated during USB reset. In Peripheral Mode, becomes valid when USB reset completes (as indicated by USB reset interrupt). In Host Mode, becomes valid when Reset bit is cleared. Remains valid for the duration of the session.
D3	Reset	This bit is set when Reset signaling is present on the bus.
D2	Resume	Set by the CPU to generate Resume signaling when the function is in Suspend mode. The CPU should clear this bit after 10 ms (a maximum of 15 ms) to end Resume signaling. In Host mode, this bit is also automatically set when Resume signaling from the target is detected while the USB controller is suspended.
D1	Suspend Mode	In Host mode, this bit is set by the CPU to enter Suspend mode. In Peripheral mode, this bit is set on entry into Suspend mode. It is cleared when the CPU reads the interrupt register, or sets the Resume bit above.

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Bit	Name	Function
D0	Enable SuspendM	Set by the CPU to enable the SUSPENDM signal.

7.6.3 INTRTX

IntrTx is a 16-bit read-only register that indicates which interrupts are currently active for Endpoint 0 and the Tx Endpoints 1–15.

Table 7-9: INTRTX

Address offset: 0x02 Reset Value:16'h0000

Bit	Name	Function
D15	EP15 Tx	Tx Endpoint 15 interrupt.
D14	EP14 Tx	Tx Endpoint 14 interrupt.
D13	EP13 Tx	Tx Endpoint 13 interrupt.
D12	EP12 Tx	Tx Endpoint 12 interrupt.
D11	EP11 Tx	Tx Endpoint 11 interrupt.
D10	EP10 Tx	Tx Endpoint 10 interrupt.
D9	EP9 Tx	Tx Endpoint 9 interrupt.
D8	EP8 Tx	Tx Endpoint 8 interrupt.
D7	EP7 Tx	Tx Endpoint 7 interrupt.
D6	EP6 Tx	Tx Endpoint 6 interrupt.
D5	EP5 Tx	Tx Endpoint 5 interrupt.
D4	EP4 Tx	Tx Endpoint 4 interrupt.
D3	EP3 Tx	Tx Endpoint 3 interrupt.
D2	EP2 Tx	Tx Endpoint 2 interrupt.
D1	EP1 Tx	Tx Endpoint 1 interrupt.
D0	EP0	Endpoint 0 interrupt.

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7.6.4 INTRRX

IntrRx is an 16-bit read-only register that indicates which of the interrupts for Rx Endpoints 1 – 15 are currently active.

Table 7-10: INTRRX

Address offset: 0x04 Reset Value:16'h0000

Bit	Name	Function
D15	EP15 Rx	Rx Endpoint 15 interrupt.
D14	EP14 Rx	Rx Endpoint 14 interrupt.
D13	EP13 Rx	Rx Endpoint 13 interrupt.
D12	EP12 Rx	Rx Endpoint 12 interrupt.
D11	EP11 Rx	Rx Endpoint 11 interrupt.
D10	EP10 Rx	Rx Endpoint 10 interrupt.
D9	EP9 Rx	Rx Endpoint 9 interrupt.
D8	EP8 Rx	Rx Endpoint 8 interrupt.
D7	EP7 Rx	Rx Endpoint 7 interrupt.
D6	EP6 Rx	Rx Endpoint 6 interrupt.
D5	EP5 Rx	Rx Endpoint 5 interrupt.
D4	EP4 Rx	Rx Endpoint 4 interrupt.
D3	EP3 Rx	Rx Endpoint 3 interrupt.
D2	EP2 Rx	Rx Endpoint 2 interrupt.
D1	EP1 Rx	Rx Endpoint 1 interrupt.
D0	Unused	Always returned 0

7.6.5 INTRTXE

IntrTxE is a 16-bit register that provides interrupt enable bits for the interrupts in IntrTx. Where a bit is set to 1, MC_NINT will be asserted on the corresponding

interrupt in the IntrTx register becoming set. Where a bit is set to 0, the interrupt in IntrTx is still set but MC_NINT is not asserted. On reset, the bits corresponding to Endpoint 0 and the Tx endpoints included in the design are set to 1, while the remaining bits are set to 0.

NOTE: Bits relating to endpoints that have not been configured will always return 0.

Table 7-11: INTRTXE

Address offset: 0x06 Reset Value:16'hFFFF

Bit	Name	Function
D15	EP15 TxE	Tx Endpoint 15 interrupt Enable.
D14	EP14 TxE	Tx Endpoint 14 interrupt Enable.
D13	EP13 TxE	Tx Endpoint 13 interrupt Enable.
D12	EP12 TxE	Tx Endpoint 12 interrupt Enable.
D11	EP11 TxE	Tx Endpoint 11 interrupt Enable.
D10	EP10 TxE	Tx Endpoint 10 interrupt Enable.
D9	EP9 TxE	Tx Endpoint 9 interrupt Enable.
D8	EP8 TxE	Tx Endpoint 8 interrupt Enable.
D7	EP7 TxE	Tx Endpoint 7 interrupt Enable.
D6	EP6 TxE	Tx Endpoint 6 interrupt Enable.

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Bit	Name	Function
D5	EP5 TxE	Tx Endpoint 5 interrupt Enable.
D4	EP4 TxE	Tx Endpoint 4 interrupt Enable.
D3	EP3 TxE	Tx Endpoint 3 interrupt Enable.
D2	EP2 TxE	Tx Endpoint 2 interrupt Enable.
D1	EP1 TxE	Tx Endpoint 1 interrupt Enable.
D0	EP0 Enable	Endpoint 0 interrupt Enable.

7.6.6 INTRRXE

IntrRxE is a 16-bit register that provides interrupt enable bits for the interrupts in IntrRx. Where a bit is set to 1, MC_NINT will be asserted on the corresponding interrupt in the IntrRx register becoming set. Where a bit is set to 0, the interrupt in IntrTx is still set but MC_NINT is not asserted. On reset, the bits corresponding to the Rx endpoints included in the design are set to 1, while the remaining bits are set to 0.

Table 7-12: INTRRXE

Address offset: 0x08 Reset Value:16'hFFFF

Bit	Name	Function
D15	EP15 RxE	Rx Endpoint 15 interrupt Enable.
D14	EP14 RxE	Rx Endpoint 14 interrupt Enable.
D13	EP13 RxE	Rx Endpoint 13 interrupt Enable.
D12	EP12 RxE	Rx Endpoint 12 interrupt Enable.
D11	EP11 RxE	Rx Endpoint 11 interrupt Enable.
D10	EP10 RxE	Rx Endpoint 10 interrupt Enable.
D9	EP9 RxE	Rx Endpoint 9 interrupt Enable.
D8	EP8 RxE	Rx Endpoint 8 interrupt Enable.
D7	EP7 RxE	Rx Endpoint 7 interrupt Enable.
D6	EP6 RxE	Rx Endpoint 6 interrupt Enable.
D5	EP5 RxE	Rx Endpoint 5 interrupt Enable.
D4	EP4 RxE	Rx Endpoint 4 interrupt Enable.
D3	EP3 RxE	Rx Endpoint 3 interrupt Enable.
D2	EP2 RxE	Rx Endpoint 2 interrupt Enable.
D1	EP1 RxE	Rx Endpoint 1 interrupt Enable.
D0	Unused	Always returned 0

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7.6.7 INTRUSB

IntrUSB is an 8-bit read-only register that indicates which USB interrupts are currently active. All active interrupts will be cleared when this register is read.

Table 7-13: INTRUSB

Address offset: 0x0A Reset Value:8'h00

Bit	Name	Function
D7	VBus Error	Set when VBus drops below the VBus Valid threshold during a session. Only valid when USB controller is 'A' device.
D6	Sess Req	Set when Session Request signaling has been detected. Only valid when USB controller is 'A' device.
D5	Discon	Set in Host mode when a device disconnect is detected. Set in Peripheral mode when a session ends. Valid at all transaction speeds.
D4	Conn	Set when a device connection is detected. Only valid in Host mode. Valid at all transaction speeds.
D3	SOF	Set when a new frame starts
D2	Reset	Set in Peripheral mode when Reset signaling is detected on the bus.
	Babble	Set in Host mode when babble is detected.
D1	Resume	Set when Resume signaling is detected on the bus while the USB controller is in Suspend mode.
D0	Suspend	Set when Suspend signaling is detected on the bus. Only valid in Peripheral mode.

7.6.8 INTRUSBE

IntrUSBE is an 8-bit register that provides interrupt enable bits for each of the interrupts in IntrUSB.

Table 7-14: INTRUSBE

Address offset: 0x0B Reset Value:8'h06

Bit	Name	Function
D7	VBus Error Enable	VBus Error Interrupt Enable
D6	Sess Req Enable	Sess Req Interrupt Enable
D5	Discon Enable	Discon Interrupt Enable
D4	Conn Enable	Conn Interrupt Enable
D3	SOF Enable	SOF Interrupt Enable
D2	Reset Enable	Reset Interrupt Enable
	Babble Enable	Babble Interrupt Enable
D1	Resume Enable	Resume Interrupt Enable
D0	Suspend Enable	Suspend Interrupt Enable

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7.6.9 FRAME

Frame is a 16-bit read-only register that holds the last received frame number.

Table 7-15: FRAME

Address offset: 0x0C Reset Value:16'h0000

Bit	Name	Function
D15:D11	Reserved	When reading always return 0
D10:D0	Frame Number	the last received frame number

7.6.10 INDEX

Each Tx endpoint and each Rx endpoint have their own set of control/status registers located between 100h – 1FFh. In addition one set of Tx control/status and one set of Rx control/status registers appear at 10h – 19h. Index is a 4-bit register that determines which endpoint control/status registers are accessed.

Table 7-16: INDEX

Address offset: 0x0E Reset Value:4'b0000

Bit	Name	Function
D3:D0	Index	Selected Endpoint

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7.6.11 TESTMODE

Testmode is an 8-bit register that is primarily used to put the USB controller into one of the four test modes for High-speed operation described in the USB 2.0 specification – in response to a SET FEATURE: TESTMODE command. It is not used in normal operation.

Table 7-17: TESTMODE

Address offset: 0x0F Reset Value:8'h00

Bit	Name	Function
D7	Force_Host	The CPU sets this bit to instruct the core to enter Host mode when the Session bit is set, regardless of whether it is connected to any peripheral. The state of the CID input, HostDisconnect and LineState signals are ignored. The core will then remain in Host mode until the Session bit is cleared, even if a device is disconnected, and if the Force_Host bit remains set, will re-enter Host mode the next time the Session bit is set. While in this mode, the status of the HOSTDISCON signal from the PHY may be read from bit 7 of the DevCtl register. The operating speed is determined from the Force_HS and Force_FS bits as follows: Force_HS Force_FS Operating Speed 0 0 Low Speed 0 1 Full Speed 1 0 High Speed 1 1 Undefined
D6	FIFO_Access	The CPU sets this bit to transfer the packet in the Endpoint 0 Tx FIFO to the Endpoint 0 Rx FIFO. It is cleared automatically.
D5	Force_FS	The CPU sets this bit either in conjunction with bit 7 above or to force the USB controller into Fullspeed mode when it receives a USB reset.
D4	Force_HS	The CPU sets this bit either in conjunction with bit 7 above or to force the USB controller into Highspeed mode when it receives a USB reset.
D3	Test_Packet	(High-speed mode) The CPU sets this bit to enter the Test_Packet test mode. In this mode, thev USB controller repetitively transmits on the bus a 53-byte test packet, the form of which is defined in the Universal Serial Bus Specification Revision 2.0, Section 7.1.20
D2	Test_K	(High-speed mode) The CPU sets this bit to enter the Test_K test mode. In this mode, the USB controller transmits a continuous K on the bus.

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Bit	Name	Function
D1	Test_J	(High-speed mode) The CPU sets this bit to enter the Test_J test mode. In this mode, the USB controller transmits a continuous J on the bus.
D0	Test_SE0_NAK	(High-speed mode) The CPU sets this bit to enter the Test_SE0_NAK test mode. In this mode, the USB controller remains in High-speed mode but responds to any valid IN token with a NAK.

7.6.12 DEVCTL

DevCtl is an 8-bit register that is used to select whether the USB controller is operating in Peripheral mode or in Host mode, and for controlling and monitoring the USB VBus line.

Table 7-18: DEVCTL

Address offset: 0x60 Reset Value:8'h80

Bit	Name	Function
D7	B-Device	This Read-only bit indicates whether the USB controller is operating as the 'A' device or the 'B' device. 0 ? 'A' device; 1 ? 'B' device. Only valid while a session is in progress.
D6	FSDev	This Read-only bit is set when a full-speed or high-speed device has been detected being connected to the port. (High-speed devices are distinguished from full-speed by checking for high-speed chirps when the device is reset.) Only valid in Host mode.
D5	LSDev	This Read-only bit is set when a low-speed device has been detected being connected to the port. Only valid in Host mode.
D4 ~ D3	VBus[1:0]	These Read-only bits encode the current VBus level as follows: D4 D3 Meaning 0 0 Below SessionEnd 0 1 Above SessionEnd, below AValid 1 0 Above AValid, below VBusValid 1 1 Above VBusValid
D2	Host Mode	This Read-only bit is set when the USB controller is acting as a Host.
D1	Host Req	When set, the USB controller will initiate the Host Negotiation when Suspend mode is entered. It is cleared when Host Negotiation is completed. See Section 15. ('B' device only)

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Bit	Name	Function
D0	Session	When operating as an 'A' device, this bit is set or cleared by the CPU to start or end a session. When operating as a 'B' device, this bit is set/cleared by the USB controller when a session starts/ends. It is also set by the CPU to initiate the Session Request Protocol, or cleared by the CPU when in Suspend mode to perform a software disconnect.

7.7 Indexed Registers

7.7.1 CSR0

CSR0 is a 16-bit register that provides control and status bits for Endpoint 0. **NOTE:** The interpretation of the register depends on whether the USB controller is acting as a peripheral or as a host. Users should also be aware that the value returned when the register is read reflects the status attained e.g. as a result of writing to the register.

Table 7-19: CSR0 - Peripheral Mode

Address offset: 0x12(with Index register set to 0) Reset Value:16'h0000

Bit	Name	Function
D15 ~ D9	-	Unused. Return 0 when read.
D8	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy bit (below) is cleared.
D7	ServicedSetupEnd	The CPU writes a 1 to this bit to clear the SetupEnd bit. It is cleared automatically.
D6	ServicedRxPktRdy	The CPU writes a 1 to this bit to clear the RxPktRdy bit. It is cleared automatically.
D5	SendStall	The CPU writes a 1 to this bit to terminate the current transaction. The STALL handshake will be transmitted and then this bit will be cleared automatically.
D4	SetupEnd	This bit will be set when a control transaction ends before the DataEnd bit has been set. An interrupt will be generated and the FIFO flushed at this time. The bit is cleared by the CPU writing a 1 to the ServicedSetupEnd bit.
D3	DataEnd	The CPU sets this bit: 1. When setting TxPktRdy for the last data packet. 2. When clearing RxPktRdy after unloading the last data packet. 3. When setting TxPktRdy for a zero length data packet. It is cleared automatically.
D2	SentStall	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
D1	TxPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when the data packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.

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Bit	Name	Function
D0	RxPktRdy	This bit is set when a data packet has been received. An interrupt is generated when this bit is set. The CPU clears this bit by setting the ServicedRxPktRdy bit.

Table 7-20: CSR0 - Host Mode

Address offset: 0x12(with Index register set to 0) Reset Value:16'h0000

Bit	Name	Function
D15 ~ D12	-	Unused. Return 0 when read.
D11	Dis Ping	The CPU writes a 1 to this bit to instruct the core not to issue PING tokens in data and status phases of a high-speed Control transfer (for use with devices that do not respond to PING).
D10 ~ D9	-	Unused. Return 0 when read.
D8	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be transmitted/read from the Endpoint 0 FIFO. The FIFO pointer is reset and the TxPktRdy/RxPktRdy bit (below) is cleared.
D7	NAK Timeout	This bit will be set when Endpoint 0 is halted following the receipt of NAK responses for longer than the time set by the NAKLimit0 register. The CPU should clear this bit to allow the endpoint to continue.
D6	StatusPkt	The CPU sets this bit at the same time as the TxPktRdy or ReqPkt bit is set, to perform a status stage transaction. Setting this bit ensures that the data toggle is set to 1 so that a DATA1 packet is used for the Status Stage transaction.
D5	ReqPkt	The CPU sets this bit to request an IN transaction. It is cleared when RxPktRdy is set.
D4	Error	This bit will be set when three attempts have been made to perform a transaction with no response from the peripheral. The CPU should clear this bit. An interrupt is generated when this bit is set.
D3	SetupPkt	The CPU sets this bit, at the same time as the TxPktRdy bit is set, to send a SETUP token instead of an OUT token for the transaction.
D2	RxStall	This bit is set when a STALL handshake is received. The CPU should clear this bit.
D1	TxPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when the data packet has been transmitted. An interrupt is generated (if enabled) when the bit is cleared.
D0	RxPktRdy	This bit is set when a data packet has been received. An interrupt is generated (if enabled) when this bit is set. The CPU should clear this bit when the packet has been read from the FIFO.

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7.7.2 COUNT0

Count0 is a 7-bit read-only register that indicates the number of received data bytes in the Endpoint 0 FIFO. The value returned changes as the contents of the FIFO change and is only valid while RxPktRdy (CSR0.D0) is set.

Table 7-21: COUNT0

Address offset: 0x18(with Index register set to 0) Reset Value:7'b0000000

Bit	Name	Function
D6 ~ D0	Count0	Endpoint 0 Rx Count

7.7.3 NAKLIMIT0 (Host Mode Only)

NAKLimit0 is a 5-bit register that sets the number of frames/microframes (High-Speed transfers) after which Endpoint 0 should timeout on receiving a stream of NAK responses.

The number of frames/microframes selected is 2(m-1) (where m is the value set in the register, valid values 2 – 16). If the host receives NAK responses from the target for more frames than the number represented by the Limit set in this register, the endpoint will be halted.

NOTE: A value of 0 or 1 disables the NAK timeout function.

Table 7-22: NAKLIMIT0 (Host Mode Only)

Address offset: 0x1B(with Index register set to 0) Reset Value:5'b00000

Bit	Name	Function
D4 ~ D0	NAK LIMIT0	Endpoint 0 NAK LIMIT0 NUMBER

7.7.4 TXMAXP

The TxMaxP register defines the maximum amount of data that can be transferred through the selected Tx endpoint in a single operation. There is a TxMaxP register for each Tx endpoint (except Endpoint 0).

Table 7-23: TXMAXP

Address offset: 0x10 Reset Value11'h0000

Bit	Name	Function
D10 ~ D0	TXMAXP	Maximum Payload/transaction

Bits 10:0 define (in bytes) the maximum payload transmitted in a single transaction. The value set can be up to 1024 bytes but is subject to the constraints placed by the USB Specification on packet sizes for Bulk, Interrupt and Isochronous transfers in Fullspeed and High-speed operations.

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7.7.5 TXCSR

TxCSR is a 16-bit register that provides control and status bits for transfers through the currently-selected Tx endpoint. There is a TxCSR register for each configured Tx endpoint (not including Endpoint 0). **NOTE:** The interpretation of the register depends on whether the USB controller is acting as a peripheral or as a host. Users should also be aware that the value returned when the register is read reflects the status attained e.g. as a result of writing to the register.

Table 7-24: TXCSR - Peripheral Mode

Address offset: 0x12 Reset Value:16'h0000

Bit	Name	Function
D15	AutoSet	If the CPU sets this bit, TxPktRdy will be automatically set when data of the maximum packet size (value in TxMaxP) is loaded into the Tx FIFO. If a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually.
D14	ISO	The CPU sets this bit to enable the Tx endpoint for Isochronous transfers, and clears it to enable the Tx endpoint for Bulk or Interrupt transfers.
D13	Mode	The CPU sets this bit to enable the endpoint direction as Tx, and clears it to enable the endpoint direction as Rx.
D12	DMAReqEnab	The CPU sets this bit to enable the DMA request for the Tx endpoint.
D11	FrcDataTog	The CPU sets this bit to force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt Tx endpoints that are used to communicate rate feedback for Isochronous endpoints.
D10	DMAReqMode	The CPU sets this bit to select DMA Mode 1 and clears this bit to select DMA Mode 0.
D9 ~ D8	-	*Unused, always return 0.*
D7	IncompTx	When the endpoint is being used for high-bandwidth Isochronous/Interrupt transfers, this bit is set to indicate where a large packet has been split into 2 or 3 packets for transmission but insufficient IN tokens have been received to send all the parts.
D6	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
D5	SentStall	This bit is set when a STALL handshake is transmitted. The FIFO is flushed and the TxPktRdy bit is cleared (see below). The CPU should clear this bit.
D4	SendStall	The CPU writes a 1 to this bit to issue a STALL handshake to an IN token. The CPU clears this bit to terminate the stall condition.
D3	FlushFIFO	The CPU writes a 1 to this bit to flush the latest packet from the endpoint Tx FIFO. The FIFO pointer is reset, the TxPktRdy bit (below) is cleared and an interrupt is generated. May be set simultaneously with TxPktRdy to

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Bit	Name	Function
		abort the packet that is currently being loaded into the FIFO
D2	UnderRun	The USB sets this bit if an IN token is received when the TxPktRdy bit not set. The CPU should clear this bit.
D1	FIFONotEmpty	The USB sets this bit when there is at least 1 packet in the Tx FIFO.
D0	TxPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (if enabled). TxPktRdy is also automatically cleared prior to loading a second packet into a double-buffered FIFO.

Table 7-25: TXCSR - Host Mode

Address offset: 0x12 Reset Value:16'h0000

Bit	Name	Function
D15	AutoSet	If the CPU sets this bit, TxPktRdy will be automatically set when data of the maximum packet size (value in TxMaxP) is loaded into the Tx FIFO. If a packet of less than the maximum packet size is loaded, then TxPktRdy will have to be set manually.
D14	-	*Unused, always returns zero.*
D13	Mode	The CPU sets this bit to enable the endpoint direction as Tx, and clears it to enable the endpoint direction as Rx.
D12	DMAReqEnab	The CPU sets this bit to enable the DMA request for the Tx endpoint.
D11	FrcDataTog	The CPU sets this bit to force the endpoint data toggle to switch and the data packet to be cleared from the FIFO, regardless of whether an ACK was received. This can be used by Interrupt Tx endpoints that are used to communicate rate feedback for Isochronous endpoints.
D10	DMAReqMode	The CPU sets this bit to select DMA Mode 1 and clears this bit to select DMA Mode 0.
D9 ~ D8	-	*Unused, always return 0.*

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Bit	Name	Function
D7	Nak Timeout	*Bulk endpoints only:* This bit will be set when the Tx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the TxInterval register. The CPU should clear this bit to allow the endpoint to continue.
	IncompTx	*High-bandwidth Interrupt endpoints only:* This bit will be set if no response is received from the device to which the packet is being sent.
D6	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
D5	RxStall	This bit is set when a STALL handshake is received. When this bit is set, any DMA request that is in progress is stopped, the FIFO is completely flushed and the TxPktRdy bit is cleared (see below). The CPU should clear this bit.
D3	FlushFIFO	The CPU writes a 1 to this bit to flush the latest packet from the endpoint Tx FIFO. The FIFO pointer is reset, the TxPktRdy bit (below) is cleared and an interrupt is generated. May be set simultaneously with TxPktRdy to abort the packet that is currently being loaded into the FIFO. **NOTE:** FlushFIFO should only be used when TxPktRdy is set. At other times, it may cause data to be corrupted. Also note that, if the FIFO is double-buffered, FlushFIFO may need to be set twice to completely clear the FIFO.
D2	Error	The USB sets this bit when 3 attempts have been made to send a packet and no handshake packet has been received. When the bit is set, an interrupt is generated, TxPktRdy is cleared and the FIFO completelyflushed. The CPU should clear this bit. *Valid only when the endpoint is operating in Bulk or Interrupt mode.*
D1	FIFONotEmpty	The USB sets this bit when there is at least 1 packet in the Tx FIFO.
D0	TxPktRdy	The CPU sets this bit after loading a data packet into the FIFO. It is cleared automatically when a data packet has been transmitted. An interrupt is also generated at this point (if enabled). TxPktRdy is also automatically cleared prior to loading a second packet into a double-buffered FIFO.

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7.7.6 RXMAXP

The RxMaxP register defines the maximum amount of data that can be transferred through the selected Rx endpoint in a single operation. There is a RxMaxP register for each Rx endpoint (except Endpoint 0).

Table 7-26: RXMAXP

Address offset: 0x14 Reset Value11'h0000

Bit	Name	Function
D10 ~ D0	RXMAXP	Maximum Payload/transaction

Bits 10:0 define (in bytes) the maximum payload transmitted in a single transaction. The value set can be up to 1024 bytes but is subject to the constraints placed by the USB Specification on packet sizes for Bulk, Interrupt and Isochronous transfers in Fullspeed and High-speed operations.

7.7.7 RXCSR

RxCSR is an 16-bit register that provides control and status bits for transfers through the currently-selected Rx endpoint. There is an RxCSR register for each configured Rx endpoint (not including Endpoint 0).

Table 7-27: RXCSR - Peripheral Mode

Address offset: 0x16 Reset Value:16'h0000

Bit	Name	Function
D15	AutoClear	If the CPU sets this bit then the RxPktRdy bit will be automatically cleared when a packet of RxMaxP bytes has been unloaded from the Rx FIFO. When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually.
D14	ISO	The CPU sets this bit to enable the Rx endpoint for Isochronous transfers, and clears it to enable the Rx endpoint for Bulk/Interrupt transfers.
D13	DMAReqEnab	The CPU sets this bit to enable the DMA request for the Rx endpoint.
D12	DisNyet PID Error	**Bulk/Interrupt Transactions:** The CPU sets this bit to disable the sending of NYET handshakes. When set, all successfully received Rx packets are ACK'd including at the point at which the FIFO becomes full. **ISO Transactions:** The core sets this bit to indicate a PID error in the received packet.
D11	DMAReqMode	The CPU sets this bit to select DMA Mode 1 and clears this bit to select DMA Mode 0.
D10 ~ D9	-	*Unused, always return 0.*

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Bit	Name	Function
D8	IncompRx	This bit is set in a high-bandwidth Isochronous/Interrupt transfer if the packet in the Rx FIFO is incomplete because parts of the data were not received. It is cleared when RxPktRdy is cleared.
D7	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
D6	SentStall	This bit is set when a STALL handshake is transmitted. The CPU should clear this bit.
D5	SendStall	The CPU writes a 1 to this bit to issue a STALL handshake. The CPU clears this bit to terminate the stall condition
D4	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared.
D3	DataError	This bit is set when RxPktRdy is set if the data packet has a CRC or bit-stuff error. It is cleared when RxPktRdy is cleared.
D2	OverRun	This bit is set if an OUT packet cannot be loaded into the Rx FIFO. The CPU should clear this bit.
D1	FIFOFull	This bit is set when no more packets can be loaded into the Rx FIFO
D0	RxPktRdy	This bit is set when a data packet has been received. The CPU should clear this bit when the packet has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

Table 7-28: RXCSR - Host Mode

Address offset: 0x16 Reset Value:16'h0000

Bit	Name	Function
D15	AutoClear	If the CPU sets this bit then the RxPktRdy bit will be automatically cleared when a packet of RxMaxP bytes has been unloaded from the Rx FIFO. When packets of less than the maximum packet size are unloaded, RxPktRdy will have to be cleared manually
D14	AutoReq	If the CPU sets this bit, the ReqPkt bit will be automatically set when the RxPktRdy bit is cleared.
D13	DMAReqEnab	The CPU sets this bit to enable the DMA request for the Rx endpoint.
D12	PID Error	**ISO Transactions Only:** The core sets this bit to indicate a PID error in the received packet. **Bulk/Interrupt Transactions:** *The setting of this bit is ignored.*

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Bit	Name	Function
D11	DMAReqMode	The CPU sets this bit to select DMA Mode 1 and clears this bit to select DMA Mode 0. **NOTE:** This bit should not be cleared in the same cycle as RxPktRdy is cleared.
D10 ~ D9	-	*Unused, always return 0.*
D8	IncompRx	This bit will be set in a high-bandwidth Isochronous/Interrupt transfer if the packet received is incomplete. It will be cleared when RxPktRdy is cleared.
D7	ClrDataTog	The CPU writes a 1 to this bit to reset the endpoint data toggle to 0.
D6	RxStall	When a STALL handshake is received, this bit is set and an interrupt is generated. The CPU should clear this bit
D5	ReqPkt	The CPU writes a 1 to this bit to request an IN transaction. It is cleared when RxPktRdy is set.
D4	FlushFIFO	The CPU writes a 1 to this bit to flush the next packet to be read from the endpoint Rx FIFO. The FIFO pointer is reset and the RxPktRdy bit (below) is cleared.
D3	DataError/ NAK Timeout	When operating in ISO mode, this bit is set when RxPktRdy is set if the data packet has a CRC or bit-stuff error and cleared when RxPktRdy is cleared. In Bulk mode, this bit will be set when the Rx endpoint is halted following the receipt of NAK responses for longer than the time set as the NAK Limit by the RxInterval register. The CPU should clear this bit to allow the endpoint to continue.
D2	Error	The USB sets this bit when 3 attempts have been made to receive a packet and no data packet has been received. The CPU should clear this bit. An interrupt is generated when the bit is set
D1	FIFOFull	This bit is set when no more packets can be loaded into the Rx FIFO.
D0	RxPktRdy	This bit is set when a data packet has been received. The CPU should clear this bit when the packet has been unloaded from the Rx FIFO. An interrupt is generated when the bit is set.

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7.7.8 RXCOUNT

RxCount is a 13-bit read-only register that holds the number of received data bytes in the packet currently in line to be read from the Rx FIFO. If the packet was transmitted as multiple bulk packets, the number given will be for the combined packet.

Table 7-29: RXCOUNT

Address offset: 0x18 Reset Value:13'h0000

Bit	Name	Function
D12 ~ D0	Rx Count	Endpointn Rx Count

7.7.9 TXTYPE (Host Mode Only)

TxType is a 6-bit register that should be written with the endpoint number to be targeted by the endpoint in the lower 4 bits, and the transaction protocol to use for the currently-selected Tx endpoint in the upper 2 bits.

Table 7-30: TXTYPE

Address offset: 0x1A Reset Value:8'h00

Bit	Name	Function
D5 ~ D4	Protocol	The CPU should set this to select the required protocol for the Tx endpoint: 00: Illegal 01: Isochronous 10: Bulk 11: Interrupt
D3 ~ D0	Target Endpoint Number	The CPU should set this value to the endpoint number contained in the Tx endpoint descriptor returned to the USB controller during device enumeration.

7.7.10 TXINTERVAL (Host Mode Only)

TxInterval is an 8-bit register that, for Interrupt and Isochronous transfers, defines the polling interval for the currently-selected Tx endpoint. For Bulk endpoints, this register sets the number of frames/microframes after which the endpoint should timeout on receiving a stream of NAK responses. There is a TxInterval register for each configured Tx endpoint (except Endpoint 0).

Table 7-31: TXINTERVAL

Address offset: 0x1B Reset Value:8'h00

Transfer Type	Speed	Valid values (m)	Interpretation
Interrupt	Low Speed or Full Speed	1 - 255	Polling interval is m frames
	High Speed	1 - 16	Polling interval is $2^{(m-1)}$
Isochronous	Full Speed or High Speed	1 - 16	Polling interval is $2^{(m-1)}$ frames / microframs

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Bulk	Full Speed or High Speed	2 - 16	NAK Limit is $2^{(m-1)}$ frams / microframs

7.7.11 RXTYPE (Host mode only)

RxType is a 6-bit register that should be written with the endpoint number to be targeted by the endpoint in the lower 4 bits, and the transaction protocol to use for the currently-selected Rx endpoint in the upper 2 bits.

Table 7-32: RXTYPE

Address offset: 0x1C Reset Value:8'h00

Bit	Name	Function
D5 ~ D4	Protocol	The CPU should set this to select the required protocol for the Rx endpoint: 00: Illegal 01: Isochronous 10: Bulk 11: Interrupt
D3 ~ D0	Tartget Endpoint Number	The CPU should set this value to the endpoint number contained in the Rx endpoint descriptor returned to the USB controller during device enumeration.

7.7.12 RXINTERVAL (Host Mode Only)

RxInterval is an 8-bit register that, for Interrupt and Isochronous transfers, defines the polling interval for the currently-selected Rx endpoint. For Bulk endpoints, this register sets the number of frames/microframes after which the endpoint should timeout on receiving a stream of NAK responses. There is a RxInterval register for each configured Rx endpoint (except Endpoint 0).

Table 7-33: RXINTERVAL

Address offset: 0x1D Reset Value:8'h00

Transfer Type	Speed	Valid Values (m)	Interpretation
Interrupt	Low Speed or Full Speed	1 - 255	Polling interval is m frames
	High Speed	1 - 16	Polling interval is $2^{(m-1)}$ microframes
Isochronous	Full Speed or High Speed	1 - 16	Polling interval is $2^{(m-1)}$ frames/microframes
Bulk	Full Speed or High Speed	2 - 16	NAK Limit is $2^{(m-1)}$ frames/microframes

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7.8 FIFOx (Addresses 20h - 5Fh)

This address range provides 16 addresses for CPU access to the FIFOs for each endpoint. Writing to these addresses loads data into the TxFIFO for the corresponding endpoint. Reading from these addresses unloads data from the RxFIFO for the corresponding endpoint.

The address range is 20h – 5Fh and the FIFOs are located on 32-bit double-word boundaries (Endpoint 0 at 20h, Endpoint 1 at 24h ... Endpoint 15 at 5Ch).

7.9 DYNAMIC FIFO SIZING

If needed, the USB controller can be configured to have a single overall FIFO size of 128, 256, 512, 1K … 64K bytes, areas of which may then be allocated to the different endpoints when the USB controller is initialized. (Users are however strongly advised to only use this feature where the USB controller is used in a device that actually requires different FIFO sizes in different contexts because of the cost in additional gate count and in complexity of the firmware that this feature imposes on the core.)

The allocation of FIFO space to the different endpoints requires the specification for each Tx and Rx endpoint of:

- The start address of the FIFO within the RAM block
- The maximum size of packet to be supported
- Whether double-buffering is required

(These last two together define the amount of space that needs to be allocated to the FIFO.)

Table 7-34: Dynamic FIFO register map

REGISTER MAP		
ADDR	**NAME**	**DESCRIPTION**
0x62	TxFIFOsz	Tx Endpoint FIFO size
0x63	RxFIFOsz	Rx Endpoint FIFO size
0x64,65	TxFIFOadd	Tx Endpoint FIFO address
0x66,67	RxFIFOadd	Rx Endpoint FIFO address

Table 7-35: TxFIFOsz / RxFIFOsz

Bit	Name	Function
D4	DPB	Defines whether double-packet buffering supported. When '1', double-packet buffering is supported. When '0', only single-packet buffering is supported.
D3 ~ D0	SZ[3:0]	SZ[3:0] Packet Size (Bytes) 0 0 0 0 8 0 0 0 1 16 0 0 1 0 32 0 0 1 1 64 0 1 0 0 128 0 1 0 1 256 0 1 1 0 512 0 1 1 1 1024

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Bit	Name	Function
		1 0 0 0 2048 1 0 0 1 4096 If DPB = 0, the FIFO will also be this size; if DPB = 1, the FIFO will be twice this size.

7.10 DMA

Table 7-36: TxFIFOadd / RxFIFOadd

Bit	Name	Function
D13	-	Reserved for future use
D12 ~ D0	AD[12:0]	Start address of the endpoint FIFO in units of 8 bytes as follows: AD[12:0] Start Address 0 0 0 0 0000 0 0 0 1 0008 0 0 0 2 0010 … … 1 F F F FFF8

7.10.1 DMA REGISTERS

The DMA Controller have 8 channels.The DMA controller has one interrupt register which indicates which channels have a pending interrupt, and a set of three control registers for each configured channel.

Table 7-37: DMA REGISTERS

ADDRESS	Register	Description
0x200	INTR	Indicates pending DMA interrupts, one bit per DMA channel implemented. D0 used for DMA Channel 1, D1 for DMA Channel 2 etc. Cleared when read.

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ADDRESS	Register	Description
0x204	CNTL(1)	DMA Channel 1 Control: D0: Enable DMA D1: Direction. 0 = DMA Write (Rx endpoint), 1 = DMA Read (Tx endpoint) D2: DMA Mode D3: Interrupt Enable D7–4: Endpoint number D8: Bus Error D10–9: Burst Mode 00 = Burst Mode 0 : Bursts of unspecified length 01 = Burst Mode 1 : INCR4 or unspecified length 10 = Burst Mode 2 : INCR8, INCR4 or unspecified length 11 = Burst Mode 3 : INCR16, INCR8, INCR4 or unspecified length
0x208	ADDR(1)	DMA Channel 1 AHB Memory Address (32 bits)
0x20c	COUNT(1)	DMA Channel 1 Byte Count (32 bits)*
0x210	-	Unused
0x214	CNTL(2)	DMA Channel 2 Control: D0: Enable DMA D1: Direction. 0 = DMA Write (Rx endpoint), 1 = DMA Read (Tx endpoint) D2: DMA Mode D3: Interrupt Enable D7–4: Endpoint number D8: Bus Error D10–9: Burst Mode 00 = Burst Mode 0 : Bursts of unspecified length 01 = Burst Mode 1 : INCR4 or unspecified length 10 = Burst Mode 2 : INCR8, INCR4 or unspecified length 11 = Burst Mode 3 : INCR16, INCR8, INCR4 or unspecified length
0x218	ADDR(2)	DMA Channel2 AHB Memory Address (32 bits)
0x21c	COUNT(2)	DMA Channel 2 Byte Count (32 bits)*
...	...	...
0x270	-	-

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ADDRESS	Register	Description
0x274	CNTL(7)	DMA Channel 7 Control: D0: Enable DMA D1: Direction. 0 = DMA Write (Rx endpoint), 1 = DMA Read (Tx endpoint) D2: DMA Mode D3: Interrupt Enable D7–4: Endpoint number D8: Bus Error D10–9: Burst Mode 00 = Burst Mode 0 : Bursts of unspecified length 01 = Burst Mode 1 : INCR4 or unspecified length 10 = Burst Mode 2 : INCR8, INCR4 or unspecified length 11 = Burst Mode 3 : INCR16, INCR8, INCR4 or unspecified length
0x278	ADDR(7)	DMA Channel 7 AHB Memory Address (32 bits)
0x27c	COUNT(7)	DMA Channel 7 Byte Count (32 bits)*

7.10.2 DMA BUS ERRORS

If a bus error occurs while the DMA controller is accessing memory on the AHB, the DMA controller will immediately terminate the DMA transfer and interrupt the processor with the Bus Error (D8) bit of the CNTL register set.

NOTE: The generation of this interrupt is not affected by the setting of the Interrupt Enable bit in the DMA CNTL register (bit 3).

The interrupt will still be generated when CNTL.D3 = 0.

7.10.3 TRANSFERRING PACKETS

Use of the built-in DMA controller to access the FIFOs requires both the DMA controller and the USBC endpoint to be appropriately programmed. Many variations are possible. The following sections detail the standard set-ups used for the basic actions of transferring individual packets and multiple packets.

7.10.3.1 INDIVIDUAL PACKET: RX ENDPOINT

The transfer of individual packets will normally be carried out using DMA Mode 0. For this, the USBC Rx endpoint should be programmed as follows:

The relevant interrupt enable bit in the IntrRxE register set to 1.

The DMAReqEnab bit (D13) of the appropriate RxCSR register set to 0. (**NOTE:** There is no need to set the USBC to generate DMA requests for this operation.) When a packet has been received by the USBC, it will generate the appropriate Endpoint interrupt. The processor should then program selected channel of the DMA controller as follows:

ADDR : Memory address to store packet

COUNT : Size of packet (determined by reading the USBC RxCount register)

CNTL : DMA Enable (D0) =1; Direction (D1) =0; DMA Mode (D2) =0; Interrupt Enable (D3) =1; Required Burst Mode (D10–9).

The DMA controller will then request bus mastership and transfer the packet to memory. When it has completed the transfer, it will generate a DMA interrupt (DMA_NINT taken low). The processor should then

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clear the RxPktRdy bit in the RxCSR register.

7.10.3.2 INDIVIDUAL PACKET: TX ENDPOINT

To carry out this operation using DMA Mode 0, an USBC Tx endpoint should be programmed as follows:

The relevant interrupt enable bit in the IntrTxE register set to 1.

The DMAReqEnab bit (D12) of the appropriate TxCSR register set to 0. (**NOTE:** There is no need to set the USBC to generate DMA requests for this operation.)

When the FIFO in the USBC becomes available, the USBC will interrupt the processor with the appropriate Tx Endpoint interrupt. The processor should then program the DMA controller as follows:

ADDR : Memory address of packet to send

COUNT : Size of packet to be sent

CNTL : DMA Enable (D0) =1; Direction (D1) =1; DMA Mode (D2) =0; Interrupt Enable (D3) =1;Required Burst Mode (D10–9).

The DMA controller will then request bus mastership and transfer the packet to the USBC FIFO. When it has completed thetransfer, it will generate a DMA interrupt. The processor should then set the TxPktRdy bit in the USBC TxCSR register.

7.10.3.3 MULTIPLE PACKETS: RX ENDPOINT

The transfer of multiple packets will normally be carried out using DMA Mode 1.

Where multiple packets are to be received using DMA Mode 1, the DMA Controller should be programmed as follows:

ADDR : Memory address of the buffer in which to store transfer

COUNT : Maximum size of data buffer
CNTL : DMA Enable (D0) =1; Direction (D1) =0; DMA Mode (D2) =1; Interrupt Enable (D3) =1;Required Burst Mode (D10–9). and the USBC Rx endpoint should be programmed as follows:

The relevant interrupt enable bit in the IntrRxE register should be set to 1.

The AutoClear (D15), DMAReqEnab (D13) and DMAReqMode (D11) bits of the appropriate RxCSR register should be set to 1. In Host mode, the AutoReq (D14) bit should also be set to 1 and the RqPktCount register should be programmed with the number of packets in the transfer.

As each packet is received by the USBC, the DMA controller will request bus mastership and transfer the packet to memory. With AutoClear set, the USBC will automatically clear the RxPktRdy bit.

In Peripheral mode or where RqPktCount is zero, this process will continue automatically until the USBC receives a ‘short packet’ (one of less than the maximum packet size for the endpoint) signifying the end of the transfer. This ‘short packet’ will not be transferred by the DMA controller: instead the USBC will interrupt the processor by generating the appropriate Endpoint interrupt. The processor can then read the USBC RxCount register to see the size of the ‘short packet’ and either unload it manually or reprogram the DMA controller in Mode 0 to unload the packet.

In Host mode with AutoReq set and RqPktCount non-zero, the core will decrement the value in the RqPktCount register following each request. When the value decrements from 1 to 0, the AutoReq bit is cleared to prevent any further transactions being attempted.

The DMA controller ADDR register will have been incremented as the packets were unloaded so the processor can determine the size of the transfer by comparing the current value of ADDR against the start address of the memory buffer.

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7.10.3.4 MULTIPLE PACKETS: TX ENDPOINT

To carry out this operation using DMA Mode 1, the DMA controller should be programmed as follows:

ADDR : Memory address of data block to send

COUNT : Size of data block

CNTL : DMA Enable (D0) =1; Direction (D1) =1; DMA Mode (D2) =1; Interrupt Enable (D3) =1; Required Burst Mode (D10–9). and the USBC Tx endpoint should be programmed as follows:

The relevant interrupt enable bit in the IntrTxE register should be set to 1 (simply so that errors can be detected).

The AutoSet (D15), DMAReqEnab (D12) and the DMAReqMode (D10) bits of the appropriate TxCSR register should be set to 1.

When the FIFO in the USBC becomes available, the DMA controller will request bus mastership and transfer a packet to the FIFO. With AutoSet set, the USBC will automatically set the TxPktRdy bit. This process will continue until the entire data block has been transferred to the USBC. The DMA controller will then interrupt the processor by taking DMA_NINT low. If the last packet to be loaded was less than the maximum packet size for the endpoint, the TxPktRdy bit will not have been set for this packet: the processor should therefore respond to the DMA interrupt by setting the TxPktRdy bit to allow the last 'short packet' to be sent. If the last packet to be loaded was of the maximum packet size, then the action to take depends on whether the transfer is under the control of an application such as the mass storage software on a Windows system that keeps count of the individual packets sent. If the transfer isn't under such control, the processor should still respond to the DMA interrupt by setting the TxPktRdy bit. This has the effect of sending a null packet for the receiving software to interpret as indicating the end of the transfer.

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7.11 Functional Description

7.11.1 Reset

When a reset condition is detected on the bus, the USBC tries to negotiate for high-speed operation if the HS Enable bit in the UCSR register is set. The device will then perform the following actions:

- Set the FAdress register to zero
- Set EINDEXR register to zero
- Flush all endpoints FIFOs
- Clear all control and status registers
- Enable all endpoint interrupts
- Generate a Reset interrupt

7.11.2 Soft Connect

The USBC can be configured to allow its connection to the USB bus to be controlled by software. The PHY used alongside the USBC can be switched between normal mode and non-driving mode by setting or clearing bit 6 (Soft Conn bit) in the UCSR register. When this bit is set to 1, the PHY is placed in its normal mode and the data lines of the USB bus are enabled. At the same time, the USBC is in POWERED state, in which it will not respond to any USB signaling except a USB reset. When this bit is set to 0, the PHY is put into non-driving mode, data lines are tri-stated and the USBC appears to other devices on the USB bus as if it has been disconnected.

7.11.3 Start Of Frame (SOF)

The USBC should receive a start of frame packet (SOF) from the host once every 1ms when the USBC is in Full-speed mode (the default).

A SOF interrupt will be generated when the SOF packet is received if the SOF interrupt is enabled. The frame number contained in the packet is written into the Frame Number register. If no SOF packet is received after 1.00385ms, it is assumed the packet has been lost. The USBC will continue to generate the SOF interrupt every millisecond, but the Frame number register is not updated.

7.11.4 Suspend/Resume

When no activity has occurred on the USB for 3ms, the USBC will enter Suspend mode and the suspend interrupt will be generated.

When Resume signaling is detected, the clock to the USBC must be restarted, and the USBC will exit Suspend mode, while the Resume interrupt will be generated.

The USBC can also support remote wakeup. The CPU can set the Resume bit in the UCSR register to get the USBC out of suspend mode and drive Resume signaling onto the bus. The CPU shoud clear this bit after approximately 10ms to end the Resume signal. At this time no Resume interrupt will be generated.

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7.11.5 In-Transaction Handling

In-Transactions are handled through the USBC Tx FIFOs.

The maximum size of data packet is programmable and determined by the value written to the TXPSZR register for the endpoint.

The maximum packet size set for any endpoint must not exceed the FIFO size. And the TXPSZR register should not be written to while there is data is in the FIFO as unexpected results may occur.

7.11.6 Transmit Packet Buffering

Only one packet can be buffered in the FIFO. As each packet to be sent is load into the Tx FIFO, the TxPktRdy bit in TxCSR needs to be set. If the AutoSet bit in TxCSR is set, the TxPktRdy bit will be automatically set when a maximum-sized packet is loaded into the FIFO. For packet sizes less than the maximum, TxPktRdy will always have to be set manually.

When the TxPktRdy bit is set, either manually or automatically, the FIFONotEmpty bit in TxCSR is also set and the packet is ready to be sent. When the packet has been successfully sent, both TxPktRdy and FIFONotEmpty will be cleared and the appropriate Tx endpoint interrupt generated. The next packet can then be loaded into the FIFO.

7.11.7 Out-Transaction Handling

Out-Transactions are handled through the USBC Rx FIFOs.

The maximum size of data packet is determined by the value written to the RXPSZR register for the endpoint.

The maximum packet size set for any endpoint must not exceed the FIFO size.

7.11.8 Receive Packet Buffering

Only one packet can be buffered in the FIFO. When a packet is received and placed in the Rx FIFO, the RxPktRdy bit and the FIFOFull bit in RxCSR are set and the appropriate Rx endpoint interrupt is generated (if enabled) to signal that a packet can now be unloaded from the FIFO.

After the packet has been unloaded, the RxPktRdy bit needs to be cleared in order to allow further packets to be received. If the AutoClear bit in RxCSR is set and a maximum-sized packet is unloaded from the FIFO, the RxPktRdy bit is cleared automatically. The FIFOFull bit is also cleared. For packet sizes less than the maximum, RxPktRdy will always have to be cleared manually.

7.11.9 External clock reference detector

The USBC will detect the CLK12_REF automatically in 20ms after RSTN be released. If there is a clock from CLK12_REF then USB will use it and power down internal RC and clock Recover, else USB use internal RC output as reference clock and recover 12MHz clock from DP/DM data frame.

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7.12 Transfer Operation

There are three basic transfer types supported by the USBC, as described below.

7.12.1 Control Transfer

Control transfers are typically used for command and status operations. They are essential to set up a USB device with all enumeration functions being performed using control transfers. In the USBC, Endpoint 0 supports this type of transfer.

A control transfer can have up to three stages :

- The Setup Stage is where a request is sent. This consists of three packets. A setup token is sent first which contains the address and endpoint number.
- The data packet is sent next and always has a PID type of Data0 and includes a setup packet which details the type of request. The setup packet is detailed later.
- The last packet is a handshake used for acknowledging successful receipt or to indicate an error. If the function successfully receives the setup data (valid CRC and PID etc) it responds with ACK, otherwise it ignores the data and doesn't send a handshake packet. The USBC cannot issue a STALL or NAK packet in response to a setup packet.

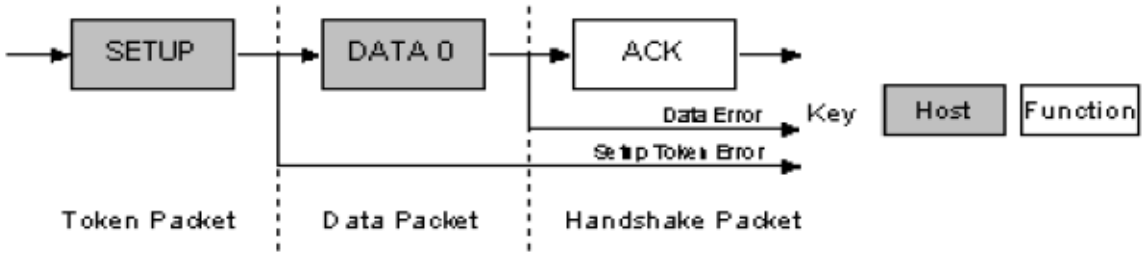


Figure 7-1: Setup Stage Format

The optional Data Stage consists of one or more IN or OUT transfers. The setup request indicates the amount of data to be transmitted in this stage. If it exceeds the maximum packet size, data will be sent in multiple transfers each being the maximum packet length except for the last packet.

The data stage has two different scenarios depending upon the direction of data transfer.

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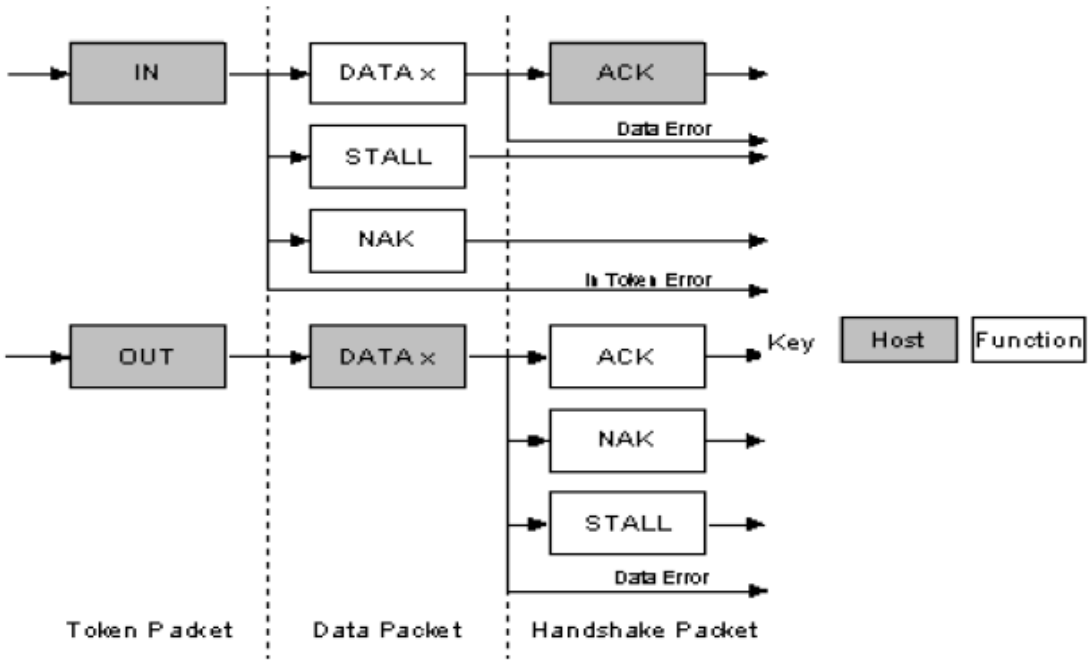


Figure 7-2: Data Stage Format

IN: When the host is ready to receive control data it issues an IN Token. If the device receives the IN token with an error e.g. the PID doesn't match the inverted PID bits, then it ignores the packet. If the token was received correctly, the device can either reply with a DATA packet containing the control data to be sent, a stall packet indicating the endpoint has had a error or a NAK packet indicating to the host that the endpoint is working, but temporary has no data to send.

OUT: When the host needs to send the device a control data packet, it issues an OUT token followed by a DATA packet containing the control data as the payload. If any part of the OUT token or DATA packet is corrupt then the device ignores the packet. If the device's endpoint buffer was empty and it has clocked the data into the endpoint buffer it issues an ACK informing the host it has successfully received the data. If the endpoint buffer was not empty due to processing of the previous packet, then the device returns a NAK. However if the endpoint has had a error and its halt bit has been set, it returns a STALL.

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The Status Stage reports the status of the overall request and this once again varies per direction of transfer. Status reporting is always performed by the device.

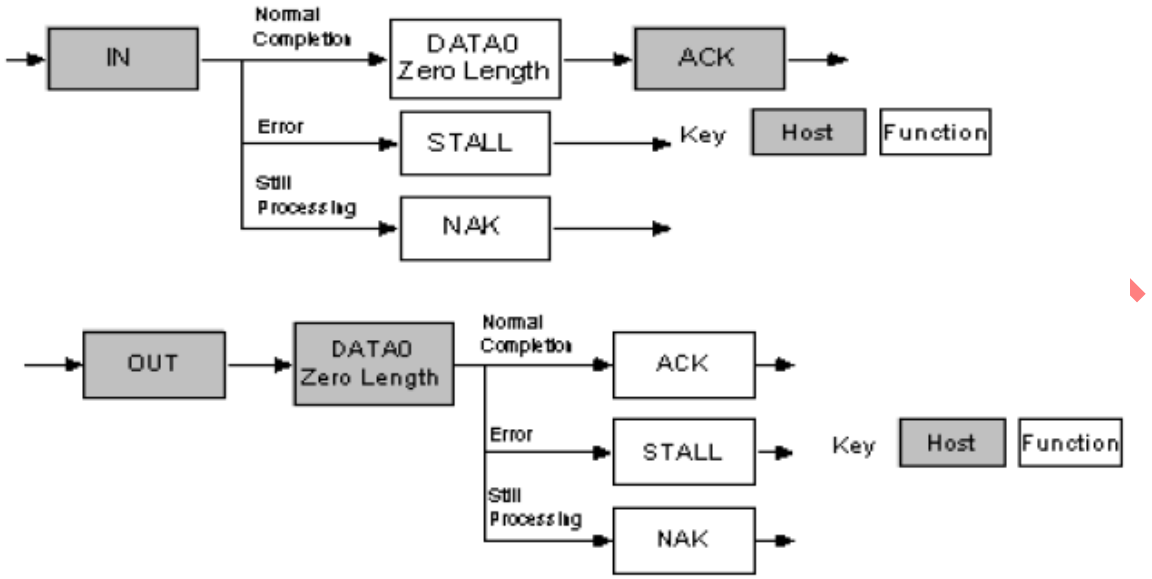


Figure 7-3: Status Stage Format

IN: If the host sent IN token(s) during the data stage to receive data, then the host must acknowledge the successful receipt of this data. This is done by the host sending an OUT token followed by a zero length data packet. The device can now report its status in the handshaking stage. An ACK indicates the device has completed the command and is now ready to accept another command. If an error occurred during the processing of this command, then the device will issue a STALL. However if the device is still processing, it returns a NAK indicating to the host to repeat the status stage later.

OUT: If the host sent OUT token(s) during the data stage to transmit data, the device will acknowledge the successful receipt of data by sending a zero length packet in response to an IN token. However if an error occurred, it should issue a STALL or if it is still busy processing data, it should issue a NAK, asking the host to retry the status phase later.

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7.12.2 Interrupt Transfer

An Interrupt request is queued by the device until the host polls the USB device asking for data. The diagram below shows the format of an Interrupt IN and Interrupt OUT transaction. In the USBC, Endpoint 1supports this type of transfer.

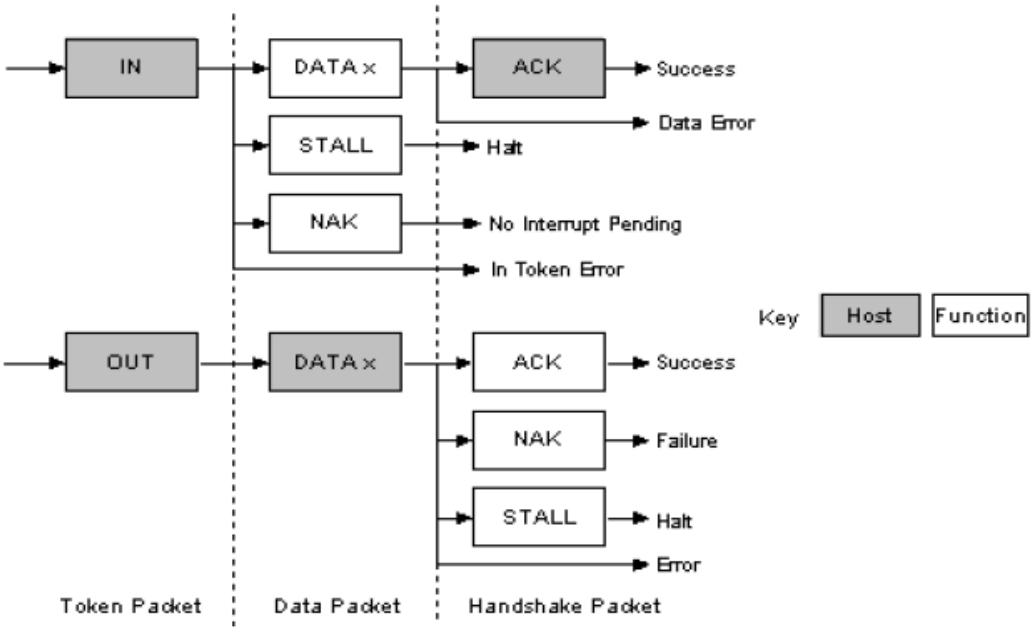


Figure 7-4: Interrupt Transfer Format

IN: The host will periodically poll the interrupt endpoint. This rate of polling is specified in the endpoint descriptor which is covered later. Each poll will involve the host sending an IN Token.

If an interrupt has been queued by the device, the device will send a data packet containing data relevant to the interrupt when it receives an IN Token. Upon successful receipt at the host, the host will return an ACK. However if the data is corrupted, the host will return no status.

If an interrupt condition was not present when the host polled the interrupt endpoint with an IN token, then the device signals this state by sending a NAK. If an error has occurred on the endpoint, a STALL is sent in reply to the IN token instead.

OUT: When the host wants to send the device interrupt data, it issues an OUT token followed by a data packet containing the interrupt data. The device will issue responses to the host based on different conditions.

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7.12.3 Bulk Transfer

Bulk transfers can be used for large volumes of data, such as a print-job sent to a printer. Bulk transfers provide error correction in the form of a CRC16 field on the data payload and error detection/re-transmission mechanism ensuring data is transmitted and received without error. In the USBC, Endpoint 1 supports this type of transfer.

For Full Speed transfers, the maximum bulk packet size is 64 bytes long. If the data payload falls short of the maximum packet size, it doesn't need to be padded with zeros. A bulk transfer is considered complete when it has transferred the exact amount of data requested, transferred a packet less than the maximum endpoint size or transferred a zero-length packet.

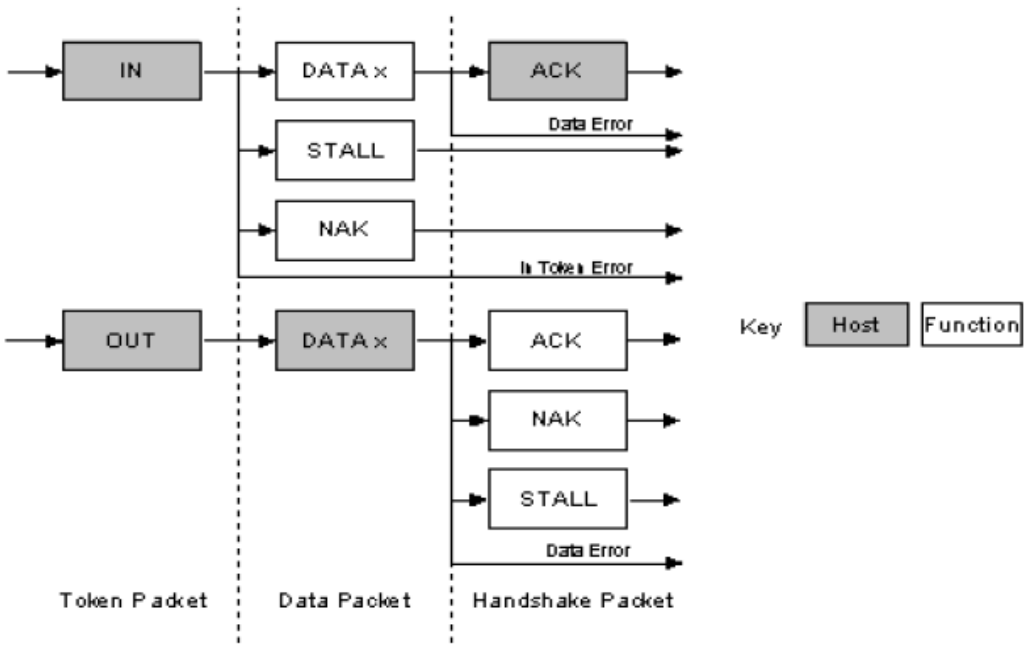


Figure 7-5: Bulk Transfer Format

The above diagram shows the format of a bulk IN and OUT transaction.

IN: When the host is ready to receive bulk data it issues an IN Token. If the device receives the IN token with an error, it ignores the packet. If the token was received correctly, the device can either reply with a DATA packet containing the bulk data to be sent, a stall packet indicating the endpoint has had a error, or a NAK packet indicating to the host that the endpoint is busy, and temporarily has no data to send.

OUT: When the host wants to send the device a bulk data packet, it issues an OUT token followed by a data packet containing the bulk data. If any part of the OUT token or data packet is corrupt, the device ignores the packet. If the device's endpoint buffer was empty and it has clocked the data into the endpoint buffer it issues an ACK, informing the host it has successfully received the data. If the endpoint buffer was not empty due to processing a previous packet, then the device returns a NAK. However if the endpoint has had an error and it's halt bit has been set, it returns a STALL.

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7.12.4 Control Transactions

EP0 is the control endpoint of the USBC. Software should handle all the Standard Device Requests that may be sent or received via EP0. The protocol for these device requests involves different numbers and types of transaction per transfer.

The Standard Device Requests required by a USB peripheral can be divided into three categories: Zero Data Requests, Write Requests, and Read Requests.

- Zero Data Requests : all the information is included in the command.
- Write Requests: the command will be followed by additional data.
- Read Requests: the device is required to send data back to the host.

7.12.4.1 Zero Data Requests

Zero data requests have all their information included in the 8-byte command and require no additional data to be transferred. For example: SET_ADDRESS, SET_CONFIGURATION, SET_INTERFACE, SET_FEATURE.

The E0CSR register is the USBC conrol transfer status register. Software should set the corresponding bit based on the different condition to handle the control transfer.

When the USBC receives an EP0 interrupt, the sequence of events will begin.

The RxPktRdy bit (E0CSR) will have been set. The 8-byte command should then be read from the Endpoint 0 FIFO, decoded, and the appropriate action taken. For example if the command is SET_ADDRESS, the 7-bit address value contained in the command should be written to the FADDRR register.

The E0CSR register should then be written to set the ServicedRxPktRdy bit (indicating that the command has been read from the FIFO) and set the DataEnd bit (indicating that no further data is expected for this request).

When the host moves to the status stage of the request, a second EP0 interrupt will be generated to indicate that the request has completed. No further action is required from Software: the second interrupt is just a confirmation that the request completed successfully.

If the command is an unrecognized command, or for some other reason cannot be executed, then the E0CSR register should be written to set the ServicedRxPktRdy bit and set the SendStall bit. When the host moves to the status stage of the request, the USBC will send a STALL to tell the host that the request was not executed. A second EP0 interrupt will be generated and the SentStall bit will be set.

If the host sends more data after the DataEnd bit has been set, then the USBC will send a STALL. An EP0 interrupt will be generated and the SentStall bit will be set.

7.12.4.2 Write Requests

Write requests involve an additional packet (or multiple packets) of data being sent from the host after the 8-byte command. For example: SET_DESCRIPTOR.

The E0CSR register is the USBC control transfer status register. Software should set corresponding bits based on different conditions to handle the control transfer.

When USBC receives an EP0 interrupt, the sequence of events will begin. The RxPktRdy bit will have been set. The 8-byte command should be read from the Endpoint0 FIFO and decoded. The E0CSR register should then be written to set the ServicedRxPktRdy bit (indicating that the command has been read from the FIFO) but in this case the DataEnd bit should not be set (indicating that more data is expected).

When a second EP0 interrupt is received, the E0CSR register should be read to check the endpoint status. The RxPktRdy bit should be set to indicate that a data packet has been received. The E0COUNTR register should then be read to determine the size of this data packet. The data packet can then be read from the Endpoint0 FIFO.

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If the length of the data associated with the request (indicated by the wLength field in the command) is greater than the maximum packet size for Endpoint0, further data packets will be sent. In this case, E0CSR should be written to set the ServicedRxPktRdy bit, but the DataEnd bit should not be set.

When all the expected data packets have been received, the E0CSR register should be written to set the ServicedRxPktRdy bit and set the DataEnd bit (indicating that no more data is expected).

When the host moves to the status stage of the request, another Endpoint0 interrupt will be generated to indicate that the request has completed. No further action is required from Software: the interrupt is just a confirmation that the request completed successfully.

If the command is an unrecognized command, or for some other reason cannot be executed, then the E0CSR register should be written to set the ServicedRxPktRdy bit and set the SendStall bit. When the host sends more data, the USBC will send a STALL to tell the host that the request was not executed. An Endpoint0 interrupt will be generated and the SentStall bit will be set. If the host sends more data after the DataEnd has been set, then the USBC will also send a STALL. An Endpoint0 interrupt will be generated and the SentStall bit will be set.

7.12.4.3 Read Requests

Read requests involve a packet (or multiple packets) of data sent from the device to the host after the 8-byte command. For example: GET_CONFIGURATION, GET_STATUS, GET_INTERFACE, GET_DESCRIPTOR.

When USBC receives an EP0 interrupt, the following sequence of events will begin: The RxPktRdy bit (E0CSR) will have been set. The 8-byte command should be read from the Endpoint0 FIFO and decoded. The E0CSR register should then be written to set the ServicedRxPktRdy bit (indicating that the command has been read from the FIFO).

The data to be sent to the host should be written to the EP0 FIFO. If the data to be sent is greater than the maximum packet size for EP0, only the maximum packet size should be written to the FIFO. The E0CSR register should then be written to set the TxPktRdy bit (indicating that there is a packet in the FIFO to be sent). When the packet has been sent to the host, another EP0 interrupt will be generated and the next data packet can be written to the FIFO. When the last data packet has been written to the FIFO, the E0CSR register should be written to set the TxPktRdy bit and to set the DataEnd bit (indicating that there is no more data after this packet).

When the host moves to the Status stage of the request, another EP0 interrupt will be generated to indicate that the request has completed. No further action is required from Software: the interrupt is just a confirmation that the request completed successfully.

If the command is an unrecognized command, or for some other reason cannot be executed, then the E0CSR register should be written to set the ServicedRxPktRdy bit and set the SendStall bit . When the host requests data, the USBC will send a STALL to tell the host that the request was not executed. An Endpoint0 interrupt will be generated and the SentStall bit will be set.

If the host requests more data after DataEnd (D3) has been set, then the USBC will send a STALL. An Endpoint 0 interrupt will be generated and the SentStall bit will be set.

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7.12.4.4 Endpoint0 States

Endpoint0 control has three modes: IDLE, TX and RX corresponding to the different phases of the control transfer and the states Endpoint0 enters for the different phases of the transfer. The default mode on power-up or reset should be IDLE.

RxPktRdy becoming set when Endpoint0 is in IDLE state indicates a new device request. Once the device request is unloaded from the FIFO, the USBC decodes the descriptor to find out whether there is a Data phase and, if so, the direction of the Data phase of the control transfer (in order to set the FIFO direction).

Depending on the direction of the Data phase, Endpoint0 goes into either TX state or RX state. If there is no Data phase, Endpoint0 remains in IDLE state, ready to accept the next device request.

The actions that the CPU needs to take at the different phases of the possible transfers (e.g. Loading the FIFO, Setting TxPktRdy) are indicated in **Figure 7-6** below. The USBC changes the FIFO direction depending on the direction of the Data phase independently of the CPU.

7.12.4.5 Endpoint0 Interrupt Service Routine

The EP0 interrupts will be generated in the following cases:

- When the core sets the RxPktRdy bit (E0CSR bit 0) after a valid token has been received and data has been written to the FIFO.
- When the core clears the TxPktRdy bit (E0CSR bit 1) after the packet of data in the FIFO has been successfully transmitted to the host.
- When the core sets the SentStall bit (E0CSR bit 2) after a control transaction is ended due to a protocol violation.
- When the core sets the SetupEnd bit (E0CSR bit 4) because a control transfer has ended before DataEnd (E0CSR bit 3) is set.

Whenever the Endpoint0 interrupt service routine is entered, Software must first check to see if the current control transfer has been ended due to either a STALL condition or a premature end of control transfer. If the control transfer ends due to a STALL condition, the SentStall bit would be set. If the control transfer ends due to a premature end of control transfer, the SetupEnd bit would be set. In either case, Software should abort processing the current control transfer and set the state to IDLE.

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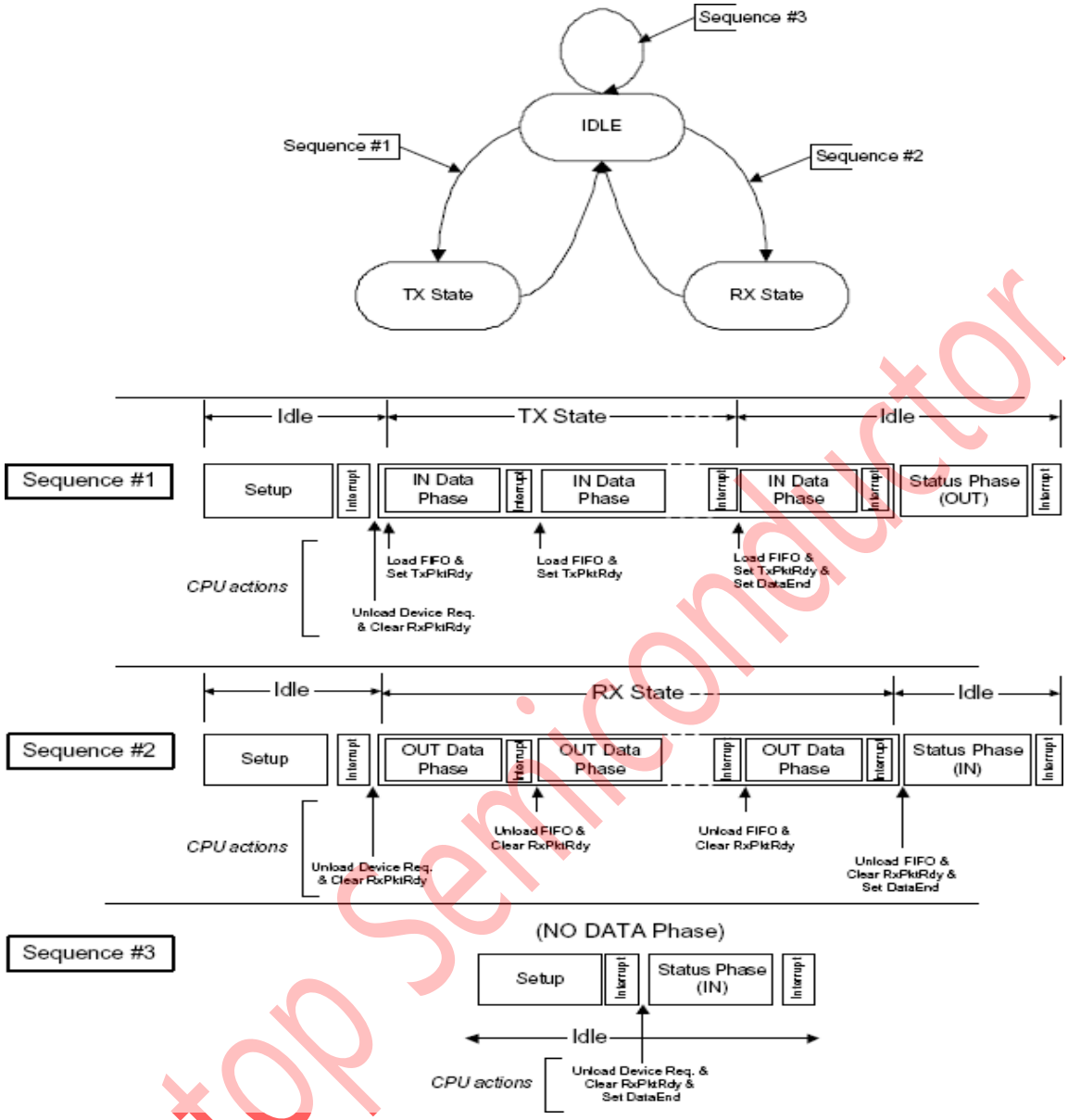


Figure 7-6: EP0 Transfer Scenarios

Once Software has determined that the interrupt was not generated by an illegal bus state, the next action taken depends on the Endpoint state.

If Endpoint0 is in IDLE state, the only valid reason an interrupt can be generated is as a result of the core receiving data from the USB bus. The service routine must check for this by testing the RxPktRdy bit.

If the RxPktRdy bit is set, then the core has received a SETUP packet, which must be unloaded from the FIFO and decoded to determine what to do next.

Depending on the command contained within the SETUP packet, Endpoint0 will enter one of three states:

- If the command is a single packet transaction without any data phase (e.g. SET_ADDRESS, SET_INTERFACE, etc.), the endpoint will remain in IDLE state.
- If the command has an OUT data phase (e.g. SET_DESCRIPTOR, etc.), the endpoint will enter RX state.

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- If the command has an IN data phase (e.g. GET_DESCRIPTOR, etc.), the endpoint will enter TX state.

If the endpoint is in TX state, the interrupt indicates that the core has received an IN token and data from the FIFO has been sent. The Software must respond to this either by placing more data in the FIFO if the host is still expecting more data, or by setting the DataEnd bit to indicate that the data phase is complete. Once the data phase of the transaction has been completed, Endpoint0 should be returned to IDLE state to await the next control transaction.

If the endpoint is in RX state, the interrupt indicates that a data packet has been received. Software must respond by unloading the received data from the FIFO. Software must then determine whether it has received all of the expected data. If it has, the Software should set the DataEnd bit and return Endpoint0 to IDLE state. If more data is expected, the Software should set the ServicedRxPktRdy bit (E0CSR bit 6) to indicate that it has read the data in the FIFO and leave the endpoint in RX state.

7.12.5 Bulk IN Transactions

7.12.5.1 Bulk IN Setup

Bulk IN transactions are used to transfer data from the device controller to the host.

The TxCSR register is the USBC Tx transfer status register. Software should set the corresponding bits based on different conditions to handle transfers to the host.

When the AutoSet bit is enabled, the TxPktRdy bit will be automatically set when a packet of TxMaxP bytes has been loaded into the FIFO.

First, the TXPSZR register must be written with the maximum packet size for the endpoint. This value should be the same as the Standard Endpoint Descriptor's wMaxPacketSize field for the endpoint. The relevant interrupt enable bit in the IntrTxE register should be set to "1" if the interrupt is required for this endpoint.

The TxCSR register should be set as follows:

- Bit15 (AutoSet): If set to 1, the AutoSet feature is enabled, otherwise the TxPktRdy bit should set manually.
- Bit13 (Mode): Set to 1 to ensure the Tx FIFO is enabled.
- Bit11 (FrcDataTog): Set to 0 to allow normal data toggle operation.

The lower byte of TxCSR should be writtern to set the ClrDataTog bit, when the endpoint is first configured. This will ensure that data toggle starts in the correct state. If there are any data packets in the FIFO, they should be flushed by setting the FlushFIFO bit.

7.12.5.2 Bulk IN Operation

When data is to be transferred over a Bulk IN pipe, a data packet needs to be loaded into the FIFO and the TxCSR register written to set the TxPktRdy bit. When the packet has been sent, the TxPktRdy bit will be cleared by the USBC and an interrupt generated so that the next packet can be loaded into the FIFO.

The packet size must not exceed the size specified by the lower 11 bits of the TXPSZR register. This part of the register defines the payload for transfers over the USB and is required by the USB Specification to be either 8, 16, 32 or 64 bits for Full-speed. If more than this amount of data is to be transferred, this needs to be sent as multiple USB packets which should all be TXPSZR[10:0] in size, except for the last packet which holds the residue.

The host determines that all the data for a transfer has been sent by comparing the actual amount of data transferred to the expected amount of data.

Alternatively, it may infer all the data has been sent when it receives a packet which is smaller than the stated

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payload. In this case, if the total size of the data is a multiple of this payload, it will be neccessary for the device to send a null packet after all the data has been sent. This is done by setting TxPktRdy when the next interrupt is received, without loading any data into the FIFO.

7.12.5.3 Bulk IN Error Handling

Software can set the SendStall bit (TxCSR) to shut down the Bulk IN pipe. After that, when the USBC receives the next IN token, it will send a STALL to the host, set the SentStall bit and generate an interrupt. When Software receives an interrupt with the SentStall bit set, it should clear the SentStall bit. It should, however, leave the SendStall bit set until it is ready to re-enable the Bulk IN pipe. When a pipe is re-enabled, the data toggle sequence should be restarted by setting the ClrDataTog bit in the TxCSR register.

7.12.6 Bulk OUT Transactions

Bulk OUT transactions are used to transfer data from the host to the device.

The RxCSR register is the USBC Rx transfer status register. Software should set the corresponding bits based on different conditions to handle transfer to the host.

When the AutoClear bit is enbaled, the RxPktRdy bit will be automatically clear when a packet of RxMaxP bytes has been unloaded from the FIFO.

7.12.6.1 Bulk OUT Setup

First, the RxMaxP register must be written with the maximum packet size for the endpoint in configuring a Rx endpoint for Bulk transactions. This value should be the same as the Standard Endpoint Descriptor's wMaxPacketSize field for the endpoint. The relevant interrupt enable bit in the IntrRxE register should be set to "1" if an interrupt is required for this endpoint.

The RxCSR regsiter should be set as follows:

- Bit15 (AutoClear): If set to 1, the AutoClear feature is enabled, otherwise the RxPktRdy bit should set manually.
- Bit12 (DisNyet): Set to 0 to allow normal PING flow control.

The lower byte of RxCSR should be written to set the ClrDataTog bit, when the endpoint is first configured. This will ensure that data toggle starts in the correct state. If there are any data packets in the FIFO, they should be flushed by setting the FlushFIFO bit.

7.12.6.2 Bulk OUT Operation

The RxPktRdy bit is set by the USBC and an interrupt is generated when a data packet is received by a Bulk Rx endpoint. Software should read the RxCount register for the endpoint to determine the size of the data packet. The data packet should be unloaded from the FIFO, and then the RxPktP bit should be cleared automaticlly or manually.

The packets received must not exceed the size specified by the lower 11 bits of the RxMaxP register. This part of the register defines the payload for transfers over the USB and is required by the USB Specification to be either 8, 16, 32 or 64 bits for Full-Speed.

When a block of data larger than wMaxPacketSize needs to be sent to the device, the host will send it as multiple packets. All the packets will be wMaxPacketSize in size, except the last packet, which will contain the residue.

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Software may use an application-specific method to determine the total size of hte block and hence when the last packet has been received. Alternatively, it may infer that the entire block has been received when it receives a packet which is less than the normal packet size. If the total size of the data block is a multiple of the wMaxPacketSize, a null data packet will be sent after the bulk data, to signify that the transfer is finished.

7.12.6.3 Bulk Out Error Handling

Software can set the SendStall bit (RxCSR) to shut down the Bulk OUT pipe. After that, when the USBC receives a next packet, it will send a STALL to the host, set the SentStall bit and generate an interrupt. When Software receives an interrupt with the SentStall bit is set, it should clear the SentStall bit. It should, however, leave the SendStall bit set until it is ready to re-enable the Bulk OUT pipe. When a pipe is re-enabled, the data toggle sequence should be re-started by setting the ClrDataTog bit in the RxCSR register.

7.12.7 Interrupt Transactions

The Interrupt IN transaction uses the same protocol as a Bulk IN transactions and can be used the same way. Similarly, an Interrupt OUT transaction uses almost the same protocol as a Bulk OUT transaction and can be used the same way.

The USB Specification defines that Interrupt transactions may support continuous toggles of the data-toggle bit, which Bulk transactions do not support. This feature is enabled by setting the FrcDataTog bit in the TxCSR register. When this bit is set to "1", the USBC will consider the packet as having been successfully sent and toggle the data bit for the endpoint, regardless of whether an ACK was received from the host.

Interrupt endpoints do not support PING flow control. This means that the USBC should never respond with a NYET handshake, only ACK/NAK/STALL. To ensure this, the DisNyet bit in the RxCSR register should be set to "1" to disable the transmission of NYET handshakes.

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8 EFlash Module

8.1 Introduction

The EFLASH(EFM) is a CMOS page-erase, 38-bit-program embedded Flash memory. 38-bit contains 32-bit data and 6 ECC bits.This memory is partitioned into two memory blocks. One is the main memory block, the other is the information block. The main memory block consists of 131072 words by 38 bits. The information block consists of 128 words by 38 bits. The information block can be used for the storage of device information. The page erase operation erases all bytes within a page. A page is composed of 2 adjacent rows for main memory block and information block. The flash erases and program with VDIO(1.5~1.98V)power supply.

8.2 Features

Features of the EFM include:

- Memory Origanization: 128Kx38+128x38
- Reads of bytes, aligned halfwords (16 bits) and aligned words (32 bits)
- Automated program and erase operation
- Optional interrupt on command completion
- Data Retention: 10 years under 85 degrees
- 0.99~1.21V/1.5~1.98V dual power supplies

8.3 Block Diagram

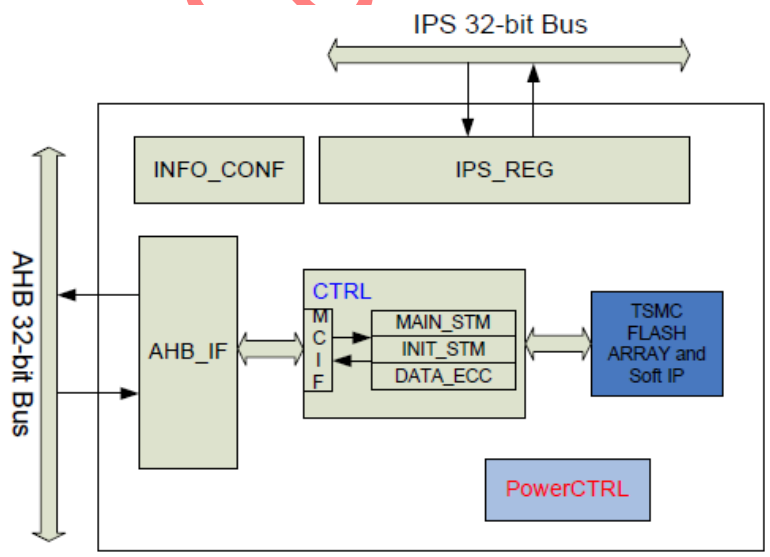


Figure 8-1: EFM Block Diagram

8.4 Modes of Operation

FLASH Normal Mode — In this mode, the EFM is programed and erased though the EFM controller designed

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by user.

8.5 Memory Map and Register

8.5.1 Memory Map

The flash block is mapped starting at address 0x0000_0000 in flash boot mode. shows the flash main blocks' address mapping.

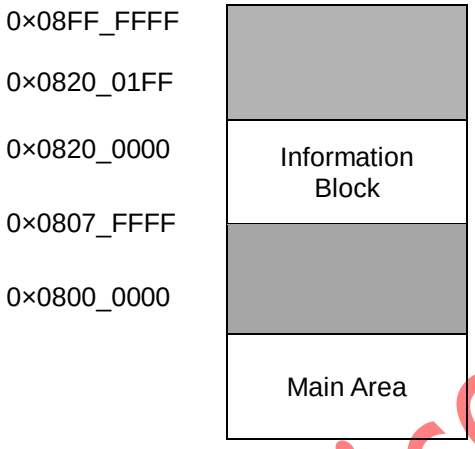


Figure 8-2: EFM Array Memory Map

The EFM module also contains a set of control and status registers. The memory map for these registers is shown in **Table 8-1**.

Table 8-1: Register Memory Map

Address Offset	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access[1,2]
0x0000	EFCR				S
0x0004	EFAPR				S
0x0008	EFSTAT				S
0x000c	EFINTM				S
0x0010	EFCMD				S
0x0014	EECCECNT				S
0x0018	ETIMBASE				S
0x001c	ETIMCFG				S
0x0020	EFPETIMER				S
0x0024	SMWOP0				S
0x0028~0x00FF	SMWOP1				S

NOTES:

- S = CPU supervisor mode access only.
- User mode accesses to supervisor-only address locations have no effect and result in a cycle termination transfer error.

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8.5.2 Register Descriptions

8.5.2.1 EFM Configuration Register(EFCR)

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000007

31	30	29	28	27	26	25	24
ECC_DIS	INFO_SWITCH	AHB_RES_CFG		Reserved	ULTRA_HIGH_SPEED	Reserved	Reserved
rw	rw	rw	rw	ro	rw	ro	

23	22	21	20	19	18	17	16
Reserved							
ro							

15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved					RWSC		
ro					rw		

Figure 8-3: EFM Configuration Register(EFCR)

ECC_DIS — ECC checking enable when read

1 = Disable ECC checking when read

0 = Enable ECC checking when read

INFO_SWITCH — When set, read/write INFO0 will switch to read/write INFO-1

1 = read/write INFO-1

0 = read/write INFO-0

AHB_ERROR_RES — When set, AHB return ERROR response when reading Flash while writting Flash

RWW_RES — When set, AHB return OK response when reading Flash while writting Flash and assert read-while-write interrupt

ULTRA_HS_ACC — When the frequency of controller works above 200MHz. This bit must be asserted.

RWSC — Configure the wait cycle number for read data valid.

NOTE: When both AHB_ERROR_RES and RWW_RES are set, AHB will return ERROR response, and no read-while-write intterrup generated.

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8.5.2.2 EFM Access Protect Register (EFAPR)

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000001

31	30	29	28	27	26	25	24
WPASSWD[31:24]							
w							
23	22	21	20	19	18	17	16
WPASSWD[23:16]							
w							
15	14	13	12	11	10	9	8
WPASSWD[15:8]							
w							
7	6	5	4	3	2	1	0
Reserved		I0AP		Reserved		M0AP	
ro		rw		ro		rw	

Figure 8-4: EFM Access Protection Register(EFAPR)

WPASSWD — Write EFAPR password.

D7-D0 can be written with IPS_WDATA[7:0] when D31-D8 equals to 24'h9786AC to config the MAIN block access protection and equals to 24'hD8D899 to config the Information block block access protection.

I0AP — Information block 0 access protection.

00: No access may be performed

01: Only read accesses may be performed

10: Only write accesses may be performed

11:Both read and write accesses may be performed

M0AP — Main block 0 access protection.

00: No access may be performed

01: Only read accesses may be performed

10: Only write accesses may be performed

11:Both read and write accesses may be performed

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8.5.2.3 EFM Status Register (EFSTAT)

Addr. Offset: 0x0008~0x000b Reset Value: 0xc0008000

31	30	29	28	27	26	25	24
PWRRDY	GRD_READ_VLD	FLASH_ERROR	SLOW_ACC_VLD	Reserved			
ro	ro	ro	ro	ro			
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
EFM_IDLE	Reserved						
ro	ro						
7	6	5	4	3	2	1	0
Reserved	FLASH_SLEEP	ECC_ERROR	RDWW	DONE	PEGOOD	SMWERR	FLASH_BUSY
ro	ro	w1c	w1c	w1c	w1c	w1c	ro

Figure 8-5: EFM Status Register(EFSTAT)

PWRRDY — Indicate Flash 0 power is ready

1 = Power is ready.

0 = Power down or power loss

GRD_READ_VLD — Indicate Chip Good record is valid

1 = Chip Good Record is valid

0 = Chip Good Record is error

FLASH_ERROR — Indicate ECC error happened when INIT Load.

SLOW_READ_FLASH — When the frequency of the clock for controller is less than 200MHz, this bit can be set. Otherwise, this bit must be cleared.

SLOW_READ_FLASH — Indicate FLASH is read at the frequency which under 200MHz

1 = Read at the frequency which under 200MHz

0 = Read at the frequency which above 200MHz

EFM_IDLE — Indicate Flash controller is idle

1 = EFM controller is idle.

0 = EFM controller isn't idle

FLASH0_SLEEP — FLASH 0 enters sleep state

1 = Enter sleep state

0 = Not in sleep state

ECC_ERROR0 — ECC error Happens

ECC error happened when read Flash0.This bit can be cleared by writing 1 to it

1 = ECC error happens

0 = No ECC error happens

RDWW0 — Read FLASH 0 while it is programming or erasing.This bit can be cleared by writing 1 to it

1 = RDWW happens

0 = No RDWW happens

DONE0 — Command done

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CPU triggered command to Flash 0 is done. When new command issued, it will be automatic cleared.This bit can be cleared by writing 1 to it

1 = Flash command is done, 0 = New command issues

PEGOOD0 — P/E operation is finshed

Indicate that whether program or erase operation has finished, it will be automatic cleared.This bit can be cleared by writing 1 to it

1 = Issued program/erase command to Flash 0 finish normally

0 = Issued program/erase command to Flash 0 time out

SMWERR0 — Smart write array0 failed

it will be automatic cleared.This bit can be cleared by writing 1 to it

1 = Smart write array 0 failed

0 = Smart write array 0 successful

FLASH_BUSY0 — Flash is busy

Indicate that whether Flash0 is ready for any operation

1 = Flash array 0 is not ready for any operation

0 = Flash array 0 is ready for operation

8.5.2.4 EFM Interrupt Mask Register (EFINTM)

Addr. Offset: 0x000c~0x000f Reset Value: 0x00000038

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved		ECC_ERR_MASK	RDWW_MASK	DONE_MASK	Reserved		
ro		rw	rw	rw	ro		

Figure 8-6: EFM Interrupt Mask(EFINTM)

ECC_ERROR0_MASK — Mask ECC_ERROR0 interrupt

1 = Mask the ECC error happened when read data checking interrupt of Flash 0

0 = ECC error happened when read data checking will trigger an interrupt of Flash 0 to system

RDWW0_MASK — MASK RDWW0 interrupt

1 = Mask reading Flash 0 while it is programming/erasing interrupt

0 = Read Flash 0 while it is programming/erasing will trigger an interrupt to system

DONE0_MASK — MASK DONE0 interrupt

1 = Mask CPU triggered command done interrupt of Flash 0

0 = Command done of Flash 0 will trigger interrupt to system

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8.5.2.5 EFM Command Register (EFCMD)

Addr. Offset: 0x0010~0x0013 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CMD_VALID	Reserved	PRGW_NUM[5:0]					
rw	ro	rw					
7	6	5	4	3	2	1	0
Reserved			CMD[4:0]				
ro			rw				

Figure 8-7: EFM Command Register(EFCMD)

CMD0_VALID — Launch the command

Write 1 to this bit trigger the command in CMD to Flash CTRL module 0. Self-cleared.

PRGW_NUM0 — Number of words to be programmed to Flash 0 PRGW_NUM+1 words will be programed, PRGW_NUM is only used in smw

program mode and is useless in normal program mode

CMD0 — Flash command

When CMD0_VALID high, CMD0 will send to CTRL module to generate corresponding command to Flash 0.

5'b0_0000: IDLE/User mode 5'b0_0001: MASS Erase

5'b0_0010: Page Erase

5'b0_0011: Program in normal mode

5'b0_0100: Smart Program 1(Program shots disabled on previously 0 bits) 5'b0_0101: Smart Erase 1(MASS Erase)

5'b0_0110: Smart Erase 2(One-shot MASS Erase) 5'b0_0111: Smart Erase 3(Page Erase)

5'b0_1001: Smart Program 2(Program shots enabled on previously 0 bits) 5'b0_1111: Enter Sleep mode

5'b1_0001: EXTC of Test mode

5'b1_0010: Erase reference cell of Test mode 5'b1_0100: Enable write IFREN-1 of Test mode 5'b1_1111: Exit Sleep mode

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8.5.2.6 EFM P/E Timing Base Register(ETIMBASE)

Addr. Offset: 0x0018~0x001b Reset Value: 0x000003C8

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				20NS_BASE_OF_CLK[2:0]			
ro				rw			
15	14	13	12	11	10	9	8
1US_BASE_OF_CLK[15:8]							
rw							
7	6	5	4	3	2	1	0
1US_BASE_OF_CLK[7:0]							
rw							

Figure 8-8: EFM P/E Timing Base Register(ETIMBASE)

20NS_BASE_OF_CLK — Configure the number of clocks to generate one 20ns-period

1US_BASE_OF_CLK — Configure the number of clocks to generate one 1us-period

8.5.2.7 EFM P/E Timing Configure Register (ETIMCFG)

Addr. Offset: 0x001c~0x001f Reset Value: 0x1C9EF877

31	30	29	28	27	26	25	24
Reserved		CFG_TNVSE[3:0]				CFG_TPWS[3:2]	
ro		rw				rw	
23	22	21	20	19	18	17	16
CFG_TPWS[1:0]		CFG_TRE[4:0]					CFG_TME[4]
rw		rw					rw
15	14	13	12	11	10	9	8
CFG_TME[3:0]				CFG_TPROG[3:0]			
rw				rw			
7	6	5	4	3	2	1	0
CFG_TNVH[3:0]				CFG_TNVSP[3:0]			
rw				rw			

Figure 8-9: EFM P/E Configure Register(ETIMCFG)

CFG_TNVSE — Configure the Tnvse(Erase to NVSTR setup time) parameter in 1us-unit

CFG_TPWS — Configure the Tpws(Positive pulse of SE) parameter in clock cycles

CFG_TRE — Configure the Tre(Reference cell erase time) parameter in 1ms-unit

CFG_TME — Configure the Tme(Mass erase time) parameter in 1ms-unit

CFG_TPROG — Configure the Tprog(Program pulse) parameter in 1us-unit

CFG_TNVH — Configure the Tnvh(NVSTR hold time for program or page erase) parameter in 1us-unit

CFG_TNVSP — Configure the Tnvsp(Program to NVSTR setup time) parameter in 1us-unit

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8.5.2.8 EFM P/E Wait Write ADDR/DATA Timer Register (EFPETIMER)

Addr. Offset: 0x0020~0x0023 Reset Value: 0x00002000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
PE_WAIT_CYCLE[23:16]							
rw							
15	14	13	12	11	10	9	8
PE_WAIT_CYCLE[15:8]							
rw							
7	6	5	4	3	2	1	0
PE_WAIT_CYCLE[7:0]							
rw							

Figure 8-10: EFM P/E Wait Write ADDR/DATA Timer Register(EFPRTIMER)

PE_WAIT_CYCLE — Configure the wait time when Program or Erase command is performing

8.5.2.9 EFM Smart Write Timing Option Register 0(SMWOP0)

Addr. Offset: 0x0024~0x0027 Reset Value: 0x015E14C8

31	30	29	28	27	26	25	24
Reserved							WMV[2]
ro							rw
23	22	21	20	19	18	17	16
WMV[1:0]		WHV[3:0]				WIPGM[1:0]	
rw		rw				rw	
15	14	13	12	11	10	9	8
Reserved		SLOW_CLK	200NS_CLK_NUM				
ro		rw	rw				
7	6	5	4	3	2	1	0
1US_CLK_NUM[7:0]							
rw							

Figure 8-11: EFM Smart Write Timing Option Register 0(SMWOP0)

WMV — Wrtie voltage select,when smart algorithm is used.During smart-program cycle, WMV[2:0] select MV voltage.

WHV — Wrtie voltage select, when smart algorithm is used.During smart-program or smart-erase cycle, WHV[3:0] select HV voltage.

SLOW_CLK_EN — Slow clock enabled, clock source: 500ns, special case for cycling support, used only on test-chip. Please set to 0 if it's not used.

200NS_CLK_NUM — define # of state required for PV/EV read cycle.

The total clocks for a PV/EV read cycle will be 2*(setting+1)*clock_cycle. Min setting: 0, user needs to add enough wait state to make sure (clock_cycle)*(2*setting+0.5) > 200NS.

1US_CLK_NUM — define # of clocks to generate a >= 1us pulse, Min: 1 when clock runs at 100NS cycle.

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8.6 Functional Description

8.6.1 Flash Initialization Flowchart

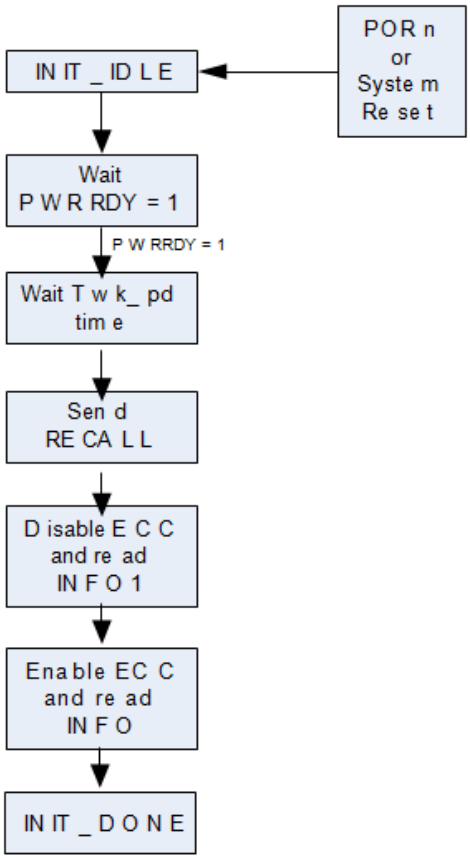


Figure 8-12: Flash Initialization Flowchart

8.6.2 Flash Valid Commands

Table 8-2: Flash Commands

EFMCMD	Meaning	Description
5'b0_0000	IDLE/User mode	Default state
5'b0_0001	Mass Erase	.Mass erase in normal mode
5'b0_0010	Page Erase	Page erase in normal mode
5'b0_0011	Program	Program in normal mode
5'b0_0100	Smart Program 1	Program in SMW mode
5'b0_0101	Smart Erase 1	Mass erase
5'b0_0110	Smart Erase 2	One-shot mass erase
5'b0_0111	Smart Erase 3	Page erase
5'b0_1001	Smart Program 2	Program in SMW mode
5'b0_1111	Enter Sleep mode	Flash sleep mode enable
5'b1_0001	EXTC	EXTC of test mode
5'b1_0010	Erase reference cell	Erase reference cell of test mode
5'b1_0100	Enable wrtie IFREN1 block	Enable wrtie IFREN1 block of test mode

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EFMCMD	Meaning	Description
5'b1_1111	Exit sleep mode	Exite sleep mode

NOTE: Make sure that the Flash is in IDLE state before issue command, otherwise the command will be ignored(except "Exit sleep mode" command).

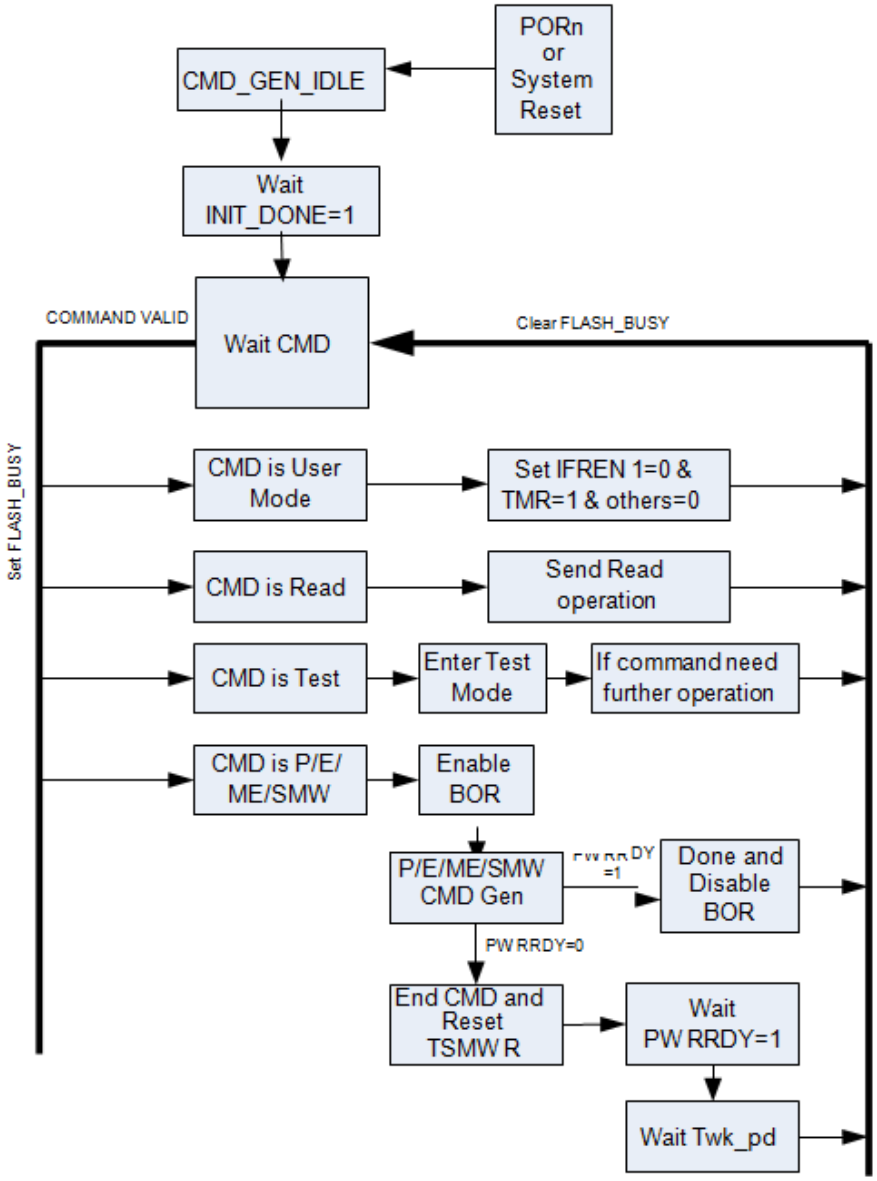


Figure 8-13: Flash Command Flowchart

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8.6.3 Program and Erase Sequences

8.6.3.1 Program Sequences

This EFM controller supports two program modes: noraml program mode and smw program mode. The program command sequence is:

- Set the flash which in need of program writeable by writing EFAPR register.
- Write CMD[4:0] in EFCMD to set flash command, and PRGW_NUM[5:0] in EFCMD to set the number of words to be programed(PRGW_NUM+1 words will be programed,PRGW_NUM is useless in normal program mode),then write CMD_VALID in EFMCMD to launch the command.
- Write the 32-bit word to be programmed to its location in the EFM array through AHB bus. The number of words should equal to PRGW_NUM+1.
- Wait for the high of EFM_IDLE in EFSTAT, the program operation complete.

8.6.3.2 Erase Sequences

This EFM controller supports two program modes: noraml erase mode and smw erase mode. The erase command sequence is:

- Set the flash which in need of erase writeable by writing EFAPR register.
- Write CMD[4:0] in EFCMD to set flash command, then write CMD_VALID in EFMCMD to launch the command.
- Write a 32-bit word(any word) into its location in the EFM array through AHB bus.
- Wait for the high of EFM_IDLE in EFSTAT, the erase operation complete.

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9 Synchronous Serial Interface (SSI)

9.1 Introduction

SSI is a programmable Synchronous Serial Interface (SSI) peripheral. This is a AMBA 2.0-compliant AHB (Advanced High-performance Bus) component .The host processor accesses data, control, and status information on the SSI through the AHB interface. The SSI may also interface with a DMA Controller using an optional set of DMA signals, which can be selected during configuration.

9.2 Features

Feathers include:

- Serial-master operation.
- DMA controller interface – Enables the SSI to interface to a DMA controller over the bus using handshaking interface for transfer requests.
- Clock stretching support in enhanced SPI transfers.
- Data item size (4 to 32 bits) – Item size of each data transfer under control of the programmer.
- FIFO depth – 8 words deep in normal transfer mode.32 words deep in Execute in Place (XIP) transfer mode.The FIFO width is fixed at 32 bits.
- Enhanced SPI support.
- Execute in Place (XIP) mode support.

9.3 Modes of Operation

The SSI functions in these three modes:

- Run mode — Run mode is the normal mode of operation.
- Doze mode — Doze mode is a configurable low-power mode.
- Stop mode — The SSI is inactive in stop mode.

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9.4 Block Diagram

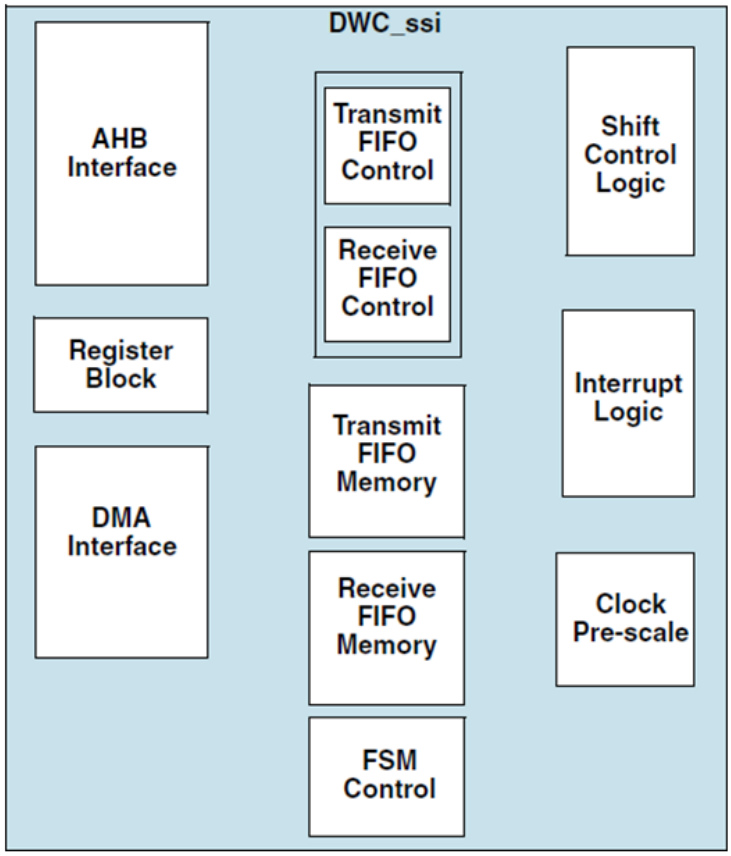


Figure 9-1: SSI Block Diagram

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9.5 Memory Map and Register

9.5.1 Memory Map

Table 9-1 shows the SSI register memory map.

Table 9-1: SSI Memory Map

Address Offset	Bits 31–0	Access
0x0000	Control Register 0(CTRLR0)	S/U
0x0004	Control Register 1 (CTRLR1)	S/U
0x0008	SSI Enable Register (SSIENR)	S/U
0x0010	Slave Select Register(SER)	S/U
0x000c	Microwire Control Register(MWCR)	S/U
0x0014	Baud Rate Select Register(BAUDR)	S/U
0x0018	Transmit FIFO Threshold Level Register(TXFTLR)	S/U
0x001c	Receive FIFO Threshold Level Register(RXFTLR)	S/U
0x0020	Transmit FIFO Level Register (TXFLR)	S/U
0x0024	Receive FIFO Level Register (RXFLR)	S/U
0x0028	Status Register (SR)	S/U
0x002c	Interrupt Mask Register (IMR)	S/U
0x0030	Interrupt Status Register (ISR)	S/U
0x0034	Raw Interrupt Status Register (RISR)	S/U
0x0038	Transmit FIFO Overflow Interrupt Clear Register(TXOICR)	S/U
0x003c	Receive FIFO Overflow Interrupt Clear Register(RXOICR)	S/U
0x0040	Receive FIFO Underflow Interrupt Clear Register(RXUICR)	S/U
0x0048	Interrupt Clear Register(ICR)	S/U
0x004c	DMA Control Register(DMACR)	S/U
0x0050	DMA Transmit Data Level Register(DMATDLR)	S/U
0x0054	DMA Receive Data Level Register(DMARDLR)	S/U
0x0060+i*0x4	SSI Data Register (DRx)	S/U
0x00f0	RX Sample Delay Register (RXSDR)	S/U
0x00f4	SPI Control Register 0(SPICTRLR0)	S/U
0x00fc	XIP Mode Bits(XIPMBR)	S/U
0x0100	XIP Incr Inst Register(XIPIIR)	S/U
0x0104	XIP Wrap Inst Register(XIPWIR)	S/U

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Address Offset	Bits 31–0	Access
0x0108	XIP Control Register(XIPCR)	S/U
0x010c	XIP Slave Enable Register(XIPSER)	S/U
0x0110	XIP Receive FIFO Overflow Interrupt Clear Register(XRXIOCR)	S/U

9.5.2 Register Descriptions

9.5.2.1 Control Register 0(SSICTRLR0)

Addr. Offset: 0x0000~0x0003 Reset Value: 0x0080481F

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
SPI_FRF		Reserved		CFS			
rw		ro		rw			
15	14	13	12	11	10	9	8
Reserved	SSTE	SRL	Reserved	TMOD		SCPOL	SCPH
ro	rw	rw	ro	rw		rw	rw
7	6	5	4	3	2	1	0
FRF		Reserved		DFS			
rw		ro		rw			

Figure 9-2: Control Register 0(SSICTRLR0)

This register controls the serial data transfer. It is impossible to write to this register when the SSI is enabled. The SSI is enabled and disabled by writing to the SSIENR.

SPI_FRF — SPI Frame Format

Selects data frame format for Transmitting/Receiving the data. Bits only valid when SSIC_SPI_MODE is either set to "Dual" or "Quad" or "Octal" mode.

0x0 (SPI_STANDARD): Standard SPI Format

0x1 (SPI_DUAL): Dual SPI Format

0x2 (SPI_QUAD): Quad SPI Format

0x3 (RESERVED): reserved

CFS — Control Frame Size

Selects the length of the control word for the Microwire frame format.

0x0 (SIZE_01_BIT): 01-bit Control Word

0x1 (SIZE_02_BIT): 02-bit Control Word

0x2 (SIZE_03_BIT): 03-bit Control Word

0x3 (SIZE_04_BIT): 04-bit Control Word

0x4 (SIZE_05_BIT): 05-bit Control Word

0x5 (SIZE_06_BIT): 06-bit Control Word

0x6 (SIZE_07_BIT): 07-bit Control Word

0x7 (SIZE_08_BIT): 08-bit Control Word

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0x8 (SIZE_09_BIT): 09-bit Control Word

0x9 (SIZE_10_BIT): 10-bit Control Word

0xA (SIZE_11_BIT): 11-bit Control Word

0xB (SIZE_12_BIT): 12-bit Control Word

0xC (SIZE_13_BIT): 13-bit Control Word

0xD (SIZE_14_BIT): 14-bit Control Word

0xE (SIZE_15_BIT): 15-bit Control Word

0xF (SIZE_16_BIT): 16-bit Control Word

SSTE — Slave Select Toggle Enable.

While operating in SPI mode with clock phase (SCPH) set to 0, this register controls the behavior of the slave select line (ss_*_n) between data frames.

1 = (TOGGLE_EN): ss_*_n line will toggle between consecutive data frames, with the serial clock (sclk) being held to its default value while ss_*_n is high

0 = (TOGGLE_DISABLE): ss_*_n will stay low and sclk will run continuously for the duration of the transfer

SRL — Shift Register Loop.

Used for testing purposes only. When internally active, connects the transmit shift register output to the receive shift register input.

1 = (TESTING_MODE): Test Mode Operation

0 = (NORMAL_MODE): Normal mode operation TMOD — Transfer Mode.

Selects the mode of transfer for serial communication. This field does not affect the transfer duplicity. Only indicates whether the receive or transmit data are valid.

0x0 (TX_AND_RX): Transmit & Receive; Not Applicable in enhanced SPI operating mode

0x1 (TX_ONLY): Transmit only mode; Or Write in enhanced SPI operating mode

0x2 (RX_ONLY): Receive only mode; Or Read in enhanced SPI operating mode

0x3 (EEPROM_READ): EEPROM Read mode; Not Applicable in enhanced SPI operating mode

SCOPL — Serial Clock Polarity.

Valid when the frame format (FRF) is set to Motorola SPI. Used to select the polarity of the inactive serial clock, which is held inactive when the SSI master is not actively transferring data on the serial bus.

1 = (INACTIVE_LOW): Inactive state of serial clock is high

0 = (INACTIVE_HIGH): Inactive state of serial clock is low

SCPH — Serial Clock Phase.

Valid when the frame format (FRF) is set to Motorola SPI. The serial clock phase selects the relationship of the serial clock with the slave select signal.

When SCPH = 0, data are captured on the first edge of the serial clock. When SCPH = 1, the serial clock starts toggling one cycle after the slave select line is activated, and data are captured on the second edge of the serial clock.

1 = (START_BIT): Serial clock toggles at start of first bit

0 = (MIDDLE_BIT): Serial clock toggles in middle of first bit

FRF — Frame Format.

Selects which serial protocol transfers the data.

0x0 (SPI): Motorola SPI Frame Format

0x1 (SSP): Texas Instruments SSP Frame Format

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0x2 (MICROWIRE): National Semiconductors Microwire Frame Format

0x3 (RESERVED): Reserved

DFS — Data Frame Size.

Selects the data frame length. When the data frame size is programmed to be less than 32 bits, the receive data is automatically right-justified by the receive logic, with the upper bits of the receive FIFO zero-padded.

You must right-justify transmit data before writing into the transmit FIFO. The transmit logic ignores the upper unused bits when transmitting the data.

0x0 (DFS_01_BIT): Reserved

0x1 (DFS_02_BIT): Reserved

0x2 (DFS_03_BIT): Reserved

0x3 (DFS_04_BIT): 04-bit serial data transfer

0x4 (DFS_05_BIT): 05-bit serial data transfer

0x5 (DFS_06_BIT): 06-bit serial data transfer

0x6 (DFS_07_BIT): 07-bit serial data transfer

0x7 (DFS_08_BIT): 08-bit serial data transfer

0x8 (DFS_09_BIT): 09-bit serial data transfer

0x9 (DFS_10_BIT): 10-bit serial data transfer

0xA (DFS_11_BIT): 11-bit serial data transfer

0xB (DFS_12_BIT): 12-bit serial data transfer

0xC (DFS_13_BIT): 13-bit serial data transfer

0xD (DFS_14_BIT): 14-bit serial data transfer

0xE (DFS_15_BIT): 15-bit serial data transfer

0xF (DFS_16_BIT): 16-bit serial data transfer

0x10 (DFS_17_BIT): 17-bit serial data transfer

0x11 (DFS_18_BIT): 18-bit serial data transfer

0x12 (DFS_19_BIT): 19-bit serial data transfer

0x13 (DFS_20_BIT): 20-bit serial data transfer

0x14 (DFS_21_BIT): 21-bit serial data transfer

0x15 (DFS_22_BIT): 22-bit serial data transfer

0x16 (DFS_23_BIT): 23-bit serial data transfer

0x17 (DFS_24_BIT): 24-bit serial data transfer

0x18 (DFS_25_BIT): 25-bit serial data transfer

0x19 (DFS_26_BIT): 26-bit serial data transfer

0x1A (DFS_27_BIT): 27-bit serial data transfer

0x1B (DFS_28_BIT): 28-bit serial data transfer

0x1C (DFS_29_BIT): 29-bit serial data transfer

0x1D (DFS_30_BIT): 30-bit serial data transfer

0x1E (DFS_31_BIT): 31-bit serial data transfer

0x1F (DFS_32_BIT): 32-bit serial data transfer

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9.5.2.2 SSI Control Register 1(SSICTRLR1)

Reading from this location returns 0. Control register 1 controls the end of serial transfers when in receive-only mode and transfer-only mode. It is impossible to write to this register when the SSI is enabled. The SSI is enabled and disabled by writing to the SSIENR register.

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
NDF							
rw							
7	6	5	4	3	2	1	0
NDF							
rw							

Figure 9-3: SSI Control Register 1(SSICTRLR1)

NDF — Number of Data Frames.

When TMOD = 01 or TMOD = 10 or TMOD = 11, this register field sets the number of data frames to be continuously received or transmitted by the SSI. The SSI continues to receive or transmit serial data until the number of data frames is equal to this register value plus 1.When TMOD = 01,CLK_STRETCH_EN in SPI_CTRLR0 must be set.

9.5.2.3 SSI Enable Register(SSIENR)

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							SSIC_EN
ro							rw

Figure 9-4: SSI Enable Register(SSIENR)

SSIC_EN — SSI Enable.

Enables and disables all SSI operations. When disabled, all serial transfers are halted immediately. Transmit and receive FIFO buffers are cleared when the device is disabled. It is impossible to program some of the SSI control registers when enabled. When disabled, the ssi sleep output is set (after delay) to inform the system that it is safe to remove the ssi_clk, thus saving power consumption in the system.

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1 = (ENABLED): Enables SSI

0 = (DISABLE): Disables SSI

9.5.2.4 SSI Microwire Control Register(MWCR)

Addr. Offset: 0x000C~0x000F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved					MHS	MDD	MWMOD
ro					rw	rw	rw

Figure 9-5: SSI Microwire Control Register(MWCR)

This register controls the direction of the data word for the half-duplex Microwire serial protocol. It is impossible to write to this register when the SSI is enabled. The SSI is enabled and disabled by writing to the SSIENR register.

MHS — Microwire Handshaking.

Used to enable and disable the busy/ready handshaking interface for the Microwire protocol. When enabled, the SSI checks for a ready status from the target slave, after the transfer of the last data/control bit, before clearing the BUSY status in the SR register.

1 = (ENABLED): handshaking interface is enabled

0 = (DISABLE): handshaking interface is disabled

MDD — Microwire Control.

Defines the direction of the data word when the Microwire serial protocol is used. When this bit is set to 0, the data word is received by the SSI MacroCell from the external serial device. When this bit is set to 1, the data word is transmitted from the SSI MacroCell to the external serial device.

1 = (TRANSMIT): SSI transmits data

0 = (RECEIVE): SSI receives data

MWMOD — Microwire Transfer Mode.

Defines whether the Microwire transfer is sequential or non-sequential. When sequential mode is used, only one control word is needed to transmit or receive a block of data words. When non-sequential mode is used, there must be a control word for each data word that is transmitted or received.

1 = (SEQUENTIAL): Sequential Transfer

0 = (NON_SEQUENTIAL): Non-Sequential Transfer

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9.5.2.5 SSI Slave Enable Register(SSISER)

Addr. Offset: 0x0010~0x0013 Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							SER
ro							rw

Figure 9-6: SSI Slave Enable Register(SSISER)

Reading from this location returns 0. The register enables the individual slave select output lines from the SSI master. You cannot write to this register when SSI is busy and when SSIC_EN = 1.

SER — Slave Select Enable Flag.

1 = Selected, 0 = Not Selected

9.5.2.6 SSI Baud Rate Select(BAUDR)

Addr. Offset: 0x0014~0x0017 Reset Value: 0x00000002

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
SCKDV							
rw							
7	6	5	4	3	2	1	0
SCKDV							
rw							

Figure 9-7: SSI Baud Rate Select(BAUDR)

SCKDV — SSI Clock Divider.

The LSB for this field is always set to 0 and is unaffected by a write operation, which ensures an even value is held in this register. If the value is 0, the serial output clock (sclk_out) is disabled. The frequency of the sclk_out is derived from the following equation:

$$Fsclk_out = Fssi_clk/SCKDV$$

where SCKDV is any even value between 2 and 65534. For example:

for Fssi_clk = 3.6864MHz and SCKDV =2 Fsclk_out = 3.6864/2 = 1.8432MHz

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9.5.2.7 SSI Transmit FIFO Threshold Level(SSITXFTLR)

Addr. Offset: 0x0018~0x001B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved			TXFTHR				
ro			rw				
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			TFT				
ro			rw				

Figure 9-8: SSI Transmit FIFO Threshold Level(SSITXFTLR)

This register controls the threshold value for the transmit FIFO memory. The SSI is enabled and disabled by writing to the SSIENR register.

TXFTHR — Transfer start FIFO level.

Used to control the level of entries in transmit FIFO above which transfer will start on serial line. This register can be used to ensure that sufficient data is present in transmit FIFO before starting a write operation on serial line.

TFT — Transmit FIFO Threshold.

Controls the level of entries (or below) at which the transmit FIFO controller triggers an interrupt. If you attempt to set this value greater than or equal to the depth of the FIFO, this field is not written and retains its current value. When the number of transmit FIFO entries is less than or equal to this value, the transmit FIFO empty interrupt is triggered.

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9.5.2.8 SSI Receive FIFO Threshold Level(SSIRXFTLR)

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			RFT				
ro			rw				

Figure 9-9: SSI Receive FIFO Threshold Level(SSIRXFTLR)

This register controls the threshold value for the receive FIFO memory. The SSI is enabled and disabled by writing to the SSIENR register.

RFT — Receive FIFO Threshold.

Controls the level of entries (or above) at which the receive FIFO controller triggers an interrupt. If you attempt to set this value greater than the depth of the FIFO, this field is not written and retains its current value. When the number of receive FIFO entries is greater than or equal to this value + 1, the receive FIFO full interrupt is triggered.This value should be set to a non-zero valve when receive data or SSI will always toggle when receive fifo is completed full.

9.5.2.9 SSI Transmit FIFO Level Register(SSITXFLR)

Addr. Offset: 0x0024~0x0027 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			RXFLR				
ro			rw				

Figure 9-10: SSI Transmit FIFO Level Register(SSITXFLR)

TXFLR — Transmit FIFO Level.

Contains the number of valid data entries in the transmit FIFO.

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9.5.2.10 SSI Receive FIFO Level Register(SSIRXFLR)

Addr. Offset: 0x0024~0x0027 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				RXFLR			
ro				rw			

Figure 9-11: SSI Receive FIFO Level Register(SSIRXFLR)

RXTFL — Receive FIFO Level.

Contains the number of valid data entries in the receive FIFO.

9.5.2.11 SSI Status Register(SSISR)

Addr. Offset: 0x0028~0x002B Reset Value: 0x00000006

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			RFF	RFNE	TFE	TFNF	BUSY
ro			ro	ro	ro	ro	ro

Figure 9-12: SSI Status Register(SSISR)

RFF — Receive FIFO Full.

When the receive FIFO is completely full, this bit is set. When the receive FIFO contains one or more empty location, this bit is cleared.

1 = (FULL): Receive FIFO is full

0 = (NOT_FULL): Receive FIFO is not full

RFNE — Receive FIFO Not Empty.

Set when the receive FIFO contains one or more entries and is cleared when the receive FIFO is empty. This bit can be polled by software to completely empty the receive FIFO.

1 = (NOT_EMPTY): Receive FIFO is not empty

0 = (EMPTY): Receive FIFO is empty

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TFE — Transmit FIFO Empty.

When the transmit FIFO is completely empty, this bit is set. When the transmit FIFO contains one or more valid entries, this bit is cleared. This bit field does not request an interrupt.

1 = (EMPTY): Transmit FIFO is empty

0 = (NOT_EMPTY): Transmit FIFO is not empty TFNF — Transmit FIFO Not Full.

Set when the transmit FIFO contains one or more empty locations, and is cleared when the FIFO is full.

1 = (NOT_FULL): Tx FIFO is not Full

0 = Tx FIFO is full

BUSY — SSI Busy Flag.

When set, indicates that a serial transfer is in progress; when cleared indicates that the SSI is idle or disabled.

1 = (ACTIVE): SSI is actively transferring data

0 = (INACTIVE): SSI is idle or disabled

9.5.2.12 SSI Interrupt Mask Register(SSIIMR)

Addr. Offset: 0x002C~0x002F Reset Value: 0x0000007F

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved	XRXOIM	Reserved	RXFIM	RXOIM	RXUIM	TXOIM	TXEIM
ro	rw	ro	rw	rw	rw	rw	rw

Figure 9-13: SSI Interrupt Mask Register(SSIIMR)

XRXOIM — XIP Receive FIFO Overflow Interrupt Mask

1 = (UNMASKED): ssi_xrxo_intr interrupt is not masked

0 = (MASKED): ssi_xrxo_intr interrupt is masked

RXFIM — Receive FIFO Full Interrupt Mask

1 = (UNMASKED): ssi_rxf_intr interrupt is not masked

0 = (MASKED): ssi_rxf_intr interrupt is masked

RXOIM — Receive FIFO Overflow Interrupt Mask

1 = (UNMASKED): ssi_rxo_intr interrupt is not masked

0 = (MASKED): ssi_rxo_intr interrupt is masked

RXUIM — Receive FIFO Underflow Interrupt Mask

1 = (UNMASKED): ssi_rxu_intr interrupt is not masked

0 = (MASKED): ssi_rxu_intr interrupt is masked

TXOIM — Transmit FIFO Overflow Interrupt Mask

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1 = (UNMASKED): ssi_txo_intr interrupt is not masked

0 = (MASKED): ssi_txo_intr interrupt is masked

TXEIM — Transmit FIFO Empty Interrupt Mask

1 = (UNMASKED): ssi_txe_intr interrupt is not masked

0 = (MASKED): ssi_txe_intr interrupt is masked

9.5.2.13 SSI Interrupt Status Register(SSIISR)

Addr. Offset: 0x0030~0x0033 Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved	XRXOIS	Reserved	RXFIS	RXOIS	RXUIS	TXOIS	TXEIS
ro	ro	ro	ro	ro	ro	ro	ro

Figure 9-14: SSI Interrupt Status Register(SSIISR)

This register reports the status of the SSI interrupts after they have been masked. XRXOIS — XIP Receive FIFO Overflow Interrupt Status

1 = (ACTIVE): ssi_xrxo_intr interrupt is active after masking

0 = (INACTIVE): ssi_xrxo_intr interrupt is not active after masking

RXFIS — Receive FIFO Full Interrupt Status

1 = (ACTIVE): ssi_rxf_intr interrupt is active after masking

0 = (INACTIVE): ssi_rxf_intr interrupt is not active after masking

RXOIS — Receive FIFO Overflow Interrupt Status

1 = (ACTIVE): ssi_rxo_intr interrupt is active after masking

0 = (INACTIVE): ssi_rxo_intr interrupt is not active after masking

RXUIS — Receive FIFO Underflow Interrupt Status

1 = (ACTIVE): ssi_rxu_intr interrupt is active after masking

0 = (INACTIVE): ssi_rxu_intr interrupt is not active after masking

TXOIS — Transmit FIFO Overflow Interrupt Status

1 = (ACTIVE): ssi_txo_intr interrupt is active after masking

0 = (INACTIVE): ssi_txo_intr interrupt is not active after masking

TXEIS — Transmit FIFO Empty Interrupt Status

1 = (ACTIVE): ssi_txe_intr interrupt is active after masking

0 = (INACTIVE): ssi_txe_intr interrupt is not active after masking

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9.5.2.14 SSI Raw Interrupt Status Register(SSIRISR)

Addr. Offset: 0x0034~0x0037 Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved	XRXOIR	Reserved	RXFIR	RXOIR	RXUIR	TXOIR	TXEIR
ro	ro	ro	ro	ro	ro	ro	ro

Figure 9-15: SSI Raw Interrupt Status Register(SSIRISR)

XRXOIR — XIP Receive FIFO Overflow Raw Interrupt Status

1 = (ACTIVE): ssi_xrxo_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_xrxo_intr interrupt is not active prior masking

RXFIR — Receive FIFO Full Raw Interrupt Status

1 = (ACTIVE): ssi_rxf_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_rxf_intr interrupt is not active prior masking

RXOIR — Receive FIFO Overflow Raw Interrupt Status

1 = (ACTIVE): ssi_rxo_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_rxo_intr interrupt is not active prior masking

RXUIR — Receive FIFO Underflow Raw Interrupt Status

1 = (ACTIVE): ssi_rxu_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_rxu_intr interrupt is not active prior masking

TXOIS — Transmit FIFO Overflow Raw Interrupt Status

1 = (ACTIVE): ssi_txo_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_txo_intr interrupt is not active prior masking

TXEIS — Transmit FIFO Empty Raw Interrupt Status

1 = (ACTIVE): ssi_txe_intr interrupt is active prior to masking

0 = (INACTIVE): ssi_txe_intr interrupt is not active prior masking

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9.5.2.15 SSI Transmit FIFO Overflow Interrupt Clear Registers.(SSI TXOICR)

Addr. Offset: 0x0038~0x003B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							TXOICR
ro							rc

Figure 9-16: SSI Transmit FIFO Overflow Interrupt Clear Registers.(SSITXOICR)

TXOICR — Clear Transmit FIFO Overflow Interrupt.

This register reflects the status of the interrupt. A read from this register clears the ssi_txo_intr interrupt; writing has no effect.

9.5.2.16 SSI Receive FIFO Overflow Interrupt Clear Register(SSIRXOICR)

Addr. Offset: 0x003C~0x003F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							RXOICR
ro							rc

Figure 9-17: SSI Receive FIFO Overflow Interrupt Clear Register(SSIRXOICR)

RXOICR — Clear Receive FIFO Overflow Interrupt.

This register reflects the status of the interrupt. A read from this register clears the ssi_rxo_intr interrupt; writing has no effect.

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9.5.2.17 SSI Receive FIFO Underflow Interrupt Clear Register(SSIRXUICR)

Addr. Offset: 0x0040~0x0043 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							RXUICR
ro							rc

Figure 9-18: SSI Receive FIFO Underflow Interrupt Clear Register(SSIRXUICR)

RXUICR — Clear Receive FIFO Underflow Interrupt.

This register reflects the status of the interrupt. A read from this register clears the ssi_rxu_intr interrupt; writing has no effect.

9.5.2.18 SSI Interrupt Clear Register(SSIICR)

Addr. Offset: 0x0048~0x004B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							ICR
ro							rw

Figure 9-19: SSI Interrupt Clear Register(SSIICR)

ICR — Clear Interrupts.

This register is set if any of the interrupts below are active. A read clears the ssi_txo_intr, ssi_rxu_intr, ssi_rxo_intr interrupts. Writing to this register has no effect.

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9.5.2.19 SSI DMA Control Register(SSIDMACR)

Addr. Offset: 0x004C~0x004F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved						TDMAE	RDMAE
ro						rw	rw

Figure 9-20: SSI DMA Control Register(SSIDMACR)

The register is used to enable the DMA Controller interface operation.

TDMAE — Transmit DMA Enable

This bit enables/disables the transmit FIFO DMA channel.

1 = (ENABLED): Transmit DMA enabled

0 = (DISABLE): Transmit DMA disabled

RDMAE — Receive DMA Enable

This bit enables/disables the receive FIFO DMA channel.

1 = (ENABLED): Receive DMA enabled

0 = (DISABLE): Receive DMA disabled

9.5.2.20 SSI DMA Transmit Data Level(SSIDMATDLR)

Addr. Offset: 0x0050~0x0053 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			DMATDL				
ro			rw				

Figure 9-21: SSI DMA Transmit Data Level(SSIDMATDLR)

DMATDL — Transmit Data Level.

This bit field controls the level at which a DMA request is made by the transmit logic. It is equal to the watermark level; that is, the dma_tx_req signal is generated when the number of valid data entries in the transmit FIFO is equal to or below this field value, and TDMAE = 1.

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9.5.2.21 SSI DMA Receive Data Level(SSIDMARDLR)

Addr. Offset: 0x0054~0x0057 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			DMARDL				
ro			rw				

Figure 9-22: SSI DMA Receive Data Level(SSIDMARDLR)

DMARDL — Receive Data Level.

This bit field controls the level at which a DMA request is made by the receive logic. The watermark level = DMARDL+1; that is, dma_rx_req is generated when the number of valid data entries in the receive FIFO is equal to or above this field value + 1, and RDMAE=1.

9.5.2.22 SSI Data Register(SSIDRx)

Addr. Offset: 0x0060+i*0x4 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DR							
rw							
23	22	21	20	19	18	17	16
DR							
rw							
15	14	13	12	11	10	9	8
DR							
rw							
7	6	5	4	3	2	1	0
DR							
rw							

Figure 9-23: SSI Data Register(SSI DRx)

The SSI data register is a 32-bit read/write buffer for the transmit/receive FIFOs. When the register is read, data in the receive FIFO buffer is accessed. When it is written to, data are moved into the transmit FIFO buffer; a write can occur only when SSIC_EN = 1. FIFOs are reset when SSIC_EN = 0.

DR — Data Register.

When writing to this register, you must right-justify the data. Read data are automatically right-justified.

Read = Receive FIFO buffer

Write = Transmit FIFO buffer.

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9.5.2.23 SSI RX Sample Delay Register (SSIRXSDR)

Addr. Offset: 0x00F0~0x00F4 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							SE
ro							rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
RSD							
rw							

Figure 9-24: SSI RX Sample Delay Register (SSIRXSDR)

SE — Receive Data (rxd) Sampling Edge.

1 = negative edge of ssi_clk will be used to sample the incoming data

0 = posedge edge of ssi_clk will be used to sample the incoming data

RSD — Receive Data (rxd) Sample Delay.

This register is used to delay the sample of the rxd input port. Each value represents a single ssi_clk delay on the sample of rxd.

9.5.2.24 SSI SPI Control Register 0(SPICTRLR0)

Addr. Offset: 0x00FC~0x00FF Reset Value: 0x00000218

31	30	29	28	27	26	25	24
Reserved	CLK_STRETCH	Reserved					
ro	rw	ro					
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
WAIT_CYCLES					Reserved	INST_L	
rw					ro	rw	
7	6	5	4	3	2	1	0
Reserved		ADDR_L				TRANS_TYPE	
ro		rw				rw	

Figure 9-25: SSI SPI Control Register 0(SPICTRLR0)

CLK_STRETCH_EN

Enables clock stretching capability in SPI transfers.

In case of write, if the FIFO becomes empty SSI will stretch the clock until FIFO has enough data to continue the transfer. In case of read, if the receive FIFO becomes full SSI will stop the clock until data has been read from the FIFO.

NOTE: Recommend always to set this bit

1 = CLK_STRETCH_ENABLE

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0 = CLK_STRETCH_DISABLE WAIT_CYCLES

Wait cycles in Dual/Quad/Octal mode between control frames transmit and data reception. Specified as number of SPI clock cycles.

INST_L — Dual/Quad/Octal mode instruction length in bits.

0x0 (INST_L0): No Instruction

0x1 (INST_L4): 4 bit Instruction length

0x2 (INST_L8): 8 bit Instruction length

0x3 (INST_L16): 16 bit Instruction length

ADDR_L — Length of Address to be transmitted

0x0 (ADDR_L0): No Address

0x1 (ADDR_L4): 4 bit Address length

0x2 (ADDR_L8): 8 bit Address length

0x3 (ADDR_L12): 12 bit Address length

0x4 (ADDR_L16): 16 bit Address length

0x5 (ADDR_L20): 20 bit Address length

0x6 (ADDR_L24): 24 bit Address length

0x7 (ADDR_L28): 28 bit Address length

0x8 (ADDR_L32): 32 bit Address length

0x9 (ADDR_L36): 36 bit Address length

0xA (ADDR_L40): 40 bit Address length

0xB (ADDR_L44): 44 bit Address length

0xC (ADDR_L48): 48 bit Address length

0xD (ADDR_L52): 52 bit Address length

0xE (ADDR_L56): 56 bit Address length

0xF (ADDR_L60): 60 bit Address length

TRANS_TYPE — Address and instruction transfer format.

Selects whether SSI will transmit instruction/address either in Standard SPI mode or the SPI mode selected in CTRLR0.SPI_FRF field.

0x0 (TT0): Instruction and Address will be sent in Standard SPI Mode.

0x1 (TT1): Instruction will be sent in Standard SPI Mode and Address will be sent in the mode specified by CTRLR0.SPI_FRF.

0x2 (TT2): Both Instruction and Address will be sent in the mode specified by SPI_FRF.

0x3 (TT3): Reserved.

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9.5.2.25 SSI XIP Mode Bits(SSIXIPMBR)

Addr. Offset: 0x00FC~0x00FF Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
XIP_MD_BITS							
rw							
7	6	5	4	3	2	1	0
XIP_MD_BITS							
rw							

Figure 9-26: SSI XIP Mode Bits(SSIXIPMBR)

This register carries the mode bits which are sent in the XIP mode of operation after address phase.This register can only be written when SSIENR register is set to 0.

XIP_MD_BITS

XIP mode bits to be sent after address phase of XIP transfer.

9.5.2.26 SSI XIP Incr Inst Register(SSIIIR)

Addr. Offset: 0x0100~0x0103 Reset Value: 0x0000006B

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
INCR_INST							
rw							
7	6	5	4	3	2	1	0
INCR_INST							
rw							

Figure 9-27: SSI XIP Incr Inst Register(SSIIIR)

This register is used to store the instruction op-code to be used in INCR transactions when the same is requested on AHB interface. It is not possible to write to this register when the SSI is enabled. The SSI is enabled and disabled by writing to the SSIENR register.

INCR_INST — XIP INCR transfer opcode.

When SPI_CTRLR0.XIP_INST_EN bit is set to 1, SSI sends instruction for all XIP transfers, this register field stores the instruction op-code to be sent when an INCR type transfer is requested on AHB bus. The number of bits to be send in instruction phase is determined by SPI_CTRL0.INST_L field.

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9.5.2.27 SSI XIP Wrap Inst Register(SSIWIR)

Addr. Offset: 0x0104~0x0107 Reset Value: 0x0000006B

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
WRAP_INST							
rw							
7	6	5	4	3	2	1	0
WRAP_INST							
rw							

Figure 9-28: SSI XIP Wrap Inst Register(SSIWIR)

This register is used to store the instruction op-code to be used in WRAP transactions when the same is requested on AHB interface. It is not possible to write to this register when the SSI is enabled (SSIC_EN=1). The SSI is enabled and disabled by writing to the SSIENR register.

WRAP_INST — XIP WRAP transfer opcode.

When SPI_CTRLR0.XIP_INST_EN bit is set to 1, SSI sends instruction for all XIP transfers, this register field stores the instruction op-code to be sent when an WRAP type transfer is requested on AHB bus. The number of bits to be send in instruction phase is determined by SPI_CTRL0.INST_L field.

9.5.2.28 SSI XIP Control Register(SSIXIPCR)

Addr. Offset: 0x0108~0x010B Reset Value: 0x28c10462

31	30	29	28	27	26	25	24
Reserved		XIP_PREFETCH_EN	Reserved	XIP_MBL		Reserved	
ro		rw	ro	rw		ro	
23	22	21	20	19	18	17	16
CONT_XFER_EN	INST_EN	Reserved			DFS_HC	WAIT_CYCLES	
rw	rw	ro			rw	rw	
15	14	13	12	11	10	9	8
WAIT_CYCLES			MD_BOTS_EN	Reserved	INST_L		Reserved
rw			rw	ro	rw		ro
7	6	5	4	3	2	1	0
ADDR_L				TRANS_TYPE		FRF	
rw				rw		rw	

Figure 9-29: SSI XIP Control Register(SSIXIPCR)

XIP_PREFETCH_EN

1 = Enables XIP pre-fetch functionality in SSI.

0 = Disables XIP pre-fetch functionality in SSI.

XIP_MBL — XIP Mode bits length.

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Sets the length of mode bits in XIP mode of operation.

0x0 (MBL_2): Mode bits length equal to 2 0x1 (MBL_4): Mode bits length equal to 4 0x2 (MBL_8): Mode bits length equal to 8 0x3 (MBL_16): Mode bits length equal to 16

CONT_XFER_EN

1 = Enable continuous transfer in XIP mode.

0 = Disables continuous transfer in XIP mode.

INST_EN

1 = XIP transfers will have instruction phase.

0 = XIP transfers will not have instruction phase.

DFS_HC — Fix DFS for XIP transfers

1 = data frame size for XIP transfers will be fixed to the programmed value in CTRLR0.DFS.

0 = data frame size will be determined by HSIZE and HBURST signals WAIT_CYCLES

Wait cycles in Dual/Quad/Octal mode between control frames transmit and data reception. Specified as number of SPI clock cycles.

MD_BITS_EN — Mode bits enable in XIP mode.

1 = insert mode bits after the address phase.

0 = no mode bits after the address phase.

INST_L — Dual/Quad/Octal mode instruction length in bits.

0x0 (INST_L0): No Instruction

0x1 (INST_L4): 4 bit Instruction length

0x2 (INST_L8): 8 bit Instruction length

0x3 (INST_L16): 16 bit Instruction length

ADDR_L

This bit defines Length of Address to be transmitted. Only after this much bits are programmed in to the FIFO the transfer can begin.

0x0 (ADDR_L0): No Address

0x1 (ADDR_L4): 4 bit Address length

0x2 (ADDR_L8): 8 bit Address length

0x3 (ADDR_L12): 12 bit Address length

0x4 (ADDR_L16): 16 bit Address length

0x5 (ADDR_L20): 20 bit Address length

0x6 (ADDR_L24): 24 bit Address length

0x7 (ADDR_L28): 28 bit Address length

0x8 (ADDR_L32): 32 bit Address length

0x9 (ADDR_L36): 36 bit Address length

0xA (ADDR_L40): 40 bit Address length

0xB (ADDR_L44): 44 bit Address length

0xC (ADDR_L48): 48 bit Address length

0xD (ADDR_L52): 52 bit Address length

0xE (ADDR_L56): 56 bit Address length

0xF (ADDR_L60): 60 bit Address length

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TRANS_TYPE

Address and instruction transfer format.

Selects whether SSI will transmit instruction/address either in Standard SPI mode or the SPI mode selected in CTRLR0.SPI_FRF field.

0x0 (TT0): Instruction and Address will be sent in Standard SPI Mode.

0x1 (TT1): Instruction will be sent in Standard SPI Mode and Address will be sent in the mode specified by XIP_CTRL.SPI_FRF.

0x2 (TT2): Both Instruction and Address will be sent in the mode specified by XIP_CTRL.SPI_FRF.

0x3 (TT3): Reserved.

FRF — SPI Frame Format

Selects data frame format for Transmitting/Receiving the data.

0x0 (RSVD): Reserved

0x1 (SPI_DUAL): Dual SPI Format

0x2 (SPI_QUAD): Quad SPI Format

0x3 (RSVD): Reserved

9.5.2.29 SSI XIP Slave Enable Register(SSIXIPSER)

Addr. Offset: 0x010C~0x010F Reset Value: 0x00000001

31	30	29	28	27	26	25	24
Reserved							
ro							

23	22	21	20	19	18	17	16
Reserved							
ro							

15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved							SER
ro							rw

Figure 9-30: SSI XIP Slave Enable Register(SSIXIPSER)

The register enables the individual slave select output lines from the SSI master for XIP mode of operation. You cannot write to this register when SSI is busy or when SSIC_EN = 1.

SER — Slave Select Enable Flag.

When a bit in this register is set (1), the corresponding slave select line from the master is activated when a XIP transfer begins. It should be noted that setting or clearing bits in this register have no effect on the corresponding slave select outputs until a XIP transfer is started.

1 = Select

0 = Not Select

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9.5.2.30 SSI XIP Receive FIFO Overflow Interrupt Clear Register(SSIXRXIOCR)

Addr. Offset: 0x0040~0x0043 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							XRXOICR
ro							rc

Figure 9-31: SSI XIP Receive FIFO Overflow Interrupt Clear Register(SSIXRXIOCR)

XRXOICR — Clear XIP Receive FIFO Overflow Interrupt.

This register reflects the status of the interrupt. A read from this register clears the ssi_xrxo_intr(_n) interrupt; writing has no effect.

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9.6 Functional Description

9.6.1 Master Mode

This mode enables serial communication with serial-slave peripheral devices. When configured as a serial-master device, the SSI initiates and controls all serial transfers. Figure 9-32 shows an example of the SSI configured as a serial master with all other devices on the serial bus configured as serial slaves.

The serial bit-rate clock, generated and controlled by the SSI, is driven out on the sclk_out line. When the SSI is disabled (SSIC_EN = 0), no serial transfers can occur and sclk_out is held in “inactive” state, as defined by the serial protocol under which it operates.

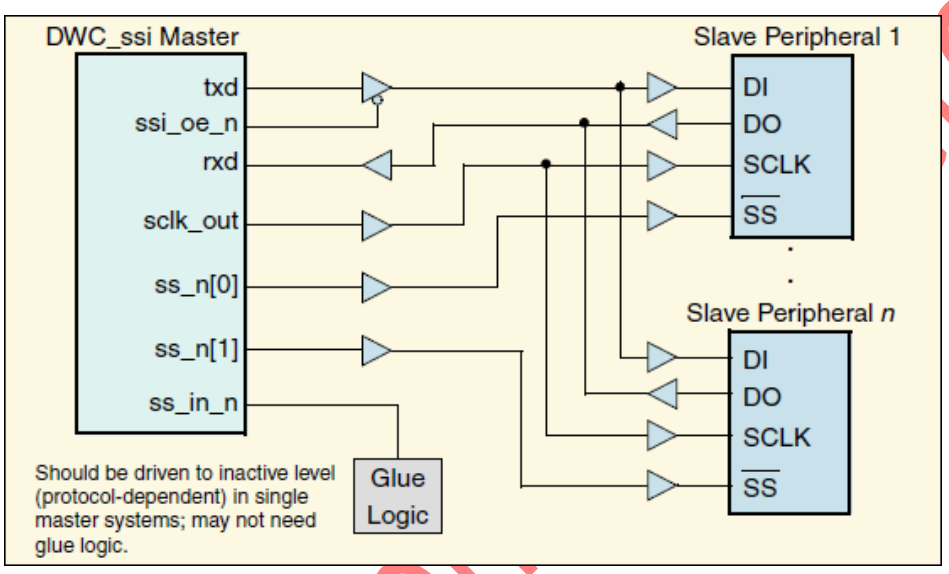


Figure 9-32: SSI Configured as Master Device

9.6.2 Clock Ratios

SSI works on an oversampling architecture. For the master mode of operation, the peripheral clock (sclk_out) period is a multiple of the internal core clock (ssi_clk).

When the SSI macrocell is configured as a master device, the maximum frequency of the bit-rate clock (sclk_out) is one-half the frequency of ssi_clk. This is to allow the shift control logic to capture data on one clock edge of sclk_out and propagate data on the opposite edge.

The frequency of sclk_out can be derived from the following equation.

$$F_{sclk_out} = \frac{F_{ssi_clk}}{SCKDV}$$

SCKDV is a programmable register holding any even value in the range 0 – 65,534. If SCKDV=0,sclk_out is disabled.

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9.6.3 Receive and Transmit FIFO Buffers

The FIFO buffers used by the SSI are internal D-type flip-flops that can be configured in depth between 8 to 256. The widths of both transmit and receive FIFO buffers are fixed at 32 bits due to the serial specifications, which state that a serial transfer (data frame) can be 4 to 32 bits in length. Data frames that are less than 32 bits in size must be right-justified when written into the transmit FIFO buffer. The shift control logic automatically right-justifies receive data in the receive FIFO buffer.

9.6.3.1 Transmit FIFO

The transmit FIFO is loaded by AHB write commands to the SSI data register (DR). Data are popped (removed) from the transmit FIFO by the shift control logic into the transmit shift register. The transmit FIFO generates a FIFO empty interrupt request (ssi_txe_intr) when the number of entries in the FIFO is less than or equal to the FIFO threshold value. The threshold value, set through the programmable register TXFTLR, determines the level of FIFO entries at which an interrupt is generated. The threshold value allows you to provide early indication to the processor that the transmit FIFO is nearly empty. A transmit FIFO overflow interrupt (ssi_txo_intr) is generated if you attempt to write data into an already full transmit FIFO.

9.6.3.2 Receive FIFO

Data are popped from the receive FIFO by AHB read commands to the SSI data register (DR). The receive FIFO is loaded from the receive shift register by the shift control logic. The receive FIFO generates a FIFO-full interrupt request (ssi_rxf_intr) when the number of entries in the FIFO is greater than or equal to the FIFO threshold value plus 1. The threshold value, set through programmable register RXFTLR, determines the level of FIFO entries at which an interrupt is generated.

The threshold value allows you to provide early indication to the processor that the receive FIFO is nearly full. A receive FIFO overrun interrupt (ssi_rxo_intr) is generated when the receive shift logic attempts to load data into a completely full receive FIFO. However, this newly received data are lost. A receive FIFO underflow interrupt (ssi_rxu_intr) is generated if you attempt to read from an empty receive FIFO. This alerts the processor that the read data are invalid.

9.6.4 DMA Operation

The SSI has optional built-in DMA capability which can be selected at configuration time; it has a handshaking interface to a DMA Controller to request and control transfers. The AHB bus is used to perform the data transfer to or from the DMA. While the SSI DMA operation is designed in a generic way to fit any DMA controller as easily as possible, it is designed to work seamlessly, and best used.To enable the DMA Controller interface on the SSI, you must write the DMA Control Register (DMACR). Writing a 1 into the TDMAE bit field of DMACR register enables the SSI transmit handshaking interface. Writing a 1 into the RDMAE bit field of the DMACR register enables the SSI receive handshaking interface.

9.6.5 SSI Interrupts

The SSI interrupts are described as follows:

Transmit FIFO Empty Interrupt (ssi_txe_intr) – Set when the transmit FIFO is equal to or below its threshold value and requires service to prevent an under-run. The threshold value, set through a software-programmable register, determines the level of transmit FIFO entries at which an interrupt is generated. This interrupt is cleared by hardware when data are written into the transmit FIFO buffer, bringing it over the threshold level.

Transmit FIFO Overflow Interrupt (ssi_txo_intr) – Set when an AHB access attempts to write into the transmit FIFO after it has been completely filled. When set, data written from the AHB is discarded. This interrupt remains set until you read the transmit FIFO overflow interrupt clear register (TXOICR).

Receive FIFO Full Interrupt (ssi_rxf_intr) – Set when the receive FIFO is equal to or above its threshold value plus 1 and requires service to prevent an overflow. The threshold value, set through a software-programmable register, determines the level of receive FIFO entries at which an interrupt is generated. This interrupt is

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cleared by hardware when data are read from the receive FIFO buffer, bringing it below the threshold level.

Receive FIFO Overflow Interrupt (ssi_rxo_intr) – Set when the receive logic attempts to place data into the receive FIFO after it has been completely filled. When set, newly received data are discarded. This interrupt remains set until you read the receive FIFO overflow interrupt clear register (RXOICR).

Receive FIFO Underflow Interrupt (ssi_rxu_intr) – Set when an AHB access attempts to read from the receive FIFO when it is empty. When set, zeros are read back from the receive FIFO. This interrupt remains set until you read the receive FIFO underflow interrupt clear register (RXUICR).

Combined Interrupt Request (ssi_intr) – OR'ed result of all the above interrupt requests after masking. To mask this interrupt signal, you must mask all other SSI interrupt requests.

9.6.6 Enhanced SPI Modes

SSI supports the dual, quad, and octal modes of SPI using the SSIC_SPI_MODE configuration parameter. The possible values for this parameter are Standard, Dual SPI, Quad SPI and Octal SPI modes. When dual, quad, or octal mode is selected for this parameter, the width of txd, rxd and ssi_oe_n signals change to 2, 4, or 8, respectively. Hence, the data is shifted out/in on more than one line, increasing the overall throughput.Dual SPI, Quad or Octal SPI modes function similarly except for the width of txd, rxd and ssi_oe_n signals. The mode of operation (write/read) can be selected using the CTRLR0.TMOD field.

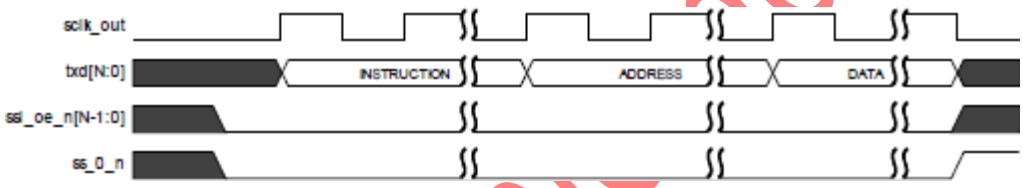


Figure 9-33: Typical Write Operation Dual/Quad/Octal SPI Mode

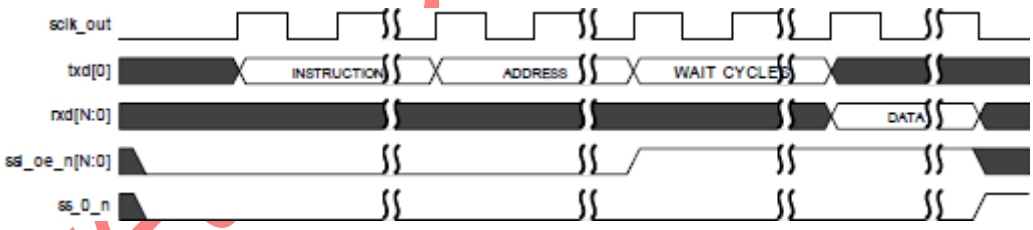


Figure 9-34: Typical Read Operation Dual/Quad/Octal SPI Mode

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9.6.7 eXecute In Place (XIP) Mode

SSI provides a function to directly perform memory read operation from AHB transaction. This is called execute in place mode, in which SSI acts as memory mapped interface to an SPI memory. The XIP mode can be enabled in SSI by selecting the configuration parameter SSIC_XIP_EN. This includes an extra sideband signal xip_en on an AHB interface. This signal level decides if the AHB transfers are register read-write or XIP reads.Only AHB READs are supported during XIP operation.If *xip_en* signal is driven to 1, then SSI expects a read request to be made on the AHB interface. This request is translated to SPI read on the serial interface. As soon as the data is received, it is returned to AHB interface in same transaction. *haddr* is used to derive the address to be sent on the SPI interface.Certain devices expect the instruction phase to exist during XIP transfers. SSI supports inclusion of some fixed instruction sets during the XIP mode of operation.

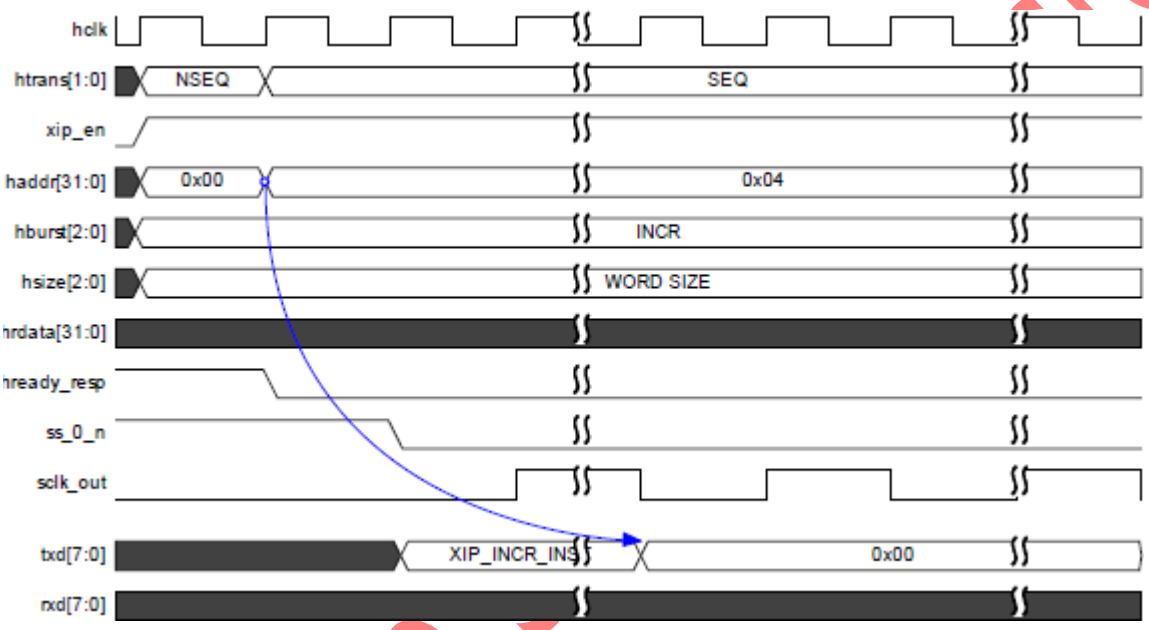


Figure 9-35: XIP Transfer with Instruction Phase

9.6.8 Continuous Transfer Mode in XIP

When SSI receives an XIP request, address from the AHB interface is transmitted onto the SPI interface directly . Each new transfer (XIP Read) on the AHB interface is treated in the same manner. Therefore, for every request, a new address must be sent to the device thereby contributing to the latency of the system.

If a memory device allows the stretching of slave select signal in between the XIP read transfers, then SSI can be programmed for continuous XIP mode to achieve higher performance. In this mode, the host fuses two or more AHB burst requests into a single SPI command by ensuring that the command and address are not retransmitted and the host controller need not wait for any dummy cycles in between these bursts.

When this function is enabled, then SSI functions in continuous XIP mode as soon as first XIP command is received. For the first XIP transfer, the address is sent on the SPI interface. After reception of requested data, SSI continues to keep the slave selected and the clock (sclk_out) remains in the default state. For subsequent XIP transfers on the AHB interface, SSI resumes the clock (sclk_out) and neither the command nor the address is transmitted onto the SPI interface and the data to be fetched from the device immediately (no dummy cycles).

- An undefined INCR (hburst = 001) burst is not supported in continuous read mode.

During the continuous transfer, a lot of power is dissipated on the slave device since the slave is selected all

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the times. To avoid such condition, SSI provides a configuration option to enable a watchdog timer to de-select the slave after the counter runs out.

SSI can de-select the slave under the following conditions:

- A non-XIP command is received on an XIP interface (effectively any AHB transaction with xip_en driven to 0).
- When the AHB transaction is to a non-consecutive address, the slave select is removed and then SSI initiates a new XIP request.
- SSI does not detect any XIP transfer on AHB interface for the time-period specified in XIP_CNT_TIME_OUT register.

9.6.9 Data Pre-fetch in XIP Operations

Using the data pre-fetch feature in SSI, the controller pre-fetches the data for the successive burst during the current XIP transaction. If the next transaction request is made to the successive address, the data can be read directly from the RX FIFO instead of waiting for a new address and data to be sent to the device. This improves the overall performance of the system.

The amount of the data to be pre-fetched should be equal to the burst length of the last AHB request or the FIFO depth (whichever is lower). For example, if AHB defines a burst length of 16 starting from address 0x00, then SSI fetches 16 beats to complete the current transfer, and then again 16 more data frames will be fetched and kept into the data register. If AHB bus again requests for data starting from end address for last transfer, then the rest of the data will be sent to the device from RX FIFO itself. In parallel, SSI again starts an XIP transfer to the device to pre-fetch the next chunk of data. If AHB master does not define the contiguous address, then current data is flushed out from the FIFO and the controller starts a fresh transaction.

When SSI completes the current burst, and is pre-fetching the data for the next burst, a new XIP request may be placed. In this case, SSI can terminate the current transfer, or increase the number to data beats to be fetched depending on the address.

- If XIP pre-fetching is enabled, AHB request for incremental transfer of undefined length (hburst = 3'b001) is not allowed.

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10 Serial Peripheral Interface Module (SPI)

10.1 Introduction

The serial peripheral interface (SPI) module allows full-duplex, synchronous, serial communication between the microcontroller unit(MCU) and peripheral devices.

Software can poll the SPI status flags or SPI operation can be interrupt driven.

10.2 Features

Features include:

- Master mode and slave mode
- Slave-select output
- Mode fault error flag with central processor unit (CPU) interrupt capability
- Control of SPI operation during doze mode
- Programmable interface operation for Freescale SPI or Texas Instruments synchronous serial interfaces
- Separate transmit and receive FIFOs, each 8 bits wide and 8 locations deep
- Programmable data frame size from 4 to 16 bits
- Internal loopback test mode for diagnostic/debug testing
- Standard FIFO-based interrupts and End-of-Transmission interrupt
- Efficient transfers with DMA interface
- Visibility into TX and RX FIFOs for ease of debugging
- High Speed Mode for transfer timing adjustment

10.3 Modes of Operation

The SPI functions in these three modes:

- Run mode — Run mode is the normal mode of operation.
- Doze mode — Doze mode is a configurable low-power mode.
- Stop mode — The SPI is inactive in stop mode.

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10.4 Block Diagram

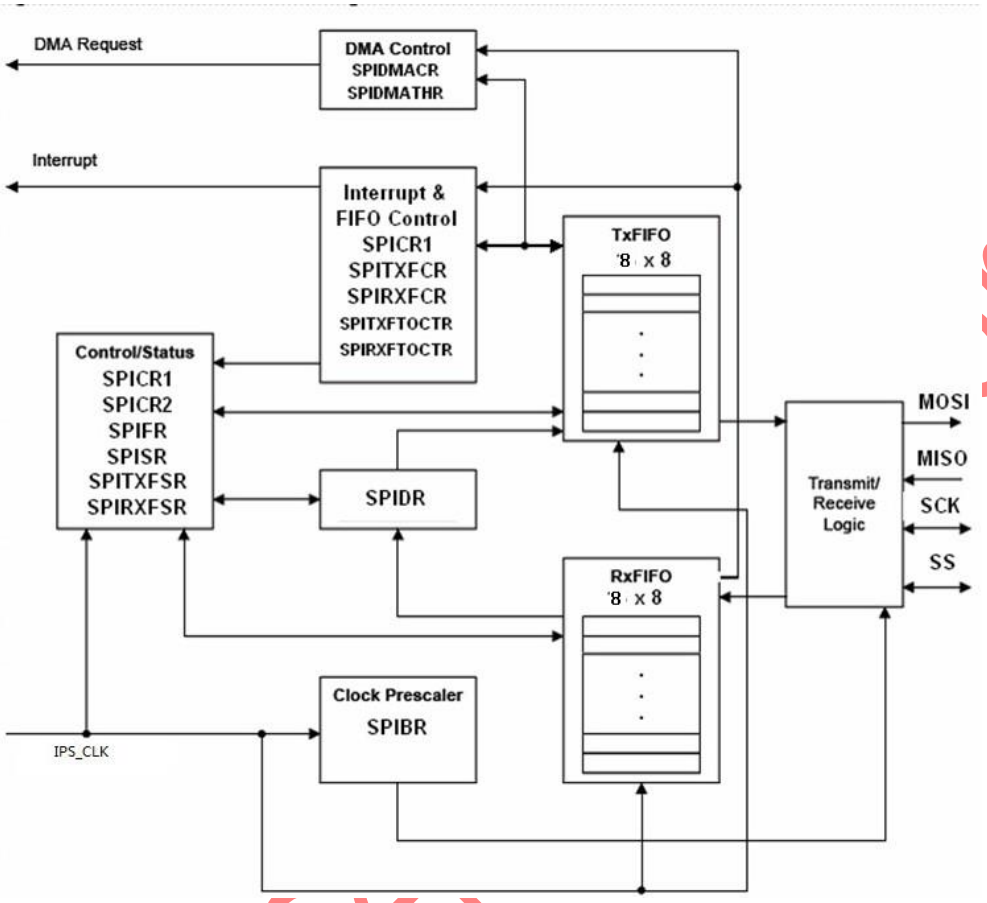


Figure 10-1: SPI Block Diagram

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10.5 Signal Description

An overview of the signals is provided in **Table 10-1**.

Table 10-1: Signal Properties

Name	Port	Function[1]	Reset State
MISO	SPIPORT[0]	Master Data In / Slave Data Out	0
MOSI	SPIPORT[1]	Master Data Out / Slave Data In	0
SCK	SPIPORT[2]	Serial Clock	0
SS	SPIPORT[3]	Slave Select	0

NOTE: The specific SPI ports (MISO,MOSI,SCK,SS) are GP I/O ports when the SPI is disabled (SPE=0).

10.5.1 MISO (Master In/Slave Out)

MISO is one of the two SPI data pins.

- In master mode, MISO is the data input.
- In slave mode, MISO is the data output and is three-stated until a master drives the SS input pin low.
- In bidirectional mode, a slave MISO pin is the SISO pin (slave in/slave out).
- In a multiple-master system, all MISO pins are tied together.

10.5.2 MOSI (Master Out/Slave In)

MOSI is one of the two SPI data pins.

- In master mode, MOSI is the data output.
- In slave mode, MOSI is the data input.
- In bidirectional mode, a master MOSI pin is the MOMI pin (master out/master in).
- In a multiple-master system, all MOSI pins are tied together.

10.5.3 SCK (Serial Clock)

The SCK pin is the serial clock pin for synchronizing transmissions between master and slave devices.

- In master mode, SCK is an output.
- In slave mode, SCK is an input.
- In a multiple-master system, all SCK pins are tied together.

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10.5.4 SS (Slave Select)

In master mode, the SS pin can be:

- A mode-fault input
- A general-purpose input
- A general-purpose output
- A slave-select output

In slave mode, the SS pin is always a slave-select input.

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10.6 Memory Map and Registers

10.6.1 Memory Map

Table 10-2 shows the SPI memory map.

Table 10-2: SPI Memory Map

Address Offset	Bits 7–0	Access
0x0001	SPI Frame Register (SPIFR)	S/U
0x0000	SPI Baud Rate Register (SPIBR)	S/U
0x0003	SPI Control Register 2 (SPICR2)	S/U
0x0002	SPI Control Register 1 (SPICR1)	S/U
0x0005	SPI TX FIFO Timeout Counter Register(SPITXFTOCTR)	S/U
0x0004	SPI RX FIFO Timeout Counter Register(SPIRXFTOCTR)	S/U
0x0007	SPI TXFIFO Control Register(SPITXFCR)	S/U
0x0006	SPI RXFIFO Control Register(SPIRXFCR)	S/U
0x0009	SPI Before SCK Delay Register (SPIBSCDR)	S/U
0x0008	SPI After SCK Delay Register (SPIASCDR)	S/U
0x000b	SPI Pullup and Reduced Drive Register (SPIPURD)	S/U
0x000a	SPI Port Data Direction Register (SPIDDR)	S/U
0x000c~d,f	SPI Transmit Counter Register (SPITCNT)	S/U
0x000e	SPI Port Data Register (SPIPORT)	S/U
0x0011	Reserved	S/U
0x0010	Interrupt Register of SS Pin (IRSP)	S/U
0x0012~13	SPI Data Register (SPIDR)	S/U
0x0015	SPI TX FIFO Status Register(SPITXFSR)	S/U
0x0014	SPI RX FIFO Status Register(SPIRXFSR)	S/U
0x0016~17	SPI Status Register(SPISR)	S/U
0x0019	SPI Interrupt Control Register(SPIICR)	S/U
0x0018	SPI FIFO Debug Control Register(SPIFDCR)	S/U
0x001b	SPI DMA Threshold Register(SPIDMATHR)	S/U
0x001a	SPI DMA Control Register(SPIDMACR)	S/U
0x001c	SPI TX FIFO Debug Register (SPITXFDBGR)	S/U
0x001e	SPI RX FIFO Debug Register (SPIRXFDBGR)	S/U

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10.6.2 Register Description

10.6.2.1 SPI Baud Rate Register

Addr. Offset: 0x0000 Reset Value: 0x00

7	6	5	4	3	2	1	0
EOTFIE	SPPR[6:4]			FIRM	SPR[2:0]		
ro	rw			ro	rw		

Figure 10-2: SPI Baud Rate Register (SPIBR)

Read: Anytime

Write: Anytime; writing to unimplemented bits has no effect

SPPR[6:4] — SPI Baud Rate Preselection Bits

The SPPR[6:4] and SPR[2:0] bits select the SPI clock divisor as shown in **Table 10-3**. Reset clears SPPR[6:4] and SPR[2:0], selecting an SPI clock divisor of 2.

SPR[2:0] — SPI Baud Rate Bits

The SPPR[6:4] and SPR[2:0] bits select the SPI clock divisor as shown in **Table 10-3**. Reset clears SPPR[6:4] and SPR[2:0], selecting an SPI clock divisor of 2.

NOTE: Writing to SPIBR during a transmission may cause spurious results.

Table 10-3: SPI Baud Rate Selection (10-MHz Module Clock)

SPPR[6:4]	SPR[2:0]	SPI Clock Divisor	Baud Rate	SPPR[6:4]	SPR[2:0]	SPI Clock Divisor	Baud Rate
000	000	2	5 MHz	100	000	10	1 MHz
000	001	4	2.5 MHz	100	001	20	0.5 MHz
000	010	8	1.25 MHz	100	010	40	0.25 MHz
000	011	16	0.625 MHz	100	011	80	125 kHz
000	100	32	0.31 MHz	100	100	160	62.5 kHz
000	101	64	156.25 kHz	100	101	320	31.25 kHz
000	110	128	78.125 kHz	100	110	640	15.625 kHz
000	111	256	39.06 kHz	100	111	1280	7.81 kHz
001	000	4	2.5 MHz	101	000	12	833.33 kHz
001	001	8	1.25 MHz	101	001	24	416.67 kHz
001	010	16	0.625 MHz	101	010	48	208.33 kHz
001	011	32	0.31 MHz	101	011	96	104.17 kHz
001	100	64	156.25 kHz	101	100	192	52.08 kHz
001	101	128	78.125 kHz	101	101	384	26.04 kHz
001	110	256	39.06 kHz	101	110	768	13.02 kHz
001	111	512	19.53 kHz	101	111	1536	6.51 kHz
010	000	6	1.67 MHz	110	000	14	714.29 kHz
010	001	12	0.83 MHz	110	001	28	357.14 kHz
010	010	24	0.42 MHz	110	010	56	178.57 kHz
010	011	48	208.33 kHz	110	011	112	89.29 kHz
010	100	96	104.17 kHz	110	100	224	44.64 kHz
010	101	192	52.08 kHz	110	101	448	22.32 kHz

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SPPR[6:4]	SPR[2:0]	SPI Clock Divisor	Baud Rate	SPPR[6:4]	SPR[2:0]	SPI Clock Divisor	Baud Rate
010	110	384	26.04 kHz	110	110	896	11.16 kHz
010	111	768	13.02 kHz	110	111	1792	5.58 kHz
011	000	8	1.25 MHz	111	000	16	0.625 MHz
011	001	16	0.625 MHz	111	001	32	0.31 MHz
011	010	32	0.31 MHz	111	010	64	156.25 kHz
011	011	64	156.25 kHz	111	011	128	78.125 kHz
011	100	128	78.125 kHz	111	100	256	39.06 kHz
011	101	256	39.06 kHz	111	101	512	19.53 kHz
011	110	512	19.53 kHz	111	110	1024	9.77 kHz
011	111	1024	9.77 kHz	111	111	2048	4.88 kHz

10.6.2.2 SPI Frame Register

Addr. Offset: 0x0001 Reset Value: 0x47

7	6	5	4	3	2	1	0
CONT	GTE	LBM	FFSEL	FMSZ[3:0]			
rw	rw	rw	rw	rw			

Figure 10-3: SPI Frame Register (SPIFR)

CONT — Continuous peripheral chip select enable

1 = Keep peripheral chip select signal low between transfers until EOTF is set

0 = Return peripheral chip select signal to high between transfers

GTE — Guard Time Enable

1 = Guard Time is Enabled, 0 = Guard Time is Disabled

LBM — Loop Back Mode

1 = Loop Back Mode, 0 = Normal Mode

FFSEL — Frame Format Select

1 = TI frame format is selected

0 = FREESCALE frame format is selected FMSZ[3:0] — Frame Size

The FMSZ[3:0] bits control the frame data length from 4 to 16bits. 0x3 sets the frame length to 4bits and 0xf sets it to 16 bits. meanwhile case 0x0~0x2 will set the frame to 4 bit long automatically.

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10.6.2.3 SPI Control Register

Addr. Offset: 0x0002 Reset Value: 0x04

7	6	5	4	3	2	1	0
SPIE	SPE	SWOM	MSTR	CPOL	CPHA	SSOE	LSBFE
rw	rw	rw	rw	rw	rw	rw	rw

Figure 10-4: SPI Control Register 1 (SPICR1)

Read: Anytime Write: Anytime

SPIE — SPI Interrupt Enable Bit

The SPIE bit enables the EOTF flag to generate interrupt requests. Reset clears SPIE.

1 = EOTF interrupt requests enabled

0 = EOTF interrupt requests disabled

SPE — SPI System Enable Bit

The SPE bit enables the SPI and dedicates SPI port pins [3:0] to SPI functions. When SPE is clear, the SPI system is initialized but in a low-power disabled state. Reset clears SPE.

1 = SPI enabled, 0 = SPI disabled

SWOM — SPI Wired-OR Mode Bit

The SWOM bit configures the output buffers of SPI port pins [3:0] as open-drain outputs. SWOM controls SPI port pins [3:0] whether they are SPI outputs or general-purpose outputs. Reset clears SWOM.

1 = Output buffers of SPI port pins [3:0] open-drain

0 = Output buffers of SPI port pins [3:0] CMOS drive

NOTE: The SWOM bit has no effect in this part.

MSTR — Master Bit

The MSTR bit selects SPI master mode or SPI slave mode operation. Reset clears MSTR.

1 = Master mode, 0 = Slave mode

CPOL — Clock Polarity Bit

The CPOL bit selects an inverted or non-inverted SPI clock. To transmit data between SPI modules, the SPI modules must have identical CPOL values.

Reset clears CPOL.

1 = Active-low clock; SCK idles high

0 = Active-high clock; SCK idles low

CPHA — Clock Phase Bit

The CPHA bit delays the first edge of the SCK clock. Reset sets CPHA.

1 = First SCK edge at start of transmission

0 = First SCK edge 1/2 cycle after start of transmission

NOTE: Any value change of CPOL or CPHA during a transmission(when SS is low) may cause spurious results.Please change CPOL or CPHA value before a transmission (when SS is high) is to be launched.

SSOE — Slave Select Output Enable Bit

The SSOE bit and the DDRSP3 bit configure the SS pin as a general-purpose input or a slave-select output. Reset clears SSOE.

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Table 10-4: SS Pin I/O Configurations

DDRSP3	SSOE	Master Mode	Slave Mode
0	0	Mode-fault input	Slave-select input
0	1	General-purpose input	Slave-select input
1	0	General-purpose output	Slave-select input
1	1	Slave-select output	Slave-select input

NOTE: *Setting the SSOE bit disables the mode fault detect function.*

LSBFE — LSB-First Enable Bit

The LSBFE enables data to be transmitted LSB first. Reset clears LSBFE.

1 = Data transmitted LSB first.

0 = Data transmitted MSB first

10.6.2.4 SPI Control Register 2

Addr. Offset: 0x0003 Reset Value: 0x00

7	6	5	4	3	2	1	0
GT[5:0]						SPISDOZ	SPC0
rw						rw	rw

Figure 10-5: SPI Control Register 2 (SPICR2)

Read: Anytime Write: Anytime;

GT — Guard Time Bits

$GT = (GT[5:3]+1)*2^{(GT[2:0]+1)}$

SPISDOZ — SPI Stop in Doze Bit

The SPIDOZ bit stops the SPI clocks when the CPU is in doze mode.

1 = SPI inactive in doze mode

0 = SPI active in doze mode

SPC0 — Serial Pin Control Bit 0

The SPC0 bit enables the bidirectional pin configurations shown in **Table 10-5**.

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Table 10-5: Bidirectional Pin Configurations

	Pin Mode	SPC0	MSTR	MISO Pin[1]	MOSI Pin[2]	SCK Pin[3]	SS Pin[4]
A	Normal	0	0	Slave data output	Slave data input	SCK input	Slave-select input
B			1	Master data input	Master data output	SCK output	MODF/GP input (DDRSP3 = 0) or GP output (DDRSP3 = 1)
C	Bidirectional	1	0	Slave data I/O	GP[5] I/O	SCK input	Slave-select input
D			1	GP I/O	Master data I/O	SCK output	MODF/GP input (DDRSP3 = 0) or GP output (DDRSP3 = 1)

NOTES:

- Slave output is enabled if SPIDDR bit 0 = 1, SS = 0, and MSTR = 0 (A, C).
- Master output is enabled if SPIDDR bit 1 = 1 and MSTR = 1 (B, D).
- SCK output is enabled if SPIDDR bit 2 = 1 and MSTR = 1 (B, D).
- SS output is enabled if SPIDDR bit 3 = 1, SPICR1 bit 1 (SSOE) = 1, and MSTR = 1 (B, D). MODF input is enabled if SPI DDR bit 3 = 0 and SSOE = 0. GP input is enabled if SPI DDR bit 3 = 0 and SSOE = 1.
- GP = General-purpose

10.6.2.5 SPI RX FIFO TimeOut Counter Register

Addr. Offset: 0x0004 Reset Value: 0x40

7	6	5	4	3	2	1	0
RXFTOIE	RXFTOE	RXFTOCNT[5:0]					
rw	rw	rw					

Figure 10-6: SPI RX FIFO TimeOut Counter Register (SPIRXFTOCTR)

Read: Anytime;

Write: Anytime;

SPIRXFTOCTR[5:0] sets the SPI RX FIFO Timeout counter number.Once RX FIFO is not empty,the counter is on.If there is no operation to RX FIFO till the counter counts down to 0,the RXF_TIMEOUT flag in SPISR will set.

RXFTOIE — RX FIFO TimeOut Interrupt Enable

1 = RX FIFO TimeOut Interrupt Enabled

0 = RX FIFO TimeOut Interrupt Disabled

RXFTOE — RX FIFO TimeOut Function Enable

1 = RX FIFO TimeOut Function Enabled

0 = RX FIFO TimeOut Function Disabled

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10.6.2.6 SPI TX FIFO TimeOut Counter Register

Addr. Offset: 0x0005 Reset Value: 0x40

7	6	5	4	3	2	1	0
TXFTOIE	TXFTOE	TXFTOCNT[5:0]					
rw	rw	rw					

Figure 10-7: SPI TX FIFO TimeOut Counter Register (SPITXFTOCTR)

Read: Anytime;

Write: Anytime;

SPITXFTOCTR[5:0] sets the SPI TX FIFO Timeout counter number.Once TX FIFO is not empty,the counter is on.If there is no operation to TX FIFO till the counter counts down to 0,the TXF_TIMEOUT flag in SPISR will set.

TXFTOIE — TX FIFO TimeOut Interrupt Enable

1 = TX FIFO TimeOut Interrupt Enabled

0 = TX FIFO TimeOut Interrupt Disabled

TXFTOE — TX FIFO TimeOut Function Enable

1 = TX FIFO TimeOut Function Enabled

0 = TX FIFO TimeOut Function Disabled

10.6.2.7 SPI RX FIFO Control Register

Addr. Offset: 0x0006 Reset Value: 0x00

7	6	5	4	3	2	1	0
RXFCLR	RXFOVIE	RXFUDIE	RXFSTHIE	Reserved	RXFSTH[2:0]		
rw	rw	rw	rw	ro	rw		

Figure 10-8: SPI RX FIFO Control Register (SPIRXFCR)

Read: Anytime Write: Anytime

RXFCLR — RX FIFO Clear

Write 1 to the bit will reset RX FIFO.

RXFOVIE — RX FIFO Overflow Interrupt Enable

1 = RX FIFO Overflow Interrupt Enabled

0 = RX FIFO Overflow Interrupt Disabled

RXFUDIE — RX FIFO Underflow Interrupt Enable

1 = RX FIFO Underflow Interrupt Enabled

0 = RX FIFO Underflow Interrupt Disabled

RXFSTHIE — RX FIFO Service Threshold Interrupt Enable

1 = RX FIFO Service Threshold Interrupt Enabled

0 = RX FIFO Service Threshold Interrupt Disabled RXFSTH[2:0] — RX FIFO Service Threshold

Once effective data number in RX FIFO is above or equal the threshold specified, RX FIFO Service Flag asserted.

data number=RXFSTH[2:0]+1

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10.6.2.8 SPI TX FIFO Control Register

Addr. Offset: 0x0007 Reset Value: 0x07

7	6	5	4	3	2	1	0
TXFCLR	TXFOVIE	TXFUDIE	TXFSTHIE	Reserved	TXFSTH[2:0]		
rw	rw	rw	rw	ro	rw		

Figure 10-9: SPI TX FIFO Control Register (SPITXFCR)

Read: Anytime Write: Anytime

TXFCLR — TX FIFO Clear

Write 1 to the bit will reset TXFIFO

TXFOVIE — TX FIFO Overflow Interrupt Enable

1 = TX FIFO Overflow Interrupt Enabled

0 = TX FIFO Overflow Interrupt Disabled

TXFUDIE — TX FIFO Underflow Interrupt Enable

1 = TX FIFO Underflow Interrupt Enabled

0 = TX FIFO Underflow Interrupt Disabled

TXFSTHIE — TX FIFO Service Threshold Interrupt Enable

1 = TX FIFO Service Threshold Interrupt Enabled

0 = TX FIFO Service Threshold Interrupt Disabled TXFSTH[2:0] — TX FIFO Service Threshold

Once effective data number in TX FIFO is below or equal the threshold specified, TX FIFO Service Flag asserted.

data number=TXFSTH[2:0]

10.6.2.9 SPI After SCK Delay Register

Addr. Offset: 0x0008 Reset Value: 0x00

7	6	5	4	3	2	1	0
ASCDE	PASCD [6:4]			Reserved	ASCD [2:0]		
rw	rw			ro	rw		

Figure 10-10: SPI After SCK Delay Register (SPIASCDR)

ASCDE — After SCK Delay Enable

1 = After SCK Delay is Enabled

0 = After SCK Delay is Disabled

PASCD[6:4] — SPI After SCK Delay Preselection Bits

The PASCD[6:4] and ASCD[2:0] bits select the SPI after SCK delay divisor.

ASCD[2:0] — SPI After SCK Delay Bits

$$ASCD = (PASCD[6:4]+1)*2^{(ASCD[2:0]+1)};\ \text{and}\ t_T = 0.5*SCK + ASCD$$

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10.6.2.10 SPI Before SCK Delay Register

Addr. Offset: 0x0009 Reset Value: 0x00

7	6	5	4	3	2	1	0
BSCDE	PBSCD [6:4]			Reserved	BSCD [2:0]		
rw	rw			ro	rw		

Figure 10-11: SPI Before SCK Delay Register (SPIBSCDR)

BSCDE — Before SCK Delay Enable

1 = Before SCK Delay is Enabled

0 = Before SCK Delay is Disabled

PBSCD[6:4] — SPI Before SCK Delay Preselection Bits

The PBSCD[6:4] and BSCD[2:0] bits select the SPI before SCK delay divisor.

BSCD[2:0] — SPI Before SCK Delay Bits

BSCD = (PBSCD[6:4]+1)*$2^{(BSCD[2:0]+1)}$; and t_L= 0.5*SCK+BSCD

10.6.2.11 SPI Port Data Direction Register

Addr. Offset: 0x000A Reset Value: 0x00

7	6	5	4	3	2	1	0
PPS[3:0]				DDRSP[3:0]			
rw				rw			

Figure 10-12: SPI Port Data Direction Register (SPIDDR)

Read: Anytime Write: Anytime

PPS[3:0] — Port Priority Select Bits

The PPS[3:0] bits select the function on priority of SPIPORT pins.

1 = GPIO function is prior to normal function

0 = normal function is prior to GPIO function

DDRSP[3:0] — Data Direction Bits

The DDRSP[3:0] bits control the data direction of SPIPORT pins. Reset clears DDRSP[3:0].

1 = Corresponding pin configured as output

0 = Corresponding pin configured as input

In slave mode, DDRSP3 has no meaning or effect. In master mode, the DDRSP3 and the SSOE bits determine whether SPI port pin 3 is a mode-fault input, a general-purpose input, a general-purpose output, or a slave-select output.

NOTE: When the SPI is enabled (SPE = 1), the MISO, MOSI, and SCK pins:

- Are inputs if their SPI functions are input functions regardless of the state of their DDRSP bits.
- Are outputs if their SPI functions are output functions only if their DDRSP bits are set

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10.6.2.12 SPI Pullup and Reduced Drive Register

Addr. Offset: 0x000B Reset Value: 0x01

7	6	5	4	3 2	1	0
HS	PSW	DBLRXDR	ENDING	MSPD[1:0]	Reserved	PUPSP
rw	rw	rw	rw	rw	ro	rw

Figure 10-13: SPI Pullup and Reduced Drive Register (SPIPURD)

Read: Anytime;

Write: Anytime; writing to unimplemented bits has no effect

HS — High Speed Mode Enable Bit

Set HS to 1 in master mode makes MSPD[1:0] to take effect.

1 = High Speed Mode is enabled

0 = High Speed Mode is disabled

PSW — Pin Switch Bit

Switch MOSI to MISO and MISO to MOSI.

1 = Switch enabled, 0 = Switch disabled

DBLTXDR — Double TX Data Reg Mode Enable Bit

Set this bit to enable double tx data reg mode to decrease the delay requirement between two frames in slave mode.In double tx data reg mode,at least one prefilled frame in TX FIFO is required.

1 = Double TX Data Reg Mode is enabled

0 = Double TX Data Reg Mode is disabled

ENDING — Ending Select Bit

1 = Big Ending, 0 = Little Ending

MSPD[1:0] — SPI Master Sample Point Delay

Specify the number of system clock cycles delayed for SCK sample edge 0x0 =no delay;0x1 = 1 cycle delay;0x2=2 cycle delay;0x3=2 cycle delay;

PUPSP — SPI Port Pullup or Pulldown Enable Bit

1 = Pullup or Pulldown devices enabled for SPIPORT bits [3:0]

0 = Pullup or Pulldown devices disabled for SPIPORT bits [3:0]

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10.6.2.13 SPI Transmit Counter Register

Addr. Offset: 0x000C~0x000D, 0x000F Reset Value: 0x000000

31	30	29	28	27	26	25	24
SPITCNT[31:24]							
rw							
23	22	21	20	19	18	17	16
SPITCNT[23:16]							
rw							
15	14	13	12	11	10	9	8
SPITCNT[15:8]							
rw							

Figure 10-14: SPI Transmit Counter Register (SPITCNT)

Read: Anytime Write: Anytime

SPITCNT[31:8] — SPI Transmit Counter Register

The counter is counted down by 1 when a frame is sent in master mode when CONT bit in SPIFR is asserted.Once the counter is counted down to 0 and another frame is sent,pin SS will set to high and the counter value will reload.

10.6.2.14 SPI Port Data Register

Addr. Offset: 0x000E Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved				PORTSP[3:0]			
ro				rw			

Figure 10-15: SPI Port Data Register (SPIPORT)

Read: Anytime Write: Anytime

RSVD[7:4] — Reserved

Writing to these read/write bits updates their values but has no effect on functionality.

PORTSP[3:0] — SPI Port Data Bits

Data written to SPIPORT drives pins only when they are configured as general-purpose outputs.

Reading an input (DDRSP bit clear) returns the pin level; reading an output (DDRSP bit set) returns the pin driver input level.

Writing to any of the PORTSP[3:0] pins does not change the pin state when the pin is configured for SPI output.

SPIPORT I/O function depends upon the state of the SPE bit in SPICR1 and the state the DDRSP bits in SPIDDR.

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10.6.2.15 Interrupt Register of SS Pin

Addr. Offset: 0x0010 Reset Value: 0x00

7	6	5	4	3	2	1	0
SSIE	Reserved		SSF	SSLPR	SSD	SSPA	
rw	ro		ro	rw	ro	ro	

Figure 10-16: Interrupt Register of SS Pin (IRSP)

SSPA[1:0] — SS Pin Assignment Select Fields

The read/write SSPA fields configure SS pins for level detection and rising and/or falling edge detection as shows.

Pins configured as level-sensitive are inverted so that a logic 0 or logic 1 on the external pin represents a valid interrupt request. Level-sensitive interrupt inputs are not latched. To guarantee that a level-sensitive interrupt request is acknowledged, the interrupt source must keep the signal asserted until acknowledged by software. Level sensitivity must be selected to bring the device out of stop mode with an SS interrupt.

Pins configured as edge-triggered are latched and need not remain asserted for interrupt generation. A pin configured for edge detection is monitored regardless of its configuration as input or output.

Table 10-6: SSPA Field Settings

SSPA[1:0]	Pin Configuration
00	Pin SS level-sensitive
01	Pin SS rising edge triggered
10	Pin SS falling edge triggered
11	Pin SS both falling edge and rising edge triggered

Interrupt requests generated in the SPI module can be masked by the interrupt controller module. SSPA functionality is independent of the selected pin direction.

Reset clears the SSPA fields.

SSIE — SS Port Interrupt Enable Bits

The read/write SSIE bits enable SS interrupt requests. If a bit in SSIE is set, SPI generates an interrupt request when:

- The corresponding bit in the SS Flag Register (SSF) is set or later becomes set, or
- The corresponding pin level is low and the pin is configured for level-sensitive operation

Clearing SSIE negates any interrupt request from the corresponding SS pin. Reset clears SSIE.

1 = Interrupt requests from SS pin enabled

0 = Interrupt requests from SS pin disabled

SSD — SS Pin Data Bit

The read-only SSD reflects the current state of the SS pins. Writing to SSD has no effect, and the write cycle terminates normally. Reset does not affect SSD.

SSLPR — SS Pin Level Polarity Bits

If SS are configured to level-sensitive, setting any bit in the SSLPR configures the corresponding pin high level-sensitive. Clearing any bit in SSLPR configures the corresponding pin low level-sensitive. Reset clears SSLPR.

1 = Corresponding SS pin configured to high level-sensitive

0 = Corresponding SS pin configured to low level-sensitive

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SSF — SS PinFlag Bits

When an SS pin is configured for edge triggering, its corresponding read/write bit in SSFR indicates that the selected edge has been detected. Reset clears SSF.

1 = Selected edge for SS pin has been detected.

0 = Selected edge for SS pin has not been detected.

Bits in this register are set when the selected edge is detected on the corresponding pin. A bit remains set until cleared by writing a 1 to it. Writing 0 has no effect. If a pin is configured as level-sensitive (SSPA = 00), pin transitions do not affect this register.

10.6.2.16 SPI Data Register

Addr. Offset: 0x0013 Reset Value: 0x00

15	14	13	12	11	10	9	8
SPIDR[15:8]							
rw							

Addr. Offset: 0x0012 Reset Value: 0x00

7	6	5	4	3	2	1	0
SPIDR[7:0]							
rw							

Figure 10-17: SPI Data Register (SPIDR)

Read: Anytime;

Write: Anytime;

SPIDR is both the input and output register for SPI data. Writing to SPIDR will fill the TX FIFO while reading from SPIDR will drain the RX FIFO.

10.6.2.17 SPI RX FIFO Status Register

Addr. Offset: 0x0014 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	RXNXTP[2:0]			RXFFCT[3:0]			
ro	ro			ro			

Figure 10-18: SPI RX FIFO Status Register (SPIRXFSR)

Read: Anytime

Write: Has no meaning or effect

RXNXTP[2:0] — RX Next Pointer

Indicates the RX FIFO pointer to the data which is to be read out.

RXFCTR[3:0] — RX FIFO Counter

Indicates the RX FIFO data counter.

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10.6.2.18 SPI TX FIFO Status Register

Addr. Offset: 0x0015 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	TXNXTP[2:0]			TXFFCT[3:0]			
ro	ro			ro			

Figure 10-19: SPI TX FIFO Status Register (SPITXFSR)

Read: Anytime

Write: Has no meaning or effect

TXNXTP[2:0] — TX Next Pointer

Indicates the TX FIFO pointer to the data which is to be transmitted.

TXFCTR[3:0] — TX FIFO Counter

Indicates the TX FIFO data counter.

10.6.2.19 SPI Status Register

Addr. Offset: 0x0017 Reset Value: 0x00

15	14	13	12	11	10	9	8
TXFTO	TXFOVF	TXFUDF	TXFSER	RXFTO	RXFOVF	RXFUDF	RXFSER
r/w1c	r/w1c	r/w1c	ro	r/w1c	r/w1c	r/w1c	ro

Addr. Offset: 0x0016 Reset Value: 0x05

7	6	5	4	3	2	1	0
SPIF	FLOAT	EOTF	MODF	TXFFULL	TXFEMP	RXFFULL	RXFEMP
ro	r/w1c	r/w1c	ro	ro	ro	ro	ro

Figure 10-20: SPI Status Register (SPISR)

Read: Anytime

Write: w1c means write 1 to the bit will clear the corresponding flag

SPIF — SPI Finish Flag

The SPIF flag is set after each single transfer.Clear SPIF by read SPIF and then access to SPIDR.

1 = a single transfer has finished

0 = a single transfer has not finished or no transfer

EOTF — End of Transmission Flag

The EOTF flag is set when all the data in TX FIFO is transmitted.

1 = End of Transmission

0 = Transmission not end or no transmission

write 1 to the bit or wirte to SPIDR will clear this flag

FLOST — Frame Lost

When SPI is in slave mode and no valid data is in the TX FIFO,if the master start a transfer at this time,then SPI will return last received data to the master and FLOST is set. If the FLOSTIE bit is also set,FLOST generates an interrupt request.

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1 = Frame lost has occured, 0 = No frame lost

write 1 to the bit will clear this flag

MODF — Mode Fault Flag

The MODF flag is set when the SS pin of a master SPI is driven low and the SS pin is configured as a mode-fault input. If the MODFIE bit is also set, MODF generates an interrupt request. A mode fault clears the SPE, MSTR, and DDRSP[2:0] bits. Clear MODF by reading SPISR with MODF set and then writing to SPICR1. Reset clears MODF.

1 = Mode fault

0 = No mode fault

TXFFULL — TX FIFO Full Flag

1 = TX FIFO Full

0 = TX FIFO not Full

TXFEMP — TX FIFO empty Flag

1 = TX FIFO Empty

0 = TX FIFO not Empty

RXFFULL — RX FIFO Full Flag

1 = TX FIFO Full

0 = TX FIFO not Full

RXFEMP — RX FIFO empty Flag

1 = RX FIFO Empty

0 = RX FIFO not Empty

TXFTO — TX FIFO TimeOut

1 = TX FIFO TimeOut has occured

0 = TX FIFO TimeOut has not occured

write 1 to the bit or wirte to SPIDR will clear this flag

TXFOVF — TX FIFO Overflow Flag

1 = TX FIFO Overflow has occured

0 = TX FIFO Overflow has not occured

TXFUDF — TX FIFO Underflow Flag

1 = TX FIFO Underflow has occured

0 = TX FIFO Underflow has not occured

TXFSER — TX FIFO Service Flag

1 = TX FIFO data number below or equal TXFSTH

0 = TX FIFO data number above TXFSTH

RXFTO — RX FIFO TimeOut

1 = RX FIFO TimeOut has occured

0 = RX FIFO TimeOut has not occured

write 1 to the bit or read from SPIDR will clear this flag

RXFOVF — RX FIFO Overflow Flag

1 = RX FIFO Overflow has occured

0 = RX FIFO Overflow has not occured

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RXFUDF — RX FIFO Underflow Flag

1 = RX FIFO Underflow has occured

0 = RX FIFO Underflow has not occured

RXFSER — RX FIFO Service Flag

1 = RX FIFO data number above RXFSTH

0 = RX FIFO data number below or equal RXFST

10.6.2.20 SPI FIFO Debug Control Register

Addr. Offset: 0x0018 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	TXFIDX [6:4]			Reserved	RXFIDX [2:0]		
ro	rw			ro	rw		

Figure 10-21: SPI FIFO Debug Control Register (SPIFDCR)

Read: Anytime;

Write: Anytime;

TXFIDX[2:0] — TX FIFO Index

Specify the data index to be read from TX FIFO to SPITXFDBGR. RXFIDX[2:0] — RX FIFO Index

Specify the data index to be read from RX FIFO to SPIRXFDBGR.

10.6.2.21 SPI Interrupt Control Register

Addr. Offset: 0x0019 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	FLOSTIE	Reserved	MODFIE	Reserved			
ro	rw	ro	rw	ro			

Figure 10-22: SPI Interrupt Control Register (SPIICR)

Read: Anytime;

Write: Anytime;

MODFIE — MODF Interrupt Enable.

1 = MODF Interrupt Enabled

0 = MODF Interrupt Disabled

FLOSTIE — FLOST Interrupt Enable.

1 = FLOST Interrupt Enabled

0 = FLOST Interrupt Disable

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10.6.2.22 SPI DMA Control Register

Addr. Offset: 0x001A Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved						TXDMAE	RXDMAE
ro						rw	rw

Figure 10-23: SPI DMA Control Register (SPIDMACR)

Read: Anytime;

Write: Anytime;

TXDMAE — TX FIFO DMA request Enable

1 = TX DMA request Enabled

0 = TX DMA request Disabled

RXDMAE — RX FIFO DMA request Enable

1 = RX DMA request Enabled

0 = RX DMA request Disabled

10.6.2.23 SPI DMA Threshold Register

Addr. Offset: 0x001B Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	TXDMATH [6:4]			Reserved	RXDMATH [2:0]		
ro	rw			ro	rw		

Figure 10-24: SPI DMA Threshold Register (SPIDMATHR)

Read: Anytime;

Write: Anytime;

TXDMATH[2:0] — TX DMA Threshold

Specify the data number threshold of TX FIFO.Once the data number in TX FIFO is less than the threshold and bit TXDMAE in SPIDMACR is set,SPI will send TX DMA request to the DMA.

data number=TXFDMATH[2:0]

RXDMATH[2:0] — RX DMA Threshold

Specify the data number threshold of RX FIFO.Once the data number in RX FIFO is more than the threshold and bit RXDMAE in SPIDMACR is set,SPI will send RX DMA request to the DMA.

data number=RXFDMATH[2:0]+1

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10.6.2.24 SPI TX FIFO Debug Register

Addr. Offset: 0x001C Reset Value: 0x00

7	6	5	4	3	2	1	0
SPITXFGBG[7:0]							
ro							

Figure 10-25: SPI TX FIFO Debug Register (SPITXFDBGR)

Read: Anytime;

Write: has no effect;

SPITXFDBGR provides visibility into the TX FIFO for debugging purposes. The register is read-only and cannot be modified. Reading the SPITXFDBGR does not alter the state of the TX FIFO.

10.6.2.25 SPI RX FIFO Debug Register

Addr. Offset: 0x001E Reset Value: 0x00

7	6	5	4	3	2	1	0
SPIRXFGBG[7:0]							
ro							

Figure 10-26: SPI RX FIFO Debug Register (SPIRXFDBGR)

Read: Anytime;

Write: has no effect;

SPIRXFDBGR provides visibility into the RX FIFO for debugging purposes. The register is read-only and cannot be modified. Reading the SPITXRDBGR does not alter the state of the RX FIFO.

10.6.2.26 SPI Config Data Register

Addr. Offset: 0x0021 Reset Value: 0x00

15	14	13	12	11	10	9	8
SPICFGDATA [15:8]							
rw							

Addr. Offset: 0x0020 Reset Value: 0x00

7	6	5	4	3	2	1	0
SPICFGDATA [7:0]							
rw							

Figure 10-27: SPI Config Data Register (SPICFGDATAR)

Bit	Name	Reset Vaule	R/W Attri.	Description
[15:0]	SPICFGDATA	0x0	RW	SPICFGDATA 对于 SPI 配置数据是输入也是输出寄存器

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10.7 Functional Description.

The SPI module allows full-duplex, synchronous, serial communication between the MCU and peripheral devices. Software can poll the SPI status flags or SPI operation can be interrupt driven.

Setting the SPE bit in SPICR1 enables the SPI and dedicates four SPI port pins to SPI functions:

- Slave select (SS)
- Serial clock (SCK)
- Master out/slave in (MOSI)
- Master in/slave out (MISO)

When the SPE bit is clear, the SS, SCK, MOSI, and MISO pins are general-purpose I/O pins controlled by SPIDDR.

The shift register in a master SPI is linked by the MOSI and MISO pins to the shift register in the slave. The linked shift registers form a distributed register. In an SPI transmission, the SCK clock from the master shifts the data,and the master and slave exchange data. Data written to the master SPIDR register is the output data to the slave. After the exchange, data read from the master SPIDR is the input data from the slave.

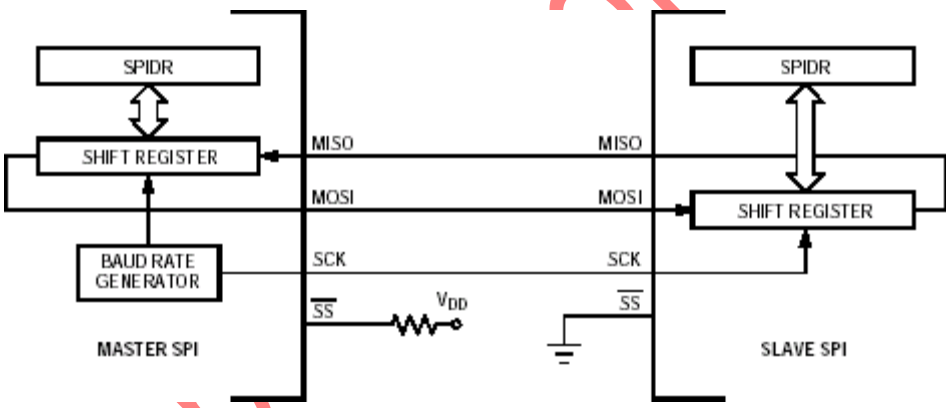


Figure 10-28: Full-Duplex Operation

10.7.1 Master Mode

Setting the MSTR bit in SPICR1 puts the SPI in master mode. Only a master SPI can initiate a transmission. Writing to the master SPIDR begins a transmission. If the shift register is empty, the byte transfers to the shift register and begins shifting out on the MOSI pin under the control of the master SCK clock. The SCK clock starts one-half SCK cycle after writing to SPIDR.

The SPR[2:0] and SPPR[6:4] bits in SPIBR control the baud rate generator and determine the speed of the shift register. The SCK pin is the SPI clock output. Through the SCK pin, the baud rate generator of the master controls the shift register of the slave.

The MSTR bit in SPICR1 and the SPC0 bit in SPICR2 control the function of the data pins, MOSI and MISO.

The SS pin is normally an input that remains in the inactive high state. Setting the DDRSP3 bit in SPIDDR configures SS as an output. The DDRSP3 bit and the SSOE bit in SPICR1 can configure SS for general-purpose I/O, mode fault detection, or slave selection. See **Table 10-5**

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The SS output goes low during each transmission and is high when the SPI is in the idle state. Driving the master SS input low sets the MODF flag in SPISR, indicating a mode fault. More than one master may be trying to drive the MOSI and SCK lines simultaneously. A mode fault clears the data direction bits of the MISO, MOSI (or MOMI), and SCK pins to make them inputs. A mode fault also clears the SPE and MSTR bits in SPICR1. If the SPIE bit is also set, the MODF flag generates an interrupt request.

10.7.2 Slave Mode

Clearing the MSTR bit in SPICR1 puts the SPI in slave mode. The SCK pin is the SPI clock input from the master, and the SS pin is the slave-select input. For a transmission to occur, the SS pin must be driven low and remain low until the transmission is complete.

The MSTR bit and the SPC0 bit in SPICR2 control the function of the data pins, MOSI and MISO. The SS input also controls the MISO pin. If SS is low, the MSB in the shift register shifts out on the MISO pin. If SS is high, the MISO pin is in a high impedance state, and the slave ignores the SCK input.

NOTE: When using peripherals with full-duplex capability, do not simultaneously enable two receivers that drive the same MISO output line.

As long as only one slave drives the master input line, it is possible for several slaves to receive the same transmission simultaneously.

If the CPHA bit in SPICR1 is clear, odd-numbered edges on the SCK input latch the data on the MOSI pin. Even-numbered edges shift the data into the LSB position of the SPI shift register and shift the MSB out to the MISO pin.

If the CPHA bit is set, even-numbered edges on the SCK input latch the data on the MOSI pin. Odd-numbered edges shift the data into the LSB position of the SPI shift register and shift the MSB out to the MISO pin.

The transmission is complete after the eighth shift. The received data transfers to SPIDR, setting the SPIF flag in SPISR.

10.7.3 FIFO Operation

When a data size of less than 16 or 8bits is selected, the user must right-justify data written to the transmit FIFO. The transmit logic ignores the unused bits. Received data less than 16 or 8 bits is automatically right-justified in the receive buffer.

10.7.3.1 Transmit FIFO

The common transmit FIFO is a 8-bit wide, 8-locations deep, first-in, first-out memory buffer. The CPU writes data to the FIFO by writing the SPI Data (SPIDR) register, and data is stored in the FIFO until it is read out by the transmission logic.

When configured as a master or a slave, parallel data is written into the transmit FIFO prior to serial conversion and transmission to the attached slave or master, respectively, through the SPI Tx pin.In slave mode, the SPI transmits data each time the master initiates a transaction. If the transmit FIFO is empty and the master initiates, the slave transmits received data in last transmit. Care should be taken to ensure that valid data is in the FIFO as needed. The SPI can be configured to generate an interrupt or a DMA request when the FIFO data number is less than the setting threshold .

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10.7.3.2 Receive FIFO

The common receive FIFO is a 8-bit wide, 8-locations deep, first-in, first-out memory buffer.Received data from the serial interface is stored in the buffer until read out by the CPU, which accesses the read FIFO by reading the SPIDR register.

When configured as a master or slave, serial data received through the SPI Rx pin is registered prior to parallel loading into the attached slave or master receive FIFO, respectively.

10.7.4 Transmission Formats

The CPHA and CPOL bits in SPICR1 select one of four combinations of serial clock phase and polarity. Clock phase and polarity must be identical for the master SPI device and the communicating slave device.

10.7.4.1 Transfer Format When CPHA = 1

Some peripherals require the first SCK edge to occur before the slave MSB becomes available at its MISO pin. When the CPHA bit is set, the master SPI waits for a synchronization delay of one-half SCK clock cycle. Then it issues the first SCK edge at the beginning of the transmission. The first edge causes the slave to transmit its MSB to the MISO pin of the master. The second edge and the following even-numbered edges latch the data. The third edge and the following

odd-numbered edges shift the latched slave data into the master shift register and shift master data out on the master MOSI pin.

After the 16th and final SCK edge:

- Data that was in the master SPIDR register is in the slave SPIDR.
- Data that was in the slave SPIDR register is in the master SPIDR.
- The SCK clock stops and the SPIF flag in SPISR is set, indicating that the transmission is complete. If the SPIE bit in SPCR1 is set, SPIF generates an interrupt request.

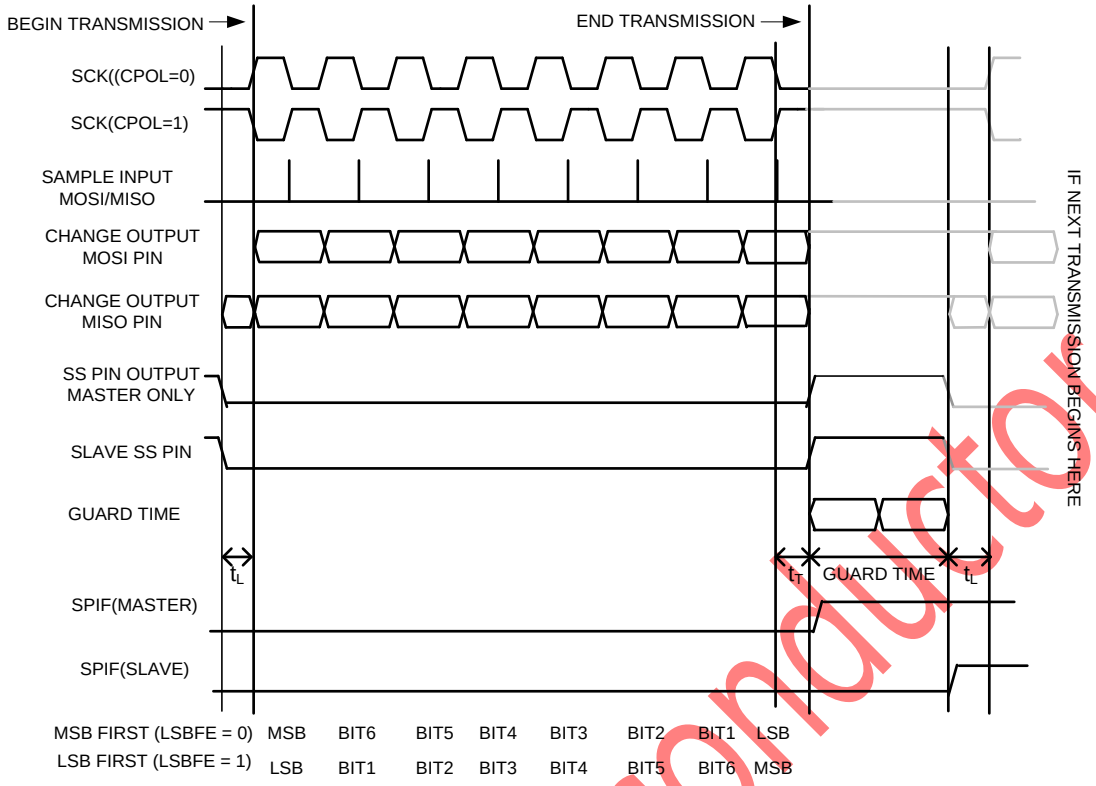
Figure 10-28 shows the timing of a transmission with the CPHA bit set. The SS pin of the master must be either high or configured as a general-purpose output not affecting the SPI.

When CPHA = 1, the slave SS line can remain low between bytes. This format is good for systems with a single master and a single slave driving the MISO data line.

The SPIF interrupt request comes at the end of each single transfer.

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Legend:

t_L = Minimum leading time before the first SCK edge

t_T = Minimum trailing time after the last SCK edge

GUARD TIME = Minimum idling time between transmissions, calculated by the formula $(GT[5:3]+1)\times 2^{(GT[2:0]+1)}$

Figure 10-29: SPI Clock Format 1 (CPHA = 1)

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10.7.4.2 Transfer Format When CPHA = 0

In some peripherals, the slave MSB is available at its MISO pin as soon as the slave is selected. When the CPHA bit is clear, the master SPI delays its first SCK edge for half a SCK cycle after the transmission starts. The first edge and all following odd-numbered edges latch the slave data. Even-numbered SCK edges shift slave data into the master shift register and shift master data out on the master MOSI pin.

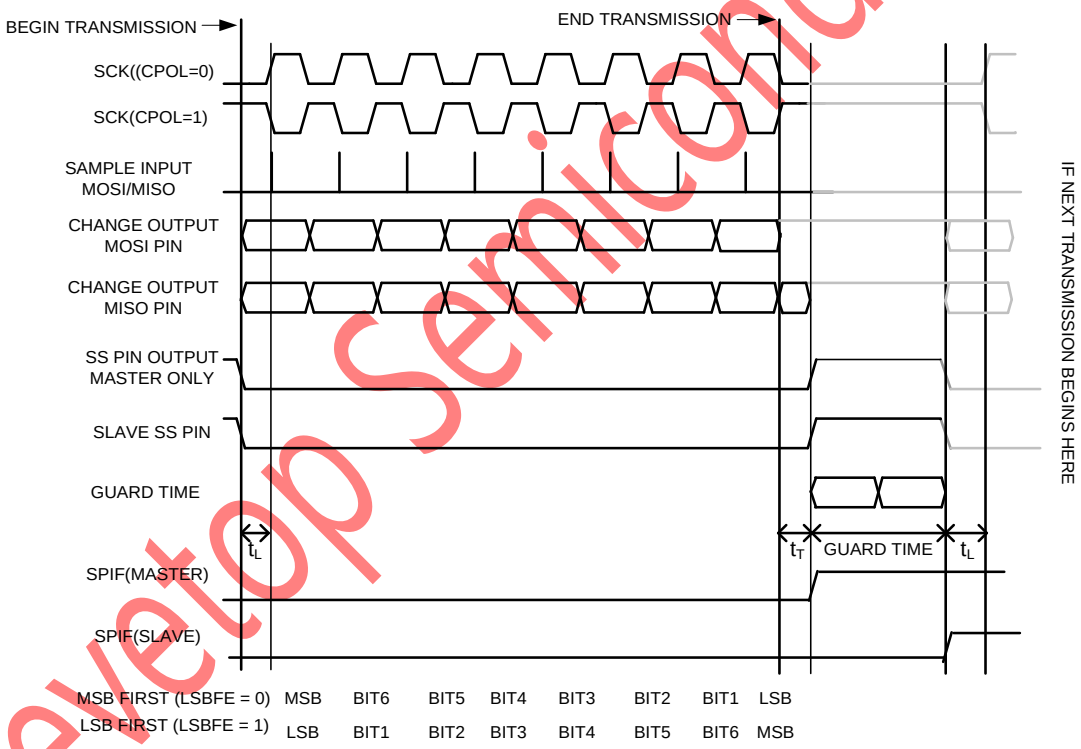
After the 16th and final SCK edge:

- Data that was in the master SPIDR is in the slave SPIDR. Data that was in the slave SPIDR is in the master SPIDR.

- The SCK clock stops and the SPIF flag in SPISR is set, indicating that the transmission is complete. If the SPIE bit in SPCR1 is set, SPIF generates an interrupt request.

Figure 10-29 shows the timing of a transmission with the CPHA bit clear. The SS pin of the master must be either high or configured as a general-purpose output not affecting the SPI.

When CPHA = 0, the slave SS pin must be negated and reasserted between bytes.



Legend:

t_L = Minimum leading time before the first SCK edge

t_T = Minimum trailing time after the last SCK edge

GUARD TIME = Minimum idling time between transmissions, calculated by the formula $(GT[5:3]+1)\times 2^{(GT[2:0]+1)}$

Figure 10-30: SPI Clock Format 0 (CPHA = 0)

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NOTE: Clock skew between the master and slave can cause data to be lost when:

- CPHA = 0, and,
- The baud rate is the SPI clock divided by two, and
- The master SCK frequency is half the slave SPI clock frequency, and
- Software writes to the slave SPIDR just before the synchronized SS signal goes low.

The synchronized SS signal is synchronized to the SPI clock. **Figure 10-30** shows an example with the synchronized SS signal almost a full SPI clock cycle late.

While the synchronized SS of the slave is high, writing is allowed even though the SS pin is already low. The write can change the MISO pin while the master is sampling the MISO line. The first bit of the transfer may not be stable when the master samples it, so the byte sent to the master may be corrupted.

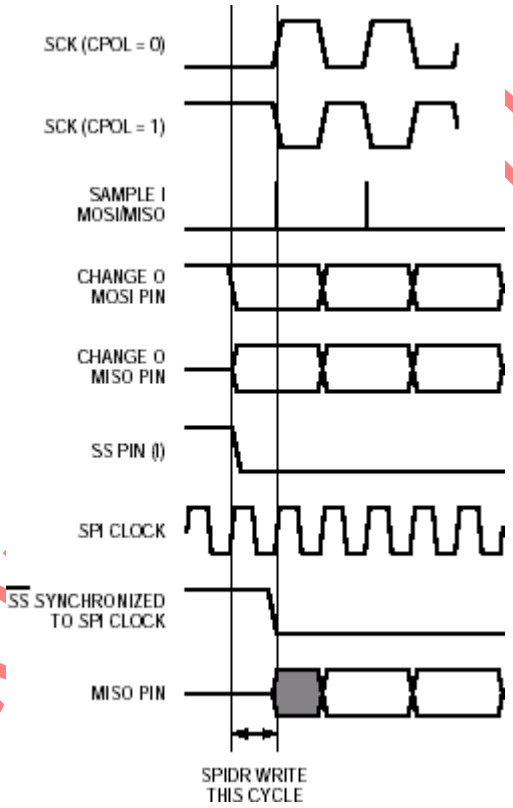


Figure 10-31: Transmission Error Due to Master/Slave Clock Skew

Also, if the slave generates a late write, its state machine may not have time to reset, causing it to incorrectly receive a byte from the master.

This error is most likely when the SCK frequency is half the slave SPI clock frequency. At other baud rates, the SCK skew is no more than one SPI clock, and there is more time between the synchronized SS signal and the first SCK edge. For example, with a SCK frequency one-fourth the slave SPI clock frequency, there are two SPI clocks between the fall of SS and the SCK edge.

As long as another late SPIDR write does not occur, the following bytes to and from the slave are correctly transmitted.

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10.7.4.3 Texas Instruments Synchronous Serial Frame Format

In this mode, SCK and SS are forced Low, and the transmit data line SPI Tx is tristated whenever the SSI is idle. Once the bottom entry of the transmit FIFO contains data, SS is pulsed High for one SCK period. The value to be transmitted is also transferred from the transmit FIFO to the serial shift register of the transmit logic. On the next rising edge of SSIClk, the MSB of the 4 to 16-bit data frame is shifted out on the SSITx pin. Likewise, the MSB of the received data is shifted onto the SSIRx pin by the off-chip serial slave device.

Both the SPI and the off-chip serial slave device then clock each data bit into their serial shifter on each falling edge of SCK. The received data is transferred from the serial shifter to the receive FIFO on the first rising edge of SCK after the LSB has been latched.

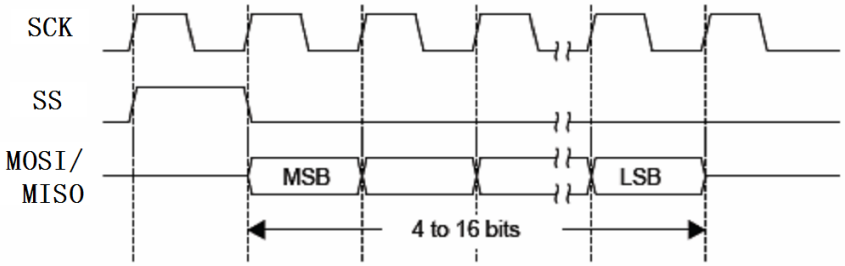


Figure 10-32: TI single tranfser

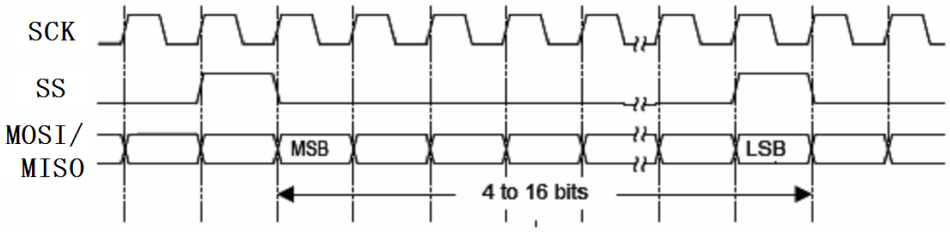


Figure 10-33: TI continuous tranfser

10.7.5 SPI Baud Rate Generation

The baud rate generator divides the SPI clock to produce the SPI baud clock. The SPPR[6:4] and SPR[2:0] bits in SPIBR select the SPI clock divisor:

$$\text{SPI clock divisor} = (\text{SPPR} + 1) \times 2^{(\text{SPR} + 1)}$$

where:

SPPR = the value written to bits SPPR[6:4]

SPR = the value written to bits SPR[2:0]

The baud rate generator is active only when the SPI is in master mode and transmitting. Otherwise, the divider is inactive to reduce IDD current.

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10.7.6 Slave-Select Output

The slave-select output feature automatically drives the SS pin low during transmission to select external devices and drives it high during idle to deselect external devices. When SS output is selected, the SS output pin is connected to the SS input pin of the external device.

In master mode only, setting the SSOE bit in SPICR1 and the DDRSP[3] bit in SPIDDR configures the SS pin as a slave-select output.

Setting the SSOE bit disables the mode fault feature.

NOTE: Be careful when using the slave-select output feature in a multimaster system. The mode fault feature is not available for detecting system errors between masters.

10.7.7 Bidirectional Mode

Setting the SPC0 bit in SPICR1 selects bidirectional mode (see **Table 10-7**). The SPI uses only one data pin for the interface with external device(s). The MSTR bit determines which pin to use. In master mode, the MOSI pin is the master out/master in pin, MOMI. In slave mode, the MISO pin is the slave out/slave in pin,

SISO. The MISO pin in master mode and MOSI pin in slave mode are general-purpose I/O pins.

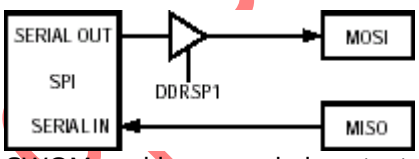
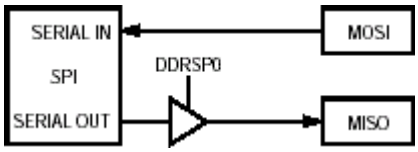
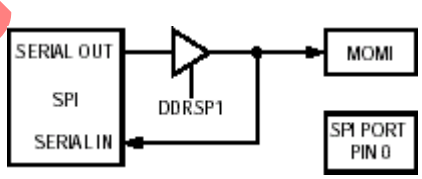
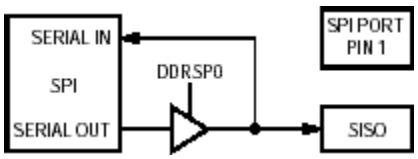
The direction of each data I/O pin depends on its data direction register bit. A pin configured as an output is the output from the shift register. A pin configured as an input is the input to the shift register, and data coming out of the shift register is discarded.

The SCK pin is an output in master mode and an input in slave mode.

The SS pin can be an input or an output in master mode, and it is always an input in slave mode.

In bidirectional mode, a mode fault does not clear DDRSP0, the data direction bit for the SISO pin.

Table 10-7: Normal Mode and Bidirectional Mode

SPE = 1	Master Mode, MSTR = 1	Slave Mode, MSTR = 0
Normal Mode SPC0 = 0	 SWOM enables open drain output.	 SWOM enables open drain output.
Bidirectional Mode SPC0 = 1	 SWOM enables open drain output. SPI port pin 0 is general-purpose I/O.	 SWOM enables open drain output. SPI port pin 1 is general-purpose I/O.

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10.7.8 DMA Operation

The SPI peripheral provides an interface to the DMA controller with separate channels for transmit and receive. The DMA operation of the SPI is enabled through the SPI DMA Control (SPIDMACR)register. When DMA operation is enabled, the SPI asserts a DMA request on the receive or transmit channel when the associated FIFO meet transfer requirements. For the receive channel, a DMA request is asserted whenever the data number in the receive FIFO is above or equal to the threshold set by SPIDMATHR or RXF timeout counter set by SPIRXFTOCTR counts down to zero. For the transmit channel, a single transfer request is asserted whenever the data number in the transmit FIFO is below or equal to the threshold set by SPIDMATHR or TXF timeout counter set by SPITXFTOCTR counts down to zero. The requests are handled automatically by the DMA controller depending how the DMA channel is configured. To enable DMA operation for the receive channel, the RXDMAE bit of the DMA Control (SPIDMACR) register should be set. To enable DMA operation for the transmit channel, the TXDMAE bit of SPIDMACR should be set.

10.7.9 High Speed Mode

In high speed mode, both the master sample later and the slave shift data out earlier in the SCK period than in normal SPI mode to allow for delays in device pads and board traces. These delays become a more significant fraction of the SCK period as the SCK period decreases with increasing baud rates.

For the master,the high speed mode is enabled by HS bit in SPIPURD register when configed as a master and the sample point delay is set by the MSPD[1:0] in SPIPURD register.

For the slave,the high speed mode is enabled by HS bit in SPIPURD register when configed as a slave. If HS is set, SPI slave will shift out data at shift-in data edge while the shift-in data timing is the same as normal mode.

For the high speed mode to operate correctly, you must thoroughly analyze the SPI link timing budget.

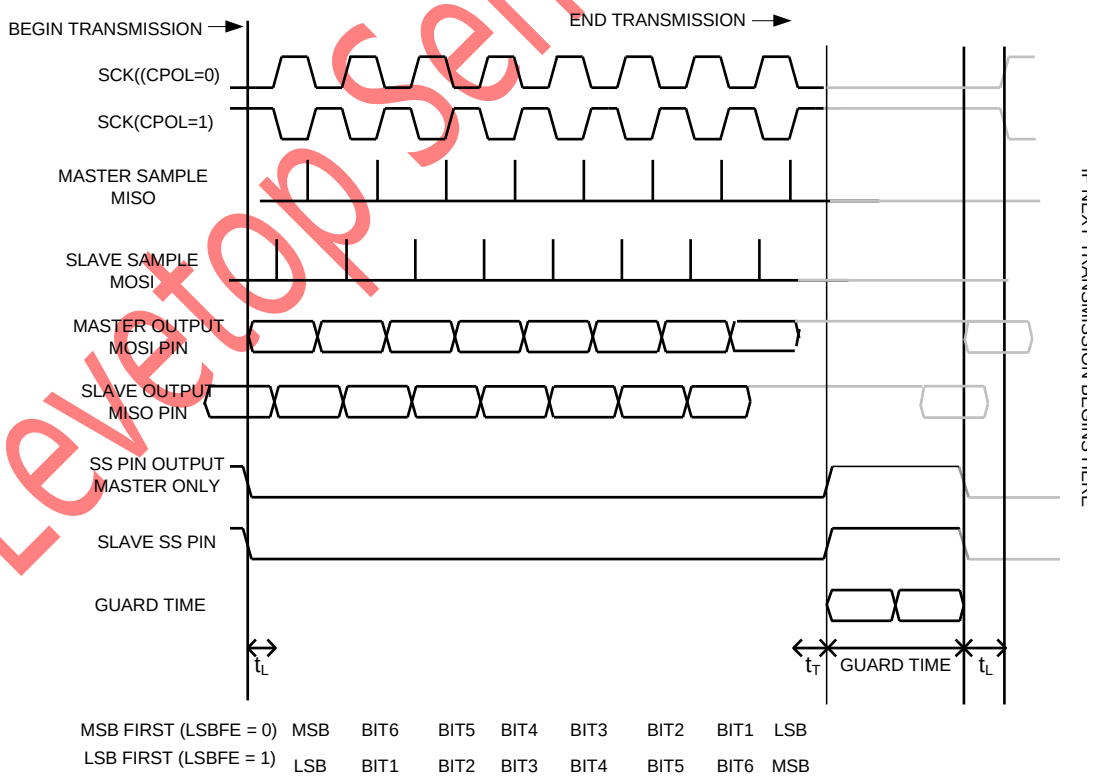


Figure 10-34: High Speed Mode (CPHA = 0)

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10.7.10 Low-Power Mode Options

This subsection describes the low-power mode options.

10.7.10.1 Run Mode

Clearing the SPE bit in SPICR1 puts the SPI in a disabled, low-power state. SPI registers are accessible, but SPI clocks are disabled.

10.7.10.2 Doze Mode

SPI operation in doze mode depends on the state of the SPISDOZ bit in SPICR2.

- If SPISDOZ is clear, the SPI operates normally in doze mode.
- If SPISDOZ is set, the SPI clock stops, and the SPI enters a low-power state in doze mode.
  - Any master transmission in progress stops at doze mode entry and resumes at doze mode exit.
  - Any slave transmission in progress continues if a master continues to drive the slave SCK pin. The slave stays synchronized to the master SCK clock.

NOTE: Although the slave shift register can receive MOSI data, it cannot transfer data to SPIDR or set the SPIF flag in doze or stop mode. If the slave enters doze mode in an idle state and exits doze mode in an idle state, SPIF remains clear and no transfer to SPIDR occurs.

10.7.10.3 Stop Mode

SPI operation in stop mode is the same as in doze mode with the SPISDOZ bit set.

10.7.11 Reset

Reset initializes the SPI registers to a known startup state as described in **10.6 Memory Map and Registers**. A transmission from a slave after reset and before writing to the SPIDR register is either indeterminate or the byte last received from the master before the reset. Reading the SPIDR after reset returns 0s.

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10.8 Interrupts

Table 10-8: SPI Interrupt Request Sources

Interrupt Request	Flag	Enable Bit
Mode fault	MODF	MODFIE
Transmission complete	EOTF	SPIE
Frame lost	FLOST	FLOSTIE
TXFIFO timeout	TXFTO	TXFTOIE
TXFIFO overflow	TXFOVF	TXFOVIE
TXFIFO underflow	TXFUDF	TXFUDIE
TXFIFO service	TXFSER	TXFSTHIE
RXFIFO timeout	RXFTO	RXFTOIE
RXFIFO overflow	RXFOVF	RXFOVIE
RXFIFO underflow	RXFUDF	RXFUDIE
RXFIFO service	RXFSER	RXFSTHIE

10.8.1 Mode Fault (MODF) Flag

MODF is set when the SS pin of a master SPI is driven low and the SS pin is configured as a mode-fault input. If the MODFIE bit is also set, MODF generates an interrupt request. A mode fault clears the SPE, MSTR, and DDRSP[2:0] bits. Clear MODF by reading SPISR with MODF set and then writing to SPICR1.

10.8.2 EOT Interrupt Flag (EOTF)

EOTF is set when all the data in TX FIFO is transmitted. If the SPIE bit is also set, EOTF generates an interrupt request. Clear EOTF by write 1 to this bit or fill TX FIFO by the CPU or DMA. Reset clears EOTF.

10.8.3 Frame Lost Interrupt Flag (FLOST)

When SPI is in slave mode and no valid data is in the TX FIFO,if the master start a transfer at this time,then SPI will return last received data to the master and FLOST is set. If the FLOSTIE bit is also set,FLOST generates an interrupt request. Clear FLOST by write 1 to this bit. Reset clears EOTF.

10.8.4 TXFIFO Timeout Interrupt Flag (TXFTO)

Once TX FIFO is not empty,the counter is on.If there is no operation to TX FIFO till the counter counts down to 0,the TXFTO is set.The counter is counted by SCK period.Clear TXFTO by write 1 to this bit.

10.8.5 TXFIFO Overflow Interrupt Flag (TXFOVF)

TXFOVF is set when data write operation is attempted to cause the data number of TX FIFO larger than 8.Clear TXFOVF by write 1 to this bit.

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10.8.6 TXFIFO Underflow Interrupt Flag (TXFUDF)

TXFUDF is set when data write operation is attempted to cause the data number of TX FIFO smaller than 0.Clear TXFUDF by write 1 to this bit.

10.8.7 TXFIFO Service Interrupt Flag (TXFSER)

TXFSER is set when data number in TX FIFO is below or equal the threshold specified in SPITXFCR.

10.8.8 RXFIFO Timeout Interrupt Flag (RXFTO)

Once RX FIFO is not empty,the counter is on.If there is no operation to RX FIFO till the counter counts down to 0,the RXFTO is set.The counter is counted by SCK period.Clear RXFTO by write 1 to this bit.

10.8.9 RXFIFO Overflow Interrupt Flag (RXFOVF)

RXFOVF is set when data write operation is attempted to cause the data number of RX FIFO larger than 8.Clear RXFOVF by write 1 to this bit.

10.8.10 RXFIFO Underflow Interrupt Flag (RXFUDF)

RXFUDF is set when data write operation is attempted to cause the data number of RX FIFO smaller than 0.Clear RXFUDF by write 1 to this bit.

10.8.11 RXFIFO Service Interrupt Flag (RXFSER)

RXFSER is set when data number in RX FIFO is above or equal the threshold specified in SPIRXFCR.

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11 Serial Communications Interface Module (SCI)

11.1 Introduction

The serial communications interface (SCI) allows asynchronous serial communications with peripheral devices and other microcontroller units (MCU).

11.2 Features

Features of each SCI module include:

- Full-duplex operation
- Standard mark/space non-return-to-zero (NRZ) format
- The baud rate divisor is a 22-bit number consisting of 16-bit integer and 6-bit fractional part
- Programmable 8-bit or 9-bit data format
- Separately enabled transmitter and receiver
- Separate receiver and transmitter central processor unit (CPU) interrupt requests
- Programmable transmitter output polarity
- Two receiver wakeup methods:
  - Idle line wakeup
  - Address mark wakeup
- Interrupt-driven operation with eight flags:
  - Transmitter empty
  - Transmission complete
  - Receiver full
  - Idle receiver input
  - Receiver overrun
  - Noise error
  - Framing error
  - Parity error
- Receiver framing error detection
- Hardware parity checking
- 1/16 bit-time noise detection
- General-purpose, I/O capability
- Serial IR interface low-speed,IrDA-compatible(up to 115.2Kbit/s)
- Separate 16x9 transmit (TX) and receive (RX) FIFOs to reduce CPU interrupt service loading
- FIFO trigger levels of 1/8, 1/4, 1/2, 3/4, and 7/8
- Efficient transfers using Direct Memory Access Controller (DMA)
- Automatic hardware flow control supported

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11.3 Block Diagram

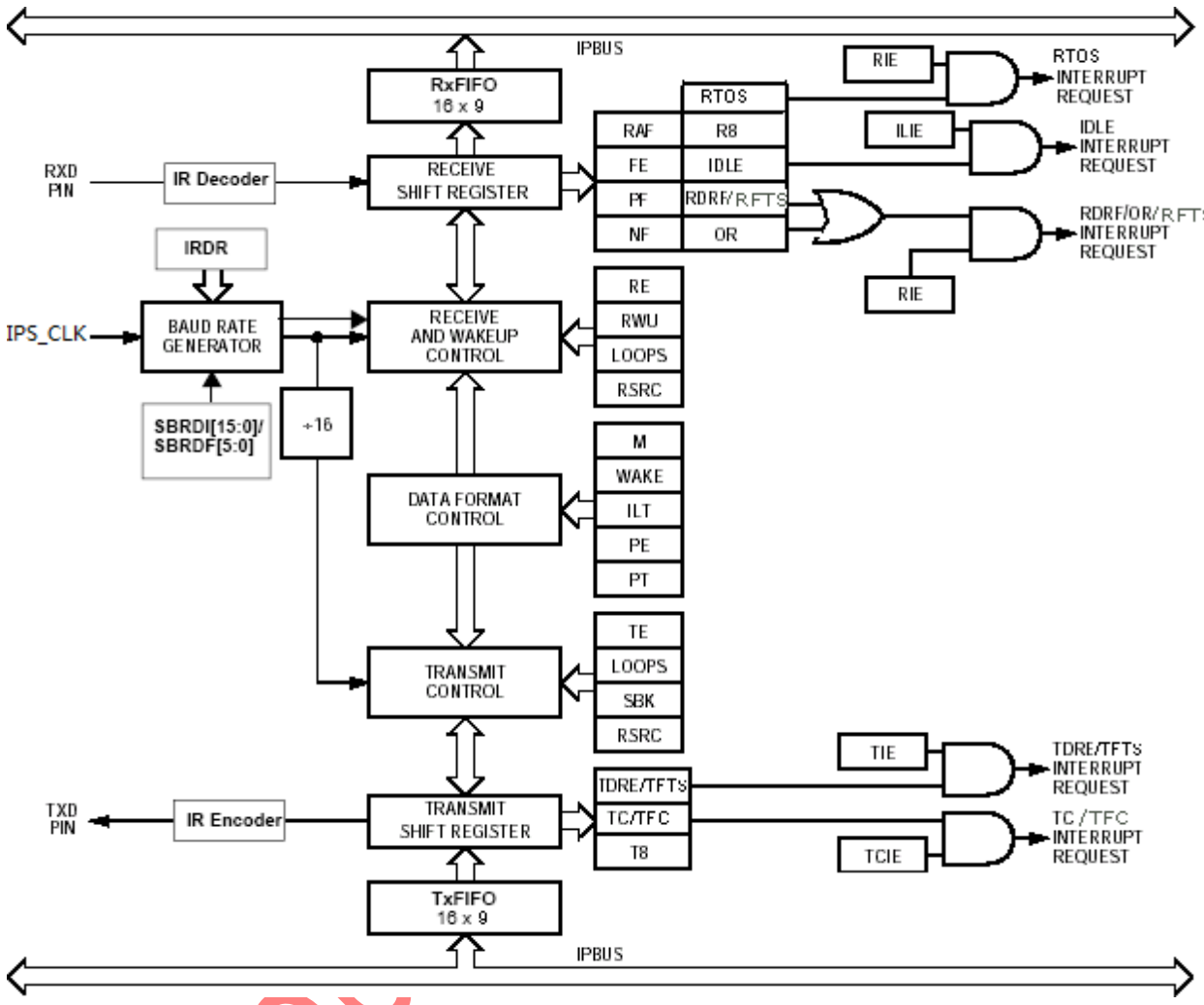


Figure 11-1: SCI Block Diagram

11.4 Modes of Operation

SCI operation is identical in run, special, and emulation modes. The SCI has two low-power modes, doze and stop.

NOTE: Run mode is the normal mode of operation and the WAIT instruction does not affect SCI operation.

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11.4.1 Doze Mode

When the SCIDOZ bit in the SCI Pullup and Reduced Drive (SCIPURD) Register is set, the DOZE instruction stops the SCI clock and puts the SCI in a low-power state. The DOZE instruction does not affect SCI register states. Any transmission or reception in progress stops at doze mode entry and resumes when an internal or external interrupt request brings the CPU out of doze mode. Exiting doze mode by reset aborts any transmission or reception in progress and resets the SCI. See

11.4.2 17.7.7 SCI Pullup and Reduced Drive Register

When the SCIDOZ bit is clear, execution of the DOZE instruction has no effect on the SCI. Normal module operation continues, allowing any SCI interrupt to bring the CPU out of doze mode.

11.4.3 Stop Mode

The STOP instruction stops the SCI clock and puts the SCI in a low-power state. The STOP instruction does not affect SCI register states. Any transmission or reception in progress halts at stop mode entry and resumes when an external interrupt request brings the CPU out of stop mode. Exiting stop mode by reset aborts any transmission or reception in progress and resets the SCI.

11.5 Signal Description

Table 11-1 gives an overview of the signals which are described here.

Table 11-1: Signal Properties

Name	Function	Port	Reset State	Default Pullup State
RXD	Receive data pin	SCIPORT0	0	Disabled
TXD	Transmit data pin	SCIPORT1	0	Disabled
RTSN	Request To Send pin	RTS	0	Disabled
CTSN	Clear To Send pin	CTS	0	Disabled

11.5.1 RXD

RXD is the SCI receiver pin. RXD is available for general-purpose I/O when it is not configured for receiver operation.

11.5.2 TXD

TXD is the SCI transmitter pin. TXD is available for general-purpose I/O when it is not configured for transmitter operation.

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11.5.3 RTSN

Request to send indicates that the SCI is ready to receive a data(when low).

11.5.4 CTSN

Clear to send blocks the data transmission at the end of the current transfer(when low).

11.6 Memory Map and Registers

11.6.1 Memory Map

Table 11-2 shows the SCI memory map.

Table 11-2: Serial Communications Interface Module Memory Map[1]

Offset Address	Bits 7–0
0x0001	SCI Baud Register High (SCIBDH)
0x0000	SCI Baud Register Low (SCIBDL)
0x0003	SCI Control Register 1 (SCICR1)
0x0002	SCI Control Register 2 (SCICR2)
0x0005	SCI Status Register 1 (SCISR1)
0x0004	SCI Status Register 2 (SCISR2)
0x0007	SCI Data Register High (SCIDRH)
0x0006	SCI Data Register Low (SCIDRL)
0x0009	SCI Pullup and Reduced Drive Register (SCIPURD)
0x0008	SCI Port Data Register (SCIPORT)
0x000b	SCI Data Direction Register (SCIDDR)
0x000a	SCI Fractional Baud-Rate Divisor Register (SCIBRDF)
0x000d	SCI Test Register (SCITR)
0x000c	SCI InfraRed Control Register (SCIIRCR)
0x000f	SCI InfraRed Divisor Register (SCIIRDR)
0x000e	SCI FIFO Control Register (SCIFCR)
0x0011	SCI FIFO Status Register (SCIFSR)
0x0010	SCI DMA Control Register (SCIDCR)
0x0013	SCI FIFO Control Register 2(SCIFCR2)
0x0012	SCI RX FIFO TimeOut Counter Register(SCIRXTOCTR)
0x0015	SCI FIFO Status Register 2(SCIFSR2)

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Offset Address	Bits 7–0
0x0014	SCI Flow Control Register

NOTE: Each module is assigned 64 Kbytes of address space, all of which may not be decoded. Accesses outside of the specified module memory map generate a bus error exception.

11.6.2 Registers Description

11.6.2.1 SCI Baud Rate Registers

Addr. Offset: 0x0001 Reset Value: 0x00

7	6	5	4	3	2	1	0
SBRDI15	SBRDI14	SBRDI13	SBRDI12	SBRDI11	SBRDI10	SBRDI9	SBRDI8
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-2:SCI Integer Baud Rate Divisor Register High (SCIBRDIH)

Addr. Offset: 0x0000 Reset Value: 0x04

7	6	5	4	3	2	1	0
SBRDI17	SBRDI6	SBRDI5	SBRDI4	SBRDI3	SBRDI2	SBRDI1	SBRDI0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-3: SCI Integer Baud Rate Divisor Register Low (SCIBRDIL)

Addr. Offset: 0x000a Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved		SBRDF5	SBRDF4	SBRDF3	SBRDF2	SBRDF1	SBRDF0
ro		rw	rw	rw	rw	rw	rw

Figure 11-4: SCI Fractional Baud Rate Divisor Register (SCIBRDF)

Read: Anytime Write: Anytime

SBRDI[15:0] — SCI Integer Baud Rate Divisor Bits

SBRDF[5:0] — SCI Fractional Baud Rate Divisor Bits

BRD = BRDI + BRDF = Fsys / (16 * UART baudrate)

SBRDI=integer(SBRD)=BRDI; SBRDF=integer(SBRDF)=BRDF;

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These read/write bits control the SCI baud rate:

UART baudrate = Fsys / (16 * SBRD)

Thereinto:

1≤SBRDI≤65535 ; 0≤SBRDF≤63

SBRD = SBRDI + (SBRDF / 64)

NOTE: The baud rate generator is disabled until the TE bit or the RE bit in SCICR2 is set for the first time after reset. The baud rate generator is disabled when SBRDI[15:0] = 0 and SBRDF[5:0] = 0.

Writing to SCIBRDH and SCIBRDF has no effect without also writing to SCIBRDL. Writing to SCIBRDH and SCIBRDF puts the data in a temporary location until data is written to SCIBRDL.

11.6.2.2 SCI Control Register 1

Addr. Offset: 0x0003 Reset Value: 0x00

7	6	5	4	3	2	1	0
LOOPS	WOMS	RSRC	M	WAKE	ILT	PE	PT
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-5: SCI Control Register 1 (SCICR1)

Read: Anytime Write: Anytime

LOOPS — Loop Select Bit

This read/write control bit switches the SCI between normal mode and loop mode. Reset clears LOOPS.

1 = Loop mode SCI operation

0 = Normal mode SCI operation

The SCI operates normally (LOOPS = 0, RSRC = X) when the output of its transmitter is connected to the TXD pin, and the input of its receiver is connected to the RXD pin.

In loop mode (LOOPS =1, RSRC = 0), the input to the SCI receiver is internally disconnected from the RXD pin logic and instead connected to the output of the SCI transmitter. The behavior of TXD is gerned by the DDRSC1 bit in SCIDDR. If DDRSC1 = 1, the TXD pin is driven with the output of the SCI transmitter. If DDRSC1 = 0, the TXD pin idles high. See **11.16 Loop Operation** for additional information. For either loop mode or single-wire mode to function, both the SCI receiver and transmitter must be enabled by setting the RE and TE bits in SCIxCR2.

NOTE: The RXD pin becomes general-purpose I/O when LOOPS = 1, regardless of the state of the RSRC bit. DDRSC0 in SCIDDR is the data direction bit for the RXD pin.

Table 11-3 shows how the LOOPS, RSRC, and DDRSC0 bits affect SCI operation and the configuration of the RXD and TXD pins.

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Table 11-3: SCI Normal, Loop, and Single-Wire Mode Pin Configurations

LOOPS	RSRC	SCI MODE	Receiver Input	RXD Pin Function	DDRSC0	Transmitter Output	TXD Pin Function
0	X	Normal	Tied to RXD input buffer	Receive pin	X	Tied to TXD output driver	Transmit pin
1	0	Loop	Tied to transmitter output	General-purpose I/O	0	Tied to receiver input only	None (idles high)
					1	Tied to receiver input and TXD output driver	Transmit pin
	1	Single-wire	Tied to TXD		0	No connection	Receive pin
					1	Tied to TXD output driver	Transmit pin

WOMS — Wired-OR Mode Select Bit

This read/write bit configures the TXD and RXD pins for open-drain operation. This allows all of the TXD pins to be tied together in a multiple-transmitter system. WOMS also affects the TXD and RXD pins when they are general-purpose outputs. External pullup resistors are necessary on open-drain outputs. Reset clears WOMS.

1 = TXD and RXD pins open-drain when outputs

0 = TXD and RXD pins CMOS drive when outputs

NOTE: This bit has no effect in this part. Thus wired-or mode is not supported.

RSRC — Receiver Source Bit

This read/write bit selects the internal feedback path to the receiver input when LOOPS = 1. Reset clears RSRC.

1 = Receiver input tied to TXD pin when LOOPS = 1

0 = Receiver input tied to transmitter output when LOOPS = 1

M — Data Format Mode Bit

This read/write bit selects 11-bit or 10-bit frames. Reset clears M.

1 = Frames have 1 start bit, 9 data bits, and 1 stop bit.

0 = Frames have 1 start bit, 8 data bits, and 1 stop bit.

WAKE — Wakeup Bit

This read/write bit selects the condition that wakes up the SCI receiver when it has been placed in a standby state by setting the RWU bit in SCICR2. When WAKE is set, a logic 1 (address mark) in the most significant bit position of a received data character wakes the receiver. An idle condition on the RXD pin does so when WAKE = 0. Reset clears WAKE.

1 = Address mark receiver wakeup

0 = Idle line receiver wakeup

ILT — Idle Line Type Bit

This read/write bit determines when the receiver starts counting logic 1s as idle character bits. The counting begins either after the start bit or after the stop bit. If the count begins after the start bit, then a string of logic 1s preceding the stop bit may cause false recognition of an idle character. Beginning the count after the stop bit avoids false idle character recognition, but requires properly synchronized transmissions. Reset clears ILT.

1 = Idle frame bit count begins after stop bit.

0 = Idle frame bit count begins after start bit.

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PE — Parity Enable Bit

This read/write bit enables the parity function. When enabled, the parity function inserts a parity bit in the most significant bit position of an SCI data word.

Reset clears PE.

1 = Parity function enabled

0 = Parity function disabled

PT — Parity Type Bit

This read/write bit selects even parity or odd parity. With even parity, an even number of 1s clears the parity bit and an odd number of 1s sets the parity bit. With odd parity, an odd number of 1s clears the parity bit and an even number of 1s sets the parity bit. Reset clears PT.

1 = Odd parity when PE = 1

0 = Even parity when PE = 1

11.6.2.3 SCI Control Register 2

Addr. Offset: 0x0002 Reset Value: 0x00

7	6	5	4	3	2	1	0
TIE	TCIE	RIE	ILIE	TE	RE	RWU	SBK
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-6: SCI Control Register 2 (SCICR2)

Read: Anytime Write: Anytime

TIE — Transmitter Interrupt Enable Bit

This read/write bit allows the TDRE/TFTS flag to generate interrupt requests. Reset clears TIE.

1 = TDRE/TFTS interrupt requests enabled

0 = TDRE/TFTS interrupt requests disabled

TCIE — Transmission Complete Interrupt Enable Bit

This read/write bit allows the TC/FTC flag to generate interrupt requests. Reset clears TCIE.

1 = TC/FTC interrupt requests enabled

0 = TC/FTC interrupt requests disabled

RIE — Receiver Interrupt Enable Bit

This read/write bit allows the RDRF/RFTS and OR/FOR and RTOS flags to generate interrupt requests. Reset clears RIE.

1 = RDRF/RFTS and OR/FOR and RTOS interrupt requests enabled

0 = RDRF/RFTS and OR/FOR and RTOS interrupt requests disabled

ILIE — Idle Line Interrupt Enable Bit

This read/write bit allows the IDLE flag to generate interrupt requests. Reset clears ILIE.

1 = IDLE interrupt requests enabled

0 = IDLE interrupt requests disabled

TE — Transmitter Enable Bit

This read/write bit enables the transmitter and configures the TXD pin as the transmitter output. Toggling TE queues an idle frame. Reset clears TE.

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1 = Transmitter enabled

0 = Transmitter disabled

RE — Receiver Enable Bit

This read/write bit enables the receiver. Reset clears RE.

1 = Receiver enabled

0 = Receiver disabled

NOTE: When LOOPS = 0 and TE = RE = 1, the RXD pin is an input and the TXD pin is an output regardless of the state of the DDRSC1 (TXD) and DDRSC0 (RXD) bits.

RWU — Receiver Wakeup Bit

This read/write bit puts the receiver in a standby state that inhibits receiver interrupt requests. The WAKE bit determines whether an idle input or an address mark wakes up the receiver and clears RWU. Reset clears RWU.

1 = Receiver asleep when RE = 1

0 = Receiver awake when RE = 1

SBK — Send Break Bit

Setting this read/write bit causes the SCI to send break frames of 10 (M = 0) or 11 (M =1) logic 0s. To send one break frame, set SBK and then clear it before the break frame is finished transmitting. As long as SBK is set, the transmitter continues to send break frames.

1 = Transmitter sends break frames.

0 = Transmitter does not send break frames.

11.6.2.4 SCI Status Register 1

Addr. Offset: 0x0005 Reset Value: 0xC0

7	6	5	4	3	2	1	0
TDRE	TC	RDRF	ILIE	OR	NF	FE	PF
ro	ro	ro	ro	ro	ro	ro	ro

Figure 11-7: SCI Status Register 1 (SCISR1)

Read: Anytime

Write: Has no meaning or effect

TDRE — Transmit Data Register Empty Flag

In character mode,The TDRE flag is set when the transmit shift register receives a word from the SCI Data Register. It signals that the SCIDRH and SCIDRL are empty and can receive new data to transmit. If the TIE bit in the SCICR2 is also set, TDRE generates an interrupt request. Clear TDRE by reading SCISR1 and then writing to SCIDRL.

In FIFO mode,The TDRE flag is set When Transmit FIFO is empty.Clear TDRE by writing data to Transmit FIFO.

Reset sets TDRE.

1 = Transmit data register empty

0 = Transmit data register not empty

TC — Transmit Complete Flag

In character mode,The TC flag is set when TDRE = 1 and no data, preamble, or break frame is being transmitted. It signals that no transmission is in progress. If the TCIE bit is set in SCICR2, TC generates an interrupt request. When TC is set, the TXD pin is idle (logic 1). TC is cleared automatically when a data, preamble, or break frame is queued. Clear TC by reading SCISR1 with TC set and then writing to SCIDRL. TC cannot be cleared while a transmission is in progress.

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In FIFO mode,The TC flag is same with character mode.

1 = No transmission in progress

0 = Transmission in progress

RDRF — Receive Data Register Full Flag

In character mode,The RDRF flag is set when the data in the receive shift register is transferred to SCIDRH and SCIDRL. It signals that the received data is available to the MCU. If the RIE bit is set in SCICR2, RDRF generates an interrupt request. Clear RDRF by reading the SCISR1 and then reading SCIDRL.

In FIFO mode,The RDRF flag is set when the data in the receive shift register,but Receive FIFO is full.Clear RDRF by reading SCIDRL then accepted next data.

Reset clears RDRF.

1 = Received data available in SCIDRH and SCIDRL

0 = Received data not available in SCIDRH and SCIDRL

IDLE — Idle Line Flag

The IDLE flag is set when 10 (if M = 0) or 11 (if M = 1) consecutive logic 1s appear on the receiver input. If the ILIE bit in SCICR2 is set, IDLE generates an interrupt request. Once IDLE is cleared, a valid frame must again set the RDRF flag before an idle condition can set the IDLE flag. Clear IDLE by reading SCISR1 and then reading SCIDRL. Reset clears IDLE.

1 = Receiver idle

0 = Receiver active or idle since reset or idle since IDLE flag last cleared

NOTE: When RWU of SCICR2 =1, an idle line condition does not set the IDLE flag.

OR — Overrun Flag

In character mode,The OR flag is set if data is not read from SCIDRL before the receive shift register receives the stop bit of the next frame. This is a receiver overrun condition. If the RIE bit in SCICR2 is set, OR generates an interrupt request. The data in the shift register is lost, but the data already in the SCIDRH and SCIDRL is not affected. Clear OR by reading SCISR1 and then reading SCIDRL.

In FIFO mode,Clear OR by reading SCIDRL. Reset clears OR.

1 = Overrun

0 = No overrun

NF — Noise Flag

In character mode,The NF flag is set when the SCI detects noise on the receiver input. NF is set during the same cycle as the RDRF flag but does not get set in the case of an overrun. Clear NF by reading SCISR1 and then reading SCIDRL.

In FIFO mode,this bit is not valid. Reset clears NF.

1 = Noise

0 = No noise

FE — Framing Error Flag

In character mode,The FE flag is set when a logic 0 is accepted as the stop bit. FE is set during the same cycle as the RDRF flag but does not get set in the case of an overrun. FE inhibits further data reception until it is cleared. Clear FE by reading SCISR1 and then reading SCIDRL.

In FIFO mode,this bit is not valid. Reset clears FE.

1 = Framing error

0 = No framing error

PF — Parity Error Flag

In character mode,The PF flag is set when PE = 1 and the parity of the received data does not match its parity bit. Clear PF by reading SCISR1 and then reading SCIDRL.

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In FIFO mode,this bit is not valid. Reset clears PF.

1 = Parity error

0 = No parity error

11.6.2.5 SCI Status Register 2

Addr. Offset: 0x0004 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved							RAF
ro							r/w1c

Figure 11-8: SCI Status Register 2 (SCISR2)

Read: Anytime

Write: Has no meaning or effect RAF — Receiver Active Flag

The RAF flag is set when the receiver detects a logic 0 during the RT1 time period of the start bit search. When the receiver detects an idle character, it clears RAF. Reset clears RAF.

1 = Reception in progress

0 = No reception in progress

11.6.2.6 SCI Data Registers

Addr. Offset: 0x0006 Reset Value: 0x00

7	6	5	4	3	2	1	0
R7/T7	R6/T6	R5/T5	R4/T4	R3/T3	R2/T2	R1/T1	R0/T0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-9: SCI Data Register Low (SCIDRL)

Addr. Offset: 0x0007 Reset Value: 0x00

7	6	5	4	3	2	1	0
R8	T8	Reserved					
ro	rw	ro					

Figure 11-10: SCI Data Register High (SCIDRH)

Read: Anytime

Write: Anytime; writing to R8 has no effect

R8 — Receive Bit 8

The R8 bit is the ninth received data bit when using the 9-bit data format (M = 1). Reset clears R8.

T8 — Transmit Bit 8

The T8 bit is the ninth transmitted data bit when using the 9-bit data format (M = 1). Reset clears T8.

R[7:0] — Receive Bits [7:0]

The R[7:0] bits are receive bits [7:0] when using the 9-bit or 8-bit data format. Reset clears R[7:0].

T[7:0] — Transmit Bits [7:0]

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The T[7:0] bits are transmit bits [7:0] when using the 9-bit or 8-bit data format. Reset clears T[7:0].

Addr. Offset: 0x0006 Reset Value: 0x00

7	6	5	4	3	2	1	0
R7/T7	R6/T6	R5/T5	R4/T4	R3/T3	R2/T2	R1/T1	R0/T0
rw	rw	rw	rw	rw	rw	rw	rw

NOTE: If the value of T8 is the same as in the previous transmission, T8 does not have to be rewritten. The same value is transmitted until T8 is rewritten.

When using the 8-bit data format, only SCIDRL needs to be accessed. When using 8-bit write instructions to transmit 9-bit data, write first to SCIDRH, then to SCIDRL.

11.6.2.7 SCI Pullup and Reduced Drive Register

Addr. Offset: 0x0009 Reset Value: 0x01

7	6	5	4	3	2	1	0
UARTDOZ	Reserved	RSVD[2:1]		Reserved		RSVD[0]	PUPUART
rw	ro	rw		ro		rw	rw

Figure 11-11: SCI Pullup and Reduced Drive Register (SCIPURD)

Write: Anytime

SCISDOZ — SCI Stop in Doze Mode Bit

The SCISDOZ bit disables the SCI in doze mode.

1 = SCI disabled in doze mode

0 = SCI enabled in doze mode

RSVD[6:1] — Reserved

Writing to these read/write bits updates their values but has no effect on functionality.

PUPSCI — Pullup Enable Bit

This read/write bit enables the pullups on pins TXD and RXD. If a pin is programmed as an output, the pullup is disabled.

1 = TXD and RXD pullups enabled

0 = TXD and RXD pullups disabled

NOTE: The PUPSCI bit has no effect in ths part. Pullups are always enabled.

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11.6.2.8 SCI Port Data Register

Addr. Offset: 0x0008 Reset Value: 0x03

7	6	5	4	3	2	1	0
RSVD[5:0]						PORTSC1	PORTSC0
rw						rw	rw

Figure 11-12: SCI Port Data Register (SCIPORT)

Read: Anytime; when DDRSCx = 0, its pin is configured as an input, and reading PORTSCx returns the pin level; when DDRSCx = 1, its pin is configured as an output, and reading PORTSCx returns the pin driver output level.

Write: Anytime; data stored in internal latch drives pin only if DDRSC bit = 1

RSVD[7:2] — Reserved

Writing to these read/write bits updates their values but has no effect on functionality.

PORTSC[1:0] — SCIPORT Data Bits

These are the read/write data bits of the SCI port.

NOTE: Writes to SCIPORT do not change the pin state when the pin is configured for SCI input.

To ensure correct reading of the SCI pin values from SCIPORT, always wait at least one cycle after writing to SCIDDR before reading SCIPORT.

11.6.2.9 SCI Data Direction Register

Addr. Offset: 0x000B Reset Value: 0x00

7	6	5	4	3	2	1	0
RSVD[5:0]						DDRSC1	DDRSC0
rw						rw	rw

Figure 11-13: SCI Data Direction Register (SCIDDR)

Read: Anytime Write: Anytime

RSVD[7:2] — Reserved

Writing to these read/write bits updates their values but has no effect on functionality.

DDRSC[1:0] — SCIPORT Data Direction Bits

These bits control the data direction of the SCIPORT pins. Reset clears DDRSC[1:0].

1 = Corresponding pin configured as output

0 = Corresponding pin configured as input

NOTE: When LOOPS = 0 and TE = RE = 1, the RXD pin is an input and the TXD pin is an output regardless of the state of the DDRSC1 (TXD) and DDRSC0 (RXD) bits.

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11.6.2.10 SCI Test Register

Addr. Offset: 0x000D Reset Value: 0x00

7	6	5	4	3	2	1	0
RSVD[1:0]		Reserved					
rw		ro					

Figure 11-14: SCI Test Register (SCITR)

The register is reserved for factory testing.

11.6.2.11 SCI InfraRed Control Register

Addr. Offset: 0x000C Reset Value: 0x90

7	6	5	4	3	2	1	0
TNUM		RNUM		TINV	RINV	IRSC	IREN
rw		rw		rw	rw	rw	rw

Figure 11-15: SCI Infrared Control Register (SCIIRCR)

Read: Anytime Write: Anytime

IREN — InfraRed Interface Enable Bit

This read/write control bit enable/disable the IR interface. Reset clears IREN.

1 = IR interface enable, 0 = IR interface disable

IRSC — InfraRed Interface Sampling Clock Select Bit

This read/write control bit select the sampling clock of the IR interface. Reset clears IRSC.

1 = IR interface use the divided clock of system clk(according to SCIIRDR)

0 = IR interface use the baudrate sampling clock(16xbaudrate)

NOTE: The value to apply to IRSC bit is based on 2 parameters: the baud rate and the Minimum Pulse Duration(MPD) of the transceiver. According to IrDA Standard Specification, for SIR (Serial IR) baud rates from 2.4 Kbit/s to 115.2 Kbit/s this nominal pulse duration is equal to 3/16 of a bit duration (at the selected baud rate). But, for all the baud rates a Minimum Pulse Duration is also specified. According to IrDA Standard,a Zero is represented by a light pulse, so the IrDA transceiver can't emit a light pulse shorter than the MPD.For SIR, the MPD is constant and equal to 1.41 µs.This means at any time, the user must insure the frequency of IR sampling clock is high enough to measure the pulse.

In normal operation, IRSC=0, the pulse must last at least 1 baudrate sampling clock cycle(IR sample clock cycle num decided by RNUM).If this condition is not fulfilled, IRSC must be set to 1.

RINV — Inverted Infrared Reception Bit

This read/write control bit determines the logic level for the detection. Reset clears RINV.

1 = Active high detection.The infrared logic block expects an active high or positive IR 3/16 pulse for 0's and active low are expected for 1's.

0 = Active low detection.The infrared logic block expects an active low or negative IR 3/16 pulse for 0's and active high are expected for 1's.

TINV — Inverted Infrared Transmission Bit

This read/write control bit sets the active level for the transmission. Reset clears TINV.

1 = Active low transmission.The infrared logic block transmits an active low or negative IR 3/16 pulse for all 0's and active high are transmitted for 1's.

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0 = Active high transmission.The infrared logic block transmits an active high or positive IR 3/16 pulse for all 0's and active low are transmitted for 1's.

RNUM[1:0] — Reception Number Bit

These read/write control bits determines the reception sample numbers for vote logic. Reset value is 2'b01.

Table 11-4: Reception Number Bit

RNUM[1:0]	Sample NUM
00	1 times
01	2 times
10	3 times
11	4 times

NOTE: The IR sampling clock is determined by IRSC.

When set RNUM[1:0]=11,users should set IRSC=1,because according to IrDA spec, 1.41us<MPD<3/16 of a bit duration.If IRSC=0,select BS_CLOCK to sample pulse for 4 times,it's impossible.

TNUM[1:0] — Transmission Number Bit

This read/write control bit determines the transmission clock cycle numbers for generate pulse. Reset value is 2'b10.

Table 11-5: Transmission Number Bit

TNUM[1:0]	Trans Time
00	1/16 baudrate
01	2/16 baudrate
10	3/16 baudrate
11	reserved

11.6.2.12 SCI InfraRed Divisor Register

Addr. Offset: 0x000F Reset Value: 0x00

7	6	5	4	3	2	1	0
IRDR[7:0]							
rw							

Figure 11-16: SCI InfraRed Divisor Register (SCIIRDR)

Read: Anytime Write: Anytime

IRDR[7:0] — SCI InfraRed Divisor Bits

Irclk = fsys / IRDR

NOTE: when the IrDA transceiver emit a light pulse which is shorter than the SCI baudrate sampling clock,the user must insure the frequency of IR sampling clock is high enough to measure the pulse. The user should set IRSC to 1,and set IRDR to get high frequency IR sampling clock which is divided by system clock .

Let's take 2 examples, with the Minimum Pulse Duration equals to the MPD of the IrDA specification (in SIR).

Example 1. Calculation of Baudrate Sampling Clock Period (BS_clock Period < 1.41us)

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The user wants to receive IrDA data at 115.2 Kbit/s. The SCIBRDI and SCIBRDF registers are set in order to create the baudrate sampling clock(BS_clock) with a frequency of 16*baud rate = 16 * 115.2 = 1.843 MHz. But at the same time,in order to correctly detect the pulse, the user must be sure that N* BS_clock period is lower than 1.41 us.(N is decided by RNUM)

Lets check:

BS_clock period = 1/1843000 = 542 ns

So 2*BS_clock period = 1.09 us < 1.41 us. It is fine.N can be 1 or 2.RNUM can be set 2'b00 or 2'b01.

Example 2. Calculation of Baudrate Sampling Clock Period (BS_clock Period >

1.41 us)

This time the user wants to receive at 19.2 Kbit/s. So, the BS_clock is set to 16*19200 = 307.2 kHz.

Let's check if N* BS_clock period < 1.41 us:

BS_clock period =1/307200 = 3.25 us,

So N*BS_clock period >> 1.41 us. It doesn't work.

So, in this case, the BS_clock can't be used to measure the pulse duration and the user must select the IR sampling clock by setting IRSC =1.

When system clock frequency is 32MHz,

SYS_clock period = 1/32000000 =0.03125us=IR_clock period/IRDR;

So when N*IR_clock period<1.41us(N<=4),only when IRDR<1.41/(0.03125*N)

when N=1,IRDR set to 8'h2d;IR_clock period=1.406us<1.41us;(N can only be 1)

when N=2,IRDR set to 8'h16;2*IR_clock period=1.375us<1.41us;(N can be 1 or 2)

when N=3,IRDR set to 8'h0f;3*IR_clock period=1.406us<1.41us;(N can be 1 or 2 or 3)

when N=4,IRDR set to 8'h0b;4*IR_clock period=1.375us<1.41us;(N can be 1 or 2 or 3 or 4)

Usually we set RNUM=2'b01(N=2) to sampling pulse.

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11.6.2.13 SCI FIFO Control Register

Addr. Offset: 0x000E Reset Value: 0x12

7	6	5	4	3	2	1	0
RFEN	TFEN	RXFLSEL			TXFLSEL		
rw	rw	rw			rw		

Figure 11-17: SCI FIFO Control Register (SCIFCR)

Read: Anytime Write: Anytime

RFEN — SCI Receive Enable FIFOs

1 = The Receive FIFO buffers are enabled (FIFO mode).

0 = The Receive FIFO are disabled (Character mode).

TFEN — SCI Transmit Enable FIFOs

1 = The Transmit FIFO buffers are enabled (FIFO mode).

0 = The Transmit FIFO are disabled (Character mode).

RXFLSEL[2:0] — SCI Receive FIFO Level Select

The trigger points for the receive FIFO are as follows:

Table 11-6: SCI Receive FIFO Level Select

Number	FIFO Level Select
0x0	RX FIFO ≥ ⅛full
0x1	RX FIFO ≥ ¼ full
0x2	RX FIFO ≥ ½ full (default)
0x3	RX FIFO ≥ ¾ full
0x4	RX FIFO ≥ ⅞full
0x5-0x7	Reserved

TXFLSEL[2:0] — SCI Transmit FIFO Level Select The trigger points for the Transmit are as follows:

Table 11-7: SCI Transmit FIFO Level Select

Number	FIFO Level Select
0x0	TX FIFO ≤ ⅞empty
0x1	TX FIFO ≤ ¾ empty
0x2	TX FIFO ≤ ½ empty (default)
0x3	TX FIFO ≤ ¼ empty
0x4	TX FIFO ≤ ⅛empty
0x5	Reserved

NOTE: The SCIFCR register is the FIFO level select register. You can use this register to define the FIFO level at which the TFTS and RFTS bits in the SCIFSR are triggered.The interrupts are generated based on a transition through a level rather than being based on the level. That is, the interrupts are generated when the fill level progresses through the trigger level.For example, if the receive trigger level is set to the half-way mark, the interrupt is triggered as the module is receiving the 8th character.
Out of reset, the TXFLSEL and RXFLSEL bits are configured so that the FIFOs trigger an interrupt at the half-way mark.

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11.6.2.14 SCI FIFO Status Register

Addr. Offset: 0x0011 Reset Value: 0xC5

7	6	5	4	3	2	1	0
TFTS	FTC	RFTS	RTOS	T_FULL	T_EMPTY	R_FULL	R_EMPTY
ro	ro	ro	ro	ro	ro	ro	ro

Figure 11-18: SCI FIFO Status Register (SCIFSR)

Read: Anytime

TFTS — SCI Transmit FIFO Trigger Status

The Transmit FIFO level has passed through the condition defined in the FCR register.If the TIE bit in the SCICR2 is also set, TFTS generates an interrupt request.

1 = SCI Transmit FIFO has passed through the FIFO level,also means that SCI Transmit FIFO level has been less than the FIFO level.

0 = SCI Transmit FIFO has no passed through the FIFO level

FTC — SCI FIFO mode Transmit Complete Flag

The bit is same defined whith TC in the SCISR1 register.

1 = No transmission in progress

0 = transmission in progress

RFTS — SCI Receive FIFO Trigger Status

The receive FIFO level has passed through the condition defined in the FCR register.If the RIE bit in the SCICR2 is also set, RFTS generates an interrupt request.

1 = SCI Receive FIFO has passed through the FIFO level,also means that SCI Receive FIFO level has been more than the FIFO level.

0 = SCI Receive FIFO has no passed through the FIFO level

RTOS — SCI Receive Time-Out Status

The RTOS is valid only when the receive FIFO is not empty.

1 = A receive time out has occurred, 0 = No effect

T_FULL — SCI Transmit FIFO Full

The meaning of this bit depends on the state of the FEN bit.

1 = If the FIFO is disabled (FEN is 0), the bit is not valid.

If the FIFO is enabled (FEN is 1), the transmit FIFO is full.

0 = The transmitter FIFO is not full.

T_EMPTY — SCI Transmit FIFO Empty

The meaning of this bit depends on the state of the FEN bit.

1 = If the FIFO is disabled (FEN is 0),the bit is not valid.

If the FIFO is enabled (FEN is 1), the transmit FIFO is empty.

0 = The transmitter FIFO has data to transmit.

R_FULL — SCI Receive FIFO Full

The meaning of this bit depends on the state of the FEN bit..

1 = If the FIFO is disabled (FEN is 0), the bit is not valid.

If the FIFO is enabled (FEN is 1), the receive FIFO is full.

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0 = The receiver FIFO can receive data.

R_EMPTY — SCI Receive FIFO Empty

The meaning of this bit depends on the state of the FEN bit.

1 = If the FIFO is disabled (FEN is 0), the bit is not valid.

If the FIFO is enabled (FEN is 1), the receive FIFO is empty.

0 = The receiver FIFO is not empty.

11.6.2.15 SCI DMA Control Register

Addr. Offset: 0x0010 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved						TXDMAE	RXDMAE
ro						rw	rw

Figure 11-19: SCI DMA Control Register (SCIDCR)

Read: Anytime Write: Anytime

TXDMAE — SCI Transmit DMA Enable

1 = DMA for the transmit FIFO is enabled.

0 = DMA for the transmit FIFO is disabled.

RXDMAE — SCI Receive DMA Enable

1 = DMA for the receive FIFO is enabled.

0 = DMA for the receive FIFO is disabled.

11.6.2.16 SCI FIFO Control Register 2

Addr. Offset: 0x0013 Reset Value: 0x04

7	6	5	4	3	2	1	0
TXFIE	TXFCIE	RXFIE	RXORIE	RXFTOIE	RXFTOE	TXFCLR	RXFCLR
rw	rw	rw	rw	rw	rw	rw	rw

Figure 11-20: SCI FIFO Control Register 2(SCIFCR2)

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Read: Anytime Write: Anytime

TXFIE — Transmit FIFO Interrupt Enable Bit

This read/write bit allows the TFTS flag to generate interrupt requests.Reset clears TXFIE.

1 = TFTS interrupt requests enabled

0 = TFTS interrupt requests disabled

TXFCIE — Transmit FIFO Complete Interrupt Enable Bit

This read/write bit allows the FTC flag to generate interrupt requests. Reset clears TXFCIE.

1 = FTC interrupt requests enabled

0 = FTC interrupt requests disabled

RXFIE — Receive FIFO Interrupt Enable Bit

This read/write bit allows the RFTS flag to generate interrupt requests. Reset clears RXFIE.

1 = RFTS interrupt requests enabled

0 = RFTS interrupt requests disabled

RXORIE — Receive FIFO overrun Interrupt Enable Bit

This read/write bit allows the RFOR flags to generate interrupt requests. Reset clears RXORIE.

1 = RFOR nterrupt requests enabled

0 = RFOR interrupt requests disabled

RXFTOIE — RX FIFO TimeOut Interrupt Enable

This read/write bit allows the RTOS flags to generate interrupt requests. Reset clears RXFTOIE.

1 = RX FIFO TimeOut Interrupt Enabled

0 = RX FIFO TimeOut Interrupt Disabled

RXFTOE — RX FIFO TimeOut Function Enable

1 = RX FIFO TimeOut Function Enabled

0 = RX FIFO TimeOut Function Disabled

TXFCLR — TX FIFO Clear

Write 1 to the bit will reset TX FIFO,read always is 0

RXFCLR — RX FIFO Clear

Write 1 to the bit will reset RX FIFO,read always is 0

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11.6.2.17 SCI RX FIFO TimeOut Counter Register

Addr. Offset: 0x0012 Reset Value: 0x28

7	6	5	4	3	2	1	0
UARTRFTOCRT							
rw							

Figure 11-21: SCI RX FIFO TimeOut Counter Register (SCIRXFTOCTR)

Read: Anytime Write: Anytime

SCIRXFTOCTR[7:0] sets the SCI RX FIFO Timeout counter number.Once RX

FIFO is not empty,the internal counter will decrease once in one bit time with the value .If there is no operation to RX FIFO till the counter counts down to 0,the RTOS flag in SCIFSR1 will set.

11.6.2.18 SCI FIFO Status Register 2

Addr. Offset: 0x0015 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved				FXOR	FXNF	FXFE	FXPF
ro				r/w1o	r/w1o	r/w1o	r/w1o

Figure 11-22: SCI FIFO Status Register 2(SCIFSR2)

Read: Anytime

Write: Has no meaning or effect FXOR — FIFO mode Overrun Flag

In FIFO mode,The FXOR flag is set when the data in the receive shift register,but Receive FIFO is full.Clear FXOR by reading SCIDRL then Write 1 to this bit.If the RXORIE bit in the SCIFCR2 is also set, FXOR generates an interrupt request.

1 = FIFO mode Overrun

0 = FIFO mode No overrun

FXNF — FIFO mode Noise Flag

In FIFO mode,When Reading data from Receving FIFO,The FXNF flag is set when the SCI detects noise on the receiver input.Clear FXNF by reading SCIDRL then Write 1 to this bit.

1 = FIFO mode Noise

0 = FIFO mode no Noise

FXFE — FIFO mode Framing Error Flag

In FIFO mode,When Reading data from Receving FIFO,The FXFE flag is set when a logic 0 is accepted as the stop bit.Clear FXFE by reading SCIDRL then Write 1 to this bit.

1 = FIFO mode Framing error

0 = FIFO mode no Framing error

FXPF — FIFO mode Parity Error Flag

In FIFO mode,When Reading data from Receving FIFO,The FXPF flag is set when PE = 1 and the parity of the received data does not match its parity bit.Clear FXPF by reading SCIDRL then Write 1 to this bit.

1 = FIFO mode Framing error

0 = FIFO mode no Framing error

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11.6.2.19 SCI Flow Control Register

Addr. Offset: 0x0014 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved				RTSE	CTSE	CTSIE	CTSIS
ro				rw	rw	rw	r/w0c

Figure 11-23: SCI Flow Control Register (SCIFCTRLR)

Read: Anytime Write: Anytime

RTSE — SCI RTS Enable

1 = SCI RTS is enabled, 0 = SCI RTS is disabled.

CTSE — SCI CTS Enable

1 = SCI CTS is enabled, 0 = SCI CTS is disabled.

CTSIE — SCI CTS Interrupt Enable

1 = SCI CTS Interrupt is enabled.

0 = SCI CTS Interrupt is disabled.

CTSIS — SCI CTS Interrupt Status

SCI CTS Interrupt Status is used to indicate CTS toggle status. This bit is cleared by writing with this bit low.

1 = CTS Interrupt effective, means CTS has toggled.

0 = CTS Interrupt not effective.

11.7 Functional Description

The SCI allows full-duplex, asynchronous, non-return-to-zero (NRZ)serial communication between the MCU and remote devices, includingother MCUs. The SCI transmitter and receiver operate independently,although they use the same baud rate generator. The CPU monitors thestatus of the SCI, writes the data to be transmitted, and processesreceived data.

The SCI module also includes a serial IR (SIR) encoder/decoder block that can be connected to an infrared transceiver to implement an IrDA SIR physical layer.

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11.7.1 Data Format

The SCI uses the standard NRZ mark/space data format shown in **Figure 17-24**

Each frame has a start bit, eight or nine data bits, and one or two stopbits. Clearing the M bit in SCCR1 configures the SCI for 10-bit frames.Setting the M bit configures the SCI for 11-bit frames.

When the SCI is configured for 9-bit data, the ninth data bit is the T8 bit in SCI Data Register high (SCIDRH). It remains unchanged after transmission and can be used repeatedly without rewriting it. A frame with nine data bits has a total of 11 bits.

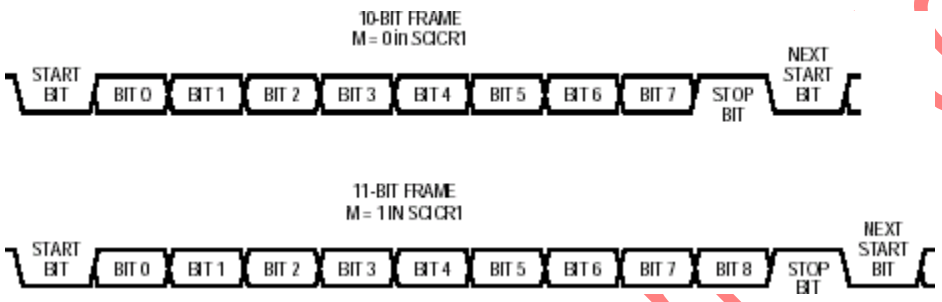


Figure 11-24: SCI Data Formats

11.7.2 Serial IR (SIR)

The SCI includes an IrDA serial-IR (SIR) encoder/decoder block. The IrDA SIR block provides functionality that converts between an asynchronous data stream and a half-duplex serial SIR interface. No analog processing is performed on-chip. The role of the SIR block is to provide a digital encoded output and decoded input to the SCI. When enabled(IREN=1), the SIR block uses the Tx and Rx pins for the

SIR protocol. These signals should be connected to an infrared transceiver to implement an IrDA SIR physical layer link. The SIR block can receive and transmit, but it is only half-duplex so it cannot do both at the same time. Transmission must be stopped before data can be received. (The IrDA SIR physical layer specifies a minimum 10-ms delay between transmission and reception.)The Infrared Interface is compatible with IrDA Serial Infrared Physical Layer Specification. In this specification, a “zero” is represented by a positive pulse, and a “one” is represented by no pulse (line remains low).

In the SCI:

In TX: For each “zero” to be transmitted, a narrow positive pulse which is 1/16~3/16 of a bit time is generated(due to TNUM).For each “one” to be transmitted no pulse is generated (output is low). External circuitry has to be provided to drive an Infrared LED.

In RX: When receiving, a narrow negative pulse is expected for each “zero” received while no pulse is expected for each “one” received (input is high).

The values of TINV and RINV depend of the IrDA transceiver connected on the TXD and RXD pins of the SCI.If this transceiver is not inverting on both paths Tx and Rx, a Zero is represented by a positive pulse and a One is represented by no pulse (line remains low).In this case, the bit TINV must be set to 0 and the bit RINV must be set to 1 (because Rx IR block expects an inverted signal).

On the contrary user must set TINV=1 and RINV=0 if both paths of the transceiver are inverting, that is,a Zero is represented as a negative pulse and a One is represented by no pulse (line remains high). The transceiver can also be inverting on only one path (Tx or Rx), in this case TINV and RINV must be together equal to 1 or to 0, depending on which path is inverted.

In **Figure 11-25**, TINV=0, RINV=0.
While if RINV=1,When receiving, a narrow positive pulse is expected for each “zero” transmitted while no

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pulse is expected for each “one” transmitted (input is low).

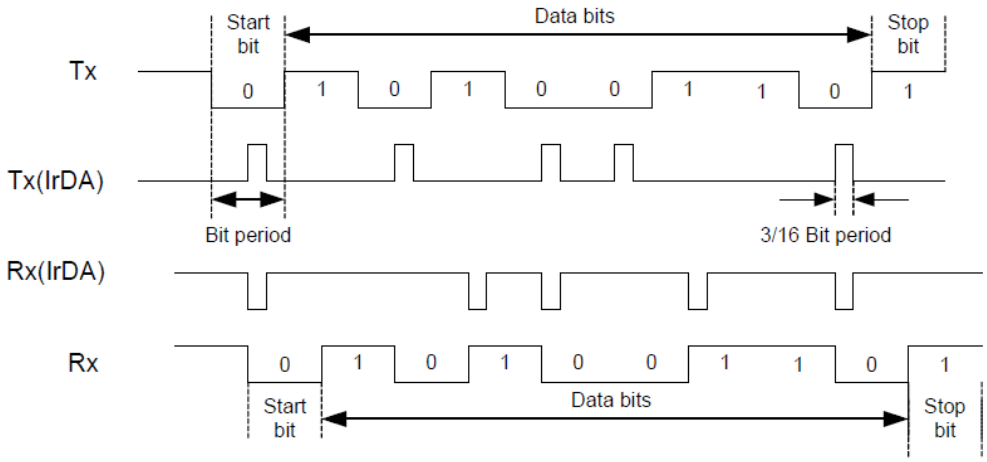


Figure 11-25: IrDA Data Modulation

11.7.3 FIFO Operation

The UART has two 16-entry FIFOs; one for transmit and one for receive. Both FIFOs are accessed via the UART Data (SCIDR) register.Read operations of the SCIDR register return a 12-bit value consisting of 9 data bits and 3 error flags while write operations place 9-bit data in the transmit FIFO.

The Transmit FIFO are enabled by setting the TFEN bit in FCR. The Receive FIFO are enabled by setting the RFEN bit in FCR.

FIFO status can be monitored via the FSR register and the SCISR2 register. Hardware monitors empty, full and overrun conditions. The FSR register contains empty and full flags (T_EMPTY, T_FULL, R_EMPTY, and R_FULL bits), and the

SCISR2 register shows overrun status via the FOR bit. The FSR register contains TFTS and RFTS flags.

The trigger points at which the FIFOs generate interrupts is controlled via the FCR register . Both FIFOs can be individually configured to trigger interrupts at different levels. Available configurations include ⅛ ¼, ½, ¾, and ⅞ For example,if the ¼ option is selected for the receive FIFO, the UART generates a receive interrupt after 4 data bytes are received. Out of reset, both FIFOs are configured to trigger an interrupt at the ½ mark.

The FSR register also contains RTOS flag.

A FIFO timeout(RTOS) will occur when at least one character is in the receiver FIFO, no serial characters have been received for four serial character time periods, and the processor has not read the FIFO for four serial character time periods. A serial character time period is:

1/(baud rate) x (# start bits + word length + # parity bits + # stop bits)

For example, the serial character time period for an 8-bit word with one parity bit and two stop bits at 56 K baud is:

1/(56000) x (1 + 8 + 1 + 2) = 214.3 us

The timeout would occur after 857.1 us,

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11.7.4 **Baud Rate** Generation

The baud rate divisor is a 22-bit number consisting of 16-bit integer and 6-bit frac- tional part.The number formed by these two values is used by the baud-rate gen- erator to derives the baud rate for both the receiver and the transmitter.The value written to SCIBRDIH,SCIBRDIL and SCIBRDF determines the system clock divi- sor. The baud rate clock is synchronized with the bus clock and drives the receiver. The baud rate clock divided by 16 drives the transmitter. The receiver acquisition rate is 16 samples per bit time.

Baud rate generation is subject to two sources of error:

- Integer division and fractional division of the module clock may not give the exact target frequency
- Synchronization with the bus clock can cause phase shift.

Table 11-8: Example Baud Rates (System Clock = 31 MHz)

SBRDI[15:0]	SBRDF[5:0]	Receiver Clock (Hz)	Transmitter Clock (Hz)	Target Baud Rate	Percent Error
0x44cd	0x29	1,760.00	110	110	0.00002%
0x193a	0x15	4,800.00	300.0002	300	0.00008%
0x0c9d	0x0b	9,599.98	599.999	600	0.00016%
0x064e	0x25	19,200.06	1,200.004	1200	0.00032%
0x0327	0x13	38,399.75	2,399.985	2400	0.00065%
0x0193	0x29	76,800.99	4,800.062	4800	0.00129%
0x00c9	0x35	153,596.04	9,599.752	9,600	0.00258%
0x0086	0x23	230,402.97	14,400.19	14,400	0.00129%
0x0064	0x3a	307,215.86	19,200.99	19,200	0.00516%
0x0032	0x1d	614,431.71	38,401.98	38,400	0.00516%
0x0022	0x26	896,115.63	56,007.23	56,000	0.01290%
0x0021	0x29	921,504.88	57,594.05	57,600	0.01032%
0x0010	0x34	1,843,866.17	115,241.6	115,200	0.03614%
0x000f	0x09	2,047,471.62	127,967	128,000	0.02580%
0x0008	0x1a	3,687,732.34	230,483.3	230,400	0.03614%
0x0004	0x000d	7,375,464.68	460,966.5	460,800	0.03614%
0x0001	0x3c	16,000,000.00	1,000,000	1,000,000	0.00000%

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11.7.5 Transmitter

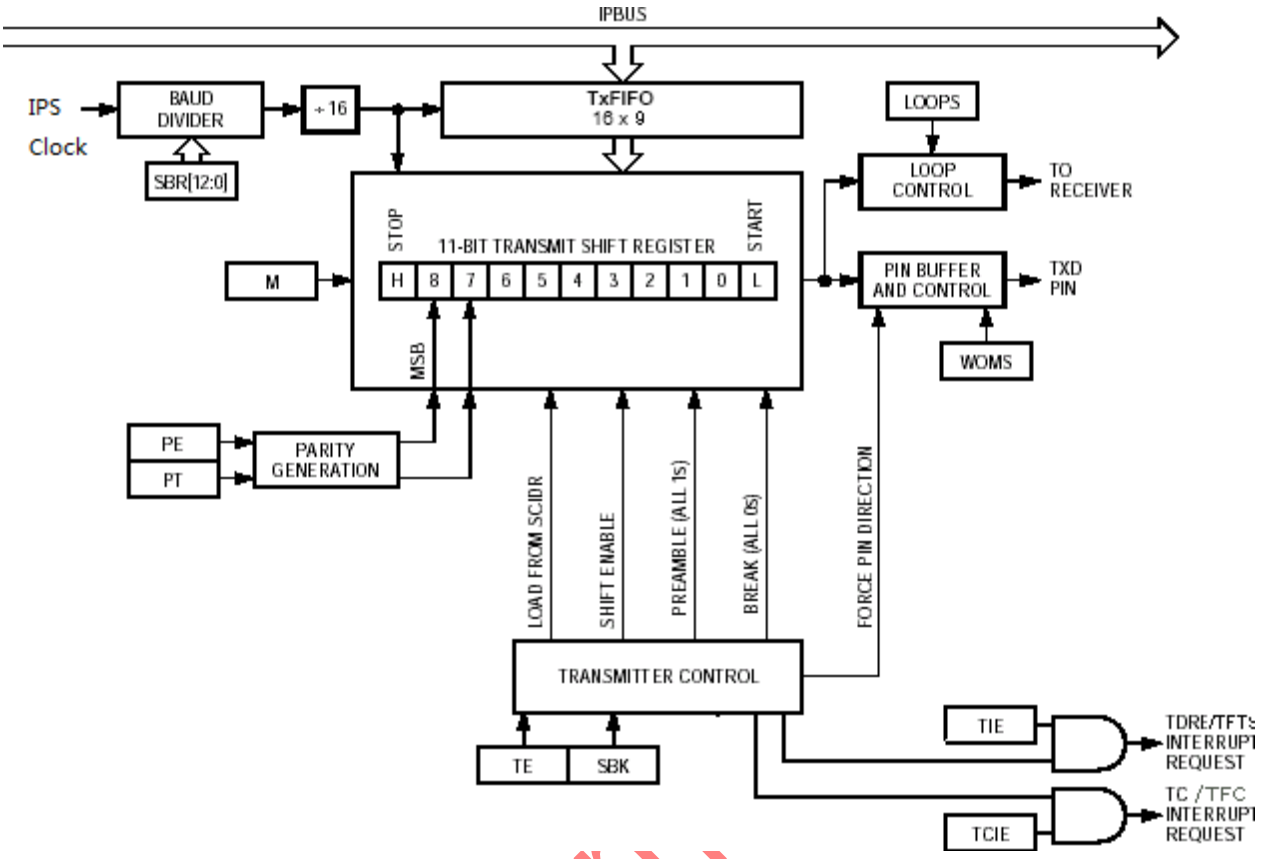


Figure 11-26: Transmitter Block Diagram

11.7.5.1 Frame Length

The transmitter can generate either 10-bit or 11-bit frames. In SCICR1, the M bit selects frame length, and the PE bit enables the parity function. One data bit may be an address mark or an extra stop bit. All frames begin with a start bit and end with one or two stop bits. When transmitting 9-bit data, bit T8 in SCI Data Register high (SCIDRH) is the ninth bit (bit 8).

Table 11-9: Example 10-Bit and 11-Bit Frames

M Bit	Frame Length	Start Bit	Data Bits	Parity Bit	Address Mark[1]	Stop Bit(s)
0	10 bits	1	8	No	No	1
		1	7	No	No	2
		1	7	No	Yes	1
		1	7	Yes	No	1
1	11 bits	1	9	No	No	1
		1	8	No	No	2
		1	8	No	Yes	1

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M Bit	Frame Length	Start Bit	Data Bits	Parity Bit	Address Mark[1]	Stop Bit(s)
		1	8	Yes	No	1
		1	7	No	Yes	2
		1	7	Yes	No	1

NOTE:

- When implementing a multidrop network using the SCI, the address mark bit is used to designate subsequent data frames as a network address and not device data.

11.7.5.2 Transmitting a Frame

To begin an SCI transmission:

- **Configure the SCI:**
  1. Write a baud rate value to SCIBDH and SCIBDL.
  2. Write to SCICR1 to:
     - Enable or disable loop mode and select the receiver feedback path
     - Select open-drain or wired-OR SCI outputs
     - Select 10-bit or 11-bit frames
     - Select the receiver wakeup condition: address mark or idle line
     - Select idle line type
     - Enable or disable the parity function and select odd or even parity
  3. Write to SCICR2 to:
     - Enable or disable TDRE, TC, RDRF, and IDLE interrupt requests
     - Enable the transmitter and queue a break frame
     - Enable or disable the receiver
     - Put the receiver in standby if required
- **Transmit a byte:**
  1. Clear the TDRE flag by reading SCISR1 and, if sending 9-bit data, write the ninth data bit to SCDRH.
  2. Write the byte to be transmitted (or low-order 8 bits if sending 9-bit data) to SCIDRL.
- Repeat step **2.** for each subsequent transmission.

Writing the TE bit from 0 to 1 loads the transmit shift register with a preamble of 10 (if M = 0) or 11 (if M = 1) logic 1s. When the preamble shifts out, the SCI transfers the data from SCIDRH and SCIDRL to the transmit shift register. The transmit shift register prefaces the data with a 0 start bit and appends the data with a 1 stop bit and begins shifting out the frame.

The SCI sets the TDRE flag every time it transfers data from SCIDRH and SCIDRL to the transmit shift register. TDRE indicates that SCIDRH and SCIDRL can accept new data. If the TIE bit is set, TDRE generates an interrupt request.

NOTE: SCIDRH and SCIDRL transfer data to the transmit shift register and sets TDRE 9/16ths of a bit time after the previous frame's stop bit starts to shift out.

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Hardware supports odd or even parity. When parity is enabled, the most significant data bit is the parity bit.

When the transmit shift register is not transmitting a frame, the TXD pin goes to the idle condition, logic 1. Clearing the TE bit while the transmitter is idle will return control of the TXD pin to the SCI data direction (SCIDDR) and SCI port (SCIPORT) registers.

If the TE bit is cleared while a transmission is in progress (while TC = 0), the frame in the transmit shift register continues to shift out. Then the TXD pin reverts to being a general-purpose I/O pin even if there is data pending in the SCI Data Register. To avoid accidentally cutting off a message, always wait until TDRE is set after the last frame before clearing TE.

To separate messages with preambles with minimum idle line time, use this sequence between messages:

- Write the last byte of the first message to SCIDRH and SCIDRL.
- Wait until the TDRE flag is set, indicating the transfer of the last frame to the transmit shift register.
- Queue a preamble by clearing and then setting the TE bit.
- Write the first byte of the second message to SCIDRH and SCIDRL.

When the SCI relinquishes the TXD pin, the SCIPORT and SCIDDR registers control the TXD pin.

To force TXD high when turning off the transmitter, set bit 1 of the SCI Port Register (SCIPORT) and bit 1 of the SCI Data Direction Register (SCIDDR). The TXD pin goes high as soon as the SCI relinquishes control of it. See **17.7.8 SCI Port Data Register** and **17.7.9 SCI Data Direction Register**

11.7.5.3 Break Frames

Setting the SBK bit in SCICR2 loads the transmit shift register with a break frame. A break frame contains all logic 0s and has no start, stop, or parity bit. Break frame length depends on the M bit in the SCICR1 register. As long as SBK is set, the SCI continuously loads break frames into the transmit shift register. After SBK is clear, the transmit shift register finishes transmitting the last break frame and then transmits at least one logic 1. The automatic logic 1 at the end of a break frame guarantees the recognition of the next start bit.

The SCI recognizes a break frame when a start bit is followed by eight or nine 0 data bits and a 0 where the stop bit should be. Receiving a break frame has these effects on SCI registers:

- Sets the FE flag
- Sets the RDRF flag
- Clears the SCIDRH and SCIDRL
- May set the OR flag, NF flag, PE flag, or the RAF flag

11.7.5.4 Idle Frames

An idle frame contains all logic 1s and has no start, stop, or parity bit. Idle frame length depends on the M bit in the SCICR1 register. The preamble is a synchronizing idle frame that begins the first transmission after writing the TE bit from 0 to 1.

If the TE bit is cleared during a transmission, the TXD pin becomes idle after completion of the transmission in progress. Clearing and then setting the TE bit during a transmission queues an idle frame to be sent after the frame currently being transmitted.

NOTE: When queueing an idle frame, return the TE bit to logic 1 before the stop bit of the current frame shifts out to the TXD pin. Setting TE after the stop bit appears on TXD causesdata previously written to SCIDRH and SCIDRL to be lost. Toggle TE to queue an idle frame, while the TDREflag is set, immediately before writing new data to SCIDRH and SCIDRL.

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11.7.6 Receiver

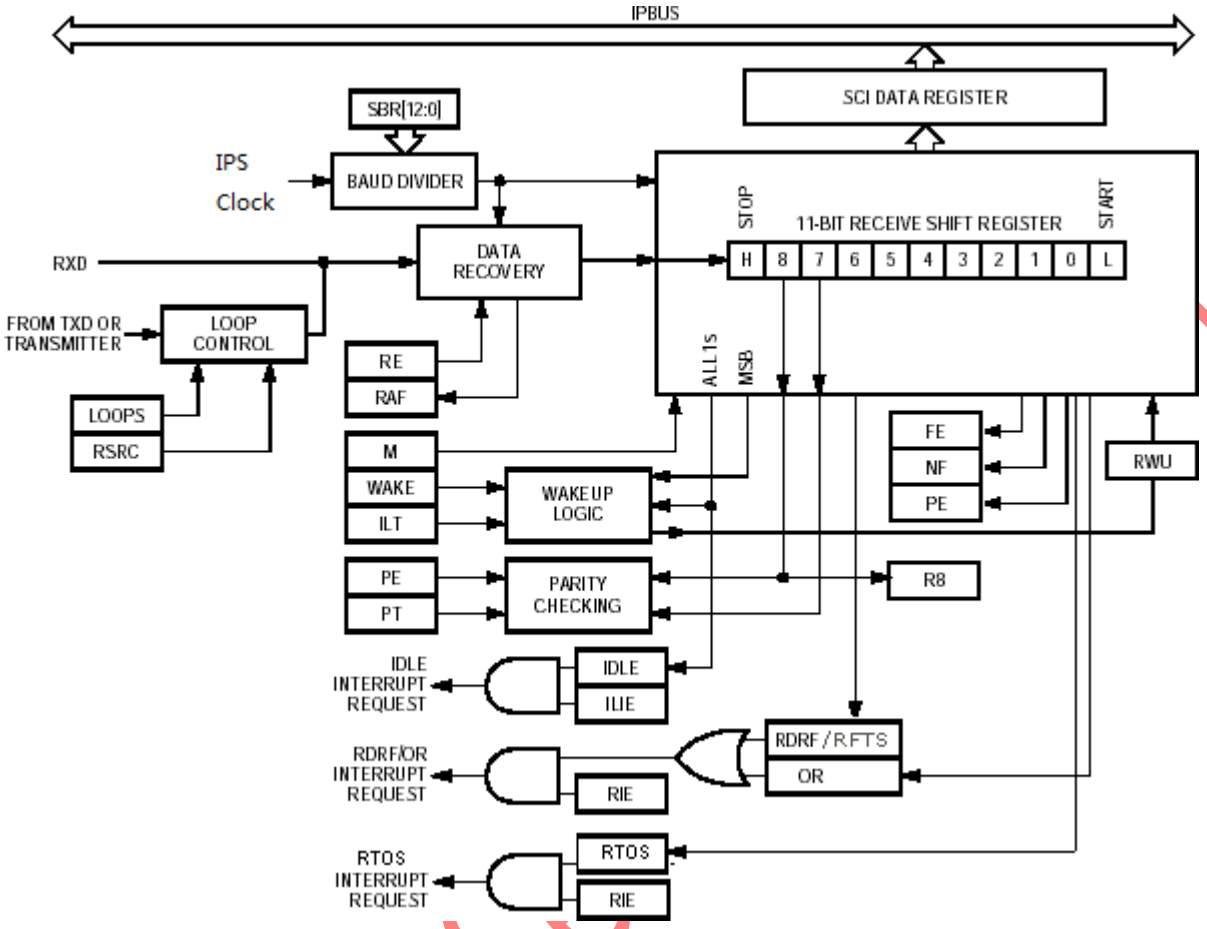


Figure 11-27: SCI Receiver Block Diagram

11.7.6.1 Frame Length

The receiver can handle either 8-bit or 9-bit data. The state of the M bit in SCICR1 selects frame length. When receiving 9-bit data, bit R8 in SCIDRH is the ninth bit (bit 8).

11.7.6.2 Receiving a Frame

When the SCI receives a frame, the receive shift register shifts the frame in from the RXD pin.

After an entire frame shifts into the receive shift register, the data portion of the frame transfers to SCIDRH and SCIDRL. The RDRF flag is set, indicating that the received data can be read. If the RIE bit is also set, RDRF generates an interrupt request.

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11.7.6.3 Data Sampling

The receiver samples the RXD pin at the RT clock rate. The RT clock is an internal signal with a frequency 16 times the baud rate. To adjust for baud rate mismatch, the RT clock resynchronizes:

- After every start bit
- After the receiver detects a data bit change from logic 1 to logic 0 (after the majority of data bit samples at RT8, RT9, and RT10 returns a valid logic 1 and the majority of the next RT8, RT9, and RT10 samples returns a valid logic 0)

To locate the start bit, data recovery logic does an asynchronous search for a 0 preceded by three 1s. When the falling edge of a possible start bit occurs, the RT clock begins to count to 16.

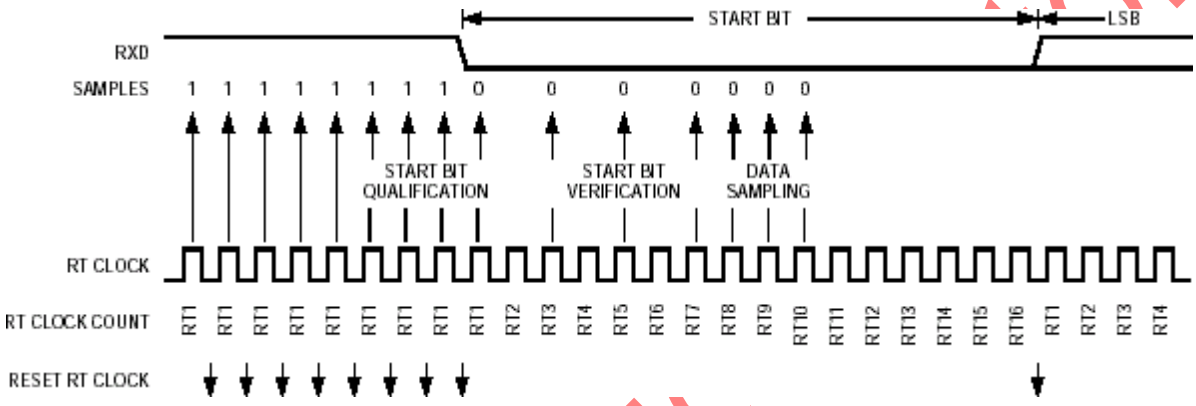


Figure 11-28: Receiver Data Sampling

To verify the start bit and to detect noise, data recovery logic takes samples at RT3, RT5, and RT7.

Table 11-10: Start Bit Verification

RT3, RT5, and RT7 Samples	Start Bit Verification	Noise Flag
000	Yes	0
001	Yes	1
010	Yes	1
011	No	0
100	Yes	1
101	No	0
110	No	0
111	No	0

If start bit verification is not successful, the RT clock is reset and a new search for a start bit begins.

To determine the value of a data bit and to detect noise, recovery logic takes samples at RT8, RT9, and RT10.

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Table 11-11: Data Bit Recovery

RT8, RT9, and RT10 Samples	Data Bit Determination	Noise Flag
000	0	0
001	0	1
010	0	1
011	1	1
100	0	1
101	1	1
110	1	1
111	1	0

NOTE: The RT8, RT9, and RT10 data samples do not affect start bit verification. If any or all of the RT8, RT9, and RT10 samples are logic 1s following a successful start bit verification, the NF flag is set and the receiver interprets the bit as a start bit (logic 0).

The RT8, RT9, and RT10 samples also verify stop bits.

Table 11-12: Stop Bit Recovery

RT8, RT9, and RT10 Samples	Framing Error Flag	Noise Flag
000	1	0
001	1	1
010	1	1
011	0	1
100	1	1
101	0	1
110	0	1
111	0	0

In **Figure 11-29** the verification samples RT3 and RT5 determine that the first low detected was noise and not the beginning of a start bit. The RT clock is reset and the start bit search begins again. The NF flag is not set because the noise occurred before the start bit was verified.

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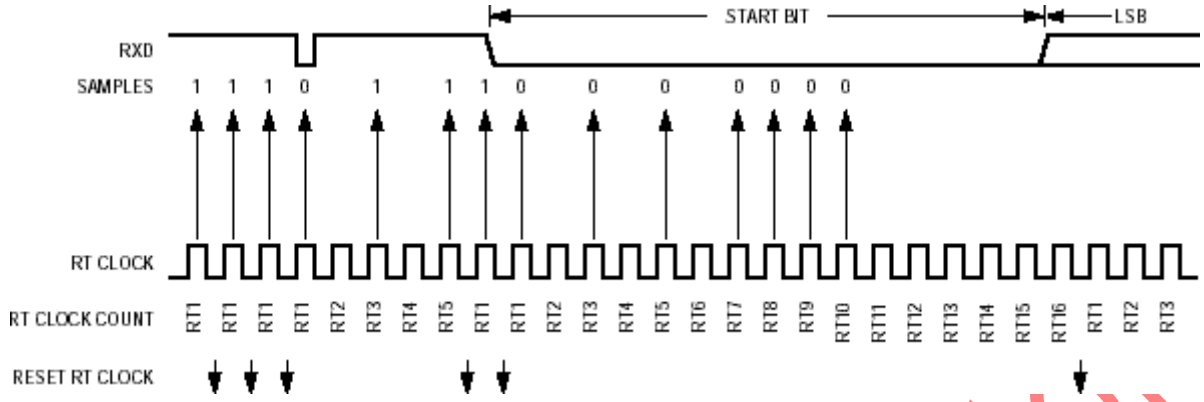


Figure 11-29: Start Bit Search Example 1

In **Figure 11-30** noise is perceived as the beginning of a start bit although the RT3 sample is high. The RT3 sample sets the noise flag. Although the perceived bit time is misaligned, the RT8, RT9, and RT10 data samples are within the bit time, and data recovery is successful.

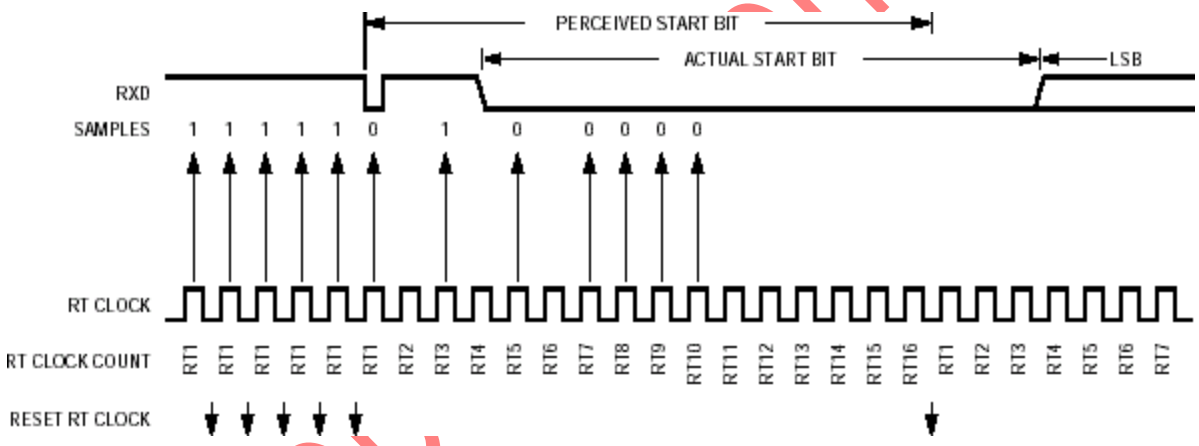


Figure 11-30: Start Bit Search Example 2

In **Figure 11-31** a large burst of noise is perceived as the beginning of a start bit, although the RT5 sample is high. The RT5 sample sets the noise flag. Although this is a worst-case misalignment of perceived bit time, the data samples RT8, RT9, and RT10 are within the bit time and data recovery is successful.

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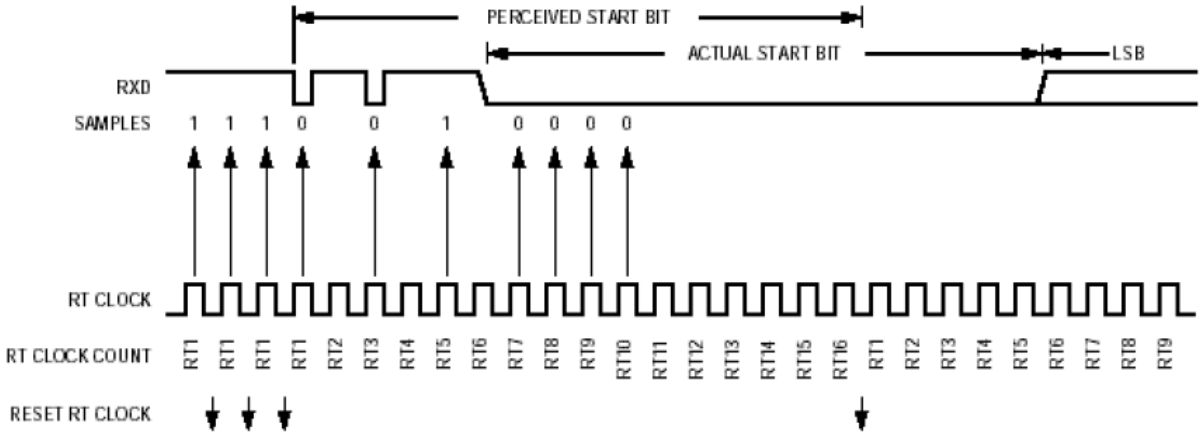


Figure 11-31: Start Bit Search Example 3

Figure 11-32 shows the effect of noise early in the start bit time. Although this noise does not affect proper synchronization with the start bit time, it does set the noise flag.

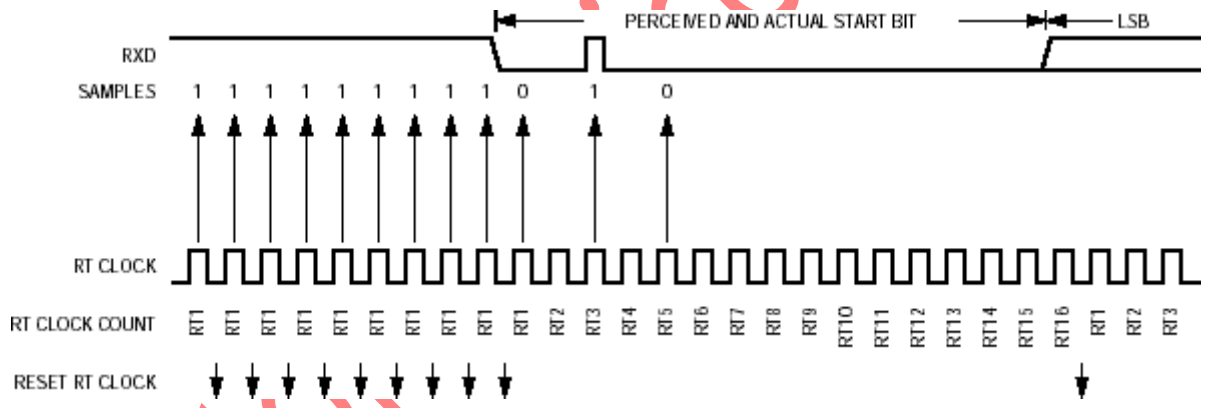


Figure 11-32: Start Bit Search Example 4

Figure 11-33 shows a burst of noise near the beginning of the start bit that resets the RT clock. The sample after the reset is low but is not preceded by three high samples that would qualify as a falling edge. Depending on the timing of the start bit search and on the data, the frame may be missed entirely or it may set the framing error flag.

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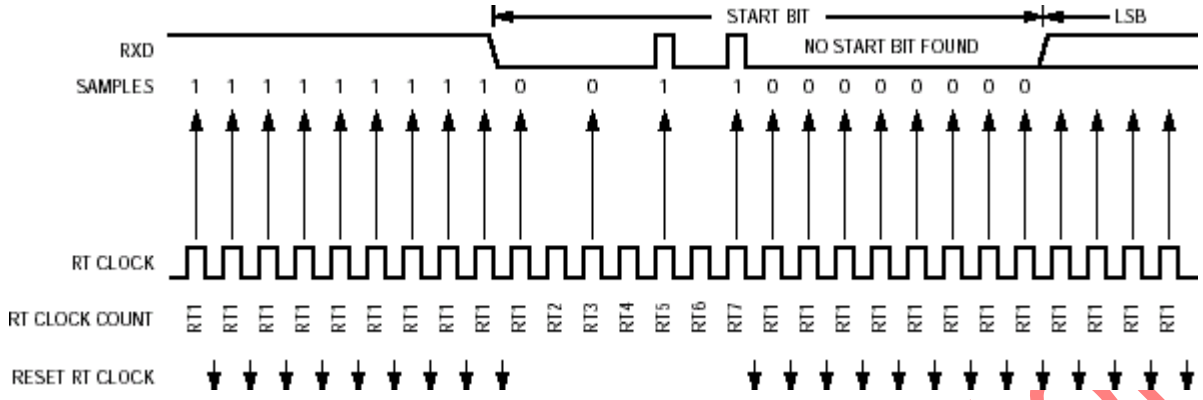


Figure 11-33: Start Bit Search Example 5

In **Figure 11-34** a noise burst makes the majority of data samples RT8, RT9, and RT10 high. This sets the noise flag but does not reset the RT clock. In start bits only, the RT8, RT9, and RT10 data samples are ignored.

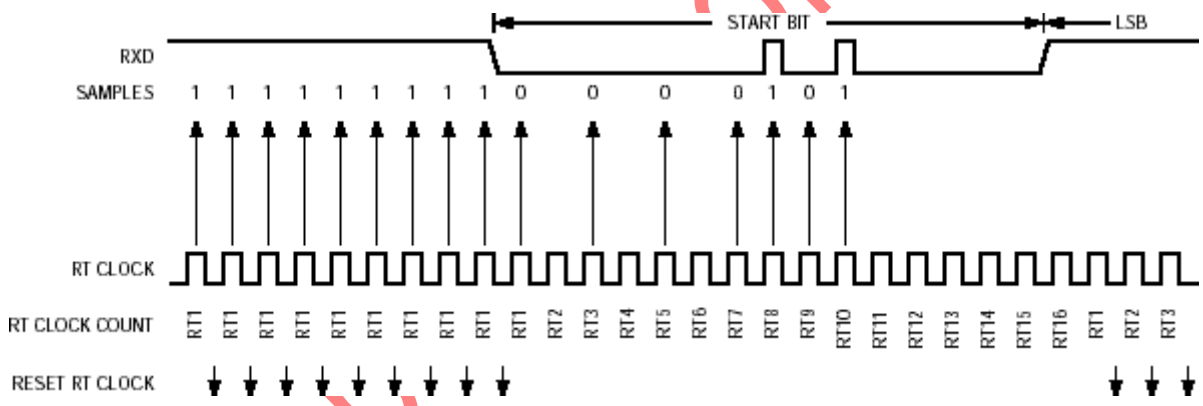


Figure 11-34: Start Bit Search Example 6

11.7.6.4 Framing Errors

If the data recovery logic does not detect a 1 where the stop bit should be in an incoming frame, it sets the FE flag in SCISR1. A break frame also sets the FE flag because a break frame has no stop bit. The FE flag is set at the same time that the RDRF flag is set.

11.7.6.5 Baud Rate Tolerance

A transmitting device may be operating at a baud rate below or above the receiver baud rate. Accumulated bit time misalignment can cause one of the RT8, RT9, and RT10 stop bit data samples to fall outside the stop bit. A noise error occurs if the samples are not all the same value. If more than one of the samples is outside the stop bit, a framing error occurs. In most applications, the baud rate tolerance is much more than the degree of misalignment that is likely to occur.

As the receiver samples an incoming frame, it resynchronizes the RT clock on any valid falling edge within the frame. Resynchronization within frames corrects misalignments between transmitter bit times and receiver bit times.

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11.7.6.5.1 Slow Data Tolerance

Figure 11-35 shows how much a slow received frame can be misaligned without causing a noise error or a framing error. The slow stop bit begins at RT8 instead of RT1 but arrives in time for the stop bit data samples at RT8, RT9, and RT10.

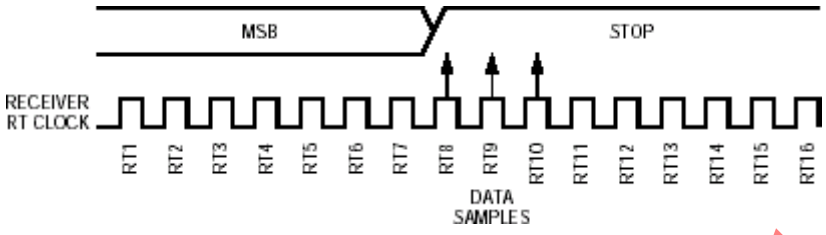


Figure 11-35: Slow Data

For 8-bit data, sampling of the stop bit takes the receiver:

9 bit times 16 RT cycles + 10 RT cycles= 154 RT cycles

With the misaligned data shown in **Figure 11-35** ,the receiver counts 154 RT cycles at the point when the count of the transmitting device is:

9 bits times x 16 RT cycles + 3 RT cycles = 147 RT cycles

The maximum percent difference between the receiver count and the transmitter count for slow 8-bit data with no errors is:

(154 – 147)/ 154 * 100% = 4.54%

For 9-bit data, sampling of the stop bit takes the receiver:

10 bits times x 16 RT cycles + 10 RT cycles = 170 RT cycles

With the misaligned data shown in **Figure 11-35**, the receiver counts 170 RT cycles at the point when the count of the transmitting device is :

10 bits times x 16 RT cycles + 3 RT cycles = 163 RT cycles

The maximum percent difference between the receiver count and the transmitter count for slow 9-bit data with no errors is:

(170 – 163)/ 170 * 100% = 4.12%

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11.7.6.5.2 Fast Data Tolerance

Figure 11-36 shows how much a fast received frame can be misaligned without causing a noise error or a framing error. The fast stop bit ends at RT10 instead of RT16 but is still sampled at RT8, RT9, and RT10.

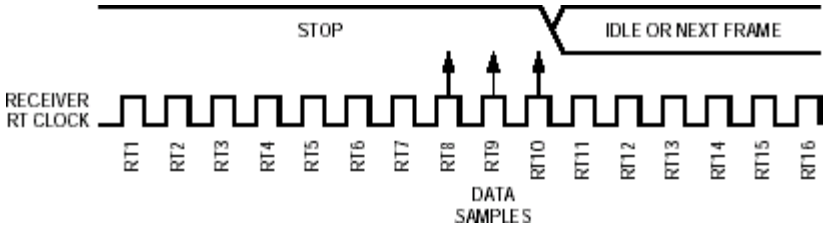


Figure 11-36: Fast Data

For 8-bit data, sampling of the stop bit takes the receiver:

9 bits times x 16 RT cycles + 10 RT cycles = 154 RT cycles

With the misaligned data shown in **Figure 11-36** , the receiver counts 154 RT cycles at the point when the count of the transmitting device is:

10 bits times x 16 RT cycles = 160 RT cycles

The maximum percent difference between the receiver count and the transmitter count for fast 8-bit data with no errors is:

(154 - 160)/ 154 * 100% = 3.90%

For 9-bit data, sampling of the stop bit takes the receiver:

10 bits times x 16 RT cycles + 10 RT cycles = 170 RT cycles

With the misaligned data shown in **Figure 11-36**, the receiver counts 170 RT cycles at the point when the count of the transmitting device is:

11 bits times x 16 RT cycles = 176 RT cycles

The maximum percent difference between the receiver count and the transmitter count for fast 9-bit data with no errors is:

(170 – 176) / 170 * 100% = 3.53%

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11.7.6.6 Receiver Wakeup

So that the SCI can ignore transmissions intended only for other devices in multiple-receiver systems, the receiver can be put into a standby state. Setting the RWU bit in SCICR2 puts the receiver into a standby state during which receiver interrupts are disabled.

The transmitting device can address messages to selected receivers by including addressing information in the initial frame or frames of each message.

The WAKE bit in SCICR1 determines how the SCI is brought out of the standby state to process an incoming message. The WAKE bit enables either idle line wakeup or address mark wakeup.

11.7.6.6.1 Idle Input Line Wakeup (WAKE = 0)

When WAKE = 0, an idle condition on the RXD pin clears the RWU bit and wakes up the receiver. The initial frame or frames of every message contain addressing information. All receivers evaluate the addressing information, and receivers for which the message is addressed process the frames that follow. Any receiver for which a message is not addressed can set its RWU bit and return to the standby state. The RWU bit remains set and the receiver remains on standby until another idle frame appears on the RXD pin.

Idle line wakeup requires that messages be separated by at least one idle frame and that no message contains idle frames.

The idle frame that wakes up the receiver does not set the IDLE flag or the RDRF flag.

The ILT bit in SCICR1 determines whether the receiver begins counting logic 1s as idle frame bits after the start bit or after the stop bit.

11.7.6.6.2 Address Mark Wakeup (WAKE = 1)

When WAKE = 1, an address mark clears the RWU bit and wakes up the receiver. An address mark is a 1 in the most significant data bit position. The receiver interprets the data as address data. When using address mark wakeup, the MSB of all non-address data must be 0. User code must compare the address data to the receiver's address and, if the addresses match, the receiver processes the frames that follow. If the addresses do not match, user code must put the receiver back to sleep by setting the RWU bit. The RWU bit remains set and the receiver remains on standby until another address frame appears on the RXD pin.

The address mark clears the RWU bit before the stop bit is received and sets the RDRF flag.

Address mark wakeup allows messages to contain idle frames but requires that the most significant byte (MSB) be reserved for address data.

NOTE: With the WAKE bit clear, setting the RWU bit after the RXD pin has been idle can cause the receiver to wake up immediately.

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11.7.7 Single-Wire Operation

Normally, the SCI uses the TXD pin for transmitting and the RXD pin for receiving (LOOPS = 0, RSRC = X). In single-wire mode, the RXD pin is disconnected from the SCI and is available as a general-purpose I/O pin. The SCI uses the TXD pin for both receiving and transmitting.

In single-wire mode (LOOPS = 1, RXRC = 1), setting the data direction bit for the TXD pin configures TXD as the output for transmitted data. Clearing the data direction bit configures TXD as the input for received data.

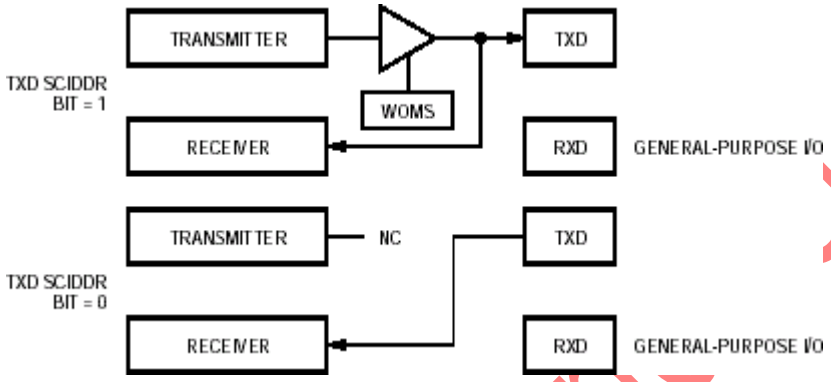


Figure 11-37: Single-Wire Operation (LOOPS = 1, RSRC = 1)

Enable single-wire operation by setting the LOOPS bit and the RSRC bit in SCICR1. Setting the LOOPS bit disables the path from the RXD pin to the receiver. Setting the RSRC bit connects the receiver input to the output of the TXD pin driver. Both the transmitter and receiver must be enabled (TE = 1 and RE = 1).

The WOMS bit in the SCICR1 register configures the TXD pin for full CMOS drive or for open-drain drive. WOMS controls the TXD pin in both normal operation and

in single-wire operation. When WOMS is set, the DDR bit for the TXD pin does not have to be cleared for transmitter to receive data.

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11.7.8 Loop Operation

In loop mode (LOOPS = 1, RSRC = 0), the transmitter output goes to the receiver input. The RXD pin is disconnected from the SCI and is available as a

general-purpose I/O pin.

Setting the DDR bit for the TXD pin connects the transmitter output to the TXD pin. Clearing the data direction bit disconnects the transmitter output from the TXD pin.

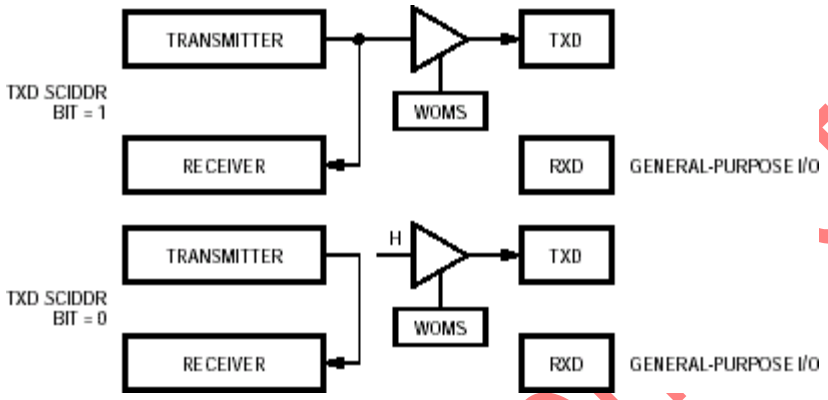


Figure 11-38: Loop Operation (LOOPS = 1, RSRC = 0)

Enable loop operation by setting the LOOPS bit and clearing the RSRC bit in SCICR1. Setting the LOOPS bit disables the path from the RXD pin to the receiver. Clearing the RSRC bit connects the transmitter output to the receiver input. Both the transmitter and receiver must be enabled (TE = 1 and RE = 1).

The WOMS bit in SCICR1 configures the TXD pin for full CMOS drive or for open-drain drive. WOMS controls the TXD pin during both normal operation and loop operation.

11.7.9 Hardware Flow Control

It is possible to control the serial data flow between 2 devices by using the CTSN input and RTSN output. RTS and CTS flow control can be enabled independently by writing respectively RTSE and CTSE to 1 (In SCIFCTRLR register).

11.7.9.1 RTS Flow Control

If the RTS flow control is enabled (RTSE=1), then RTSN is asserted (low) as long as the SCI receiver is ready to receive new data. When the receive register is full (The condition is the same with RDRF flag in character mode, and same with RFTS), RTSN is deasserted, indicating that the transmission is expected to stop at the end of the current frame.

11.7.9.2 CTS Flow Control

If the CTS flow control is enabled (CTSE=1), then the transmitter checks the CTSN input before transmitting the next frame. If CTSN is asserted (low), the the next data is transmitted (assuming that a data is to be transmitted), else the transmission does not occur. When CTSN is deasserted during a transmission, the current transmission is completed before the transmitter stops.

When CTSE=1, the CTSIS flag bit is automatically set by hardware as soon as the CTSN input toggles. It indicates when the receiver becomes ready or not ready for communication. An interrupt is generated if the CTSIE bit is set.

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11.7.10 I/O Ports

The SCIPORT register is associated with two pins:

- The TXD pin is connected to SCIPORT1.
- The RXD pin is connected to SCIPORT0.

The SCI Data Direction Register (SCIDDR) configures the pins as inputs or outputs (see **17.7.9 SCI Data Direction Register**).

11.7.11 Reset

Reset initializes the SCI registers to a known startup state as described in **17.7 Memory Map and Registers**

11.8 Interrupts

lists the five interrupt requests associated with each SCI module.

Table 11-13: SCI Interrupt Request Sources

Source	Flag	Enable Bit
Transmitter	TDRE	TIE
	TC	TCIE
	TFTS	TIE/TXFIE
	FTC	TCIE/TXFCIE
Receiver	RDRF	RIE
	OR	RIE
	RFTS	RIE/RXFIE
	FOR	RIE/RXORIE
	IDLE	ILIE
	RTOS	RIE/RXFTOIE
Flow Control	CTSIS	CTSIE

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12 Time Counter

12.1 Introduction

The time counter is a 16-bit timer used to help software recover from runaway code or give an interrupt when the operation has run longer than expected. The real time counter has a free-running down-counter that generates a reset or interrupt on underflow. To prevent a reset, software must periodically restart the countdown by servicing the watchdog.

12.2 Modes of Operation

This subsection describes the operation of the real time counter in low-power modes and debug mode of operation.

12.2.1 Wait Mode

In wait mode with the WAIT bit set in the TC Control Register (TCCR), time counter operation stops. In wait mode with the WAIT bit clear, the time counter continues to operate normally.

12.2.2 Doze Mode

In doze mode with the DOZE bit set in TCCR, time counter module operation stops. In doze mode with the DOZE bit clear, the time counter continues to operate normally.

12.2.3 Stop Mode

In stop mode with the STOP bit set in TCCR, time counter operation stops in stop mode. When stop mode is exited, the time counter operation continues operation from the state it was in prior to entering stop mode. In stop mode with the STOP bit clear, the time counter continues to operate normally.

12.2.4 Debug Mode

In debug mode with the DBG bit set in TCCR, time counter module operation stops. In debug mode with the DBG bit clear, the time counter continues to operate normally. When debug mode is exited, time counter operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain.

12.3 Block Diagram

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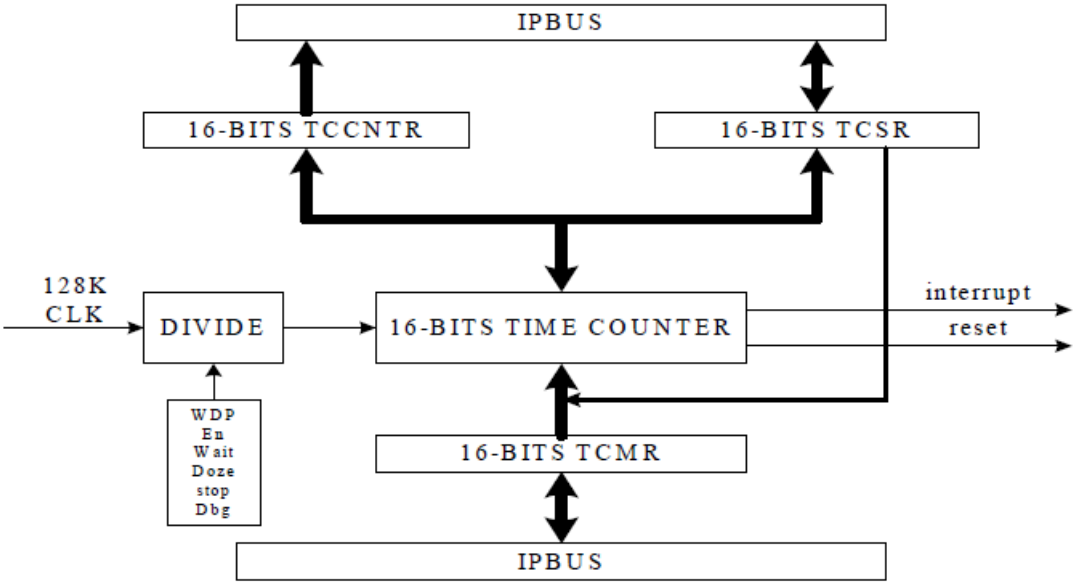


Figure 12-1: Time Counter Block Diagram

12.4 Signals

The time counter module has no off-chip signals.

12.5 Memory Map and Registers

This subsection describes the memory map and registers for the time counter.

12.5.1 Memory Map

Reger **Table 12-1** to for an overview of the time counter memory map.

Table 12-1: Real Time Counter Module Memory Map

Address Offset	Bits 15-8	Bits 7-0	Access(1)
0x0000	TC Control Register (TCCR)		S
0x0002	TC Modulus Register (TCMR)		S
0x0004	TC Count Register (TCCNTR)		S/U
0x0006	TC Service Register (TCSR)		S/U

NOTE:

- S = CPU supervisor mode access only. S/U = CPU supervisor or user mode access. User mode accesses to supervisor only addresses have no effect and result in a cycle termination transfer error.

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12.5.2 Registers

The time counter programming model consists of these registers:

- The TC Control Register (TCCR) configures watchdog timer operation. See 12.5.2.1 TC Control Register.
- The TC Modulus Register (TCMR) determines the timer modulus reload value. See 12.5.2.2 TC Modules Register.
- The TC Count Register (TCCNTR) provides visibility to the watchdog counter value. See 18.5.2.3 TC Count Register.
- The TC Service Register (TCSR) requires a service sequence to prevent reset. See 12.5.2.4 TC Service Register.

12.5.2.1 TC Control Register

The 16-bit read/write TC Control Register (TCCR) configures time counter operation.

Addr. Offset: 0x0000-0x0001 Reset Value: 0x0F75

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	WAIT	DOZE	STOP	DBG
ro	ro	ro	ro	rw	rw	rw	rw

7	6	5	4	3	2	1	0
IS	WDP[2:0]			IF	IE	CU	RN
ro	rw	rw	rw	r/w1C	rw	r/w10	rw

Figure 12-2: TC Control Register (TCCR)

WAIT — Wait Mode Bit

WAIT bit controls the function of the time counter in wait mode. Reset sets WAIT.

1 = Time counter stopped in wait mode

0 = Time counter not affected in wait mode DOZE — Doze Mode Bit

DOZE bit controls the function of the time counter in doze mode. Reset sets DOZE.

1 = Time counter stopped in doze mode

0 = Time counter not affected in doze mode STOP — STOP Mode Bit

STOP bit controls the function of the time counter in stop mode. Reset sets STOP.

1 = Time counter stopped in stop mode

0 = Time counter not affected in stop mode

DBG — Debug Mode Bit

DBG bit controls the function of the time counter in debug mode. During debug mode, time counter registers can be written and read normally.

When debug mode is exited, timer operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain.

1 = Time counter stopped in debug mode

0 = Time counter not affected in debug mode

NOTE: Changing the DBG bit from 1 to 0 during debug mode starts the time counter. Changing the DBG bit

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from 0 to 1 during debug mode stops the time counter.
RN — TC Reset Enable Bit

RN bit enables the time counter for reset.

1 = Time counter enabled for reset

0 = Time counter disabled for reset

CU — TC Change Update Bit

Write One to CU bit update the WDP[2:0] and WMR to the work latch.

IE — TC Interrupt Enable Bit

IE bit enables the time counter interrupt mode. Once interrupt generated and the RN bit is 1, this bit will be auto clear.

1 = Time counter interrupt mode enabled

0 = Time counter interrupt mode disabled

IF — TC Interrupt Flag Bit

Write One to this bit will clear the flag.

IS — TC Clock Domain Interrupt Status Bit

This bit is read-only, if this bit is 1'b1, the status of watchdog clock domain is not cleared, so if CPU want to sleep or stop, it will be wakeup again. So before CPU want to sleep or stop, please check this bit first.

WDP[2:0] — TC Timer Prescaler

The WDP[2:0] bits determine the time counter prescaling when the time counter is runnung. The different prescaling values and their corresponding time-out periods are shown in **Table 12-2 TC Timer Prescaler**.

Table 12-2: TC Timer Prescaler

WDP[2:0]	Preacaler
000	2048
001	1024
010	512
011	256
100	128
101	64
110	32
111	16

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12.5.2.2 TC Modules Register

Addr. Offset: 0x0002-0x0003 Reset Value: 0xFFFF

15	14	13	12	11	10	9	8
TCMR[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
TCMR[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 12-3: TC Modulus Register (TCMR)

TCMR[15:0] — TC Modulus Field

TCMR[15:0] field contains the modulus that is reloaded into the time counter by a service sequence. Writing to TCMR immediately loads the new modulus value into the time counter. The new value is also used at the next and all subsequent reloads.

Reading TCMR returns the value in the modulus register. Reset initializes the TCMR[15:0] field to 0xFFFF.

12.5.2.3 TC Count Register

Addr. Offset: 0x0004-0x0005 Reset Value: 0xFFFF

15	14	13	12	11	10	9	8
TCCNTR[15:8]							
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
TCCNTR[7:0]							
ro	ro	ro	ro	ro	ro	ro	ro

Figure 12-4: TC Count Register (TCCNTR)

TCCNTR[15:0] — TC Count Field

The read-only TCCNTR[15:0] field reflects the current value in the time counter. Reading the 16-bit TCCNTR with two 8-bit reads is not guaranteed to return a coherent value. Writing to TCCNTR has no effect, and write cycles are terminated normally. This register is for TC work domain, so the read value maybe not stabilization, please read it time after time continuously.

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12.5.2.4 TC Service Register

When the time counter is enabled, writing 0x5555 and then 0xAAAA to the TC Service Register (TCSR) before the time counter times out prevents a reset. If TCSR is not serviced before the timeout, the time counter sends a signal to the reset controller or interrupt controller module and asserts a system reset or interrupt.

Both writes must occur in the order listed before the timeout, but any number of instructions can be executed between the two writes. However, writing any value other than 0x5555 or 0xAAAA to TCSR resets the servicing sequence, requiring both values to be written to keep the time counter from causing a reset.

Addr. Offset: 0x0006-0x0007 Reset Value: 0x0000

15	14	13	12	11	10	9	8
TCSR[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
TCSR[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 12-5: TC Service Register (TCSR)

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13 I2C

13.1 Introduction

I2C is a two-wire, bidirectional serial bus that provides a simple, efficient method of data exchange, minimizing the interconnection between devices. This bus is suitable for applications requiring occasional communications over a short distance between many devices. The flexible I2C allows additional devices to be connected to the bus for expansion and system development.

The two bus lines provide data transfer rates up to 100Kbits/s IN Standard Mode, data rates up to 400Kbits/s in Fast Mode and data rates up to 3.4Mbits/s in

High-Speed Mode..

The I2C system is a true multiple-master bus including arbitration and collision detection that prevents data corruption if multiple devices attempt to control the bus simultaneously. This feature supports complex applications with multiprocessor control and can be used for rapid testing and alignment of end products through external connections to an assembly-line computer.

13.2 Features

- Supports 10 bit addressing.
- Supports Standard Mode, Fast Mode and High-Speed Mode
- Software option to select between High-Speed mode and Standard/Fast mode
- Compatibility with standard and fast-mode of I2C bus version 2.1 standard.
- Multiple-master operation.
- Software-programmable for one of 64 different serial clock frequencies.
- Software-selectable acknowledge bit.
- Interrupt-driven, byte-by-byte data transfer.
- Arbitration-lost interrupt with automatic mode switching from master to slave.
- Transfer completion and read configure interrupt.
- Start and stop signal generation/detection.
- Repeated START signal generation.
- Acknowledge bit generation/detection.
- Bus-busy detection.
- Option slave address receiving enable when system clock stop mode
- SCL or SDA line gpio function supported

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13.3 System and Block Diagram

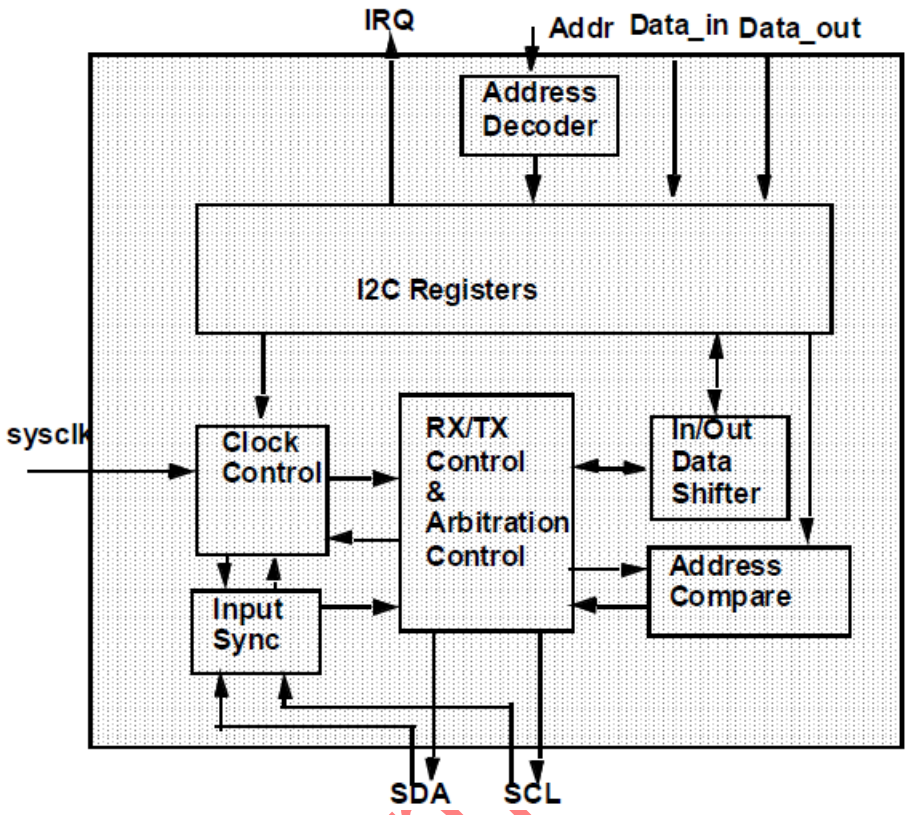


Figure 13-1: I2C Block Diagram

13.4 Memory Map and Registers

13.4.1 Memory Map

There are five registers in the I2C memory map, as shown in **Table 13-1**.

Table 13-1: I2C Memory Map

Offset Ad- dress	Bits 7- 0	Access
0x0000	I2C Slave Address Register High bit(I2CSAH)	S/U
0x0001	I2C Slave Address Register Low bit(I2CSAL)	S/U
0x0002	I2C Control Register (I2CC)	S/U
0x0003	I2C Clock Prescaler Register (I2CP)	S/U
0x0004	I2C Status Register (I2CS)	S/U
0x0005	I2C Data Register (I2CD)	S/U
0x0006	I2C slave SDA hold time Register	S/U
0x0007	I2C slave high-speed indicator reg- ister	S/U
0x0008	I2C Port Control Register (I2CPCR)	S/U

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Offset Ad- dress	Bits 7- 0	Access
0x0009	I2C Port Data Register (I2CPDR)	S/U
0x000a	I2C Port Direction Register (I2CDDR)	S/U
0x000b	I2C filter and current source test register (I2CFCTR)	S/U
0x000c	I2C 10ns filter trimming value regis- ter (I2C10NSFTVR)	S/U
0x000d	I2C 50ns filter trimming value regis- ter (I2C50NSFTVR)	S/U

13.4.2 Register Description

13.4.2.1 I2C Slave Address Register High Bit(I2CSAH)

Addr. Offset: 0x0000 Reset Value: 0x40

7	6	5	4	3	2	1	0
ADDR7	ADDR6	ADDR5	ADDR4	ADDR3	ADDR2	ADDR1	Reserved
rw	rw	rw	rw	rw	rw	rw	ro

The I2CSAH stores the high 7 bit of the address when the I2C works as slave and responds to the address sent by a master. When the I2C works as slave and support for 10 bit address,the I2CSAH is 11110XX.The first seven bits of the first byte are the combination11110XX of which the last two bits (XX) are the two most-significant bits (MSBs) of the 10-bit address; the eighth bit of the first byte is the R/W bit that determines the direction of the message.

13.4.2.2 I2C Slave Address Register Low Bit(I2CSAL)

The I2CSAL stores the low 8 bit of the address when the I2C works as slave and support for 10 bit address.In 7 bit address mode,the I2CSAL is useless.

Addr. Offset: 0x0001 Reset Value: 0x00

7	6	5	4	3	2	1	0
ADDR7	ADDR6	ADDR5	ADDR4	ADDR3	ADDR2	ADDR1	ADDR0
rw	rw	rw	rw	rw	rw	rw	rw

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13.4.2.3 I2C Control Register (I2CC)

Addr. Offset: 0x0002 Reset Value: 0x08

7	6	5	4	3	2	1	0
SLV_HSIE	HMS_EN	AMIE	REPSTA	ACKEN	MSMOD	IEN	EN
rw	rw	rw	rw	rw	rw	rw	rw

I2CC is used to control the working of I2C module.

EN - I2C enable/disable control.

1 = module is enabled.

0 = module is disabled.

It enables/disables the module. Also controls the software reset of the entire I2C module. Setting the bit generates an internal reset to the module which gets asserted after 2 clock of setting the bit and remain asserted for 3 clocks. Thus reset gets negated after 5 clocks of setting EN bit.

If the module is enabled in the middle of a byte transfer, slave mode ignores the current I2C bus transfer and starts operating when the next start condition is detected. Master mode is not aware that the bus is busy; so initiating a start cycle may corrupt the current bus cycle, ultimately causing either the current master or the I2C module to lose arbitration, after which bus operation returns to normal.

IEN - I2C interrupt enable control.

It enables the I2C interrupt when it is set.

1 = I2C interrupt enabled, 0 = I2C interrupt disabled.

MSMOD - I2C master/slave mode selection control.

Changing MSMOD from 1 to 0 generates a STOP on the bus and selects slave mode. Changing MSMOD from 0 to 1 generates a START on the bus and selects master mode.

1 = Master mode.

0 = Slave mode.

ACKEN - Acknowledge enable control.

Specifies the value driven onto SDA during acknowledge cycles for both master and slave receivers. Note that ACKEN bit applies only when the I2C bus receives data byte. During receiving address, if the received address is matched with the slave address, the Acknowledge bit will be sent regardless of the ACKEN bit.

1 = An acknowledge signal is sent to the SDA at the ninth clock bit after receiving one byte of data.

0 = No acknowledge signal is sent to the SDA at the ninth clock bit after receiving one byte of data.

REPSTA - Repeat Start

In these case: 1) The master receiver wasn't acknowledged after sending slave address or data byte, 2) the master transmitter has send a address or data byte regardless of acknowledged or not, the master can repeat the START signal, followed by a new slave address. The repeat start bit will be send when configure slave address(by writing I2CD) after REPSTA bit set.

1 = Generate repeat start, 0 = No repeat start.

AMIE - Address Match Interrupt Enable

Selects whether Slave address match will generate interrupt request if I2C interrupt enable control is set to 1. AMIE should be set before entering stop mode to wakeup the system when slave address matched and must be clear when normal work mode.

1 = Enable address match interrupt request.

0 = Disable address match interrupt request.

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HSM_EN - High Speed Mode Enable

Selects the High Speed mode or Fast/Standard mode operation for the module in master mode. The HSM_EN bit should be set to select the High Speed mode operation. After the transmission of the master code in F/S mode and the HS_I2C does not loose the arbitration, HSM_EN should be set by software, and if you prepare to clear MSMOD to transfer STOP, HSM_EN should be clear by software before clear MSMOD.

0 = Fast/Standard mode operation (Default)

1 = High-Speed Mode operation

SLV_HSIE - Slave High-Speed Mode Interrupt Enable

Selects whether Slave High-Speed Mode status will generate interrupt request if 2C interrupt enable control is set to 1.

1 = Enable Slave High-Speed Mode status request.

0 = Disable Slave High-Speed Mode status request.

13.4.2.4 I2C Clock Prescalar Register (I2CP)

Addr. Offset: 0x0003 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved	TEST	PRE5	PRE4	PRE3	PRE2	PRE1	PRE0
ro	rw	rw	rw	rw	rw	rw	rw

I2CP is a prescaler to generate a bit-rate clock for the data transceiver. Due to potentially slow SCL and SDA rise and fall times, bus signals are sampled at the prescaler frequency. The serial bit clock frequency is equal to the system clock divided by the divider.

PRE[5:0] - Prescaler Divider Value TEST - Clock Test Enable

It enables test mode for the I2C clock.In normal mode, its frequency is fsys/(396x(PRE[5:0]+1)+1). In test mode, the frequency is fsys/(8x(PRE[5:0]+1)+1).

1 = Clock Test Mode Enable, 0 = Normal Mode

13.4.2.5 I2C Status Register(I2CS)

Addr. Offset: 0x0004 Reset Value: 0x00

7	6	5	4	3	2	1	0
AACK	DACK	RXTX	ARBL	BBUSY	AASLV	RC	TF
ro	ro	ro	ro	ro	ro	ro	ro

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The I2CS register shows the status of I2C module.

TF - Transfer Complete Flag

Indicates there is data transmitted or received. For receiver, It is set by the falling edge of the ninth clock of a data(not address) byte received regardless of acknowledge bit detected or not. For master transmitter, It is set by the falling edge of the ninth clock of a data or address byte send regardless of acknowledge bit detected or not. For slave transmitter, it is set by the falling edge of the ninth clock of address or data byte transfered and acknowledge bit must be detected. If the IEN bit in I2CC is also set, an interrupt will be generated. Clear TF by reading I2CS with TF set and then accessing I2CD or master-transmit mode writing I2CC.

0 = meaningless,

1 = Data or address transfer complete.

RC - Receive Complete

Indicates it is time to configure the receiver. For master receiver, It is set by the falling edge of the ninth clock of data or address byte transfered regardless of acknowledge bit detected or not. For slave receiver, it is set by the falling edge of the ninth clock of address or data byte received and acknowledgement bit must be received. If the IEN bit in I2CC is also set, a interrupt will be generated. Clear RC by reading I2CS with RC set and then writing I2CC.

0 = meaningless,

1 = Indicates it is time to configure the receiver.

AASLV - Addressed as a slave

It is set by the falling edge of the eigth clock if I2C module is addressed as a slave and its own slave address matches the calling address received on SDL.It is clear by START or STOP bit detected

0 = Not as a slave.

1 = Addressed as a slave.

BBUSY - I2C bus busy Shows the bus status.

0 = Bus is idle. It is cleared from STOP bit.

1 = Bus is busy. It is set from START bit.

ARBL - Arbitration lost

Shows the arbitration status of the bus. It will be set in the following cases during SCL high:

- SDA is sampled low when the master drives high during START condition, address cycle, data-transmit cycle or STOP condition.
- SDA is sampled low when the master drives high during the acknowledge bit of a data-receive cycle.
- A start cycle is attempted when the bus is busy.

ARBL must be cleared by software by reading the I2CS register.

0 = Arbitration not lost.

1 = Arbitration lost.

RXTX - Receive or transmit.

Indicates the I2C module function as receiver or transmitter. It is valid of falling edge of the eigth clock.

0 = Receive, receive data.

1 = Transmit, transmit data.

DACK - Data acknowledge received.

Indicates whether the acknowledge bit is detected during address or data transfer. It is valid of rising edge of the ninth clock.

0 = No acknowledge bit received.

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1 = Acknowledge bit received.

AACK - Address acknowledge error.

Indicates whether the acknowldege bit was detected or not by master during address phase. It is set by the rising edge of ninth clock of the address phase and clear by changing MSMOD from 1 to 0 or repeat start condition.

0 = No address acknowledge error.

1 = Address acknowledge error.

13.4.2.6 I2C Data Register(I2CD)

Addr. Offset: 0x0005 Reset Value: 0x00

7	6	5	4	3	2	1	0
R/T[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

The I2CD register holds the data to be transmitted (next byte) or data received. In master mode it also holds the slave address and transfer direction to be transmitted. Bits [7:1] form the slave address and bit [0] is the transfer direction (R/W). In master-receiver mode, reading I2CD allows a read to occur and initiates next byte data receiving. In master-transmit mode, writing I2CD will store the byte of the next transmit. In slave mode, the same function is available after it is addressed.

R[7:0] - Receive Bits [7:0] T[7:0] - Transmit Bits [7:0]

13.4.2.7 I2C slave SDA hold time register(I2CSHT)

Addr. Offset: 0x0006 Reset Value: 0x09

7	6	5	4	3	2	1	0
SCL_FILTER_EN	SDA_FILTER_EN	SLVHT					
rw	rw	rw	rw	rw	rw	rw	rw

SCL_FILTER_EN - SCL Filter enable.

If SCL filter is enabled, when the HS mode transfer is ongoing, the 10ns pulse occured on SCL line will be filter out. If the F/S mode transfer is ongoing, the 50ns pulse occured on SCL line will be filter on.

SDA_FILTER_EN - SDA Filter enable.

If SDA filter is enabled, when the HS mode transfer is ongoing, the 10ns pulse occured on SDA line will be filter out. If the F/S mode transfer is ongoing, the 50ns pulse occured on SDA line will be filter on.

SLVHT - slave SDA line hold time configuration.

when I2C work as slave output mode, the data will be changed after SCL falling edge and the value of internal SDA hold register equal to SLVHT.

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13.4.2.8 I2C slave high-speed mode indicator register(I2CSHIR)

Addr. Offset: 0x0007 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved							SLV_HS
ro	ro	ro	ro	ro	ro	ro	r/w1c

SLV_HS -Slave high speed mode.

When I2C is selected as slave, this bit indicates whether the module is selected for High-Speed mode or Fast/Standard mode data transfer. This bit is set on the ninth SCL rising edge of Master code and Not Acknowledgement bit is received. This bit must be cleared by writing 1 to it otherwise the SCL line will be force to LOW state.

0 = I2C is selected for Fast/Standard data transfer (Default).

1 = I2C is selected for High-Speed mode data transfer.

13.4.2.9 I2C Port Control Register (I2CPCR)

Addr. Offset: 0x0008 Reset Value: 0x03

7	6	5	4	3	2	1	0
SDAPA	SCLPA	WOMI2C[1:0]		PDI2C[1:0]		PUI2C[1:0]	
rw	rw	rw	rw	rw	rw	rw	rw

SDAPA - SDA Port Assignment Bit.

The read/write bit selects the SDA's function mode.

1 = Pin configured as GPIO.

0 = Pin configured as primary function.

SCLPA - SCL Port Assignment Bit.

The read/write bit selects the SCL's function mode.

1 = Pin configured as GPIO.

0 = Pin configured as primary function.

WOMI2C[1:0] — Wired-OR mode Bits

The read/write bits set the corresponding I2C pins to open-drain drive mode. WOMI2C[1] is for SDA pin and WOMI2C[0] is for SCL pin.These bits are available only in GPIO mode

1 = Open-drain when output, 0 = CMOS drive when output.

PUI2C[1:0] — Pullup or Pulldown Enable Bits

The read/write bits enable the pullup or pulldown of corresponding I2C pins. PUI2C[1] is for SDA pin and PUI2C[0] is for SCL pin.These bits are available only in GPIO mode.

1 = Pullup or Pulldown Enabled, 0 = Pullup or Pulldown Disabled.

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13.4.2.10 I2C Port Data Register (I2CPDR)

Addr. Offset: 0x0009 Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved						PORTI2C[1:0]	
ro	ro	ro	ro	ro	ro	rw	rw

PORTI2C[1:0] — I2C Port Data Bits

Writes to these bits set the output data for the corresponding I2C pins configured as GPIO. Reading these bits return the I2C pins' level.

13.4.2.11 I2C Port Direction Register (I2CDDR)

Addr. Offset: 0x000A Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved						DDRI2C[1:0]	
ro	ro	ro	ro	ro	ro	rw	rw

DDRI2C[1:0] — I2C Data Direction Bits

The read/write bits control the data direction of the I2C pins. These bits are available only in GPIO mode

1 = Corresponding pin configured as output

0 = Corresponding pin configured as input

13.4.2.12 I2C Filter and Current Source Test Register (I2CFCTR)

Addr. Offset: 0x000B Reset Value: 0x00

7	6	5	4	3	2	1	0
Reserved				I2CCSTE	I2CFTE	SCL10NSTE	SDA10NSTE
ro	ro	ro	ro	rw	rw	rw	rw

I2CCSTE — I2C Current Source Test Enable Bit

1 = I2C Current Source Test Enabled.

0 = I2C Current Source Test Disabled.

I2CFTE — I2C Filter Test Enable Bit

1 = I2C Filter Test Enabled.

0 = I2C Filter Test Disabled.

SCL10NSTE — I2C SCL 10ns Filter Test Enable Bit

1 = I2C SCL line 10ns Filter Test Enabled if I2CFTE is set.

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0 = I2C SCL line 50ns Filter Test Enabled if I2CFTE is set.

SDA10NSTE — I2C SDA 10ns Filter Test Enable Bit

1 = I2C SDA line 10ns Filter Test Enabled if I2CFTE is set.

0 = I2C SDA line 50ns Filter Test Enabled if I2CFTE is set.

13.4.2.13 I2C 10ns Filter trimming value Register (I2C10NSFTVR)

Addr. Offset: 0x000C Reset Value: 0x88

7	6	5	4	3	2	1	0
SCL10F[3:0]				SDA10F[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw

The value of this register can be written by CPU if the value in the flash information area does not match the key value. If the the key value matched, the value is overwritten by the value from flash information area.

SCL10F[3:0] — I2C SCL line 10ns Filter Trimming value SDA10F[3:0] — I2C SDA line 10ns Filter Trimming value

13.4.2.14 I2C 50ns Filter trimming value Register (I2C50NSFTVR)

Addr. Offset: 0x000D Reset Value: 0x66

7	6	5	4	3	2	1	0
SCL50F[3:0]				SDA50F[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw

The value of this register can be written by CPU if the value in the flash information area does not match the key value. If the the key value matched, the value is overwritten by the value from flash information area.

SCL50F[3:0] — I2C SCL line 50ns Filter Trimming value SDA50F[3:0] — I2C SDA line 50ns Filter Trimming value

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13.5 Functional Description

The I2C module is a two-wire, bidirectional serial bus that provides a simple, efficient method of data exchange, minimizing the interconnection between devices. Software can poll the I2C status flags or I2C operation can be interrupt driven. When a byte is received or to be transmitted, the TF bit in I2CS will be set and if the IEN bit in I2CC is also set, an interrupt will be generated.

If arbitration is lost during it's transfer, the ARBL will be set and if the IEN bit in I2CC is also set, an interrupt will be generated. An interrupt can also be generated if there is no address acknowledge from the slave (AACK set).

The module can clear ACKEN if it does not want to receive next byte.

13.5.1 Master Mode

The I2C module may initial a transfer if the bus is free (the BBUSY bit of I2CS is clear) when it works as a master. Changing the MSMOD bit from 0 to 1 generates a START on the bus and selects master mode. The master controls the direction of transfer, namely the R/W bit. The first byte of a transfer sent by the master is the slave address, the following byte is data. Change the MSMOD bit from 1 to 0 to generate a STOP on the bus if no acknowledge is received after each ninth cycle of clock. The PRE[5:0] bits in I2CP control the bit-rate clock of I2C-bus.

By configuring slave address after setting the REPSTA bit, the master can repeat the START signal instead of signalling a STOP.

13.5.2 Slave Mode

If the MSMOD bit is cleared the module is a slave and can be addressed by other masters. When a winning master is trying to address it, it will release the SDA line and switch over immediately to its slave mode.

NOTE: The I2C can't work in slave mode and master mode at the same time.

13.5.3 Protocol

The I2C communication protocol consists of six components: START, Data Source/Recipient, Data Direction, Slave Acknowledge, Data, Data Acknowledge and STOP. These are shown in **Figure 13-2** and described in the following text.

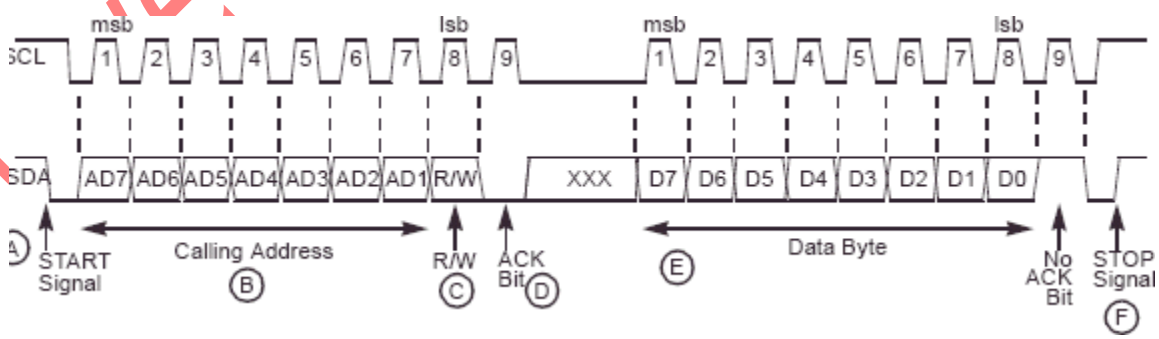


Figure 13-2: I2C Communication Protocol

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● START signal — When no other device is bus master (both SCL and SDA lines are at logic high), a device can initiate communication by sending a START signal (see A in **Figure 13-2**). A START signal is defined as a high-to-low transition of SDA while SCL is high. This signal denotes the beginning of a data transfer (each data transfer can be several bytes long) and awakens all slaves.

● Slave address transmission — The master sends the slave address in the first byte after the START signal (B). After the seven-bit calling address, it sends the R/W bit (C), which tells the slave the data transfer direction.
Each slave must have a unique address. An I2C master must not transmit an address that is the same as its own slave address; it cannot be master and slave at the same time. The slave whose address matches that sent by the master pulls SDA low at the ninth clock (D) to return an acknowledge bit.

● Data transfer — When successful slave addressing is achieved, the data transfer can proceed (E) on a byte-by-byte basis in the direction specified by the R/W bit sent by the calling master.
Data can be changed only while SCL is low and must be held stable while SCL is high, as **Figure 13-2** shows. SCL is pulsed once for each data bit, with the MSB being sent first. The receiving device must acknowledge each byte by pulling SDA low at the ninth clock; therefore, a data byte transfer takes nine clock pulses.

If it does not acknowledge the master, the slave receiver must leave SDA high. The master can then generate a STOP signal to abort the data transfer or generate a START signal (repeated start, shown in **Figure 13-3**) to start a new calling sequence.

If the master receiver does not acknowledge the slave transmitter after a byte transmission, it means end-of-data to the slave. The slave releases SDA for the master to generate a STOP or START signal.

● STOP signal — The master can terminate communication by generating a STOP signal to free the bus. A STOP signal is defined as a low-to-high transition of SDA while SCL is at logical high (F). Note that a master can generate a STOP even if the slave has made an acknowledgment, at which point the slave must release the bus.

● Instead of signalling a STOP, the master can repeat the START signal, followed by a calling command, (A in **Figure 13-3**). A repeated START occurs when a START signal is generated without first generating a STOP signal to end the communication. The master uses a repeated START to communicate with another slave or with the same slave in a different mode (transmit/receive mode), without releasing the bus.

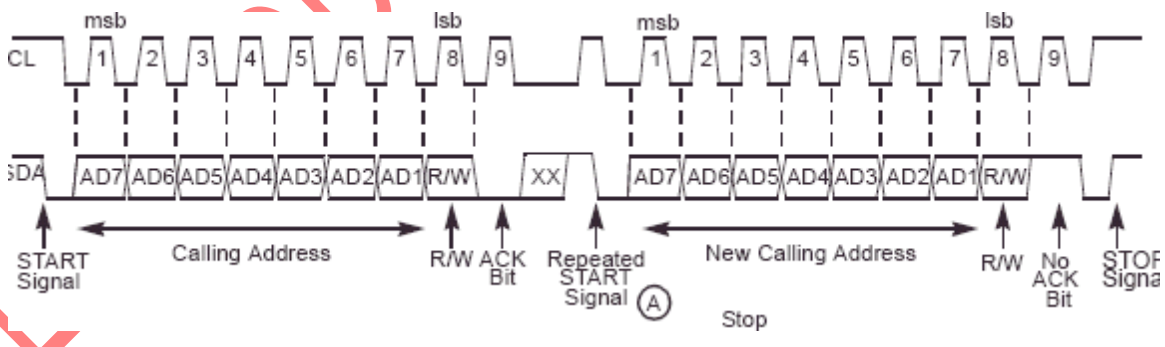


Figure 13-3: Repeat Start of I2C protocol

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13.5.4 Arbitration Procedure

If multiple devices simultaneously request the bus, the bus clock is determined by a synchronization procedure in which the low period equals the longest clock-low period among the devices and the high period equals the shortest. A data arbitration procedure determines the relative priority of competing devices. A device loses arbitration if it sends logic high while another sends logic low; it immediately switches to slave-receive mode and stops driving SDA.

A master that loses the arbitration can generate clock pulses until the end of the byte in which it loses the arbitration. It's possible that the winning master is trying to address it. The losing master must therefore switch over immediately to its slave mode.

In this case, the transition from master to slave mode does not generate a STOP condition. Meanwhile, hardware sets the ARBL bit of I2CSR to indicate loss of arbitration.

13.5.5 Clock Synchronization

Because wired-AND logic is used, a high-to-low transition on SCL affects all devices connected to the bus. Devices start counting their low period when the master drives SCL low. When a device clock goes low, it holds SCL low until the clock high state is reached. However, the low-to-high change in this device clock may not change the state of SCL if another device clock is still in its low period.

Therefore, the device with the longest low period holds the synchronized clock SCL low. Devices with shorter low periods enter a high wait state during this time (See **Figure 13-4**). When all devices involved have counted off their low period, the synchronized clock SCL is released and pulled high.

There is then no difference between device clocks and the state of SCL, so all of the devices start counting their high periods. The first device to complete its high period pulls SCL low again.

13.5.6 Handshaking

The clock synchronization mechanism can be used as a handshake in data transfers. Slave devices can hold SCL low after completing one byte transfer (9 bits). In such a case, the clock mechanism halts the bus clock and forces the master clock into wait states until the slave releases SCL.

13.5.7 Clock Stretching

Slaves can use the clock synchronization mechanism to slow down the transfer bit rate. After the master has driven SCL low, the slave can drive SCL low for the required period and then release it. If the slave SCL low period is longer than the master SCL low period, the resulting SCL bus signal low period is stretched.

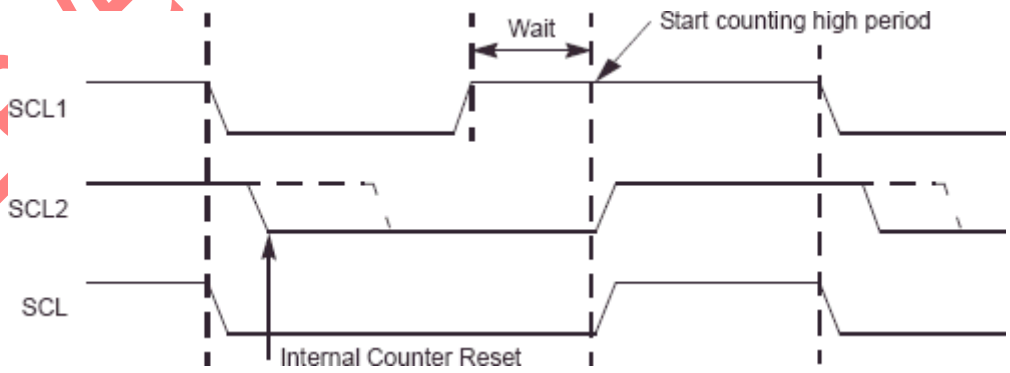


Figure 13-4: SCL Synchronization

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13.5.8 High-Speed Mode operation

The I2C module can transfer information at bit rates of up to 3.4 Mbits/s in

HS-mode, yet it remains fully downward compatible with Fast or Standard-mode (F/S-mode) devices for bi-directional communication in a mixed-speed bus system. With the exception that arbitration and clock synchronization is not performed during the HS-mode transfer, the same serial bus protocol and data format is maintained as with the F/S-mode system.

Serial data transfer format in HS-mode meets the Standard-mode I2C-bus specification. HS-mode can only commence after the following conditions are met (all of which occur while in F/S-mode):

- START condition (S)
- 8-bit master code (00001XXX)
- not-acknowledge bit (A)

(see **Figure 13-5 Data transfer format in HS mode**) and (see **Figure 13-6 A Complet HS mode transfer**) show this in more detail. The HS master code has two main functions:

- It allows arbitration and synchronization between competing masters at F/S-mode speeds, resulting in one winning master.
- It indicates the beginning of an HS-mode transfer.

HS-mode master codes are reserved 8-bit codes, which are not used for slave addressing or other purposes. Furthermore, as each master has its own unique master code, up to eight HS-mode masters can be present on the one I2C-bus system (although master code 0000 1000 should be reserved for test and diagnostic purposes). The master code for an I2C module is software programmable. Arbitration and clock synchronization only take place during the transmission of the master code and not-acknowledge bit (A), after which one winning master remains active. The master code indicates to other devices that an HS-mode transfer is to begin and the connected devices must meet the HS-mode specification. As no device is allowed to acknowledge the master code, the master code is followed by a not-acknowledge (A). After the not-acknowledge bit (A), and the SCL line has been pulled-up to a HIGH level, the active master switches to HS-mode and enables (at time tH, refer to (see **Figure 13-6 A Complet HS mode transfer**)) the current-source pull-up circuit for the SCL signal. As other devices can delay the serial transfer before tH by stretching the LOW period of the SCL signal, the active master will enable its current-source pull-up circuit when all devices have released the SCL line and the SCLsignal has reached a HIGH level, thus speeding up the last part of the rise time of the SCL signal. The active master then sends a repeated START condition (Sr)

followed by a 7-bit slave address with a R/W bit address, and receives an acknowledge bit (A) from the selected slave.

After a repeated START condition and after each acknowledge bit (A) or not-acknowledge bit (A), the active master disables its current-source pull-up cir- cuit. This enables other devices to delay the serial transfer by stretching the LOW period of the SCL signal. The active master re-enables its current-source

pull-up circuit again when all devices have released and the SCL signal reaches a HIGH level, and so speeds up the last part of the SCL signal s rise time.

Data transfer continues in Hs-mode after the next repeated START (Sr), and only switches back to F/S-mode after a STOP condition (P). To reduce the overhead of the master code, it s possible that a master links a number of HS-mode transfers, separated by repeated START conditions (Sr).

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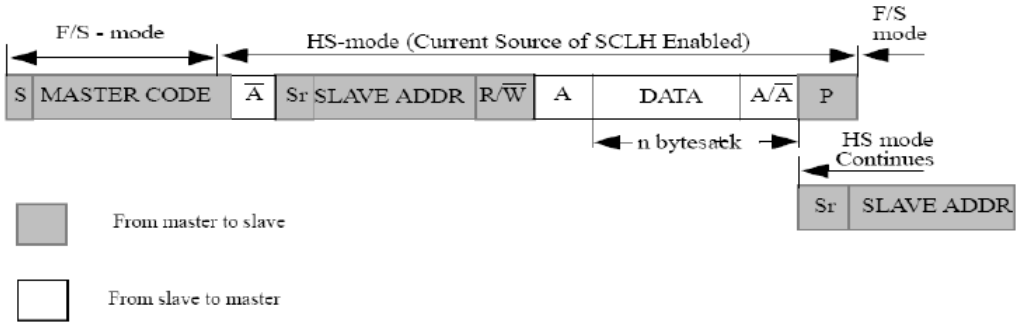


Figure 13-5: Data transfer format in HS mode

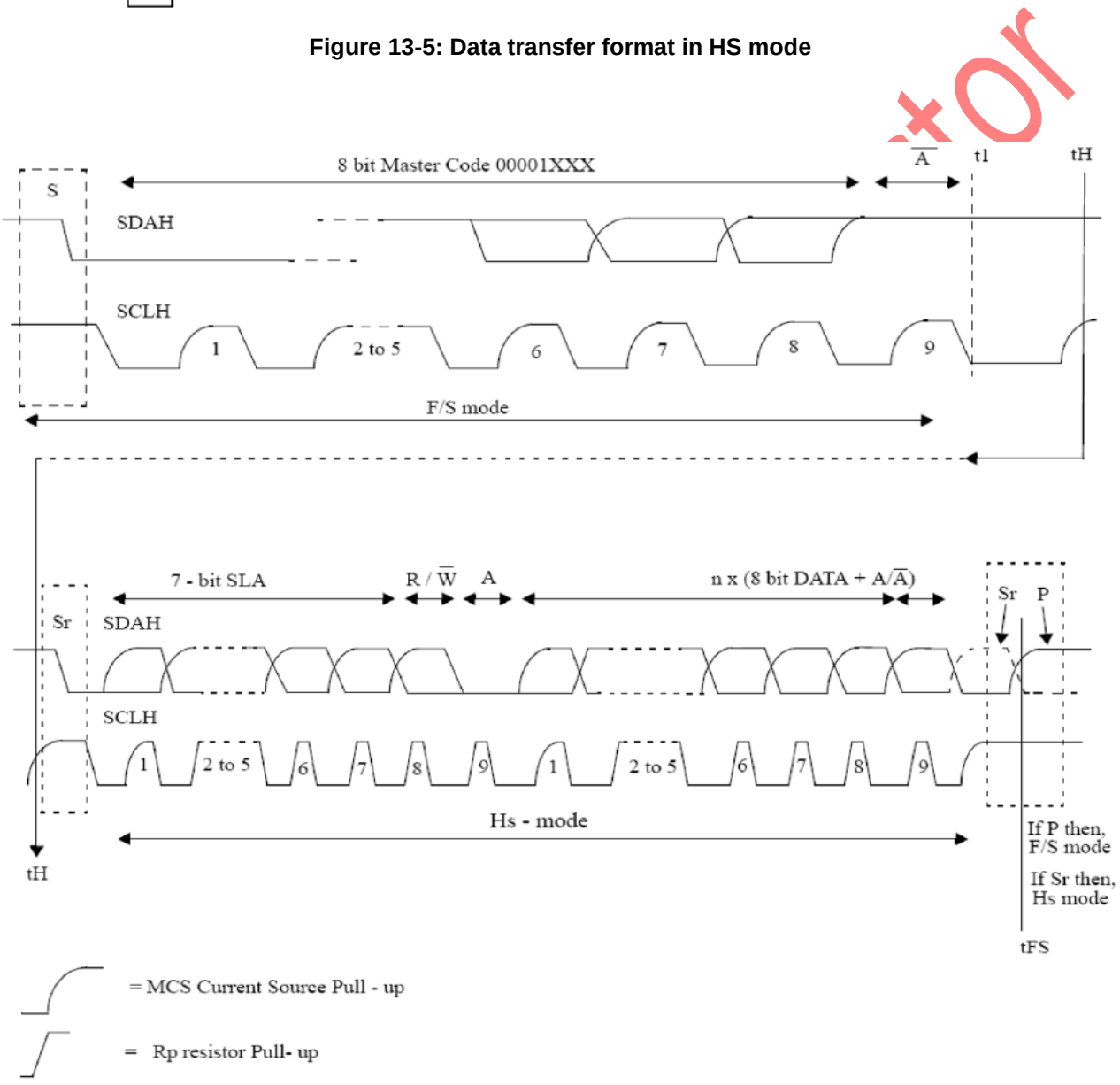


Figure 13-6: A Complet HS mode transfer

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13.5.9 Bit Addressing

The 10-bit slave address is formed from the first two bytes following a START condition (S) or a repeated START condition (Sr).The first seven bits of the first byte are the combination 11110XX of which the last two bits (XX) are the two most-significant bits (MSBs) of the 10-bit address; the eighth bit of the first byte is the R/W bit that determines the direction of the message. A ‘zero’ in the least significant position of the first byte means that the master will write information to a selected slave. A ‘one’ in this position means that the master will read information from the slave. If the R/W bit is ‘zero’, then the second byte contains the remaining 8 bits (XXXXXXXX) of the 10-bit address. If the R/W bit is ‘one’, then the next byte contains data transmitted from a slave to a master.

Various combinations of read/write formats are possible within a transfer that includes 10-bit addressing. Possible data transfer formats are:

- Master-transmitter transmits to slave-receiver with a 10-bit slave address.

The transfer direction is not changed (see Fig21-7). When a 10-bit address follows a START condition, each slave compares the first seven bits of the first byte of the slave address (11110XX) with its own address and tests if the eighth bit (R/W direction bit) is 0. It is possible that more than one device will find a match and generate an acknowledge (A1). All slaves that found a match will compare the eight bits of the second byte of the slave address (XXXXXXXX) with their own addresses, but only one slave will find a match and generate an acknowledge (A2). The matching slave will remain addressed by the master until it receives a STOPcondition (P) or a repeated START condition (Sr) followed by a different slave address.

- Master-receiver reads slave- transmitter with a 10-bit slave address.

The transfer direction is changed after the second R/W bit (Fig21-8). Up to and including acknowledge bit A2, the procedure is the same as that described for a master-transmitter addressing a slave-receiver. After the repeated START condition (Sr), a matching slave remembers that it was addressed before. This slave then checks if the first seven bits of the first byte of the slave address following Sr are the same as they were after the START condition (S), and tests if the eighth (R/W) bit is 1. If there is a match, the slave considers that it has been addressed as a transmitter and generates acknowledge A3. The slave-transmitter remains addressed until it receives a STOP condition (P) or until it receives another repeated START condition (Sr) followed by a different slave address. After a repeated START condition (Sr), all the other slave devices will also compare the first seven bits of the first byte of the slave address (11110XX) with their own addresses and test the eighth (R/W) bit. However, none of them will be addressed because R/W = 1 (for 10-bit devices), or the 11110XX slave address (for 7-bit devices) does not match.

- Combined format. A master transmits data to a slave and then reads data from the same slave (Fig21-9). The same master occupies the bus all the time. The transfer direction is changed after the second R/W bit.
- Combined format. A master transmits data to one slave and then transmits data to another slave (Fig21-10). The same master occupies the bus all the time.
- Combined format. 10-bit and 7-bit addressing combined in one serial transfer (Figure 13-11). After each START condition (S), or each repeated START condition (Sr), a 10-bit or 7-bit slave address can be transmitted. Figure 13-11 shows how a master transmits data to a slave with a 7-bit address and then transmits data to a second slave with a 10-bit address. The same master occupies the bus all the time.

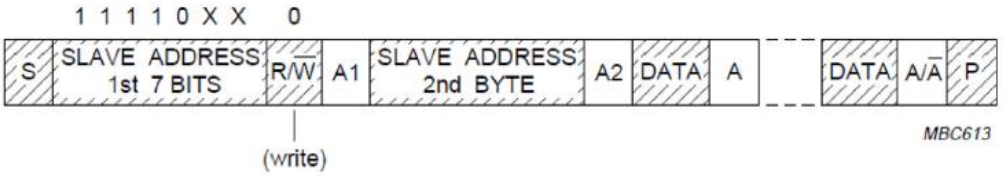


Figure 13-7: A master-transmitter addresses a slave-receiver with a 10-bit address

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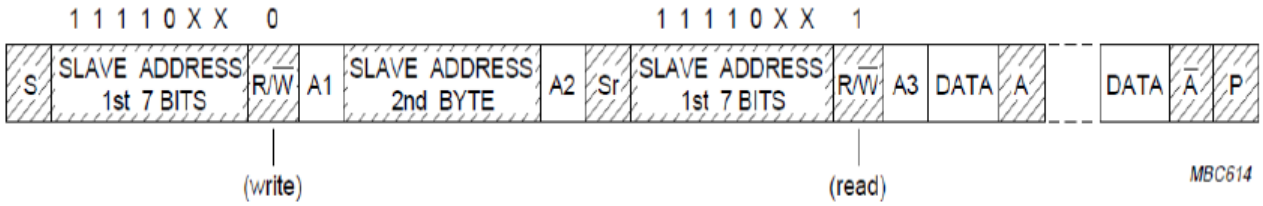


Figure 13-8: A master-receiver addresses a slave-transmitter with a 10-bit address

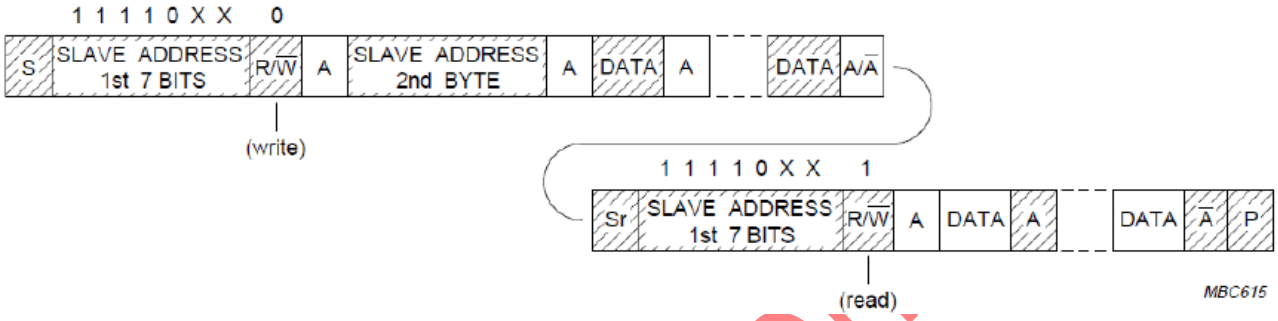


Figure 13-9: Combined format. A master addresses a slave with a 10-bit address,then transmits data to this slave and reads data from this slave

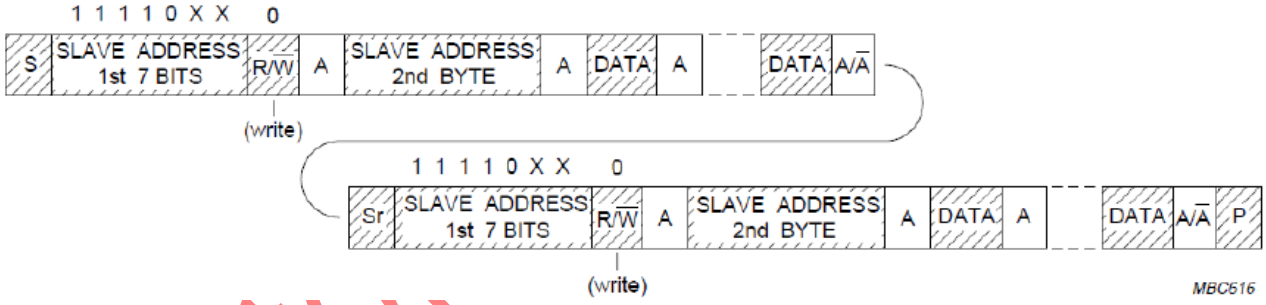


Figure 13-10: Combined format. A master transmits data to two slaves, both with 10-bit addresses

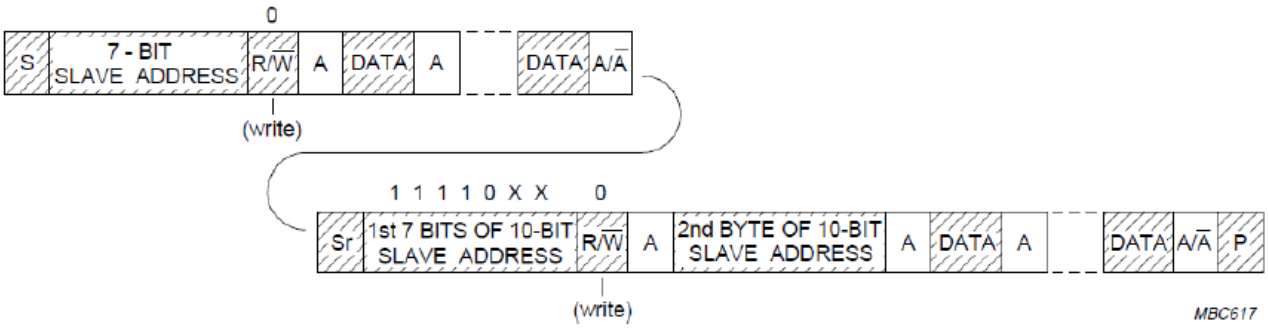


Figure 13-11: Combined format. A master transmits data to two slaves, one with a 7-bit address,and one with a 10-bit address

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13.5.10 Software Transaction Flow Diagrams

- Initialization.

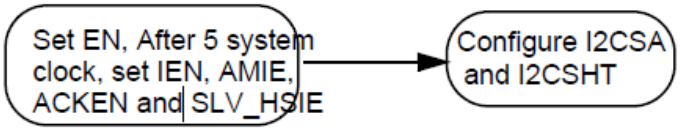


Figure 13-12: Slave Mode Initialization

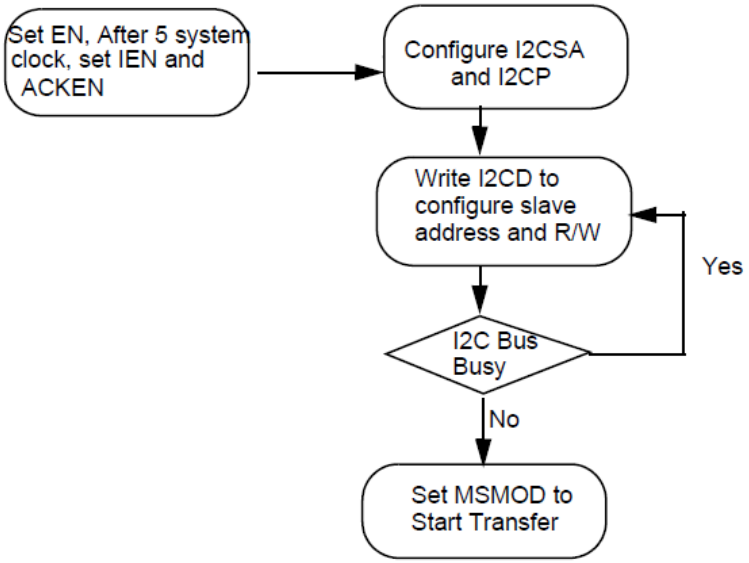


Figure 13-13: Master Mode Initialization

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- Interrupt transaction

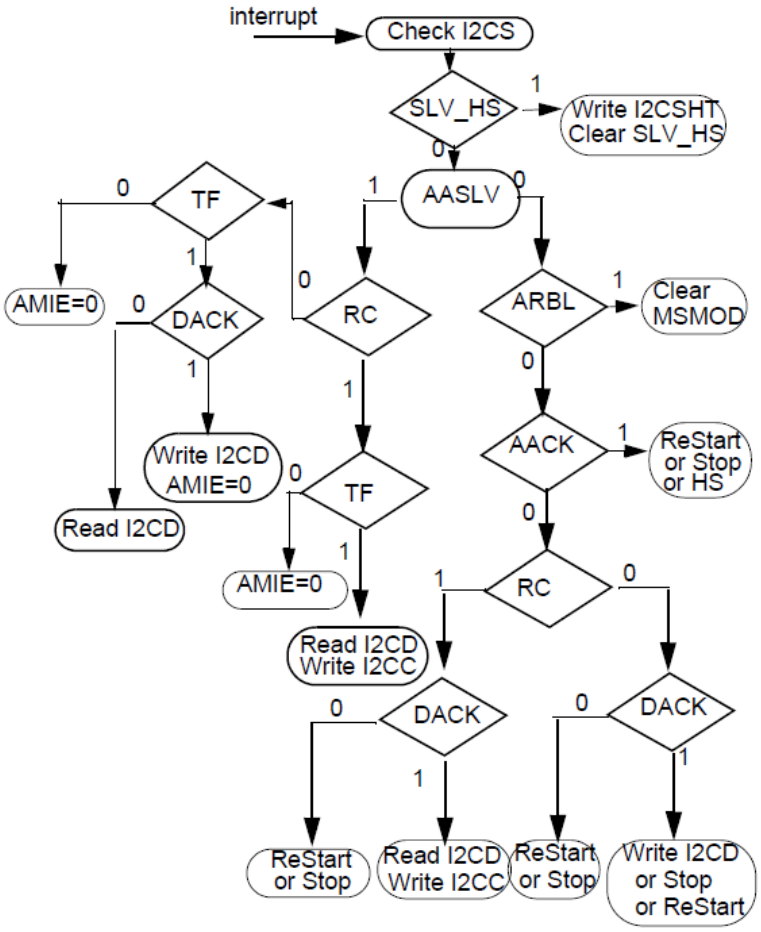


Figure 13-14: Interrupt Transaction

If there is a interruption comes from I2C,first check I2CS to confirm whether the I2C is in the master or slave mode.

First check SLV_HS status, if SLV_HS is setted, then write I2CSHT to configure High-Speed Timing and write 1 to I2CSHIR to clear SLV_HS, otherwise:

In the slave mode, if RC has been setted,means I2C is in the slave-receiver mode then check TF,if TF has also been setted,means data(not address)has been received, then read I2CD and write I2CC to clear RC and TF to receive next byte data.If TF has not been setted, means only slave address has been received,then write I2CC to clear RC and AMIE then receive next byte data.

In the slave mode,if RC has not been setted,then check TF, if TF has been setted, then check DACK,if DACK has also been setted,means I2C is in the slave-transmit mode, then write I2CD to clear TF, write I2CC to clear AMIE and transmit next byte data.if DACK has not been setted, then read I2CD for last received byte to clear TF.

In the slave mode, if both the TF and RC are not setted, then write I2CC to clear AMIE.

In the master mode,if ARBL has been setted,means arbitration lost has been occurred, then write I2CC to clear MSMOD.if ARBL has not been setted then check AACK,if AACK has been setted,means address acknowledge error,then write I2CC to repeat start or stop.

In the master mode, if ARBL and AACK has not been setted, then check RC,if RC has been setted,means I2C is in the mast-receiver mode, then check DACK,if DACK has also been setted,then read I2CD to clear RC and write I2CC to choice whether acknowledge the data.if DACK has not been setted,write I2CC to repeat start or

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stop.

In the master mode,if ARBL and AACK has not been setted, then check RC,if RC has not been setted,means I2C is in the master-transmit mode,then check DACK,if DACK has been setted, write I2CD to clear TF and transmit next byte data or write I2CC to repeat start or stop.if DACK has not been setted,then write I2CC to repeat start or stop.

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14 Reset Controller Module

14.1 Overview

The reset controller is provided to determine the cause of reset, assert the appropriate reset signals to the system, and then to keep a history of what caused the reset.The low-voltage detect (LVD) and high-voltage detect(HVD) control and status bits are implemented in the reset module.

14.2 Features

Module features include:

- Six sources of reset:
- Power on reset
- Software
- Timer overflow
- USI Reset
- Watchdog Timer Reset
- Low and High Voltage Detect Reset
- Low and High Frequency Detect Reset
- Software-assertable RSTOUT pin independant of chip reset state
- Software-readable status flags indicating the cause of the last reset

14.3 Block Diagram

Figure 14-1 illustrates the reset controller .

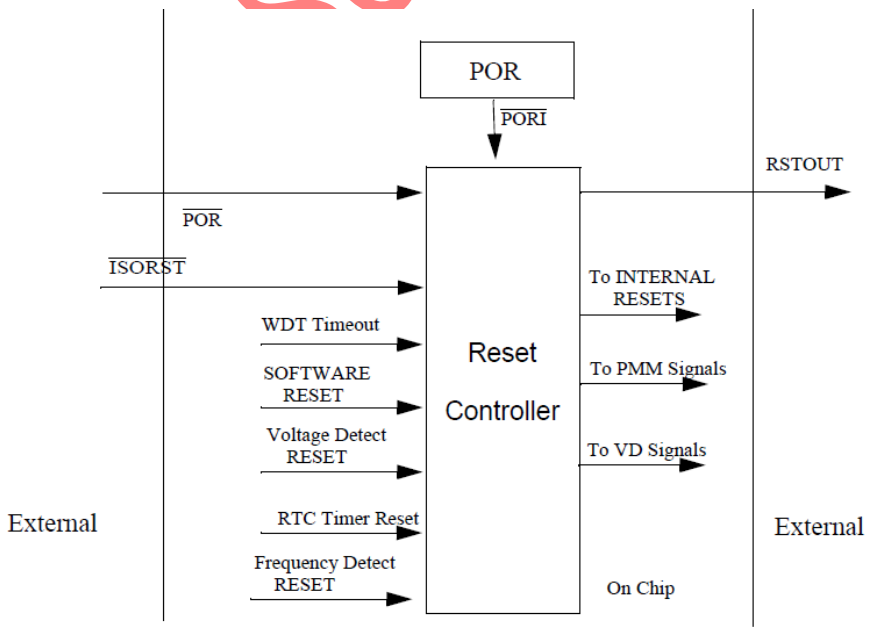


Figure 14-1: Reset Controller Block Diagram

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14.4 Signals

Table 14-1 provides a summary of the reset controller signal properties. The signals are described in the following paragraphs.

Table 14-1: Reset Controller Signal Properties

Name	Direction	Input Hysteresis	Input Synchronization
POR pin	I	Y	N
RSTOUT pin	O	—	—

14.4.1 POR

Asserting the external POR pin will causes the system to reset immediately.

14.4.2 RSTOUT

This active-low output signal is driven low when the internal reset controller module resets the chip. When RSTOUT is active, the user can drive override options on the data bus.

14.5 Memory Map and Registers

The reset controller programming model consists of these registers:

- Reset Control Register (RCR) Selects reset controller functions
- Reset Status Register (RSR) reflects the state of the last reset source

See **Table 14-2** for the address map and the following paragraphs for a description of the registers.

Table 14-2: Reset Controller Address Map

Address Offset	Bit 7:0	Access [1]
0x0000	RCR — Reset Control Register	S/U
0x0007	RSR — Reset Status Register	S/U
0x0006	RTR — Reset Test Register	S/U

NOTE: S/U = supervisor or user mode access.

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14.5.1 Reset Control Register

The Reset Control Register (RCR) allows software control for requesting a reset, for independently asserting the external RSTOUT pin, and for controlling voltage detect (VD) and frequency detect(FD) functions, and SIM card reset enable.

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
SOFTRST	FRCRSTOUT	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
CRE	CRWE	Reserved					
rw	rw	ro					
15	14	13	12	11	10	9	8
Reserved				Reserved	Reserved	Reserved	Reserved
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
Reserved				Reserved	Reserved	Reserved	Reserved
ro	ro	ro	ro	ro	ro	ro	ro

Figure 14-2: Reset Control Register(RCR)

NOTE: Only Power-On Reset can reset LVDRE, LVDE, HVDRE, HVDE, LFDRE, LFDE, HFDRE, HFDE, CRE, CRWE, USIPORTCS[3:0], SCKPORTCS[1:0] bits.

SOFTRST — Software Reset Request

The SOFTRST bit allows software to request a reset. The reset caused by setting this bit clears this bit.

1 = Software reset request

0 = No software reset request

FRCRSTOUT — Force RSTOUT Pin

The FRCRSTOUT bit allows software to assert or negate the external RSTOUT pin.

1 = Assert RSTOUT pin

0 = Negate RSTOUT pin

CRE — SIM Card Reset Enable

The CRE bit allows SIM card reset pin to request a reset.

1 = SIM Card Reset Enable

0 = SIM Card Reset Disable

CRWE — SIM Card Reset Wakeup Enable

The CRWE bit allows SIM card reset pin to wakeup system.

1 = SIM Card Reset wakeup Enable

0 = SIM Card Reset wakeup Disable

NOTE: CRE and CRWE is only active for USI1.

LVDF — LVD Flag

The LVDF bit indicates the low-voltage detect status if LVDE is set.

1 = Low voltage has been detected.

0 = Low voltage has not been detected.

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NOTE: The setting of this flag causes an LVD interrupt if LVDE and LVDIE bits are set and LVDRE is cleared when the analog supply voltage AV_{DD} drops below V_{DD} (minimum).

LVDIE — LVD Interrupt Enable

The LVDIE bit controls the LVD interrupt if LVDE is set. This bit has no effect if the LVDE bit is a logic 0.

1 = LVD interrupt enabled

0 = LVD interrupt disabled

LVDRE — LVD Reset Enable

The LVDRE bit controls the LVD reset if LVDE is set. This bit has no effect if the LVDE bit is a logic 0. LVD reset has priority over LVD interrupt., if both are enabled.

1 = LVD reset enabled

0 = LVD reset disabled

LVDE — LVD Enable

The LVDE bit controls whether the LVD is enabled.

1 = LVD is enabled

0 = LVD is disabled

HVDF — HVD Flag

The HVDF bit indicates the high-voltage detect status if HVDE is set.

1 = High voltage has been detected.

0 = High voltage has not been detected.

NOTE: The setting of this flag causes an HVD interrupt if HVDE and HVDIE bits are set and HVDRE is cleared when the analog supply voltage AV_{DD} rise above V_{DD} (maximum).

HVDIE — HVD Interrupt Enable

Teh HVDIE bit controls the HVD interrupt if HVDE is set. This bit has no effect if the HVDE bit is a logic 0.

1 = HVD interrupt enabled

0 = HVD interrupt disabled

HVDRE — HVD Reset Enable

The HVDRE bit controls the HVD reset if HVDE is set. This bit has no effect if the HVDE bit is a logic 0. HVD reset has priority over HVD interrupt., if both are enabled.

1 = HVD reset enabled

0 = HVD reset disabled

HVDE — HVD Enable

The HVDE bit controls whether the HVD is enabled.

1 = HVD is enabled

0 = HVD is disabled

LFDF — LFD Flag

The LFDF bit indicates the low-frequency detect status if LFDE is set.

1 = Low frequency has been detected.

0 = Low frequency has not been detected.

NOTE: The setting of this flag causes an LFD interrupt if LFDE and LFDIE bits are set and LFDRE is cleared when the card clock frequency lower than 100KHz.

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LFDIE — LFD Interrupt Enable

Teh LFDIE bit controls the LFD interrupt if LFDE is set. This bit has no effect if the LFDE bit is a logic 0.

1 = LFD interrupt enabled

0 = LFD interrupt disabled

LFDRE — LFD Reset Enable

The LFDRE bit controls the LFD reset if LFDE is set. This bit has no effect if the LFDE bit is a logic 0. LFD reset has priority over LFD interrupt., if both are enabled.

1 = LFD reset enabled

0 = LFD reset disabled

LFDE — LFD Enable

The LFDE bit controls whether the LFD is enabled.

1 = LFD is enabled

0 = LFD is disabled

HFDF — HFD Flag

The HFDF bit indicates the high-frequency detect status if HFDE is set.

1 = High frequency has been detected.

0 = High frequency has not been detected.

NOTE: The setting of this flag causes an HFD interrupt if HFDE and HFDIE bits are set and HFDRE is cleared when the card clock frequency higher than 10MHz.

HFDIE — HFD Interrupt Enable

Teh HFDIE bit controls the HFD interrupt if HFDE is set. This bit has no effect if the HFDE bit is a logic 0.

1 = HFD interrupt enabled

0 = HFD interrupt disabled

HFDRE — HFD Reset Enable

The HFDRE bit controls the HFD reset if HFDE is set. This bit has no effect if the HFDE bit is a logic 0. HFD reset has priority over HFD interrupt., if both are enabled.

1 = HFD reset enabled

0 = HFD reset disabled

HFDE — HFD Enable

The HFDE bit controls whether the HFD is enabled.

1 = HFD is enabled

0 = HFD is disabled

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14.5.2 Reset Status Register

The Reset Status Register(RSR) contains a status bit for every reset source. When reset is entered, the cause of the reset condition is latched along with a value of 0 for the other reset sources that were not pending at the time of the reset condition. These values are then reflected in RSR. One or more status bits may be set at the same time. The cause of any subsequent reset is also recored in the register, overwriting status from the previous reset condition.

RSR can be read at any time. Writing to RSR has no effect.

Addr. Offset: 0x0007 Reset Value: 0x00

7	6	5	4	3	2	1	0
TCR	Reserved	SOFT	WDR	POR	Reserved	CR	Reserved
ro	ro	ro	ro	ro	ro	ro	ro

Figure 14-3: Reset Status Register(RSR)

TCR — TC Timer Reset Flag

This bit indicates that the last reset state was caused by a TC timer timeout.

1 = Last reset state was caused by TC timer timeout.

0 = Last reset state was not caused by TC timer timeout.

VD — Voltage Detect

This bit indicates that the last reset state was caused by an VD reset.

1 = Last reset state was caused by an VD reset

0 = Last reset state was not caused by an VD reset

FD — Frequency Detect

This bit indicates that the last reset state was caused by an FD reset.

1 = Last reset state was caused by an FD reset

0 = Last reset state was not caused by an FD reset

CR — SIM Card Reset

This bit indicates that the last reset state was caused by an SIM Card reset.

1 = Last reset state was caused by SIM Card reset

0 = Last reset state was not caused by SIM Card reset

SOFT — Software Reset Flag

This bit indicates that the last reset state was caused by software.

1 = Last reset state was caused by software.

0 = Last reset state was not caused by software.

WDR — Watchdog Timer Reset Flag

This bit indicates that the last reset state was caused by a watchdog timer timeout.

1 = Last reset state was caused by watchdog timer timeout.

0 = Last reset state was not caused by watchdog timer timeout.

POR — External Power-On Reset Flag

This bit indicates that the last reset state was caused by a external power-on reset.

1 = Last reset state was caused by power-on reset.

0 = Last reset state was not caused by power-on reset.

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14.5.3 Reset Test Register

The Reset Test Register(RTR) is only for factory testing and is read-only when not in test mode.

Addr. Offset: 0x0006 Reset Value: 0x00

7	6	5	4	3	2	1	0
FPOR	Reserved						Reserved
rw	ro						ro

Figure 14-4: Reset Test Register(RTR)

FPOR — Force Power On Reset

Write the bit will result in sytem power on reset.

14.6 Functional Description

14.6.1 Reset Sources

Table 14-5 defines the sources of reset and the signals driven by the reset controller.

Table 14-3: Reset Source Summary

Source	Type
POR pin	Asynchronous
ISORST pin	Asynchronous
Watchdog timer	Synchronous
TC timer	Asynchronous
Software	Synchronous
HVD	Aynchronous
LVD	Aynchronous

To protect data integrity, a synchronous reset source is not acted upon by the reset control logic until the end of the current bus cycle. Reset is then asserted on the next rising edge of the system clock after the cycle is terminated. Whenever the reset control logic must synchronize reset to the end of the bus cycle, the internal bus monitor is automatically enabled regardless of the BME bit state in the chip configuration module CCR register. Then, if the current bus cycle is not terminated normally the bus monitor terminates the cycle based on the length of time programmed in the BMT field of the CCR register.

Internal single-byte, half-word, or word writes are guaranteed to complete without data corruption when a synchronous reset occurs. External writes, including word writes to 16-bit ports, are also guaranteed to complete.

Asynchronous reset sources usually indicate a catastrophic failure. Therefore, the reset control logic does not wait for the current bus cycle to complete. Reset is asserted immediately to the system.

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14.6.1.1 Power-On Reset (POR pin)

At power up, the reset controller asserts RSTOUT. RSTOUT continues to be asserted until POR has reached a minimum acceptable level. Then after many cycles, RSTOUT is negated and the part begins operation.

14.6.1.2 SIM Card Reset

A SIM Card Reset pin causes resert request to be recognized and latched. The bus monitor is enabled and the current busy cycle is completed. If the LRESET pin is negated, the reset controller asserts RSTOUT many cycles. Then the part exits resert and begins operation.

14.6.1.3 Watchdog Timer Reset

A watchdog timer timeout causes timer resert request to be recognized and latched. The bus monitor is enabled and the current busy cycle is completed. If the LRESET pin is negated, the reset controller asserts RSTOUT many cycles. Then the part exits resert and begins operation.

14.6.1.4 TC Timer Reset

A TC timer timeout causes timer resert request to be recognized and latched. The bus monitor is enabled and the current busy cycle is completed. If the LRESET pin is negated, the reset controller asserts RSTOUT many cycles. Then the part exits resert and begins operation.

14.6.1.5 Software Reset

A software reset occurs when the SOFTRST bit is set. If the RESET pin is negated, the reset controller asserts RSTOUT many cycles. Then the part exits reset and resumes operation.

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14.6.2 Reset Control Flow

The reset logic control flow is shown in **Figure 14-7**. All cycle counts given are approximate.

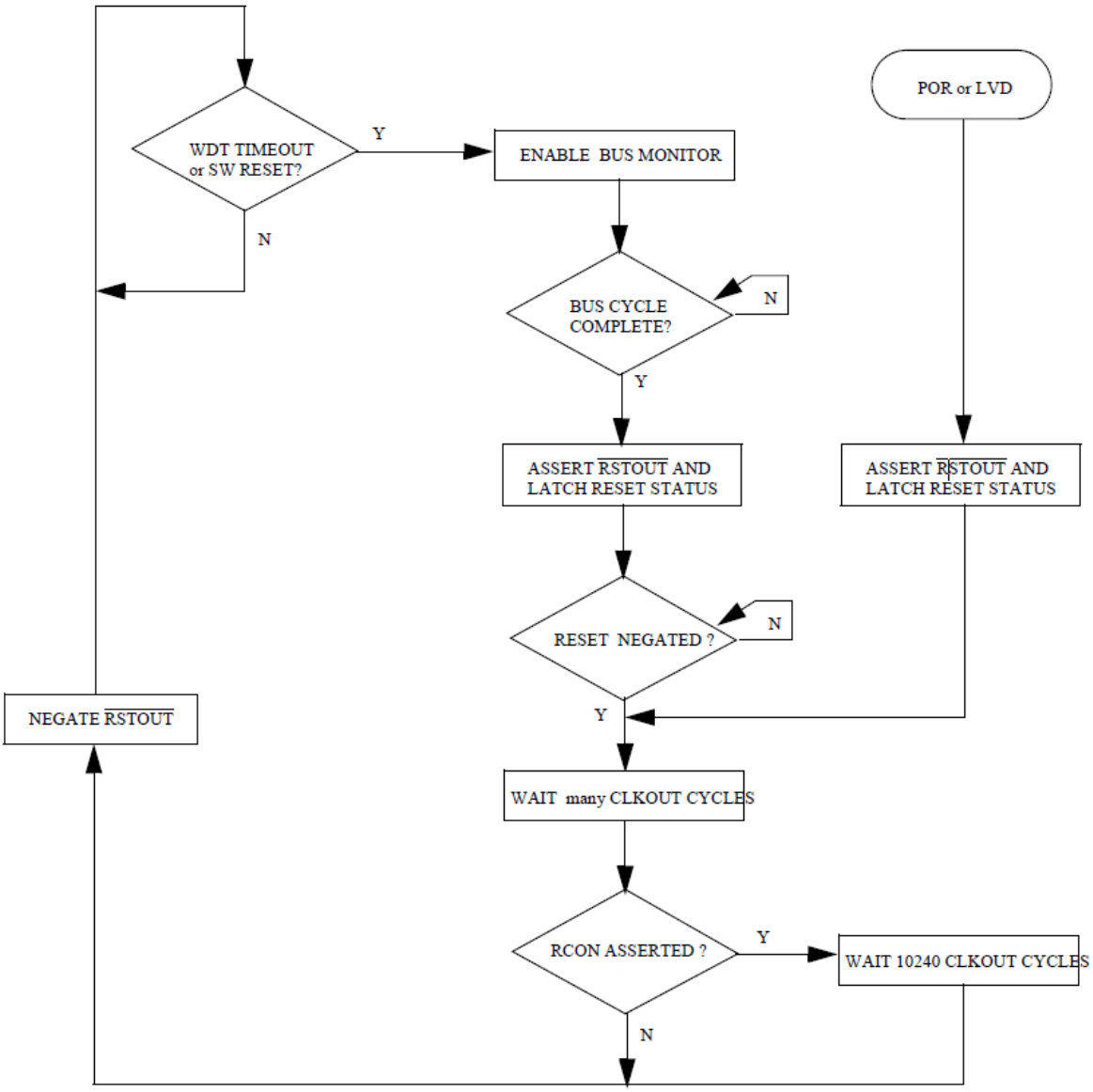


Figure 14-5: Reset Control Flow

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15 Analog-to-digital converter (ADC)

15.1 Introduction

The 12-bit ADC is a successive approximation analog-to-digital converter. It has up to 9 channels allowing it to measure signals from 8 external and 1 internal sources. A/D conversion of the various channels can be performed in single, continuous, scan or discontinuous mode. The results of the ADC are stored in a 12bit x 8depth FIFO, and the data format can be left-aligned or right-aligned.

The analog watchdog feature allows the application to detect if the input voltage goes outside the user-defined higher or lower thresholds.

An efficient low power mode is implemented to allow very low consumption at low frequency.

15.2 ADC main features

- High performance
  - 12-bit, 10-bit, 8-bit or 6-bit configurable resolution
  - ADC conversion time: 1.0 μs for 12-bit resolution (1 MHz), 0.88 μs conversion time for 10 bit resolution, faster conversion times can be obtained by lowering resolution.
  - Programmable sampling time
  - Data alignment with built-in data coherency
  - DMA support
- Low power
  - Application can reduce PLCK frequency for low power operation while still keeping optimum ADC performance. For example, 1.0 μs conversion time is kept, whatever the frequency of PCLK.
  - Wait mode: prevents ADC overrun in applications with low frequency PCLK
  - Auto off mode: ADC is automatically powered off except during the active conversion phase. This dramatically reduces the power consumption of the ADC.
- Analog input channels
  - 8 external analog inputs
  - 1 channel for internal vbattery detect
- Start-of-conversion can be initiated:
  - By software
  - By hardware trigger with configurable polarity
- Conversion modes
  - Can convert a single channel or can scan a sequence of channels.
  - Single mode converts selected inputs once per trigger
  - Continuous mode converts selected inputs continuously
  - Discontinuous mode
- Interrupt generation at the end of sampling, end of conversion, end of sequence conversion, and in case of analog watchdog or overrun events.
- Analog watchdog
- Single-ended and differential-input configurations
- Converter uses an internal reference or an external reference
- Data gather function compatible with DMA

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15.3 ADC functional description

Figure 15-1shows the ADC block diagram.

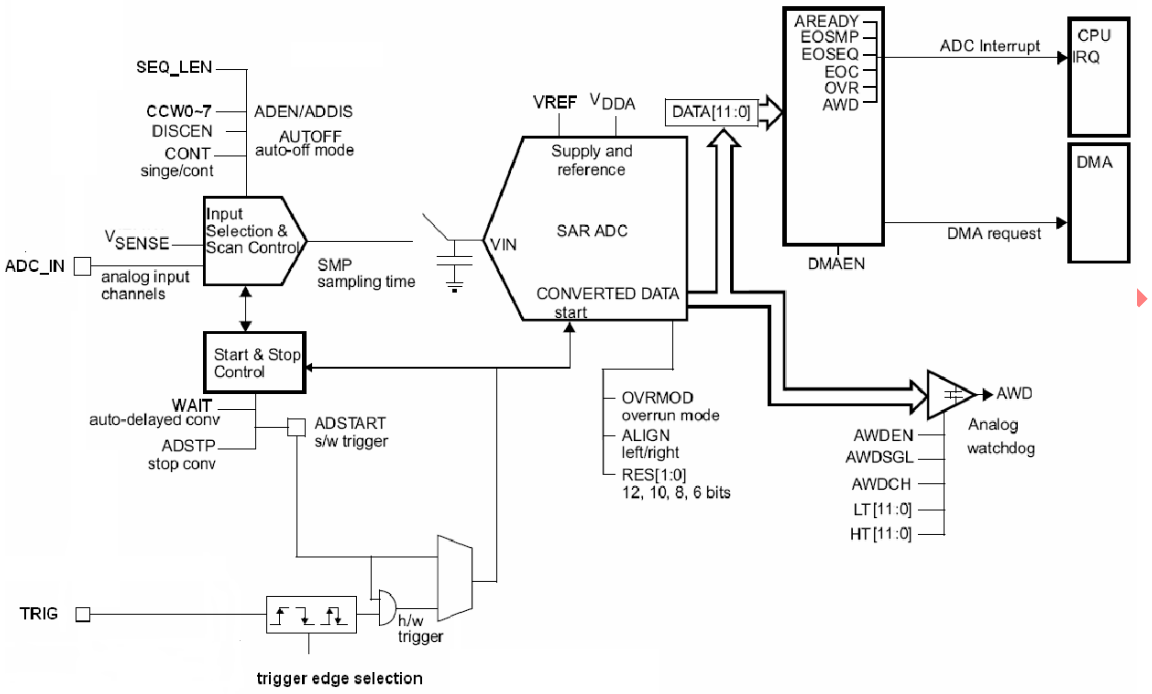


Figure 15-1: ADC block diagram

15.3.1 ADC on-off control (ADEN, ADDIS, ADRDY)

At MCU power-up, the ADC is disabled and put in power-down mode (ADEN=0). As shown in **Figure 15-2**, the ADC needs a stabilization time of tSTAB(~2.0 μs) before it starts converting accurately.

Two control bits are used to enable or disable the ADC:

- Set ADEN=1 to enable the ADC. The ADRDY flag is set as soon as the ADC is ready for operation.
- Set ADDIS=1 to disable the ADC and put the ADC in power down mode. The ADEN and ADDIS bits are then automatically cleared by hardware as soon as the ADC is fully disabled.

Conversion can then start either by setting ADSTART=1 or when an external trigger event occurs if triggers are enabled.

Follow this procedure to enable the ADC:

- Set ADEN=1 in the ADC_CR register.
- Wait until ADRDY=1 in the ADC_ISR register (ADRDY is set after the ADC startup time). This can be handled by interrupt if the interrupt is enabled by setting the ADRDYIE bit in the ADC_IER register.

Follow this procedure to disable the ADC:

- Check that ADSTART=0 in the ADC_CR register to ensure that no conversion is ongoing. If required, stop any ongoing conversion by writing 1 to the ADSTP bit in the the ADC_CR register and waiting

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until this bit is read at 0.

- Set ADDIS=1 in the ADC_CR register.
- If required by the application, wait until ADEN=0 in the ADC_CR register, indicating that the ADC is fully disabled (ADDIS is automatically reset once ADEN=0).

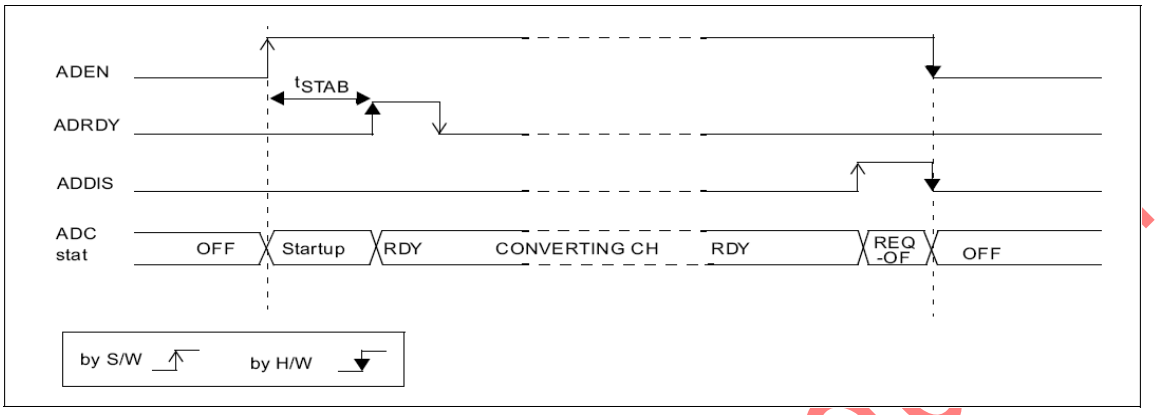


Figure 15-2: Enabling/disabling the ADC

NOTE: In auto-off mode (AUTOFF=1) the power-on/off phases are performed automatically, by hardware and the ADRDY flag is not set.

15.3.2 ADC clock

The ADC has a dual clock-domain architecture, as show in **Figure 15-3**, this has the advantage of reaching the maximum ADC clock frequency whatever the IPG clock scheme selected.

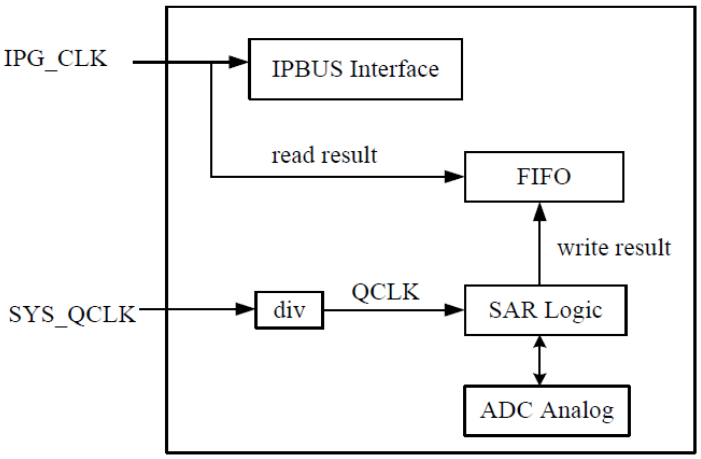


Figure 15-3: ADC clock scheme

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15.3.3 Configuring the ADC

Software must write to the ADEN bit in the ADC_CR register if the ADC is disabled (ADEN must be 0).

Software must only write to the ADSTART and ADDIS bits in the ADC_CR register only if the ADC is enabled and there is no pending request to disable the ADC (ADEN = 1 and ADDIS = 0).

For all the other control bits in the ADC_IER, ADC_CFGRi, ADC_SMPR, ADC_TR, ADC_CHSELRi and ADC_WDG registers, software must only write to the configuration control bits if there is no conversion ongoing (ADSTART = 0).

Software must only write to the ADSTP bit in the ADC_CR register if the ADC is enabled (and possibly converting) and there is no pending request to disable the ADC (ADSTART = 1 and ADDIS = 0) .

NOTE: There is no hardware protection preventing software from making write operations forbidden by the above rules. If such a forbidden write access occurs, the ADC may enter an undefined state. To recover correct operation in this case, the ADC must be disabled(ADDIS = 1).

15.3.4 Channel selection (CCWi)

There are up to 17 multiplexed channels:

- 16 analog inputs from pins (ADC_IN0...ADC_IN15)
- 1 internal analog input (Temperature Sensor)

It is possible to convert a single channel or to automatically scan a sequence of channels.

The sequence of the channels to be converted is organized as CCW[0], then CCW[1], ..., then CCW[7]. The CCWi are programed in ADC_CHSELRi. The sequence length is programmed in SEQ_LEN[2:0] of ADC_CFGR1. For example, if sequence length is set as 3, then the sequence is organized as CCW[0], then CCW[1], then CCW[2].

The channel decode is shown in **Table 15-1**.

Table 15-1: Channel decode

CCWi[4:0]	channel select
5'b00000	ADC_IN0
5'b00001	ADC_IN1
5'b00010	ADC_IN2
5'b00011	ADC_IN3
5'b00100	ADC_IN4
5'b00101	ADC_IN5
5'b00110	ADC_IN6
5'b00111	ADC_IN7
5'b01000	ADC_IN8
5'b01001	ADC_IN9
5'b01010	ADC_IN10
5'b01011	ADC_IN11
5'b01100	ADC_IN12
5'b01101	ADC_IN13
5'b01110	ADC_IN14

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CCWi[4:0]	channel select
5'b01111	ADC_IN15
5'b10000	VBATTERY DETECT

15.3.5 Programmable sampling time (SMP)

Before starting a conversion, the ADC needs to establish a direct connection between the voltage source to be measured and the embedded sampling capacitor of the ADC. This sampling time must be enough for the input voltage source to charge the sample and hold capacitor to the input voltage level.

Having a programmable sampling time allows to trim the conversion speed according to the input resistance of the input voltage source. The ADC samples the input voltage for a number of ADC clock cycles that can be modified using the SMP[3:0] bits in the ADC_SMPR register. This programmable sampling time is common to all channels.

The ADC indicates the end of the sampling phase by setting the EOSMP flag.

15.3.6 Single conversion mode (CONT=0)

In Single conversion mode, the ADC converts the sequence once. This mode is selected when CONT=0 in the ADC_CFGR1 register. Conversion is started by either:

- Setting the ADSTART bit in the ADC_CR register
- Hardware trigger event

Inside the sequence, after each conversion is complete:

- The converted data are stored in the FIFO
- The EOC (end of conversion) flag is set

An interrupt is generated if the EOCIE bit is set After the sequence of conversions is complete:

- The EOSEQ (end of sequence) flag is set
- An interrupt is generated if the EOSEQIE bit is set

Then the ADC stops until a new external trigger event occurs or the ADSTART bit is set again.

NOTE: To convert a single channel once, program a sequence with a length of 1.

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15.3.7 Continuous conversion mode (CONT=1)

In continuous conversion mode, when a software or hardware trigger event occurs, the ADC performs a sequence of conversions, converting the sequence once and then automatically re-starts and continuously performs the same sequence of conversions. This mode is selected when CONT=1 in the ADC_CFGR1 register. Conversion is started by either:

- Setting the ADSTART bit in the ADC_CR register
- Hardware trigger event

Inside the sequence, after each conversion is complete:

- The converted data are stored in the FIFO
- The EOC (end of conversion) flag is set

An interrupt is generated if the EOCIE bit is set After the sequence of conversions is complete:

- The EOSEQ (end of sequence) flag is set
- An interrupt is generated if the EOSEQIE bit is set

Then, a new sequence restarts immediately and the ADC continuously repeats the conversion sequence.

NOTE: It is not possible to have both discontinuous mode and continuous mode enabled: it is forbidden to set both bits DISCEN=1 and CONT=1.

15.3.8 Starting conversions (ADSTART)

Software starts ADC conversions by setting ADSTART=1. When ADSTART is set, the conversion:

- Starts immediately if TRIGMODE = 0x0 (software trigger)
- At the next active edge of the selected hardware trigger if TRIGMODE ≠ 0x0

The ADSTART bit is also used to indicate whether an ADC operation is currently ongoing. It is possible to re-configure the ADC while ADSTART=0, indicating that the ADC is idle.

The ADSTART bit is cleared by hardware:

- In single mode with software trigger
- At any end of conversion sequence (EOSEQ=1)
- In discontinuous mode with software trigger
- At any end of conversion
- In all cases
- After execution of the ADSTP procedure invoked by software

NOTE:

- In continuous mode (CONT=1), the ADSTART bit is not cleared by hardware when the EOSEQ flag is set because the sequence is automatically relaunched.
- When hardware trigger is selected in single mode, ADSTART is not cleared by hardware when the EOSEQ flag is set. This avoids the need for software having to set the ADSTART bit again and ensures the next trigger event is not missed.

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15.3.9 Timings

The elapsed time between the start of a conversion and the end of conversion is the sum of the configured sampling time plus the successive approximation time depending on data resolution:

tADC = tSMPL + tSAR = [4|min + 12|12bit] x tQCLK = 1µs |min (for fQCLK = 16 MHz)

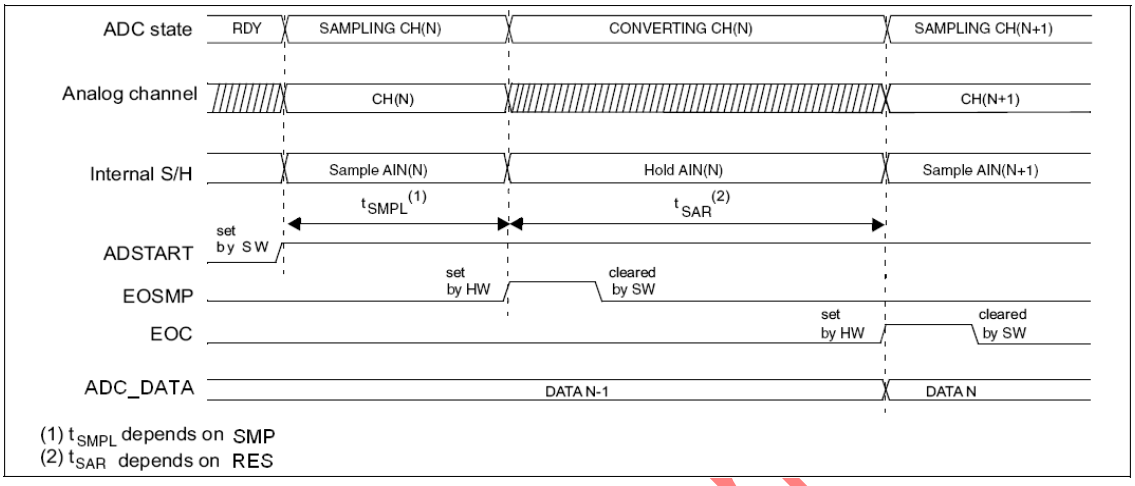


Figure 15-4: Analog to digital conversion time

15.3.10 Stopping an ongoing conversion (ADSTP)

The software can decide to stop any ongoing conversions by setting ADSTP=1 in the ADC_CR register.

This will reset the ADC operation and the ADC will be idle, ready for a new operation.

When the ADSTP bit is set by software, any ongoing conversion is aborted and the result is discarded (FIFO is not updated with the current conversion). The scan sequence is also aborted and reset (meaning that restarting the ADC would re- start a new sequence).

Once this procedure is complete, the ADSTP and ADSTART bits are both cleared by hardware and the software must wait until ADSTART=0 before starting new conversions.

NOTE: The flags in QADC_ISR are not cleared by STOP command, and the datas in FIFO are not lost.

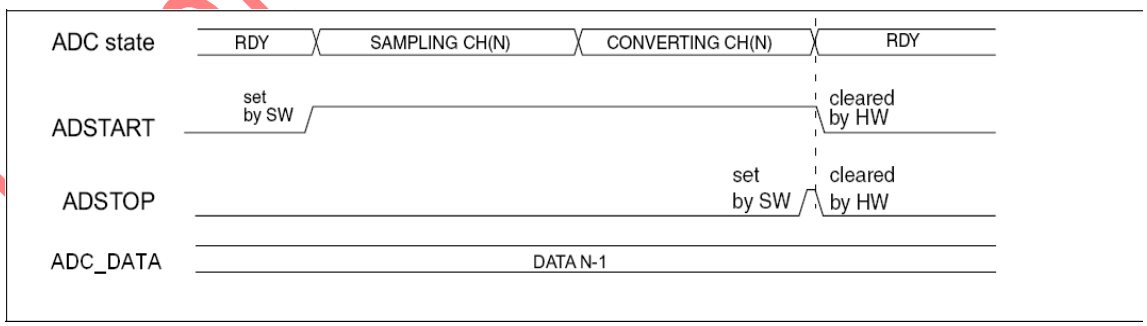


Figure 15-5: Stopping an ongoing conversion

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15.4 Conversion on external trigger and trigger polarity (TRIGMODE, TRIGSCR)

A conversion or a sequence of conversion can be triggered either by software or by an external event. If the TRIGMODE control bits are not equal to “0”, then external events are able to trigger a conversion with the selected polarity. The trigger selection is effective once software has set bit ADSTART=1.

Any hardware triggers which occur while a conversion is ongoing are ignored. If bit ADSTART=0, any hardware triggers which occur are ignored.

Table 15-2 provides the correspondence between the TRIGMODE values and the trigger polarity.

Table 15-2: Configuring the trigger polarity

TRIGMODE[2:0]	Source
3’b000	Trigger detection disabled, software trigger
3’b001	Detection on rising edge
3’b010	Detection on falling edge
3’b011	Detection on both rising and falling edges
3’b100	Detection on high level voltage
3’b101	Detection on low level votage
3’b110	Detection on one shot trigger
3’b111	Reserved

NOTE: The polarity of the external trigger can be changed only when the ADC is not converting (ADSTART= 0).

15.4.1 Discontinuous mode (DISCEN)

This mode is enabled by setting the DISCEN bit in the ADC_CFGR1 register.

In this mode (DISCEN=1), a hardware or software trigger event is required to start each conversion defined in the sequence. On the contrary, if DISCEN=0, a single hardware or software trigger event successively starts all the conversions defined in the sequence.

Example:

- DISCEN=1, channels to be converted = 0, 3, 7, 10
  - 1st trigger: channel 0 is converted and an EOC event is generated
  - 2nd trigger: channel 3 is converted and an EOC event is generated
  - 3rd trigger: channel 7 is converted and an EOC event is generated
  - 4th trigger: channel 10 is converted and both EOC and EOSEQ events are generated.
  - 5th trigger: channel 0 is converted an EOC event is generated
  - 6th trigger: channel 3 is converted and an EOC event is generated
- DISCEN=0, channels to be converted = 0, 3, 7, 10
  - 1st trigger: the complete sequence is converted: channel 0, then 3, 7 and 10. Each conversion generates an EOC event and the last one also generates an EOSEQ event.
  - Any subsequent trigger events will restart the complete sequence.

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15.4.2 Programmable resolution (RES) - fast conversion mode

It is possible to obtain faster conversion times (tSAR) by reducing the ADC resolution. The resolution can be configured to be either 12, 10, 8, or 6 bits by programming the RES[1:0] bits in the ADC_CFGR1 register. Lower resolution allows faster conversion times for applications where high data precision is not required.

NOTE: The RES[1:0] bit must only be changed when the ADEN bit is reset.

The result of the conversion is always 13 bits wide and any unused LSB bits are read as zeroes.

Lower resolution reduces the conversion time needed for the successive approximation steps.

15.4.3 End of conversion, end of sampling phase (EOC, EOSMP flags)

The ADC indicates each end of conversion (EOC) event.

The ADC sets the EOC flag in the ADC_ISR register as soon as a new conversion data result is available. An interrupt can be generated if the EOCIE bit is set in the ADC_IER register. The EOC flag is cleared by software either by writing 1 to it, or by reading the FIFO.

The ADC also indicates the end of sampling phase by setting the EOSMP flag in the ADC_ISR register. The EOSMP flag is cleared by software by writing1 to it. An interrupt can be generated if the EOSMPIE bit is set in the ADC_IER register.

15.4.4 End of conversion sequence (EOSEQ flag)

The ADC notifies the application of each end of sequence (EOSEQ) event.

The ADC sets the EOSEQ flag in the ADC_ISR register as soon as the last data result of a conversion sequence is available in the FIFO. An interrupt can be generated if the EOSEQIE bit is set in the ADC_IER register. The EOSEQ flag is cleared by software by writing 1 to it.

15.4.5 Example timing diagrams (single/continuous modes hardware/software triggers)

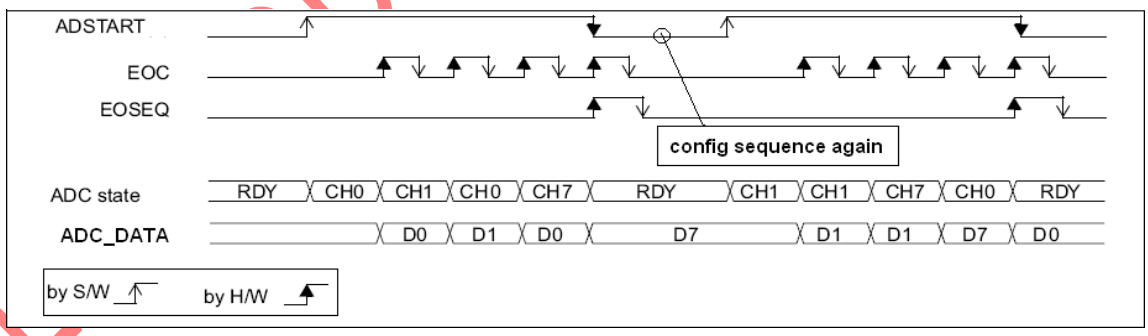


Figure 15-6: Single conversions of a sequence, software trigger

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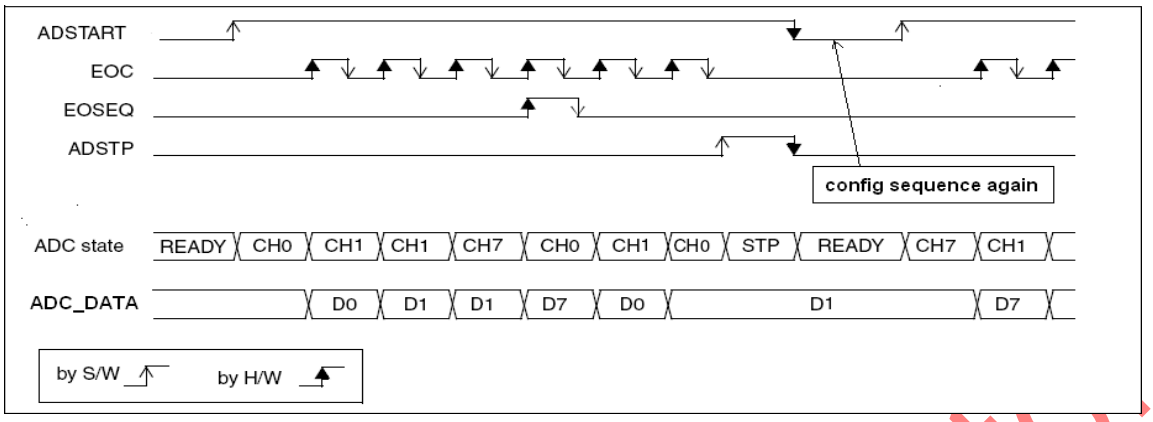


Figure 15-7: Continuous conversion of a sequence, software trigger

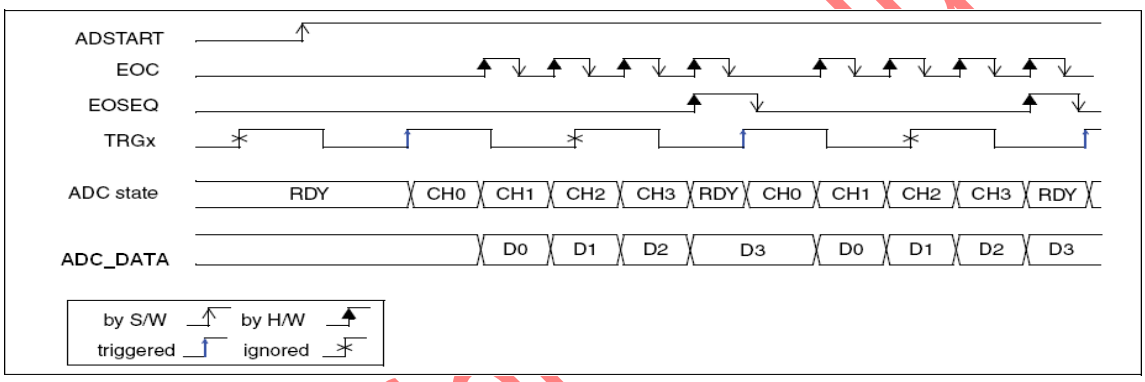


Figure 15-8: Single conversions of a sequence, hardware trigger

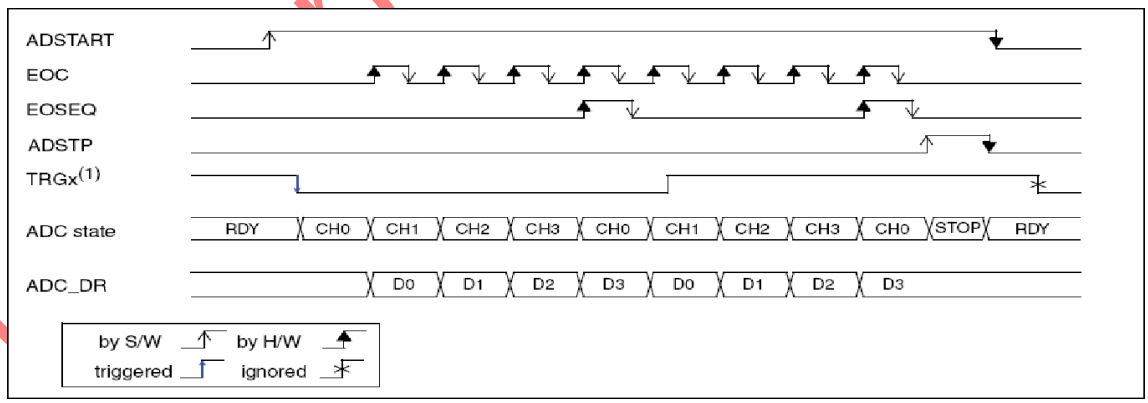


Figure 15-9: Continuous conversions of a sequence, hardware trigger

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15.5 Data management

15.5.1 Data FIFO & data alignment (ADC_FIFO, ALIGN)

At the end of each conversion (when an EOC event occurs), the result of the converted data is stored in the ADC_FIFO which is 13-bit wide x 8 depth.

The format of the read out data depends on the configured data alignment and resolution.

The ALIGN bit in the ADC_CFGR1 register selects the alignment of the data stored after conversion. Data can be right-aligned (ALIGN=0) or left-aligned (ALIGN=1) as shown in **Figure 15-10**.

ALIGN	RES[1:0]	31	30	…	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	0x0			…					data[11:0]											
	0x1			…							data[9:0]									
	0x2			…									data[7:0]							
	0x3			…											data[5:0]					
1	0x0			…	data[11:0]															
	0x1			…	data[9:0]															
	0x2			…									data[7:0]							
	0x3			…									data[5:0]							

Figure 15-10: Data alignment and resolution

The FIFO supports byte, half-word and word read, but the address offset should be always 0x4c.

For different data alignments and resolutions, users should note:

- If read by word, then the data format is as data[31:0] as **Figure 15-10** shows
- if read by half word, then the data format is as data[15:0] as **Figure 15-10** shows
- If read by byte when the data is longer than 8bit, then the high byte is first read out, the low byte should read again.
- If read by byte when the data is not longer than 8bit, then the data format is as data[7:0] as **Figure 15-10** shows

15.5.2 ADC overrun (OVR, OVRMOD)

The overrun flag (OVR) indicates a data overrun event, when the converted data was not read in time by the CPU or the DMA, before the FIFO is full.

The OVR flag is set in the ADC_ISR register if the FULL flag is still at '1' at the time when a new conversion completes. An interrupt can be generated if the OVRIE bit is set in the ADC_IER register.

When an overrun condition occurs, the ADC keeps operating and can continue to convert unless the software decides to stop and reset the sequence by setting the ADSTP bit in the ADC_CR register.

The OVR flag is cleared by software by writing 1 to it.

It is possible to configure if the data is preserved or overwritten when an overrun event occurs by programming the OVRMOD bit in the ADC_CFGR1 register:

- OVRMOD=0
  _ An overrun event preserves the data register from being overwritten: the old data is maintained and the new conversion is discarded. If OVR remains at 1, further conversions can be performed but the resulting data is discarded.
- OVRMOD=1
  _ The data register is overwritten with the last conversion result. If OVR remains at 1, further

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conversions can be performed and the FIFO always contains the data from the latest conversion.

15.5.3 Managing a sequence of data converted without using the DMA

If the conversions are slow enough, the conversion sequence can be handled by software. In this case the software can use the EOC flag and its associated interrupt to handle each data result. Each time a conversion is complete, the EOC bit is set in the ADC_ISR register and the FIFO register can be read.

Software can also use FIFO EMPTY flag to handle each data result. If EMPTY is not "0", this means FIFO has new data.

The OVRMOD bit in the ADC_CFGR1 register should be configured to 0 to manage overrun events as an error.

15.5.4 Managing converted data without using the DMA without overrun

It may be useful to let the ADC convert one or more channels without reading the data after each conversion. In this case, the OVRMOD bit must be configured at 1 and the OVR flag should be ignored by the software. When OVRMOD=1, an overrun event does not prevent the ADC from continuing to convert and the FIFO always contains the latest conversion data.

15.5.5 Managing converted data using the DMA

Once the data number in the FIFO is not empty and bit DMAEN is set, QADC will send request to the DMA. This allows the transfer of the converted data from the FIFO to the destination location selected by the software.

Despite this, if an overrun occurs (OVR=1) because the DMA could not serve the DMA transfer request in time, the ADC stops generating DMA requests and the data corresponding to the new conversion is not transferred by the DMA. Which means that all the data transferred to the RAM can be considered as valid.

Depending on the configuration of OVRMOD bit, the data is either preserved or overwritten.

The DMA transfer requests are blocked until the software clears the OVR bit.

15.6 Low power features

15.6.1 Wait mode conversion

Wait mode conversion can be used to simplify the software as well as optimizing the performance of applications clocked at low frequency where there might be a risk of ADC overrun occurring. When the WAIT bit is set to 1 in the ADC_CFGR1 register, a new conversion can start only if the FIFO is not full.

This is a way to automatically adapt the speed of the ADC to the speed of the system that reads the data.

NOTE: Any hardware triggers which occur while a conversion is ongoing or during the wait time preceding the read access are ignored.

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15.6.2 Auto-off mode (AUTOFF)

The ADC has an automatic power management feature which is called auto-off mode, and is enabled by setting AUTOFF=1 in the ADC_CFGR1 register.

When AUTOFF=1, the ADC is always powered off when not converting and automatically wakes-up when a conversion is started (by software or hardware trigger). A startup-time is automatically inserted between the trigger event which starts the conversion and the sampling time of the ADC. The ADC is then automatically disabled once the sequence of conversions is complete.

Auto-off mode can cause a dramatic reduction in the power consumption of applications which need relatively few conversions or when conversion requests are timed far enough apart (for example with a low frequency hardware trigger) to justify the extra power and extra time used for switching the ADC on and off.

Auto-off mode can be combined with the wait mode conversion (WAIT=1) for applications clocked at low frequency. This combination can provide significant power savings if the ADC is automatically powered-off during the wait phase and restarted as soon as the FIFO is read by the application

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15.7 Analog window watchdog (AWDEN, AWDSGL, AWDCH, AWD_HTR/LTR, AWD)

The AWD analog watchdog feature is enabled by setting the AWDEN bit in the ADC_WDG register. It is used to monitor that either one selected channel or all enabled channels remain within a configured voltage range (window) as shown in **Figure 15-11**.

The AWD analog watchdog status bit is set if the analog voltage converted by the ADC is below a lower threshold or above a higher threshold. These thresholds are programmed in the ADC_TR register. An interrupt can be enabled by setting the AWDIE bit in the ADC_IER register.

The AWD flag is cleared by software by writing 1 to it.

When converting a data with a resolution of less than 12-bit (according to bits RES[1:0]), the LSB of the programmed thresholds must be kept cleared because the internal comparison is always performed on the full 12-bit raw converted data (left aligned).

Table 15-3 shows how to configure the AWDSGL and AWDEN bits in the ADC_WDG register to enable the analog watchdog on one or more channels.

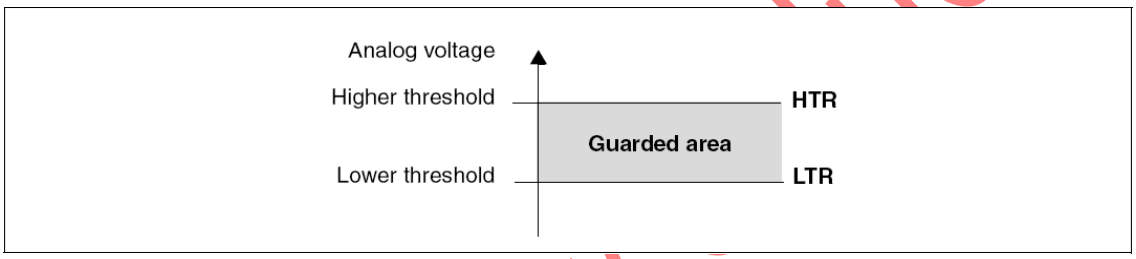


Figure 15-11: Analog watchdog guarded area

Table 15-3: Analog watchdog channel selection

Channels guarded by the analog watchdog	AWDSGL bit	AWDEN bit
None	x	0
All channels	0	1
Single(1) channel	1	1

1. Selected by the AWDCH

15.8 Temperature sensor

The temperature sensor is internally connected to the TS input channel which is used to convert the sensor's output voltage to a digital value.

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15.9 Data Gather

The data gather function is used to gather and join data from QADC result FIFO together automatically.To use this feature,you should:

Config **DGAT_WRTH[3:0]** in **ADC_DGATR** to specify the result data number written into the data buffer[95:0]. The bit width of every result data is specified by **RES[1:0]** in **ADC_CFGR1**.

Config **DGAT_RDTH[3:0]** in **ADC_DGATR** to specify the gathered data number read from the data buffer[95:0]. The bit width of every gathered data is specified by **DGAT_DSIZ[1:0]** in **ADC_DGATR**.

Set **DMAEN** in **ADC_CFGR1** to set up a link between data buffer[95:0] and DMA.Or leave **DMAEN** cleared to access the data buffer by reading **ADC_DBUFR** when **DGAT_DBRDY** flag in **ADC_ISR2** is set.

Set **DGAT_EN** in **ADC_DGATR** to enable this feature and start a conversion.

15.10 ADC interrupts

An interrupt can be generated by any of the following events:

- ADC power-up, when the ADC is ready (ADRDY flag)
- End of any conversion (EOC flag)
- End of a sequence of conversions (EOSEQ flag)
- When an analog watchdog detection occurs (AWD flag)
- When the end of sampling phase occurs (EOSMP flag)
- When a data overrun occurs (OVR flag) Separate interrupt enable bits are available for flexibility.
- When the result FIFO timeout event(FIFO_TOF flag) happened.
- When the data buffer timeout event(DBUF_TOF flag) happened.
- When the data buffer of data gather function is ready(DGAT_DBRDY flag).

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15.11 Memory Map and Registers

This subsection describes the memory map and register structure.

15.11.1 Memory Map

Refer to **Table 15-4** for a description of the QADC memory map.

Table 15-4: QADC Memory Map

Address	31:16	15:0	Access[1]
0x0000	ADC_ISR		S
0x0004	ADC_IER		S
0x0008	ADC_CR		S
0x000c	ADC_CFGR1		S
0x0010	ADC_CFGR2		S
0x0014	ADC_SMPR		S
0x0018	ADC_WDGR		S
0x001c	ADC_TR		S
0x0020~28	Reserved		S
0x002c	ADC_CHSELR1		S
0x0030	ADC_CHSELR2		S
0x0034~48	Reserved		S
0x004c	ADC_FIFOR		S
0x0050	Reserved		S
0x0054	ADC_ISR2		S
0x0058	ADC_DGATR		S
0x005c	ADC_DBUFR		S
0x0060	ADC_FIFOTOR		S
0x0064~7c	Reserved		S
0x0080	ADC_DFT3		S
0x0084	ADC_DFT2		S
0x0088	ADC_DFT1		S
0x008c	ADC_DFT0		S
0x0090	ADC_DFT7		S
0x0094	ADC_DFT6		S
0x0098	ADC_DFT5		S
0x009c	ADC_DFT4		S
0x00a0	ADC_DFT8		S
0x00a4	ADC_CHSELR3		S

NOTE:

- S = supervisor-only access. User mode accesses to supervisor only address locations have no effect and result in a cycle termination transfer error.

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15.11.2 Registers

15.11.2.1 ADC interrupt and status register (ADC_ISR)

Addr. Offset: 0x0000 Reset Value: 0x00000040

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
AWD	EMPTY	FULL	OVR	EOSEQ	EOC	EOSMP	ADRDY
r/w1c	ro	ro	r/w1c	r/w1c	r/w1c	r/w1c	r/w1c

Figure 15-12: ADC interrupt and status register (ADC_ISR)

Read: Anytime Write: Anytime

AWD — Analog watchdog flag

This bit is set by hardware when the converted voltage crosses the values programmed in the ADC_TR register. It is cleared by software writing 1 to it.

1 = Analog watchdog event occurred

0 = No analog watchdog event occurred (or the flag event was already acknowledged and cleared by software)

EMPTY — FIFO empty status
This bit is set by hardware when the FIFO is empty. It is cleared by hardware when FIFO is not empty.

1 = FIFO is empty

0 = FIFO is not empty

FULL — FIFO full status
This bit is set by hardware when the FIFO is full. It is cleared by hardware when FIFO is not full.

1 = FIFO is full

0 = FIFO is not full

OVR — ADC overrun
This bit is set by hardware when an overrun occurs, meaning that a new conversion has complete while the FULL flag was already set. It is cleared by software writing 1 to it.

1 = Overrun has occurred

0 = No overrun occurred (or the flag event was already acknowledged and cleared by software)

EOSEQ — End of sequence flag
This bit is set by hardware at the end of the conversion of a sequence. It is cleared by software writing 1 to it.

1 = Conversion sequence complete

0 = Conversion sequence not complete (or the flag event was already acknowledged and cleared by software)

EOC — End of conversion flag

This bit is set by hardware at the end of each conversion of a channel when a new data result is available in the ADC_FIFO register. It is cleared by software writing 1 to it or by reading the ADC_FIFO register.

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1 = Channel conversion complete

0 = Channel conversion not complete (or the flag event was already acknowledged and cleared by software)

EOSMP — End of sampling flag
This bit is set by hardware during the conversion, at the end of the sampling phase.

1 = End of sampling phase reached

0 = Not at the end of the sampling phase (or the flag event was already acknowledged and cleared by software)

ADRDY — ADC ready
This bit is set by hardware after the ADC has been enabled (bit ADEN=1) and when the ADC reaches a state where it is ready to accept conversion requests. It is cleared by software writing 1 to it.

1 = ADC is ready to start conversion

0 = ADC not yet ready to start conversion (or the flag event was already acknowledged and cleared by software)

15.11.2.2 ADC interrupt enable register (ADC_IER)

Addr. Offset: 0x0004 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
AWDIE	Reserved		OVRIE	EOSEQIE	EOCIE	EOSMPIE	ADRDYIE
rw	ro		rw	rw	rw	rw	rw

Figure 15-13: ADC interrupt enable register (ADC_IER)

Read: Anytime

Write: Before ADC start

AWDIE — Analog watchdog interrupt enable

This bit is set and cleared by software to enable/disable the analog watchdog interrupt.

1 = Analog watchdog interrupt enabled

0 = Analog watchdog interrupt disabled

OVRIE — Overrun interrupt enable

This bit is set and cleared by software to enable/disable the overrun interrupt.

1 = Overrun interrupt enabled. An interrupt is generated when the OVR bit is set.

0 = Overrun interrupt disabled

EOSEQIE — End of conversion sequence interrupt enable

This bit is set and cleared by software to enable/disable the end of sequence of conversions interrupt.

1 = EOSEQ interrupt enabled. An interrupt is generated when the EOSEQ bit is set.

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0 = EOSEQ interrupt disabled

EOCIE — End of conversion interrupt enable

This bit is set and cleared by software to enable/disable the end of conversion interrupt.

1 = EOC interrupt enabled. An interrupt is generated when the EOC bit is set.

0 = EOC interrupt disabled

EOSMPIE — End of sampling flag interrupt enable

This bit is set and cleared by software to enable/disable the end of the sampling phase interrupt.

1 = EOSMP interrupt enabled. An interrupt is generated when the EOSMP bit is set.

0 = EOSMP interrupt disabled.

ADRDYIE — ADC ready interrupt enable
This bit is set and cleared by software to enable/disable the ADC Ready interrupt.

1 = ADRDY interrupt enabled. An interrupt is generated when the ADRDY bit is set.

0 = ADRDY interrupt disabled.

15.11.2.3 ADC control register (ADC_CR)

Addr. Offset: 0x0008 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				ADSTP	ADSTART	ADDIS	ADEN
ro				r/w1	r/w1	r/w1	r/w1

Figure 15-14: ADC control register (ADC_CR)

Read: Anytime

Write: See each bit description

ADSTP — ADC stop conversion command

This bit is set by software to stop and discard an ongoing conversion (ADSTP Command). It is cleared by hardware when the conversion is effectively discarded and the ADC is ready to accept a new start conversion command.

1 = Write 1 to stop the ADC. Read 1 means that an ADSTP command is in progress.

0 = No ADC stop conversion command ongoing

NOTE: Software is allowed to set ADSTP only when ADSTART=1 and ADDIS=0 (ADC is enabled and may be converting and there is no pending request to disable the ADC)

ADSTART — ADC start conversion command
This bit is set by software to start ADC conversion. Depending on the TRIGMODE configuration bits, a conversion either starts immediately (software trigger configuration) or once a hardware trigger event occurs (hardware trigger configuration).

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It is cleared by hardware:

- In single conversion mode when software trigger is selected : at the assertion of the End of Conversion Sequence (EOSEQ) flag.
- In discontinued conversion mode when software trigger is selected: at the assertion of the End of Conversion (EOC) flag.
- In all cases: after the execution of the ADSTP command, at the same time as the ADSTP bit is cleared by hardware.

1 = Write 1 to start the ADC. Read 1 means that the ADC is operating and may be converting.

0 = No ADC conversion is ongoing.

NOTE: Software is allowed to set ADSTART only when ADEN=1 and ADDIS=0 (ADC is enabled and there is no pending request to disable the ADC)

ADDIS — ADC disable command
This bit is set by software to disable the ADC (ADDIS command) and put it into power-down state (OFF state). It is cleared by hardware once the ADC is effectively disabled (ADEN is also cleared by hardware at this time).

1 = Write 1 to disable the ADC. Read 1 means that an ADDIS command is in progress.

0 = No ADDIS command ongoing

NOTE: Software is allowed to set ADDIS only when ADEN=1 and ADSTART=0 (which ensures that no conversion is ongoing)

ADEN — ADC enable command

This bit is set by software to enable the ADC. The ADC will be effectively ready to operate once the ADRDY flag has been set. It is cleared by hardware when the ADC is disabled, after the execution of the ADDIS command.

1 = Write 1 to enable the ADC.

0 = ADC is disabled (OFF state) .

NOTE: Software is allowed to set ADEN only when all bits of ADC_CR registers are 0 (ADSTP=0, ADSTART=0, ADDIS=0 and ADEN=0)

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15.11.2.4 ADC configuration register 1 (ADC_CFGR1)

Addr. Offset: 0x000C Reset Value: 0x07008000

31	30	29	28	27	26	25	24
DIFF	OVRMOD	QCLK_DIS	SMP_OBE	Reserved	SEQ_LEN[2:0]		
rw	rw	Rw	rw	ro	rw		
23	22	21	20	19	18	17	16
DISCEN	AUTOFF	WAIT	CONT	Reserved			
rw	rw	Rw	rw	ro			
15	14	13	12	11	10	9	8
SEL_EXT_VREF	Reserved	TRIGMODE[2:0]			ALIGN	RES[1:0]	
rw	ro	rw			rw	rw	
7	6	5	4	3	2	1	0
Reserved	DMATH[2:0]			Reserved			DMAEN
ro	rw			ro			rw

Figure 15-15: ADC configuration register 1 (ADC_CFGR1)

Read: Anytime

Write: Before ADC start

DIFF — Select differential-input

This bit determines that if the input is single-ended or differential-input.

1 = The analog input is differentially sampled.

0 = The analog input is single sampled

OVRMOD — Overrun management mode
This bit is set and cleared by software and configure the way data overruns are managed.

1 = ADC_DR register is overwritten with the last conversion result when an overrun is detected.

0 = ADC_DR register is preserved with the old data when an overrun is detected.

QCLK_DIS — QADC Clock Disable Specify if the QADC clock is disabled.

1 = disable

0 = not disable

SMP_OBE — Sample signal output buffer enable

This bit determines that if the sample signal output buffer is enabled.NOTE: This bit is not implemented in this project, that is this bit is reserved and write data to this bit has no effect.

1 = Enabled

0 = Disabled

SEQ_LEN[2:0] — Sequence length
These bits define the length of sequence. The sequence length=SEQ_LEN+1. For example, SEQ_LEN=7 means sequence length is 8; SEQ_LEN=0 means sequence length is 1.

DISCEN — Discontinuous mode
This bit is set and cleared by software to enable/disable discontinuous mode.

1 = Discontinuous mode enabled

0 = Discontinuous mode disabled

AUTOFF — Auto-off mode
This bit is set and cleared by software to enable/disable auto-off mode..

1 = Auto-off mode enabled

0 = Auto-off mode disabled

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WAIT — Wait conversion mode

This bit is set and cleared by software to enable/disable wait conversion mode..

1 = Wait conversion mode on

0 = Wait conversion mode off

CONT — Single / continuous conversion mode

This bit is set and cleared by software. If it is set, conversion takes place continuously until it is cleared.

1 = Continuous conversion mode

0 = Single conversion mode

SEL_EXT_VREF — Select external vref
This bit is set and cleared by software to select vref for adc analog.

1 = Select external vref

0 = Select internal vref

TRIGMOD[2:0] — Trigger mode select
These bits are used to select software trigger mode or external trigger polarity, see Table 15-2.

ALIGN — Data alignment
This bit is set and cleared by software to select right or left alignment. Refer to Figure 15-16.

1 = Left alignment

0 = Right alignment

RES[1:0] — Data resolution
These bits are written by software to select the resolution of the conversion. Refer to Figure 15-10.

DMATH[2:0] — Direct memory access Threshold
Specify the data number threshold of result FIFO.Once the data number in result FIFO is more than the threshold and **DMAEN** is set,QADC will send a transfer request to the DMA.

DMAEN — Direct memory access enable

This bit is set and cleared by software to enable the generation of DMA requests. This allows to use the DMA controller to manage automatically the converted data.

1 = DMA enabled

0 = DMA disabled

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15.11.2.5 ADC configuration register 2 (ADC_CFGR2)

Addr. Offset: 0x0010 Reset Value: 0x00000220

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
ADC_BYPASS	Reserved			QPR[3:0]			
rw	ro			rw			
7	6	5	4	3	2	1	0
STCNT[7:0]							
rw							

Figure 15-16: ADC configuration register 2 (ADC_CFGR2)

Read: Anytime

Write: Before ADC enable

ADC_BYPASS — ADC analog input bypass
0 = normal adc
1 = adc analog input bypass, data is input from ADC_DFT* for test.

QPR[3:0] — Prescaler Clock Divider Bits
These bits select the system clock divisor to generate the QADC clock as follows:

$$F_{QCLK} = F_{sys_QCLK} / (QPR[3:0] + 1)$$

where:

$$0 <= QPR[3:0] <= 15.$$

STCNT[7:0] — ADC startup counter bits

The ADC needs a stabilization time of tSTAB(~2us) before it starts converting accurately. This time is calculated by counting QCLK cycles until the internal counter reachs the STCNT[7:0]. So user should set these bits before ADC enable. For example, if the QCLK = 16MHz, then should set the STCNT[7:0] = 2000/(1000/16)=32.

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15.11.2.6 ADC sampling time register (ADC_SMPR)

Addr. Offset: 0x0014 Reset Value: 0x00000002

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
SMP[7:0]							
rw							

Figure 15-17: ADC sampling time register (ADC_SMPR)

Read: Anytime

Write: Before ADC start

SMP[7:0] — Sampling time selection
These bits are written by software to select the sampling time that applies to all channels. The sample time is calculated as (SMP[7:0]+2) QCLKs

Example : SMP[7:0] = 0x2 means the sample time is 4 QCLKs

15.11.2.7 ADC watch dog register (ADC_WDG)

Addr. Offset: 0x0018 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
AWDEN	AWDSGL	Reserved	AWDCH[4:0]				
rw	rw	ro	rw				

Figure 15-18: ADC watch dog register (ADC_WDG)

Read: Anytime

Write: Before ADC start

AWDEN — Analog watchdog enable This bit is set and cleared by software.

1 = Analog watchdog enabled

0 = Analog watchdog disabled

AWDSGL — Enable the watchdog on a single channel or on all channels

This bit is set and cleared by software to enable the analog watchdog on the channel identified by the

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AWDCH[4:0] bits or on all the channels

1 = Analog watchdog enabled on a single channel

0 = Analog watchdog enabled on all channels

AWDCH[4:0] — Analog watchdog channel selection
These bits are set and cleared by software. They select the input channel to be guarded by the analog watchdog.

- 00000: ADC analog input Channel 0 monitored by AWD
- 00001: ADC analog input Channel 1 monitored by AWD
- .....
- 01111: ADC analog input Channel 15 monitored by AWD
- 10000: VBattery Detect monitored by AWD
- other values: Reserved, must not be used

15.11.2.8 ADC watchdog threshold register (ADC_TR)

Addr. Offset: 0x001C Reset Value: 0x1FFF0000

31	30	29	28	27	26	25	24
Reserved			NOTE	HT[11:8]			
	ro		rw	rw			
23	22	21	20	19	18	17	16
HT[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved			NOTE	LT[11:8]			
	ro		rw	rw			
7	6	5	4	3	2	1	0
LT[7:0]							
rw							

Figure 15-19: ADC watchdog threshold register (ADC_TR)

Read: Anytime

Write: Before ADC start

NOTE: Please clear this bit when config HT[11:0] & LT[11:0].

HT[11:0] — Analog watchdog higher threshold
These bits are written by software to define the higher threshold for the analog watchdog.

LT[11:0] — Analog watchdog lower threshold
These bits are written by software to define the lower threshold for the analog watchdog.

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15.11.2.9 ADC channel selection register i(ADC_CHSELR1, ADC_CHSELR2)

Addr. Offset: 0x002C Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved			CCW3[4:0]				
ro			rw				
23	22	21	20	19	18	17	16
Reserved			CCW2[4:0]				
ro			rw				
15	14	13	12	11	10	9	8
Reserved			CCW1[4:0]				
ro			rw				
7	6	5	4	3	2	1	0
Reserved			CCW0[4:0]				
ro			rw				

Figure 15-20: ADC channel selection register 1(ADC_CHSELR1)

Addr. Offset: 0x0030 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved			CCW7[4:0]				
ro			rw				
23	22	21	20	19	18	17	16
Reserved			CCW6[4:0]				
ro			rw				
15	14	13	12	11	10	9	8
Reserved			CCW5[4:0]				
ro			rw				
7	6	5	4	3	2	1	0
Reserved			CCW4[4:0]				
ro			rw				

Figure 15-21: ADC channel selection register 2(ADC_CHSELR2)

Read: Anytime

Write: Before ADC start

CCWi[4:0] — The number i conversion select channel, refer to **Table 15-1**.

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15.11.2.10 ADC FIFO access register (ADC_FIFO)

Addr. Offset: 0x004C Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
DATA[15:8]							
ro							
7	6	5	4	3	2	1	0
DATA[7:0]							
ro							

Figure 15-22: ADC FIFO access register (ADC_FIFO)

Read: Anytime Write: Never

DATA[15:0]: Converted data Refer to **Figure 15-10**.

15.11.2.11 ADC interrupt and status register 2 (ADC_ISR2)

Addr. Offset: 0x0054 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
FIFO_TOF	Reserved			FIFO_CNT[3:0]			
r/w1c	ro			ro			
23	22	21	20	19	18	17	16
Reserved				AWD_ERRCH[3:0]			
ro				ro			
15	14	13	12	11	10	9	8
DBUF_TOF	Reserved		DGAT_DBRDY	DGAT_RDCNT[3:0]			
r/w1c	ro		ro	ro			
7	6	5	4	3	2	1	0
DGAT_WRCNT[3:0]				Reserved			
ro				ro			

Figure 15-23: ADC interrupt and status register 2 (ADC_ISR2)

Read: Anytime Write: Anytime

FIFO_TOF — FIFO TimeOut Flag
This bit is set by hardware when the result FIFO timeout event happened. It is cleared by software writing 1 to it.

1 = FIFO timeout event happened

0 = FIFO timeout event not happened

FIFO_CNT[3:0] — FIFO counter
Indicate the data number of result FIFO.

AWD_ERRCH[3:0] — Analog Watch Dog Error Channel Indicate the error channel of last AWD event.

DBUF_TOF — Data Buffer TimeOut Flag
This bit is set by hardware when the data buffer timeout event happened. It is cleared by software writing 1 to

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it.

1 = data buffer timeout event happened

0 = data buffer timeout event not happened

DGAT_DBRDY — Data Gather Data Buffer Ready
This bit is set by hardware when the data buffer of data gather function is ready. It is cleared by software writing 1 to it.

1 = Data buffer ready

0 = Data buffer not ready

DGAT_RDCNT[3:0] — Data Gather Read Counter

Indicate the read counter number of data gather function.When the counter reaches **DGAT_RDTH[3:0]** in **ADC_DGATR**,**DGAT_DBRDY** is to be cleared when another read access to **ADC_DBUFR** is happened.

DGAT_WRCNT[3:0] — Data Gather Write Counter

Indicate the write counter number of data gather function.When the counter reaches **DGAT_WRTH[3:0]** in **ADC_DGATR**,**DGAT_DBRDY** is to be set when another write access to **ADC_DBUFR** is happened.

15.11.2.12 ADC Data Gather register (ADC_DGATR)

Addr. Offset: 0x0058 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DBUF_TOVAL[15:8]							
rw							
23	22	21	20	19	18	17	16
DBUF_TOVAL[7:0]							
rw							
15	14	13	12	11	10	9	8
DBUF_TOIE	DBUF_TOEN	Reserved	DGAT_DBRDYIE	DGAT_RDTH[3:0]			
rw	rw	ro	rw	rw			
7	6	5	4	3	2	1	0
DGAT_WRTH[3:0]				DGAT_DSIZ[1:0]		Reserved	DGAT_EN
rw				rw		ro	rw

Figure 15-24: ADC Data Gather register (ADC_DGATR)

Read: Anytime Write: Anytime

DBUF_TOVAL[15:0] — Data Buffer TimeOut Value
Specify the data buffer timeout trigger value. Once the data buffer is not empty and **DBUF_TOEN** is set,the data buffer timeout counter is turned on.If there is no operation to the data buffer until the counter reaches the trigger value,the **DBUF_TOF** in **ADC_ISR2** will be set.

DBUF_TOIE — Data Buffer TimeOut Interrupt Enable
This bit is set and cleared by software to enable/disable the data buffer timeout interrupt.

1 = Enabled

0 = Disabled

DBUF_TOEN — Data Buffer TimeOut Function Enable
This bit is set and cleared by software to enable/disable the data buffer timeout function.

1 = Enabled

0 = Disabled

DGAT_DBRDYIE — Data Gather Data Buffer Ready Interrupt Enable

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This bit is set and cleared by software to enable/disable the data gather data buffer ready interrupt.

1 = Enabled

0 = Disabled

DGAT_RDTH[3:0] — Data Gather Read Threshold

Specify the read counter threshold of data gather function.When the **DGAT_RDCNT[3:0]** in **ADC_ISR2** reaches the threshold.**DGAT_DBRDY** is to be cleared when another read access to **ADC_DBUFR** is happened.

DGAT_WRTH[3:0] — Data Gather Write Threshold

Specify the write counter threshold of data gather function.When the **DGAT_WRCNT[3:0]** in **ADC_ISR2** reaches the threshold.**DGAT_DBRDY** is to be set when another write access to **ADC_DBUFR** is happened.

DGAT_DSIZ[1:0] — Data Gather read Data Size

Specify the read data size of data gather function when a read access to the data buffer is executed by software or DMA.

Table 15-5: Data Gather read Data Size

DGAT_DSIZ[1:0]	Read Data Size
2'b00	Word
2'b01	Halfword
2'b10	Byte
2'b11	Reserved

DGAT_EN — Data Gather Function Enable
This bit is set and cleared by software to enable/disable the data gather function.

1 = Enabled

0 = Disabled

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15.11.2.13 ADC Data Buffer Register (ADC_DBUFR)

Addr. Offset: 0x005C Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DBUF_DATA[31:24]							
ro							
23	22	21	20	19	18	17	16
DBUF_DATA[23:16]							
ro							
15	14	13	12	11	10	9	8
DBUF_DATA[15:8]							
ro							
7	6	5	4	3	2	1	0
DBUF_DATA[7:0]							
ro							

Figure 15-25: ADC Data Buffer register (ADC_DBUFR)

Read: Anytime Write: Anytime

DBUF_DATA[31:0] — Data Buffer Data
Read the data buffer register with **DGAT_EN** set returns gathered data.Note that the software read access type should be consistent with the one configed by **DGAT_DSIZ[1:0]** in **ADC_DGATR**.

15.11.2.14 ADC FIFO TimeOut register (ADC_FIFOTOR)

Addr. Offset: 0x0060 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
FIFO_TOVAL[15:8]							
rw							
23	22	21	20	19	18	17	16
FIFO_TOVAL[7:0]							
rw							
15	14	13	12	11	10	9	8
FIFO_TOIE	FIFO_TOEN	Reserved					
rw	rw	ro					
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 15-26: ADC FIFO TimeOut register (ADC_FIFOTOR)

Read: Anytime Write: Anytime

FIFO_TOVAL[15:0] — Result FIFO TimeOut Value
Specify the result FIFO timeout trigger value. Once the result FIFO is not empty and **FIFO_TOEN** is set,the result FIFO timeout counter is turned on.If there is no operation to the result FIFO until the counter reaches the trigger value,the **FIFO_TOF** in **ADC_ISR2** will be set.

FIFO_TOIE — Result FIFO TimeOut Interrupt Enable
This bit is set and cleared by software to enable/disable the result FIFO timeout interrupt.

1 = Enabled

0 = Disabled

FIFO_TOEN — Result FIFO TimeOut Function Enable
This bit is set and cleared by software to enable/disable the result FIFO timeout function.

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1 = Enabled

0 = Disabled

15.11.2.15 ADC Data For Test3 (ADC_DFT3)

Addr. Offset: 0x0080 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL7[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL7[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL6[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL6[7:0]							
rw							

Figure 15-27: ADC Data For Test3 (ADC_DFT3)

Read: Anytime Write: Anytime

CHANEL7[11:0] — channel7 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL6[11:0] — channel6 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

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15.11.2.16 ADC Data For Test2 (ADC_DFT2)

Addr. Offset: 0x0084 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL5[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL5[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL4[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL4[7:0]							
rw							

Figure 15-28: ADC Data For Test2 (ADC_DFT2)

CHANEL5[11:0] — channel5 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL4[11:0] — channel4 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

15.11.2.17 ADC Data For Test1 (ADC_DFT1)

Addr. Offset: 0x0088 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL3[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL3[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL2[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL2[7:0]							
rw							

Figure 15-29: ADC Data For Test1 (ADC_DFT1)

Read: Anytime Write: Anytime

CHANEL3[11:0] — channel3 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL2[11:0] — channel2 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

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15.11.2.18 ADC Data For Test0 (ADC_DFT0)

Addr. Offset: 0x008C Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL1[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL1[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL0[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL0[7:0]							
rw							

Figure 15-30: ADC Data For Test0 (ADC_DFT0)

Read: Anytime Write: Anytime

CHANEL1[11:0] — channel1 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL0[11:0] — channel0 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

15.11.2.19 ADC Data For Test7 (ADC_DFT7)

Addr. Offset: 0x0090 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL15[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL15[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL14[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL14[7:0]							
rw							

Figure 15-31: ADC Data For Test7 (ADC_DFT7)

Read: Anytime Write: Anytime

CHANEL15[11:0] — channel15 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL14[11:0] — channel14 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

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15.11.2.20 ADC Data For Test6 (ADC_DFT6)

Addr. Offset: 0x0094 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL13[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL13[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL12[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL12[7:0]							
rw							

Figure 15-32: ADC Data For Test6 (ADC_DFT6)

Read: Anytime Write: Anytime

CHANEL13[11:0] — channel13 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL12[11:0] — channel12 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

15.11.2.21 ADC Data For Test5 (ADC_DFT5)

Addr. Offset: 0x0098 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL11[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL11[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL10[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL10[7:0]							
rw							

Figure 15-33: ADC Data For Test5 (ADC_DFT5)

Read: Anytime Write: Anytime

CHANEL11[11:0] — channel11 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL10[11:0] — channel10 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

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15.11.2.22 ADC Data For Test4 (ADC_DFT4)

Addr. Offset: 0x009c Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CHANEL9[11:8]			
ro				rw			
23	22	21	20	19	18	17	16
CHANEL9[7:0]							
rw							
15	14	13	12	11	10	9	8
Reserved				CHANEL8[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL8[7:0]							
rw							

Figure 15-34: ADC Data For Test4 (ADC_DFT4)

Read: Anytime Write: Anytime

CHANEL9[11:0] — channe9 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

CHANEL8[11:0] — channel8 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

15.11.2.23 ADC Data For Test8 (ADC_DFT8)

Addr. Offset: 0x00A0 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				CHANEL16[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
CHANEL16[7:0]							
rw							

Figure 15-35: ADC Data For Test8 (ADC_DFT8)

Read: Anytime Write: Anytime

CHANEL16[11:0] — channe16 input data of ADC for test It works only when ADC_CFGR2[ADC_BYPASS] = 1.

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15.11.2.24 ADC channel selection register3 (ADC_CHSELR3)

Addr. Offset: 0x00A4 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CHSEL[15:8]							
rw							
7	6	5	4	3	2	1	0
CHSEL[7:0]							
rw							

Figure 15-36: ADC channel selection register3 (ADC_CHSELR3)

Read: Anytime Write: Anytime

CHSEL16[15:0] — ADC channel selected.
These bits are used for pad reuse, the pads of ADC channel[15:12] and channel[7:4] are not used as ADC input pads as default. When needed, channel selected in CCWi should be set in corresponding bit in CHSEL16[15:0]. For example, if data in CCW1 is 0x1a3c, it means channel1, 10, 3, 12 are selected as ADC input channel, so bit1, 10, 3, 12 in ADC_CHSELR3 should be set. The following table shows the concrete pad reuse.

NOTE: Read **ADC_CHSELR3** will return 0, single bit operation is not advised, word write operation of ADC_CHSELR3 is recommended.

Table 15-6: PAD reuse of ADC

bits	value	pad reuse
CHSEL[0]	0	adc_in[0]
	1	
CHSEL[1]	0	adc_in[2]
	1	
CHSEL[2]	0	vout1(inside input from mcc)
	1	
CHSEL[3]	0	vout3(inside input from mcc)
	1	
CHSEL[4]	0	gint[26]
	1	adc_ch_in[0]
CHSEL[5]	0	gint[28]
	1	adc_ch_in[2]
CHSEL[6]	0	gint[22]
	1	adc_ch_in[4]
CHSEL[7]	0	gint[24]
	1	adc_ch_in[6]
CHSEL[8]	0	adc_in[1]
	1	
CHSEL[9]	0	vref1v(inside input)
	1	

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bits	value	pad reuse
CHSEL[10]	0	vout2(inside input from mcc)
	1	
CHSEL[11]	0	dac_out(inside input from dac)
	1	
CHSEL[12]	0	gint[27]
	1	adc_ch_in[1]
CHSEL[13]	0	gint[29]
	1	adc_ch_in[3]
CHSEL[14]	0	gint[23]
	1	adc_ch_in[5]
CHSEL[15]	0	gint[25]
	1	adc_ch_in[7]

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16 Pulse Width Modulator (PWM)

16.1 Introduction

There are 4 PWM-Timers enclosed. The 4 PWM-Timers has 2 Pre-scale, 2 clock divider, 4 clock selectors, 4 16-bit counters, 4 16-bit comparators, 2 Dead-Zone generators.Each can be used as a timer and issues interrupt independently.

Each two PWM-Timers share the same pre-scale. Clock divider provides each timer with 5 clock sources (1, 1/2, 1/4, 1/8, 1/16). The 16-bit counter in each timer receive clock signal from clock selector and can be used to handle one PWM period. The 16-bit comparator compares number in counter with threshold number in register loaded previously to generate PWM duty cycle. The clock signal from clock divider is called PWM clock. Dead-Zone generator utilize PWM clock as clock source. Once Dead-Zone generator is enabled, output of two PWM-Timers are blocked. Two output pin are all used as Dead-Zone generator output signal to control off-chip power device. The value of comparator is used for pulse width modulation. The counter control logic changes the output level when down-counter value matches the value of compare register.

Each PWM-Timer includes a capture channel. The Capture 0 and PWM 0 share a timer that included in PWM 0; and the Capture 1 and PWM 1 share another timer, and etc.Therefore user must setup the PWM-Timer before turn on Capture feature. After enabling capture feature, the capture always latched PWM-counter to CRLR when input channel has a rising transition and latched PWM-counter to CFLR when input channel has a falling transition. Capture channel 0 interrupt is programmable by setting CCR0[1] (Rising latch Interrupt enable) and CCR0[2] (Falling latch Interrupt enable) to decide the condition of interrupt occur. Capture channel1&2&3 has the same feature.Whenever Capture issues Interrupt 0/1/2/3, the PWM counter 0/1/2/3 will be reload at this moment. The maximal capture frequency should be decided by interrupt process time.If interrupt precess time is T0,then the capture channel input signal should not change in T0,the maximal capture frequency is 1/T0.

There are only four interrupts from PWM to interrupt controller (INTC). PWM 0 and Capture 0 share the same interrupt; PWM1 and Capture 1 share the same interrupt and so on. Therefore, PWM function and Capture function in the same channel cannot be used at the same time.

16.2 Features

The Pulse Width Modulator includes these distinctive features:

- Programmable period
- Programmable duty cycle
- Two Dead-Zone generator
- Capture function
- Pins can be configured as general-purpose I/O

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16.3 Block Diagram

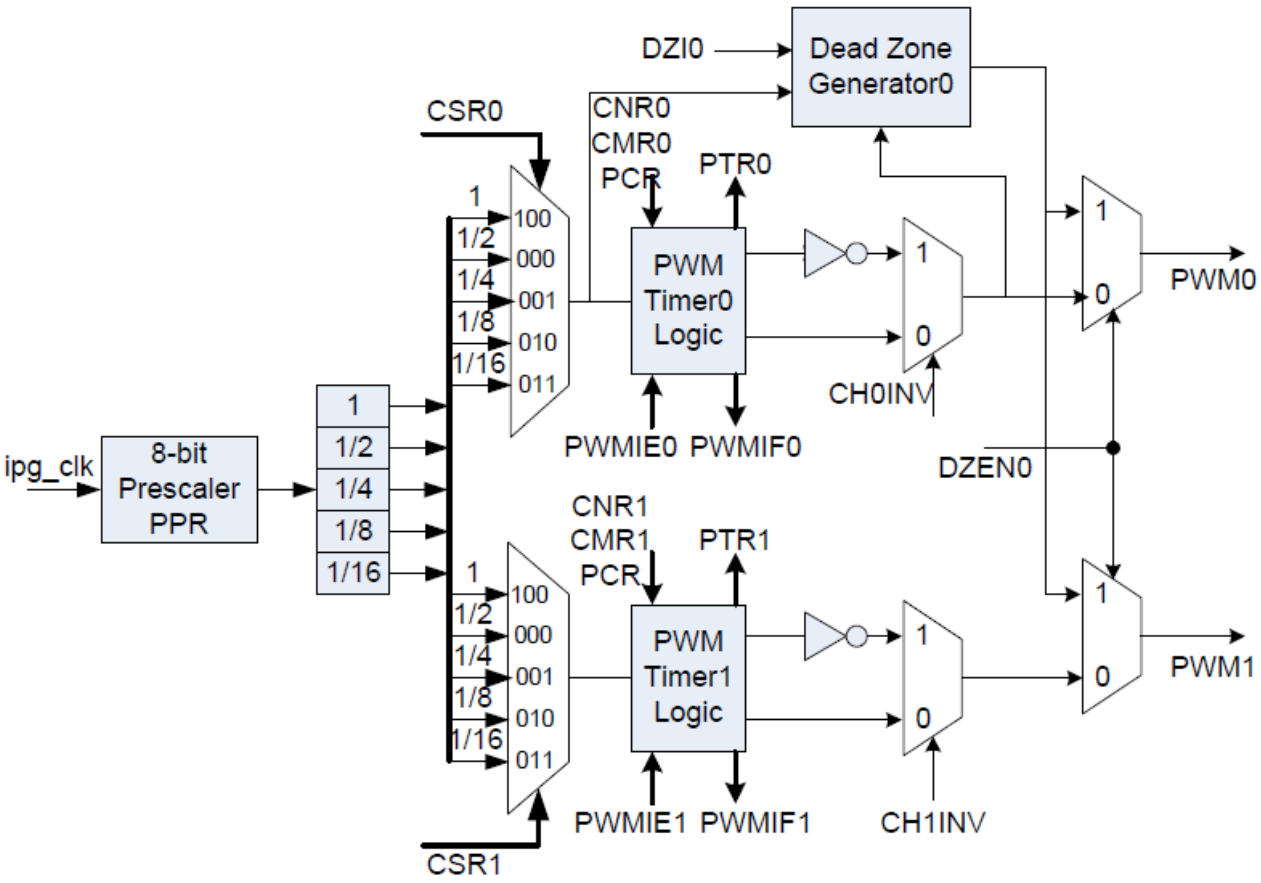


Figure 16-1: PWM Block Diagram

16.4 Signal Description

Table 16-1: PWM Signal Description

Name	I/O	Width	Reset State	Description
PWM0	I/O	1	0	PWM0 pin
PWM1	I/O	1	0	PWM1 pin
PWM2	I/O	1	0	PWM2 pin
PWM3	I/O	1	0	PWM3 pin

PWMx are used as a general-purpose input/output,and also used as the pwm send output or capture input. In default state, it is used as general-purpose input port.

NOTE: PWMx_EN should be set at ccm register PCFG3

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16.5 Memory Map and Registers

This subsection describes the memory map and register structure.

16.5.1 Memory Map

Table 16-2: Module Memory Map

Address	Bit[15:8]	Bit[7:0]	Access(1)
0x0000	PWM Pre-scale Register (PPR)		S/U
0x0004	PWM Clock Select Register (PCSR)		S/U
0x0008	PWM Control Register (PCR)		S/U
0x000C	PWM Counter Register0 (PCNR0)		S/U
0x0010	PWM Comparator Register0 (PCMR0)		S/U
0x0014	PWM Timer Register0 (PTR0)		S/U
0x0018	PWM Counter Register1 (PCNR1)		S/U
0x001C	PWM Comparator Register1 (PCMR1)		S/U
0x0020	PWM Timer Register1 (PTR1)		S/U
0x0024	PWM Counter Register2 (PCNR2)		S/U
0x0028	PWM Comparator Register2 (PCMR2)		S/U
0x002C	PWM Timer Register2 (PTR2)		S/U
0x0030	PWM Counter Register3 (PCNR3)		S/U
0x0034	PWM Comparator Register3 (PCMR3)		S/U
0x0038	PWM Timer Register3 (PTR3)		S/U
0x003C	PWM Interrupt Enable Register (PIER)		S/U
0x0040	PWM Interrupt FlagRegister (PIFR)		S/U
0x0044	PWM Capture Control Register0 (PCCR0)		S/U
0x0048	PWM Capture Control Register1 (PCCR1)		S/U
0x004C	PWM Capture Rising Latch Register0 (PCRLR0)		S/U
0x0050	PWM Capture Falling Latch Register0 (PCFLR0)		S/U
0x0054	PWM Capture Rising Latch Register1 (PCRLR1)		S/U
0x0058	PWM Capture Falling Latch Register1 (PCFLR1)		S/U
0x005C	PWM Capture Rising Latch Register2 (PCRLR2)		S/U
0x0060	PWM Capture Falling Latch Register2 (PCFLR2)		S/U
0x0064	PWM Capture Rising Latch Register3 (PCRLR3)		S/U
0x0068	PWM Capture Falling Latch Register3 (PCFLR3)		S/U
0x006C	PWM Port Control Register (PPCR)		S/U

NOTE: S/U = CPU supervisor or user mode access.

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16.5.2 Registers

16.5.2.1 PWM Pre-scale Register (PPR)

The register*(PPR)* is used to set prescaler and set dead zone length.

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DZI1[7:0]							
rw							
23	22	21	20	19	18	17	16
DZI0[7:0]							
rw							
15	14	13	12	11	10	9	8
CP1[7:0]							
rw							
7	6	5	4	3	2	1	0
CP0[7:0]							
rw							

Figure 16-2: PWM Pre-scale Register (PPR)

DZI1[7:0] — Dead zone interval register 1 for PWM2 and PWM3

These 8-bit determine dead zone length. The 1 unit time of dead zone length is received from clock selector 1.

DZI0[7:0] — Dead zone interval register 0 for PWM0 and PWM1

These 8-bit determine dead zone length. The 1 unit time of dead zone length is received from clock selector 0.

CP1[7:0] — Clock pre-scale 1 for PWM Timer 2 & 3

Clock input is divided by (CP1 + 1) before it is fed to the counter of PWM Timer 2 & 3.If CP1 = 0, then the pre-scale 1 output clock will be stopped.

CP0[7:0] — Clock pre-scale 0 for PWM Timer 0 & 1

Clock input is divided by (CP0 + 1) before it is fed to the counter of PWM Timer 0 & 1.If CP0 =0, then the pre-scale 0 output clock will be stopped.

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16.5.2.2 PWM Clock Select Register (PCSR)

Clock divider provides each timer with 5 clock sources (1, 1/2, 1/4, 1/8, 1/16). Each timer receives its own clock signal from clock divider which receives clock from

8-bit pre-scale.

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved	CSR3			Reserved	CSR2		
ro	rw			ro	rw		
7	6	5	4	3	2	1	0
Reserved	CSR1			Reserved	CSR0		
ro	rw			ro	rw		

Figure 16-3: PWM Clock Select Register(PCSR)

CSR3[2:0] — Timer 3 Clock Source Selection Select clock input for timer 3.

Table 16-3: Clock Source Select

CSR3[2:0]	Input clock devided by
100~111	1
011	16
010	8
001	4
000	2

CSR2[2:0] — Timer 2 Clock Source Selection Select clock input for timer 2.Same as CSR3. CSR1[2:0] — Timer 1 Clock Source Selection

Select clock input for timer 1.Same as CSR3. CSR0[2:0] — Timer 0 Clock Source Selection Select clock input for timer 0.Same as CSR3.

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16.5.2.3 PWM Control Register (PCR)

This register is the PWM control register.

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved				CH3MOD	CH3INV	Reserved	CH3EN
ro				rw	rw	ro	rw
23	22	21	20	19	18	17	16
Reserved				CH2MOD	CH2INV	Reserved	CH2EN
ro				rw	rw	ro	rw
15	14	13	12	11	10	9	8
Reserved				CH1MOD	CH1INV	Reserved	CH1EN
ro				rw	rw	ro	rw
7	6	5	4	3	2	1	0
Reserved		DZEN1	DZEN0	CH0MOD	CH0INV	Reserved	CH0EN
ro		rw	rw	rw	rw	ro	rw

Figure 16-4: PWM Control Register(PCR)

CH3MOD — Timer 3 Auto-load/One-Shot Mode

1 = Auto-load Mode

0 = One-Shot Mode

NOTE: If there is a rising or falling transition at this bit, it will cause CNR3 and CMR3 be clear.

CH3INV — Timer 3 Inverter ON/OFF

1 = Inverter ON

0 = Inverter OFF

CH3EN — Timer 3 Enable/Disable

1 = Enable

0 = Disable

CH2MOD — Timer 2 Auto-load/One-Shot Mode

1 = Auto-load Mode

0 = One-Shot Mode

NOTE: If there is a rising or falling transition at this bit, it will cause CNR2 and CMR2 be clear.

CH2INV — Timer 2 Inverter ON/OFF

1 = Inverter ON

0 = Inverter OFF

CH2EN — Timer 2 Enable/Disable

1 = Enable

0 = Disable

CH1MOD — Timer 1 Auto-load/One-Shot Mode

1 = Auto-load Mode

0 = One-Shot Mode

NOTE: If there is a rising or falling transition at this bit, it will cause CNR1 and CMR1 be clear.

CH1INV — Timer 1 Inverter ON/OFF

1 = Inverter ON

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0 = Inverter OFF

CH1EN — Timer 1 Enable/Disable

1 = Enable

0 = Disable

CH0MOD — Timer 0 Auto-load/One-Shot Mode

1 = Auto-load Mode

0 = One-Shot Mode

NOTE: If there is a rising or falling transition at this bit, it will cause CNR0 and CMR0 be clear.

CH0INV — Timer 0 Inverter ON/OFF

1 = Inverter ON

0 = Inverter OFF

CH0EN — Timer 0 Enable/Disable

1 = Enable

0 = Disable

DZEN1 — Dead-Zone 1 Generator Enable/Disable

1 = Enable

0 = Disable

NOTE: When DZEN1 is enable,CH3EN should be set disable. Because channel3 and channel2 outputs both decided by channel2.

DZEN0 — Dead-Zone 0 Generator Enable/Disable

1 = Enable

0 = Disable

NOTE: When DZEN0 is enable,CH1EN should be set disable. Because channel1 and channel0 outputs both decided by channel0.

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16.5.2.4 PWM Counter Register (PCNR0/1/2/3)

These registers control the period of the PWM by defining the number of the count_pulse in the period.

Addr. Offset: 0x000C~0x000F/0x0018~0x001B/ 0x0024~0x0027/0x0030~0x0033 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CNR0/1/2/3[15:8]							
rw							
7	6	5	4	3	2	1	0
CNR0/1/2/3[7:0]							
rw							

CNR[15:0] — PWM Counter/Timer Loaded Value

Inserted data range : 65535~0 (Unit : 1 PWM clock cycle)

Note 1: One PWM cycle width = CNR + 1. If CNR equal zero, PWM counter/timer will be stopped.

Note 2: Programmer can feel free to write a data to CNR at any time, and it will take effect in next PWM Counter cycle.

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16.5.2.5 PWM Comparator Register (PCMR0/1/2/3)

These registers define the width of the pulse. When the counter matches the value in this register, the output is reset for the duration of the period.

Addr. Offset: 0x0010~0x0013/0x001C~0x001F/ 0x0028~0x002B/0x0034~0x0037 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CMR0/1/2/3[15:8]							
rw							
7	6	5	4	3	2	1	0
CMR0/1/2/3[7:0]							
rw							

Figure 16-5: PWM Comparator Register (PCMR)

CMR[15:0] — PWM Comparator Register

Inserted data range : 65535~0 (Unit : 1 PWM clock cycle)

CMR are used to determine PWM output duty ratio.

Assumption : PWM output initial : high

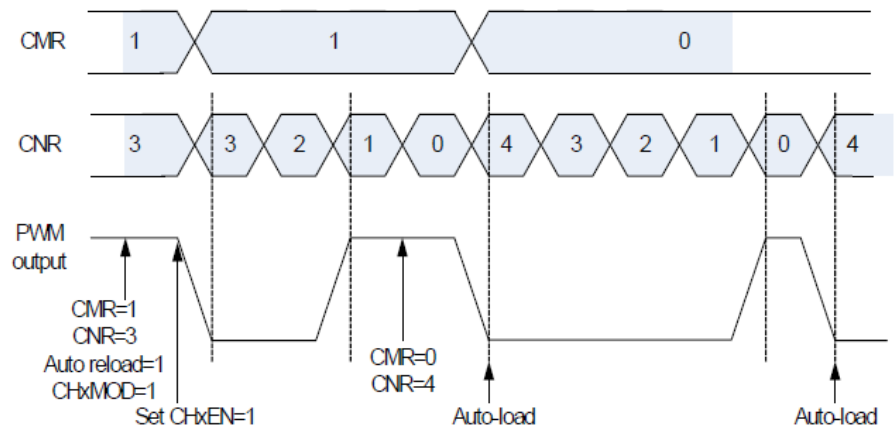
CMR >= CNR : PWM output is always high

CMR < CNR : PWM output high = (CMR + 1) unit

CMR = 0 : PWM output high = 1 unit

Note 1: PWM duty = CMR + 1. If CMR equal zero, PWM duty = 1

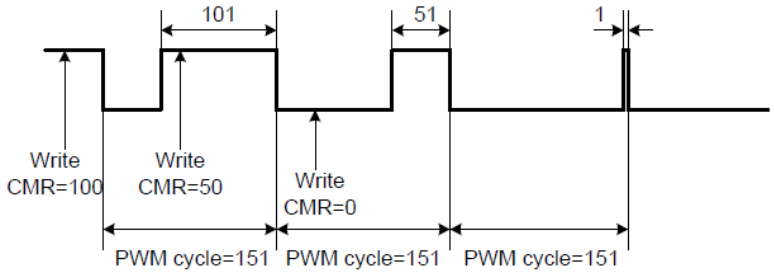
Note 2: Programmer can feel free to write a data to CMR at any time, and it will take effect in next PWM Counter cycle.



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Modulate PWM Controller output duty ratio (CNR=150)



16.5.2.6 PWM Timer Register (PTR0/1/2/3)

The read-only PWM timer registers hold the current count value. It can be read at any time without disturbing the counter.

Addr. Offset: 0x0014~0x0017/0x0020~0x0023/ 0x002C~0x002F/0x0038~0x003B

Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
PTR0/1/2/3[15:8]							
ro							
7	6	5	4	3	2	1	0
PTR0/1/2/3[7:0]							
ro							

Figure 16-6: PWM Timer Register (PTR)

PTR[15:0] — PWM Timer value

The read-only PTR bits holds the current count value. User can monitor PTR to know current value in 16-bit down counter.

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16.5.2.7 PWM Interrupt Enable Register (PIER)

This register is used to enable PWM timer interrupt.

Addr. Offset: 0x003C~0x003F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				PIER3	PIER2	PIER1	PIER0
ro				rw	rw	rw	rw

Figure 16-7: PWM Interrupt Enable Register (PIER)

PIER3 — PWM Timer 3 Interrupt Enable

1 = Enable, 0 = Disable

PIER2 — PWM Timer 2 Interrupt Enable

1 = Enable, 0 = Disable

PIER1 — PWM Timer 1 Interrupt Enable

1 = Enable, 0 = Disable

PIER0 — PWM Timer 0 Interrupt Enable

1 = Enable, 0 = Disable

16.5.2.8 PWM Interrupt Flag Register (PIFR)

This register is used to indicate PWM timer interrupt flag.

Addr. Offset: 0x0040~0x0043 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				PIFR3	PIFR2	PIFR1	PIFR0
ro				rw	rw	rw	rw

Figure 16-8: PWM Interrupt Flag Register (PIFR)

PIFR3 — PWM Timer 3 Interrupt Flag.

When PWM timer3 count to 0,and PIER3 =1,PIFR3 will be set to 1.Write 1 to this bit will clear PIFR3.

1 = Interrupt Flag on

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0 = Interrupt Flag off

PIFR2 — PWM Timer 2 Interrupt Flag.

When PWM timer2 count to 0,and PIER2 =1,PIFR2 will be set to 1.Write 1 to this bit will clear PIFR2.

1 = Interrupt Flag on

0 = Interrupt Flag off

PIFR1 — PWM Timer 1 Interrupt Flag.

When PWM timer1 count to 0,and PIER1 =1,PIFR1 will be set to 1.Write 1 to this bit will clear PIFR1.

1 = Interrupt Flag on

0 = Interrupt Flag off

PIFR0 — PWM Timer 0 Interrupt Flag.

When PWM timer0 count to 0,and PIER0 =1,PIFR0 will be set to 1.Write 1 to this bit will clear PIFR0.

1 = Interrupt Flag on

0 = Interrupt Flag off

16.5.2.9 PWM Capture Control Register (PCCR0/1)

These registers are used to control capture function.

Addr. Offset: 0x0044~0x0047 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
CFLRD1	CRLRD1	Reserved	CAPIF1	CAPCH1EN	FL_IE1	RL_IE1	INV1
rw	rw	ro	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
CFLRD0	CRLRD0	Reserved	CAPIF0	CAPCH0EN	FL_IE0	RL_IE0	INV0
ro	rw	ro	rw	rw	rw	rw	rw

Figure 16-9: PWM Capture Control Register (PCCR0/1)

CFLRDx — Capture Falling Latch Register load flag

1 = When input channel x has a falling transition, CFLRx was updated and this bit was “1”.

0 = When input channel x doesn’t have a falling transition. write 1 clear this bit.

CRLRDx — Capture Rising Latch Register load flag

1 = When input channel x has a rising transition, CRLRx was updated and this bit was “1”.

0 = When input channel x doesn’t have a rising transition. write 1 clear this bit.

CAPIFx — Capture Channel x interrupt flag

1 = When input channel x has a falling transition, and FL_IEx bit enable, this interrupt flag set. When input channel x has a rising transition, and RL_IEx bit enable, this interrupt flag also set.

0 = Capture channel x interrupt flag not set. write 1 clear this bit.

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CAPCHxEN — Capture Channel x Enable/Disable

1 = Enable, 0 = Disable

When Enable, Capture latched the PMW-counter and saved to CRLR (Rising latch) and CFLR (Falling latch). When Disable, Capture does not update CRLR and CFLR, and disable Channel x Interrupt.

FL_IEx — Channel x Falling Interrupt Enable ON/OFF

1 = Enable, 0 = Disable

When Enable, if Capture detects Channel x has falling transition, Capture issues an Interrupt.

RL_IEx — Channel x Rising Interrupt Enable ON/OFF

1 = Enable, 0 = Disable

When Enable, if Capture detects Channel x has rising transition, Capture issues an Interrupt.

INVx — Channel x Inverter ON/OFF

1 = Inverter ON, 0 = Inverter OFF

16.5.2.10 PWM Capture Rising Latch Register (PCRLR0/1/2/3)

These registers are used to latch the PWM counter when capture rising transition.

Addr. Offset: 0x004C~0x004F/0x0054~0x0057/ 0x005C~0x005F/0x0064~0x0067 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CRLR0/1/2/3[15:8]							
ro							
7	6	5	4	3	2	1	0
CRLR0/1/2/3[7:0]							
ro							

Figure 16-10: PWM Capture Rising Latch Register (PCRLR0/1/2/3)

CRLRx[15:0] — Capture Rising Latch Registerx

Latch the PWM counter when Channel x has rising transition.

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16.5.2.11 PWM Capture Falling Latch Register (PCFLR0/1/2/3)

These registers are used to latch the PWM counter when capture falling transition.

Addr. Offset: 0x0050~0x0053/0x0058~0x005B/ 0x0060~0x0063/0x0068~0x006B

Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
CFLR0/1/2/3[15:8]							
ro							
7	6	5	4	3	2	1	0
CFLR0/1/2/3[7:0]							
ro							

Figure 16-11: PWM Capture Falling Latch Register (PCFLR0/1/2/3)

CFLRx[15:0] — Capture Falling Latch Registerx

Latch the PWM counter when Channel x has falling transition.

16.5.2.12 PWM Port Control Register (PPCR)

The register*(PPCR)* is used to control PWMx pin direction and pin status.

Addr. Offset: 0x006C~0x006F

Reset Value: 0x00000F00

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved				PDDR[3:0]			
ro				rw			
15	14	13	12	11	10	9	8
Reserved				PULLUP_EN[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				PDR[3:0]			
ro				rw			

Figure 16-12: PWM Port Control Register (PPCR)

PDDR[3:0] — Port Data Direction Register

The PDDR[3:0] bits control the direction of PWM Pins. Reset clear PDDR[3:0].

1 = Corresponding pin configured as output

0 = Corresponding pin configured as input

PULLUP_EN[3:0] — Port Pull up Enable

The PULLUP_EN[3:0] bits control the pull-up characteristic of PWM Pins.

1 = Enable pull up

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0 = Disable pull up

PDR[3:0] — Port Data Register

Data written to PDR[3:0] drives pins only when they are configured as general-purpose outputs. Reading an input (PDDR bit clear) returns the pin level.

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16.6 Functional Descriptions

This subsection describes the PWM functional operation.

16.6.1 29.6.1 PWM Double Buffering and Automatic Reload

PWM-Timers have a double buffering function, enabling the reload value changed for next timer operation without stopping current timer operation. Although new timer value is set, current timer operation still operate successfully. The counter value can be written into CNR0~3 and current counter value can be read from PTR0~3. The auto-reload operation will copy from CNR0~3 to down-counter when down-counter reaches zero. If CNR0~3 are set as zero, counter will be halt when counter count to zero. If auto-reload bit is set as zero, counter will be stopped immediately.

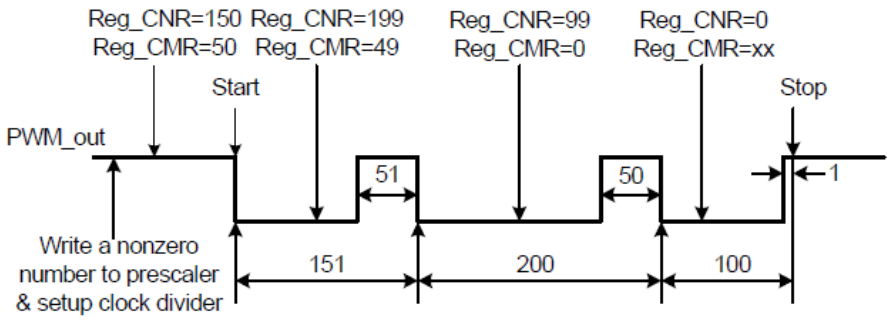


Figure 16-13: PWM Double Buffering Illustration

16.6.2 Modulate Duty Ratio

The double buffering function allows CMR written at any point in current cycle. The loaded value will take effect from next cycle.

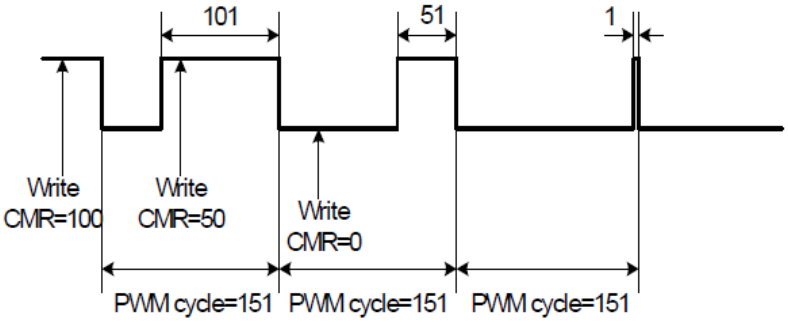


Figure 16-14: PWM Controller Output Duty Ratio

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16.6.3 Dead-Zone Generator

PWM is implemented with Dead Zone generator. They are built for power device protection. This function enables generation of a programmable time gap at the rising of PWM output waveform. User can program PPR [31:24] and PPR [23:16] to determine the two Dead Zone interval respectively.

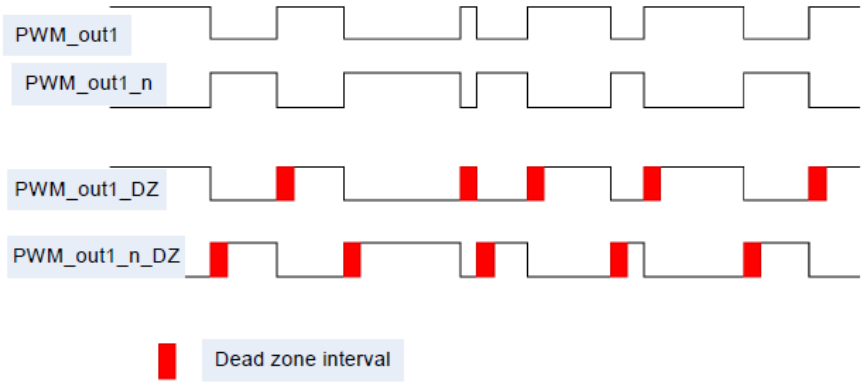


Figure 16-15: Dead Zone Generation Operation

16.6.4 PWM Timer Start Procedure

- Setup clock selector (CSR)
- Setup prescale & dead zone interval (PPR)
- Setup inverter on/off, dead zone generator on/off, toggle mode /one-shot mode, and PWM timer off. (PCR)
- Setup the comparator register (CMR)
- Setup the counter register (CNR)
- Setup the interrupt enable register (PIER)
- Setup PWMx as output pin (PPCR)
- Enable PWM timer (PCR)

16.6.5 PWM Timer Stop Procedure

Method 1:

Set 16-bit down counter (CNR) as 0, and monitor PTR. When PTR reaches to 0, disable PWM timer (PCR). (Recommended)

Method 2:

Set 16-bit down counter (CNR) as 0. When interrupt request happen, disable PWM timer (PCR). (Recommended)

Method 3:

Disable PWM timer directly (PCR). (Not recommended)

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16.6.6 Capture Start Procedure

- Setup clock selector (CSR)
- Setup pre-scale (PPR)
- Setup inverter on/off, dead zone generator on/off, auto-load mode/one-shot mode, and PWM timer off. (PCR)
- Setup the counter register (CNR)
- Setup the capture register (CCR)
- Setup PWMx as input pin (PPCR)
- Enable PWM timer (PCR)

16.6.7 Capture Basic Timer Operation

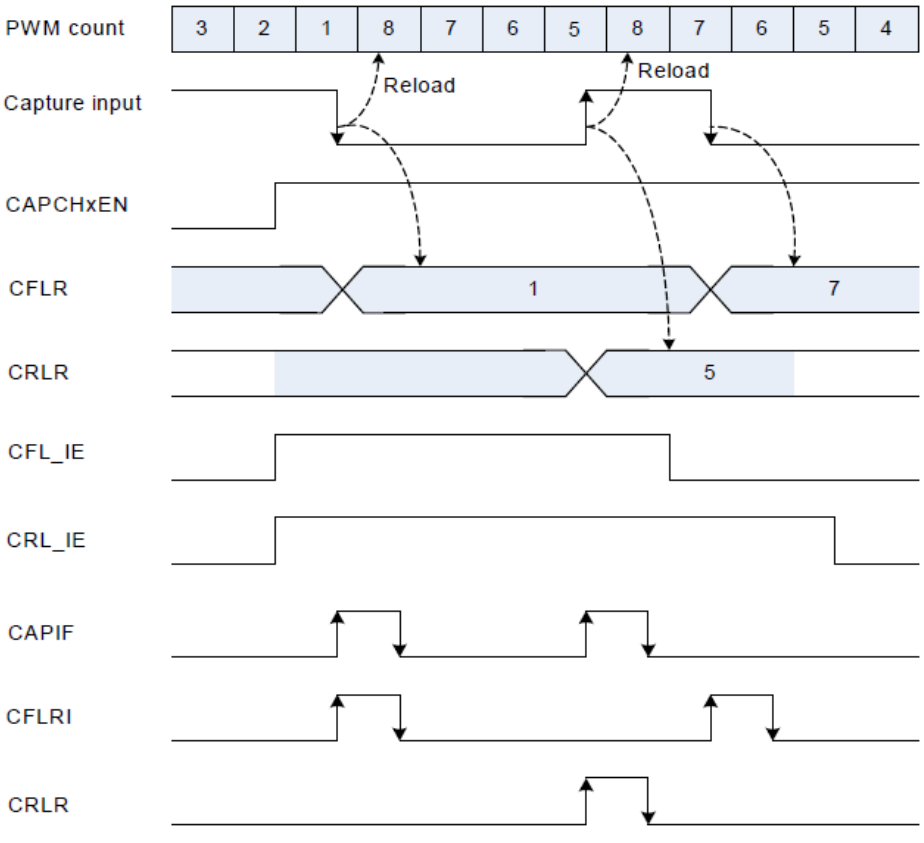


Figure 16-16: Capture Basic Timer Operation

At this case, the CNR is 8:

- When CAPIFx set 1, the PWM counter CNRx will be reload.
- The channel low pulse width is (CNR +1 – CRLR).
- The channel high pulse width is (CNR +1 - CFLR).

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17 PMU_RTC Controller

17.1 Introduction

This module is a controller which controls a comprehensive PMU with RTC function. It can control loading time to RTC, setting alarms and generating time/alarm interrupts.

17.2 Feature

The main features of the module:

- Load time data to and read time data from seconds, minutes, hours and days counters
- Support alarm settings
- Interrupt sources:second, minute, hour,day interrupts,programmable alarm interrupts, 1KHZ/32KHZ periodic interrupts .

17.3 Test Mode

In test mode, we can config the RTC by CPU or by chip pins, and we can directly check the RTC status and clk signals by chip pins.

17.4 Block Diagram

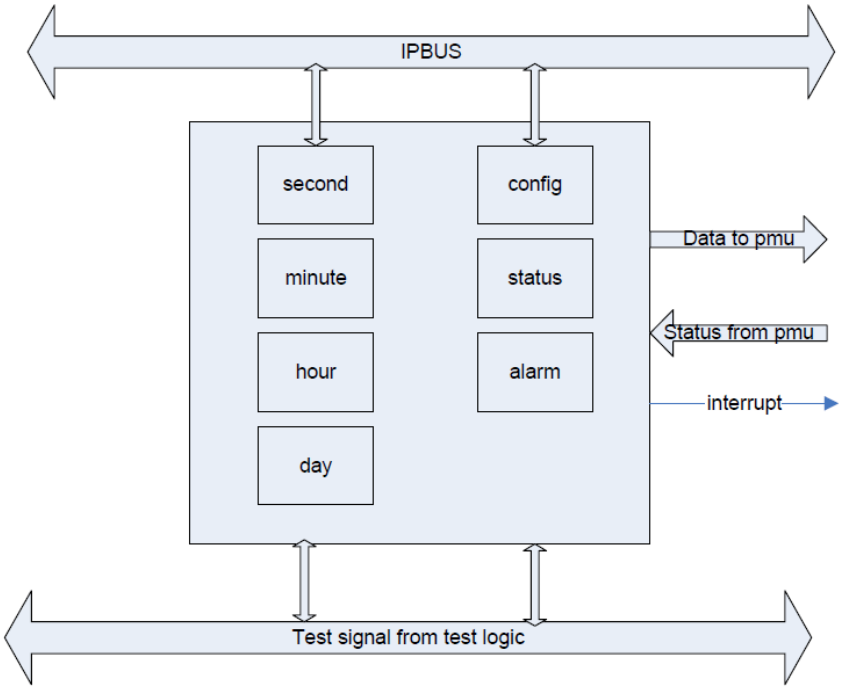


Figure 17-1: Module Block Diagram

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17.5 Signals

The module has no off-chip signals.

17.6 Memory Map and Registers

This subsection describes the memory map and registers for the time counter.

17.6.1 Memory Map

Refer to for an overview of the time counter memory map.

Table 17-1: Module Memory Map

Address Offset	Bits 31-0	Access(1)
0x0000	PMU_RTC Timer1 Register (PRT1R)	S/U
0x0004	PMU_RTC Timer2 Register (PRT2R)	S/U
0x0008	PMU_RTC Alarm1 Register (PRA1R)	S/U
0x000c	PMU_RTC Alarm2 Register (PRA2R)	S/U
0x0010	PMU_RTC Time Counter Register (PRTCR)	S/U
0x0014	PMU_RTC Control and Status Register (PRCSR)	S/U
0x0018	PMU_RTC Enable Register (PRENR)	S/U
0x001c	PMU_RTC Key Register (PRKEYR)	S/U

NOTE: S = CPU supervisor mode access only. S/U = CPU supervisor or user mode access. User mode accesses to supervisor only addresses have no effect and result in a cycle termination transfer error.

17.6.2 Registers

The module consists of these registers:

- PMU_RTC Timer1 Register (PRT1R) can configure and read preload value of days counter.
- PMU_RTC Timer2 Register (PRT2R) can configure and read preload value of seconds,minutes,hours counter.
- PMU_RTC Alarm1 Register (PRA1R) can configure alarm value of days counter.
- PMU_RTC Alarm2 Register (PRA2R) can configure alarm value of seconds,minutes,hours counter.
- PMU_RTC Timer Counter Register (PRTCR) can reflect counter value of seconds,minutes,hours,days counter.
- The PMU_RTC Control and Status Register (PRCSR) can control RTC operation and indicate interrupt flags.
- The PMU_RTC Enable Register (PRENR) can control RTC enable and disable.
- The PMU_RTC Key Register (PRKEYR) can advoid the PMU_RTC change when the voltage is unstable.

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17.6.2.1 PMU_RTC Timer1 Register

PMU_RTC Timer1 Register (PRT1R) can configure and read preload value of days counter.

Addr. Offset: 0x0000~0x0003 Reset Value: 0x0000XXXX

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved	Days[14:8]						
ro	rw						
7	6	5	4	3	2	1	0
Days[7:0]							
rw							

Figure 17-2: PMU_RTC Timer1 Register (PRT1R)

Days[14:0] — RTC days counter

When writing the PRT1R register, Days value will be preloaded to RTC days counter. And when reading the PRT1R register, the value of the register will reflect the value of RTC days counter.

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17.6.2.2 PMU_RTC Timer2 Register

PMU_RTC Timer2 Register (PRT2R) can configure and read preload value of seconds, minutes, hours counter.

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00XXXXXX

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved			Hours[4:0]				
ro			rw				
15	14	13	12	11	10	9	8
Reserved		Minutes[5:0]					
ro		rw					
7	6	5	4	3	2	1	0
Reserved		Seconds[5:0]					
ro		rw					

Figure 17-3: PMU_RTC Timer2 Register (PRT2R)

Seconds — RTC seconds counter

When writing the PRT2R register, seconds value will be preloaded to RTC seconds counter. And when reading the PRT2R register, the value of the register will reflect the value of RTC seconds counter. The setting value should be 0~59.

Minutes — RTC minutes counter

When writing the PRT2R register, minutes value will be preloaded to RTC minutes counter. And when reading the PRT2R register, the value of the register will reflect the value of RTC minutes counter. The setting value should be 0~59.

Hours — RTC hours counter

When writing the PRT2R register, hours value will be preloaded to RTC hours counter. And when reading the PRT2R register, the value of the register will reflect the value of RTC hours counter. The setting value should be 0~23.

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17.6.2.3 PMU_RTC Alarm1 Register

PMU_RTC Alarm1 Register (PRA1R) can configure alarm value of days .

Addr. Offset: 0x0008~0x000b Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
AlaD_en	Alarm_days[14:8]						
rw	rw						
7	6	5	4	3	2	1	0
Alarm_days[7:0]							
rw							

Figure 17-4: PMU_RTC Alarm1 Register (PRA1R)

AlaD_en — Day alarm enable

1 = Enable day alarm function.

0 = Disable day alarm function.

Alarm_Days — RTC day alarm time setting

Alarm_days value configures day alarm compare time.

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17.6.2.4 PMU_RTC Alarm2 Register

PMU_RTC Alarm2 Register (PRA2R) can configure alarm value of seconds, minutes, hours.

Addr. Offset: 0x000c~0x000f Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
AlaH_en	Reserved			Alarm_Hours[4:0]			
rw	ro			rw			
15	14	13	12	11	10	9	8
AlaM_en	Reserved	Alarm_Minutes[5:0]					
rw	ro	rw					
7	6	5	4	3	2	1	0
AlaS_en	Reserved	Alarm_Seconds[5:0]					
rw	ro	rw					

Figure 17-5: PMU_RTC Alarm2 Register (PRA2R)

Alarm_Seconds — RTC seconds alarm time setting Alarm_seconds value configures seconds alarm compare time.

Alarm_Minutes — RTC minute alarm time setting Alarm_minutes value configures minute alarm compare time.

Alarm_Hours — RTC hour alarm time setting

Alarm_hours value configures hour alarm compare time.

AlaS_en — Seconds alarm enable

1 = Enable seconds alarm function.

0 = Disable seconds alarm function.

AlaM_en — Minute alarm enable

1 = Enable minute alarm function.

0 = Disable minute alarm function.

AlaH_en — Hour alarm enable

1 = Enable hour alarm function.

0 = Disable hour alarm function.

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17.6.2.5 PMU_RTC Timer Counter Register

PMU_RTC *Timer Counter* Register can reflect the value of RTC timer counter., which is read only.

Addr. Offset: 0x0010~0x0013 Reset Value: 0xXXXXXXXX

31	30	29	28	27	26	25	24
Days[14:7]							
ro							
23	22	21	20	19	18	17	16
Days[6:0]							Hours[4]
ro							ro
15	14	13	12	11	10	9	8
Hours[3:0]				Minutes[5:2]			
ro				ro			
7	6	5	4	3	2	1	0
Minutes[1:0]		Seconds[5:0]					
ro		ro					

Figure 17-6: PMU_RTC Timer Counter Register (PRTCR)

17.6.2.6 PMU_RTC Control and Status Register

PMU_RTC Control and Status Register can control the RTC operation and indicate interrupt flags

Addr. Offset: 0x0014~0x0017 Reset Value: 0x00000004

31	30	29	28	27	26	25	24
Reserved	Day_IEN	Hou_IEN	Min_IEN	Sec_IEN	Ala_IEN	1KHz_IEN	32KHz_IEN
ro	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved	Day_intf	Hou_intf	Min_intf	Sec_intf	Ala_intf	1KHz_intf	32KHz_intf
ro	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved						int_type[1:0]	
ro						rw	
7	6	5	4	3	2	1	0
Wclk_Div[4:0]					CntUpdate_en	RCNT_WEN	Dir
rw					ro	rw	rw

Figure 17-7: PMU_RTC Control and Status Register (PRCSR)

Day_IEN — Enable/Disable interrupt bit of RTC day pulse

1 = Enable

0 = Disable

Hou_IEN — Enable interrupt bit of RTC hour pulse

1 = Enable

0 = Disable

Min_IEN — Enable interrupt bit of RTC minute pulse

1 = Enable

0 = Disable

Sec_IEN — Enable interrupt bit of RTC seconds pulse

1 = Enable

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0 = Disable

Ala_IEN — Enable interrupt bit of Alarm

1 = Enable

0 = Disable

1KHz_IEN — Enable interrupt bit of RTC 1KHz pulse

1 = Enable

0 = Disable

32KHz_IEN — Enable interrupt bit of RTC 32KHz pulse

1 = Enable

0 = Disable

Day_intf — Day pulse flag

Hou_intf — Hour pulse flag

Min_intf — Minute pulse flag

Sec_intf — Seconds pulse flag

Ala_intf — Alarm flag

1KHz_intf — 1KHz pulse flag

32KHz_intf — 32KHz pulse flag

int_type[1:0] — Interrupt and flag generate type.

00: low level generate interrupt

01: rise edge generate interrupt

10: fall edge generate interrupt

11: both rise and fall edge generate interrupt

Dir — Direct control enable bit

1 = RCNT_WEN bit can directly control PRT1/2R value loading to RTC (see Update RTC Counter Flow (test case Dir=1))

0 = RCNT_WEN bit can't directly control PRT1/2R value loading to RTC, when write PRT1/2R the value loading to RTC automatically, only need to check Cntupdate_en bit(see Update RTC Counter Flow (function case Dir=0))

RCNT_WEN — PRT1/2R value loading to RTC counter write signal(when Dir=1)

1 = Enable

0 = Disable

Wclk_Div[4:0] — Used to divide system clk to low frequency (<10MHz) for loading PRT1/2R value to RTC automatically

divided clk frequency=system_clk/(wclk_div+1) for example:

when system_clk frequency is 50MHz,wclk_div=3'b100,then the divided counter clk is 10MHz.

CntUpdate_en — preload counter value ready bit

1 = Ready

0 = Not ready

NOTE1: When preload counter value by dir=0,we must set both PRT1R and PRT2R.

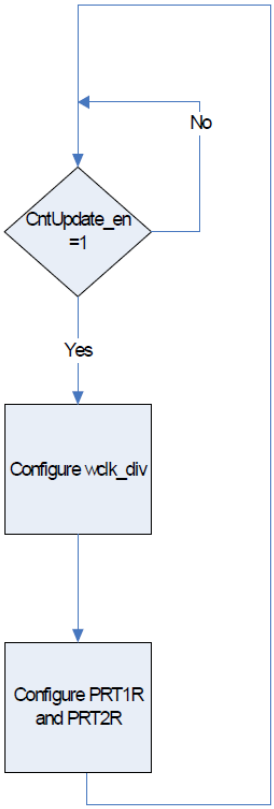
NOTE2: When chip enter stop mode, we must make sure the preload counter value be ready (cntupdate_en=1).

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NOTE3: When preload counter value by dir=1, we must set rtc_en_interface=1(in RCFG12) before set PRT1R and PRT2R.

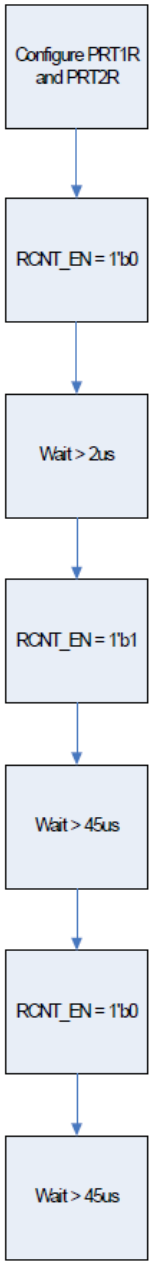
Update RTC Counter Flow (function case Dir=0)



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Update RTC Counter Flow (test case Dir=1)



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17.6.2.7 PMU_RTC Enable Register

PMU_RTC Enable Register can control the RTC enable and disable

Addr. Offset: 0x0018~0x001b Reset Value: 0x00000401

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved					RTC_EN_CntUpdate_en	RTC_EN_RCNT_WEN	RTC_EN_Dir
ro					ro	rw	rw
7	6	5	4	3	2	1	0
Reserved							RTC_EN
ro							rw

Figure 17-8: PMU_RTC Enable Register (PRENR)

RTC_EN — Enable/Disable RTC

1 = Enable

0 = Disable

RTC_EN_Dir — RTC_EN Direct control enable bit

1 = RTC_EN_RCNT_WEN bit can directly control RTC_EN value loading to RTC (see Update RTC_EN Flow (test case Dir=1))

0 = RTC_EN_RCNT_WEN bit can't directly control RTC_EN value loading to RTC, when write RTC_EN the value loading to RTC automatically, only need to check Cntupdate_en bit(see Update RTC_EN Flow (function case Dir=0))

RTC_EN_RCNT_WEN — RTC_EN value loading to RTC_EN write signal(when Dir=1)

1 = Enable

0 = Disable

RTC_EN_CntUpdate_en — RTC_EN value update ready bit

1 = Ready

0 = Not ready

NOTE1: Wclk_div is the same as PRCSR.

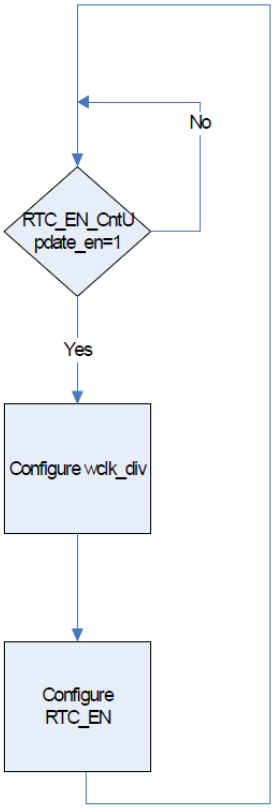
NOTE2: When chip enter stop mode, we must make sure the RTC_EN value be ready (RTC_EN_cntupdate_en=1).

NOTE3: When set RTC_EN value by dir=1,we must set rtc_en_interface=1(in RCFG12) before set RTC_EN.

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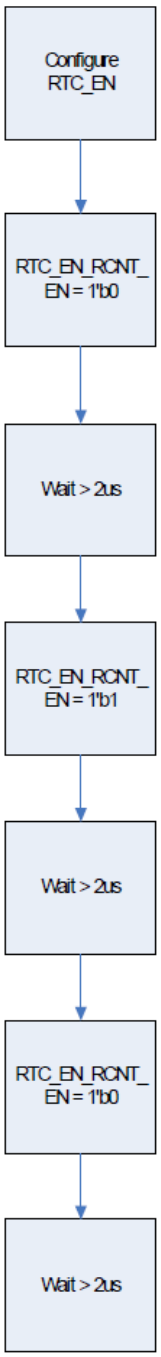
Update RTC_EN Flow (function case Dir=0)



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Update RTC_EN Flow (test case Dir=1)



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17.6.2.8 PMU_RTC Key Register

PMU_RTC Key Register can avoid the PMU_RTC change when the voltage is unstable.

Addr. Offset: 0x001c~0x001f Reset Value: 0x00000000

31	30	29	28	27	26	25	24
rtc_key[31:24]							
rw							
23	22	21	20	19	18	17	16
rtc_key[23:16]							
rw							
15	14	13	12	11	10	9	8
rtc_key[15:8]							
rw							
7	6	5	4	3	2	1	0
rtc_key[7:0]							
rw							

Figure 17-9: PMU_RTC Key Register (PRKEYR)

NOTE: rtc_key should be set to 32'h5aa55aa5.

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18 Programmable 32bit Interrupt Timer Modules(PIT32)

18.1 Introduction

The programmable 32 bit interrupt timer (PIT32) is a 32-bit timer that provides precise interrupts at regular intervals with minimal processor intervention. The timer can either count down from the value written in the modulus latch, or it can be a free-running down-counter.

18.2 Block Diagram

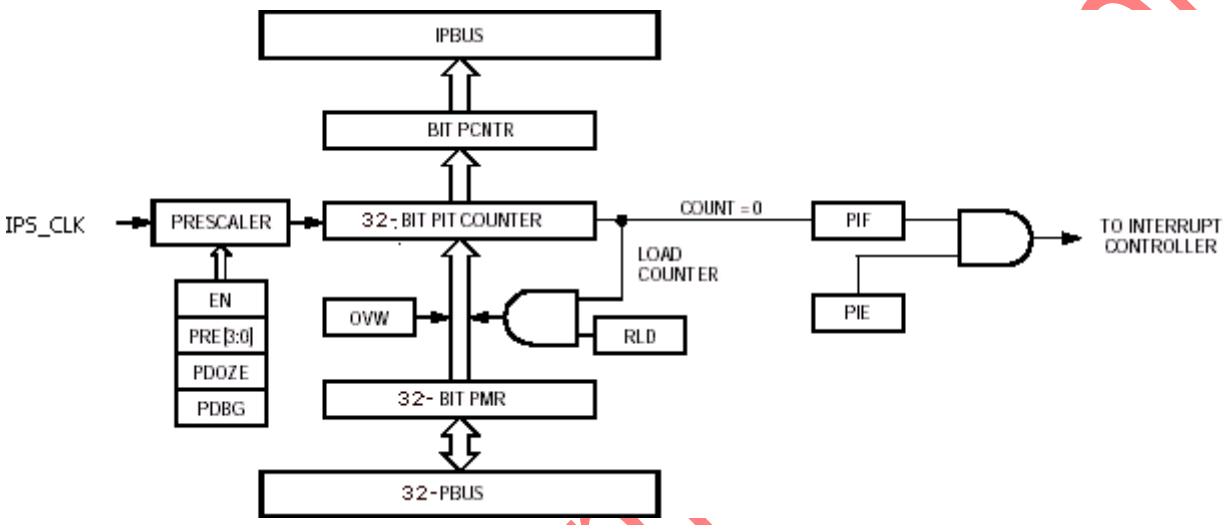


Figure 18-1: PIT Block Diagram

18.3 Modes of Operation

This subsection describes the three low-power modes and the debug mode.

18.3.1 Wait Mode

In wait mode, the PIT32 module continues to operate normally and can be configured to exit the low-power mode by generating an interrupt request.

18.3.2 Doze Mode

In doze mode with the PDOZE bit set in the PIT32 Control and Status Register (PCSR), PIT32 module operation stops. In doze mode with the PDOZE bit clear, doze mode does not affect PIT operation. When doze mode is exited, PIT operation continues from the state it was in before entering doze mode.

18.3.3 Stop Mode

In stop mode, the ips clock is absent, and PIT32 module operation stops.

18.3.4 Debug Mode

In debug mode with the PDBG bit set in PCSR, PIT32 module operation stops. In debug mode with the PDBG

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bit clear, debug mode does not affect PIT operation. When debug mode is exited, PIT operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain.

18.4 Signals

The PIT32 module has no off-chip signals.

18.5 Memory Map and Registers

This subsection describes the memory map and register structure for PIT32.

18.5.1 Memory Map

Refer to **Table 18-1** for a description of the memory map.

Table 18-1: Programmable Interrupt Timer Module Memory Map

Address Offset	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access(1)
0x0000	reserved		PIT32 Control and Status Register (PCSR)		S
0x0004	PIT32 Modulus Register (PMR)				S
0x0008	PIT32 Count Register (PCNTR)				S/U
0x000c	Unimplemented(2)				—

NOTES:

- S = CPU supervisor mode access only. S/U = CPU supervisor or user mode access. User mode accesses to su- pervisor only addresses have no effect and result in a cycle termination transfer error.
- Accesses to unimplemented address locations have no effect and result in a cycle termination transfer error.

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18.5.2 Registers

The PIT programming model consists of these registers:

- The PIT Control and Status Register (PCSR) configures the timer's operation. See 18.5.2.1 PIT Control and Status Register.
- The PIT Modulus Register (PMR) determines the timer modulus reload value. See 18.5.2.2 PIT Modulus Register.
- The PIT Count Register (PCNTR) provides visibility to the counter value. See 18.5.2.3 PIT Count Register.

18.5.2.1 PIT Control and Status Register

Addr. Offset: 0x0000~0x0001 Reset Value: 0x0000

15	14	13	12	11	10	9	8
Reserved				PRE3	PRE2	PRE1	PRE0
ro				rw	rw	rw	rw

7	6	5	4	3	2	1	0
Reserved	PDOZE	0	0	0	PIF	RLD	EN
ro	rw	rw	rw	rw	rw	rw	rw

Figure 18-2: PIT Control and Status Register (PCSR)

PRE[3:0] — Prescaler Bits

The read/write PRE[3:0] bits select the ips clock divisor to generate the PIT clock as **Table 18-2** shows.

To accurately predict the timing of the next count, change the PRE[3:0] bits only when the enable bit (EN) is clear. Changing the PRE[3:0] resets the prescaler counter. System reset and the loading of a new value into the counter also reset the prescaler counter. Setting the EN bit and writing to PRE[3:0] can be done in this same write cycle. Clearing the EN bit stops the prescaler counter.

Table 18-2: Prescaler Select Encoding

PRE[3:0]	Divisor
0000	1
0001	2
0010	4
0011	8
0100	16
0101	32
0110	64
0111	128
1000	256
1001	512
1010	1,024
1011	2,048
1100	4,096
1101	8,192
1110	16,384
1111	32,768

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PDOZE — Doze Mode Bit

The read/write PDOZE bit controls the function of the PIT in doze mode. Reset clears PDOZE.

1 = PIT function stopped in doze mode

0 = PIT function not affected in doze mode

When doze mode is exited, timer operation continues from the state it was in before entering doze mode.

PDBG — Debug Mode Bit

The read/write PDBG bit controls the function of the PIT in debug mode. Reset clears PDBG.

1 = PIT function stopped in debug mode

0 = PIT function not affected in debug mode

During debug mode, register read and write accesses function normally. When debug mode is exited, timer operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain.

NOTE: Changing the PDBG bit from 1 to 0 during debug mode starts the PIT timer. Likewise, changing the PDBG bit from 0 to 1 during debug mode stops the PIT timer.

OVW — Overwrite Bit

The read/write OVW bit enables writing to PMR to immediately overwrite the value in the PIT counter.

1 = Writing PMR immediately replaces value in PIT counter.

0 = Value in PMR replaces value in PIT counter when count reaches 0x0000.

PIE — PIT Interrupt Enable Bit

The read/write PIE bit enables the PIF flag to generate interrupt requests.

1 = PIF interrupt requests enabled

0 = PIF interrupt requests disabled

PIF — PIT Interrupt Flag

The read/write PIF flag is set when the PIT counter reaches 0x0000. Clear PIF by writing a 1 to it or by writing to PMR. Writing 0 has no effect. Reset clears PIF.

1 = PIT count has reached 0x0000.

0 = PIT count has not reached 0x0000.

RLD — Reload Bit

The read/write RLD bit enables loading the value of PMR into the PIT counter when the count reaches 0x00000000.

1 = Counter reloaded from PMR on count of 0x00000000

0 = Counter rolls over to 0xFFFFFFFF on count of 0x00000000

EN — PIT Enable Bit

The read/write EN bit enables PIT operation. When the PIT is disabled, the counter and prescaler are held in a stopped state.

1 = PIT enabled, 0 = PIT disabled

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18.5.2.2 PIT Modulus Register

The 32-bit read/write PIT Modulus Register (PMR) contains the timer modulus value for loading into the PIT counter when the count reaches 0x0000 and the RLD bit is set.

When the OVW bit is set, PMR is transparent, and the value written to PMR is immediately loaded into the PIT counter. The prescaler counter is reset anytime a new value is loaded into the PIT counter and also during reset. Reading the PMR returns the value written in the modulus latch. Reset initializes PMR to 0xFFFFFFFF.

Addr. Offset: 0x0004~0x0007 Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
PM31	PM30	PM29	PM28	PM27	PM26	PM25	PM24
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
PM23	PM22	PM21	PM20	PM19	PM18	PM17	PM16
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
PM15	PM14	PM13	PM12	PM11	PM10	PM9	PM8
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
PM7	PM6	PM5	PM4	PM3	PM2	PM1	PM0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 18-3: PIT32 Modulus Register (PMR)

18.5.2.3 PIT Count Register

The 32-bit, read-only PIT Control Register (PCNTR) contains the counter value. Reading the 32-bit counter with two 8-bit reads is not guaranteed to be coherent. Writing to PCNTR has no effect, and write cycles are terminated normally.

Addr. Offset: 0x0008~0x000B Reset Value: 0xFFFFFFFF

31	30	29	28	27	26	25	24
PC31	PC30	PC29	PC28	PC27	PC26	PC25	PC24
ro	ro	ro	ro	ro	ro	ro	ro
23	22	21	20	19	18	17	16
PC23	PC22	PC21	PC20	PC19	PC18	PC17	PC16
ro	ro	ro	ro	ro	ro	ro	ro
15	14	13	12	11	10	9	8
PC15	PC14	PC13	PC12	PC11	PC10	PC9	PC8
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
PC7	PC6	PC5	PC4	PC3	PC2	PC1	PC0
ro	ro	ro	ro	ro	ro	ro	ro

Figure 18-4: PIT32 Count Register (PCNTR)

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18.6 Functional Description

This subsection describes the PIT32 functional operation.

18.6.1 Set-and-Forget Timer Operation

This mode of operation is selected when the RLD bit in the PCSR register is set.

When the PIT counter reaches a count of 0x0000, the PIF flag is set in PCSR. The value in the modulus latch is loaded into the counter, and the counter begins decrementing toward 0x0000. If the PIE bit is set in PCSR, the PIF flag issues an interrupt request to the CPU.

When the OVW bit is set in PCSR, the counter can be directly initialized by writing to PMR without having to wait for the count to reach 0x0000.

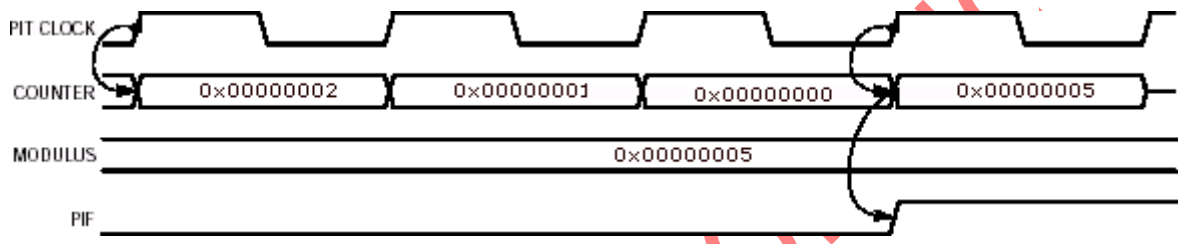


Figure 18-5: Counter Reloading from the Modulus latch

18.6.2 Free-Running Timer Operation

This mode of operation is selected when the RLD bit in PCSR is clear. In this mode, the counter rolls over from 0x0000 to 0xFFFFFFFF without reloading from the modulus latch and continues to decrement.

When the counter reaches a count of 0x0000, the PIF flag is set in PCSR. If the PIE bit is set in PCSR, the PIF flag issues an interrupt request to the CPU.

When the OVW bit is set in PCSR, the counter can be directly initialized by writing to PMR without having to wait for the count to reach 0x0000.

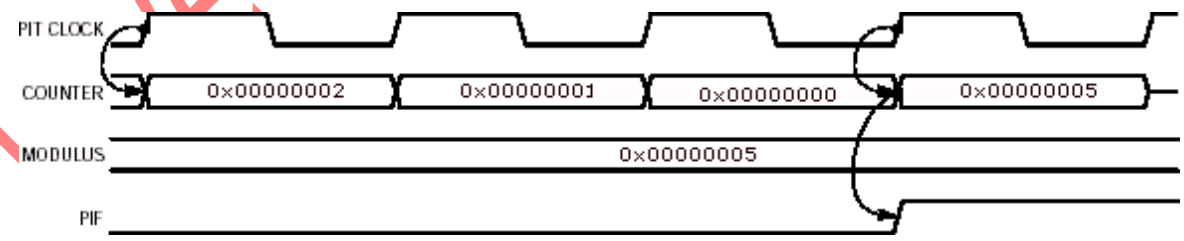


Figure 18-6: Counter in Free-Running Mode

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18.6.3 Timeout Specifications

The 32-bit PIT counter and prescaler supports different timeout periods. The prescaler divides the ips clock as selected by the PRE[3:0] bits in PCSR. The PM[31:0] bits in PMR select the timeout period.

$$\text{timeout period} = 2^{PRE[3:0]} (PM[31:0] + 1) \text{ clocks}$$

18.7 Interrupt Operation

Table 18-3 lists the interrupt requests generated by the PIT.

Table 18-3: PIT Interrupt Requests

Interrupt Request	Flag	Enable Bit
Timeout	PIF	PIE

The PIF flag is set when the PIT counter reaches 0x0000. The PIE bit enables the PIF flag to generate interrupt requests. Clear PIF by writing a 1 to it or by writing to the PMR.

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19 Nested Vectored Interrupt Controller

19.1 Introduction

This document describes the interrupt controller.

19.2 Features

The main features of the NVIC are as follows:

- 64 maskable interrupt channels (not including the 16 interrupt lines of Cortex-M4 with FPU)
- 8 programmable priority levels (3 bits of interrupt priority are used)
- low-latency exception and interrupt handling
- power management control
- implementation of system control registers

The NVIC and the processor core interface are closely coupled, which enables low latency interrupt processing and efficient processing of late arriving interrupts. All interrupts including the core exceptions are managed by the NVIC.

19.3 Interrupt and exception vectors

The Cortex-M4 processor provides a feature-packed exception architecture that supports a number of system exceptions and external interrupts. Exceptions are numbered 1 to 15 for system exceptions and 16 and above for interrupt inputs (inputs to the processor, but not necessarily accessible on the I/O pins of the package). Most of the exceptions, including all interrupts, have programmable priorities, and a few system exceptions have fixed priority.

Table 19-1: Vector table

Posi tion	Priority	Type of priority	Source	Source Description	Address
	-	-	-	Reserved	0x00000000
	-3(highest)	fixed	Reset	Reset	0x00000004
	-2	fixed	NMI	Non maskable interrupt.	0x00000008
	-1	fixed	HardFault	All class of fault	0x0000000C
	-	settable	MemManage	Memory management	0x00000010
	-	settable	BusFault	Pre-fetch fault, memory access fault	0x00000014
	-	settable	UsageFault	Undefined instruction or illegal state	0x00000018
	-	-	-	Reserved	0x0000001C~ 0x0000002B
	-	settable	SVCall	System service call via SWI instruction	0x0000002C
	-	-	-	Reserved	0x00000030
	-	-	-	Reserved	0x00000034
	-	settable	PendSV	Pendable request for system service	0x00000038
	-	settable	SysTick	System tick timer	0x0000003C

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Posi tion	Priority	Type of priority	Source	Source Description	Address
0	-	settable	TFM	Tfm flash interrupt	0x00000040
1	-	settable	PMU	vcc_lvdt5v_int vcc_lvdt1p8v_int card0/1_lvd_int card0/1_en_fail_int	0x00000044
2	-	settable	TC	Time counter interrupt	0x00000048
3	-	settable	PIT1	Pit1 interrupt	0x0000004C
4	-	settable	PIT2	Pit2 interrupt	0x00000050
5	-	settable	EDMAC1	Edmac1 interrupt	0x00000054
6	-	settable	EDMAC2	Edmac2 interrupt	0x00000058
7	-	settable	DMAC1	Dmac1 interrupt	0x0000005C
8	-	settable	DMAC2	Dmac2 interrupt	0x00000060
9	-	settable	RESERVED	RESERVED	0x00000064
10	-	settable	TRNG	Trng interrupt	0x00000068
11	-	settable	PGD/LD/FD/SEC_DET	pgd_int ld_int hfd_int lfd_int shield_fail shield7_fail shield17_fail	0x0000006C
12	-	settable	PCI	wkp_pci_det_valid Pci_detect interrupt	0x00000070
13	-	settable	PCI	wkp_timer0_ov_flag Async timer interrupt	0x00000074
14	-	settable	PCI/SSF33	wkp_WAKEUP_IN Pad wakeup interrupt SSF33 done interrupt	0x00000078
15	-	settable	PMU_RTC	rims_day_int rims_hour_int rims_minute_int rims_second_int Day_int Hour_int Minute_intt Second_int Alarm_intt 1khz_int 32khz_int	0x0000007C
16	-	settable	RSA	RSA done interrupt	0x00000080
17	-	settable	SHA	SHA done interrupt	0x00000084
18	-	settable	RESERVED	RESERVED	0x00000088
19	-	settable	SMS4/AES	SMS4 interrupt AES interrupt	0x0000008C
20	-	settable	QADC	adrdy_int eosmp_int eoc_int eoseq_int ovr_int ewd_int dbrdy_int dgatto_int fifoto_int	0x00000090
21	-	settable	DAC	DAC interrupt	0x00000094
22	-	settable	MCC	MCC interrupt	0x00000098
23	-	settable	TSI	TSI interrupt	0x0000009C

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Posi tion	Priority	Type of priority	Source	Source Description	Address
24	-	settable	USBC	wkp_USBDET_IN wkp_ipi_int_usbc Usbc_int Otg_dma_int	0x000000A0
25	-	settable	Reserved	Reserved	0x000000A4
26	-	settable	SPI1	SPIF_int MODF_int SSF_int	0x000000A8
27	-	settable	SPI2	SPIF_int MODF_int SSF_int	0x000000AC
28	-	settable	SPI3	wkp_ipp_ind_pl2_ss3 SPIF_int MODF_int SSF_int	0x000000B0
29	-	settable	SPIM1	SS1 interrupt	0x000000B4
30	-	settable	SPIM2	SS2 interrupt	0x000000B8
31	-	settable	Reserved	Reserved	0x000000BC
32	-	settable	SCI1	TDRE int TC int RDRF int TIMOUT int OR int IDLE int	0x000000C0
33	-	settable	SCI2	TDRE int TC int RDRF int TIMOUT int OR int IDLE int	0x000000C4
34	-	settable	USI2	ATRI int ERI int REI int TRI int	0x000000C8
35	-	settable	Reserved	Reserved	0x000000CC
36	-	settable	I2C1	I2C1 interrupt	0x000000D0
37	-	settable	PWM	PWM[0] interrupt	0x000000D4
38	-	settable	PWM	PWM[1] interrupt	0x000000D8
39	-	settable	PWM	PWM[2] interrupt	0x000000DC
40	-	settable	PWM	PWM[3] interrupt	0x000000E0
41	-	settable	EPORT1/EPORT3	EPF0 interrupt EPF16 interrupt	0x000000E4
42	-	settable	EPORT1/EPORT3	EPF1 interrupt EPF17 interrupt	0x000000E8
43	-	settable	EPORT1/EPORT3	EPF2 interrupt EPF18 interrupt	0x000000EC
44	-	settable	EPORT1/EPORT3	EPF3 interrupt EPF19 interrupt	0x000000F0
45	-	settable	EPORT1/EPORT3	EPF4 interrupt EPF20 interrupt	0x000000F4
46	-	settable	EPORT1/EPORT3	EPF5 interrupt EPF21 interrupt	0x000000F8
47	-	settable	EPORT1/EPORT3	EPF6 interrupt EPF22 interrupt	0x000000FC
48	-	settable	EPORT1/EPORT3	EPF7 interrupt EPF23 interrupt	0x00000100

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Posi tion	Priority	Type of priority	Source	Source Description	Address
49	-	settable	EPORT2/EPORT4/EPORT5	EPF8 interrupt EPF24 interrupt EPF32 interrupt	0x00000104
50	-	settable	EPORT2/EPORT4/EPORT5	EPF9 interrupt EPF25 interrupt EPF33 interrupt	0x00000108
51	-	settable	EPORT2/EPORT4/EPORT5	EPF10 interrupt EPF26 interrupt EPF34 interrupt	0x0000010C
52	-	settable	EPORT2/EPORT4/EPORT5	EPF11 interrupt EPF27 interrupt EPF35 interrupt	0x00000110
53	-	settable	EPORT2/EPORT4/EPORT5	EPF12 interrupt EPF28 interrupt EPF36 interrupt	0x00000114
54	-	settable	EPORT2/EPORT4/EPORT5	EPF13 interrupt EPF29 interrupt EPF37 interrupt	0x00000118
55	-	settable	EPORT2/EPORT4/EPORT5	EPF14 interrupt EPF30 interrupt EPF38 interrupt	0x0000011C
56	-	settable	EPORT2/EPORT4/EPORT5	EPF15 interrupt EPF31 interrupt EPF39 interrupt	0x00000120
57	-	settable	Reserved	Reserved	0x00000124
58	-	settable	C0_SUB	sub_sys_interrupt wkp_sub_sys_interrupt	0x00000128
59	-	settable	I2C2	I2C2 interrupt	0x0000012C
60	-	settable	I2C3	I2C3 interrupt	0x00000130
61	-	settable	SCI3	TDRE int TC int RDRF int TIMOUT int OR int IDLE int	0x00000134
62	-	settable	Reserved	Reserved	0x00000138
63	-	settable	USI1	ATRI int ERI int REI int TRI int	0x0000013C

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19.4 Registers

19.4.1 System Control Block(SCB) Registers about interrupt.

Base address:0xE000ED00

19.4.1.1 Interrupt Control and State Register

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
NMIPENDSET	Reserved		PENDSVSET	PENDSVCLR	PENDSTSET	PENDSTCLR	Reserved
rw	ro	ro	rw	wo	rw	wo	ro
23	22	21	20	19	18	17	16
ISRPRE-EMPT	ISRPENDING	VECTPENDING					
ro	ro	ro	ro	ro	ro	ro	ro
15	14	13	12	11	10	9	8
VECTPENDING				RETTOBASE	Reserved	VECTACTIVE	
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
VECTACTIVE							
ro	ro	ro	ro	ro	ro	ro	ro

Figure 19-1: Interrupt Control and State Register (ICSR)

NMIPENDSET-NMI pend set, read value indicates NMI pending status

1 = write 1 to pend NMI.

0 = write 0 has no effect.

PENDSVSET-System call pend set, read value indicates SV pending status

1 = write 1 to pend SVI.

0 = write 0 has no effect.

PENDSVCLR-System call pend clear

1 = write 1 to clear SV pending.

0 = write 0 has no effect.

PENDSTSET-SYSTICK pend set, read value indicates SYSTICK pending status

1 = write 1 to pend SYSTICK.

0 = write 0 has no effect.

PENDSTCLR-SYSTICK pend clear

1 = write 1 to clear SYSTICK.

0 = write 0 has no effect.

ISRPRE-EMPT-Indicates that a pending interrupt is going to be active in the next step (for debug).Read-only register.

1 = a pending interrupt is going to be active in the next step.

0 = no pending interrupt is going to be active in the next step.

ISRPENDING-External interrupt pending(excluding system exceptions such as NMI for fault).Read-only register.

1 = external interrupt is pending.

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0 = no external interrupt is pending.

VECTPENDING[9:0]- Pending ISR number RETTOBASE-Return to Thread. Read-only register.

1 = Set to 1 when the processor is running an exception handler; will return to Thread level if interrupt return and no other exceptions pending.

0 = the processor is not running an exception handler.

VECTACTIVE[9:0]- Current running interrupt service routine number.

19.4.1.2 Vector Table Offset Register

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
TBL_OFF							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
TBL_OFF							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
TBL_OFF							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
TBL_OFF	Reserved						
rw	ro	ro	ro	ro	ro	ro	ro

Figure 19-2: Vector Table Offset Register(VTOR)

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19.4.1.3 Application Interrupt and Reset Control Register

Addr. Offset: 0x000C~0x000F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
VECTKEY[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
VECTKEY[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
ENDIANESS	Reserved				PRIGROUP[2:0]		
ro	ro	ro	ro	ro	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved					SYS RESET REQ	VECT CLR ACTIVE	VECT RESET
ro	ro	ro	ro	ro	rw	rw	rw

Figure 19-3: Application Interrupt and Reset Control Register(AIRCR)

VECTKEY[15:0]- Access Key.

Access key:0x05FA must be written to this field to write to this register, otherwise the write will be ignored; the read-back value of the upper half-word is 0xFA05.

ENDIANESS- Indicates endianness for data. This can only change after a reset.

1 = big endian, 0 = little endian

PRIGROUP- Priority group.

The PRIGROUP field indicates the position of the binary point that splits the PRI_*n* fields in the Interrupt Priority Registers into separate *group priority* and *subpriority* fields.

SYSRESETREQ- System reset request

1 = Asserts a signal to the system that requests a reset.

0 = No system reset request

This is intended to force a large system reset of all major components except for debug. This bit reads as 0.

VECTCLRACTIVE- Clears all active state information for exceptions.

This bit is typically used in debug or OS to allow system to recover from system error.

VECTRESET- Resets the Cortex-M3/M4 processor (except debug logic).

This will not reset circuits outside the processor. This is intended for debug operations. Do not use this at the same time as SYSRESETREQ.

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19.4.1.4 System control register (SCR)

Addr. Offset: 0x0010~0x0013 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved			SEVONPEND	Reserved	SLEEPDEEP	SLEEPONEXIT	Reserved
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-4: System control register (SCR)

SEVEONPEND - Send Event on Pending bit.

1 = Enabled events and all interrupts, including disabled interrupts, can wakeup the processor.

0 = Only enabled interrupts or events can wakeup the processor, disabled interrupts are excluded

When an event or interrupt enters pending state, the event signal wakes up the processor from WFE. If the processor is not waiting for an event, the event is registered and affects the next WFE. The processor also wakes up on execution of an SEV instruction or an external event .

SLEEPDEEP - Controls whether the processor uses sleep or deep sleep as its low power mode.

1 = Deep sleep, 0 = Sleep

SLEEPONEXIT - Configures sleep-on-exit when returning from Handler mode to Thread mode.

1 = Enter sleep, or deep sleep, on return from an interrupt service routine.

0 = Do not sleep when returning to Thread mode.

Setting this bit to 1 enables an interrupt-driven application to avoid returning to an empty main application.

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19.4.1.5 Configuration and control register (CCR)

Addr. Offset: 0x0014~0x0017 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved						STKALIGN	BFHFNMIGN
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved			DIV_0_TRP	UNALIGN_TRP	Reserved	USERSETPEND	NONBASETHRDENA
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-5: Configuration and control register (CCR)

STKALIGN - Configures stack alignment on exception entry. On exception entry, the processor uses bit 9 of the stacked PSR to indicate the stack alignment. On return from the exception it uses this stacked bit to restore the correct stack alignment.

1 = 8-byte aligned

0 = 4-byte aligned

BFHFNMIGN - Enables handlers with priority -1 or -2 to ignore data bus faults caused by load and store instructions. This applies to the hard fault, NMI, and FAULTMASK escalated handlers. Set this bit to 1 only when the handler and its data are in absolutely safe memory. The normal use of this bit is to probe system devices and bridges to detect control path problems and fix them.

1 = Handlers running at priority -1 and -2 ignore data bus faults caused by load and store instructions

0 = Data bus faults caused by load and store instructions cause a lock-up

DIV_0_TRP - Enables faulting or halting when the processor executes an SDIV or UDIV instruction with a divisor of 0.

1 = Trap divide by 0, 0 = Do not trap divide by 0

UNALIGN_TRP - Enables unaligned access traps .

1 = Trap unaligned halfword and word accesses

0 = Do not trap unaligned halfword and word accesses

If this bit is set to 1, an unaligned access generates a usage fault. Unaligned LDM, STM, LDRD, and STRD instructions always fault irrespective of whether UNALIGN_TRP is set to 1.

USERSETMPEND - Enables unprivileged software access to the STIR.

1 = Enable, 0 = Disable

NONBASETHRDENA - Configures how the processor enters Thread mode.

1 = Processor can enter Thread mode from any level under the control of an EXC_RETURN value

0 = Processor can enter Thread mode only when no exception is active

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19.4.1.6 System handler priority registers

Addr. Offset: 0x0018+I (i = 0~11) Reset Value: 0x00

31	30	29	28	27	26	25	24
SHP[i][7:5]			0	0	0	0	0
rw	rw	rw	ro	ro	ro	ro	ro

Figure 19-6: System handler priority registers(SHP)

SHP-System handler priority

SHP[0]:MemManage Fault Priority Level SHP[1]:Bus Fault Priority Level SHP[2]:Usage Fault Priority Level SHP[7]:SVC Priority Level SHP[10]: PendSV Priority Level SHP[11]: SysTick Priority Level SHP[3]/[4]/[5]/[6]/[8]/[9] not implemented.

19.4.1.7 System handler control and state register

Addr. Offset: 0x0024~0x0027 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro	ro	ro	ro	ro	ro	ro	ro
23	22	21	20	19	18	17	16
Reserved					USGFAULTENA	BUSFAULTENA	MEMFAULTENA
ro	ro	ro	ro	ro	rw	rw	rw
15	14	13	12	11	10	9	8
SVCALLPENDED	BUSFAULTPENDED	MEMFAULTPENDED	USGFAULTPENDED	SYSTICKACT	PENDSVACT	Reserved	
rw	rw	rw	rw	rw	rw	ro	ro
7	6	5	4	3	2	1	0
SVCALLACT	Reserved			USGFAULTACT	Reserved	BUSFAULTACT	MEMFAULTACT
rw	ro	ro	ro	rw	ro	rw	rw

Figure 19-7: System Handler Control and State Register (SHCSR)

USGFAULTENA-Usage fault handler enable

1 = handler enable.

0 = handler disable.

BUSFAULTENA-Bus fault handler enable

1 = handler enable.

0 = handler disable.

MEMFAULTENA-Memory Management fault handler enable

1 = handler enable.

0 = handler disable.

SVCALLPENDED-SVC pended. SVCall was started but was replaced by a higher priority exception.

1 = handler pended.

0 = handler not pended

BUSFAULTPENDED-Bus fault pended. Bus fault was started but was replaced by a higher priority exception.

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1 = handler pended.

0 = handler not pended

MEMFAULTPENDED-Memory Management fault pended. Memory Management fault was started but was replaced by a higher priority exception.

1 = handler pended.

0 = handler not pended

USGFAULTPENDED-Usage fault pended. Usage fault was started but was replaced by a higher priority exception.

1 = handler pended.

0 = handler not pended

SYSTICKACT-SYSTICK is active

1 = handler is active.

0 = handler is not active.

PENDSVACT-PendSV is active

1 = handler is active.

0 = handler is not active.

SVCALLACT-SVCall is active

1 = handler is active.

0 = handler is not active.

USGFAULTACT-USGFAULT is active

1 = handler is active.

0 = handler is not active.

BUSFAULTACT-BUSFAULTis active

1 = handler is active.

0 = handler is not active.

MEMFAULTACT-MEMFAULT is active

1 = handler is active.

0 = handler is not active.

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19.4.2 NVIC Registers.

Base address:0xE000E000

19.4.2.1 Interrupt Set Enable Registers

Addr. Offset: 0x0100+0x4*i (i = 0~7) Reset Value: 0x00000000

31	30	29	28	27	26	25	24
ISE[31:24]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
ISE[23:16]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
ISE[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
ISE[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-8: Interrupt Set Enable Registers (ISER)

ISE-Interrupt Set enable

1 = write 1 to set interrupt enable

0 = write 0 has no effect.

19.4.2.2 Interrupt Clear Enable Registers

Addr. Offset: 0x0180+0x4*i (i = 0~7) Reset Value: 0x00000000

31	30	29	28	27	26	25	24
ICE[31:24]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
ICE[23:16]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
ICE[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
ICE[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-9: Interrupt Clear Enable Registers (ICER)

ICE-Interrupt Clear enable

1 = write 1 to clear interrupt enable

0 = write 0 has no effect.

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19.4.2.3 Interrupt Set Pending Registers

Addr. Offset: 0x0200+0x4*i (i = 0~1) Reset Value: 0x00000000

31	30	29	28	27	26	25	24
ISP[31:24]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
ISP[23:16]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
ISP[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
ISP[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-10: Interrupt Set Pending Registers (ISPR)

ISP-Interrupt Set pending status

1 = write 1 to set interrupt pending status

0 = write 0 has no effect.

19.4.2.4 Interrupt Clear Pending Registers

Addr. Offset: 0x0280+0x4*i (i = 0~1) Reset Value: 0x00000000

31	30	29	28	27	26	25	24
ICP[31:24]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
ICP[23:16]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
ICP[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
ICP[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-11: Interrupt Set Pending Registers (ICPR)

ICP-Interrupt Clear pending status

1 = write 1 to clear interrupt pending status

0 = write 0 has no effect.

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19.4.2.5 Interrupt Active Bit Registers

Addr. Offset: 0x0300+0x4*i (i = 0~1) Reset Value: 0x00000000

31	30	29	28	27	26	25	24
IAB[31:24]							
ro	ro	ro	ro	ro	ro	ro	ro
23	22	21	20	19	18	17	16
IAB[23:16]							
ro	ro	ro	ro	ro	ro	ro	ro
15	14	13	12	11	10	9	8
IAB[15:8]							
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
IAB[7:0]							
ro	ro	ro	ro	ro	ro	ro	ro

Figure 19-12: Interrupt Set Pending Registers (ISPR)

IAB-Interrupt Active status bit. Read-only register.

1 = interrupt active status set.

0 = interrupt active status not set.

19.4.2.6 Interrupt Priority Registers

Addr. Offset: 0x0400+i (i = 0~239) Reset Value: 0x0000

31	30	29	28	27	26	25	24
IP[i][7:5]			Reserved				
rw	rw	rw	ro	ro	ro	ro	ro

Figure 19-13: Interrupt Priority Registers (IPR)

IP-Interrupt Priority level(8bit wide) for each interrupt.

Table 19-2: Priority Grouping

	Interrupt Priority level IP[7:5]		Number of	
PRIGROUP [2:0]	**Group priority bits**	**Subpriority bits**	**Group priorities**	**Sub priorities**
0b111	None	[7:5]	None	8
0b110	[7]	[6:5]	2	4
0b101	[7:6]	[5]	4	2
other	[7:5]	None	8	None

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19.4.2.7 Software Trigger Interrupt Register

Addr. Offset: 0x0F00~0x0F03 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
STI[31:24]							
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
STI[23:16]							
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
STI[15:8]							
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
STI[7:0]							
rw	rw	rw	rw	rw	rw	rw	rw

Figure 19-14: Software Trigger Interrupt Register (STIR)

STI-Software Trigger Interrupt set. Write an interrupt number to set its pending status.

1 = write 1 to set interrupt pending status

0 = write 0 has no effect.

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20 IO Control Module

20.1 Introduction

The IO control module is responsible for controlling the signal of the IO PAD.

20.2 Block Diagram

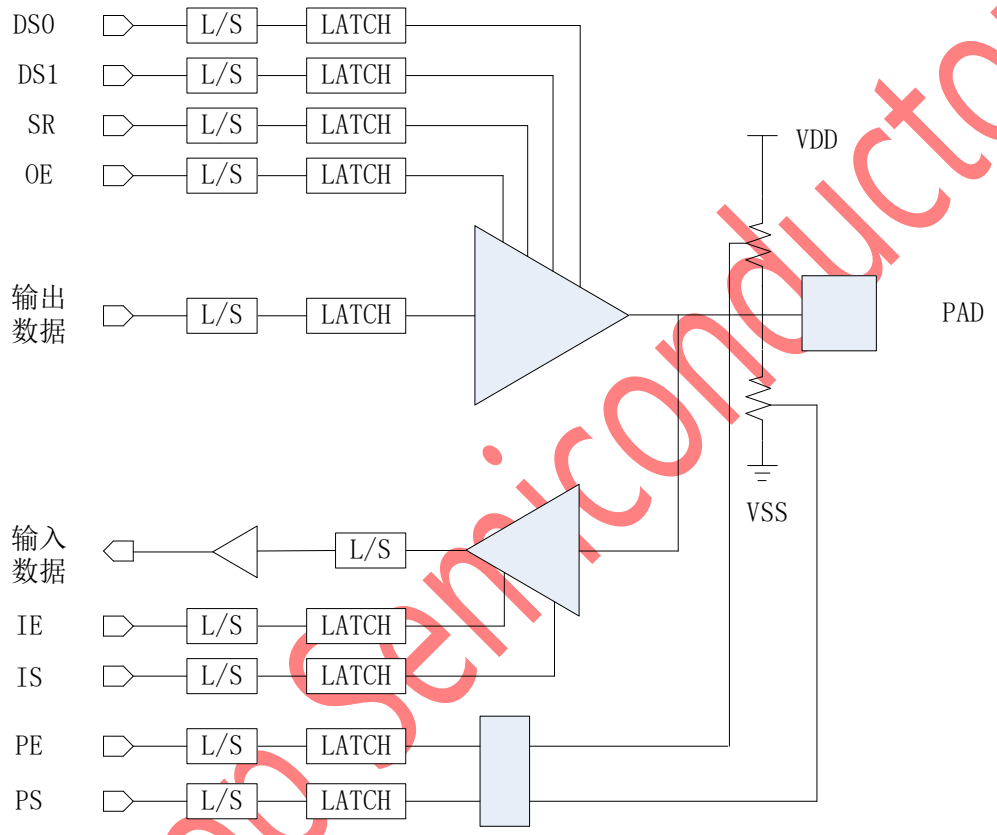


Figure 20-1: IO_CTRL Block Diagram

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20.3 Module Memory Map and Register

20.3.1 Memory Map

Table 20-1 shows the register memory map.

Table 20-1: Register Memory Map

Address Offset	Bits 31-16	Bits 15-0	Access[1,2]
0x0000	SPI Control Register		S
0x0004	USI Control Register		S
0x0008	I2C Control Register		S
0x000c	SCI Control Register		S
0x0010	GINTL Control Register		S
0x0014	GINTH Control Register		S
0x0018	Reserved[3]		S
0x001c	SWAP Control Register		S
0x0020	SPIM1 Control Register		S
0x0024	SPIM2 Control Register		S
0x0028	Reserved		S
0x002C	GINT[31:30] Control Register		S
0x0030	Reserved		S
0x0034	Reserved		S
0x0038	GINT[39:32] Control Register		S
0x003c	GINT[29:22] Control Register		S
0x0040	Reserved		S
0x0044	CLKOUT/RSTOUT Control Register		S
0x0048	Reserved		S
0x004c	Reserved		S
0x0050	Reserved		S

NOTES:

- S = CPU supervisor mode access only.
- User mode accesses to supervisor-only address locations have no effect and result in a cycle termination transfer error.
- Reserved register can't be written.If user config it, transfer error will generate.

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20.3.2 Register Descriptions

20.3.2.1 SPI Control Register

Addr. Offset: 0x0000~0x0003 Reset Value: 0xFFFFFF05

31	30	29	28	27	26	25	24
MOSI3_IE	MISO3_IE	SCK3_IE	SS3_IE	MOSI3_PS	MISO3_PS	SCK3_PS	SS3_PS
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
MOSI2_IE	MISO2_IE	SCK2_IE	SS2_IE	MOSI2_PS	MISO2_PS	SCK2_PS	SS2_PS
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
MOSI1_IE	MISO1_IE	SCK1_IE	SS1_IE	MOSI1_PS	MISO1_PS	SCK1_PS	SS1_PS
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	ro	

Figure 20-2: SPI Control Register (SPICR)

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IE-Input enable.

0 = input disable, 1 = input enable

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-2: SPI DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

20.3.2.2

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20.3.2.3 USI Control Register

Addr. Offset: 0x0004~0x0007 Reset Value: 0x707007C5

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved	ISOCLK2_DIEN	ISODAT2_DIEN	ISORST2_DIEN	Reserved	ISOCLK2_PUEN	ISODAT2_PUEN	ISORST2_PUEN
ro	rw	rw	rw	ro	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved					ISOCLK1_PS	ISODAT1_PS	ISORST1_PS
ro					rw	rw	rw
7	6	5	4	3	2	1	0
Reserved	USI2_DREN	Reserved		IS	SR	DS[1:0]	
ro	rw	ro		rw	rw	ro	

Figure 20-3: USI Control Register (USICR)

DIEN-Input enable,

0 = input disable, 1 = input enable

PUEN-Pull up enable,

0 = pull up disable, 1 = pull up enable

USI2_DREN-USI2 Drive strength enable.

0 = drive strength disable, 1 = drive strength enable

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-3: USI1 DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.4 I2C Control Register

Addr. Offset: 0x0008~0x000B

Reset Value: 0x33333305

31	30	29	28	27	26	25	24
Reserved		SDA3_IE	SCL3_IE	Reserved		SDA3_PS	SCL3_PS
ro		rw	rw	ro		rw	rw
23	22	21	20	19	18	17	16
Reserved		SDA2_IE	SCL2_IE	Reserved		SDA2_PS	SCL2_PS
ro		rw	rw	ro		rw	rw
15	14	13	12	11	10	9	8
Reserved		SDA_ IE	SCL_ IE	Reserved		SDA_ PS	SCL_ PS
ro		rw	rw	ro		rw	rw
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-4: I2C Control Register (I2CCR)

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IE-Input enable.

0 = input disable, 1 = input enable

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select., 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-4: I2C DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.5 SCI Control Register

Addr. Offset: 0x000C~0x000F

Reset Value: 0x803BBB05

31	30	29	28	27	26	25	24
SCI_GINT_SWAP_LOAD_EN	Reserved	SCI_GINT_SWAP[5:0]					
rw	ro	rw					
23	22	21	20	19	18	17	16
Reserved				SCI2_CTS_PUE	Reserved	RXD2_PS	TXD2_PS
ro				rw	ro	rw	rw
15	14	13	12	11	10	9	8
SCI3_CTS_PUE	Reserved	RXD3_PS	TXD3_PS	SCI1_CTS_PUE	Reserved	RXD_PS	TXD_PS
rw	ro	rw	rw	rw	ro	rw	rw
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-5: SCI Control Register (SCICR)

SCI_GINT_SWAP_LOAD_EN- SCI/GINT swap control load from eflash enable signal.

0 = swap control not load from eflash

1 = swap control load from eflash

SCI_GINT_SWAP[5:0] - SCI/GINT swap enable signals.

0 = swap disable(GINT function)

1 = swap enable(SCI function)

SCI_GINT_SWAP[0] — sci1 rxd

SCI_GINT_SWAP[1] — sci3 txd

SCI_GINT_SWAP[2] — sci3 rxd

SCI_GINT_SWAP[3] — sci2 rxd

SCI_GINT_SWAP[4] — sci1 txd

SCI_GINT_SWAP[5] — sci2 txd

CTS_PUE- SCI control flow signal pull up enable.

0 = pull down select, 1 = pull up select

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

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Table 20-5: SCI DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

20.3.2.6 GINTL Control Register

Addr. Offset: 0x0010~0x0013 Reset Value: 0xFFFF0005

31	30	29	28	27	26	25	24
GINT7_PS	GINT6_PS	GINT5_PS	GINT4_PS	GINT3_PS	GINT2_PS	GINT1_PS	GINT0_PS
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
GINT7_IE	GINT6_IE	GINT5_IE	GINT4_IE	GINT3_IE	GINT2_IE	GINT1_IE	GINT0_IE
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-6: GINTL Control Register (GINTLCR)

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IE-Input enable.

0 = input disable, 1 = input enable

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-6: GINT[7:0] DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.7 GINTH Control Register

Addr. Offset: 0x0014~0x0017 Reset Value: 0xFFFF0005

31	30	29	28	27	26	25	24
GINT15_PS	GINT14_PS	GINT13_PS	GINT12_PS	GINT11_PS	GINT10_PS	GINT9_PS	GINT8_PS
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
GINT15_IE	GINT14_IE	GINT13_IE	GINT12_IE	GINT11_IE	GINT10_IE	GINT9_ IE	GINT8_ IE
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-7: GINTH Control Register (GINTHCR)

PS-Pull up or down select.

0 = pull down select

1 = pull up select

IE-Input enable.

0 = input disable

1 = input enable

IS-Input select.

0 = CMOS input select

1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select

1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-7: GINT[15:8] DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.8 SWAP Control Register

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000002

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
SWAP[23:16]							
rw							
15	14	13	12	11	10	9	8
SWAP[15:8]							
rw							
7	6	5	4	3	2	1	0
SWAP[7:0]							
rw							

Figure 20-8: SWAP Control Register (SWAPCR)

SWAP[0] — {sck3,miso3,mosi3}/gint[9:11]

0 = swap[0] disable.(spi3 function enable.)

1 = swap[0] enable.(gint[9:11] enable.)

SWAP[1] — gint[15:14] Control.

0 = swap[1] disable.(gint[15:14] enable.)

1 = swap[1] enable.(gint[15:14] disable.)

SWAP[2] — gint[39:38]

0 = swap[2] disable.(not enable.)

1 = swap[2] enable.(gint[39:38] enable.)

SWAP[3] — {ss1,sck1,mosi1,miso1}/{ss_spi5,sck_spi5,spi5[0],spi5[1]}

0 = swap[3] disable.(spi1 function enable.)

1 = swap[3] enable.(spi5 function enable.)

SWAP[4] — gint[7:6]/{spi5[2],spi5[3]}

0 = swap[4] disable.(gint[7:6] enable.)

1 = swap[4] enable.(spi5 function enable.)

SWAP[5] — usb DRVVBUS/ss_spi4 swap

0 = swap[5] disable.(spi4 function enable.)

1 = swap[5] enable.(USB function enable.)

SWAP[6] — usb DTO/sck_spi4 swap

0 = swap[6] disable.(spi4 function enable.)

1 = swap[6] enable.(USB Test function enable.)

SWAP[7] — gint[37:32]

0 = swap[7] disable.(not enable.)

1 = swap[7] enable.(gint[37:32] enable.)

SWAP[8] — ss2 swap

0 = swap[8] disable.(spi2 function enable.)

1 = swap[8] enable.(not enable.)

SWAP[9] — sck2 swap

0 = swap[9] disable.(spi2 function enable.)

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1 = swap[9] enable.(not enable.)

SWAP[10] — mosi2 swap

0 = swap[10] disable.(spi2 function enable.)

1 = swap[10] enable.(not enable.)

SWAP[11] — miso2 swap

0 = swap[11] disable.(spi2 function enable.)

1 = swap[11] enable.(not enable.)

SWAP[12] — gint[31:26]

0 = swap[12] disable.(not enable.)

1 = swap[12] enable.(gint[31:26] enable.)

SWAP[13] — gint[25:22]

0 = swap[13] disable.(not enable.)

1 = swap[13] enable.(gint[25:22] enable.)

SWAP[14] — {ss_spi4,sck_spi4,spi4[1:0]}/gint[21:18]

0 = swap[14] disable.(spi4 function enable.)

1 = swap[14] enable.(gint[21:18] enable.)

SWAP[15] — spi4[3:2]/gint[17:16]

0 = swap[15] disable.(spi4 function enable.)

1 = swap[15] enable.(gint[17:16] enable.)

SWAP[16] — fls_ctrl_pin2pad[6:0]

0 = swap[16] disable.(not enable.)

1 = swap[16] enable.(flash test pin enable.)

SWAP[17] — gint[13]/swv

0 = swap[17] disable.(gint function enable.)

1 = swap[17] enable.(trace function enable.)

SWAP[18] — gint[14]/sci1_rts

0 = swap[18] disable.(gint function enable.)

1 = swap[18] enable.(sci1 function enable.)

SWAP[19] — gint[15]/sci1_cts

0 = swap[19] disable.(gint function enable.)

1 = swap[19] enable.(sci1 function enable.)

SWAP[20] — gint[12]/sci2_rts

0 = swap[20] disable.(gint function enable.)

1 = swap[20] enable.(sci2 function enable.)

SWAP[21] — gint[13]/sci2_cts

0 = swap[21] disable.(gint function enable.)

1 = swap[21] enable.(sci2 function enable.)

SWAP[22] — gint[6]/sci3_rts

0 = swap[22] disable.(gint function enable.)

1 = swap[22] enable.(sci3 function enable.)

SWAP[23] — gint[7]/sci3_cts

0 = swap[23] disable.(gint function enable.)

1 = swap[23] enable.(sci3 function enable.)

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20.3.2.9 SPIM1 Control Register

Addr. Offset: 0x0020~0x0023 Reset Value: 0x3F3F0015

31	30	29	28	27	26	25	24
Reserved		D3_IE	D2_IE	D1_IE	D0_IE	SCK_IE	SS_IE
ro		rw	rw	rw	Rw	rw	rw
23	22	21	20	19	18	17	16
Reserved		D3_PS	D2_PS	D1_PS	D0_PS	SCK_PS	SS_PS
ro		rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved		ODE	PUE	IS	SR	DS[1:0]	
ro		rw	rw	rw	rw	rw	

Figure 20-9: SPIM1 Control Register (SPIM1CR)

ODE-Open-drain output enable.

0 = CMOS output enable, 1 = Open-drain output enable

PUE-Pull up or down enable.

0 = pull up or down disable, 1 = pull up or down enable

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IE-Input enable.

0 = input disable, 1 = input enable

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-8: SPIM1 DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.10 SPIM2 Control Register

Addr. Offset: 0x0024~0x0027 Reset Value: 0x3F3F0015

31	30	29	28	27	26	25	24
Reserved		D3_IE	D2_IE	D1_IE	D0_IE	SCK_IE	SS_IE
ro		rw	rw	rw	Rw	rw	rw
23	22	21	20	19	18	17	16
Reserved		D3_PS	D2_PS	D1_PS	D0_PS	SCK_PS	SS_PS
ro		rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved		ODE	PUE	IS	SR	DS[1:0]	
ro		rw	rw	rw	rw	rw	

Figure 20-10: SPIM2 Control Register (SPIM2CR)

ODE-Open-drain output enable.

0 = CMOS output enable

1 = Open-drain output enable

PUE-Pull up or down enable.

0 = pull up or down disable, 1 = pull up or down enable

PS-Pull up or down select.

0 = pull down select, 1 = pull up select

IE-Input enable.

0 = input disable, 1 = input enable

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-9: SPIM2 DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.11 GINT[31:30] Control Register

Addr. Offset: 0x002C~0x002F Reset Value: 0x3F3F0012

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro		rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-11: GINT[31:30] Control Register (GINT31:30CR)

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-10: DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.12 GINT[39:32] Control Register

Addr. Offset: 0x0038~0x003B Reset Value: 0xFFFF0312

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro		rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-12: GINT[39:32] Control Register (GINT39:32CR)

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-11: DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.13 GINT[29:22] Control Register

Addr. Offset: 0x003C~0x003F Reset Value: 0xFFFFFF01

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro		rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
PS[7:0]							
rw							
7	6	5	4	3	2	1	0
Reserved				IS	SR	DS[1:0]	
ro				rw	rw	rw	

Figure 20-13: GINT[29:22] Control Register (GINT29:22CR)

PS-Pull up/down select

0 = pull down

1 = pull up

IS-Input select.

0 = CMOS input select, 1 = Schmitt input select

SR-Slew rate select.

0 = fast slew rate select, 1 = slow slew rate select

DS1/DS0-Output driver strength select.

Table 20-12: DS1/DS0

DS0	DS1	Driving capability
0	0	2mA
0	1	4mA
1	0	8mA
1	1	12mA

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20.3.2.14 CLKOUT/RSTOUT Control Register

Addr. Offset: 0x0044~0x0047

Reset Value: 0x3F070501

31	30	29	28	27	26	25	24
SWAP[3:0]				Reserved			
rw				ro			
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 20-14: CLKOUT/RSTOUT Control Register (CLKRSTCR)

swap[0] — clkout swap

0 = swap disable.(not enable.)

1 = swap enable.(swap function enable.)

swap[1] — rstout swap

0 = swap disable.(not enable.)

1 = swap enable.(swap function enable.)

swap[2] — clkout_gp swap

0 = swap disable.(not enable.)

1 = swap enable.(swap function enable.)

swap[3] — rstout_gp swap

0 = swap disable.(not enable.)

1 = swap enable.(swap function enable.)

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21 Edge Port Module (EPORT)

21.1 Introduction

The edge port module (EPORT) has eight external interrupt pins. Each pin can be configured individually as a low level-sensitive interrupt pin, an edge-detecting interrupt pin (rising edge, falling edge, or both), or a general-purpose input/output/ (I/O) pin. See **Figure 21-1**.

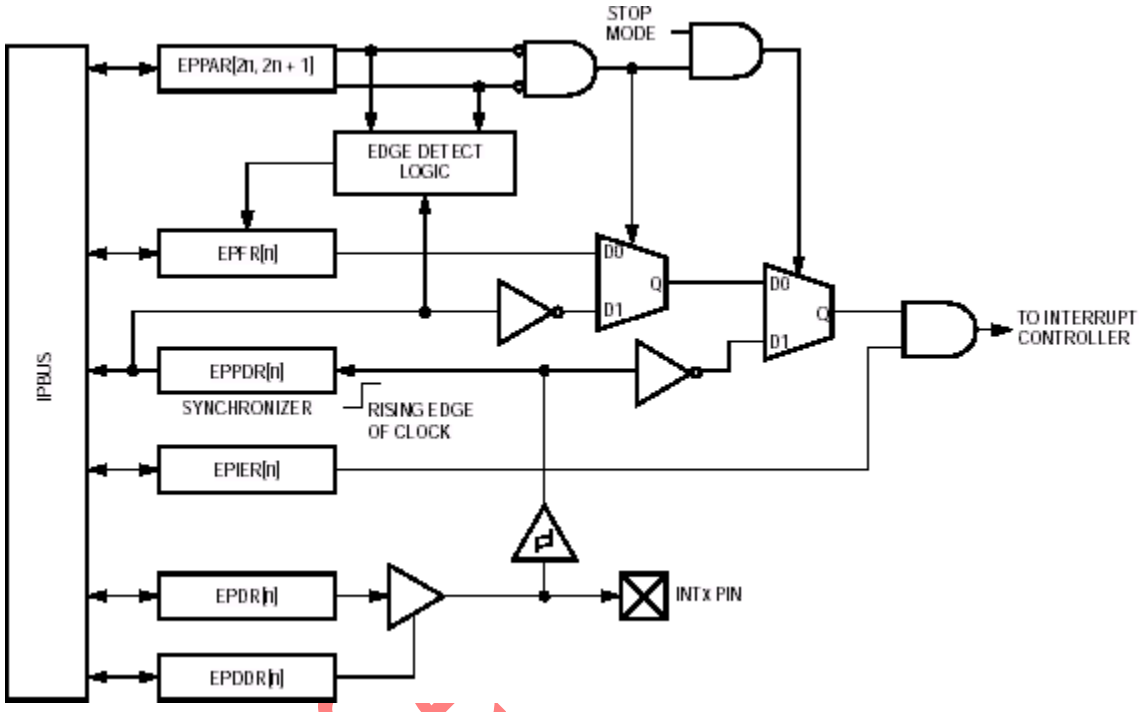


Figure 21-1: EPORT Block Diagram

21.2 Low-Power Mode Operation

This subsection describes the operation of the EPORT module in low-power modes.

21.2.1 Wait and Doze Modes

In wait and doze modes, the EPORT module continues to operate normally and may be configured to exit the low-power modes by generating an interrupt request on either a selected edge or a low level on an external pin.

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21.2.2 Stop Mode

In stop mode, there are no clocks available to perform the edge-detect function. Only the level-detect logic is active (if configured) to allow any low level on the external interrupt pin to generate an interrupt (if enabled) to exit stop mode.

NOTE: The input pin synchronizer is bypassed for the level-detect logic since no clocks are available.

21.3 Interrupt/General-Purpose I/O Pin Descriptions

All pins default to general-purpose input pins at reset. The pin value is synchronized to the rising edge of CLKOUT when read from the EPORT Pin Data Register (EPPDR). The values used in the edge/level detect logic are also synchronized to the rising edge of CLKOUT. These pins use Schmitt triggered input buffers which have built in hysteresis designed to decrease the probability of generating false edge-triggered interrupts for slow rising and falling input signals.

21.4 Memory Map and Registers

This subsection describes the memory map and register structure.

21.4.1 Memory Map

Refer to **Table 21-1** for a description of the EPORT memory map. The EPORT has a base address of 0x00cd_0000.

Table 21-1: Module Memory Map

Address Offset	Bits 15-8	Bits 7-0	Access(1)
0x0000	EPORT Pin Assignment Register (EPPAR)		S
0x0002	EPORT Data Direction Register (EPDDR)	EPORT Interrupt Enable Register (EPIER)	S
0x0004	EPORT Data Register (EPDR)	EPORT Pin Data Register (EPPDR)	S/U
0x0006	EPORT Flag Register (EPFR)	EPORT Pin Pull-up enable Register (EPPUE)	S/U
0x0008	EPORT Level Polarity Register (EPLPR)	EPORT Open Drain Register (EPODE)	S

NOTE:

- S = CPU supervisor mode access only. S/U = CPU supervisor or user mode access. User mode accesses to supervisor only addresses have no effect and result in a cycle termination transfer error.

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21.4.2 Registers

The EPORT programming model consists of these registers:

- The EPORT Pin Assignment Register (EPPAR) controls the function of each pin individually.
- The EPORT Data Direction Register (EPDDR) controls the direction of each one of the pins individually.
- The EPORT Interrupt Enable Register (EPIER) enables interrupt requests for each pin individually.
- The EPORT Data Register (EPDR) holds the data to be driven to the pins.
- The EPORT Pin Data Register (EPPDR) reflects the current stateof the pins.
- The EPORT Flag Register (EPFR) individually latches EPORT edge events.
- The EPORT Pin Pull-up Enable Register (EPPUE) controls the pull-up of each one of the pins individually.
- The EPORT Level Polarity Register (EPLPR) controls the level polarity of each one of the pins for level-sensitive.
- The EPORT Open Drain Enable Register (EPODE) controls the Open Drain of each one of the pins for output individually.

21.4.2.1 EPORT Pin Assignment Register

Addr. Offset: 0x0000~0x0001 Reset Value: 0x0000

15	14	13	12	11	10	9	8
EPPA7		EPPA6		EPPA5		EPPA4	
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
EPPA3		EPPA2		EPPA1		EPPA0	
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-2: EPORT Pin Assignment Register (EPPAR)

EPPA[7:0] — EPORT Pin Assignment Select Fields

The read/write EPPAx fields configure EPORT pins for level detection and rising and/or falling edge detection as shows.

Pins configured as level-sensitive are inverted so that a logic 0 or logic 1 on the external pin represents a valid interrupt request. Level-sensitive interrupt inputs are not latched. To guarantee that a level-sensitive interrupt request is acknowledged, the interrupt source must keep the signal asserted until acknowledged by software. Level sensitivity must be selected to bring the device out of stop mode with an INTx interrupt.

Pins configured as edge-triggered are latched and need not remain asserted for interrupt generation. A pin configured for edge detection is monitored regardless of its configuration as input or output.

Table 21-2: EPPAx Field Settings

EPPAx	Pin Configuration
00	Pin INTx level-sensitive
01	Pin INTx rising edge triggered
10	Pin INTx falling edge triggered
11	Pin INTx both falling edge and rising edge triggered

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Interrupt requests generated in the EPORT module can be masked by the interrupt controller module. EPPAR functionality is independent of the selected pin direction.

Reset clears the EPPAx fields.

21.4.2.2 Edge Port Interrupt Enable Register

Addr. Offset: 0x0002 Reset Value: 0x00

7	6	5	4	3	2	1	0
EPIE7	EPIE6	EPIE5	EPIE4	EPIE3	EPIE2	EPIE1	EPIE0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-3: EPORT Port Interrupt Enable Register (EPIER)

EPIE[7:0] — Edge Port Interrupt Enable Bits

The read/write EPIE[7:0] bits enable EPORT interrupt requests. If a bit in EPIER is set, EPORT generates an interrupt request when:

- The corresponding bit in the EPORT Flag Register (EPFR) is set or later becomes set, or
- The corresponding pin level is low and the pin is configured for level-sensitive operation

Clearing a bit in EPIER negates any interrupt request from the corresponding EPORT pin. Reset clears EPIE[7:0].

1 = Interrupt requests from corresponding EPORT pin enabled

0 = Interrupt requests from corresponding EPORT pin disabled

21.4.2.3 EPORT Data Direction Register

Addr. Offset: 0x0003 Reset Value: 0x00

7	6	5	4	3	2	1	0
EPDD7	EPDD6	EPDD5	EPDD4	EPDD3	EPDD2	EPDD1	EPDD0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-4: EPORT Data Direction Register (EPDDR)

EPDD[7:0] — Edge Port Data Direction Bits

Setting any bit in the EPDDR configures the corresponding pin as an output. Clearing any bit in EPDDR configures the corresponding pin as an input. Pin direction is independent of the level/edge detection configuration. Reset clears EPDD[7:0].

To use an EPORT pin as an external interrupt request source, its corresponding bit in EPDDR must be clear. Software can generate interrupt requests by programming the EPORT Data Register when the EPDDR selects output.

1 = Corresponding EPORT pin configured as output

0 = Corresponding EPORT pin configured as input

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21.4.2.4 Edge Port Pin Data Register

Addr. Offset: 0x0004 Reset Value: 当前管脚状态

7	6	5	4	3	2	1	0
EPPD7	EPPD6	EPPD5	EPPD4	EPPD3	EPPD2	EPPD1	EPPD0
ro	ro	ro	ro	ro	ro	ro	ro

Figure 21-5: EPORT Port Pin Data Register (EPPDR)

EPPD[7:0] — Edge Port Pin Data Bits

The read-only EPPDR reflects the current state of the EPORT pins. Writing to EPPDR has no effect, and the write cycle terminates normally. Reset does not affect EPPDR.

21.4.2.5 Edge Port Data Register

Addr. Offset: 0x0005 Reset Value: 0xFF

7	6	5	4	3	2	1	0
EPD7	EPD6	EPD5	EPD4	EPD3	EPD2	EPD1	EPD0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-6: EPORT Port Data Register (EPDR)

EPD[7:0] — Edge Port Data Bits

Data written to EPDR is stored in an internal register; if any pin of the port is configured as an output, the bit stored for that pin is driven onto the pin. Reading EDPR returns the data stored in the register. Reset sets EPD[7:0].

21.4.2.6 EPORT Pin Pull-up enable Register

Addr. Offset: 0x0006 Reset Value: 0xFF

7	6	5	4	3	2	1	0
EPPUE7	EPPUE6	EPPUE5	EPPUE4	EPPUE3	EPPUE2	EPPUE1	EPPUE0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-7: EPORT Pin Pull-up enable Register (EPPUE)

EPPUE[7:0] — Edge Port Pin Pull-up enable Bits

Setting any bit in the EPPUE configures the corresponding pin to enable pull-up. Clearing any bit in EPPUE configures the corresponding pin disable pull-up.

Reset sets EPPUE[7:0].

1 = Corresponding EPORT pin configured to enable pull-up

0 = Corresponding EPORT pin configured to disable pull-up

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21.4.2.7 Edge Port Flag Register

Addr. Offset: 0x0007 Reset Value: 0x00

7	6	5	4	3	2	1	0
EPFR7	EPFR6	EPFR5	EPFR4	EPFR3	EPFR2	EPFR1	EPFR0
ro	ro	ro	ro	ro	ro	ro	ro

Figure 21-8: EPORT Port Flag Register (EPFR)

EPF[7:0] — Edge Port Flag Bits

When an EPORT pin is configured for edge triggering, its corresponding read/write bit in EPFR indicates that the selected edge has been detected. Reset clears EPF[7:0].

1 = Selected edge for INTx pin has been detected.

0 = Selected edge for INTx pin has not been detected.

Bits in this register are set when the selected edge is detected on the corresponding pin. A bit remains set until cleared by writing a 1 to it. Writing 0 has no effect. If a pin is configured as level-sensitive (EPPARx = 00), pin transitions do not affect this register.

21.4.2.8 EPORT Open Drain enable Register

Addr. Offset: 0x0008 Reset Value: 0x00

7	6	5	4	3	2	1	0
EPODE7	EPODE6	EPODE5	EPODE4	EPODE3	EPODE2	EPODE1	EPODE0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-9: EPORT Open Drain enable Register (EPODE)

EPODE[7:0] — Edge Port Open Drain enable Bits

If EPORT are configured to output, setting any bit in the EPODE configures the corresponding pin to Open Drain output. Clearing any bit in EPODE configures the corresponding pin CMOS output. Reset clears EPODE[7:0].

1 = Corresponding EPORT pin configured to Open Drain output

0 = Corresponding EPORT pin configured to CMOS output

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21.4.2.9 EPORT Level Polarity Register

Addr. Offset: 0x0009 Reset Value: 0x00

7	6	5	4	3	2	1	0
EPLPR7	EPLPR6	EPLPR5	EPLPR4	EPLPR3	EPLPR2	EPLPR1	EPLPR0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 21-10: EPORT Level Polarity Register (EPLPR)

EPLPR[7:0] — Edge Port Level Polarity Bits

If EPORT are configured to level-sensitive, setting any bit in the EPLPR configures the corresponding pin high level-sensitive. Clearing any bit in EPLPR configures the corresponding pin low level-sensitive. Reset clears EPLPR[7:0].

1 = Corresponding EPORT pin configured to high level-sensitive

0 = Corresponding EPORT pin configured to low level-sensitive

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22 Enhanced DMA Controller (EDMAC)

22.1 Introduction

The EDMAC is the controller which sends data from SRAM to peripheral or from peripheral to SRAM.

22.2 Features

Features of the EDMAC include:

- Support two channels
- Programmable transfer total number
- Programmable read buffer address and write buffer address
- Multiple periperal select
- Support read, write and write then read transfer

22.3 Block Diagram

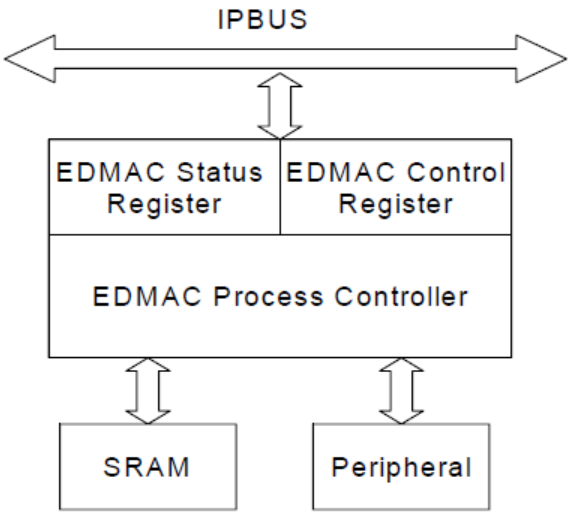


Figure 22-1: EDMAC Diagram

22.4 Modes of Operation

The EDMAC can transfer data from SRAM to peripheral or from peripheral to SRAM. The peripheral can be SPI1, SPI2, SPI3, USI2, USI3, AES, DES, SCI1, SMS4, SHA, RSA(CRYPTO). The read or write buffer of EDMAC shares with system SRAM.

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22.5 Memory Map and Registers

22.5.1 Memory Map

Table 22-1: EDMAC Memory Map

Ad- dress Offset	Bits 31-16	Bits 15-0	Access (1)
0x0000	EDMAC Control Register of Channel 0 (EDMACCR0)		S/U
0x0004	EDMAC Status Register of Channel 0 (EDMACSR0)		S/U
0x0008	EDMAC Read Buffer Address Register of Channel 0 (EDMACRBAR0)		S/U
0x000c	EDMAC Write Buffer Address Register of Channel 0 (EDMACWBAR0)		S/U
0x0010	EDMAC Minor Sum Register of Channel 0 (EDMACMINSUMR0)		S/U
0x0014	EDMAC Minor Counter Register of Channel 0 (EDMACMINCNTR0)		S/U
0x0018	EDMAC Major Sum Register of Channel 0 (EDMACMAJSUMR0)		S/U
0x001c	EDMAC Major Counter Register of Channel 0 (EDMACMAJCNTR0)		S/U
0x0020	EDMAC Special Peripheral Address Register of Channel 0 (EDMACSPAR0)		S/U
0x0024	EDMAC Read Buffer Address Step Register of Channel 0 (EDMACRBARSTEP0)		S/U
0x0028	EDMAC Write Buffer Address Step Register of Channel 0 (EDMACWBARSTEP0)		S/U
0x002c	EDMAC Last Minor Sum Register of Channel 0 (EDMACLASTMINSUMR0)		S/U
	RESERVED		
0x0040	EDMAC Control Register of Channel 1 (EDMACCR1)		S/U
0x0044	EDMAC Status Register of Channel 1 (EDMACSR1)		S/U
0x0048	EDMAC Read Buffer Address Register of Channel 1 (EDMACRBAR1)		S/U
0x004c	EDMAC Write Buffer Address Register of Channel 1 (EDMACWBAR1)		S/U
0x0050	EDMAC Minor Sum Register of Channel 1 (EDMACMINSUMR1)		S/U
0x0054	EDMAC Minor Counter Register of Channel 1 (EDMACMINCNTR1)		S/U
0x0058	EDMAC Major Sum Register of Channel 1 (EDMACMAJSUMR1)		S/U
0x005c	EDMAC Major Counter Register of Channel 1 (EDMACMAJCNTR1)		S/U
0x0060	EDMAC Special Peripheral Address Register of Channel 1 (EDMACSPAR1)		S/U
0x0064	EDMAC Read Buffer Address Step Register of Channel 1 (EDMACRBARSTEP1)		S/U
0x0068	EDMAC Write Buffer Address Step Register of Channel 1 (EDMACWBARSTEP1)		S/U

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Ad- dress Offset	Bits 31-16	Bits 15-0	Access (1)
0x006c	EDMAC Last Minor Sum Register of Channel 1 (EDMACLASTMINSUMR1)		S/U

Note: S = CPU supervisor mode access only. U=User mode accesses to supervisor-only address locations have no effect and result in a cycle termination transfer error.

22.5.2 Register Description

22.5.2.1 EDMAC Control Register of Channel 0/1 (EDMACCR0/1)

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
PIPELINE	SPI_HW	PRIOR_CFG	PRIOR	SPI_WROPT	INFINITY	SPI_FIFO_PRE_LOAD[3]	MAJ_DONEIE
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
MAJ_CRC_SEL	CRC_CHEN	SEND_CRC_CHEN	WRITE_CRC_CHEN	LINK_SRC_SEL[2:0]			LINK
rw	rw	rw	rw	rw			rw
15	14	13	12	11	10	9	8
COMPARE_EN	COMPARE_SKIP	PRELOAD_DIS	MEM-TRANS CRC	PN[3:0]			
rw	rw	rw	rw	rw			
7	6	5	4	3	2	1	0
STARTIE	TTYPE[1:0]		SPI_FIFO_PRELOAD[2:0]			VALID	MIN_DONEIE
rw	rw		rw			rw	rw

Figure 22-2: EDMAC Control Register of Channel 0 (EDMACCR0)

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Addr. Offset: 0x0040~0x0043 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
PIPELINE	SPI_HW	PRIOR_CFG	PRIOR	SPI_WROPT	INFINITY	SPI_FIFO_PRE_LOAD[3]	MAJ_DONEIE
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
MAJ_CRC_SEL	CRC_CHEN	SEND_CRC_CHEN	WRITE_CRC_CHEN	LINK_SRC_SEL[2:0]			LINK
rw	rw	rw	rw	rw			rw
15	14	13	12	11	10	9	8
COMPARE_EN	COMPARE_SKIP	PRELOAD_DIS	MEM-TRANS CRC	PN[3:0]			
rw	rw	rw	rw	rw			
7	6	5	4	3	2	1	0
STARTIE	TTYPE[1:0]		SPI_FIFO_PRELOAD[2:0]			VALID	MIN_DONEIE
rw	rw		rw			rw	rw

Figure 22-3: EDMAC Control Register of Channel 1 (EDMACCR1)

PIPELINE — Pipeline transfer mode

1 = Pipeline transfer mode enable

0 = Pipeline transfer mode not enable

SPI_HW — SPI halfword operation

In SPI halfword operation mode, please config **TXFSTH** in **SPITXFCR** to 6 and **RXFSTH** in **SPIRXFCR** to 1.For details please refer to SPI section.

1 = SPI halfword operation, 0 = SPI byte operation

PRIOR_CHG — Priority Change Enable,

1 = Enabled

0 = Disabled

PRIOR — Channel Priority

Set **PRIOR_CHG** before config this bit.

1 = Channel1 is prior

0 = Channel0 is prior

INFINITY — Channel infinity transfer,

1 = infinity

0 = finity

NOTE: When write 1 to INFINITY, MINOR_SUM keep valid and MAJOR_SUM invalid, MAJOR_SUM not restrict data transmit.

SPI_WROPT — SPI Write Optimize when TTYPE is 2'b11 in pipeline mode

1 = Write SPI timing optimize is enabled

0 = Write SPI timing optimize is not enabled

MAJ_CRC_SEL — Send or Write CRC code operation timing select

1 = Operation takes place when major loop ends

0 = Operation takes place when minor loop ends

CRC_CHEN — CRC channel enable

1 = Enable edmac operation for CRC

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0 = Not enable edmac operation for CRC

SEND_CRC_CHEN — Send crc code to peripheral IP channel enable

1 = Send the crc code to peripheral IP

0 = Not Send the crc code to peripheral IP

WRITE_CRC_CHEN — Write crc code to sram channel enable

1 = Write the received crc code to sram

0 = Not write the received crc code to sram

NOTE: Send or Write crc code operation is enabled only when channel enable and mod- ule enable(please refer to module CRC section) both asserted.

COMPARE_EN — Compare data function enable(used in MPT).

1 = Enable the compare data function.

0 = Disable the compare data function.

COMPARE_SKIP — Compare data skip function(used in MPT).

1 = Once compare data fail, skip the compare operation, set done and fail flag.

0 = Once compare data fail, not stop the compare operation, set fail flag but not set done flag until all data compare finish.

PRELOAD_DIS — Preload data to internal buffer disable function(used in MPT).

1 = When we want to generate random data by LBA(used in MPT),we should disable preload data function by set PRELOAD_DIS=1.

0 = In normal mode, preload data function enable, PRELOAD_DIS=0.

MEM-TRANS CRC — CRC check between memory transfer

1 = CRC check When transfer type is memory transfer

0 = Not CRC check When transfer type is memory transfer

PN[3:0] — Channel Peripheral Number

The PN field determines which peripheral will be selected.

NOTE: When PN[3:0]=8/13/11(sci1/2/3),we only support M=0 transfer, and before transfer please check sci status register first. when PN[3:0]=2/3(usi1/2), before transfer please check usi status register first. when PN[3:0]=0/1/14(spi1/2/3), before transfer please check spi status register first.

Table 22-2: Peripheral Number VS Peripheral

PN[3:0]	Peripheral(EDMAC0)	Peripheral(EDMAC1)
0	Reserved	SPI1
1	Reserved	SPI2
2	Reserved	USI1
3	Reserved	USI2
4	Reserved	Reserved
5	AES	Reserved
6	DES	Reserved
7	Reserved	Reserved
8	Reserved	SCI1
9	SMS4	Reserved
10	SHA	Reserved
11	Reserved	SCI3

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PN[3:0]	Peripheral(EDMAC0)	Peripheral(EDMAC1)
12	RSA(CRYPTO)	Reserved
13	Reserved	SCI2
14	Reserved	SPI3
15	ZUC	Reserved

TTYPE[1:0] — Channel Transfer Type

The TTYPE field determines which transfer type of the channel will be selected. When PN[3:0] select SHA or RSA,TTYPE[1:0] can only set 01 or 10;

Table 22-3: Transfer Type description

TTYPE[1:0]	Peripheral
2'b00	From SRAM to SRAM
2'b01	From Peripheral to SRAM
2'b10	From SRAM to Peripheral
2'b11	From SRAM to Peripheral Then From Peripheral to SRAM

SPI_FIFO_PRELOAD[3:0] — Preload data to spi write fifo

The field determine how many data preload to spi send fifo. When edmac select SPI1/2/3 and transfer type is 2'b11 or 2'b10, this field is valid. When **SPI_HW** is enabled, please config SPI_FIFO_PRELOAD to an even value.

Table 22-4: SPI FIFO Preload

SPI fifo preload[3:0]	Preload data number (byte)
4'b0000	Not preload
4'b0001	1
4'b0010	2
4'b0011	3
4'b0100	4
4'b0101	5
4'b0110	6
4'b0111	7
4'b1000	8
4'b1001	9
4'b1010	10
4'b1011	11
4'b1100	12
4'b1101	13
4'b1110	14
4'b1111	15

VALID — Channel Valid Bit

This Valid bit signals the channel is available. When the channel is accepted and start a block of transfer, this bit will be cleared to 0.

1 = The channel is available

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0 = The channel is not available

STARTIE — START Interrupt Enable Bit

This STARTIE bit enable START flag of the channel to generate interrupt request.

1 = START interrupt request enabled

0 = START interrupt request disabled

*_DONEIE — DONE Interrupt Enable Bit

This *_DONEIE bit enable DONE flag of the channel to generate interrupt request.

1 = DONE interrupt request enabled

0 = DONE interrupt request disabled

LINK — Link function enable.

The VALID bit of the channel can be asserted by hardware event selected by **LINK_SRC_SEL[1:0]** when LINK function enabled.

1 = Enable the link function, 0 = Disable the link function.

LINK_SRC_SEL[2:0] — Link Source Select

Table 22-5: Link Source

LINK_SRC_ SEL[2:0]	Hardware Event
3'b000	EDMAC0 CH0 MINOR DONE
3'b001	EDMAC0 CH1 MINOR DONE
3'b010	EDMAC1 CH0 MINOR DONE
3'b011	EDMAC1 CH1 MINOR DONE
3'b100	Reserved
3'b101	Reserved
3'b110	Reserved
3'b111	Reserved

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22.5.2.2 EDMAC Control State Register of Channel 0/1 (EDMACCSR0/1)

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
SSF_SPI1	SSF_SPI2	SSF_SPI3	SSF_SPI4	Reserved			Reserved
r/w1c	r/w1c	r/w1c	r/w1c	ro			rw
23	22	21	20	19	18	17	16
Reserved							EDMACEN
ro							rw
15	14	13	12	11	10	9	8
FAIL	Reserved						
ro	ro						
7	6	5	4	3	2	1	0
SCHNUM	DCHNUM	Reserved		MAJ_DONE	START	MIN_DONE	BUSY
ro	ro	ro		r/w1c	ro	r/w1c	ro

Figure 22-4: EDMAC Control State Register of Channel 0 (EDMACCSR0)

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							EDMACEN
ro							rw
15	14	13	12	11	10	9	8
FAIL	Reserved						
ro	ro						
7	6	5	4	3	2	1	0
SCHNUM	DCHNUM	Reserved		MAJ_DONE	START	MIN_DONE	BUSY
ro	ro	ro		r/w1c	ro	r/w1c	ro

Figure 22-5: EDMAC Control State Register of Channel 1 (EDMACCSR1)

SSF_SPI* — SPI* SS flag bit. The bit is only read, write 1 clear 0.

EDMACEN — EDMAC function enable bit

The EDMACEN bit enables the EDMAC function.

1 = EDMAC enable, 0 = EDMAC disable

NOTE: EDMACEN can be cleared automatically when COMPARE_SKIP=1,and compare fail happen.

NOTE: When using EDMAC control peripheral transfer, disable EDMAC, then enable EDMAC. Before enable EDMAC, should reset peripheral first(for example, SPI Clear SPE then enable it again).

FAIL — The compare data operation of channel fail

FAIL is set when channel compare data operation fail. Clear FAIL by writing "1" to DONE.

START — The transfer of channel start

START is set when channel starts a block of transfer. START will generate interrupt if the STARTIE is set in EDMACCR. Clear START by writing "1" to it.

1 = Channel starts a block of transfer

0 = Channel does not start a block of transfer

MIN_DONE — The minor transfer of channel is done

MIN_DONE is set when channel finishes a block of minor transfer. MIN_DONE will generate interrupt if the

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MIN_DONEIE is set in EDMACCR. Clear MIN_DONE by writing "1" to it.

1 = Channel finish a block of minor transfer

0 = Channel not finish a block of minor transfer

BUSY — Channel is busy transferring

Busy is set when channel starts a block of minor transfer. It will be cleared automatically when the block of minor transfer is done.

1 = A block of minor transfer is processing

0 = No block of minor transfer is process

MAJ_DONE — The major transfer of channel is done

MAJ_DONE is set when channel finishes a block of major transfer. MAJ_DONE will generate interrupt if the MAJ_DONEIE is set in EDMACCR. Clear MAJ_DONE by writing "1" to it.

1 = Channel finish a block of transfer

0 = Channel not finish a block of transfer

DCHNUM — The channel number of which last finished a block of transfer

1 = Channel 1 last finished a block of transfer

0 = Channel 0 last finished a block of transfer

SCHNUM — The channel number of which last start a block of transfer

1 = Channel 1 last started a block of transfer

0 = Channel 0 last started a block of transfer

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22.5.2.3 EDMAC Read Buffer Address Register of Channel 0/1 (EDMACRBAR 0/1)

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
RNONINC	RDEC	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		EDMACRBAR[21:16]					
ro		rw					
15	14	13	12	11	10	9	8
EDMACRBAR[15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACRBAR[7:0]							
rw							

Figure 22-6: EDMAC Read Buffer Address Register of Channel 0 (EDMACRBAR0)

Addr. Offset: 0x0048~0x004B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
RNONINC	RDEC	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		EDMACRBAR[21:16]					
ro		rw					
15	14	13	12	11	10	9	8
EDMACRBAR[15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACRBAR[7:0]							
rw							

Figure 22-7: EDMAC Read Buffer Address Register of Channel 1 (EDMACRBAR1)

RNONINC — EDMAC Read Buffer Address Incrementor Disable

1 = EDMAC Read Buffer Address Incrementor is disable

0 = EDMAC Read Buffer Address Incrementor is enable

RDEC — EDMAC Read Buffer Address decrease

1 = EDMAC Read Buffer Address decrease is enable

0 = EDMAC Read Buffer Address depend on RNONINC

EDMACRBAR — Select EDMAC Read Buffer Base Address

The EDMACRBAR field selects Read Buffer Address of the channel

NOTE:

- When RNONINC=1,EDMACRBAR[1:0] must be set 00(word align).
- When NFRAND module enable,EDMACRBAR[1:0] must be set 00(word align).

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22.5.2.4 EDMAC Write Buffer Address Register of Channel 0/1 (EDMACWBAR 0/1)

Addr. Offset: 0x000C~0x000F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
WNONINC	WDEC	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		EDMACWBAR[21:16]					
ro		rw					
15	14	13	12	11	10	9	8
EDMACWBAR[15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACWBAR[7:0]							
rw							

Figure 22-8: EDMAC Write Buffer Address Register of Channel 0 (EDMACWBAR0)

Addr. Offset: 0x004C~0x004F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
WNONINC	WDEC	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		EDMACWBAR[21:16]					
ro		rw					
15	14	13	12	11	10	9	8
EDMACWBAR[15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACWBAR[7:0]							
rw							

Figure 22-9: EDMAC Write Buffer Address Register of Channel 1 (EDMACWBAR1)

WNONINC — EDMAC Write Buffer Address Incrementor Disable

1 = EDMAC Write Buffer Address Incrementor is disable

0 = EDMAC Write Buffer Address Incrementor is enable

WDEC — EDMAC Write Buffer Address decrease

1 = EDMAC Write Buffer Address decrease is enable

0 = EDMAC Write Buffer Address depend on WNONINC

EDMACWBAR — EDMAC Write Buffer Address

The EMACWBAR field selects Write Buffer Bass Address of the channel

NOTE:

- When WNONINC=1,EDMACWBAR[1:0] must be set 00(word align).
- When NFRAND module enable,EDMACWBAR[1:0] must be set 00(word align).

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22.5.2.5 EDMAC Minor Sum Register of Channel 0/1 (EDMACMINSUMR 0/1)

Addr. Offset: 0x0010~0x0013 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MINOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MINOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
MINOR_SUM [7:0]							
rw							

Figure 22-10: EDMAC Minor Sum Register of Channel 0 (EDMACMINSUMR0)

Addr. Offset: 0x0050~0x0053 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MINOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MINOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
MINOR_SUM [7:0]							
rw							

Figure 22-11: EDMAC Minor Sum Register of Channel 1 (EDMACMINSUMR1)

MINOR_SUM — Channel Minor Transfer Data Sum

The MINOR_SUM field controls the data sum(in byte) transmitted in one minor block transfer of the channel.

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22.5.2.6 EDMAC Minor Transfer Counter Register of Channel 0/1 (EDMACMINCNTR 0/1)

Addr. Offset: 0x0014~0x0017 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MINOR_CNT [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MINOR_CNT [15:8]							
rw							
7	6	5	4	3	2	1	0
MINOR_CNT [7:0]							
rw							

Figure 22-12: EDMAC Minor Transfer Counter Register of Channel 0 (EDMACMINCNTR0)

Addr. Offset: 0x0054~0x0057 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MINOR_CNT [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MINOR_CNT [15:8]							
rw							
7	6	5	4	3	2	1	0
MINOR_CNT [7:0]							
rw							

Figure 22-13: EDMAC Minor Transfer Counter Register of Channel 1 (EDMACMINCNTR1)

MINOR_CNT — EDMAC Minor Transfer Counter

The counter is used to signal the total bytes which have been written to the edmac write buffer in a minor loop.

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22.5.2.7 EDMAC Major Sum Register of Channel 0/1 (EDMACMAJSUMR 0/1)

Addr. Offset: 0x0018~0x001B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MAJOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MAJOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
MAJOR_SUM [7:0]							
rw							

Figure 22-14: EDMAC Major Sum Register of Channel 0 (EDMACMAJSUMR0)

Addr. Offset: 0x0058~0x005B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MAJOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MAJOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
MAJOR_SUM [7:0]							
rw							

Figure 22-15: EDMAC Major Sum Register of Channel 1 (EDMACMAJSUMR1)

MAJOR_SUM — Channel Major Transfer Data Sum

The MAJOR_SUM field controls the minor block sum transmitted in one major block transfer of the channel.

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22.5.2.8 EDMAC Major Transfer Counter Register of Channel 0/1 (EDMACMINCNTR 0/1)

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MAJOR_CNT [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MAJOR_CNT [15:8]							
rw							
7	6	5	4	3	2	1	0
MAJOR_CNT [7:0]							
rw							

Figure 22-16: EDMAC Major Transfer Counter Register of Channel 0 (EDMACMAJCNTR0)

Addr. Offset: 0x001C~0x001F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved		MAJOR_CNT [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
MAJOR_CNT [15:8]							
rw							
7	6	5	4	3	2	1	0
MAJOR_CNT [7:0]							
rw							

Figure 22-17: EDMAC Major Transfer Counter Register of Channel 1 (EDMACMAJCNTR1)

MAJOR_CNT — EDMAC Major Transfer Counter

The counter is used to signal the total minor block counts which have been done in a major loop.

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22.5.2.9 EDMAC Special Peripheral Address Register of Channel 0/1 (EDMACSPAR 0/1)

Addr. Offset: 0x0020~0x0023 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
EDMACSPAR [15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACSPAR [7:0]							
rw							

Figure 22-18: EDMAC Special Peripheral Address Register of Channel 0 (EDMACSPAR0)

Addr. Offset: 0x0060~0x0063 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
EDMACSPAR [15:8]							
rw							
7	6	5	4	3	2	1	0
EDMACSPAR [7:0]							
rw							

Figure 22-19: EDMAC Special Peripheral Address Register of Channel 1 (EDMACSPAR1)

EDMACSPAR — EDMAC Special Peripheral Address

When peripheral has several data registers such as RSA, please config this register to select different peripheral data registers.

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22.5.2.10 EDMAC Read Buffer Address Step Register of Channel 0/1 (EDMACRBARSTEP 0/1)

Addr. Offset: 0x0024~0x0027 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
RBAR_STEP_EN	RBAR_STEP_DIR	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		RBAR_STEP [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
RBAR_STEP [15:8]							
rw							
7	6	5	4	3	2	1	0
RBAR_STEP [7:0]							
rw							

Figure 22-20: EDMAC Read Buffer Address Step Register of Channel 0 (EDMACRBARSTEP0)

Addr. Offset: 0x0064~0x0067 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
RBAR_STEP_EN	RBAR_STEP_DIR	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		RBAR_STEP [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
RBAR_STEP [15:8]							
rw							
7	6	5	4	3	2	1	0
RBAR_STEP [7:0]							
rw							

Figure 22-21: EDMAC Read Buffer Address Step Register of Channel 1 (EDMACRBARSTEP1)

RBAR_STEP_EN — EDMAC Read Buffer Address Step Enable

1 = Enable

0 = Not enable

RBAR_STEP_DIR — EDMAC Read Buffer Address Step Direction

1 = EDMAC Read Buffer Address decrease

0 = EDMAC Read Buffer Address increase

RBAR_STEP — EDMAC Read Buffer Base Address Step

The EDMACRBAR field updates with RBAR_STEP when a minor loop is start(except the first minor loop).

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22.5.2.11 EDMAC Write Buffer Address Step Register of Channel 0/1 (EDMACWBARSTEP 0/1)

Addr. Offset: 0x0028~0x002B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
WBAR_STEP_EN	WBAR_STEP_DIR	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		WBAR_STEP [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
WBAR_STEP [15:8]							
rw							
7	6	5	4	3	2	1	0
WBAR_STEP [7:0]							
rw							

Figure 22-22: EDMAC Write Buffer Address Step Register of Channel 0 (EDMACWBARSTEP0)

Addr. Offset: 0x0068~0x006B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
WBAR_STEP_EN	WBAR_STEP_DIR	Reserved					
rw	rw	ro					
23	22	21	20	19	18	17	16
Reserved		WBAR_STEP [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
WBAR_STEP [15:8]							
rw							
7	6	5	4	3	2	1	0
WBAR_STEP [7:0]							
rw							

Figure 22-23: EDMAC Write Buffer Address Step Register of Channel 1 (EDMACWBARSTEP1)

WBAR_STEP_EN — EDMAC Write Buffer Address Step Enable

1 = Enable, 0 = Not enable

WBAR_STEP_DIR — EDMAC Write Buffer Address Step Direction

1 = EDMAC Write Buffer Address decrease

0 = EDMAC Write Buffer Address increase

WBAR_STEP — EDMAC Write Buffer Base Address Step

The EDMACWBAR field updates with WBAR_STEP when a minor loop is start(except the first minor loop).

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22.5.2.12 EDMAC Last Minor Sum Register of Channel 0/1 (EDMACLASTMINSUMR 0/1)

Addr. Offset: 0x002C~0x002F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
LASTMINSUM_EN	Reserved						
rw	ro						
23	22	21	20	19	18	17	16
Reserved		LAST_MINOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
LAST_MINOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
LAST_MINOR_SUM [7:0]							
rw							

Figure 22-24: EDMAC Last Minor Sum Register of Channel 0 (EDMACLASTMINSUMR0)

Addr. Offset: 0x006C~0x006F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
LASTMINSUM_EN	Reserved						
rw	ro						
23	22	21	20	19	18	17	16
Reserved		LAST_MINOR_SUM [21:16]					
ro		rw					
15	14	13	12	11	10	9	8
LAST_MINOR_SUM [15:8]							
rw							
7	6	5	4	3	2	1	0
LAST_MINOR_SUM [7:0]							
rw							

Figure 22-25: EDMAC Last Minor Sum Register of Channel 1 (EDMACLASTMINSUMR1)

LASTMINSUM_EN — Last Minor Sum Function Enable

1 = Enable, 0 = Not enable

LAST_MINOR_SUM — Channel Last Minor Transfer Data Sum

The LAST_MINOR_SUM field controls the data sum(in byte) transmitted in the last minor block transfer of the channel when LASTMINSUM_EN is set.

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22.6 Function Description

Enhance Direct Memory Access Controller (EDMAC) supports transfer data between SRAM and peripherals. It includes two channels: channel 0 and channel

1. Each channel can support different peripherals and different transfer types. And edmac will exchange priority after the priority channel starts a block transfer.

22.6.1 Transfer Type

22.6.1.1 Transfer data from SRAM to Peripheral then from Peripheral to SRAM

In the mode the edmac will:

- Read a data from SRAM
- Write the data to Peripheral
- Polling whether the peripheral is ready
- Read a data from peripheral
- Write the data to SRAM
- If transfer is not done, repeat 1~5
- Transfer data From SRAM to Periperal

In the mode the edmac will:

- Read a data from SRAM
- Write the data to Peripheral
- Polling whether the peripheral is ready
- If transfer is not done, repeat 1~3

22.6.1.2 Transfer data From Peripheral to SRAM

In the mode the edmac will:

- Polling whether the peripheral is ready
- Read a data from peripheral
- Write the data to SRAM
- If transfer is not done, repeat 1~3

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22.6.1.3 Transfer data From SRAM to SRAM

- Read a data from SRAM
- Write the data to SRAM
- If transfer is not done, repeat 1~2

22.6.2 Two channel configuration

Before the channel n of the edmac start a block of transfer, software has to configurate read buffer address (EDMACRABARn), write buffer address (EDMACWBARn), transfer number (SUM), transfer type (TTYPE), peripheral number (PN) and VALID bit. If interrupt request is needed, interrupt enable is also needed to be set.

When EDMACEN is set and one of the two channels' valid, a block of transfer will be started.

And the system resets or EDMACEN is clear, channel 0 has higher priority. After channel 0 starts a block transfer, channel 1 will get higher priority. When channel 0 finish the block of transfer, channel 1 start a block transfer if its VALID bit is set and then channel 0 will get higher priority.

In the following cases are possible when a block of transfer is being finished during EDMACEN:

- Channel 0 has higher priority and both two channels are valid. Channel 0 will start a block of transfer first and channel 1 get higher priority. After channel 0 finishes the block of tranfer, channel 1 will start a block of transfer and channel 0 gets higher priority.
- Channel 0 has higher priority and only channel 0 is valid. Channel 0 will start a block of transfer and channel 1 gets high priority.
- Channel 0 has higher priority and only channel 1 is valid. Channel 1 will start a block of transfer and channel 0 remains high priority.
- Channel 1 has higher priority and both two channels are valid. Channel 1 will start a block of transfer first and channel 0 get higher priority. After channel 1 finishes the block of tranfer, channel 0 will start a block of transfer and channel 1 gets higher priority.
- Channel 1 has higher priority and only channel 1 is valid. Channel 1 will start a block of transfer and channel 0 gets high priority.
- Channel 1 has higher priority and only channel 0 is valid. Channel 0 will start a block of transfer and channel 1 remains high priority.

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22.7 Interrupts

Table 22-6: EDMAC Interrupt Request Sources

Interrupt Request	Flag	Enable Bit
Channel 0 trasnfer done	MIN/MAJ_DONE0	MIN/MAJ_DONE0IE
Channel 1 transfer done	MIN/MAJ_DONE1	MIN/MAJ_DONE1IE
Channel 0 transfer start	START0	START0IE
Channel 1 transfer start	START1	START1IE

22.7.1 EDMAC Channel 0 Transfer Done (MIN/MAJ_DONE0)

DONE0 is set after the block of transfer is finished. Writing 1 to the bit will clear the flag during the flag is set.

22.7.2 EDMAC Channel 1 Transfer Done (MIN/MAJ_DONE1)

DONE1 is set after the block of transfer is finished. Writing 1 to the bit will clear the flag during the flag is set.

22.7.3 EDMAC Channel 0 Transfer Start (START0)

START0 is set after the block of transfer is started. Writing 1 to the bit will clear the flag during the flag is set.

22.7.4 EDMAC Channel 1 Transfer Start (START1)

DONE1 is set after the block of transfer is started. Writing 1 to the bit will clear the flag during the flag is set.

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23 DMA Controller

23.1 Introduction

The DMA controller is one of the master that can transfer data with minimal CPU interaction. The DMA module provides four channels that allow byte, halfword, and word transfers.

23.2 Features

The DMA controller features include:

- Four independently programmable DMA controller channels
- Data transfers in 8, 16, 32bits
- Support single transfer, Burst 4, 8, 16 transfer, and burst always under a speical case.
- Support LLI transfer mode
- Follow a configurate priority rule
- Support halt operation for a channel
- Support Peripheral transfer.

23.3 Low-Power Mode Operation

The DMA controller is not affected by any low-power modes. CPU can stop DMAC by setting the corresponding module stop bit in Clock Module.

23.4 Block Diagram

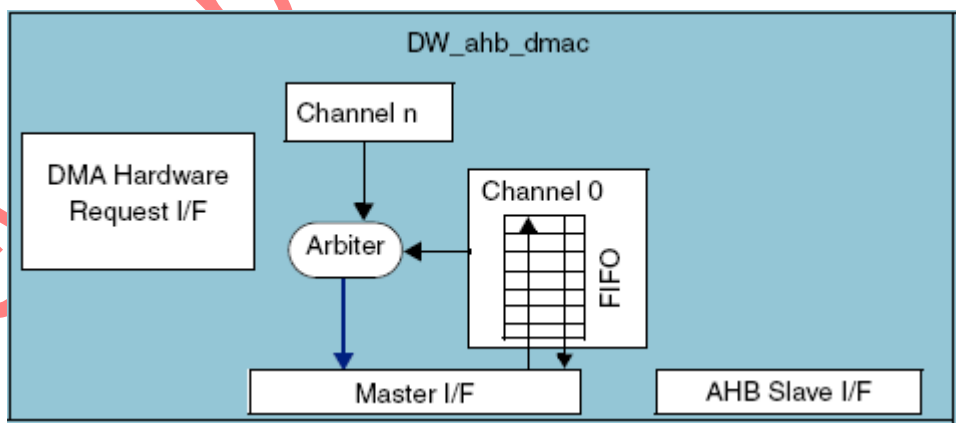


Figure 23-1: DMAC Block Diagram

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23.5 Module Memory Map and Register

23.5.1 Module Memory Map

Table 23-1 shows the DMAC Register Memory Map.

Table 23-1: Register Memory Map

Address Offset[3]	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access[1,2]
0x0	DMAC Source Address Register 0(DMA_SADDR0)				S
0x8	DMAC Destination Address Register 0 (DMA_DADDR0)				S
0x10	DMAC Linked List Pointer 0 (DMA_LLP0)				S
0x18	DMAC Control Register 0(DMA_CTRL0)				S
0x1C	DMAC Control Higher Register 0(DMA_CTRL_HIGH0)				S
0x40	DMAC Configuration Register 0 (DMA_CFG0)				S
0x44	DMAC Configuration higher Register 0 (DMA_CFG_HIGH0)				S
0x58	DMAC Source Address Register 1(DMA_SADDR1)				S
0x60	DMAC Destination Address Register 1 (DMA_DADDR1)				S
0x68	DMAC Linked List Pointer 1 (DMA_LLP1)				S
0x70	DMAC Control Register 1(DMA_CTRL1)				S
0x74	DMAC Control Higher Register 1(DMA_CTRL_HIGH1)				S
0x98	DMAC Configuration Register 1(DMA_CFG1)				S
0x9C	DMAC Configuration higher Register 1(DMA_CFG_HIGH1)				S
0xB0	DMAC Source Address Register 2(DMA_SADDR2)				
0xB8	DMAC Destination Address Register 2(DMA_DADDR2)				
0xC0	DMAC Linked List Pointer 2 (DMA_LLP2)				
0xC8	DMAC Control Register 2(DMA_CTRL2)				
0xCC	DMAC Control Higher Register 2(DMA_CTRL_HIGH2)				
0xF0	DMAC Configuration Register 2(DMA_CFG2)				
0xF4	DMAC Configuration higher Register 2(DMA_CFG_HIGH2)				
0x108	DMAC Source Address Register 3(DMA_SADDR3)				
0x110	DMAC Destination Address Register 3 (DMA_DADDR3)				
0x118	DMAC Linked List Pointer 3 (DMA_LLP3)				
0x120	DMAC Control Register 3(DMA_CTRL3)				
0x124	DMAC Control Higher Register 3(DMA_CTRL_HIGH3)				
0x148	DMAC Configuration Register 3(DMA_CFG3)				
0x14C	DMAC Configuration higher Register 3(DMA_CFG_HIGH3)				
0x2C0	Raw Status for IntTfr Interrupt(DMA_RAWTFR)				
0x2C8	Raw Status forIntBlock Interrupt(DMA_RAWBLOCK)				
0x2D0	Raw Status for IntSrcTran Interrupt(DMA_RAWSRCTRAN)				
0x2D8	Raw Status for IntDstTran Interrupt(DMA_RAWDSTTRAN)				
0x2E0	Raw Status for IntErr Interrupt(DMA_RAWERR)				
0x2E8	Status for IntTfr Interrupt(DMA_STATTFR)				
0x2F0	Status for IntBlock Interrupt(DMA_STATBLOCK)				
0x2F8	Status for IntSrcTran Interrupt(DMA_STATSRC)				
0x300	Status for IntDstTran Interrupt(DMA_STATDST)				
0x308	Status for IntErr Interrupt(DMA_STATERR)				

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Address Offset[3]	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access[1,2]
0x310	Mask for IntTfr Interrupt(DMA_MASKTFR)				
0x318	Mask for IntBlock Interrupt(DMA_MASKBLOCK)				
0x320	Mask for IntSrcTran Interrupt(DMA_MASKSRC)				
0x328	Mask for IntDstTran Interrupt(DMA_MASKDST)				
0x330	Mask for IntErr Interrupt(DMA_MASKERR)				
0x338	Clear for IntTfr Interrupt(DMA_CLRTFR)				
0x340	Clear for IntBlock Interrupt(DMA_CLRBLOCK)				
0x348	Clear for IntSrcTran Interrupt(DMA_CLRSRC)				
0x350	Clear for IntDstTran Interrupt(DMA_CLRDST)				
0x358	Clear for IntErr Interrupt(DMA_CLRERR)				
0x360	Status for each interrupt type(DMA_STATUSINT)				
0x368	Source Software Transaction Request Register(DMA_SRCREQ)				
0x370	Destination Software Transaction Request Register(DMA_DSTREQ)				
0x378	Single Source Transaction Request Register(DMA_SINGLESRC)				
0x380	Single Destination Transaction Request Register(DMA_SINGLEDST)				
0x388	Last Source Transaction Request Register(DMA_LASTSRC)				
0x390	Last Destination Transaction Request Register(DMA_LASTDST)				
0x398	DMA Configuration Register(DMA_CONFIG)				
0x3A0	DMA Channel Enable Register(DMA_CHEN)				

NOTES:

- S = CPU supervisor mode access only.
- User mode accesses to supervisor-only address locations have no effect and result in a cycle termination transfer error.
- n=0~5.

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23.5.2 Register Descriptions

23.5.2.1 DMAC Source Address Register n(DMA_SADDRx)

DMA_SADDRx contains the memory address pointing to the source data.

Addr. Offset: 0x0000, 0x0058, 0x00B0, 0x0108 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DMA_SADDR[31:24]							
rw							
23	22	21	20	19	18	17	16
DMA_SADDR[23:16]							
rw							
15	14	13	12	11	10	9	8
DMA_SADDR[15:8]							
rw							
7	6	5	4	3	2	1	0
DMA_SADDR[7:0]							
rw							

Figure 23-2: DMAC Source Address Register n (DMA_SADDRx)

DMA_SADDR[31:0] — Memory address pointing to the source data

23.5.2.2 DMAC Destination Address Register n(DMA_DADDRx)

DMA_DADDRx contains the memory address pointing to the destination data.

Addr. Offset: 0x0008, 0x0060, 0x00B8, 0x0110 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
DMA_DADDR[31:24]							
rw							
23	22	21	20	19	18	17	16
DMA_DADDR[23:16]							
rw							
15	14	13	12	11	10	9	8
DMA_DADDR[15:8]							
rw							
7	6	5	4	3	2	1	0
DMA_DADDR[7:0]							
rw							

Figure 23-3: DMAC Destination Address Register n(DMA_DADDRx)

DMA_DADDR[31:0] — Memory address pointing to the destination data

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23.5.2.3 DMAC Linked List Register n(DMA_LLPx)

Addr. Offset: 0x0010, 0x0068, 0x00C0, 0x0118 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
LOC[31:24]							
rw							
23	22	21	20	19	18	17	16
LOC[23:16]							
rw							
15	14	13	12	11	10	9	8
LOC[15:8]							
rw							
7	6	5	4	3	2	1	0
LOC[7:2]						Reserved	
rw						ro	

Figure 23-4: DMAC Linked List Register n(DMA_LLPx)

LOC[31:2] — Starting Address In Memory of LLI

Starting Address In Memory of LLI. Note that the two LSBs of the starting address are not stored because the address is assumed to be aligned to a 32-bit boundary. LLI accesses are always 32-bit accesses (Hsize = 2) aligned to 32-bit boundaries and cannot be changed or programmed to anything other than 32- bit.

23.5.2.4 DMAC Control Register (DMA_CTRLx)

Addr. Offset: 0x0010, 0x0068, 0x00C0, 0x0118 Reset Value: 0x00304801

31	30	29	28	27	26	25	24
Reserved			LLP_SRC_EN	LLP_DST_EN	SMS[1:0]		DMS[1]
ro			rw	rw	rw		rw
23	22	21	20	19	18	17	16
DMS[[0]	TT_FC[2:0]			Reserved			SRC_MSIZE[2]
rw	rw			ro			rw
15	14	13	12	11	10	9	8
SRC_MSIZE[1:0]		DEST_MSIZE[2:0]			SINC[1: 0]		DINC[1]
rw		rw			rw		rw
7	6	5	4	3	2	1	0
DINC[0]	SRC_TR_WIDTH[2:0]			DST_TR_WIDTH[2:0]			INT_EN
rw	rw			rw			rw

Figure 23-5: DMAC Control Register (DMA_CTRLx)

LLP_SRC_EN — Source Side LLP enable

1 = DMAC source side use LLI operate

0 = DMAC source side not use LLI operate

LLP_DST_EN — Destination Side LLP enable

1 = DMAC destination side use LLI operate

0 = DMAC destination side not use LLI operate

DMS — Source Side Master select

In this DMAC, these bits is fixed to 2'b00

SMS — Destination Side Master select

In this DMAC, these bits is fixed to 2'b00

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TT_FC[2:0] — Transfer Type and Flow Control

000 = memory to memory, dmac controller

001 = memory to peripheral, dmac controller

010 = peripheral to memory, dmac controller

011 = peripheral to peripheral, dmac controller

100 = peripheral to memory, peripheral controller

101 = peripheral to peripheral, source peripheral controller

110 = memory to peripheral, peripheral controller

111 = peripheral to peripheral, destination peripheral controller

SRC_MSIZE[2:0] — Source Burst Transaction Length

000 = 1

001 = 4

010 = 8

011 = 16

100 = 32

101 = 64

110 = 128

111 = 256

DEST_MSIZE[2:0] — Destination Burst Transaction Length

000 = 1

001 = 4

010 = 8

011 = 16

100 = 32

101 = 64

110 = 128

111 = 256

SINC[1:0] — DMAC Source Address Increment

00 = increment

01 = decrement

1x = no change

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DINC[1:0] — DMAC Destination Address Increment

00 = increment

01 = decrement

1x = no change

SRC_TR_WIDTH[2:0] — DMAC source transfer width

000 = 8 bits

001 = 16 bits

010 = 32 bits

011 = 64 bits

100 = 128 bits

101 = 256 bits

11x = 256 bits

DST_TR_WIDTH[2:0] — DMAC destination transfer width

000 = 8 bits

001 = 16 bits

010 = 32 bits

011 = 64 bits

100 = 128 bits

101 = 256 bits

11x = 256 bits

INT_EN — DMAC Interrupt enable bit

1 = DMAC interrupt enable

0 = DMAC interrupt disable

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23.5.2.5 DMAC Control Register Higher(DMAC_CTRL_HIGHn)

Addr. Offset: 0x001C, 0x0074, 0x00CC, 0x0124 Reset Value: 0x00000002

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				BLOCK_TS[11:8]			
ro				rw			
7	6	5	4	3	2	1	0
BLOCK_TS[7:0]							
rw							

Figure 23-6: DMAC Control Register (DMA_CTRL_HIGHn)

BLOCK_TS[11:0] — Block Transfer Size

When the DMAC is the flow controller, the user writes this field before the channel is enabled in order to indicate the block size. The number programmed into BLOCK_TS indicates the total number of single transactions to perform for every block transfer; a single transaction is mapped to a single AMBA beat.

Width: The width of the single transaction is determined by CTLx.SRC_TR_WIDTH.

Once the transfer starts, the read-back value is the total number of data items already read from the source peripheral, regardless of what is the flow controller. When the source or destination.

When the source or destination peripheral is assigned as the flow controller, then the maximum block size that can be read back may not right.

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23.5.2.6 DMAC Configuration Register x (DMA_CFGx)

Addr. Offset: 0x0040, 0x0098, 0x00F0, 0x0148 Reset Value: 0x00000C00

31	30	29	28	27	26	25	24
Reserved				MAX_ABRST[9:4]			
ro				rw			
23	22	21	20	19	18	17	16
MAX_ABRST[3:0]				Reserved			
rw				ro			
15	14	13	12	11	10	9	8
Reserved				HS_SEL_SRC	HS_SEL_DST	FIFO_EMPTY	CH_SUSP
ro				rw	rw	rw	rw
7	6	5	4	3	2	1	0
CH_PRIOR[2:0]				Reserved			
rw				ro			

Figure 23-7: DMAC Configuration Register (DMA_CFGx)

FC_MODE — Flow Control Mode

Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller.

1 = Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.

0 = Source transaction requests are serviced when they occur. Data pre-fetching is enabled.

MAX_ABRST — Maximum AMBA Burst Length

Maximum AMBA burst length that is used for DMA transfers on this channel.

A value of 0 indicates that software is not limiting the maximum AMBA burst length for DMA transfers on this channel.

HS_SEL_SRC — Source Software or Hardware Handshaking Select

This register selects which of the handshaking interfaces – hardware or software

– is active for source requests on this channel.

1 = Software handshaking interface. Hardware-initiated transaction requests are ignored. If the source peripheral is memory, then this bit is ignored.

0 = Hardware handshaking interface. Software-initiated transaction requests are ignored.

HS_SEL_DST — Destination Software or Hardware Handshaking Select

This register selects which of the handshaking interfaces – hardware or software

– is active for destination requests on this channel.

1 = Software handshaking interface. Hardware-initiated transaction requests are ignored. If the destination peripheral is memory, then this bit is ignored.

0 = Hardware handshaking interface. Software-initiated transaction requests are ignored.

FIFO_EMPTY — Indicate if there is data left in the channel FIFO

This register selects which of the handshaking interfaces – hardware or software

– is active for destination requests on this channel.

1 = Channel FIFO empty.

0 = Channel FIFO not empty.

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CH_SUSP — Channel Suspend

Suspends all DMA data transfers from the source until this bit is cleared. There is no guarantee that the current transaction will complete. Can also be used in conjunction with CFG*x*.FIFO_EMPTY to cleanly disable a channel without losing any data.

1 = Suspend DMA transfer from the source.

0 = Not suspended.

CH_PRIOR — Channel priority

A priority of 3 is the highest priority, and 0 is the lowest. This field must be programmed within 0-3. for channel 0, default value is 0, channel 1 is 1, etc.

23.5.2.7 DMAC Configuration Higher Register x (DMA_CFG_HIGHx)

Addr. Offset: 0x0040, 0x0098, 0x00F0, 0x0148 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved	DST_PER[3:0]				SRC_PER[3:1]		
ro	rw				rw		
7	6	5	4	3	2	1	0
SRC_PER[0]	Reserved						FC_MODE
rw	ro						rw

Figure 23-8: DMAC Configuration higher Register (DMA_CFG_HIGHx)

FC_MODE — Flow Control Mode

Determines when source transaction requests are serviced when the Destination Peripheral is the flow controller.

1 = Source transaction requests are not serviced until a destination transaction request occurs. In this mode, the amount of data transferred from the source is limited so that it is guaranteed to be transferred to the destination prior to block termination by the destination. Data pre-fetching is disabled.

0 = Source transaction requests are serviced when they occur. Data pre-fetching is enabled.

SRC_PER[3:0] — Assigns a hardware handshaking interface for source

For dmac1:

0000: spi1_tx_dma_request

0001: spi2_tx_dma_request

0010: ss2_tx_dma_request

0011: spi1_rx_dma_request

0100: spi2_rx_dma_request

0101: ss2_rx_dma_request

0110: qadc_dma_request

0111: mcc1_dma_request

1000: mcc2_dma_request

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1001: mcc3_dma_request

1010: sci1_tx_dma_request

1011: sci1_rx_dma_request

1100: sci3_tx_dma_request

1101: sci3_rx_dma_request

others: reserved

For dmac2:

0000: sci2_rx_dma_request

0001: sci2_tx_dma_request

0010: ssi1_rx_dma_request

0011: ssi1_tx_dma_request

0100: spi3_rx_dma_request

0101: spi3_tx_dma_request

0110: reserved

0111: reserved

1000: dac_dma_request

1001: reserved

1010: reserved

1011: reserved

1100: sddc_dma_req

others: reserved

DST_PER[3:0] — Assigns a hardware handshaking interface for destination same as SRC_PER

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23.5.2.8 DMAC Raw Status for IntTfr/RawBlock/RawSRC/RawDst/RawErr Interrupt Register (DMA_RAWTFR / DMA_RAWBLOCK / DMAC_RAWSRC / DMAC_RAWDST / DMAC_RAWERR)

Addr. Offset: 0x02C0, 0x02C8, 0x02D0, 0x02D8, 0x02E0

Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							

23	22	21	20	19	18	17	16
Reserved							
ro							

15	14	13	12	11	10	9	8
Reserved							
ro							

7	6	5	4	3	2	1	0
Reserved				RAW[3:0]			
ro				ro			

RAW[3:0] — Raw Interrupt status

Interrupt events are stored in these Raw Interrupt Status registers before masking: RawBlock, RawDstTran, RawErr, RawSrcTran, and RawTfr. Each Raw Interrupt Status register has a bit allocated per channel; for example, RawTfr[2] is the Channel 2 raw transfer complete interrupt. Each bit in these registers is cleared by writing a 1 to the corresponding location in the ClearTfr, ClearBlock, ClearSrcTran, ClearDstTran, ClearErr registers.

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23.5.2.9 DMAC Interrupt Status for STATTFR / STATBlock / STATSRC / STATDst / STATErr Register (DMA_STATTFR / DMA_STATBLOCK / DMAC_STATSRC / DMAC_STATDST / DMAC_STATERR)

Addr. Offset: 0x02E8, 0x02F0, 0x02F8, 0x0300, 0x0308 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				STATUS[3:0]			
ro				ro			

All interrupt events from all channels are stored in these Interrupt Status registers after masking: StatusBlock, StatusDstTran, StatusErr, StatusSrcTran, and StatusTfr. Each Interrupt Status register has a bit allocated per channel; for example, StatusTfr[2] is the Channel 2 status transfer complete interrupt. The contents of these registers are used to generate the interrupt signals .

23.5.2.10 DMAC Interrupt Mask Registers (DMA_MASKTFR / DMA_MASKBLOCK / DMAC_MASKSRC / DMAC_MASKDST / DMAC_MAS KERR)

Addr. Offset: 0x0310, 0x0318, 0x0320, 0x0328, 0x0330 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				INT_MASK_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				INT_MASK [3:0]			
ro				rw			

The contents of the Raw Status registers are masked with the contents of the Mask registers: MaskBlock, MaskDstTran, MaskErr, MaskSrcTran, and MaskTfr. Each Interrupt Mask register has a bit allocated per channel; for example, MaskTfr[2] is the mask bit for the Channel 2 transfer complete interrupt. When the source peripheral of DMA channel i is memory, then the source transaction complete interrupt, MaskSrcTran[i], must be masked to prevent an erroneous triggering of an interrupt on the int_combined signal.

Similarly, when the destination peripheral of DMA channel i is memory, then the destination transaction complete interrupt, MaskDstTran[i], must be masked to prevent an erroneous triggering of an interrupt on the int_combined(_n) signal. A channel INT_MASK bit will be written only if the corresponding mask write enable bit in the INT_MASK_WE field is asserted on the same AHB write transfer. This allows software to set a mask bit without performing a read-modified write operation. Writing a 1 to any bit in these registers unmasks the corresponding interrupt, thus allowing the DW_ahb_dmac to set the appropriate bit in the Status registers and interrupt port signals.

INT_MASK_WE — Interrupt Mask Write Enable

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1 = Write enable, 0 = Write disable.

INT_MASK — Interrrupt Mask

1 = unmasked, 0 = masked.

23.5.2.11 DMAC Interrupt Clear Registers (DMA_CLRTFR / DMA_CLRBLOCK / DMAC_CLRSRC / DMAC_CLRDST / DMAC_CLRERR)

Addr. Offset: 0x0338, 0x0340, 0x0348, 0x0350, 0x0358 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved				CLEAR[3:0]			
ro				wo			

Each bit in the Raw Status and Status registers is cleared on the same cycle by writing a 1 to the corresponding location in the Clear registers: ClearBlock, ClearDstTran, ClearErr, ClearSrcTran, and ClearTfr. Each Interrupt Clear register has a bit allocated per channel; for example, ClearTfr[2] is the clear bit for the Channel 2 transfer complete interrupt. Writing a 0 has no effect. These registers are not readable.

23.5.2.12 DMAC Combined Interrupt Status Register(DMA_STATE)

Addr. Offset: 0x0360 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved			ERR	DSTT	SRCT	BLOCK	TRR
ro			ro	ro	ro	ro	ro

Figure 23-9: DMAC Int Status Register(DMA_INT)

The contents of each of the five Status registers – StatusTfr, StatusBlock, StatusSrcTran, StatusDstTran, StatusErr – is ORed to produce a single bit for each interrupt type in the Combined Status register (StatusInt). This register is read-only.

ERR — Or of the contents of StatusErr register.

1 = error

0 = no error

DSTT — Or of the contents of StatusDst register.

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1 = complete

0 = not complete

SRCT — Or of the contents of StatusSrc register.

1 = complete

0 = not complete

BLOCK — Or of the contents of StatusBlock register.

1 = complete

1 = not complete

TFR — Or of the contents of StatusTfr register.

1 = all transfer completed

0 = not completed

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23.5.2.13 Source Software Transaction Request Register(DMA_SRCREQ)

Addr. Offset: 0x0368 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				SRC_REQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				SRC_REQ[3:0]			
ro				rw			

SRC_REQ_WE — source request write enable.

1 = write enable, 0 = write disable

SRC_REQ — source request.

23.5.2.14 Destination Software Transaction Request Register(DMA_DSTREQ)

Addr. Offset: 0x0370 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				DST_REQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				DST_REQ[3:0]			
ro				rw			

DST_REQ_WE — source request write enable.

1 = write enable, 0 = write disable

DST_REQ — source request.

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23.5.2.15 Single Source Transaction Request Register(DMA_SGLSRCREQ)

Addr. Offset: 0x0378 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				SRC_SGLREQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				SRC_SGLREQ[3:0]			
ro				rw			

SRC_SGLREQ_WE — single source request write enable.

1 = write enable, 0 = write disable

SRC_SGLREQ — single source request.

23.5.2.16 Single Destination Transaction Request Register(DMA_SGLDSTREQ)

Addr. Offset: 0x0380 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				DST_SGLREQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				DST_SGLREQ[3:0]			
ro				rw			

DST_SGLREQ_WE — single source request write enable.

1 = write enable, 0 = write disable

DST_SGLREQ — single source request.

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23.5.2.17 Last Source Transaction Request Register(DMA_LASTSRC)

Addr. Offset: 0x0388 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				SRC_LSTREQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				SRC_LSTREQ[3:0]			
ro				rw			

SRC_LSTREQ_WE — single source request write enable.

1 = write enable, 0 = write disable

SRC_LSTREQ — single source request.

23.5.2.18 Last Destination Transaction Request Register(DMA_LASTDST)

Addr. Offset: 0x0390 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				DST_LSTREQ_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				DST_LSTREQ[3:0]			
ro				rw			

DST_LSTREQ_WE — single source request write enable.

1 = write enable, 0 = write disable

DST_LSTREQ — single source request.

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23.5.2.19 DMA Configuration Register(DMA_CONFIG)

Addr. Offset: 0x0398 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							DMA_EN
ro							rw

DMA_EN — DMAC enable bit.

1 = dmac enable, 0 = dmac disable

23.5.2.20 DMA Channel Enable Register(DMA_CHEN)

Addr. Offset: 0x03A0 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				CH_EN_WE[3:0]			
ro				rw			
7	6	5	4	3	2	1	0
Reserved				CH_EN[3:0]			
ro				rw			

CH_EN_WE — DMAC channel write enable bit.

1 = CH_EN write enable, 0 = CH_EN write disable

CH_EN — DMAC channel enable bit.

1 = channel enable, 0 = channel disable

The ChEnRegCH_EN bit is automatically cleared by hardware to disable the channel after the last AMBA transfer of the DMA transfer to the destination has completed. Software can therefore poll this bit to determine when this channel is free for a new DMA transfer.

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23.6 Function Descriptions

23.6.1 Single-block Mode Configuration Procedure

- Read the Channel Enable register to choose a free (disabled) channel;
- Clear any pending interrupts on the channel from the previous DMA transfer by writing to the Interrupt Clear registers: ClearTfr, ClearBlock, ClearSrcTran, ClearDstTran, and ClearErr. Reading the Interrupt Raw Status and Interrupt Status registers confirms that all interrupts have been cleared.
- Program the following channel registers:

  A. Write the starting source address in the SARx register for channel x .

  B. Write the starting destination address in the DARx register for channel x .

  C. Program CTLx and CFGx . Program the LLPx register with 0.

  D. Write the control information for the DMA transfer in the CTLx register for channel x . For example, in the register, you can program the following:

  1. Set up the transfer type (memory or non-memory peripheral for source and destination) and flow control device by programming the TT_FC of the CTLx register.
  2. Set up the transfer characteristics, such as:
     - Transfer width for the source in the SRC_TR_WIDTH field.
     - Transfer width for the destination in the DST_TR_WIDTH field.
     - Source master layer in the SMS field where the source resides.
     - Destination master layer in the DMS field where the destination resides.
     - Incrementing/decrementing or fixed address for the source in the SINC field.
     - Incrementing/decrementing or fixed address for the destination in the DINC field.

  E. Write the channel configuration information into the CFGx register for channel x

  1. Designate the handshaking interface type (hardware or software) for the source and destination peripherals; this is not required for memory. This step requires programming the HS_SEL_SRC/HS_SEL_DST bits, respectively. Writing a 0 activates the hardware handshaking interface to handle source/destination requests. Writing a 1 activates the software handshaking interface to handle source and destination requests.
  2. If the hardware handshaking interface is activated for the source or destination peripheral, assign a handshaking interface to the source and destination peripheral; this requires programming the SRC_PER and DEST_PER bits, respectively.

- Ensure that bit 0 of the DmaCfgReg register is enabled before writing to ChEnReg.
- Source and destination request single and burst DMA transactions in order to transfer the block of data (assuming non-memory peripherals). The DW_ahb_dmac acknowledges at the completion of every transaction (burst and single) in the block and carries out the block transfer.
- Once the transfer completes, hardware sets the interrupts and disables the channel. At this time, you can respond to either the Block Complete or Transfer Complete interrupts, or poll for the transfer complete raw interrupt status register (RawTfr[n], n = channel number) until it is set by hardware, in

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order to detect when the transfer is complete. Note that if this polling is used, the software must ensure that the transfer complete interrupt is cleared by writing to the Interrupt Clear register, ClearTfr[n], before the channel is enabled.

23.6.2 Multi-Block Transfer with Linked List for Source and Linked List for Destination

- Read the Channel Enable register to choose a free (disabled) channel.
- Set up the chain of Linked List Items (otherwise known as block descriptors) in memory. Write the control information in the LLI.CTLx register location of the block descriptor for each LLI in memory for channel x. For example, in the register, you can program the following:
  - A. Set up the transfer type (memory or non-memory peripheral for source and destination) and flow control device by programming the TT_FC of the CTLx register.
  - B. Set up the transfer characteristics, such as:
    1. Transfer width for the source in the SRC_TR_WIDTH field. .
    2. Transfer width for the destination in the DST_TR_WIDTH field.
    3. Incrementing/decrementing or fixed address for the source in the SINC field.
    4. Incrementing/decrementing or fixed address for the destination in the DINC field.
- Write the channel configuration information into the CFGx register for channel x.
  - A. Designate the handshaking interface type (hardware or software) for the source and destination peripherals; this is not required for memory. This step requires programming the HS_SEL_SRC/HS_SEL_DST bits, respectively. Writing a 0 activates the hardware handshaking interface to handle source/destination requests for the specific channel. Writing a 1 activates the software handshaking interface to handle source/destination requests.
  - B. If the hardware handshaking interface is activated for the source or destination peripheral, assign the handshaking interface to the source and destination peripheral. This requires programming the SRC_PER and DEST_PER bits, respectively.
- Make sure that the LLI.CTLx register locations of all LLI entries in memory (except the last) are set .
- Make sure that the LLI.LLPx register locations of all LLI entries in memory (except the last) are non-zero and point to the base address of the next Linked List Item.
- Make sure that the LLI.SARx/LLI.DARx register locations of all LLI entries in memory point to the start source/destination block address preceding that LLI fetch.
- Clear any pending interrupts on the channel from the previous DMA transfer by writing to the Interrupt Clear registers: ClearTfr, ClearBlock, ClearSrcTran, ClearDstTran, and ClearErr. Reading the Interrupt Raw Status and Interrupt Status registers confirms that all interrupts have been cleared.
- Program the CTLx and CFGx registers.
- Program the LLPx register with LLP(0), the pointer to the first linked list item.
- Finally, enable the channel by writing a 1 to the ChEnReg.CH_EN bit; the transfer is performed.
- The DW_ahb_dmac fetches the first LLI from the location pointed to by LLPx(0).
- Source and destination request single and burst DMA transactions to transfer the block of data (assuming non-memory peripheral). The DW_ahb_dmac acknowledges at the completion of every transaction (burst and single) in the block and carries out the block transfer.
- To check the RAWTFR for the final transfer complete, RAWBLOCK for the block transfer complete. and if MASKTFR and MASKBLOCK is set, STATTFR/STATBLOCK will set and the interrupt will happen(if

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interrupt enable).

- The DW_ahb_dmac does not wait for the block interrupt to be cleared, but continues fetching the next LLI from the memory location pointed to by the current LLPx register and automatically reprograms the SARx, DARx, LLPx, and CTLx channel registers. The DMA transfer continues until the DW_ahb_dmac determines that the CTLx and LLPx registers at the end of a block transfer. The DW_ahb_dmac then knows that the previously transferred block was the last block in the DMA transfer.

23.6.3 Disabling a Channel Prior to Transfer Completion

Under normal operation, software enables a channel by writing a 1 to the channel enable register, ChEnReg.CH_EN, and hardware disables a channel on transfer completion by clearing the ChEnReg.CH_EN register bit.

The recommended way for software to disable a channel without losing data is to use the CH_SUSP bit in conjunction with the FIFO_EMPTY bit in the Channel Configuration Register (CFGx).

- If software wishes to disable a channel prior to the DMA transfer completion, then it can set the CFGx.CH_SUSP bit to tell the DW_ahb_dmac to halt all transfers from the source peripheral. Therefore, the channel FIFO receives no new data.
- Software can now poll the CFGx.FIFO_EMPTY bit until it indicates that the channel FIFO is empty.
- When CTLx.SRC_TR_WIDTH < CTLx.DST_TR_WIDTH and the CFGx.CH_SUSP bit is high, the CFGx.FIFO_EMPTY is asserted once the contents of the FIFO do not permit a single word of CTLx.DST_TR_WIDTH to be formed. However, there may still be data in the channel FIFO, but not enough to form a single transfer of CTLx.DST_TR_WIDTH. In this scenario, once the channel is disabled, the remaining data in the channel FIFO is not transferred to the destination peripheral.
- If want to continue this channel's transfer,then to remove the channel from the suspension state by writing a 0 to the CFGx.CH_SUSP register. The DMA transfer completes in the normal manner.

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24 Digital-to-analog converter (DAC)

24.1 Introduction

The DAC module is a 12-bit, voltage output digital-to-analog converter. The DAC can be configured in 8- or 12-bit mode and may be used in conjunction with the DMA controller. The data could be left- or right-aligned. An input reference voltage, Vrefh is available. The output can optionally be buffered for higher current drive.

24.2 DAC main features

- Left or right data alignment
- DMA capability
- External triggers for conversion
- Programmable internal buffer
- Input voltage reference, Vrefh
- FIFO Based operation

24.3 DAC functional description

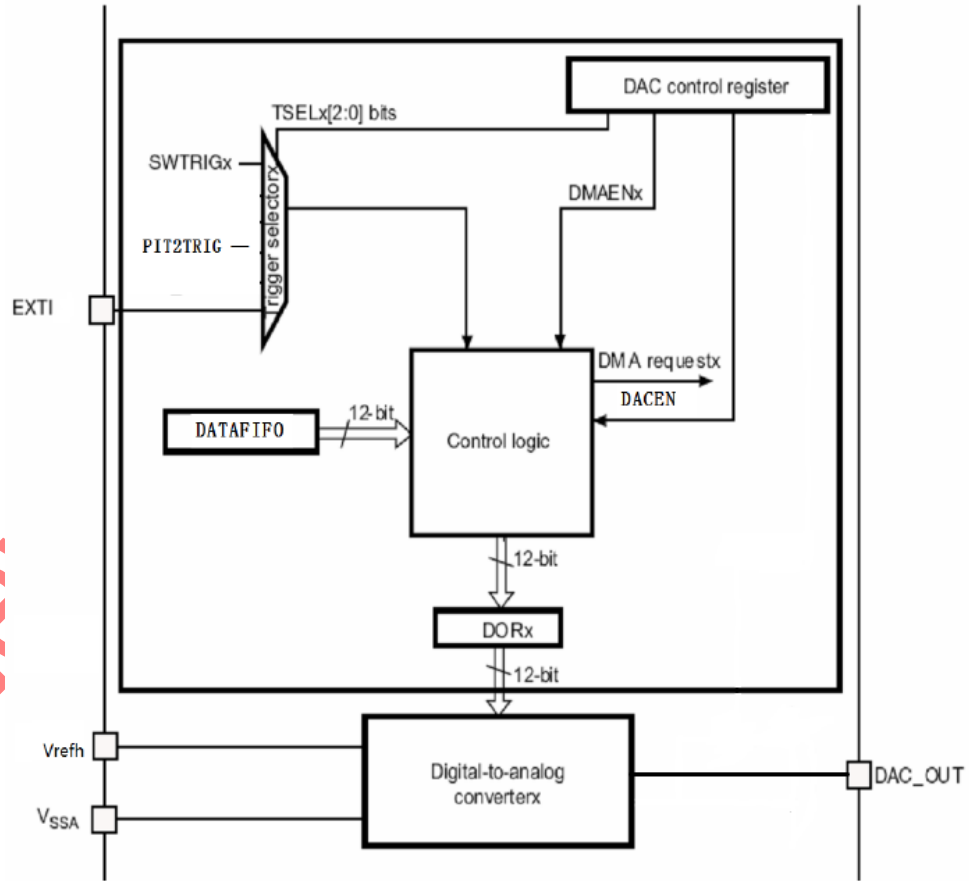


Figure 24-1: DAC block diagram

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24.3.1 DAC enable

The DAC can be powered on by setting the DACEN bit in the DAC_CR register. The DAC is then enabled after a startup time tWAKEUP.

24.3.2 DAC data format

Depending on the RES and ALIGN BIT in DAC_CR, the data written into the DAC_DR will be re-arranged and store into the DAC DATA FIFO.

Table 24-1: Data Format

CONFIG	RES	ALIGN
8-BIT, RIGHT-ALIGNED	0	0
8-BIT, LEFT-ALIGNED	0	1
12-BIT, RIGHT-ALIGNED	1	0
12-BIT, LEFT-ALIGNED	1	1

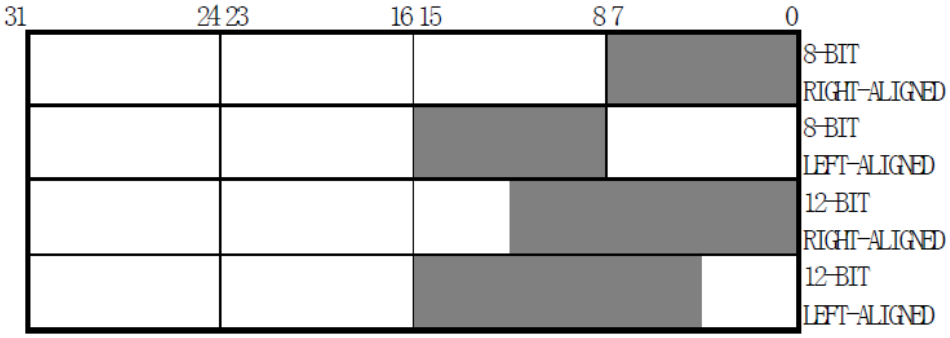


Figure 24-2: Effective Data to be store into FIFO

24.3.3 DAC conversion

The DAC_DOR cannot be written directly and any data transfer to the DAC must be performed by loading the DAC DATA FIFO (write to DAC_DR).

Data stored in the DAC DATA FIFO are automatically transferred to the DAC_DOR register when a trigger is detected.

When DAC_DOR is loaded with the DAC DATA FIFO contents, the analog output voltage becomes available after a time tSETTLING that depends on the power supply voltage and the analog output load.

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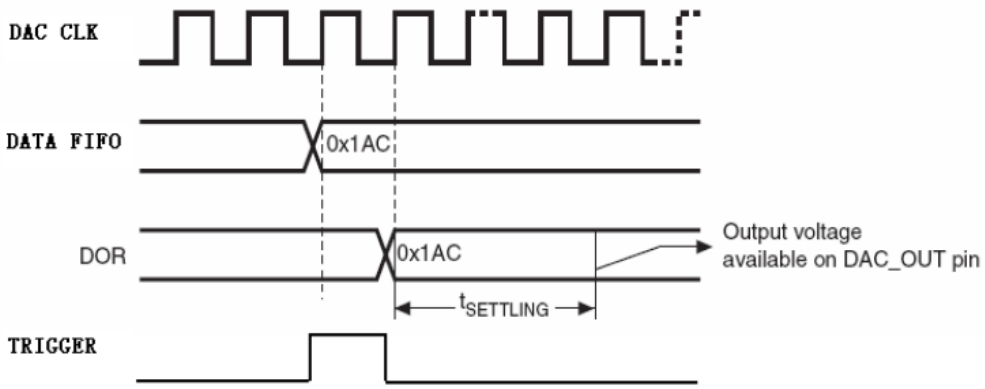


Figure 24-3: Timing Diagram

24.3.4 DAC output voltage

Digital inputs are converted to output voltages on a linear conversion between 0 and Vrefh.

The analog output voltages on each DAC channel pin are determined by the following equation:

DAC output =Vrefh*DOR/4095

24.3.5 DMA request

Once the effective data number in DAC FIFO is less than or equal to the threshold specified by DMATH when DMAEN is set, DAC will send DMA request to the DMA.

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24.4 Memory Map and Registers

This subsection describes the memory map and register structure.

24.4.1 Memory Map

Refer to **Table 24-2** for a description of the DAC memory map.

Table 24-2: DAC Memory Map

Address Offset	Bits 31-24	Bits 23-16	Bits 15-8	Bits 7-0	Access[1,2]
0x0000	DAC Control Register (DAC_CR)				S/U
0x0004	DAC DATA Register(DAC_DR)				S/U
0x0008	DAC Software Trigger Register(DAC_SWTR)				S/U
0x000C	DAC DATA Output Register(DAC_DOR)				S/U
0x0010	DAC FIFO Status Register(DAC_FSR)				S/U
0x0014	DAC Trim Register(DAC_TRIMR)				S/U
0x0018	Reserved				S/U
0x001c	Reserved				S/U

NOTES:

- S = CPU supervisor mode access only.
- User mode accesses to supervisor-only address locations have no effect and result in a cycle termination transfer error.

24.4.2 Registers

24.4.2.1 DAC control register (DAC_CR)

Addr. Offset: 0x0000~0x0003 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
FCLR	OVFIE	UDFIE	STHIE	Reserved			STH
rw	rw	rw	rw	ro			rw
23	22	21	20	19	18	17	16
DMAEN	Reserved						DMATH
rw	ro						rw
15	14	13	12	11	10	9	8
Reserved	EXT_TMOD[2:0]			Reserved	TSEL[1:0]		Reserved
ro	rw	rw	rw	ro	rw	rw	ro
7	6	5	4	3	2	1	0
Reserved				RES	ALIGN	BUFDIS	DACEN
ro				rw	rw	rw	rw

Figure 24-4: DAC control register (DAC_CR)

FCLR — DAC FIFO Clear
Write 1 to the bit will reset DAC FIFO

OVFIE — FIFO Overflow Interrupt Enable

1 = FIFO Overflow Interrupt Enabled

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0 = FIFO Overflow Interrupt Disabled

UDFIE — FIFO Underflow Interrupt Enable

1 = FIFO Underflow Interrupt Enabled

0 = FIFO Underflow Interrupt Disabled

STH — FIFO Service Threshold

Once the effective data number in DAC FIFO is less than or equal to the threshold specified, DAC FIFO Service Flag in DAC_FSR is asserted.

DMAEN — DMA Enable

1 = DMA is Enabled, 0 = DMA is Disabled

DMATH — DMA Threshold
Once the effective data number in DAC FIFO is less than or equal to the threshold when DMAEN is set, DAC will send DMA request to the DMA.

EXT_TMOD[2:0] — External Trigger Mode
These bits are used to select external trigger polarity

Table 24-3: Configure the trigger polarity

EXT_TMOD[2:0]	Trigger polarity
3'b000	Detection on rising edge
3'b001	Detection on falling edge
3'b010	Detection on both rising and falling edges
3'b011	Detection on high level voltage
3'b100	Detection on low level votage
3'b101	Reserved
3'b110	Reserved
3'b111	Reserved

TSEL[1:0] — Trigger Source Select
These bits are used to select trigger source of DAC conversion.

Table 24-4: Configure the trigger source

TSEL[1:0]	Trigger Source
2'b00	Software trigger
2'b01	External trigger[1]
2'b10	Pit trigger[2]
2'b11	Reserved

NOTES:

- External trigger means trigger on ss2 pad.
- Pit trigger means pit2 trigger.

RES — DAC Data Resolution

1 = 12bit, 0 = 8bit

ALIGN — DAC Data Align

1 = left aligned, 0 = right aligned

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BUFDIS — DAC Output Buffer Disable

1 = Buffer Disabled

0 = Buffer Enabled

DACEN — DAC Enable

1 = DAC Enabled, 0 = DAC Disabled

24.4.2.2 DAC DATA register (DAC_DR)

Addr. Offset: 0x0004~0x0007 Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
DATA[15:8]							
rw							
7	6	5	4	3	2	1	0
DATA[7:0]							
rw							

Figure 24-5: DAC DATA register (DAC_DR)

Write to DAC_DR will fill the DAC DATA FIFO. Read from DAC_DR returns 0.

24.4.2.3 DAC Software Trigger register (DAC_SWTR)

Addr. Offset: 0x0008~0x000B Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							SWTRIG
ro							rw

Figure 24-6: DAC Software Trigger register (DAC_SWTR)

SWTRIG — Software Trigger
Set this bit will trig the DAC conversion data load from DAC DATA FIFO once.

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24.4.2.4 DAC DATA Output register (DAC_DOR)

Addr. Offset: 0x000C~0x000F Reset Value: 0x00000000

31	30	29	28	27	26	25	24
Reserved							
ro							
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved				DAC_DOR[11:8]			
ro				ro			
7	6	5	4	3	2	1	0
DAC_DOR[7:0]							
ro							

Figure 24-7: DAC DATA Output register (DAC_DOR)

Read from DAC_DOR will return the data on conversion by the DAC.

24.4.2.5 DAC FIFO Status Register (DAC_FSR)

Addr. Offset: 0x0010~0x0013 Reset Value: 0x10000000

31	30	29	28	27	26	25	24
LD_DONE	OVF	UDF	SER	Reserved			FCNT[1:0]
rw	rw	rw	ro	ro			
23	22	21	20	19	18	17	16
Reserved							
ro							
15	14	13	12	11	10	9	8
Reserved							
ro							
7	6	5	4	3	2	1	0
Reserved							
ro							

Figure 24-8: DAC FIFO Status register (DAC_FSR)

LD_DONE — Load Done Flag

1 = Dac Data Load Done, 0 = Dac Data Load not Done

OVF — FIFO Overflow Flag

1 = FIFO Overflow has occurred, 0 = FIFO Overflow has not occured

UDF — FIFO Underflow Flag

1 = FIFO Underflow has occurred, 0 = FIFO Underflow has not occured

SER — FIFO Service Flag

1 = FIFO DATA number is less than or equal to the threshold

0 = FIFO DATA number is more than the threshold

FCNT[1:0] — FIFO DATA Counter

Indicates the FIFO DATA number.

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24.4.2.6 DAC Trim Register (DAC_TRIMR)

Addr. Offset: 0x0014~0x0017 Reset Value: 0x01000410

31	30	29	28	27	26	25	24
Reserved							EXT_VREF_EN
rw	rw	rw	rw	rw	rw	rw	rw
23	22	21	20	19	18	17	16
Reserved							HD_EN
rw	rw	rw	rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8
TEST_BIAS	Reserved				BIAS_TRIM[2:0]		
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
TRIM_VOS_EN	Reserved		TRIM_VOS[4:0]				
rw	rw	rw	rw	rw	rw	rw	rw

Figure 24-9: DAC Trim register (DAC_TRIMR)

NOTE: bit0-15 is not open to user, just for test.

EXT_VREF_EN — External Vref Enable

1 = Enable, 0 = Disable

HD_EN — High Driver Enable

1 = Enable, 0 = Disable

TEST_BIAS — Bias Current Output Enable

1 = Enable, 0 = Disable

BIAS_TRIM — Bias Current trim.

TRIM_VOS_EN — Voltage offset trim enable.

1 = Enable, 0 = Disable

TRIM_VOS — Voltage offset trim.

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25 Watchdog Timer Module

25.1 Introduction

The watchdog timer is a 16-bit timer used to help software recover from runaway code. The watchdog timer has a free-running down-counter (watchdog counter) that generates a reset on underflow. To prevent a reset, software must periodically restart the countdown by servicing the watchdog.

25.2 Modes of Operation

This subsection describes the operation of the watchdog timer in low-power modes and debug mode of operation.

25.2.1 Wait Mode

In wait mode with the WAIT bit set in the Watchdog Control Register (WCR), watchdog timer operation stops. In wait mode with the WAIT bit clear, the watchdog timer continues to operate normally.

25.2.2 Doze Mode

In doze mode with the DOZE bit set in WCR, watchdog timer module operation stops. In doze mode with the DOZE bit clear, the watchdog timer continues to operate normally.

25.2.3 Stop Mode

The watchdog operation stops in stop mode. When stop mode is exited, the watchdog operation continues operation from the state it was in prior to entering stop mode.

25.2.4 Debug Mode

In debug mode with the DBG bit set in WCR, watchdog timer module operation stops. In debug mode with the DBG bit clear, the watchdog timer continues to operate normally. When debug mode is exited, watchdog timer operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain.

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25.3 Block Diagram

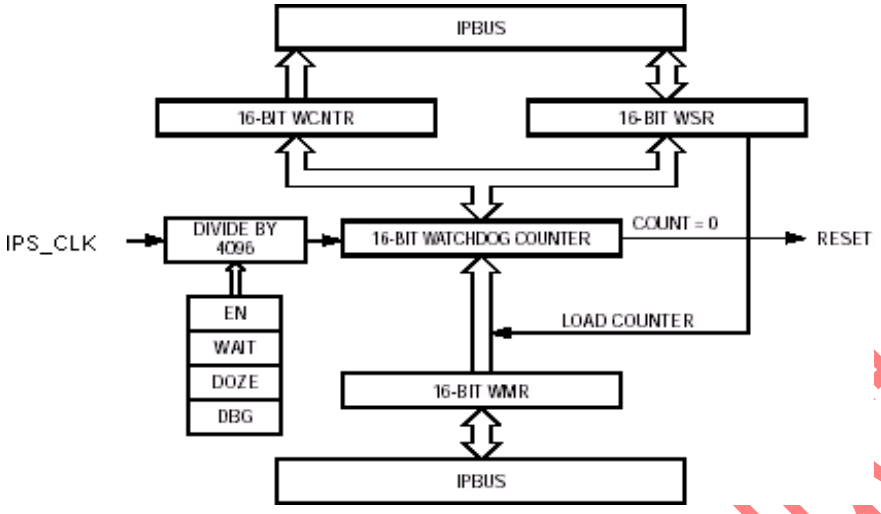


Figure 25-1: Watchdog Timer Block Diagram

25.4 Signals

The watchdog timer module has no off-chip signals.

25.5 Memory Map and Registers

This subsection describes the memory map and registers for the watchdog timer. The watchdog timer has a base address of 0x00c6_0000.

25.5.1 Memory Map

Refer **Table 25-1** to for an overview of the watchdog memory map.

Table 25-1: Watchdog Timer Module Memory Map

Address Offset	Bits 15-8	Bits 7-0	Access(1)
0x0000	Watchdog Control Register (WCR)		S
0x0002	Watchdog Modulus Register (WMR)		S
0x0004	Watchdog Count Register (WCNTR)		S/U
0x0006	Watchdog Service Register (WSR)		S/U

NOTE:

- S = CPU supervisor mode access only. S/U = CPU supervisor or user mode access. User mode accesses to supervisor only addresses have no effect and result in a cycle termination transfer error.

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25.5.2 Registers

The watchdog timer programming model consists of these registers:

- The Watchdog Control Register (WCR) configures watchdog timer operation. See 25.5.2.1 Watchdog Control Register.
- The Watchdog Modulus Register (WMR) determines the timer modulus reload value. See 25.5.2.2 Watchdog Modules Register.
- The Watchdog Count Register (WCNTR) provides visibility to the watchdog counter value. See 25.5.2.3 Watchdog Count Register.
- The Watchdog Service Register (WSR) requires a service sequence to prevent reset. See 25.5.2.4 Watchdog Service Register.

25.5.2.1 Watchdog Control Register

The 16-bit read/write Watchdog Control Register (WCR) configures watchdog timer operation.

Addr. Offset: 0x0000~0x0001 Reset Value: 0x0000

15	14	13	12	11	10	9	8
Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved	Reserved
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
Reserved	Reserved	Reserved	Reserved	WAIT	DOZE	DBG	EN
ro	ro	ro	ro	rw	rw	rw	rw

Figure 25-2: Watchdog Control Register (WCR)

WAIT — Wait Mode Bit

The read-always, write-once WAIT bit controls the function of the watchdog timer in wait mode. Once written, the WAIT bit is not affected by further writes except in debug mode. Reset sets WAIT.

1 = Watchdog timer stopped in wait mode

0 = Watchdog timer not affected in wait mode

DOZE — Doze Mode Bit

The read-always, write-once DOZE bit controls the function of the watchdog timer in doze mode. Once written, the DOZE bit is not affected by further writes except in debug mode. Reset sets DOZE.

1 = Watchdog timer stopped in doze mode

0 = Watchdog timer not affected in doze mode

DBG — Debug Mode Bit

The read-always, write-once DBG bit controls the function of the watchdog timer in debug mode. Once written, the DBG bit is not affected by further writes except in debug mode.

During debug mode, watchdog timer registers can be written and read normally. When debug mode is exited, timer operation continues from the state it was in before entering debug mode, but any updates made in debug mode remain. If a write-once register is written for the first time in debug mode, the register is still writable when debug mode is exited.

1 = Watchdog timer stopped in debug mode

0 = Watchdog timer not affected in debug mode

NOTE: Changing the DBG bit from 1 to 0 during debug mode starts the watchdog timer. Changing the DBG bit from 0 to 1 during debug mode stops the watchdog timer.

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EN — Watchdog Enable Bit

The read-always, write-once EN bit enables the watchdog timer. Once written, the EN bit is not affected by further writes except in debug mode. When the watchdog timer is disabled, the watchdog counter and prescaler counter are held in a stopped state.

1 = Watchdog timer enabled

0 = Watchdog timer disabled

25.5.2.2 Watchdog Modules Register

Addr. Offset: 0x0002~0x0003 Reset Value: 0xFFFF

15	14	13	12	11	10	9	8
WM15	WM14	WM13	WM12	WM11	WM 10	WM 9	WM8
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
WM7	WM6	WM5	WM4	WM3	WM2	WM1	WM0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 25-3: Watchdog Modulus Register (WMR)

WM[15:0] — Watchdog Modulus Field

The read-always, write-once WM[15:0] field contains the modulus that is reloaded into the watchdog counter by a service sequence. Once written, the WM[15:0] field is not affected by further writes except in debug mode. Writing to WMR immediately loads the new modulus value into the watchdog counter. The new value is also used at the next and all subsequent reloads.

Reading WMR returns the value in the modulus register. Reset initializes the WM[15:0] field to 0xFFFF.

NOTE: The prescaler counter is reset anytime a new value is loaded into the watchdog counter and also during reset.

25.5.2.3 Watchdog Count Register

Addr. Offset: 0x0004~0x0005 Reset Value: 0xFFFF

15	14	13	12	11	10	9	8
WC15	WC14	WC13	WC12	WC11	WC10	WC8	WC8
ro	ro	ro	ro	ro	ro	ro	ro
7	6	5	4	3	2	1	0
WC7	WC6	WC5	WC4	WC3	WC2	WC1	WC0
ro	ro	ro	ro	ro	ro	ro	ro

Figure 25-4: Watchdog Count Register (WCNTR)

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WC[15:0] — Watchdog Count Field

The read-only WC[15:0] field reflects the current value in the watchdog counter. Reading the 16-bit WCNTR with two 8-bit reads is not guaranteed to return a coherent value. Writing to WCNTR has no effect, and write cycles are terminated normally.

25.5.2.4 Watchdog Service Register

When the watchdog timer is enabled, writing 0x5555 and then 0xAAAA to the Watchdog Service Register (WSR) before the watchdog counter times out prevents a reset. If WSR is not serviced before the timeout, the watchdog timer sends a signal to the reset controller module which sets the WDR bit and asserts a system reset.

Both writes must occur in the order listed before the timeout, but any number of instructions can be executed between the two writes. However, writing any value other than 0x5555 or 0xAAAA to WSR resets the servicing sequence, requiring both values to be written to keep the watchdog timer from causing a reset.

Addr. Offset: 0x0006~0x0007 Reset Value: 0x0000

15	14	13	12	11	10	9	8
WS15	WS14	WS13	WS12	WS11	WS10	WS8	WS8
rw	rw	rw	rw	rw	rw	rw	rw
7	6	5	4	3	2	1	0
WS7	WS6	WS5	WS4	WS3	WS2	WS1	WS0
rw	rw	rw	rw	rw	rw	rw	rw

Figure 25-5: Watchdog Service Register (WSR)

Advance Information — INTRO

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26 Electrical Characteristic

26.1 General

This chapter provides the electrical characteristic parameters and limits for LT32U03.

26.2 Absolute Maximum Ratings

If the loading to the chip exceeds the absolute maximum rating listed in Table 26-1, it may result in permanent damage to the chip. Although the chip contains circuitry to resist damage from high quiescent voltages, do not apply more voltages on the chip than is rated in the table. These values are only ratings and do not mean that the chip functions properly under these conditions.

Table 26-1: Absolute Maximum Rating

No.	Item	Symbol	Range	Unit
1	Operation temperature	T_{OPT}	-40 ~ 85	℃
2	Store temperature	T_{STG}	-40 ~ 125	℃

26.3 Electrostatic Discharge (ESD) Protection

Table 26-2: Electrostatic Discharge Protection Characteristic

ESD target for human body model	Symbol	Max	Unit	Reference Standard
CDM	HBM	2000	V	ANSI/ESDA/JEDEC JS-001-2014
Latch Up	CDM	500	V	JEDEC EIA/JESD22-C101F
ESD target for human body model	LATCH UP	200	mA	JEDEC STANDARD NO. 78D NOVEMBER 2011

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26.4 DC Electrical Specifications

Table 26-3: IO Static Characteristic (3.3V)

Item	Symbol	Min	Typical	Max	Unit
IO Supply Power	VDD33	2.97	3.3	3.63	V
Input High Voltage	V_{IH}	2	-	VDD33	V
Input Low Voltage	V_{IL}	0	-	0.8	V
Output High Voltage	V_{OH}	2.4	-	VDD33	V
Output Low Voltage	V_{OL}	0	-	0.4	V
Input Leakage Current	I_{IN}	-	-	1	uA
Pull-Up Resistor	RPU	9	-	19.4	KΩ
Pull-Down Resistor	RPD	6.7	-	16	KΩ

Table 26-4: Power Characteristic

Item	Symbol	Min	Typical	Max	Unit
LDO Power Supply	VCC	3.0	3.3	5.5	V
Chip Power	VDD33	3.0	3.3	3.63	V
IO Power	VDDIO	3.0	3.3	3.63	V
ADC / DAC Power I/P	AVDD	3.0	3.3	3.63	V
Chip Core Power (LDO O/P)	LDO3_V12	1.1	1.2	1.3	V
EFLASH Core Power (LDO O/P)	LDO1_V11	0.99	1.1	1.21	V
USBPHY Power Input	VCC33	2.97	3.3	3.63	V
USBPHY Power (LDO O/P)	LDO2_V11	0.99	1.1	1.21	V
RTC Power	VBAT	2.7	3.3	3.63	V

Table 26-5: Power Comsumption

Item	Symbol	Min	Typical	Max	Unit
Lower Power Mode	I_{LP}	-	60	-	uA
Stop Mode Power	I_{RET}	-	8	-	uA
Deep Sleep Mode	I_{DS}	-	6	-	uA
Sleep Mode	I_{HIBER}	-	0.3	-	uA
Oeration Mode	I_{RUN}	-	15	-	mA
RTC Operation Mode	I_{RTC}	-	3	-	uA

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Table 26-6: Wake up Timing Characteristic

Item	Symbol	Min	Typical	Max	Unit
Power on Reset Time	T_{POR}	-	800	-	uS
Lower Power Mode Wake up Time	T_{LP}	-	66	-	uS
Stop Mode Wake up Time	T_{RET}	-	100	-	uS
Deep Sleep Wake up Time	T_{DS}	-	100	-	uS
Sleep Mode Wake up Time	T_{HIBER}	-	660	-	uS

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27 Package

27.1 LT32U03A (QFN-48pin)

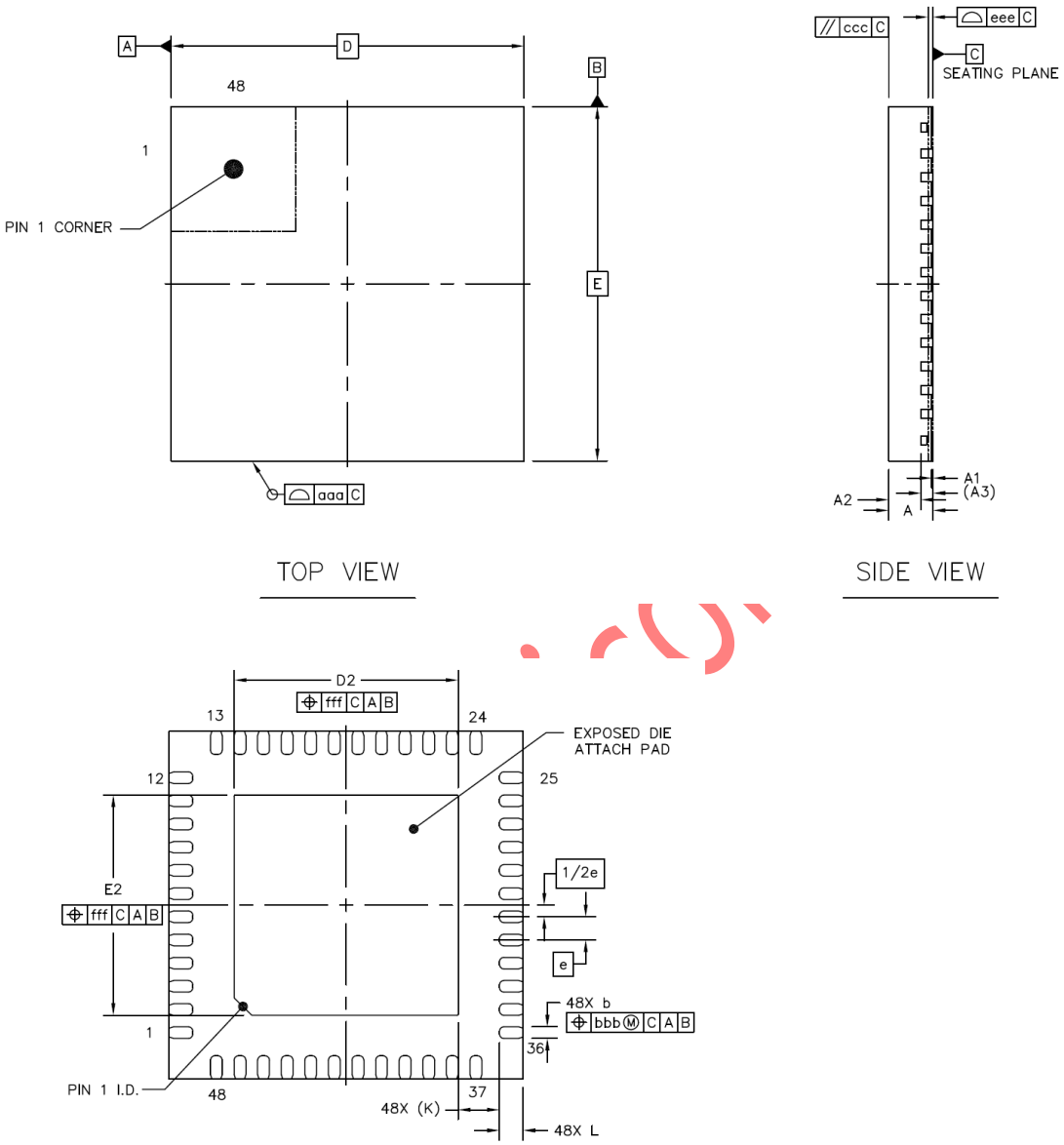


Figure 27-1: LT32U03A Package Overview

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Table 27-1: LT32U03A Package Parameter

		SYMBOL	MIN	NOM	MAX
TOTAL THICKNESS		A	0.7	0.75	0.8
STAND OFF		A1	0	0.02	0.05
MOLD THICKNESS		A2	---	0.55	---
L/F THICKNESS		A3		0.203 REF	
LEAD WIDTH		b	0.15	0.2	0.25
BODY SIZE	X	D		6 BSC	
	Y	E		6 BSC	
LEAD PITCH		e		0.4 BSC	
EP SIZE	X	D2	3.7	3.8	3.9
	Y	E2	3.7	3.8	3.9
LEAD LENGTH		L	0.3	0.4	0.5
LEAD TIP TO EXPOSED PAD EDGE		K		0.7 REF	
PACKAGE EDGE TOLERANCE		aaa		0.1	
MOLD FLATNESS		ccc		0.1	
COPLANARITY		eee		0.08	
LEAD OFFSET		bbb		0.07	
EXPOSED PAD OFFSET		fff		0.1	

NOTE: When layout PCB, LT268C Thermal Pad Zone must connect to ground.

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27.2 LT32U03B (QFN-68pin)

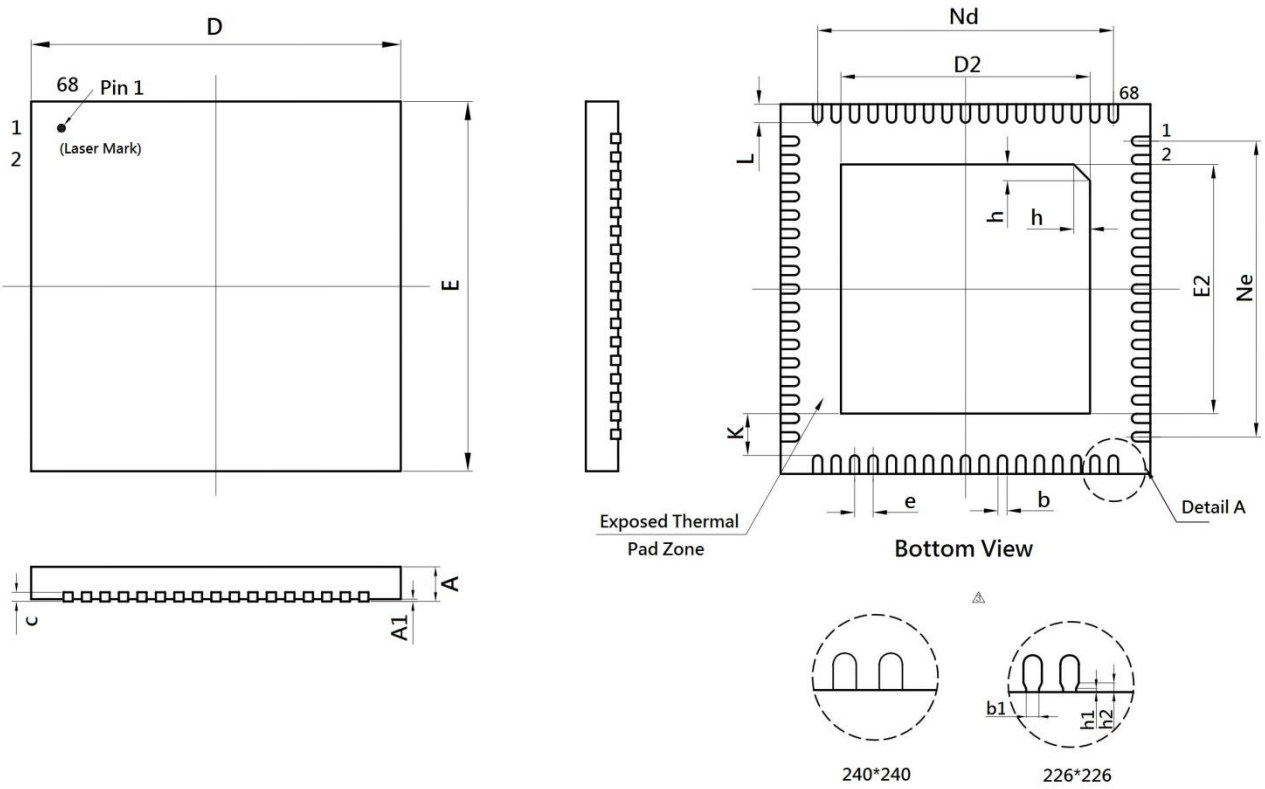


Figure 27-2: LT32U03B Package Overview

NOTE: When layout PCB, LT268C Thermal Pad Zone must connect to ground.

Table 27-2: LT32U03B Package Parameter

Symbol	Millimeter			Symbol	Millimeter		
	Min.	Nom.	Max		Min.	Nom.	Max
A	0.70	0.75	0.8	Ne	6.40BSC		
A1	-	0.02	0.05	L	0.35	0.40	0.45
b	0.15	0.20	0.25	K	0.20	-	-
b1	0.14REF			h	0.30	0.35	0.40
c	0.18	0.20	0.25	h1	0.04REF		
D	7.90	8.00	8.10	h2	0.10REF		
e	0.40BSC			D2	5.39	5.49	5.59
Nd	6.40BSC			E2	5.39	5.49	5.59
E	7.9	8.0	8.10				

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27.3 LT32U03C (LQFP-100pin)

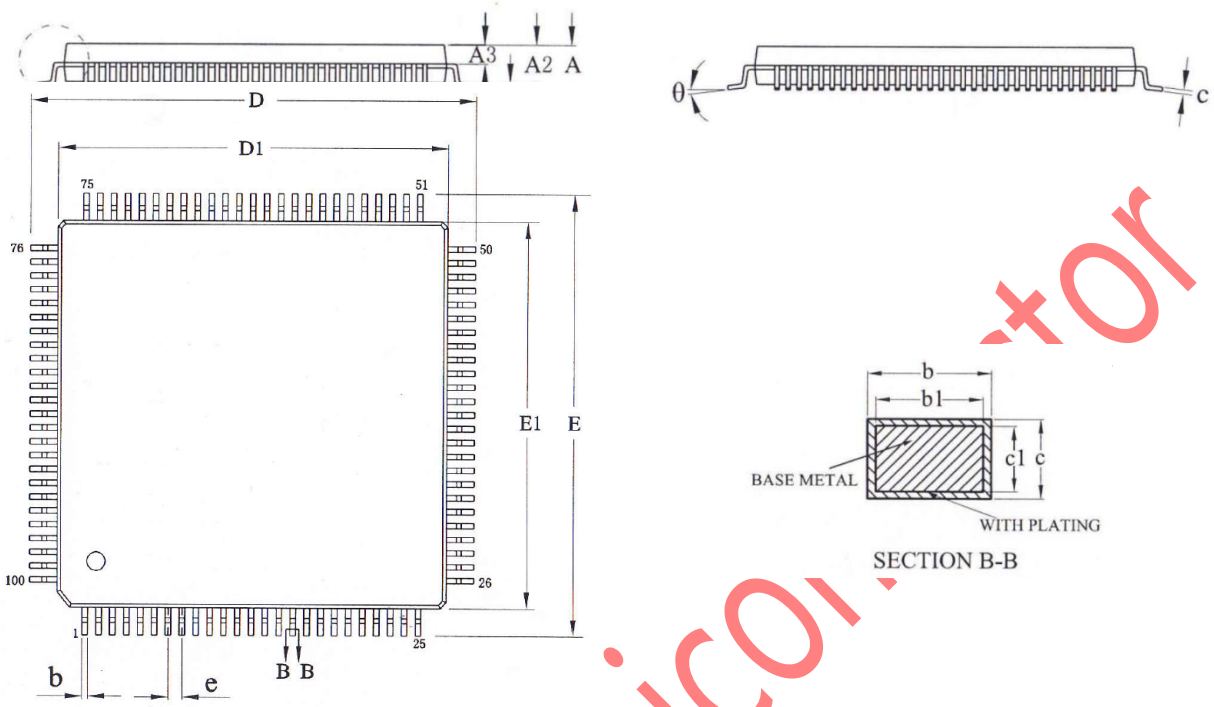


Figure 27-3: LT32U03C Package Overview

Table 27-3: LT32U03C Package Parameter

Symbol	Millimeter			Symbol	Millimeter		
	Min.	Nom.	Max		Min.	Nom.	Max
A	-	-	1.60	D1	13.9	14.0	14.1
A1	0.05	-	0.15	E	15.8	16.0	16.2
A2	1.35	1.40	1.45	E1	13.9	14.0	14.1
A3	0.59	0.64	0.69	eB	15.05	-	15.35
b	0.18	-	0.26	e	0.50BSC		
b1	0.17	0.20	0.23	L	0.45	-	0.75
c	0.13	-	0.17	L1	1.00REF		
c1	0.12	0.13	0.14	θ	0		7
D	15.8	16.00	16.2				

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28 Version History

Table 28-1: Version History

Version	Date	Version Description
V1.0	2020/8/3	LT32U03 Preliminary Release
V1.1	2022/04/02	Update Section 11.6.2.2

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